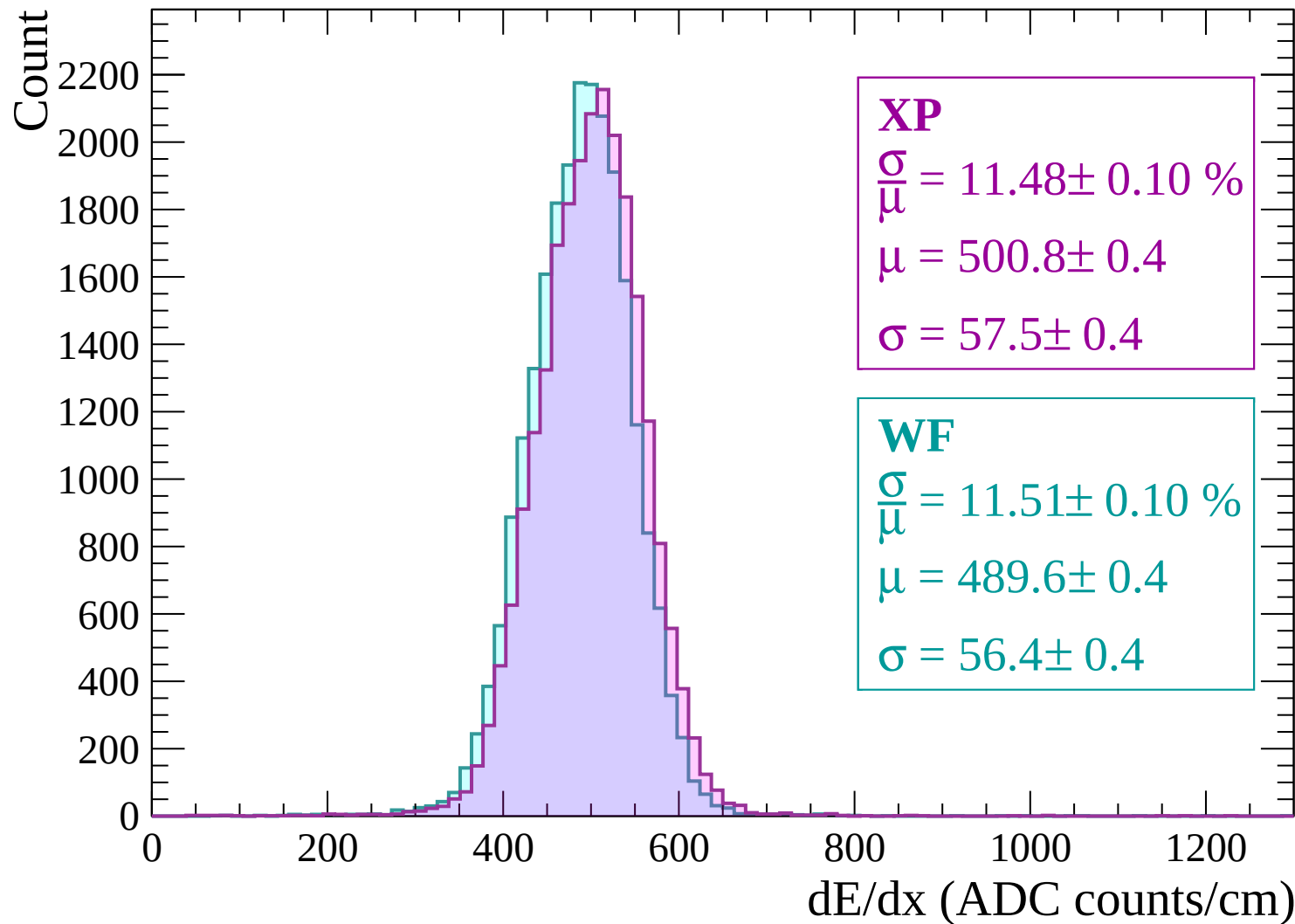
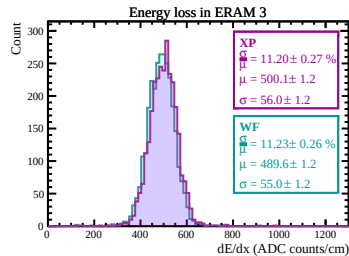
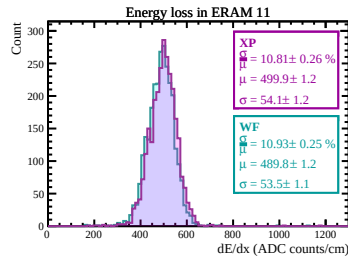
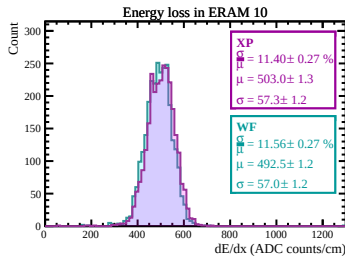
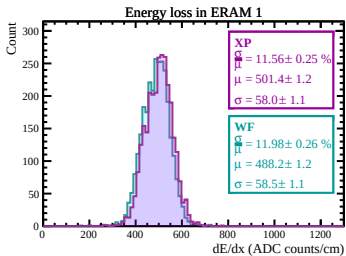
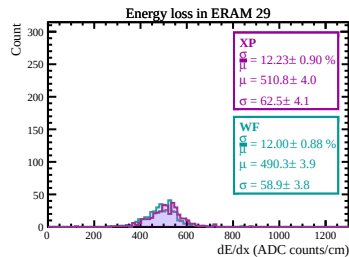
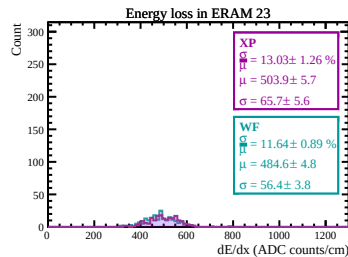
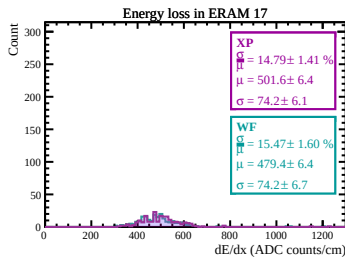
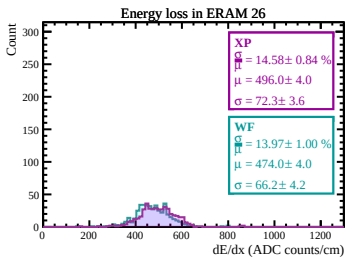
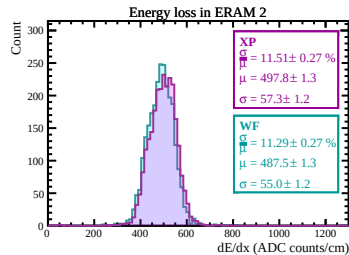
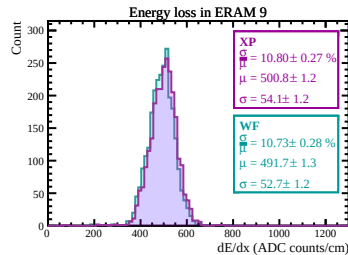
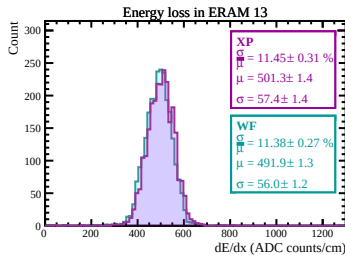
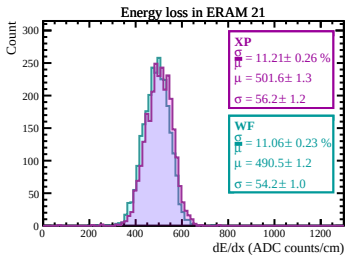
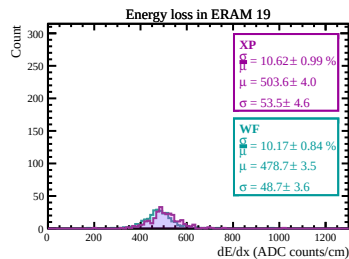
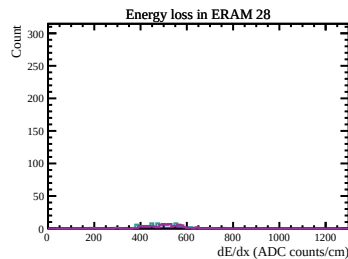
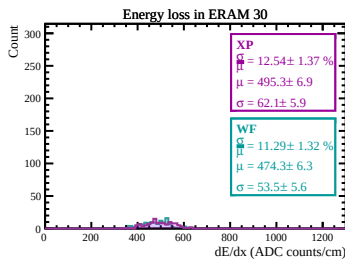
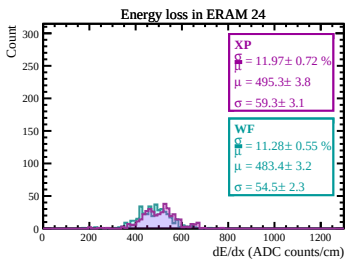
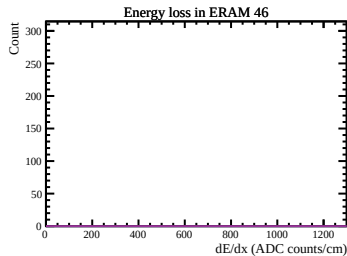
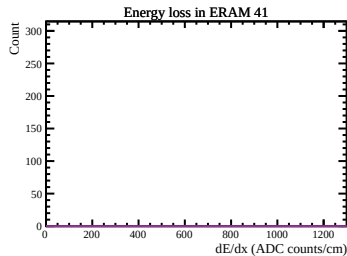
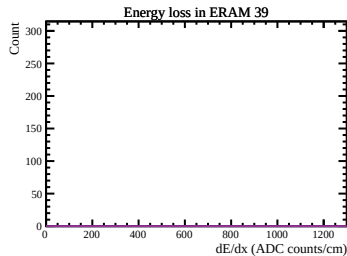
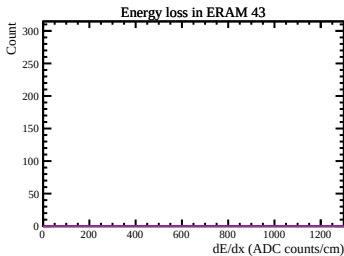
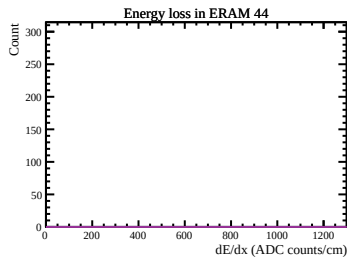
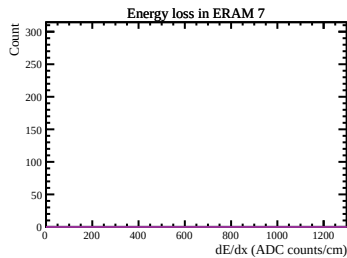
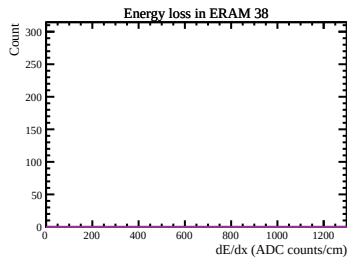
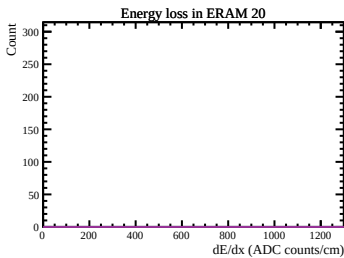
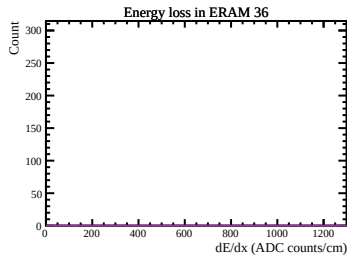
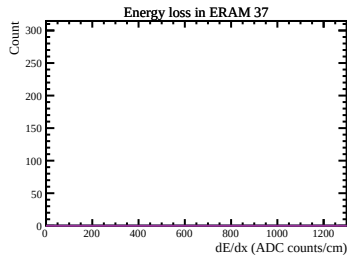
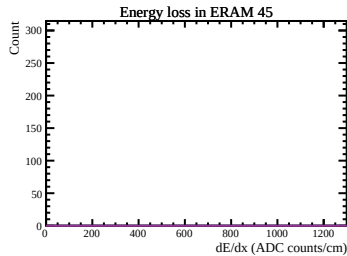
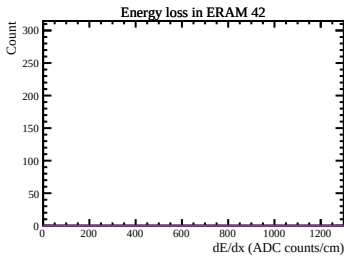
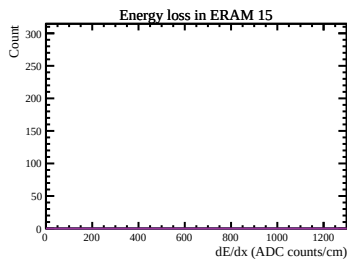
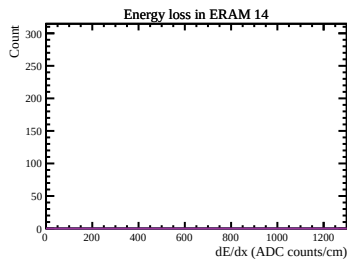
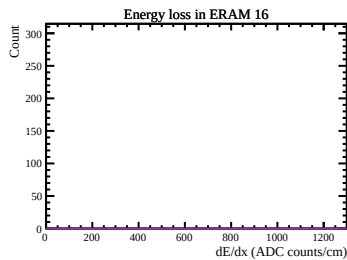
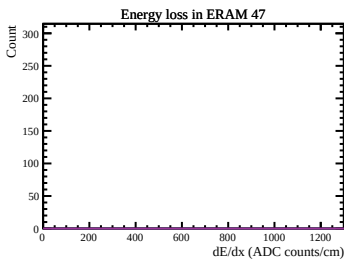
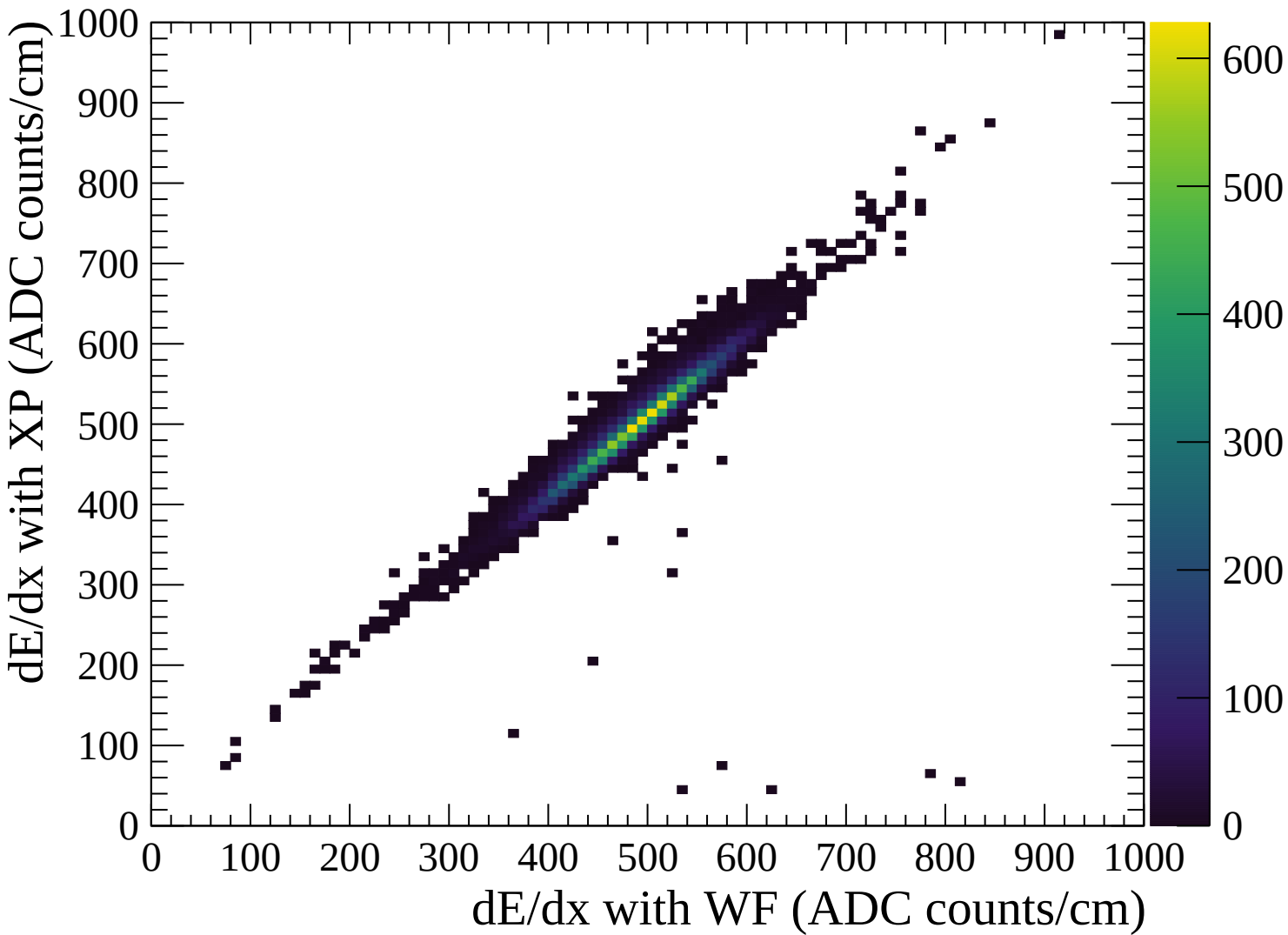
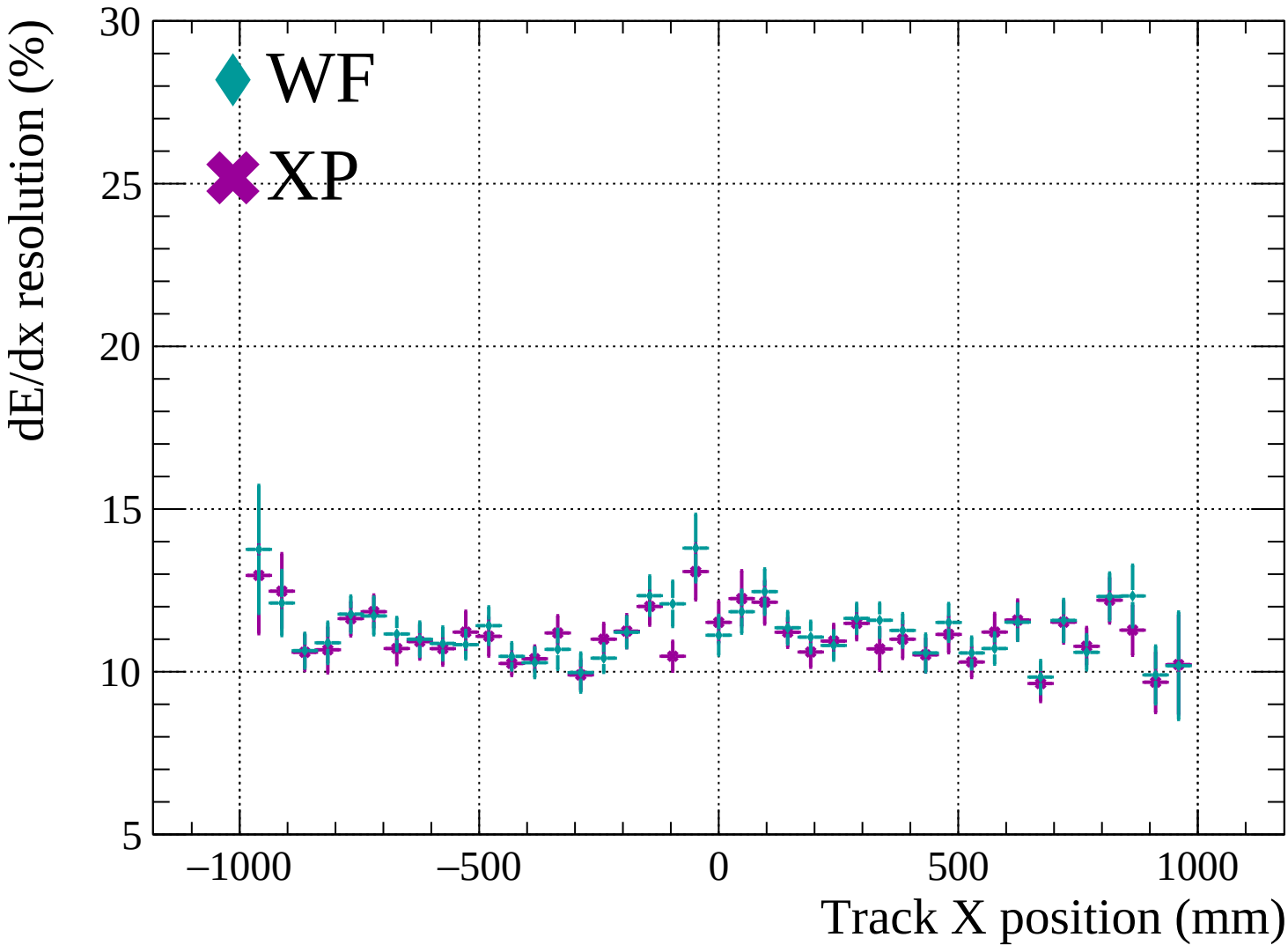


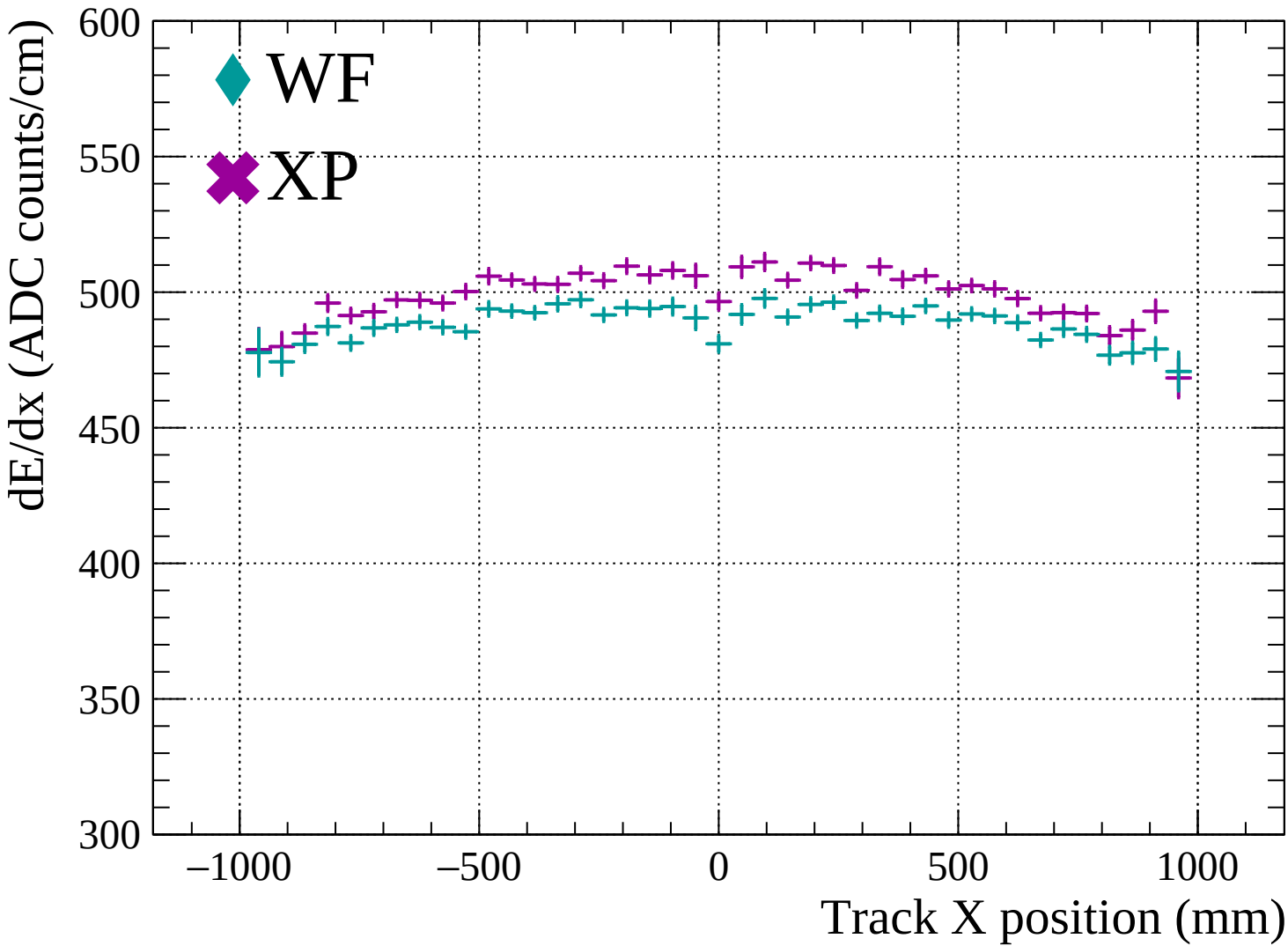


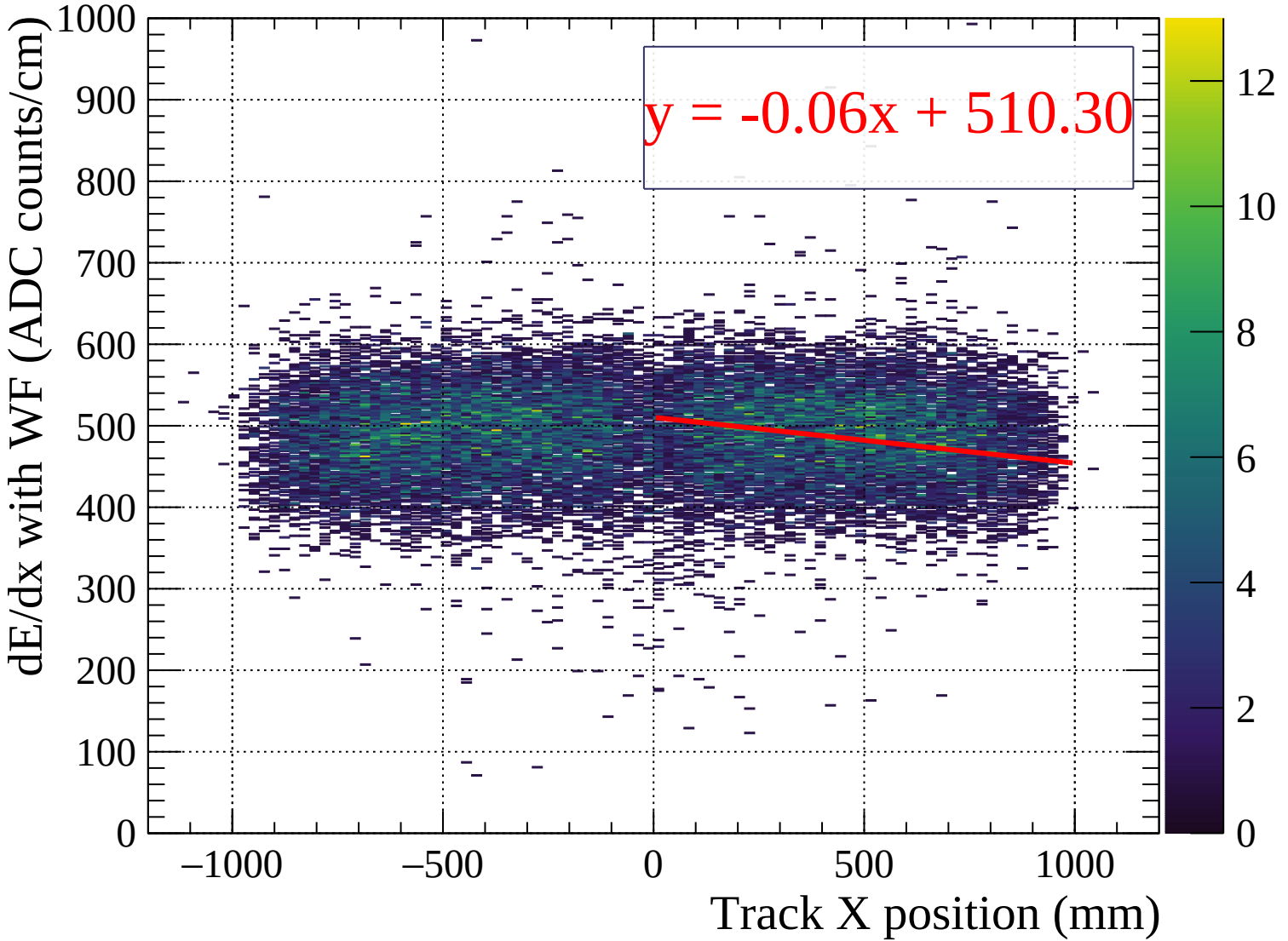


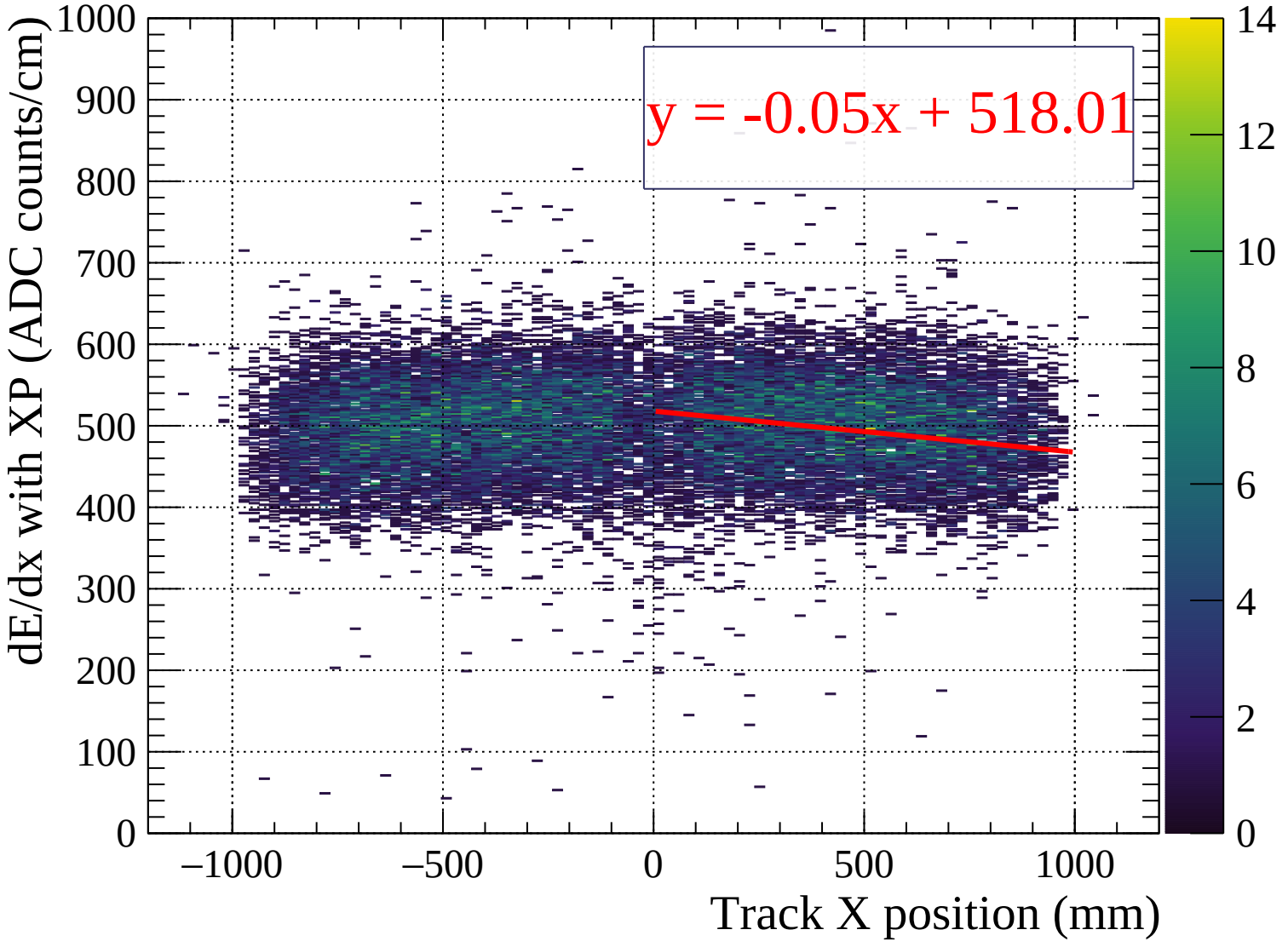


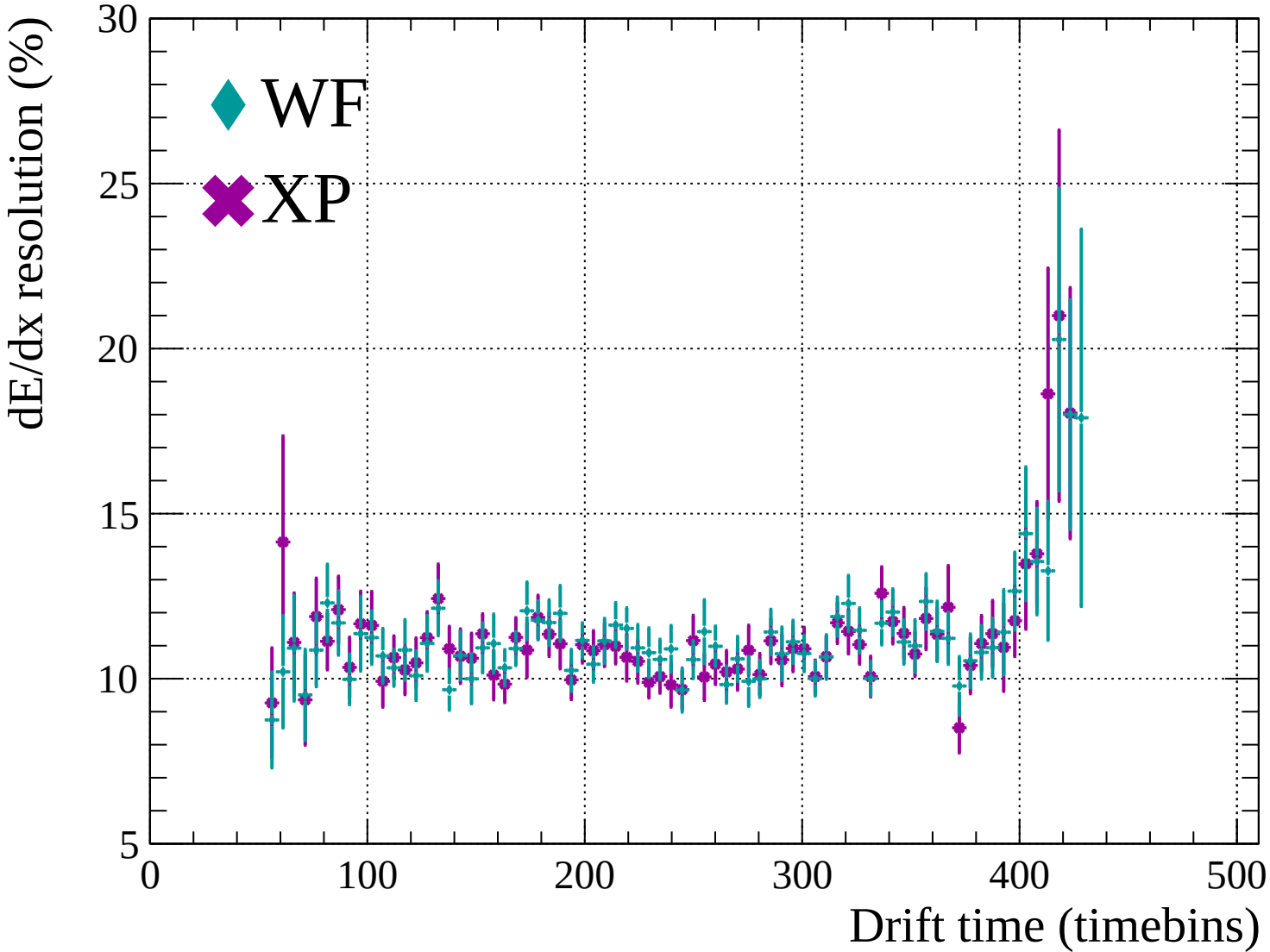


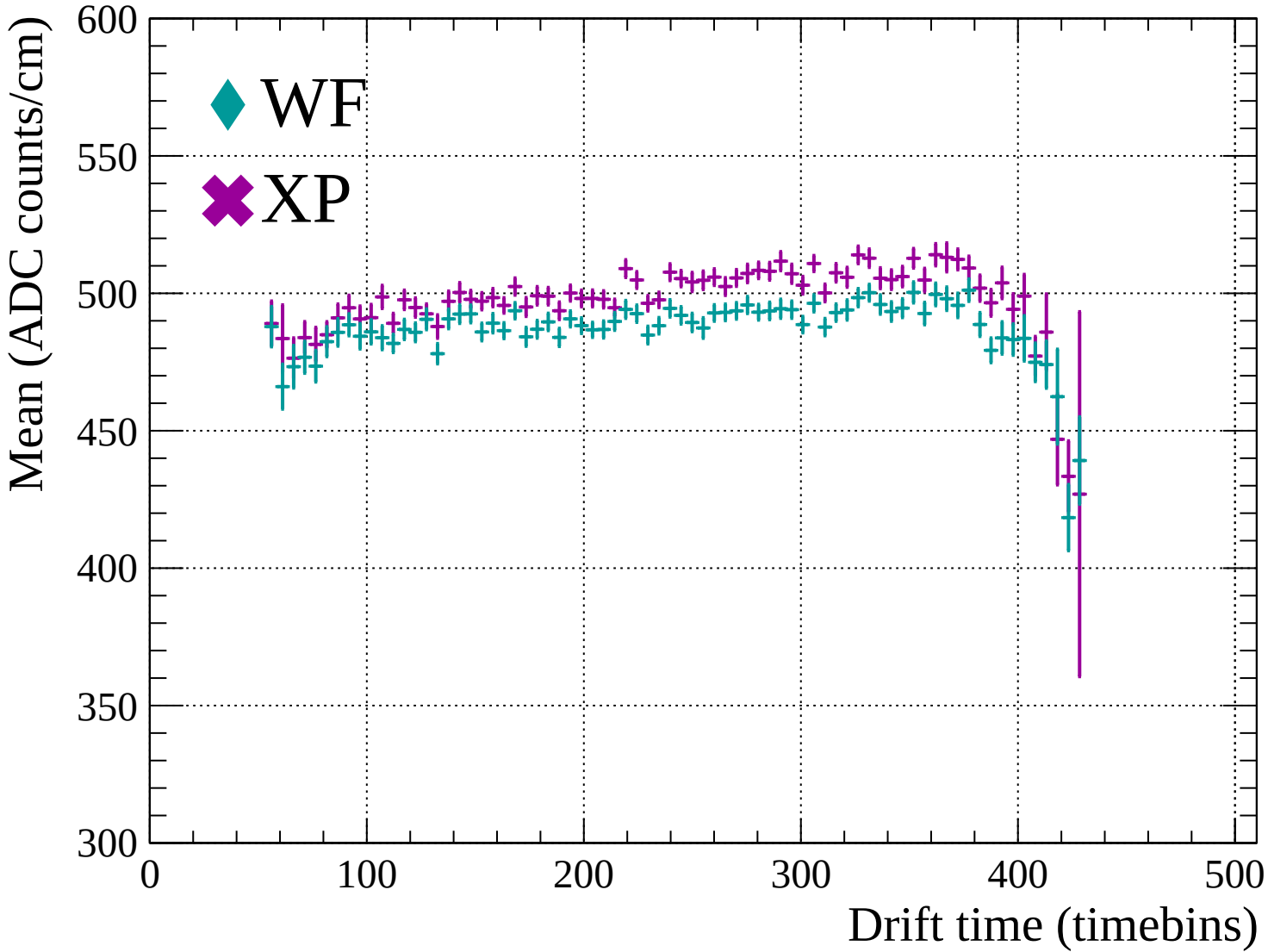


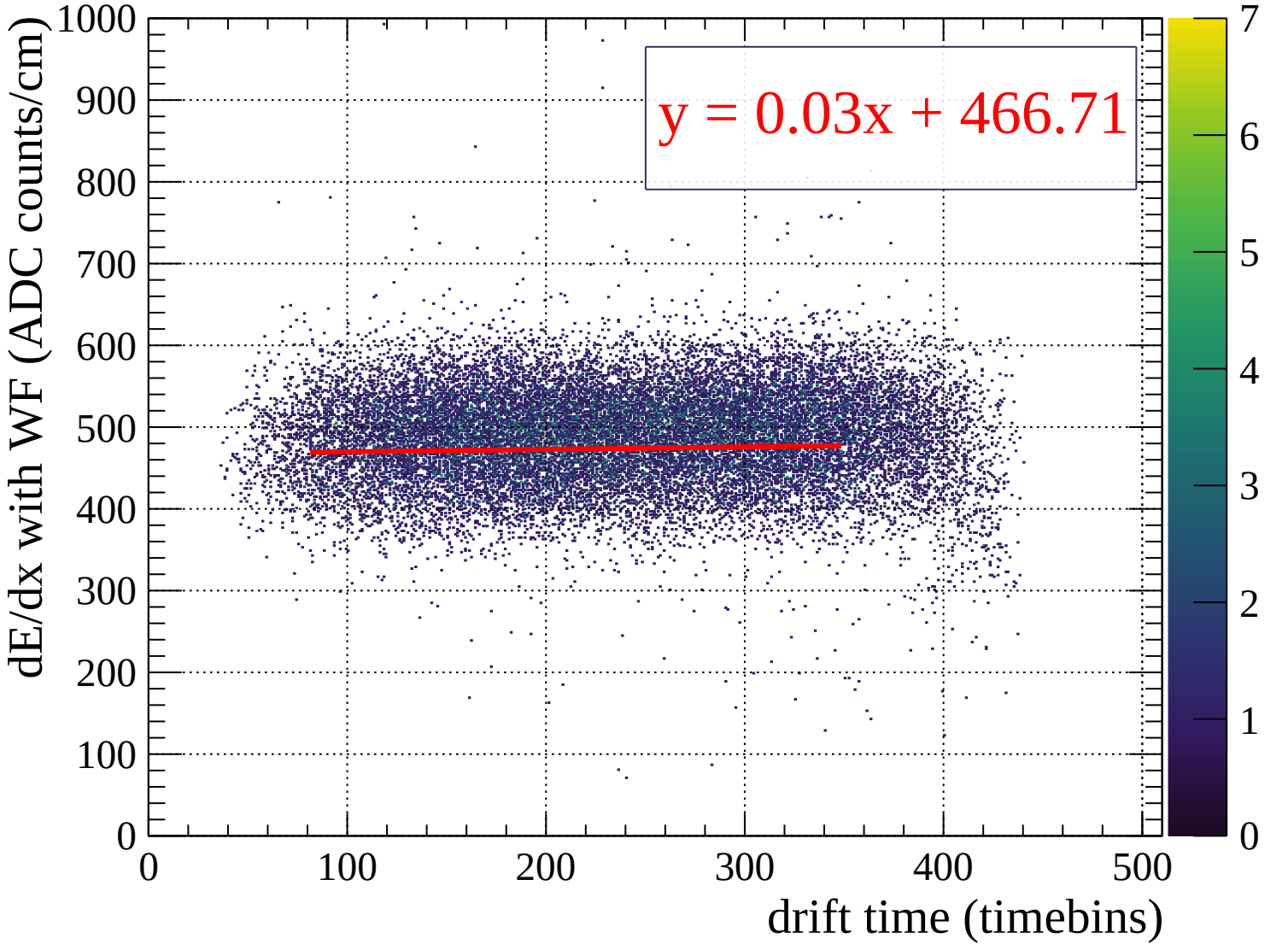


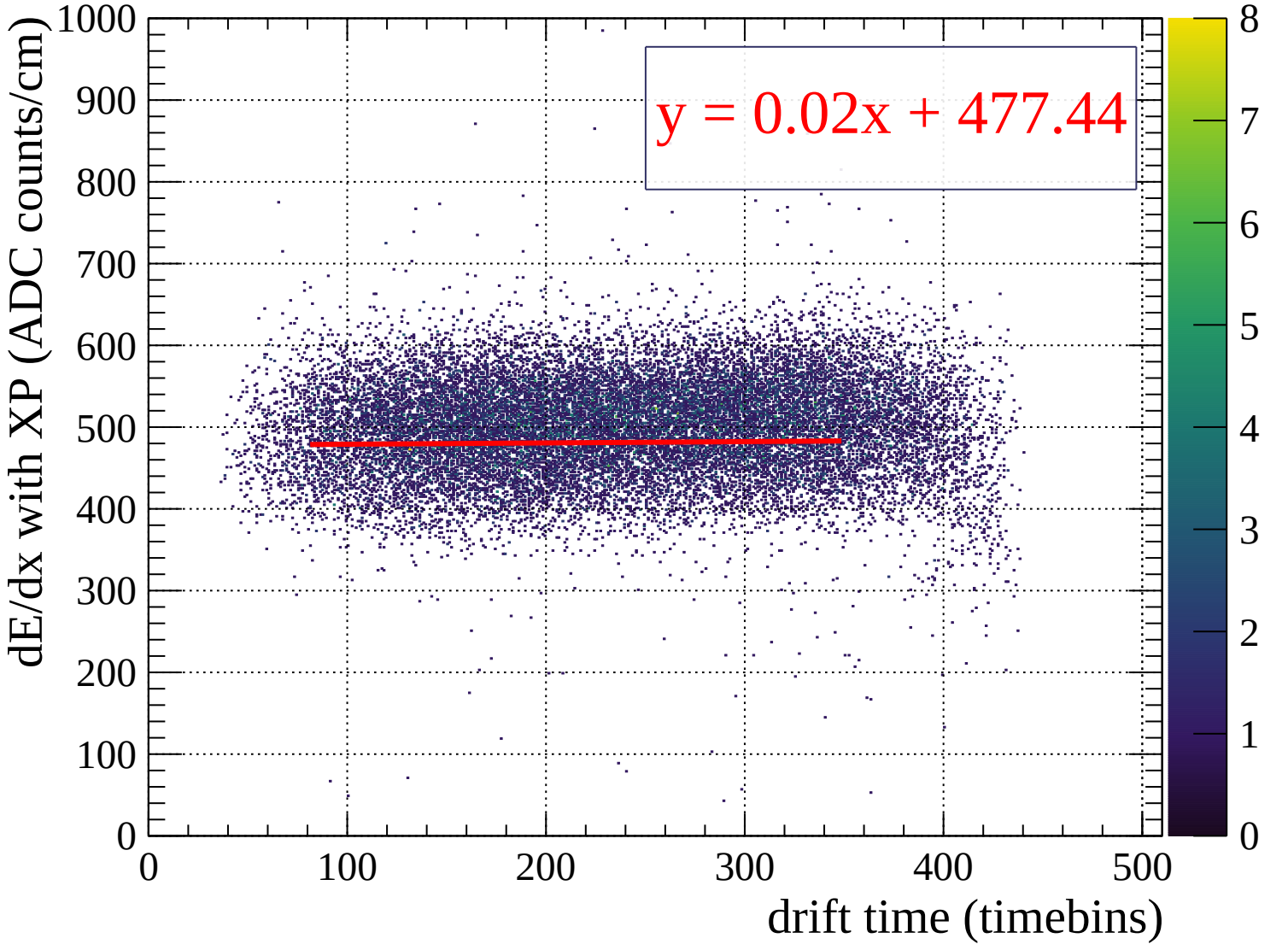


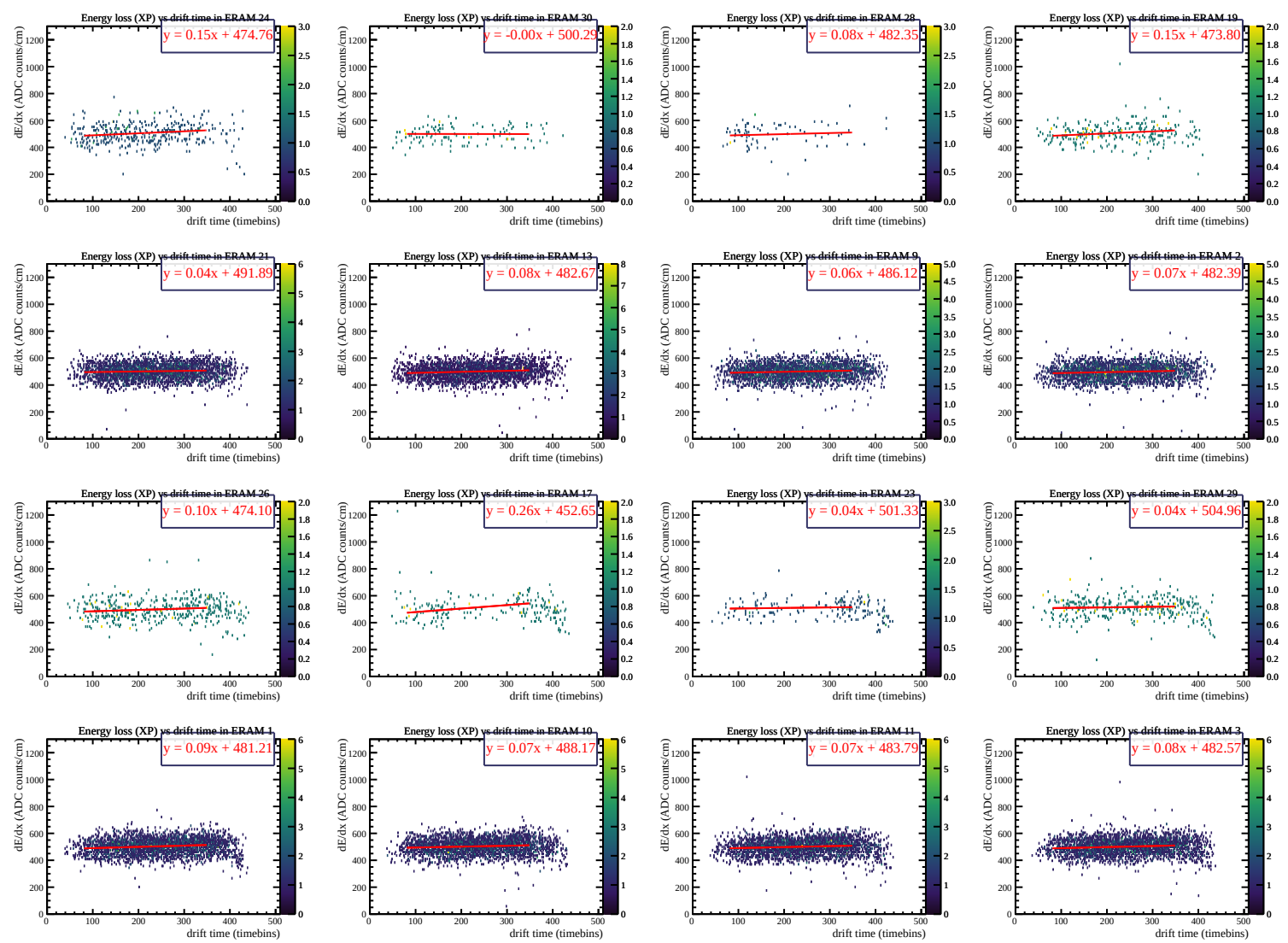


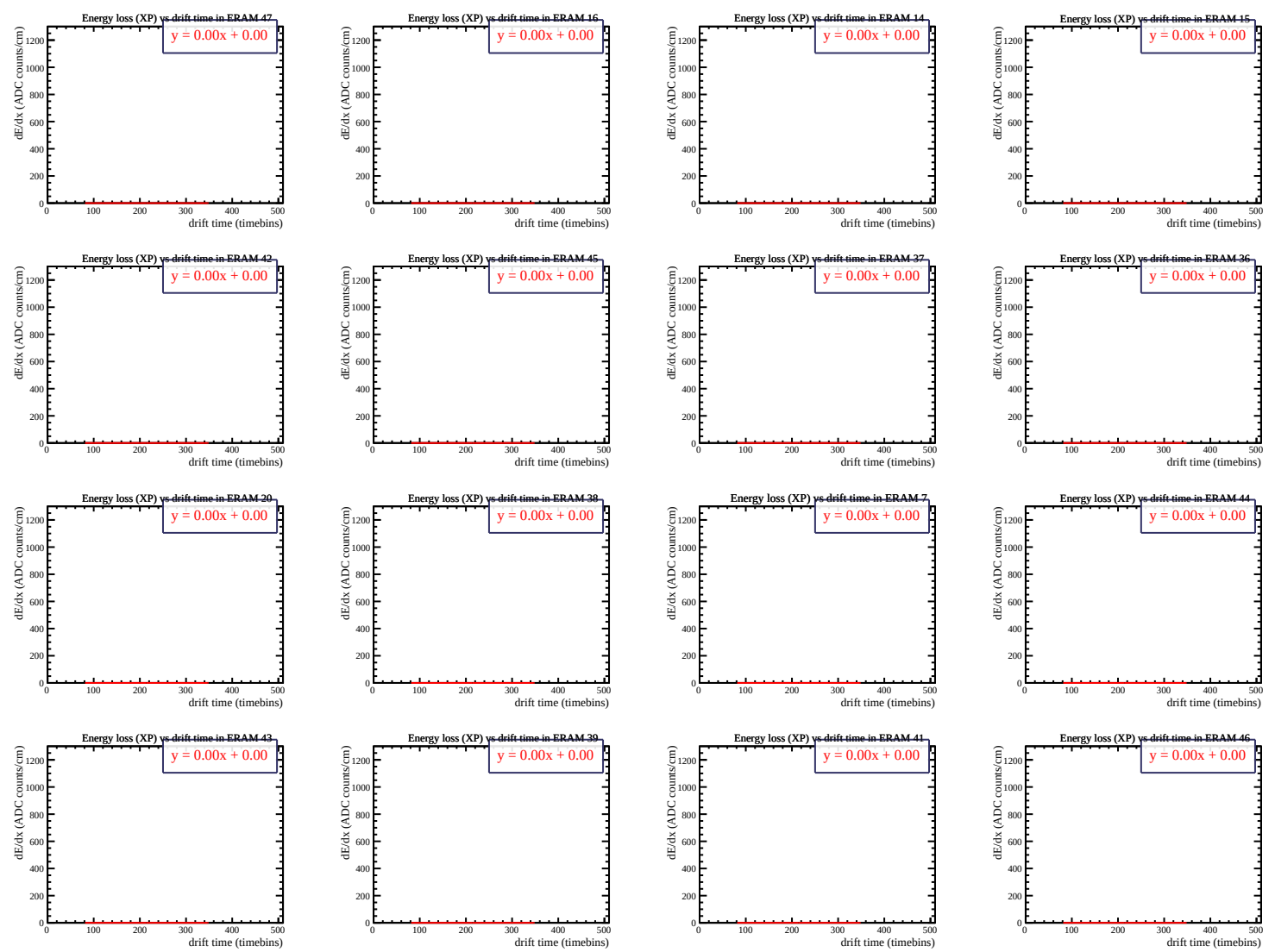


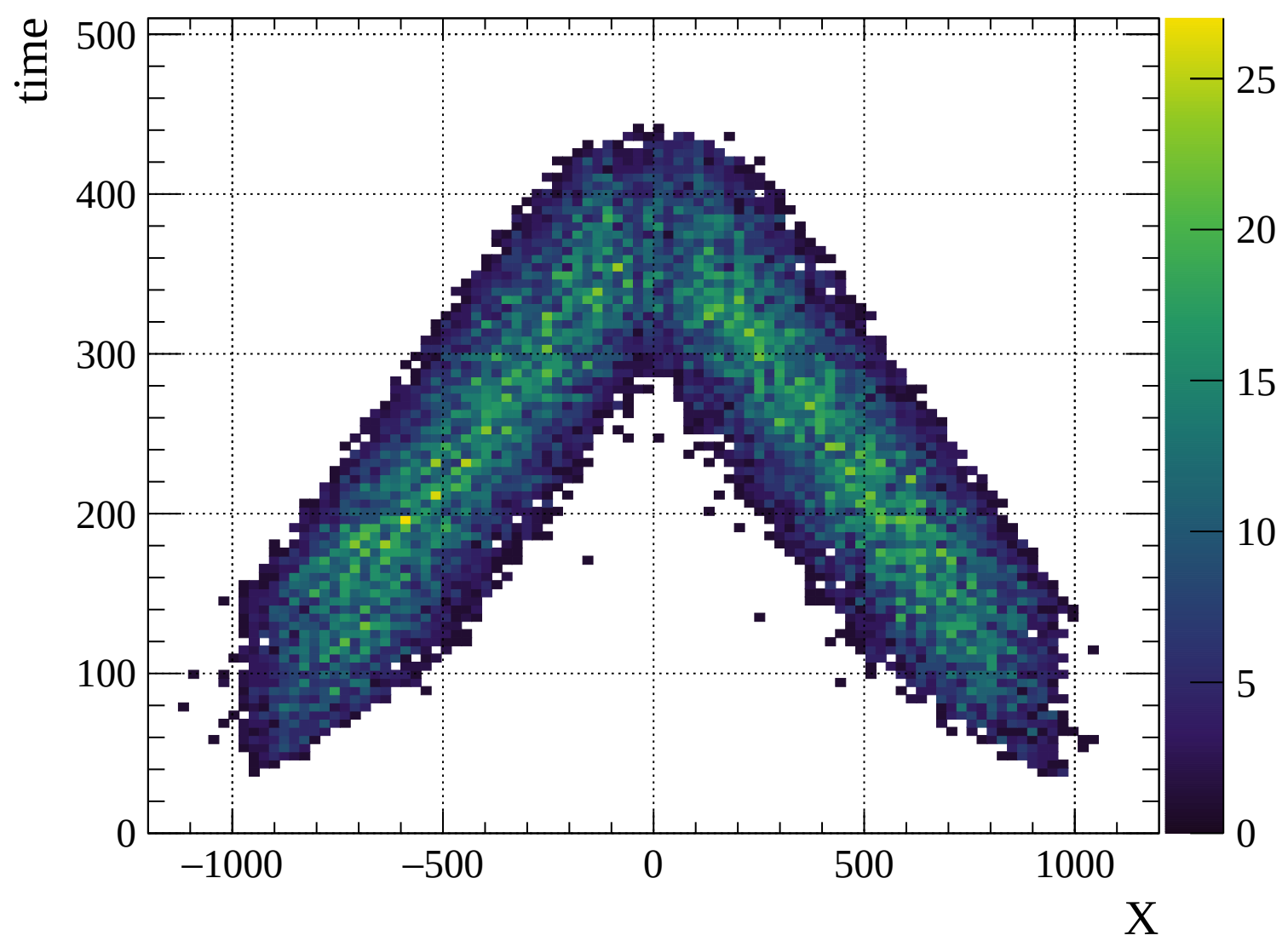


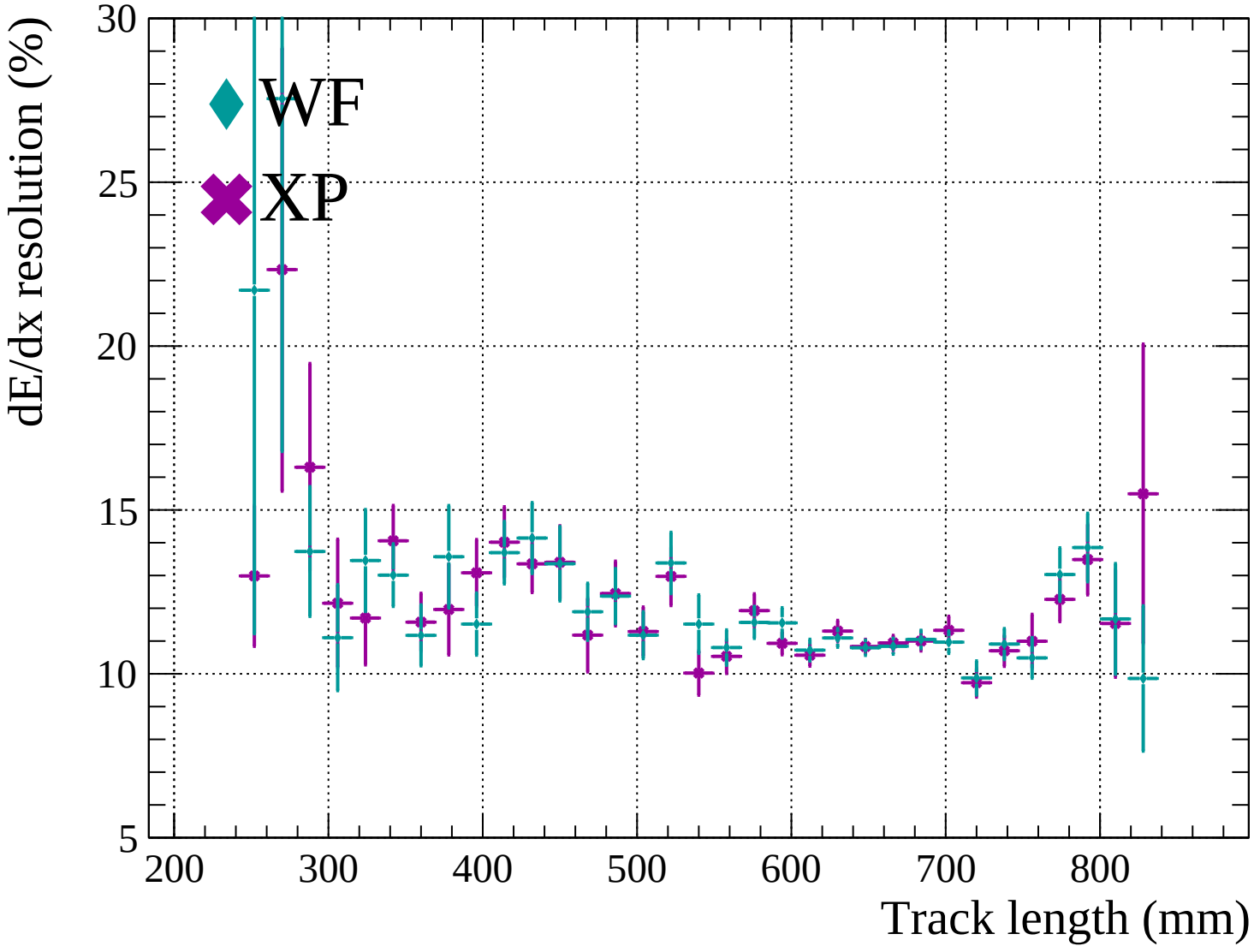


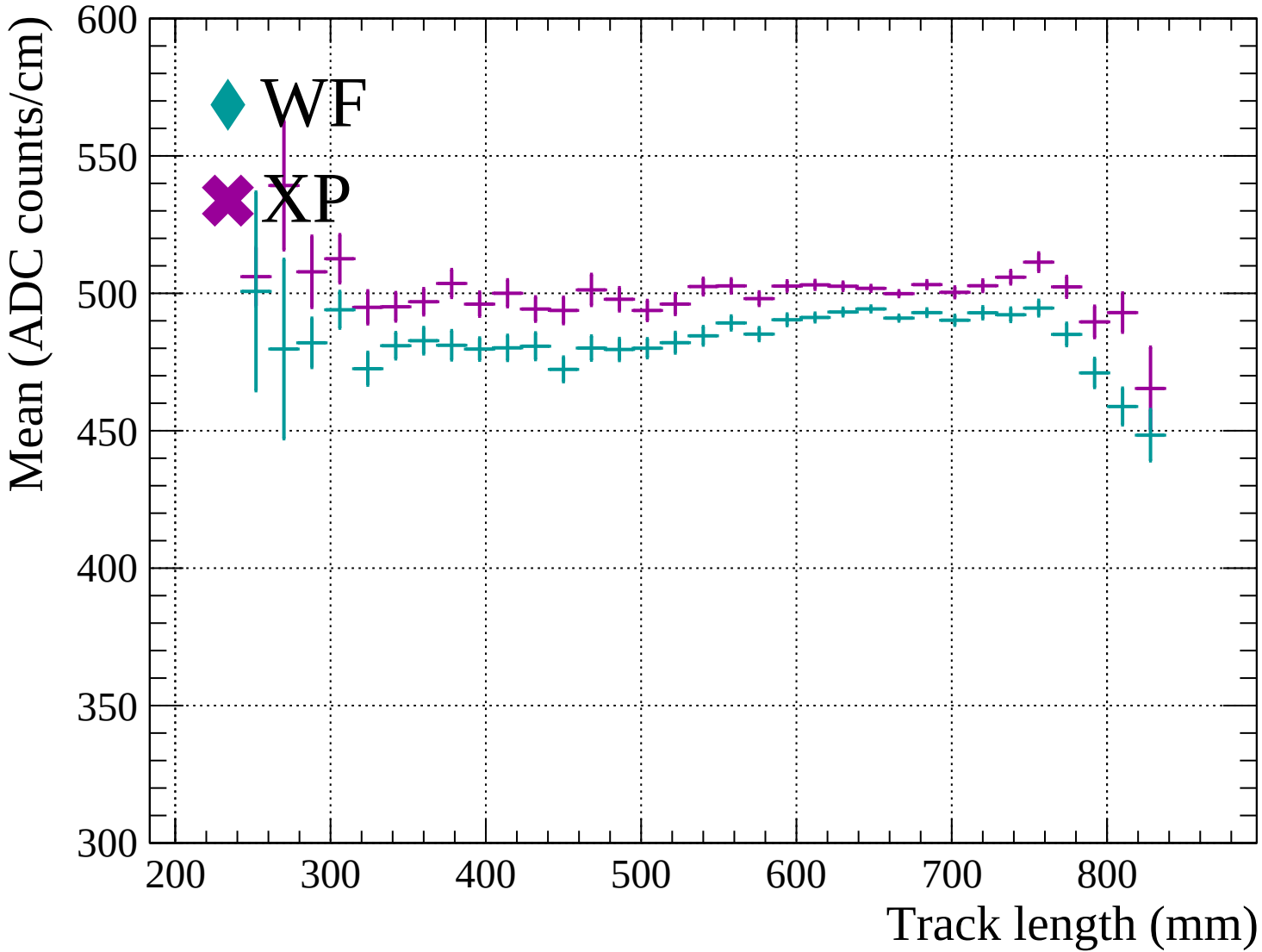


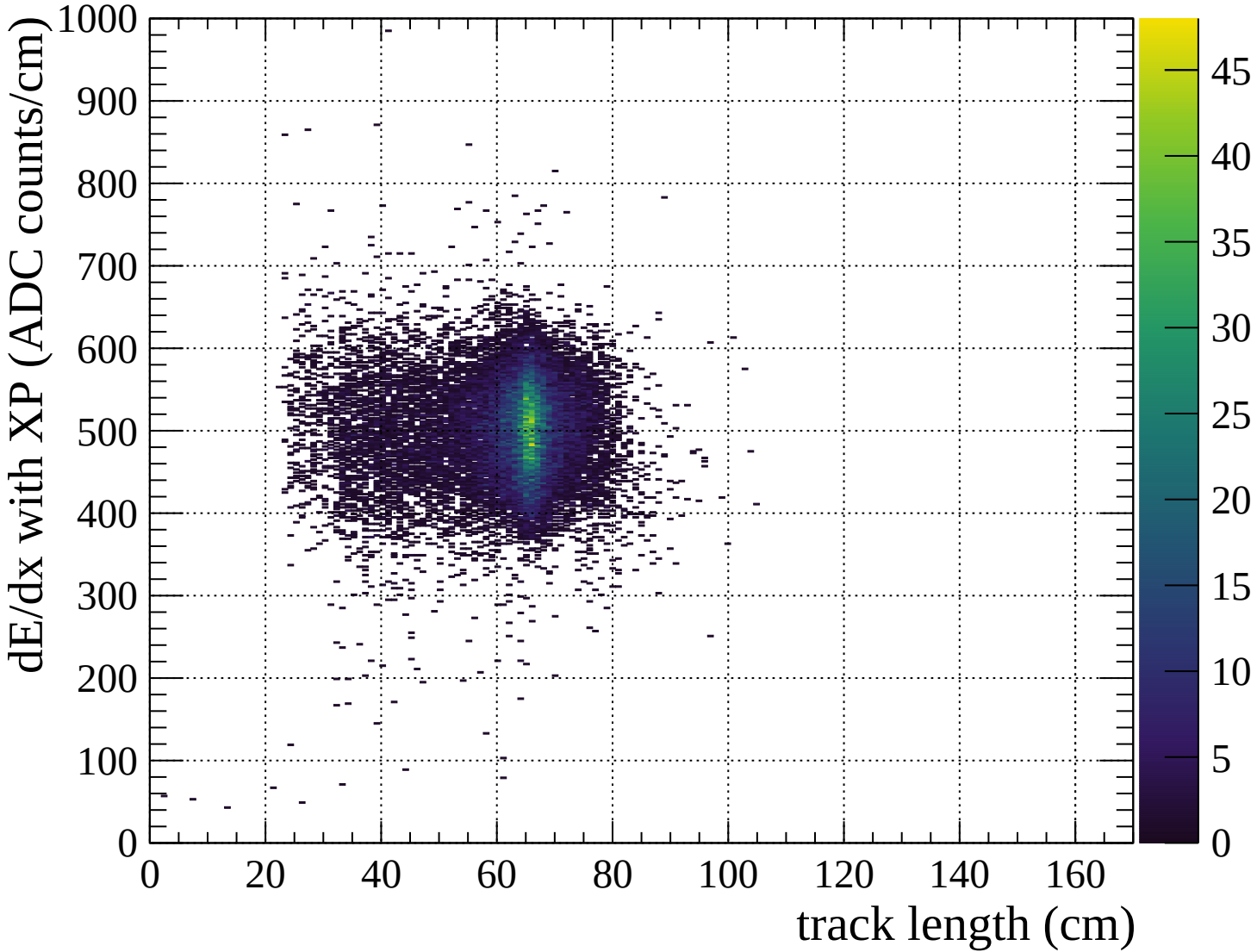


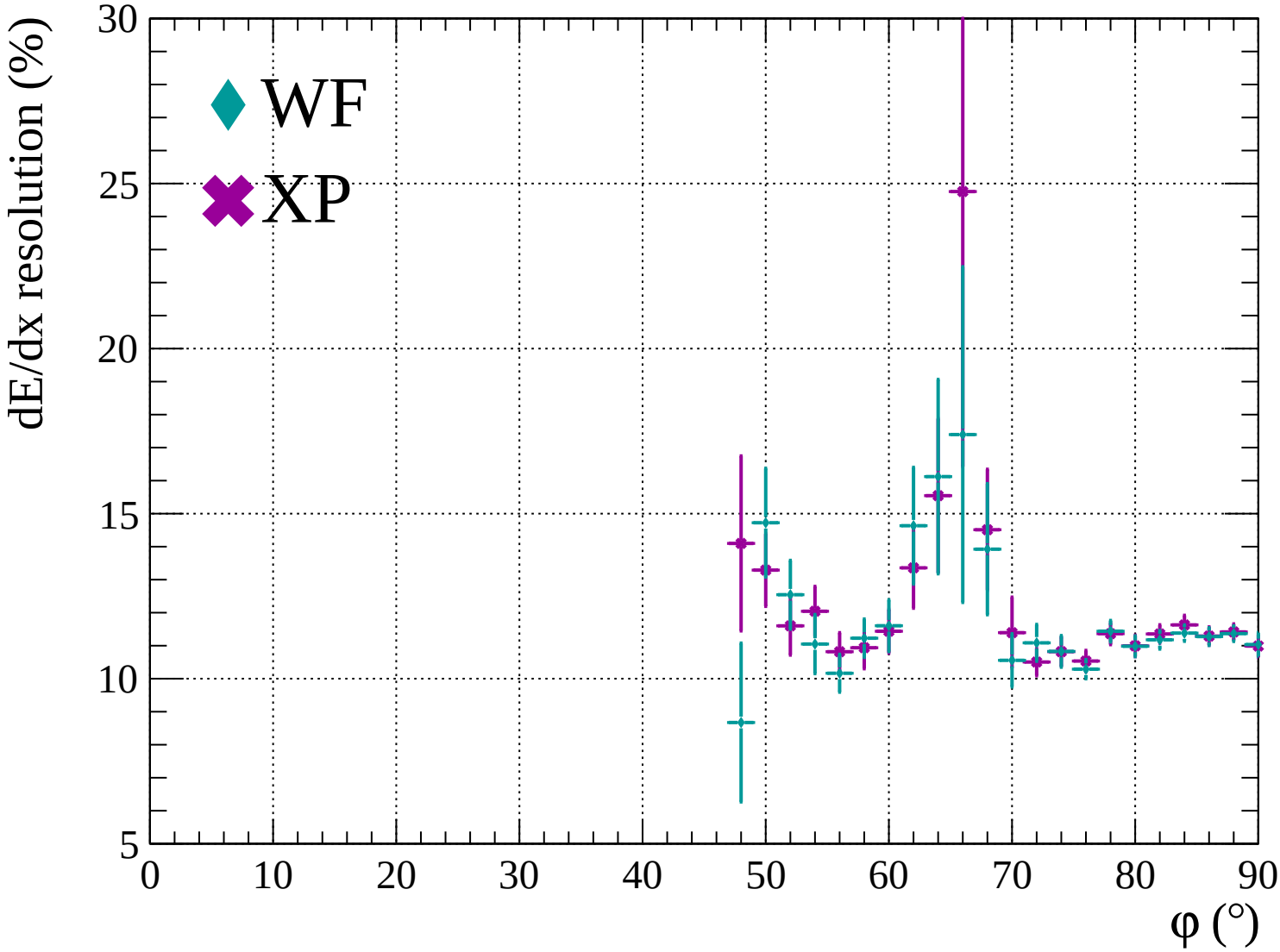


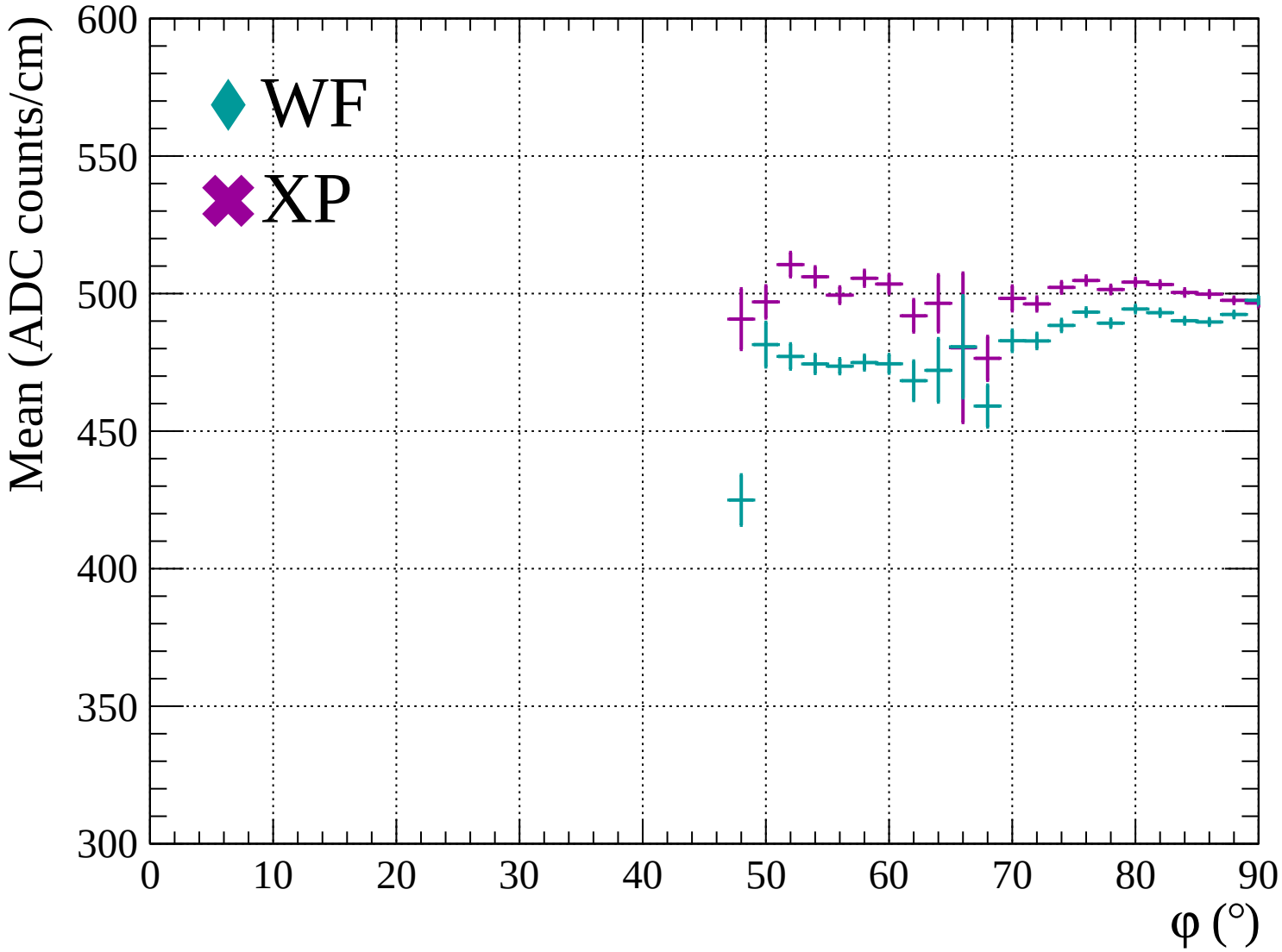


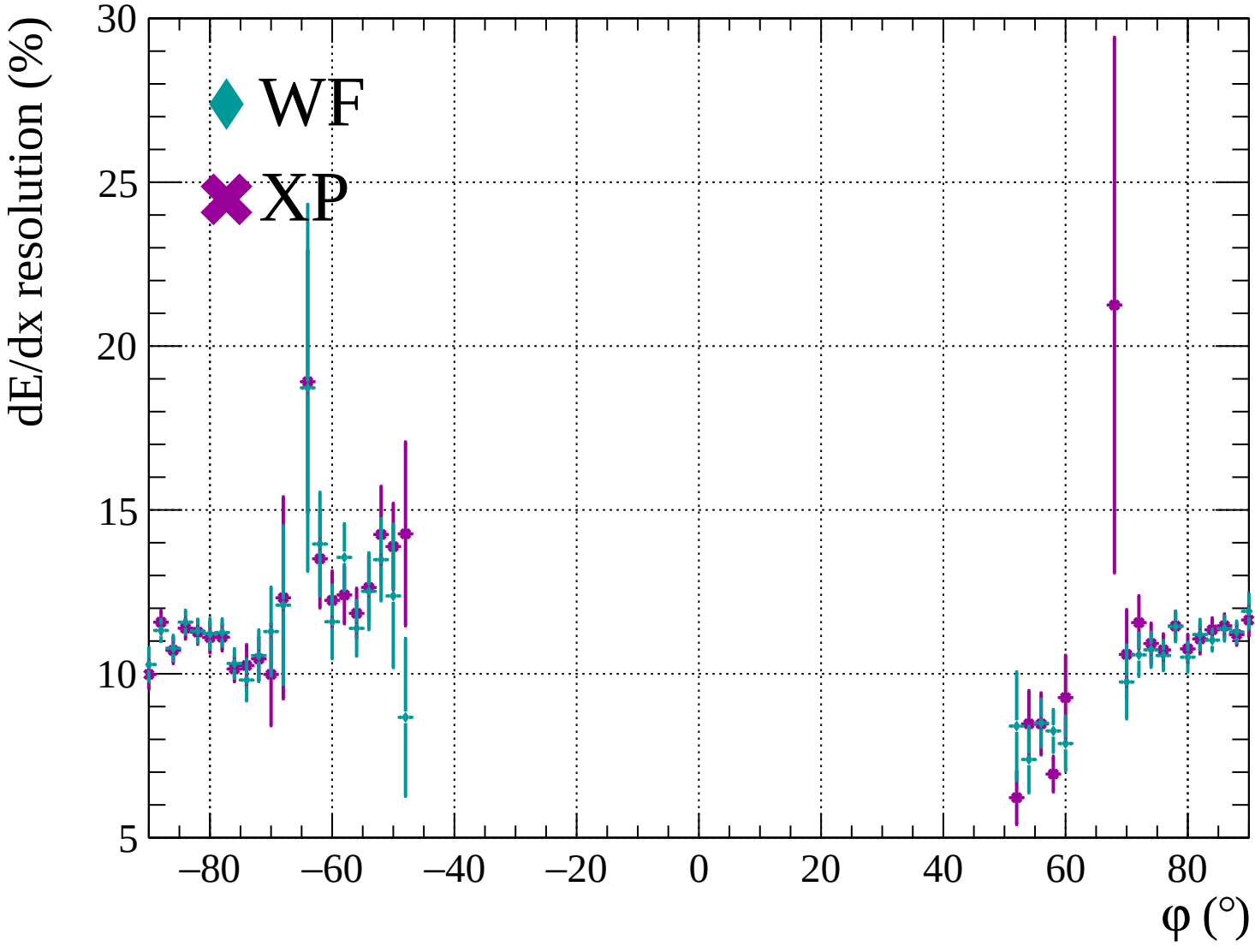


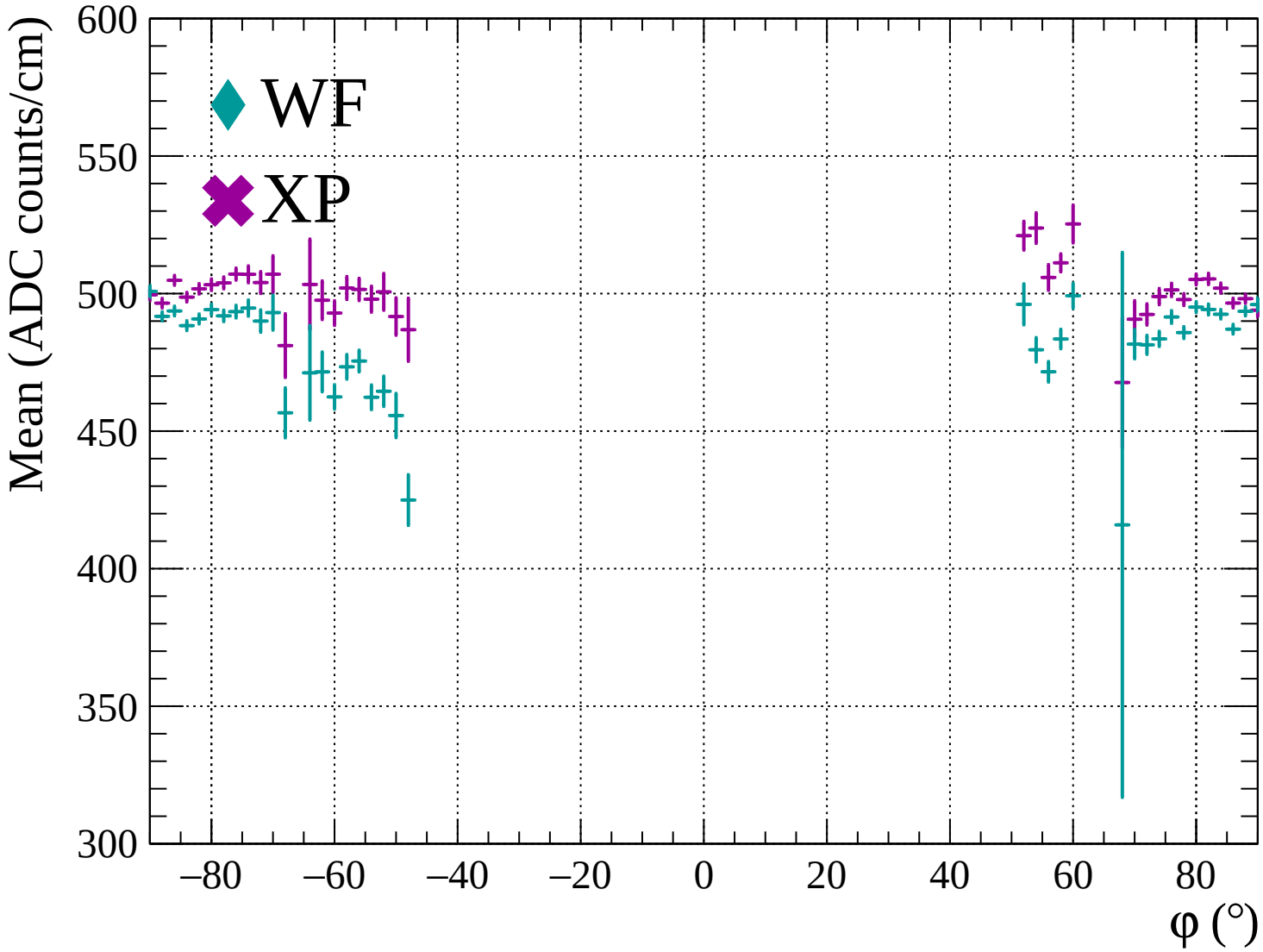


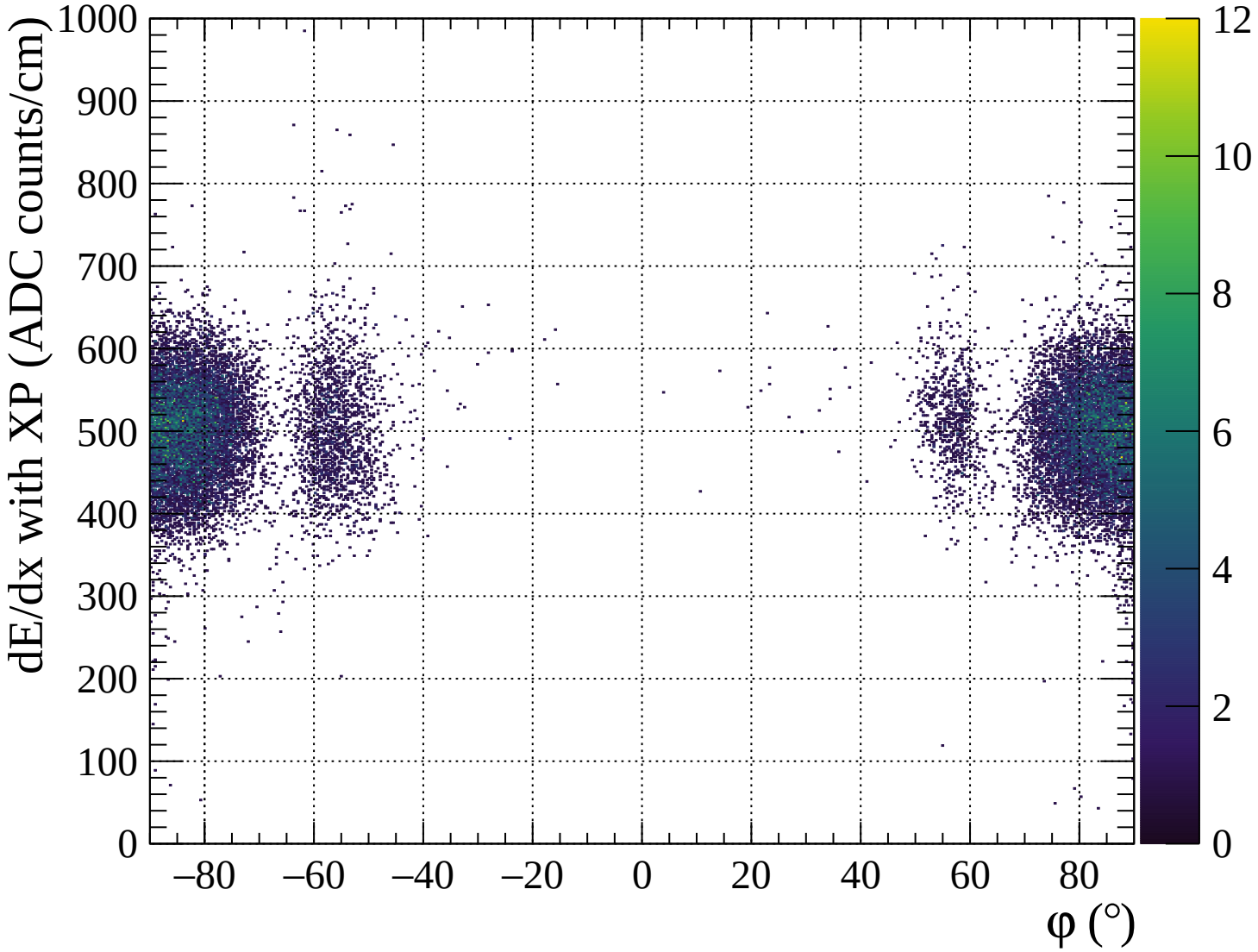


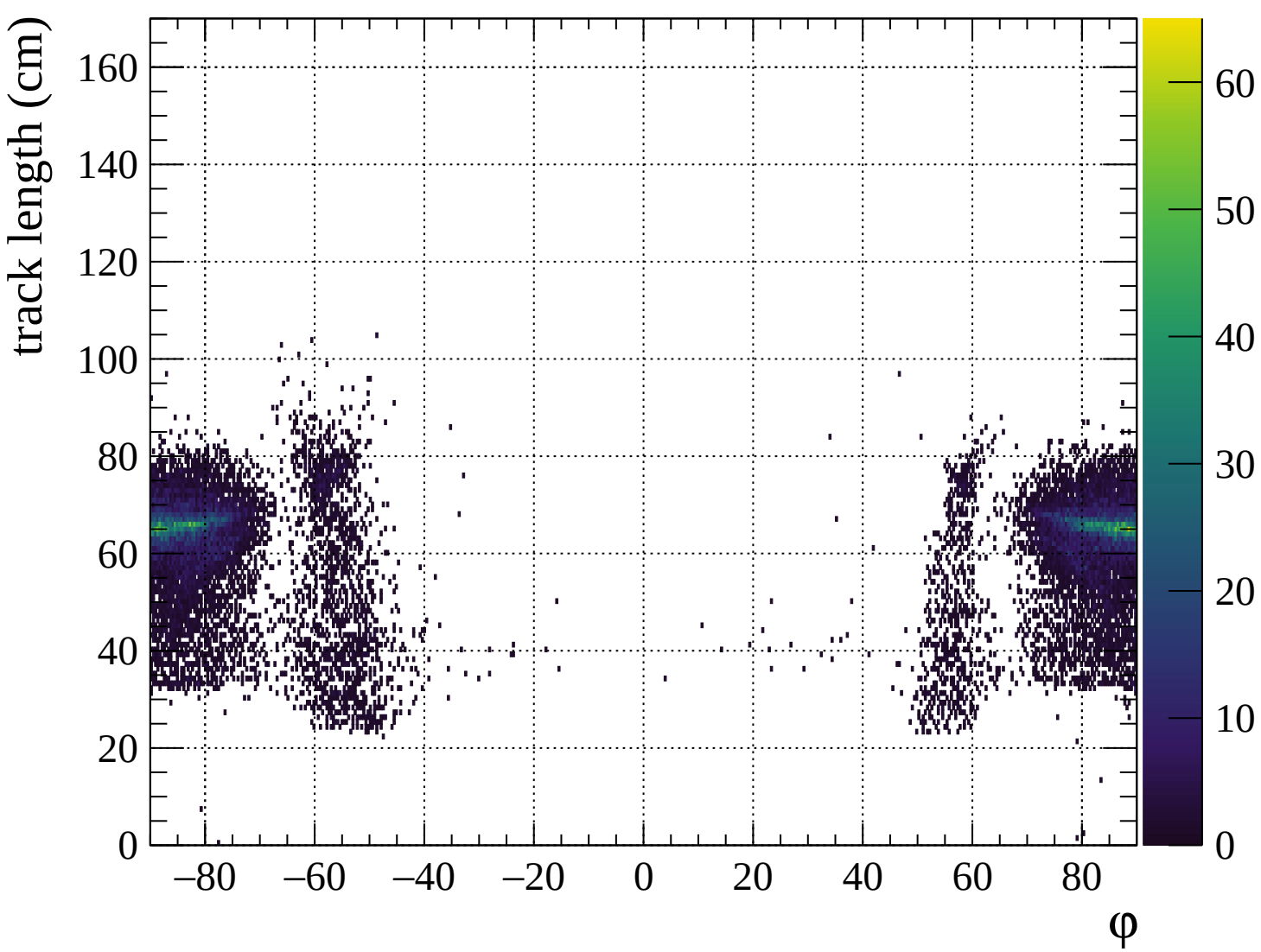


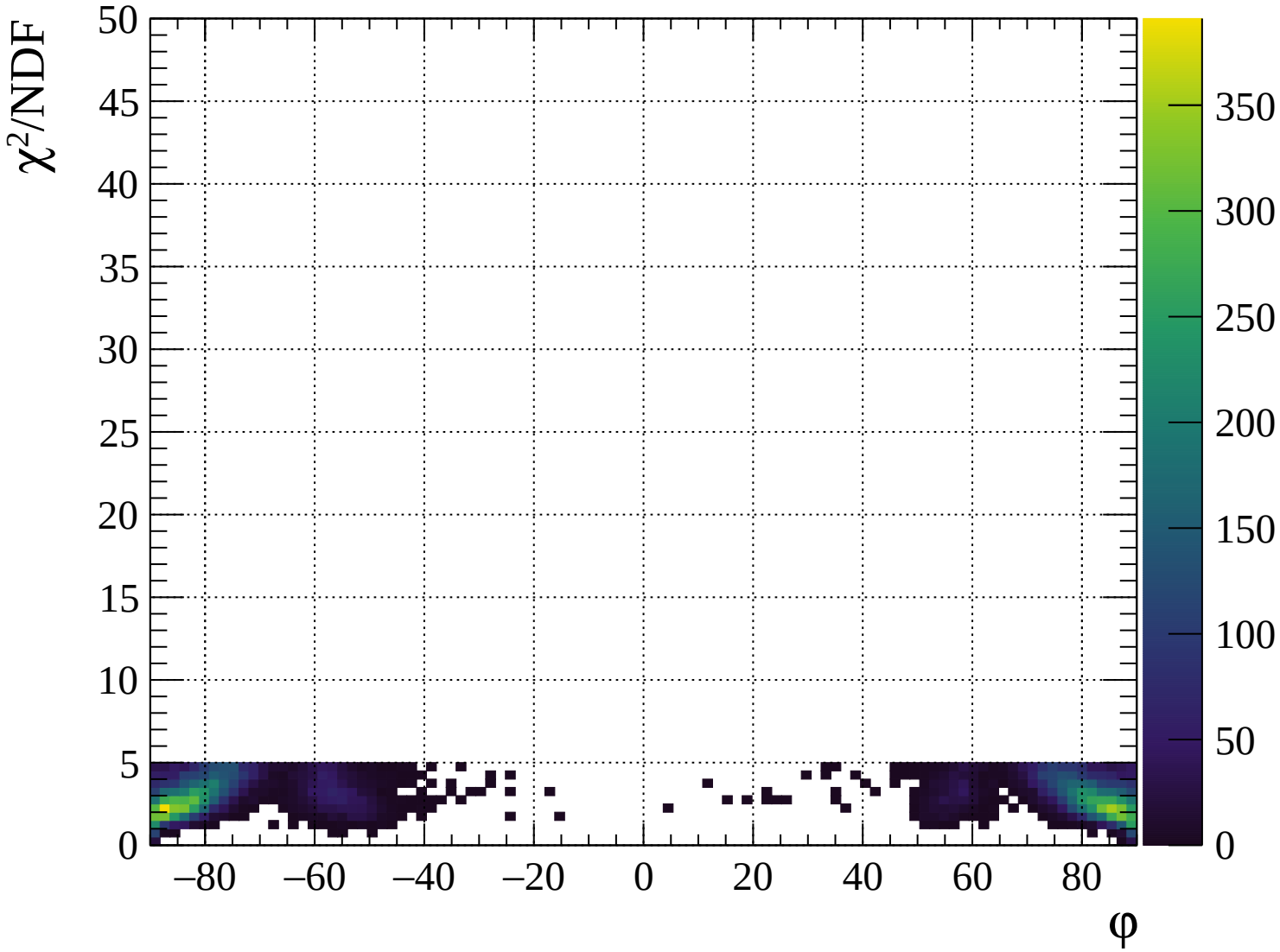


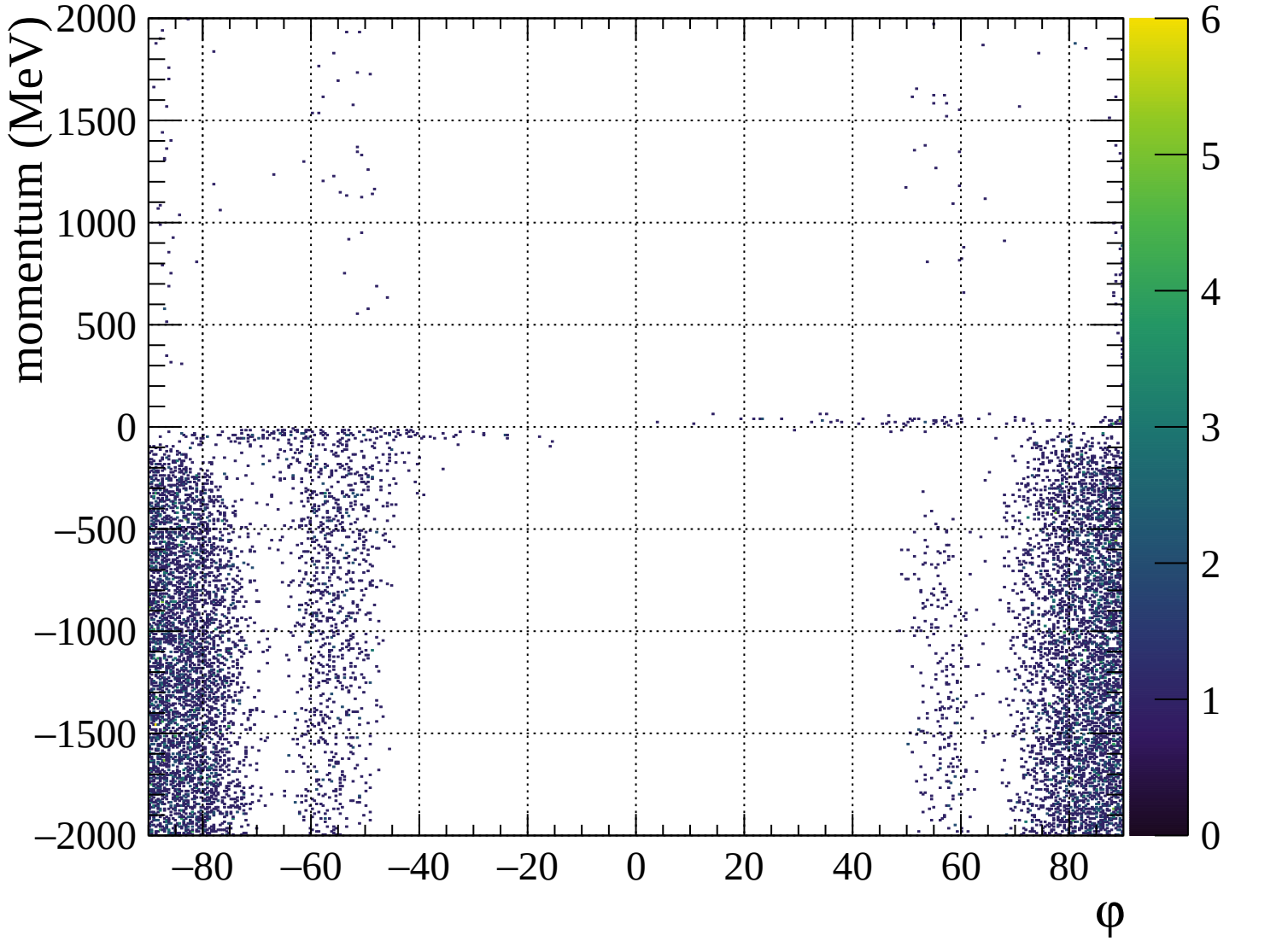


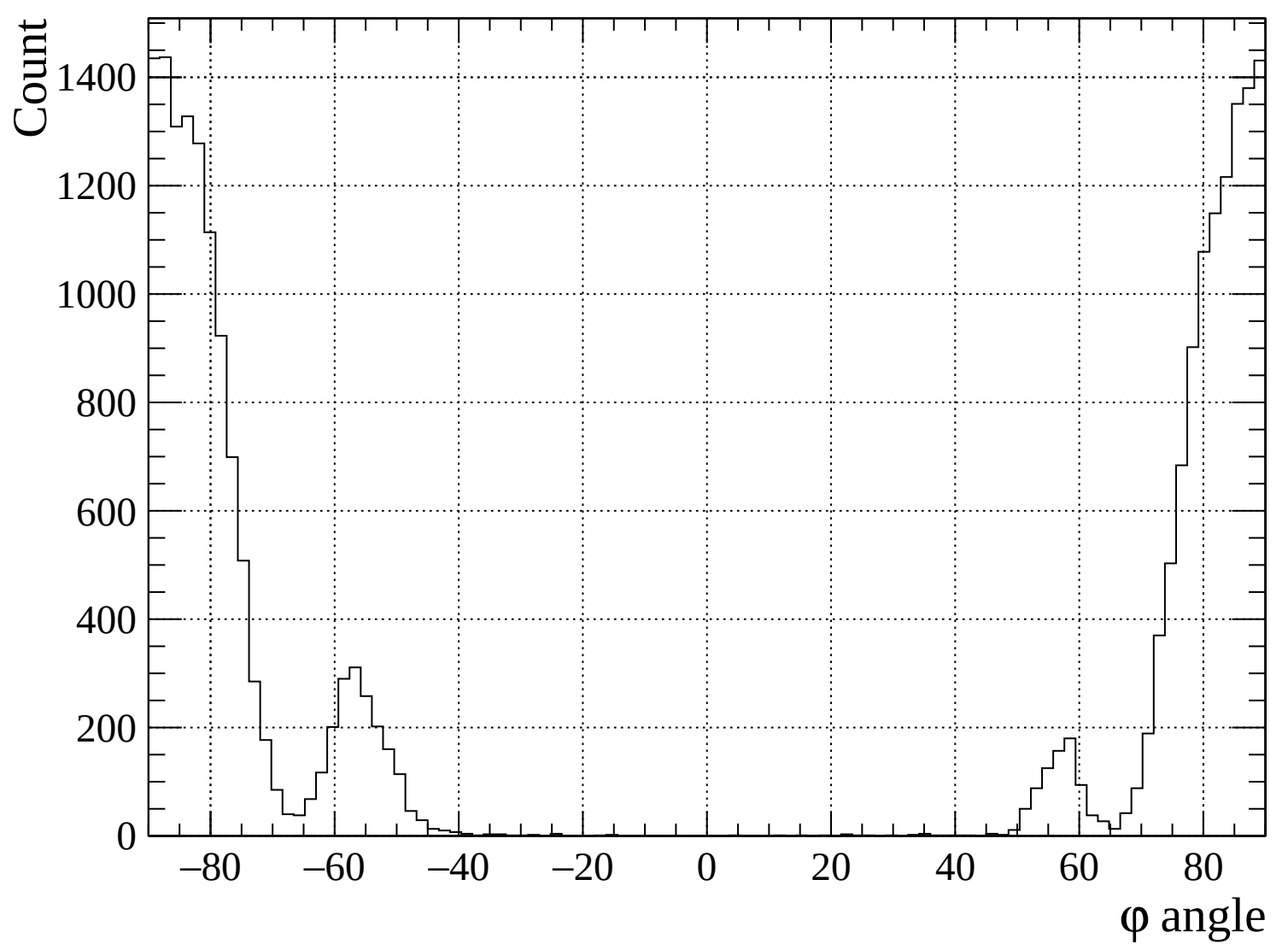


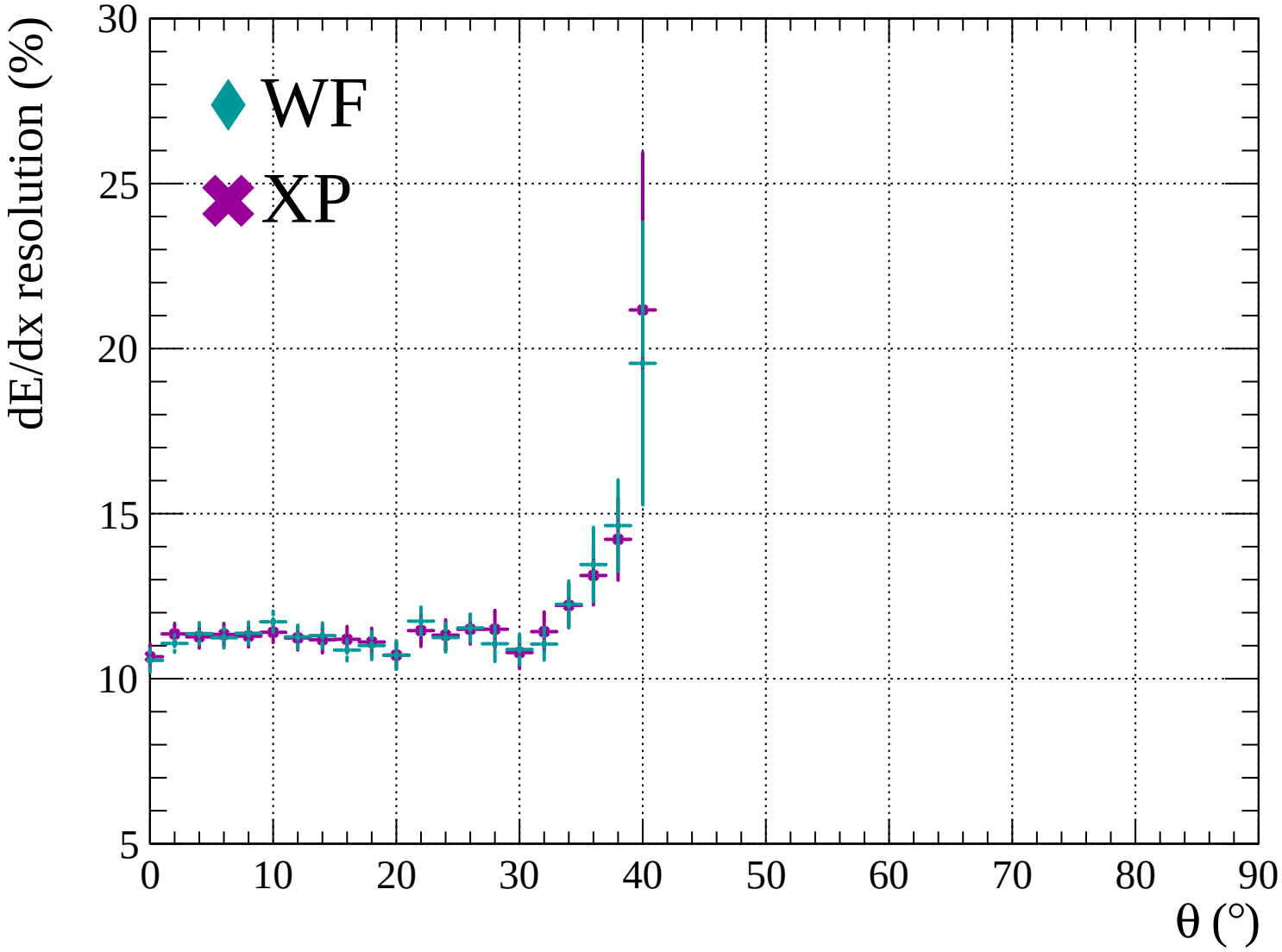


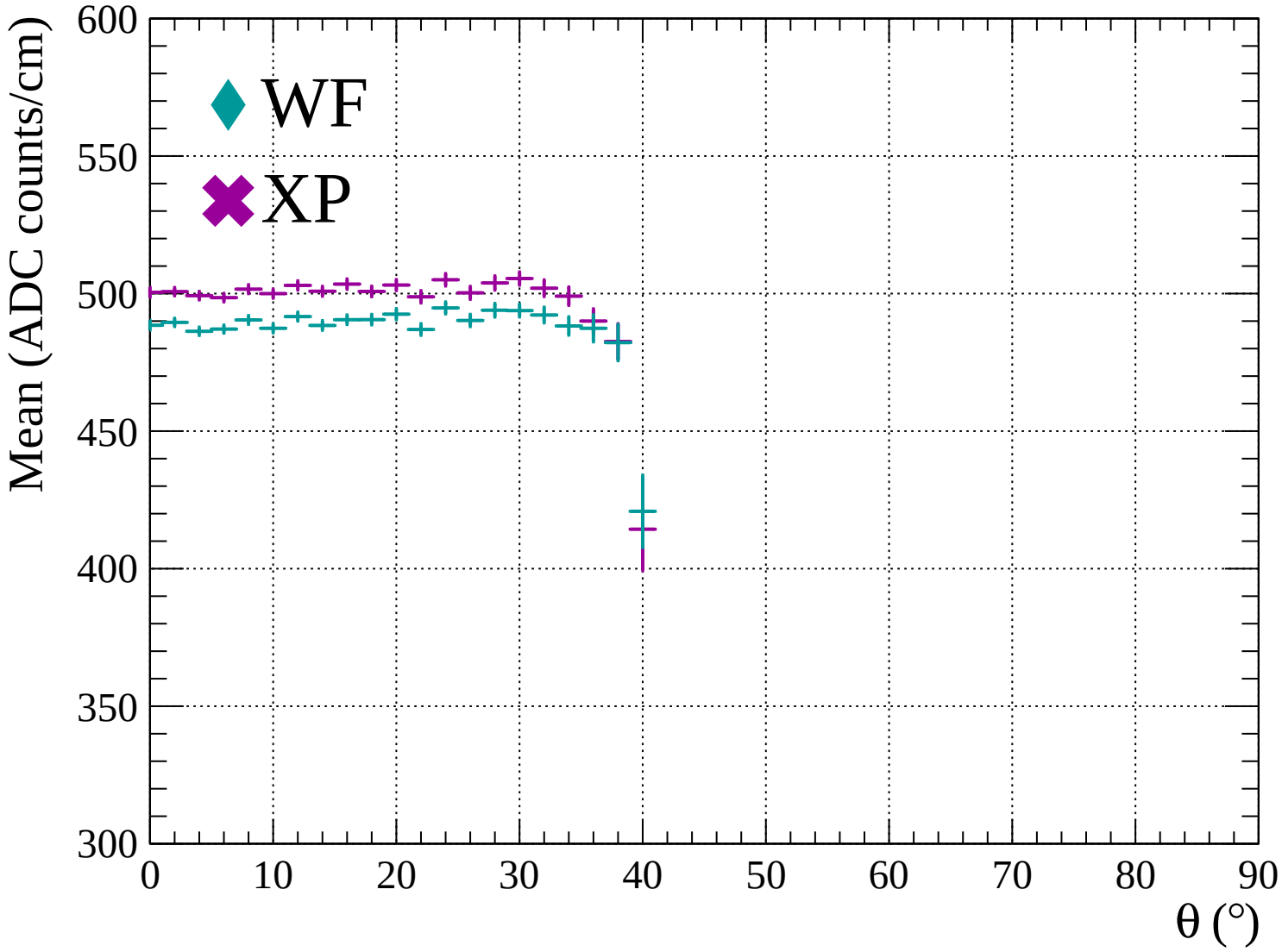


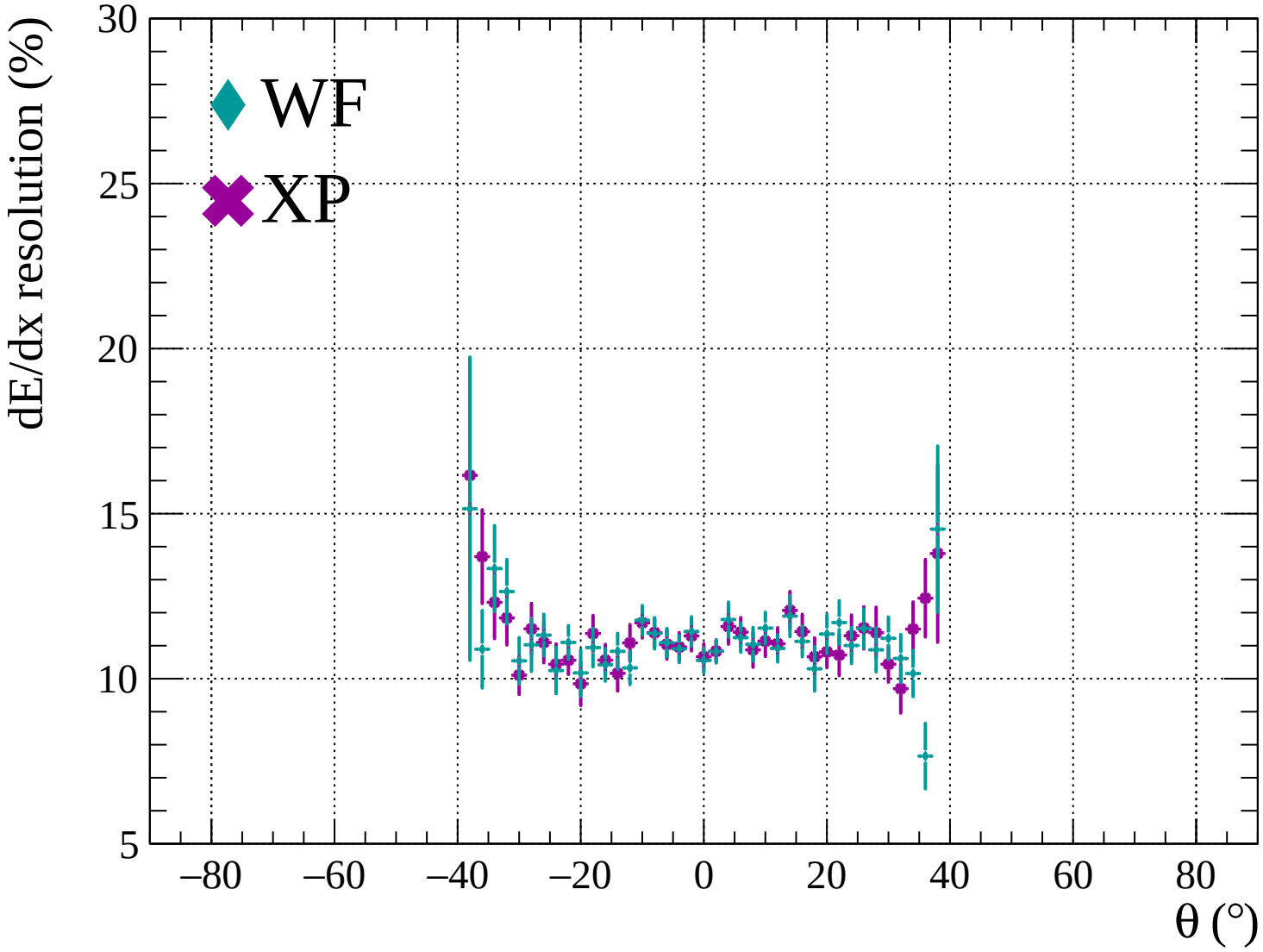


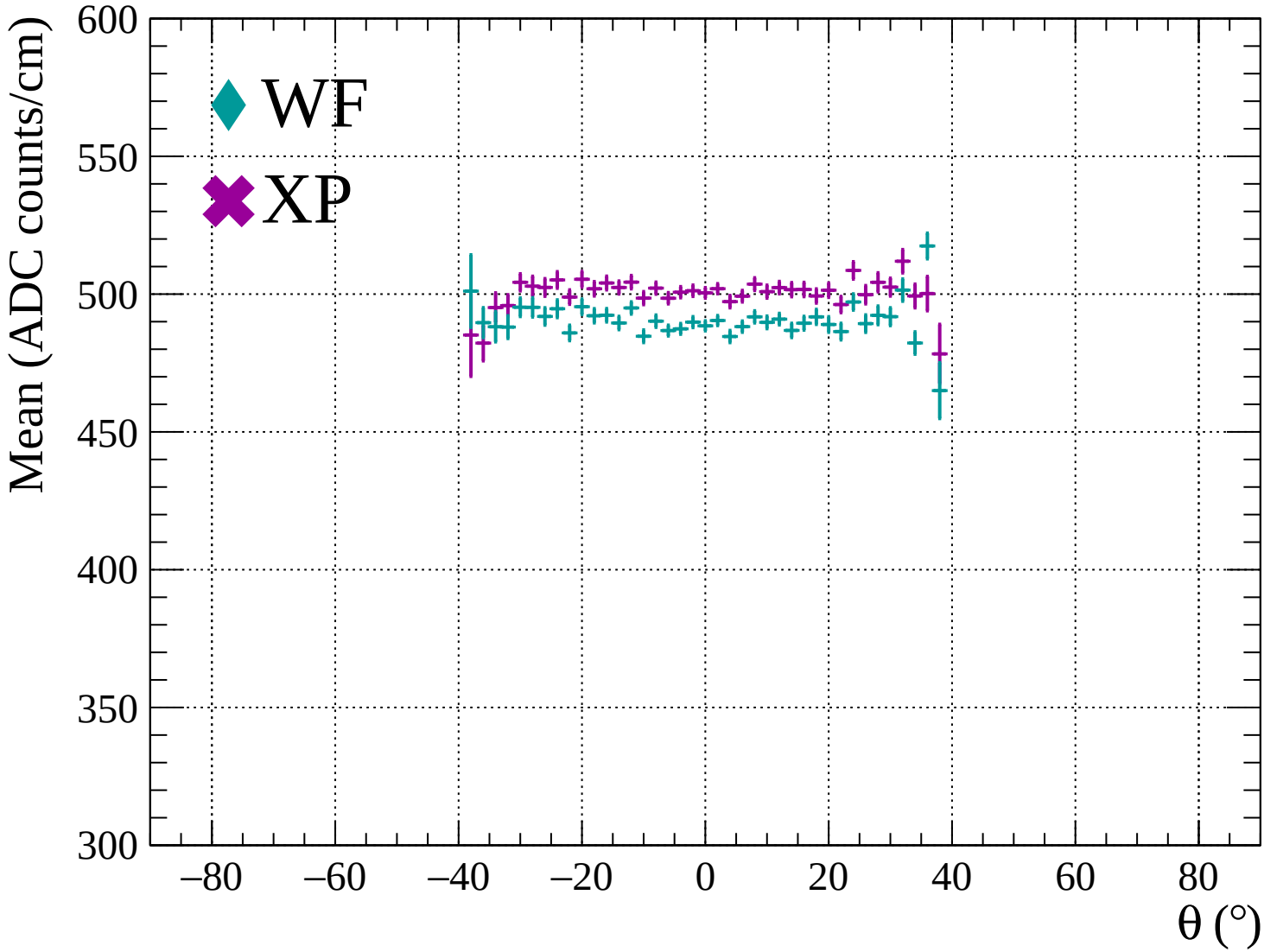


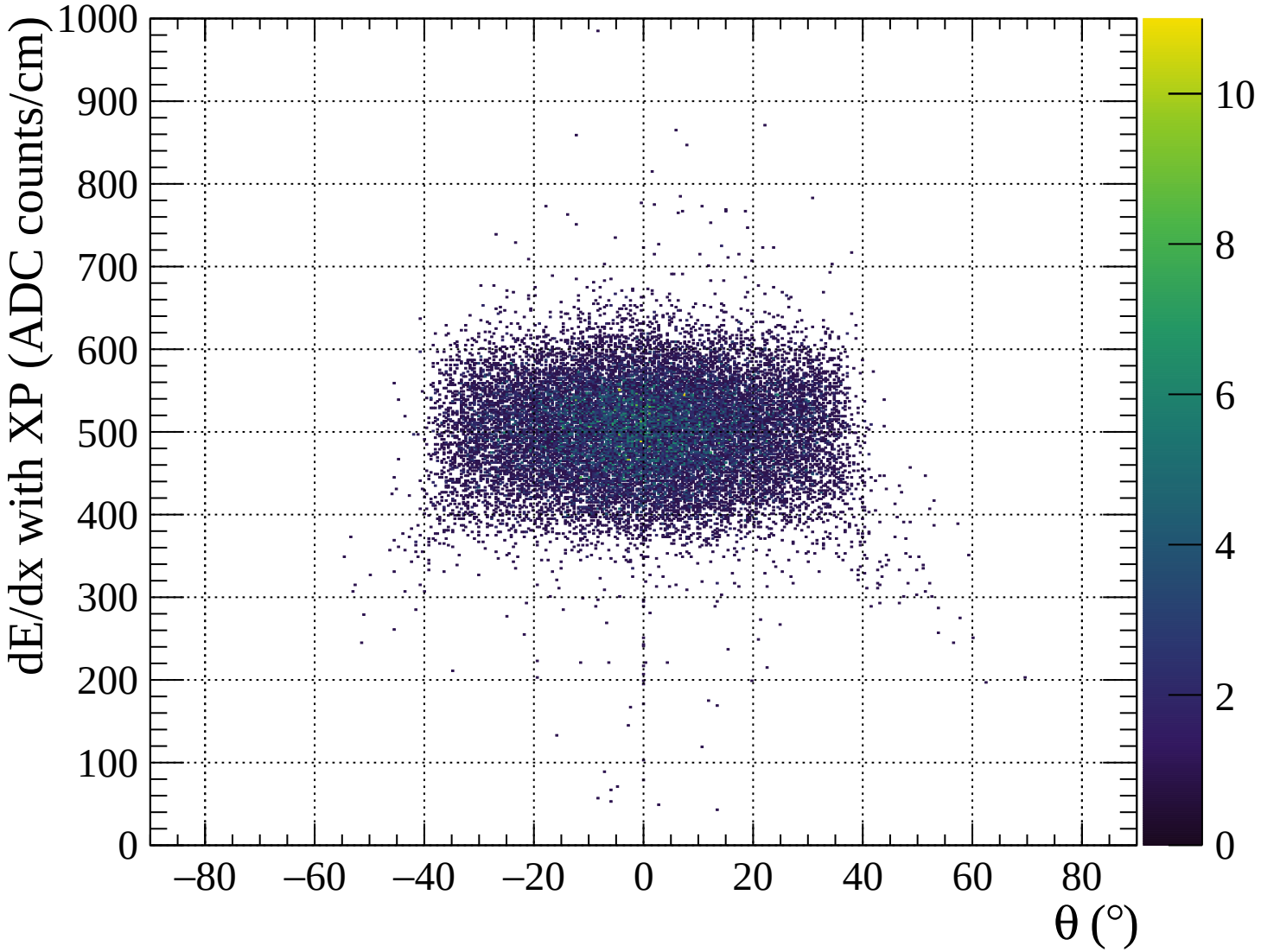


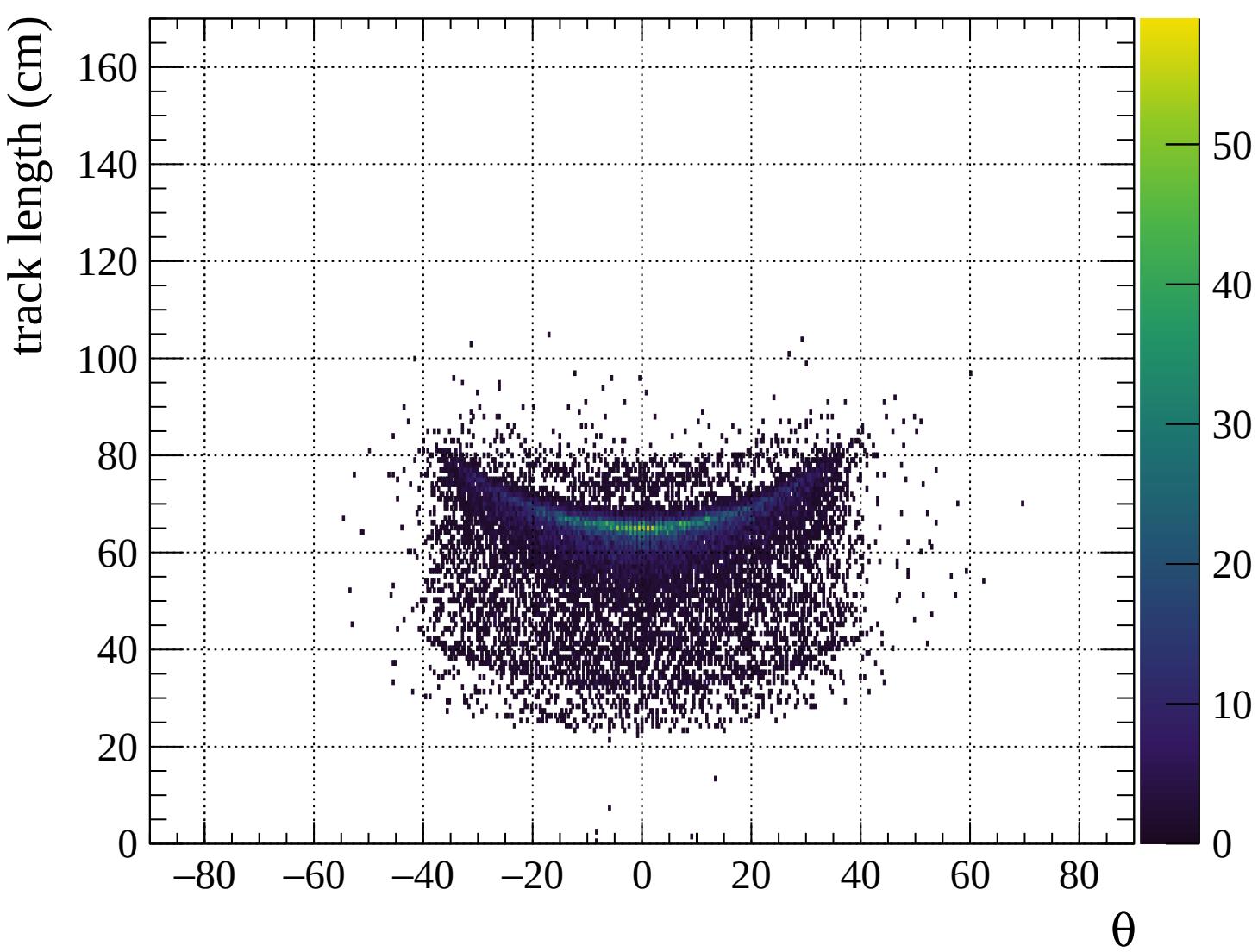


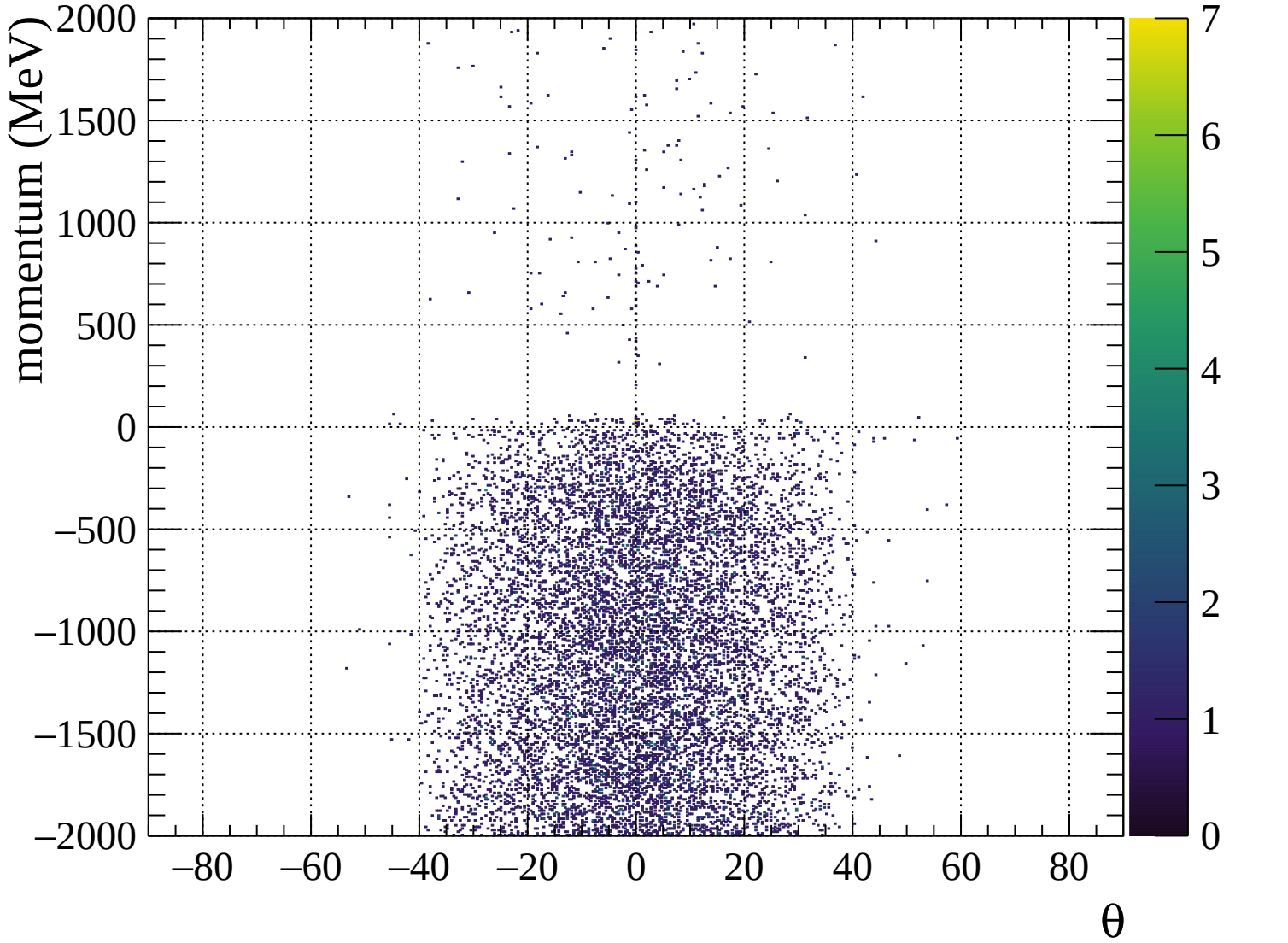


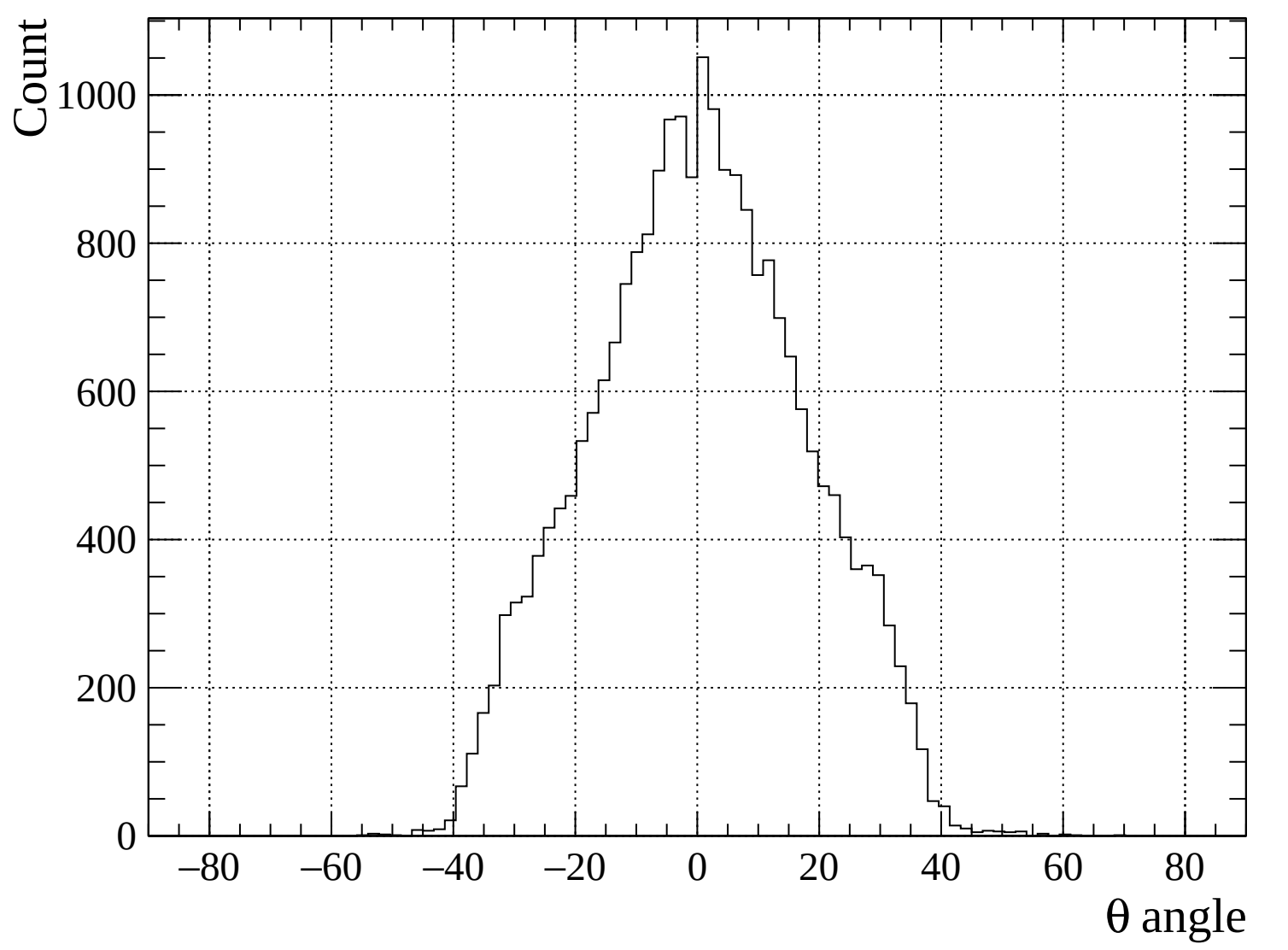


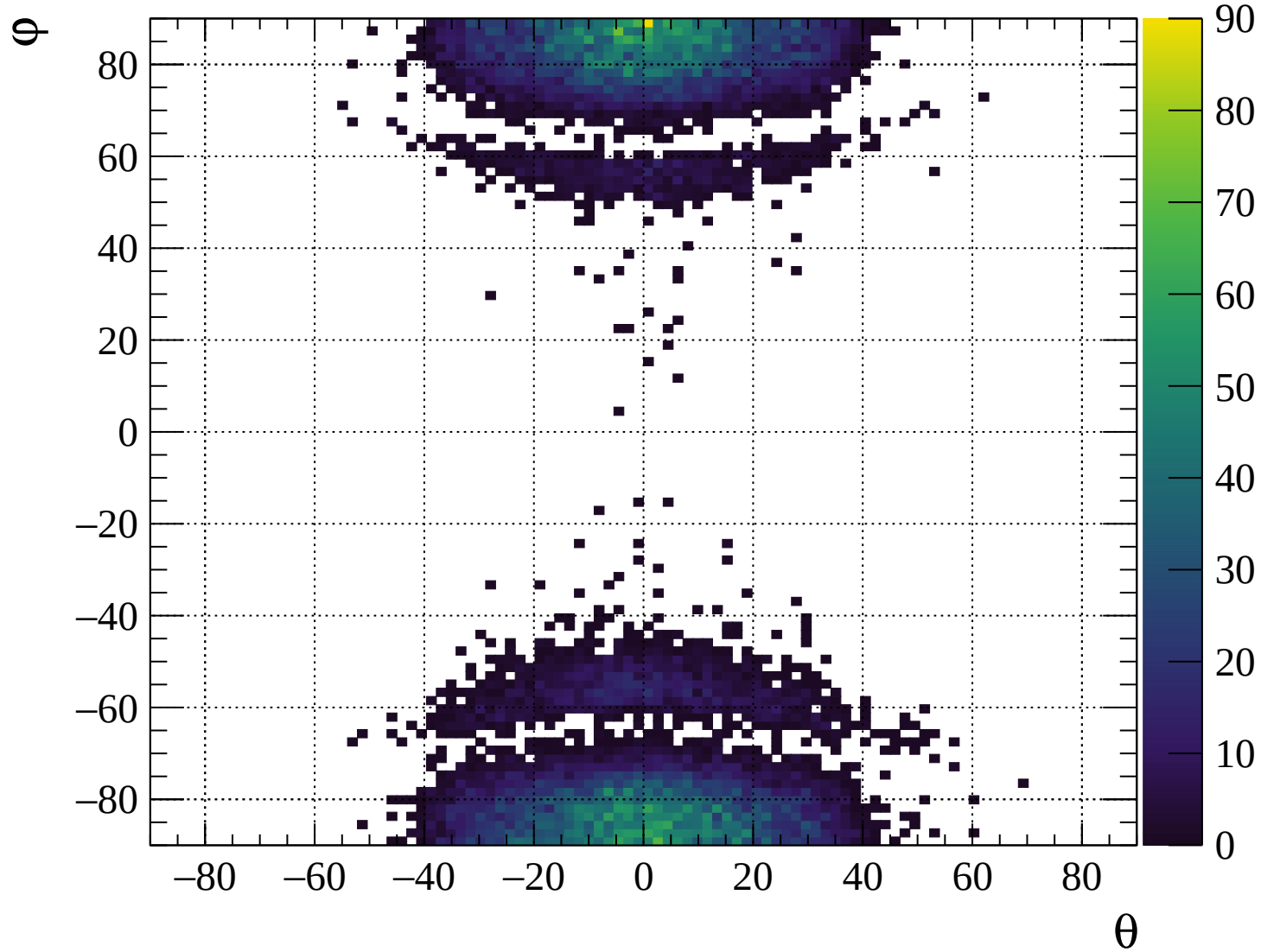


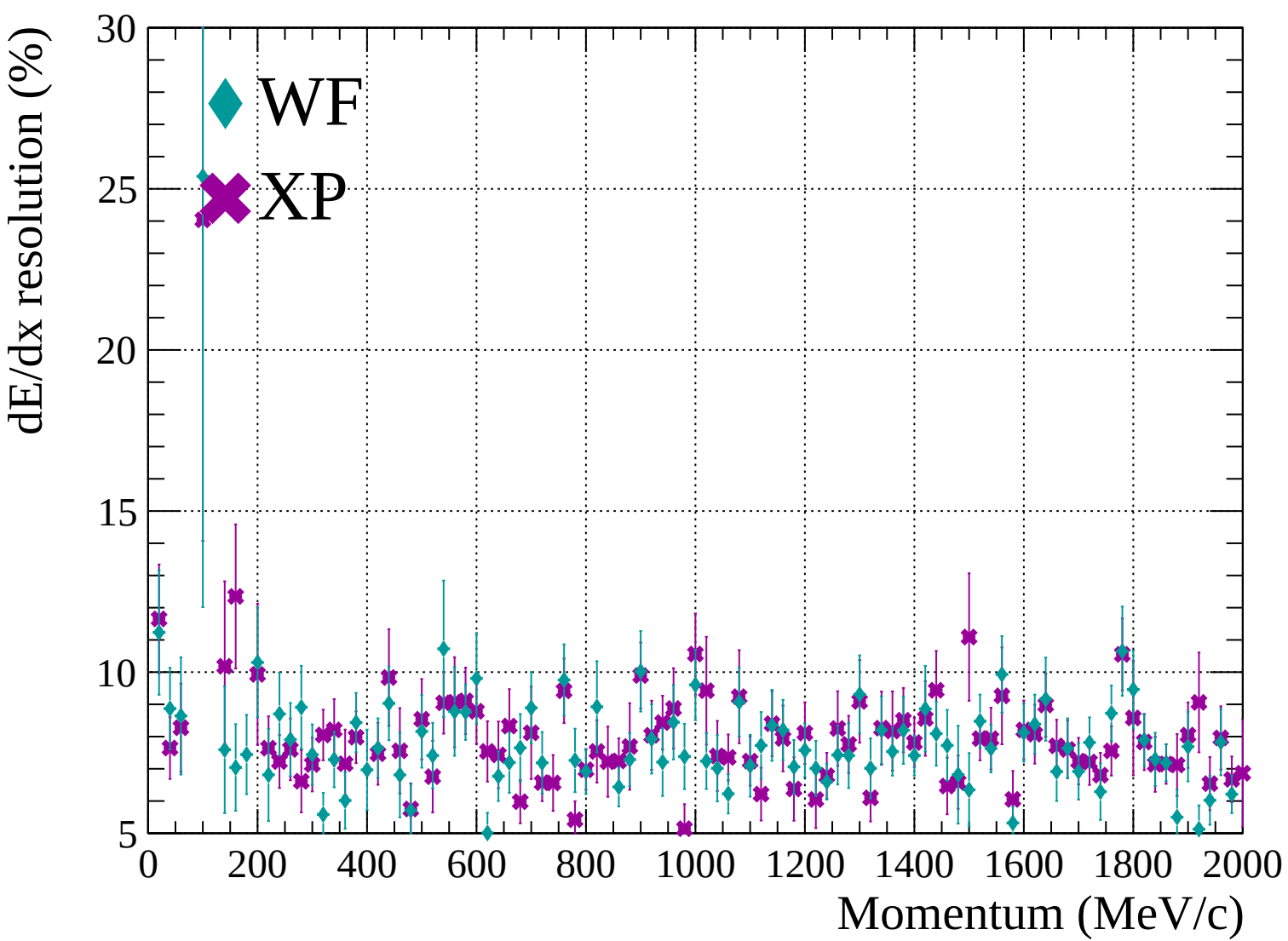


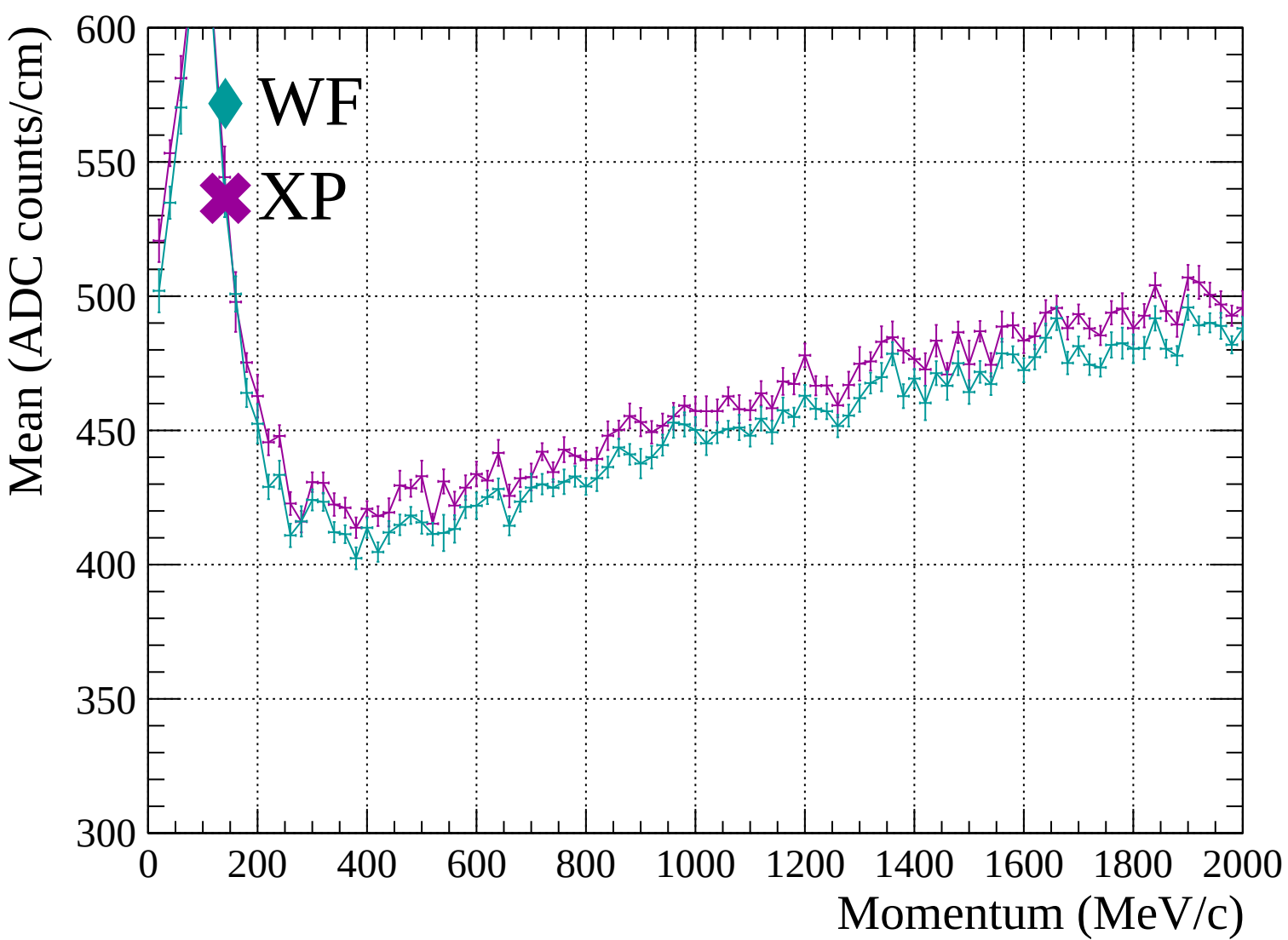


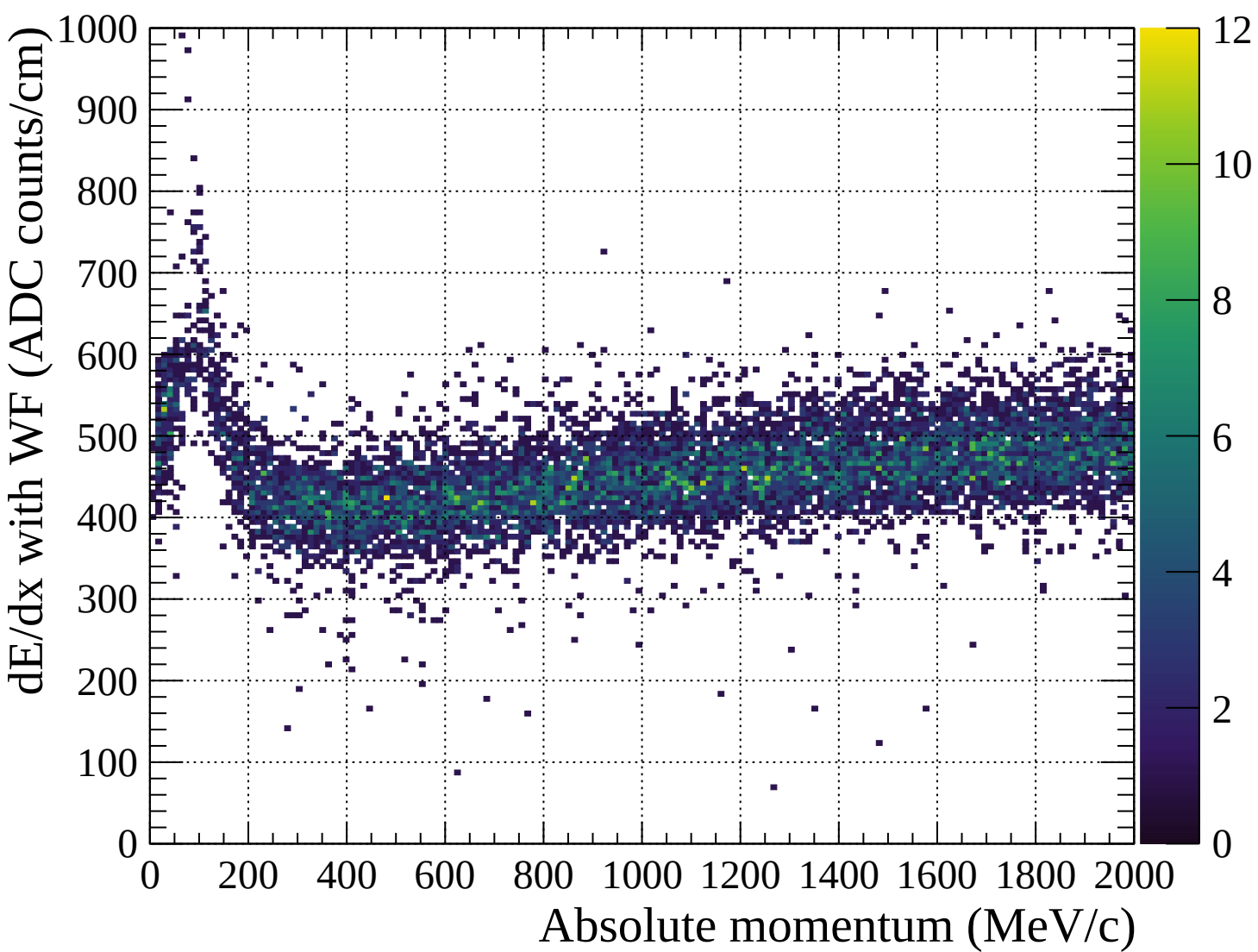


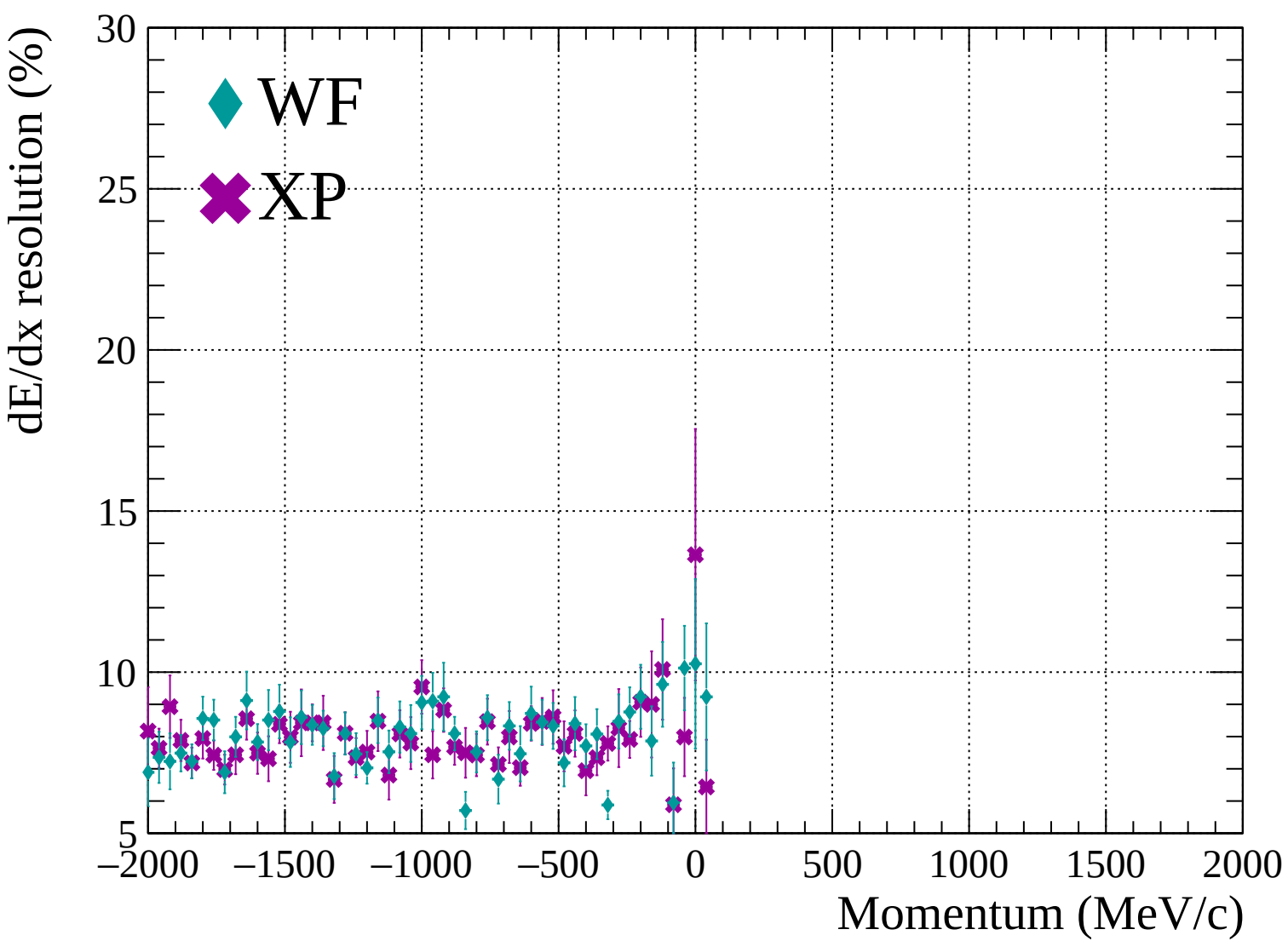


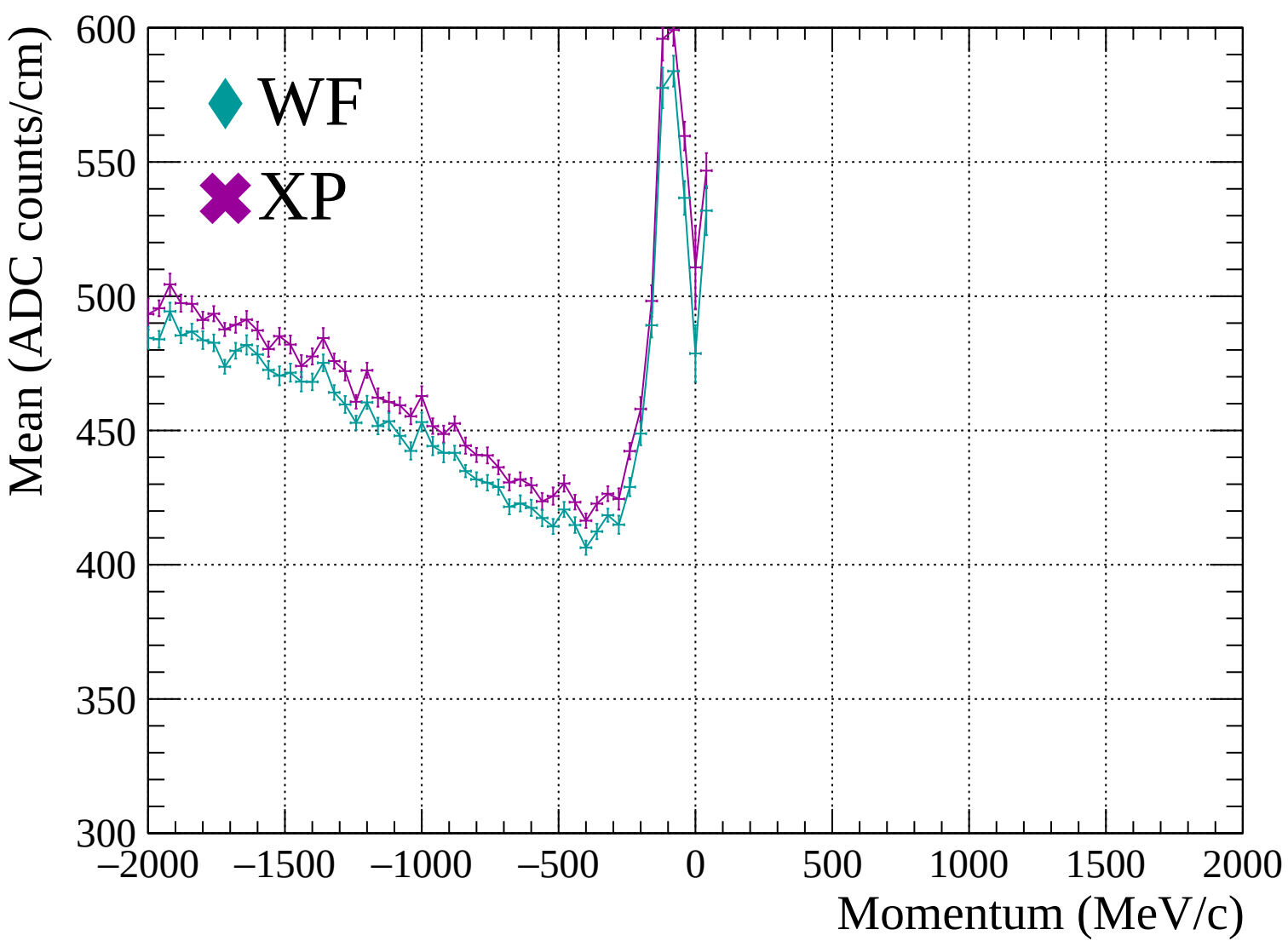


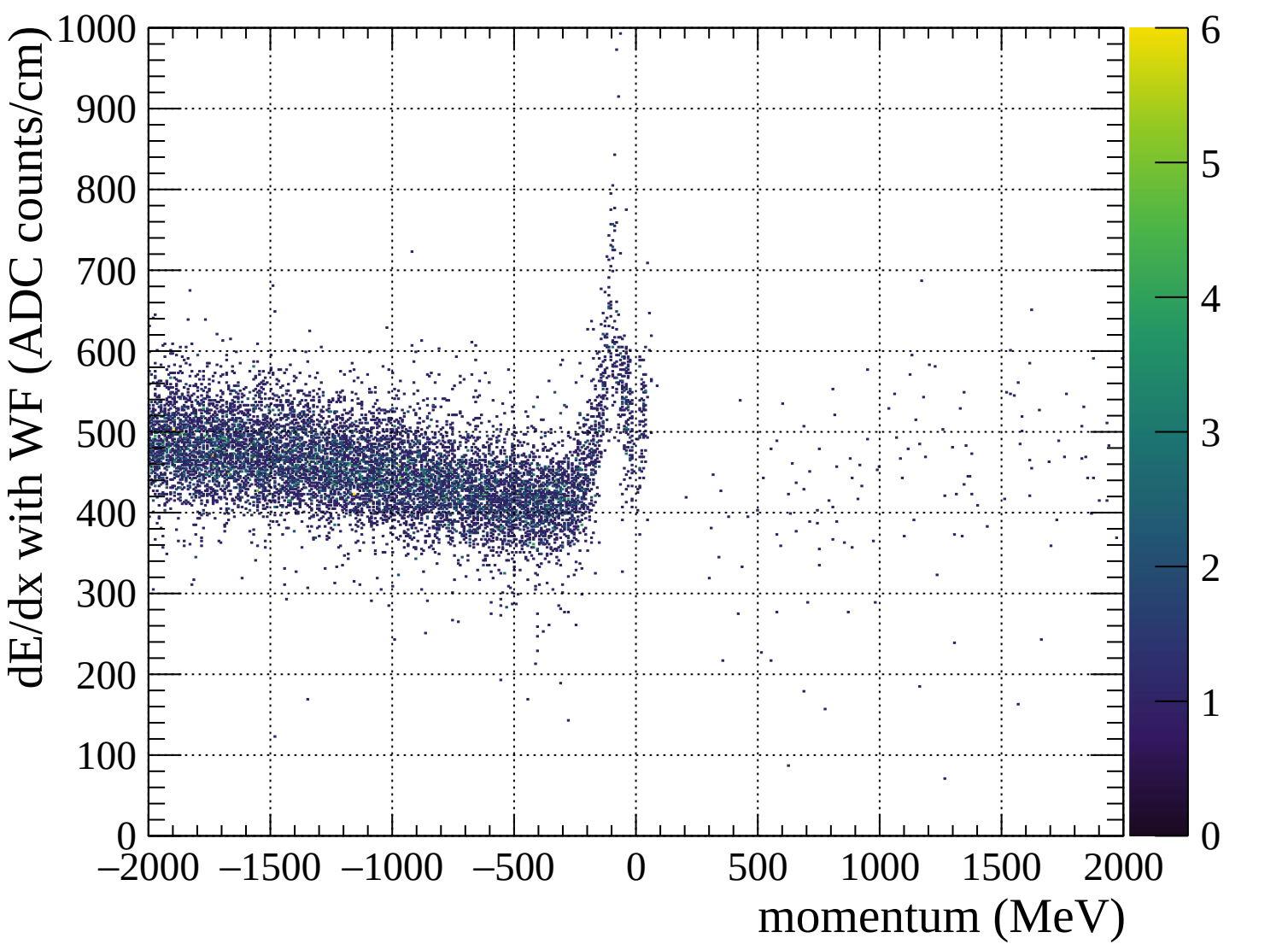


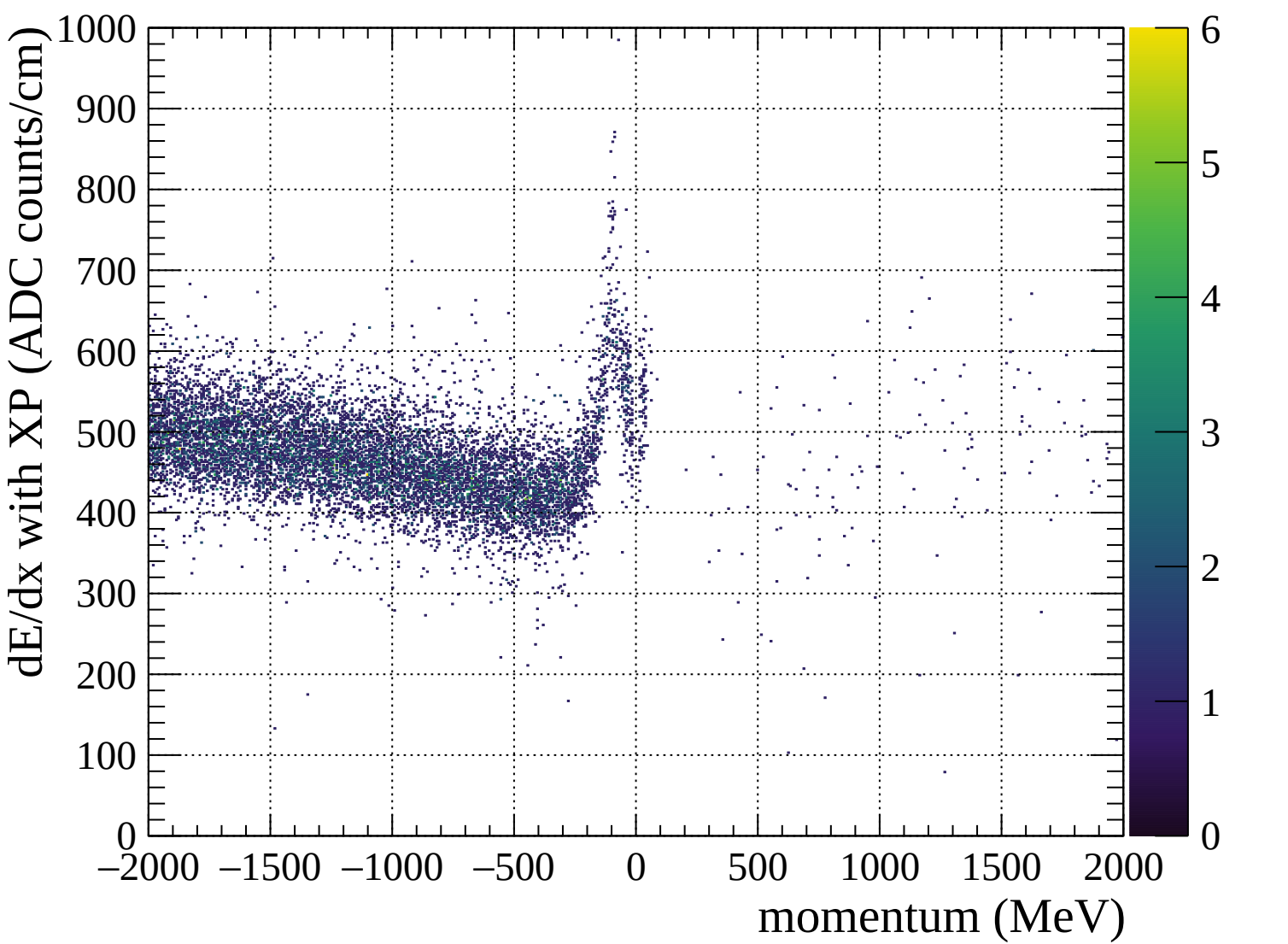


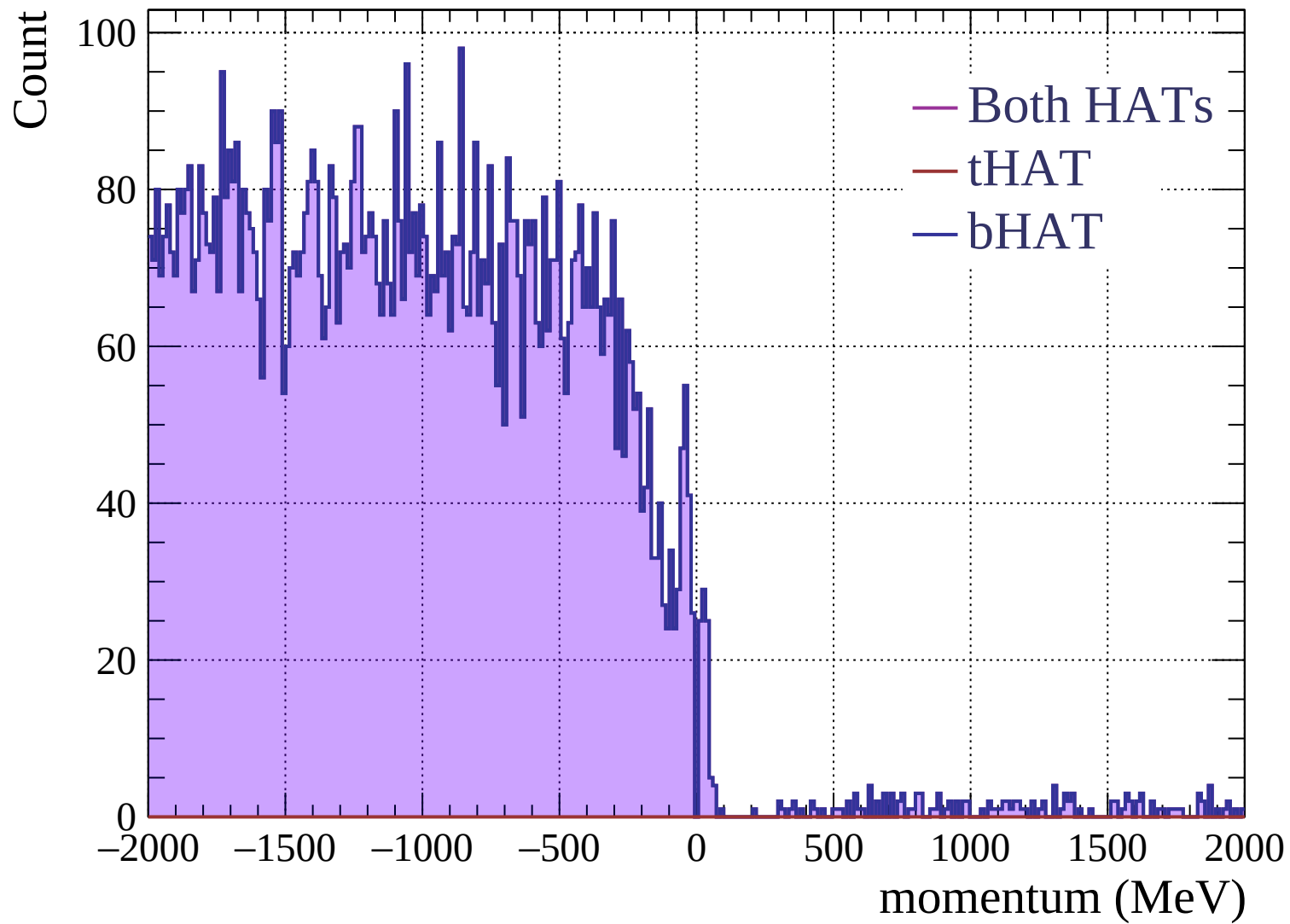


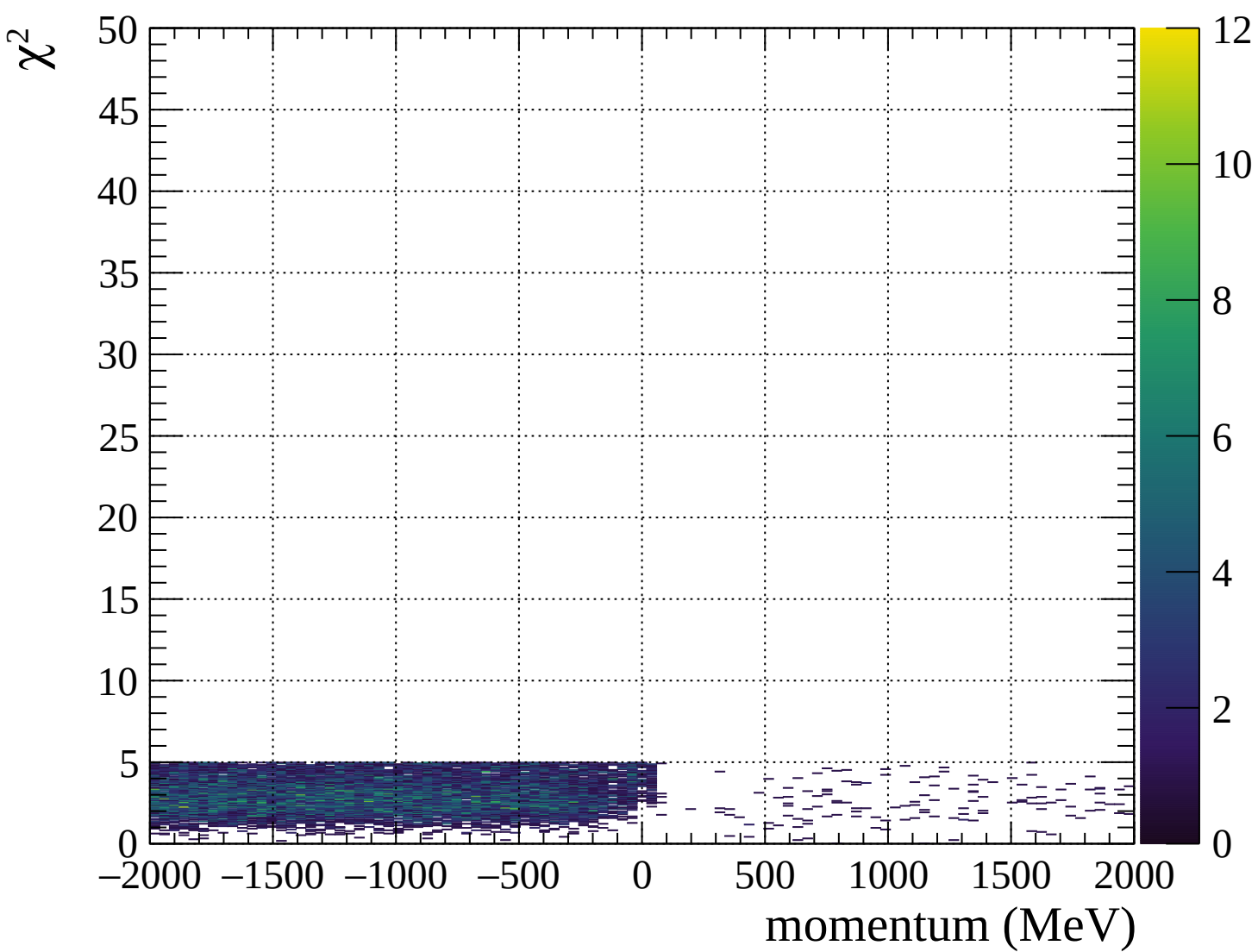












Pulls standard deviation

1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0

-80

-60

-40

-20

0

20

40

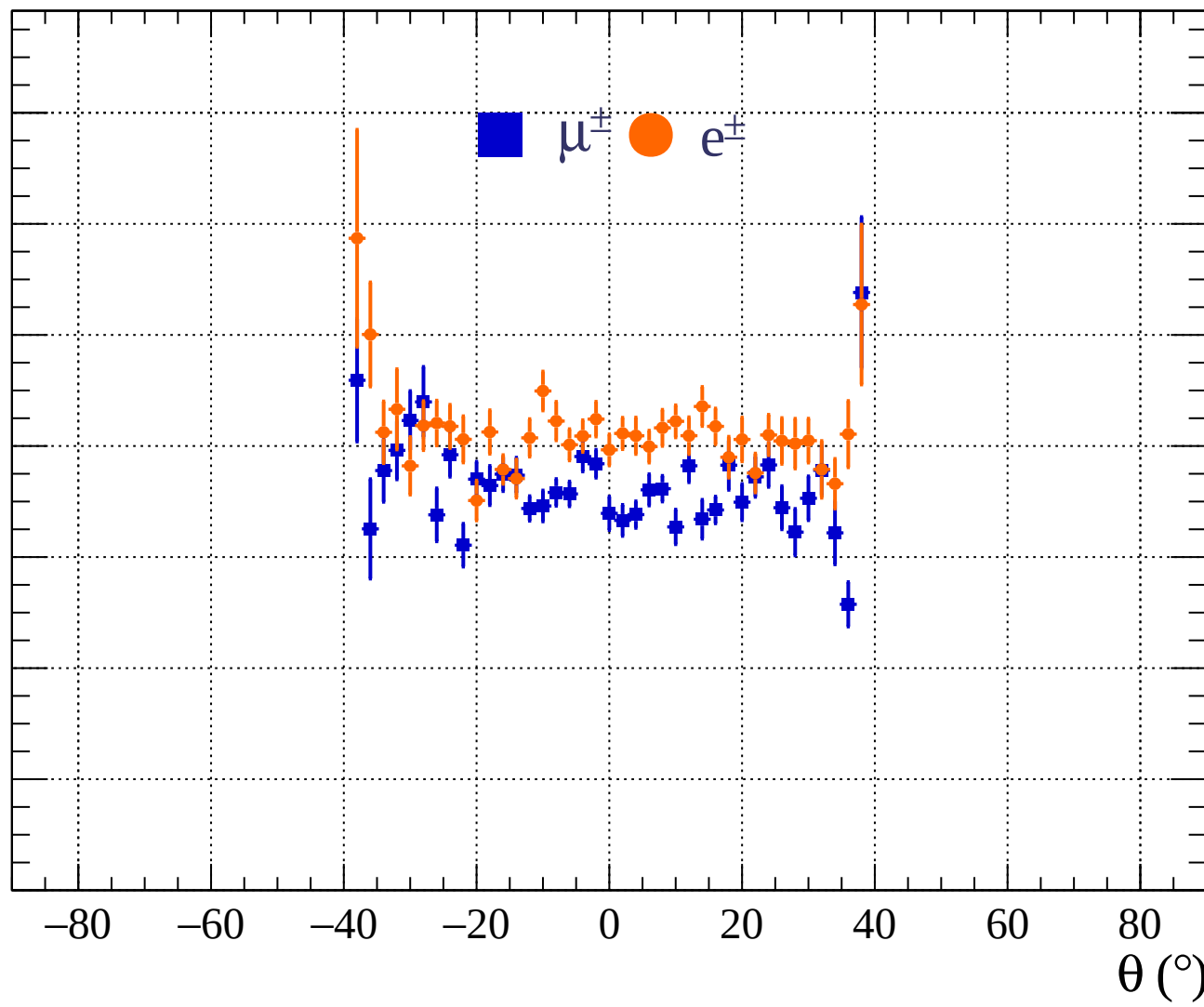
60

80

$\theta (^{\circ})$

$\mu^{\pm}$

$e^{\pm}$



Pulls mean

0.5
0.0
-0.5
-1.0
-1.5
-2.0
-2.5

-80

-60

-40

-20

0

20

40

60

80

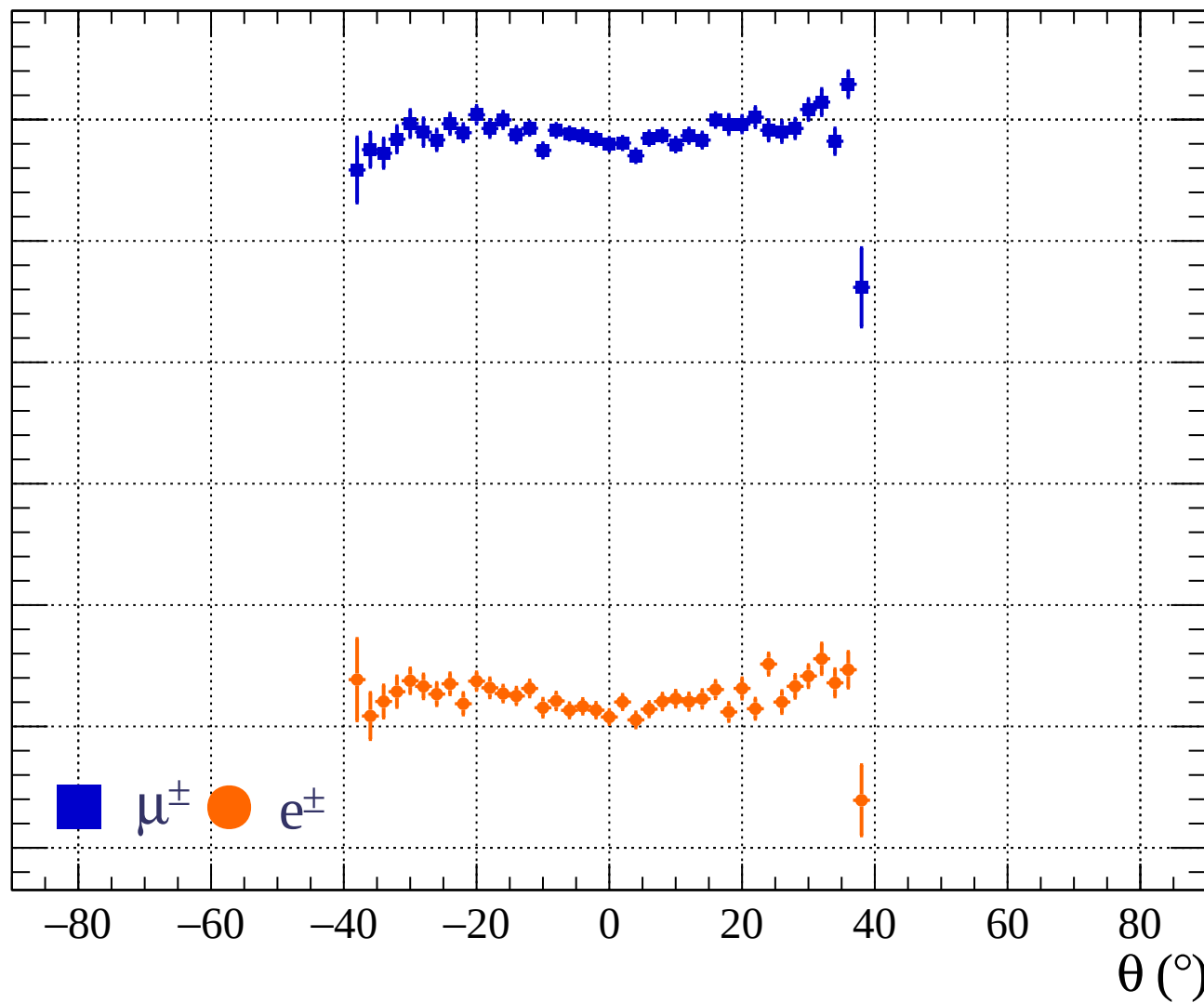
$\theta (^{\circ})$



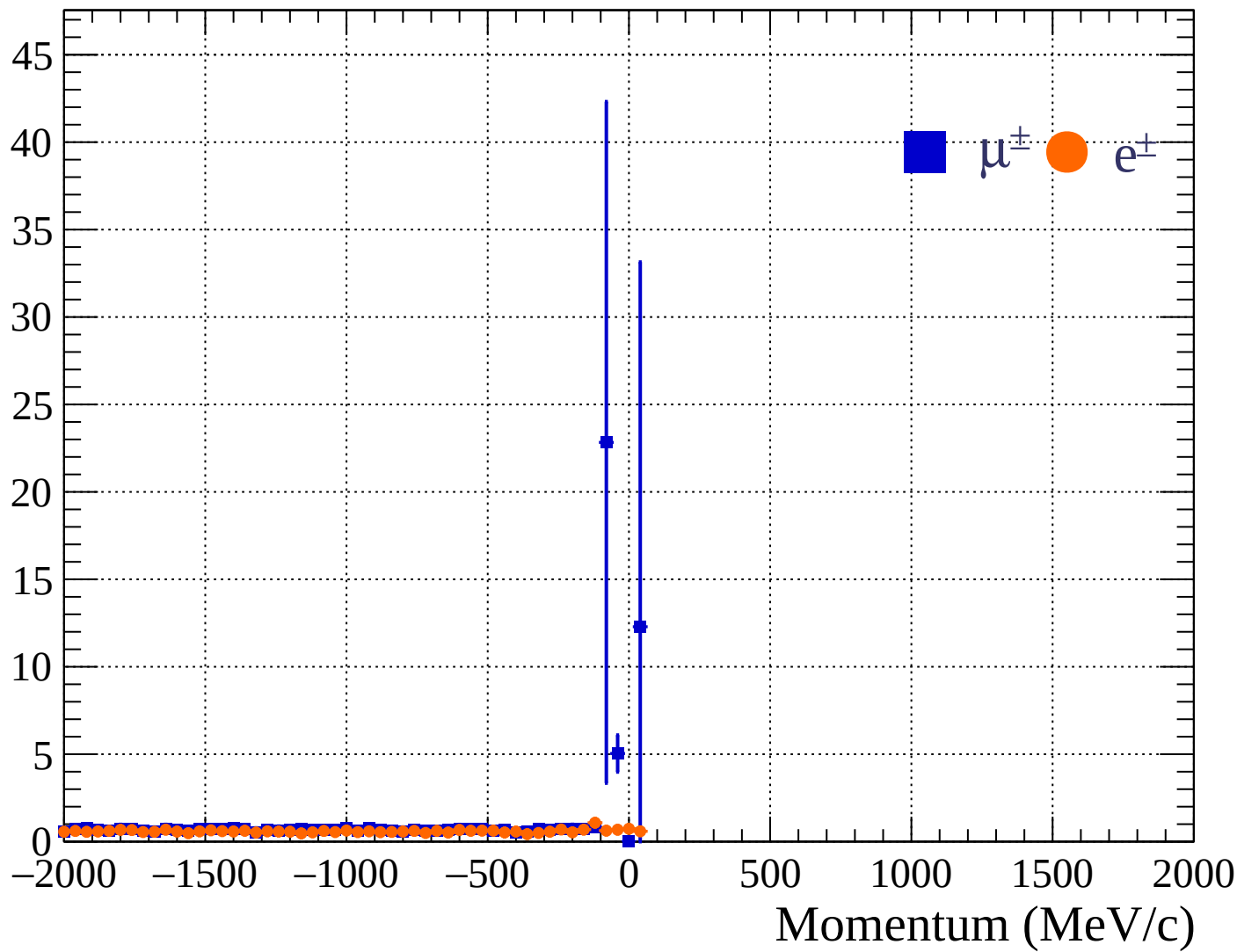
$\mu^{\pm}$

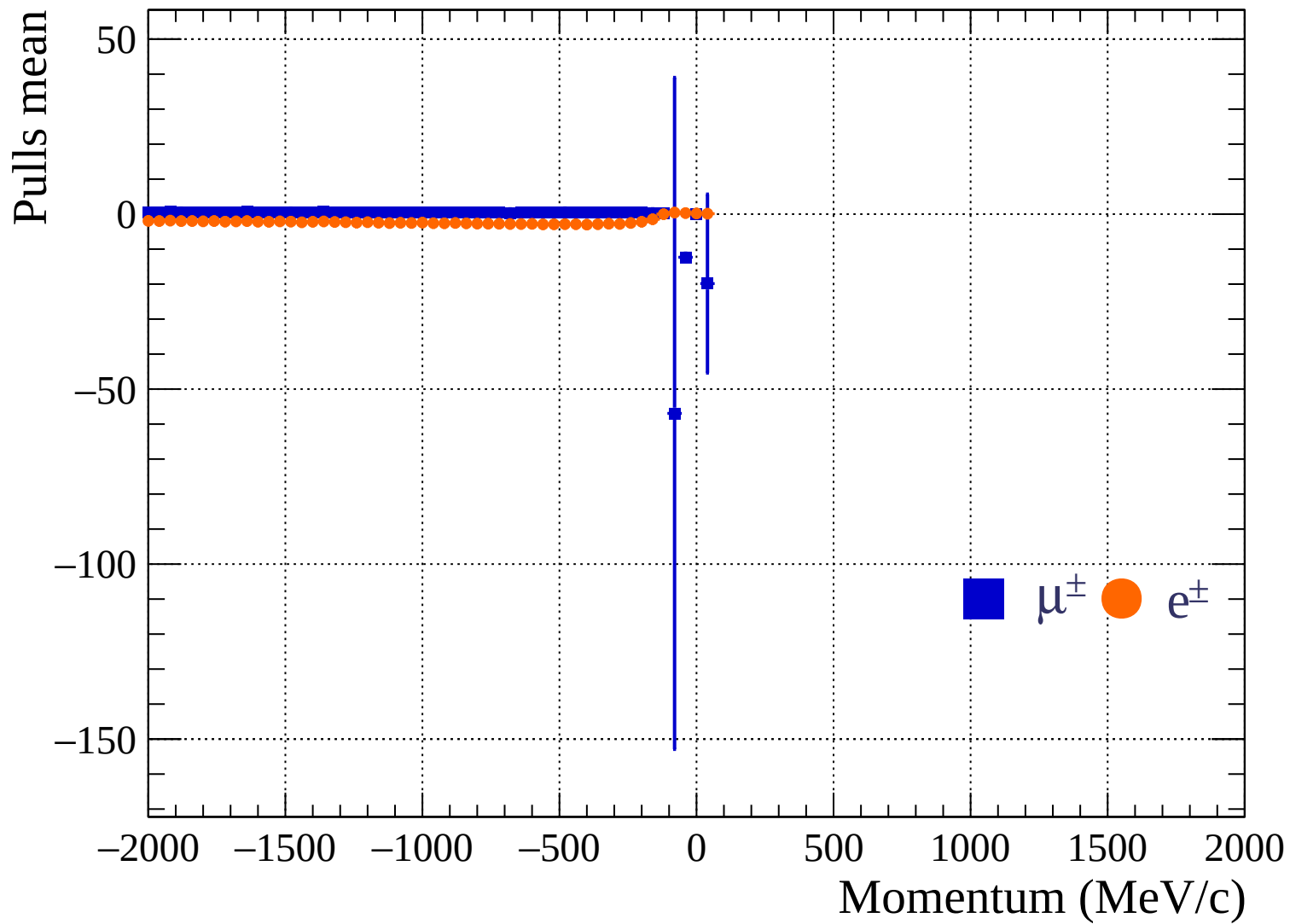


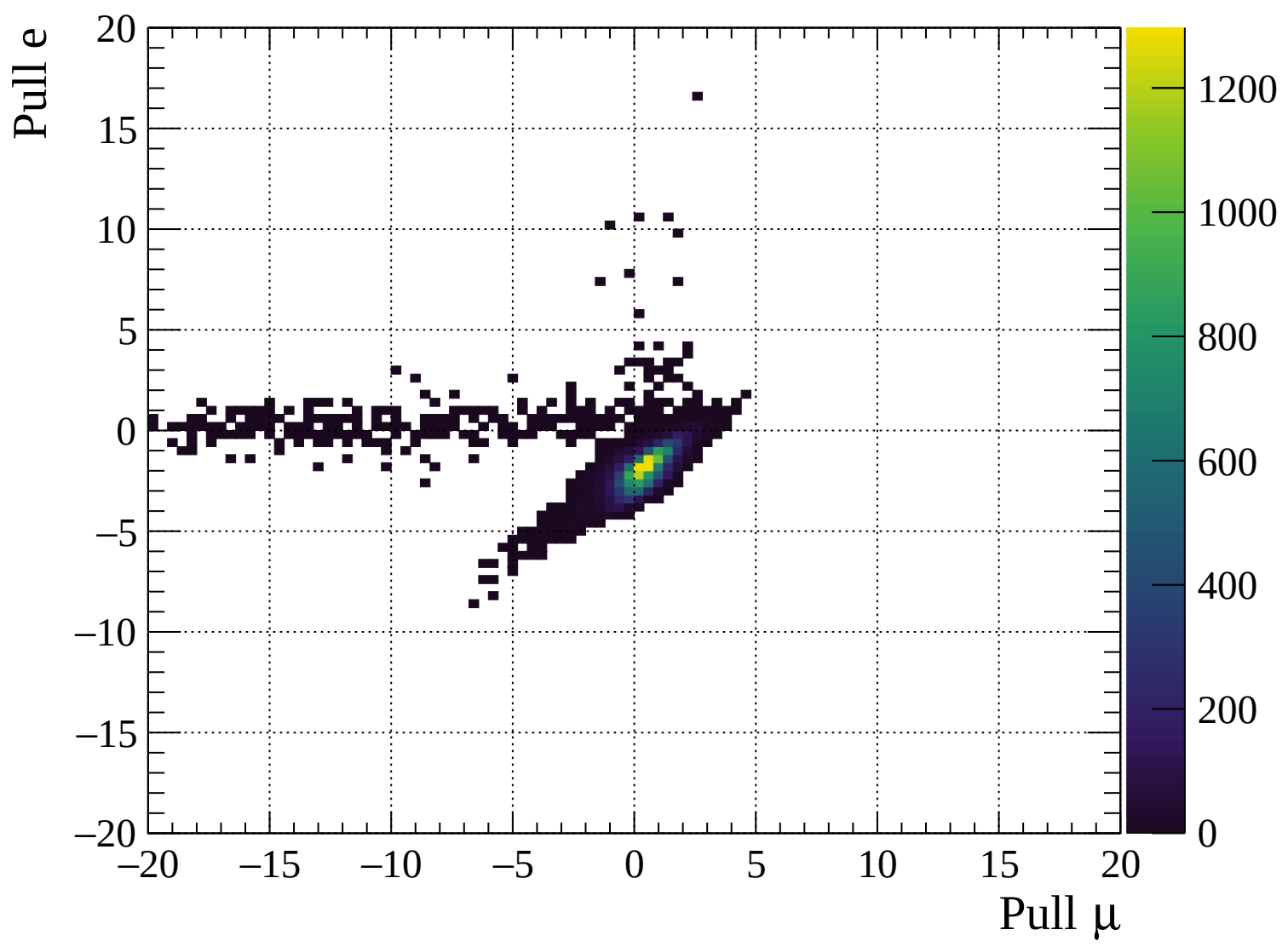
$e^{\pm}$

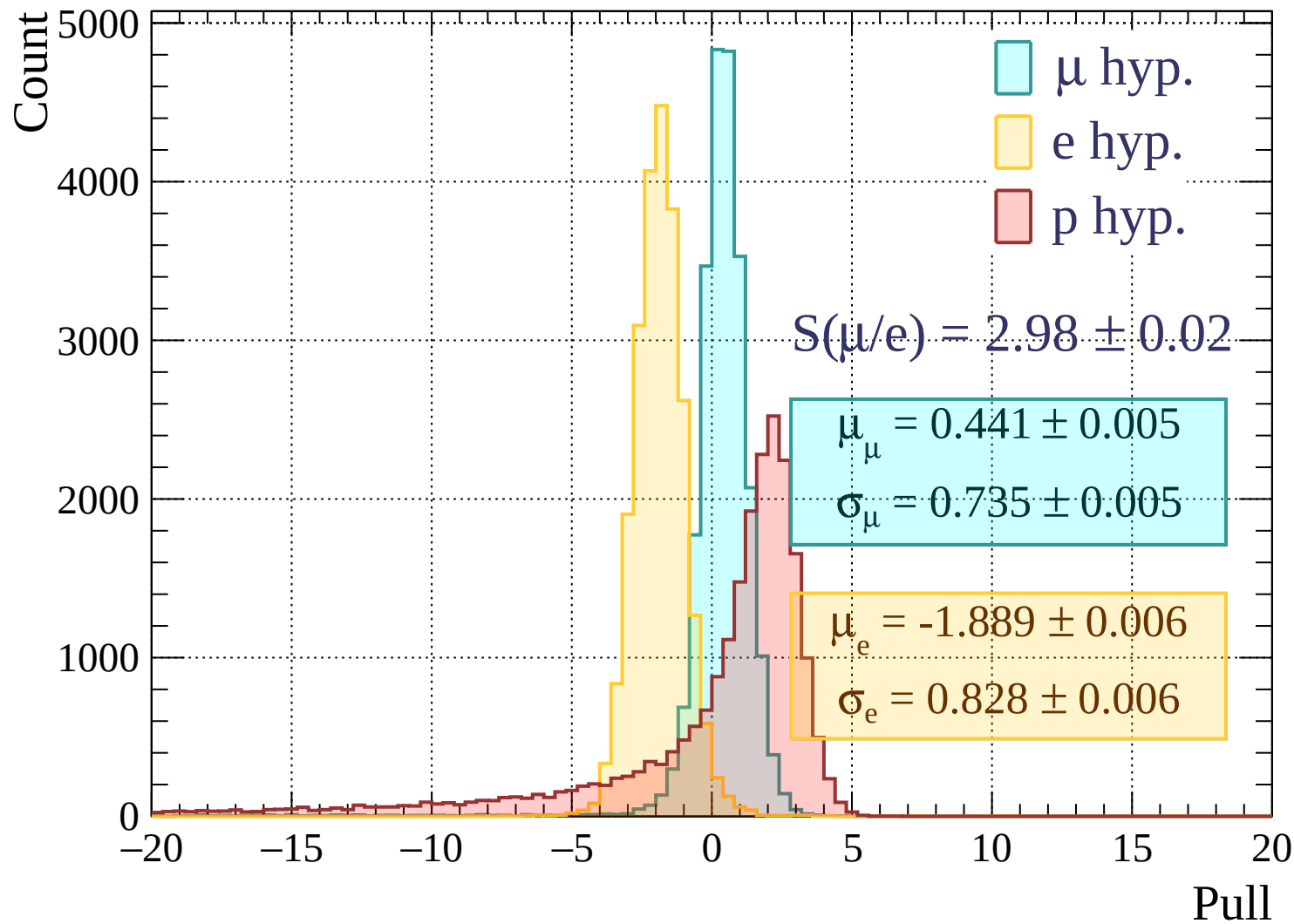


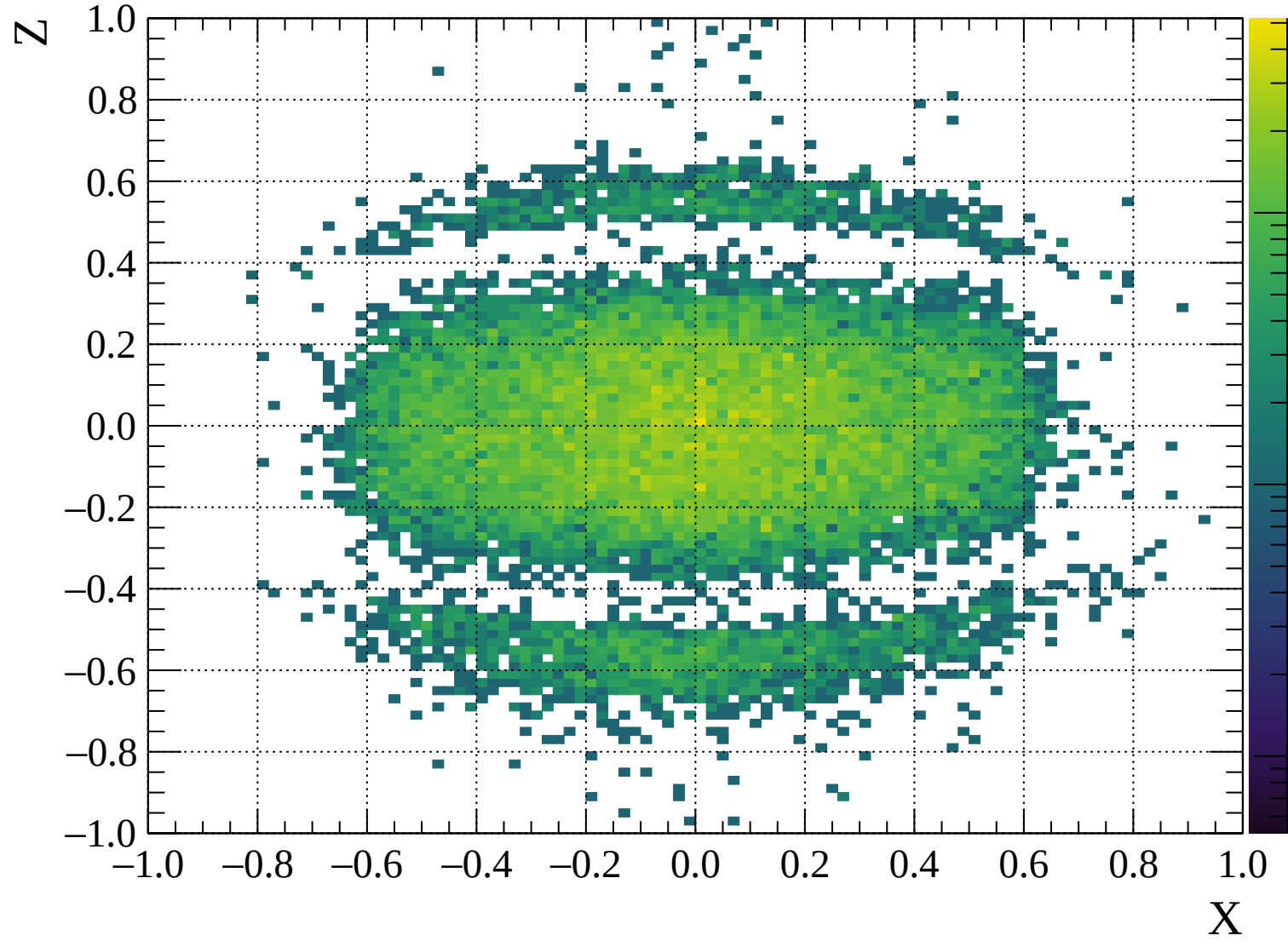
Pulls standard deviation

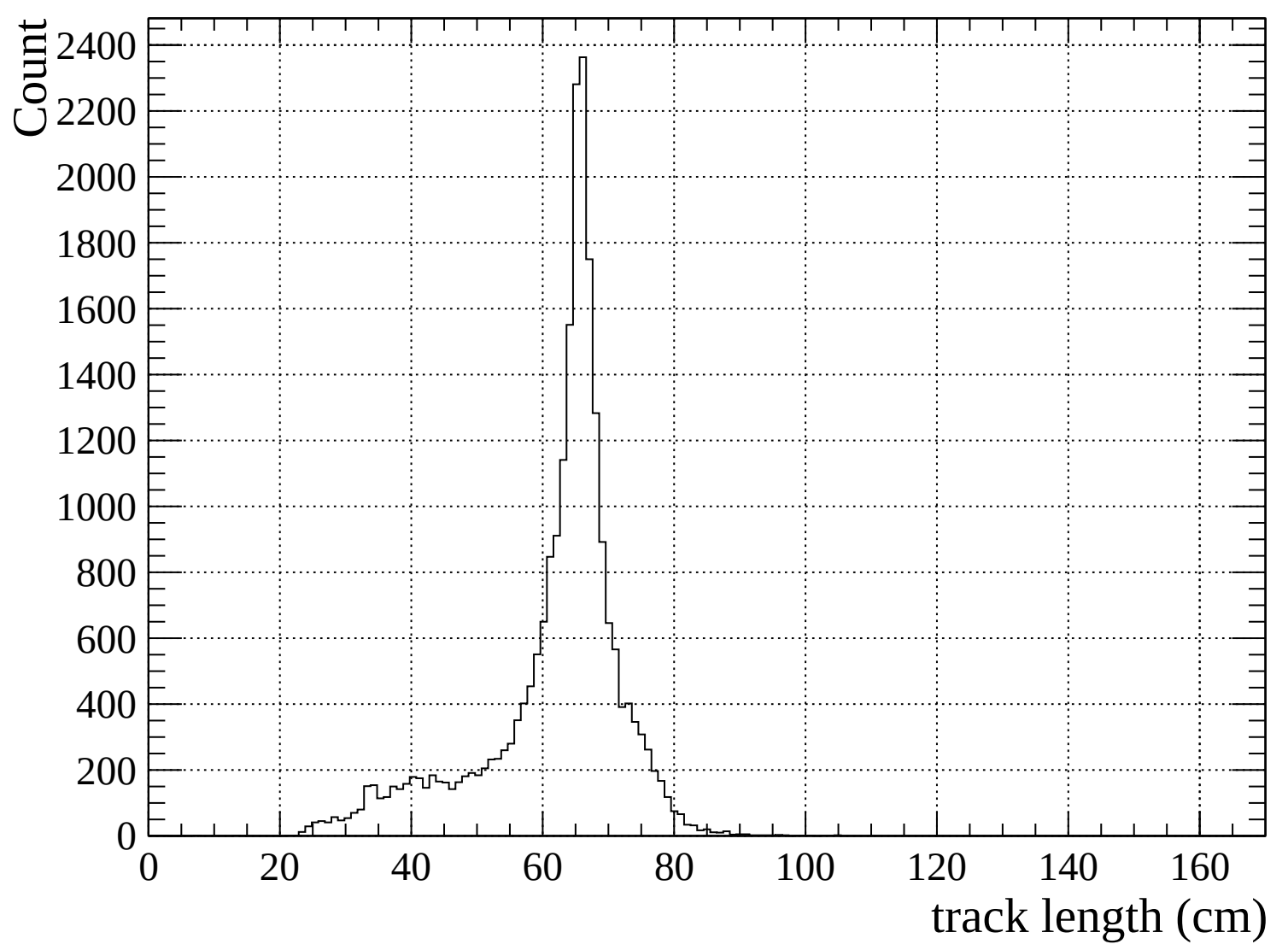


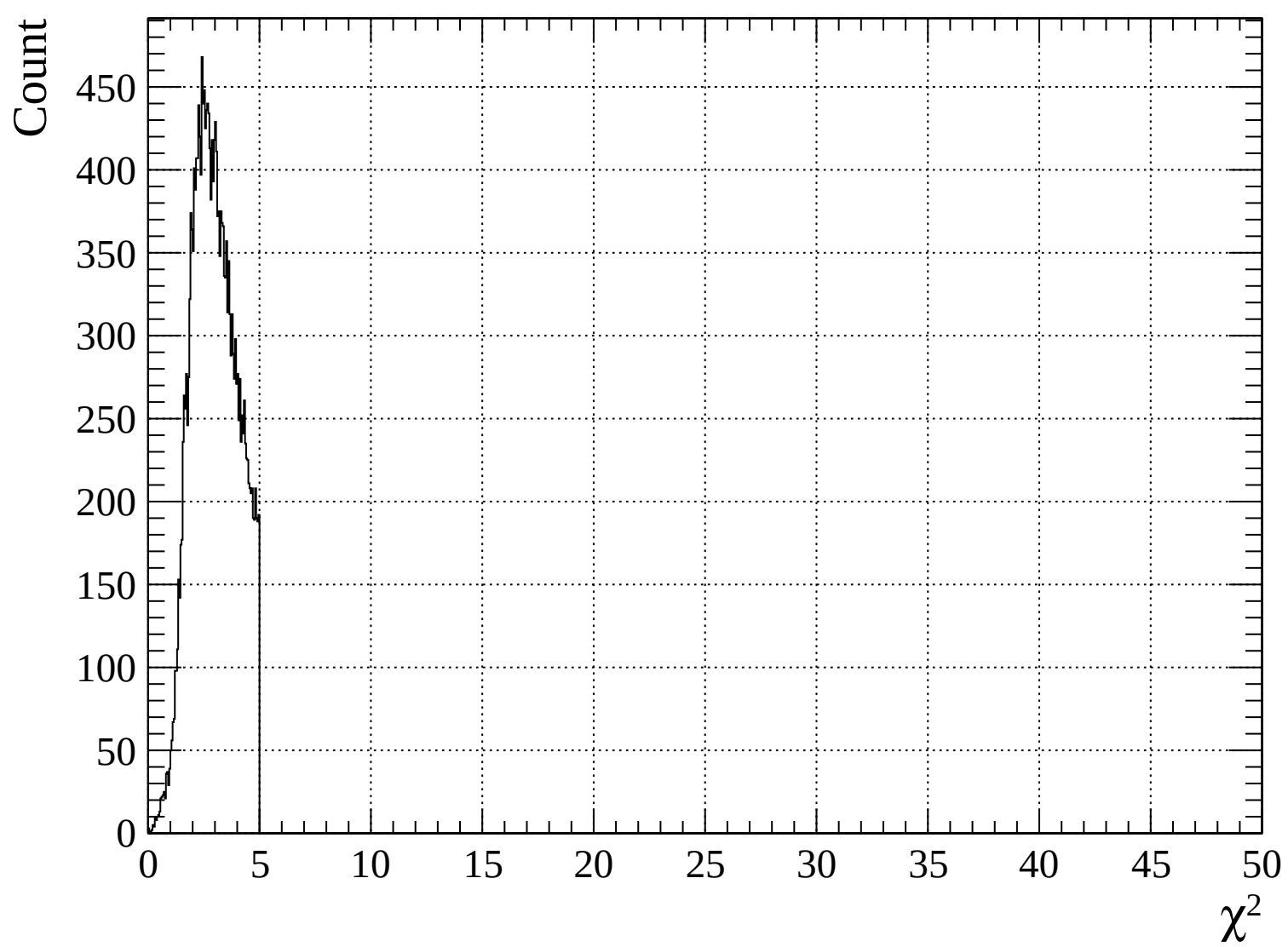


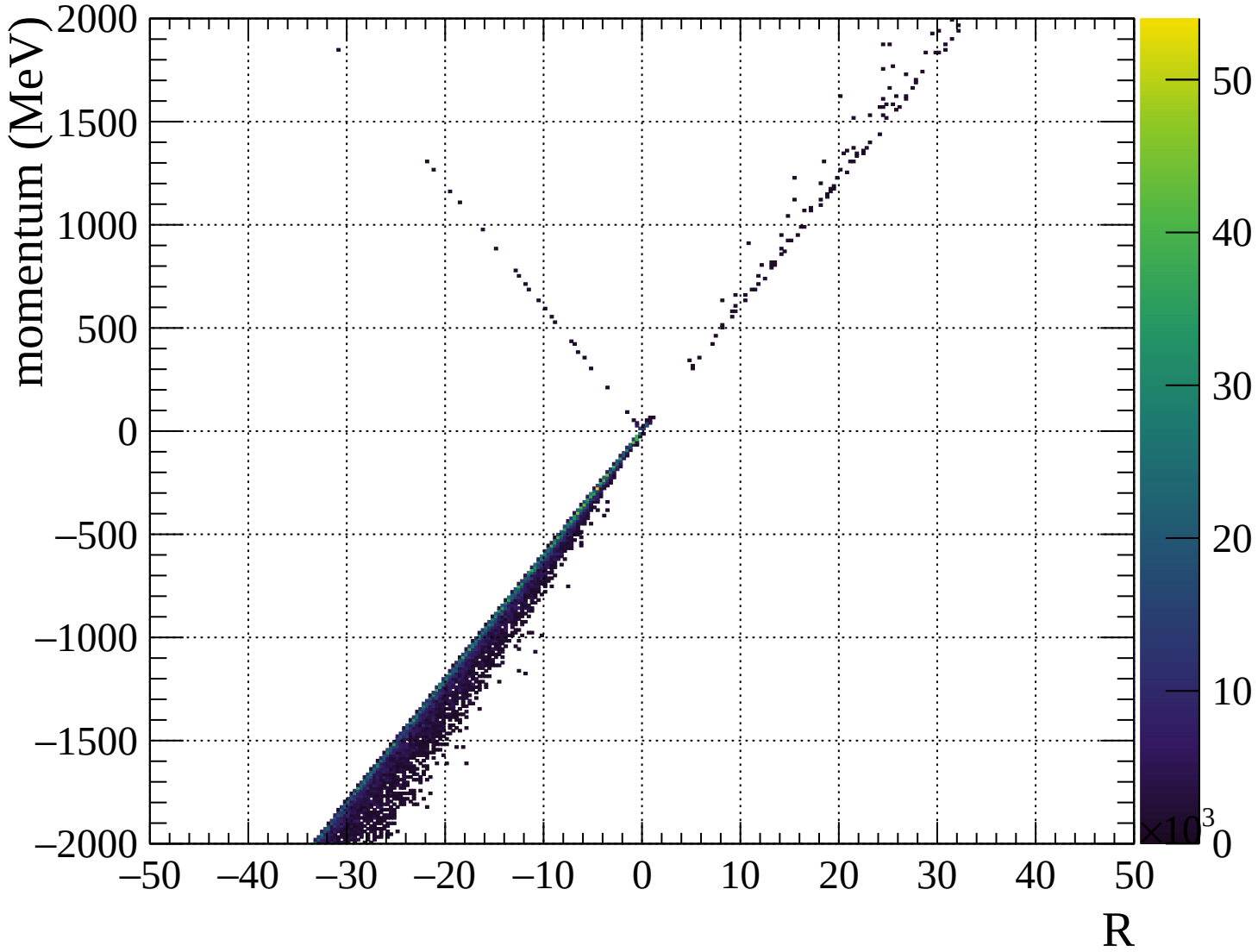


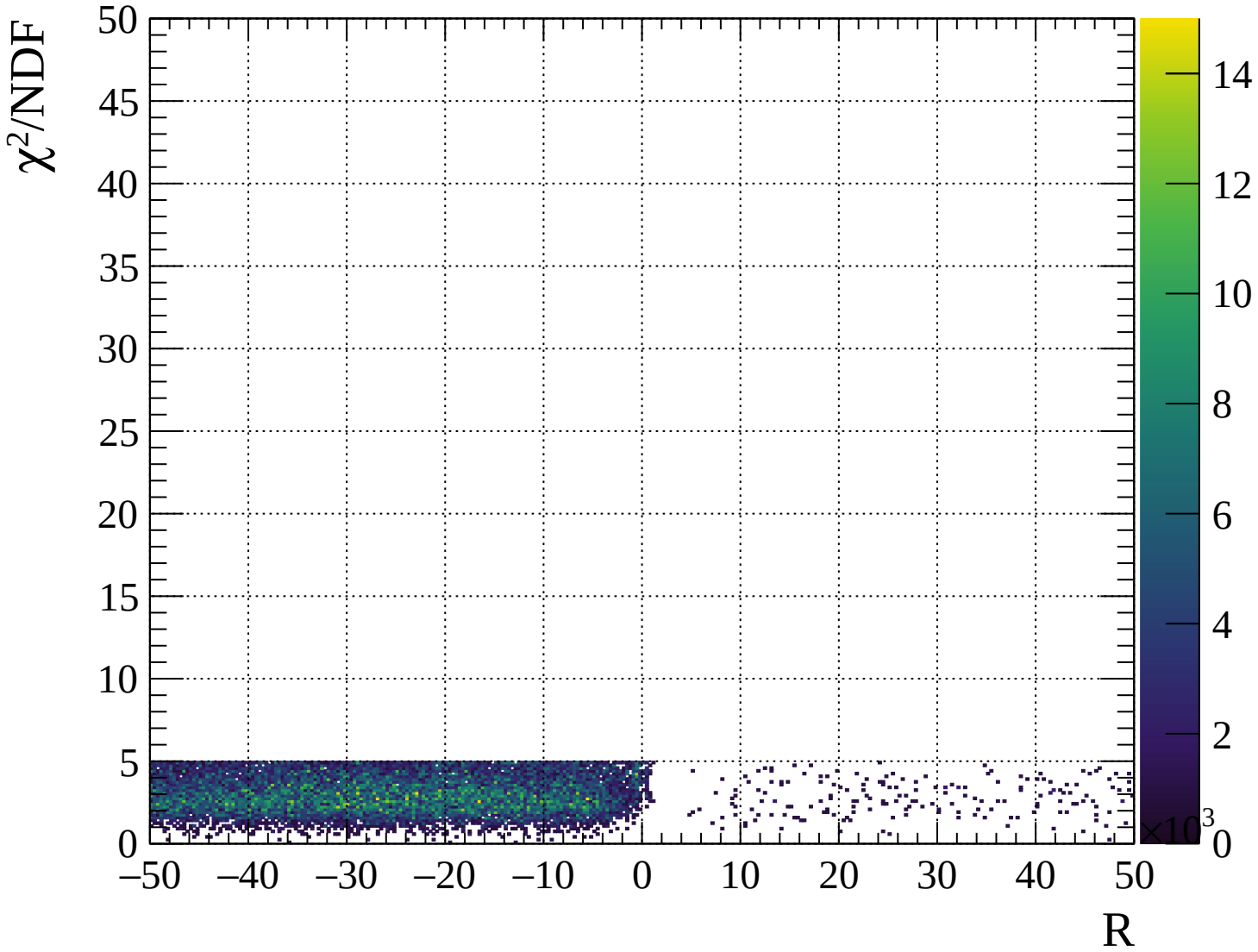






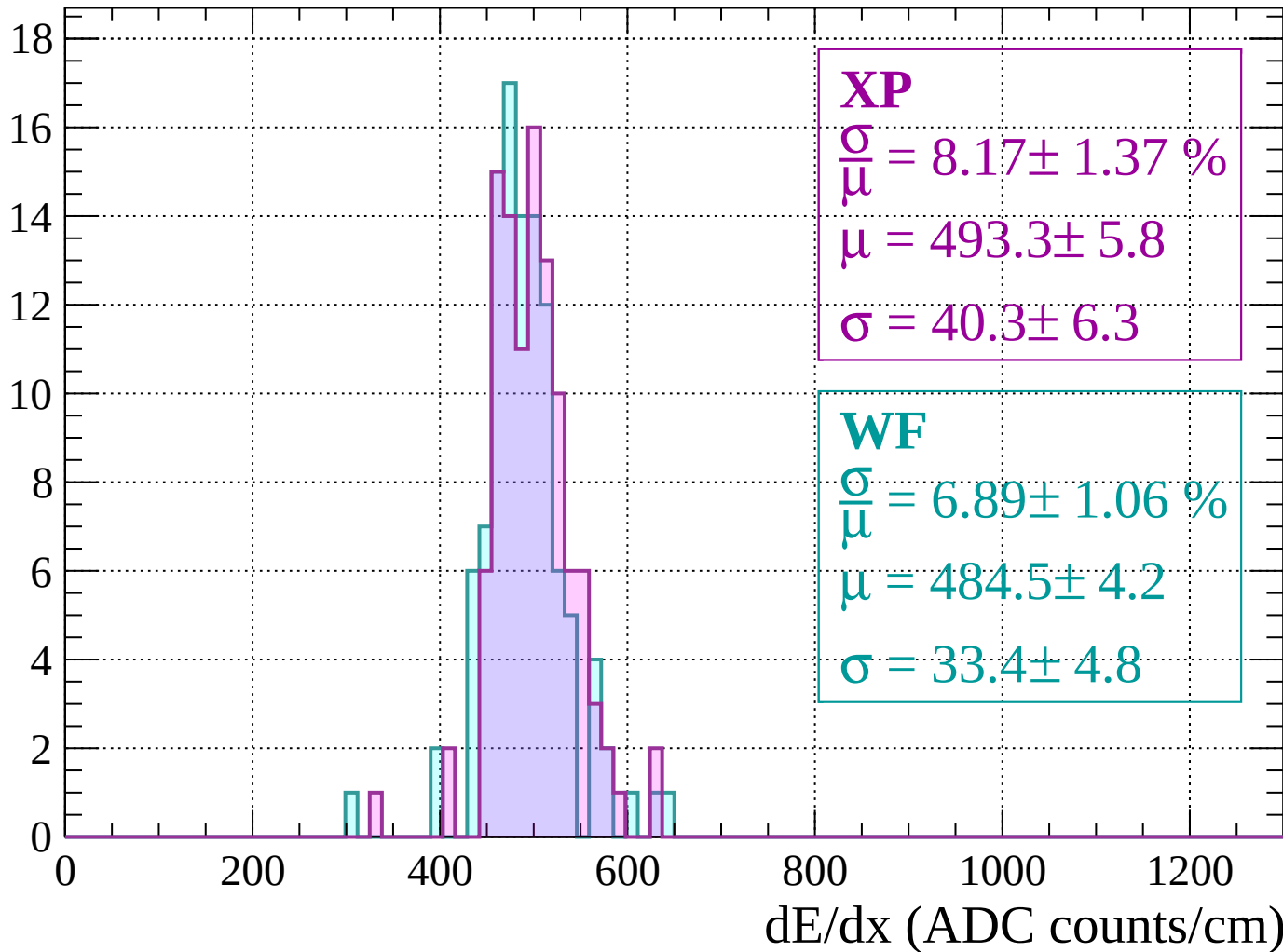






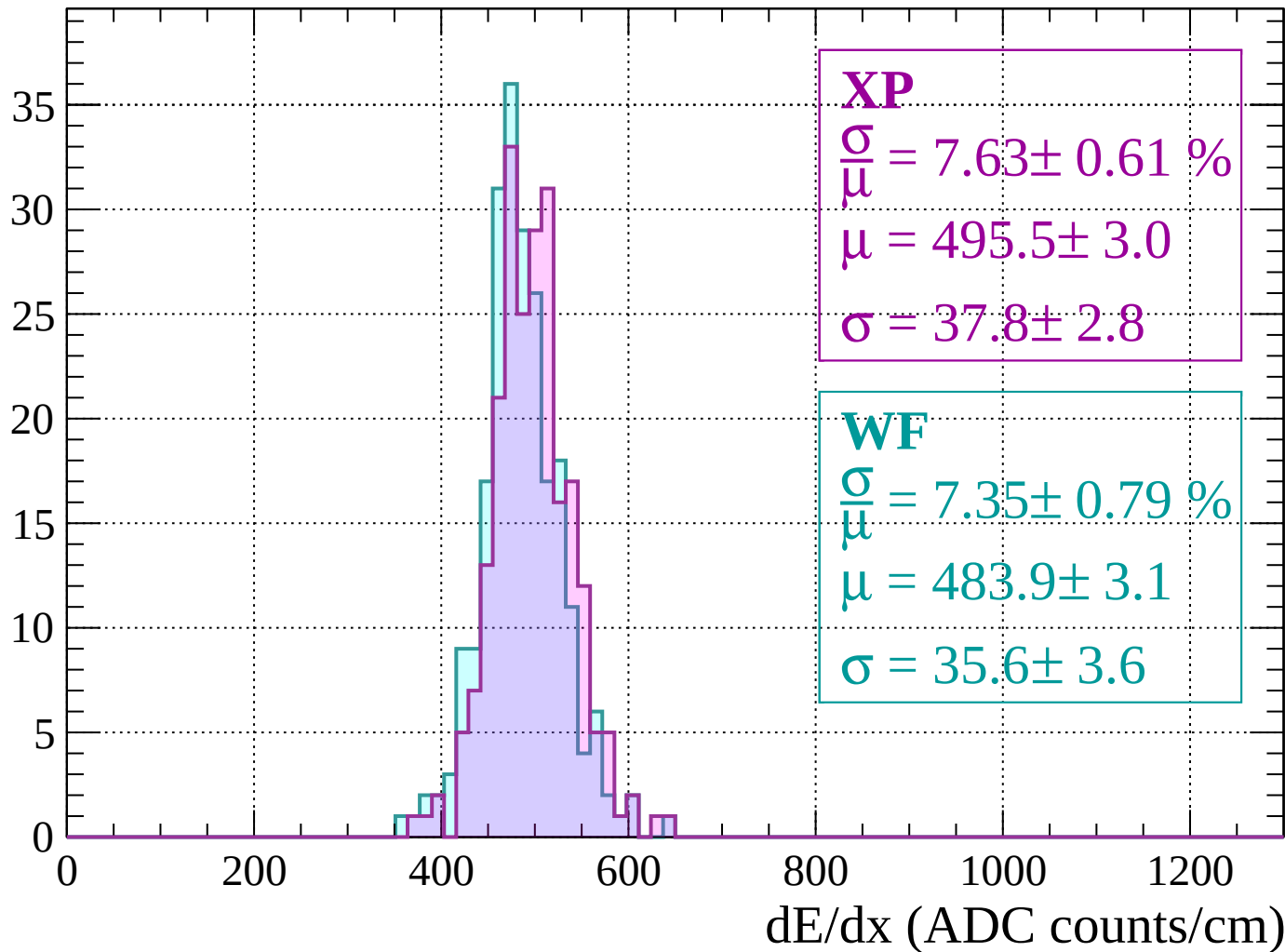
Energy loss | $-2000 < p < -1960$

Count



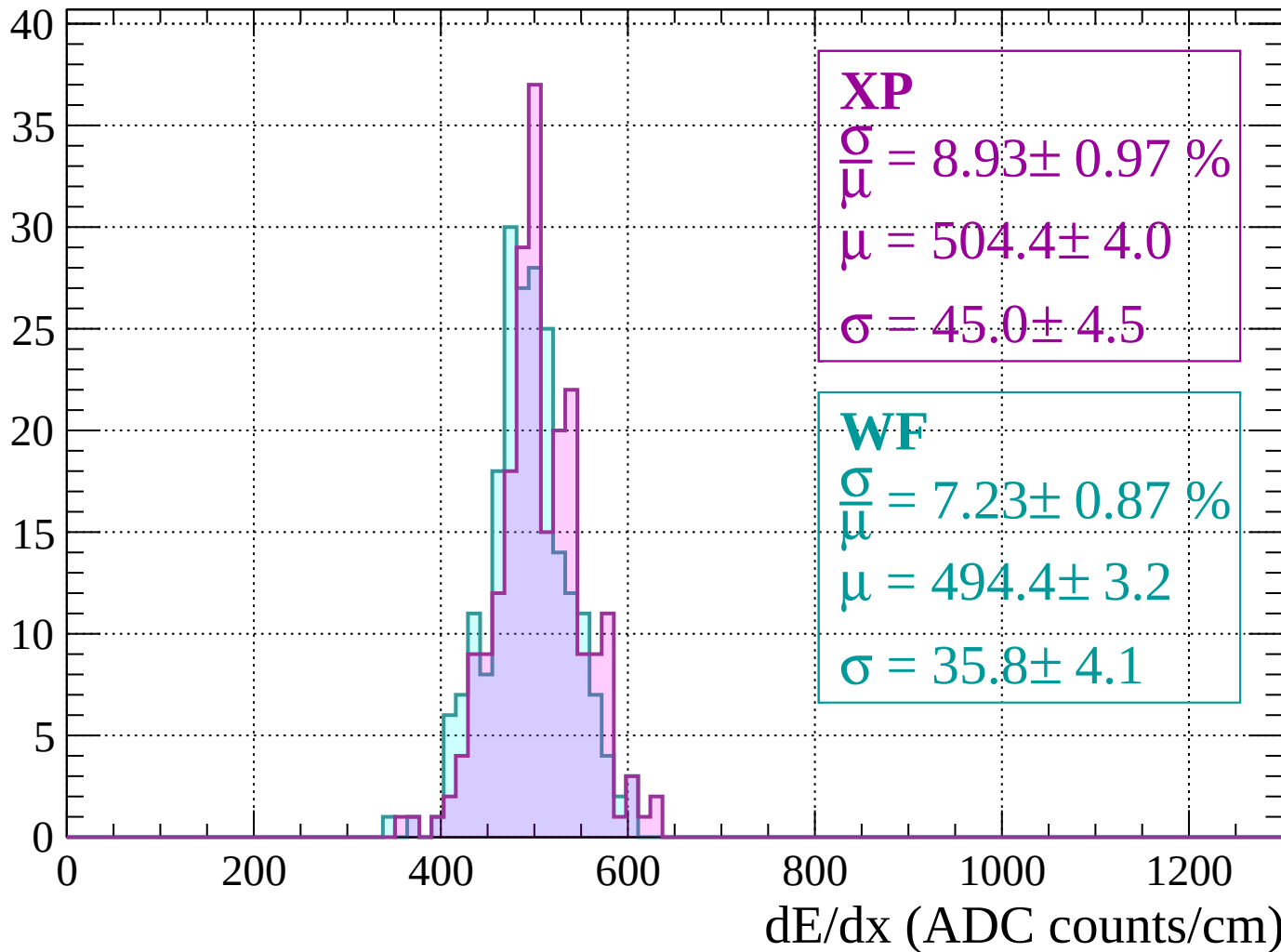
Energy loss | -1960 < p < -1920

Count



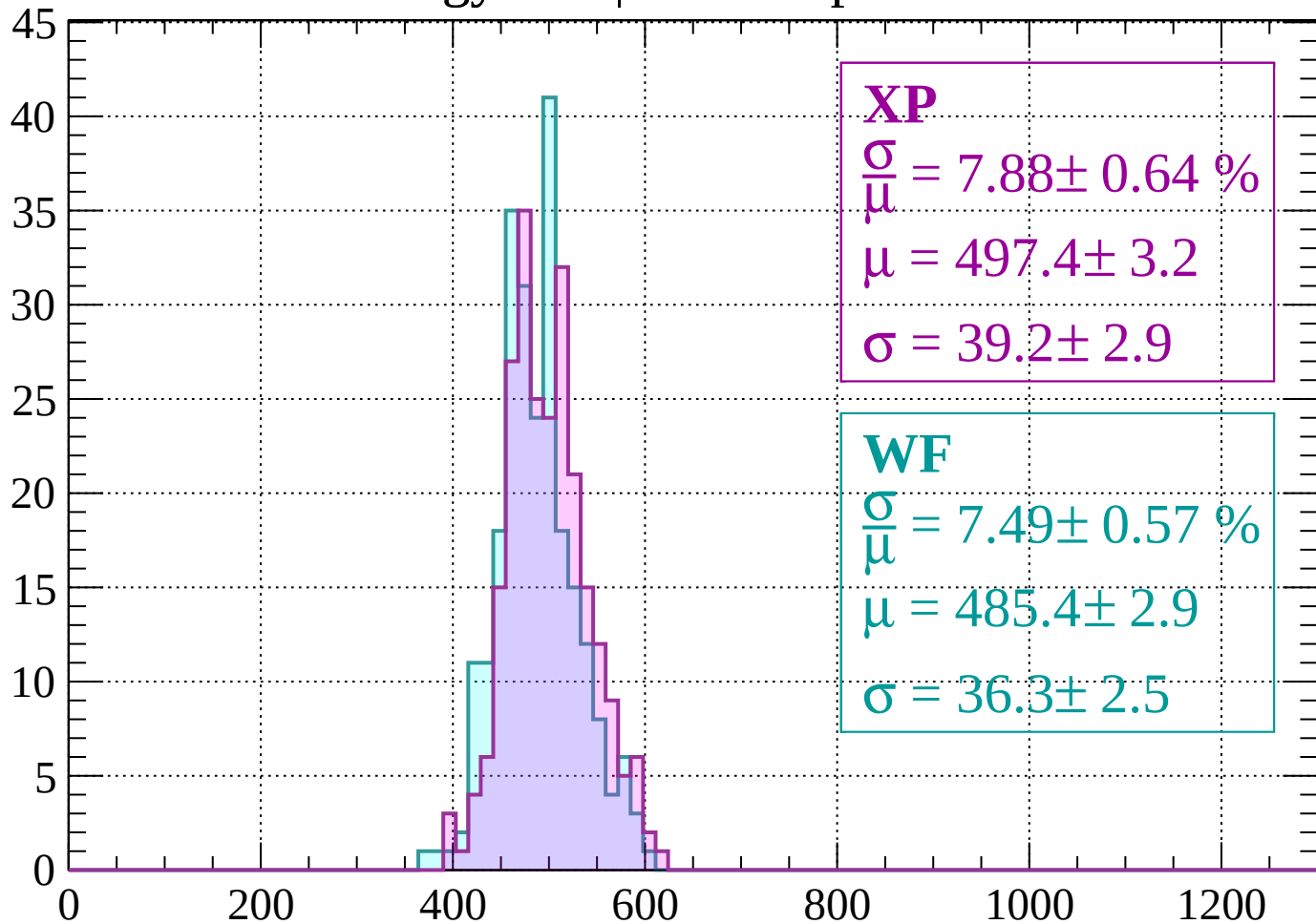
Energy loss | -1920 < p < -1880

Count



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.88 \pm 0.64 \%$$

$$\mu = 497.4 \pm 3.2$$

$$\sigma = 39.2 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 7.49 \pm 0.57 \%$$

$$\mu = 485.4 \pm 2.9$$

$$\sigma = 36.3 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count

XP

$$\frac{\sigma}{\mu} = 7.18 \pm 0.48 \%$$

$$\mu = 497.2 \pm 2.8$$

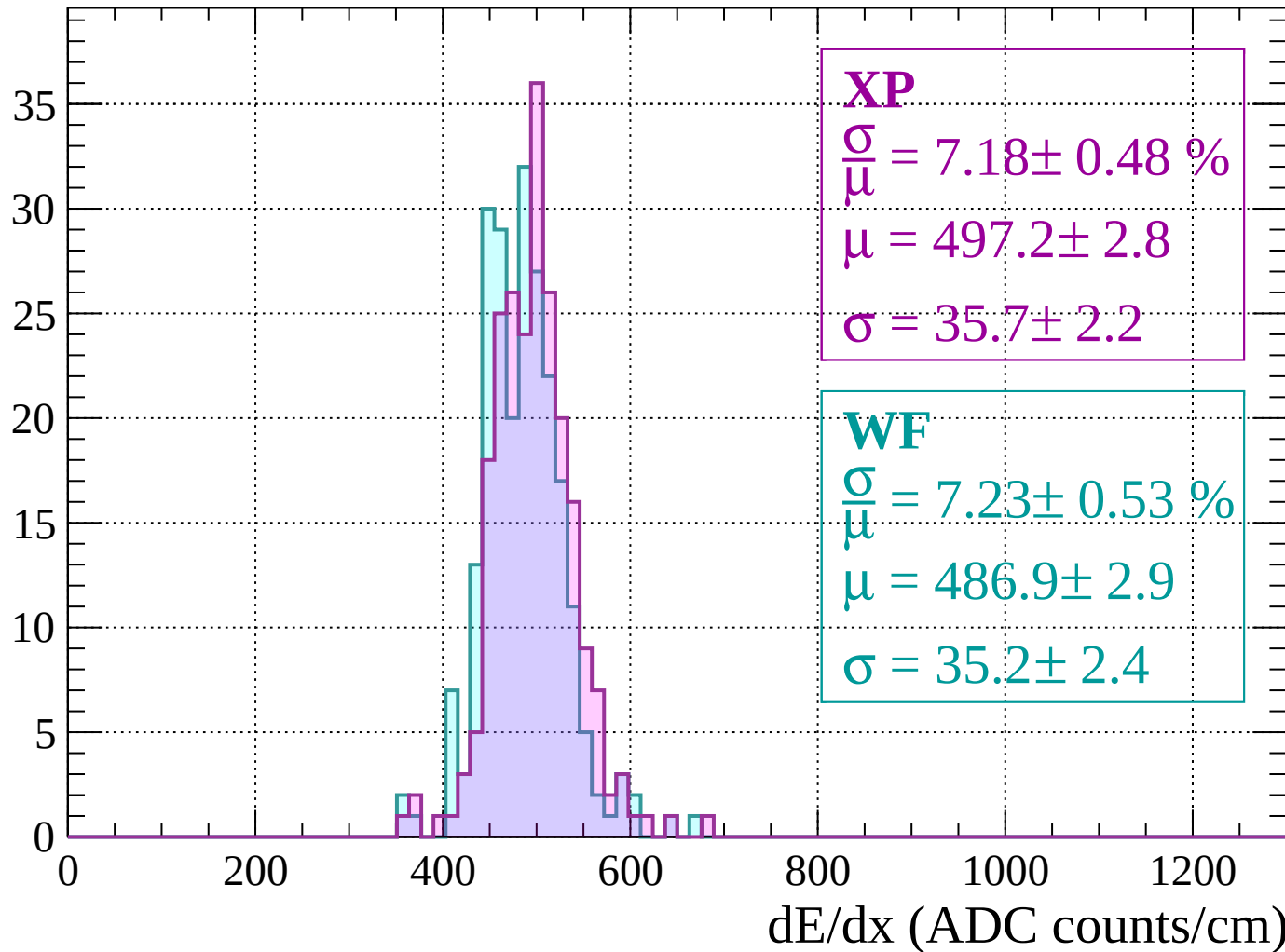
$$\sigma = 35.7 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.23 \pm 0.53 \%$$

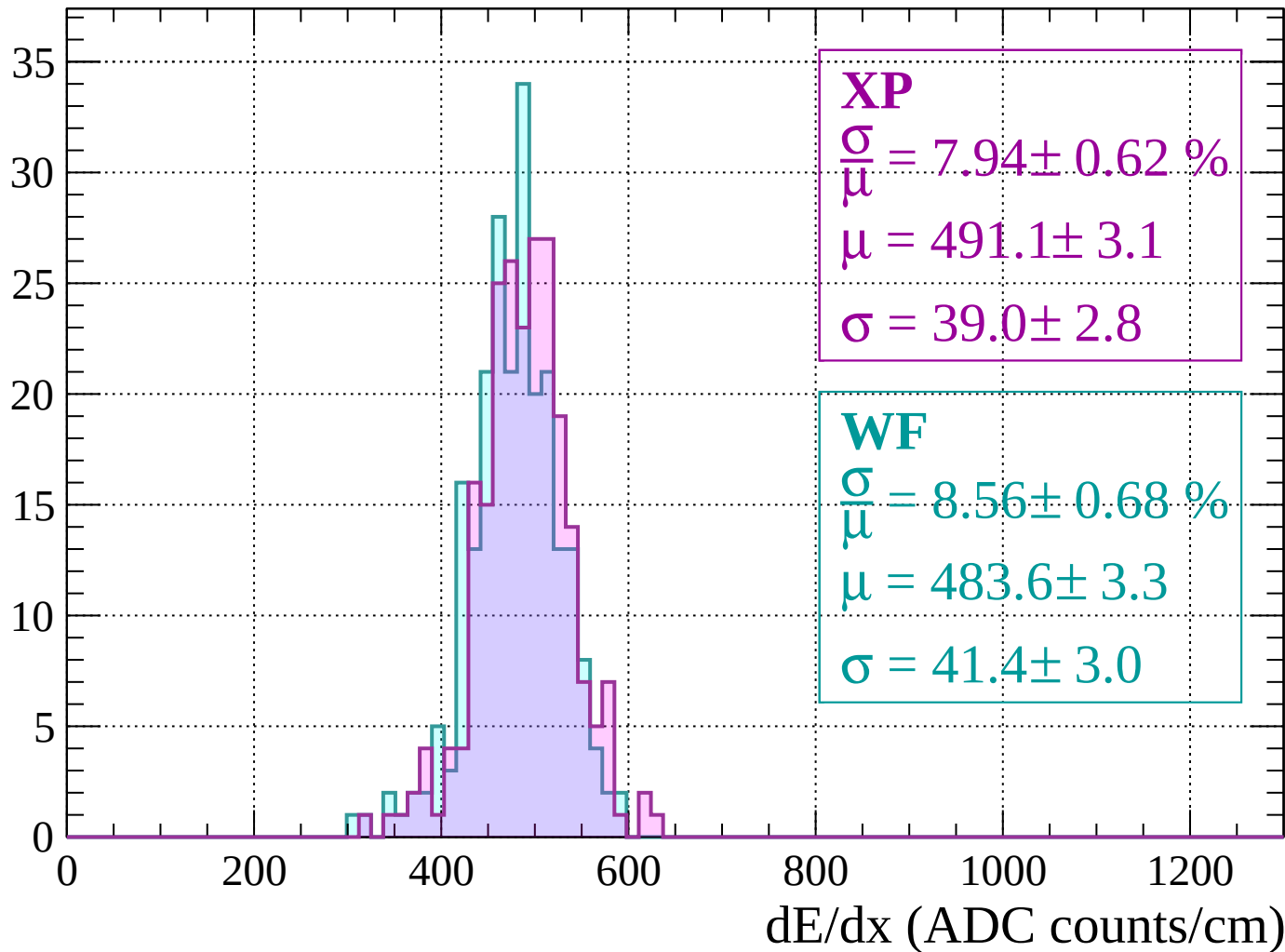
$$\mu = 486.9 \pm 2.9$$

$$\sigma = 35.2 \pm 2.4$$



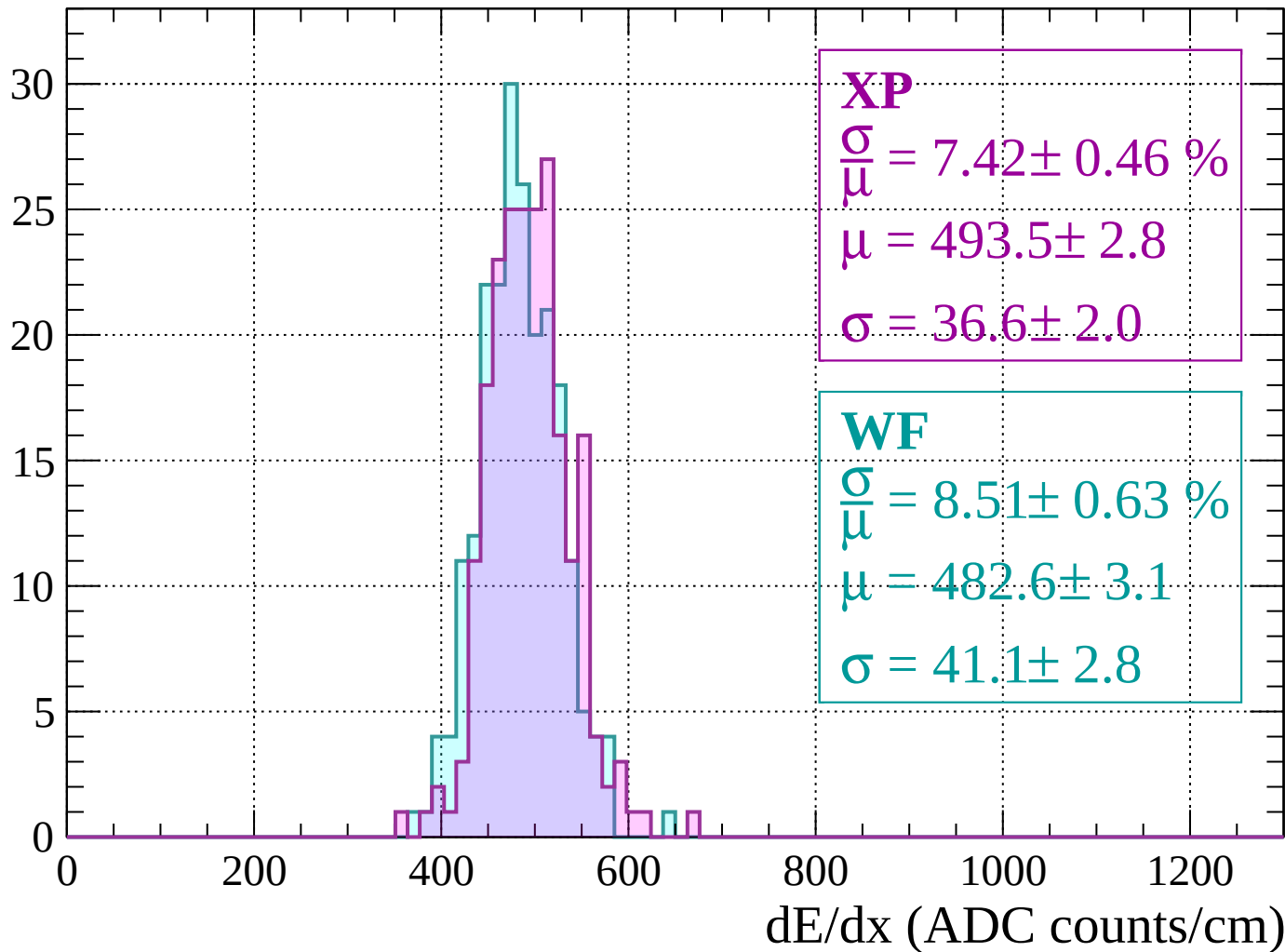
Energy loss | $-1800 < p < -1760$

Count



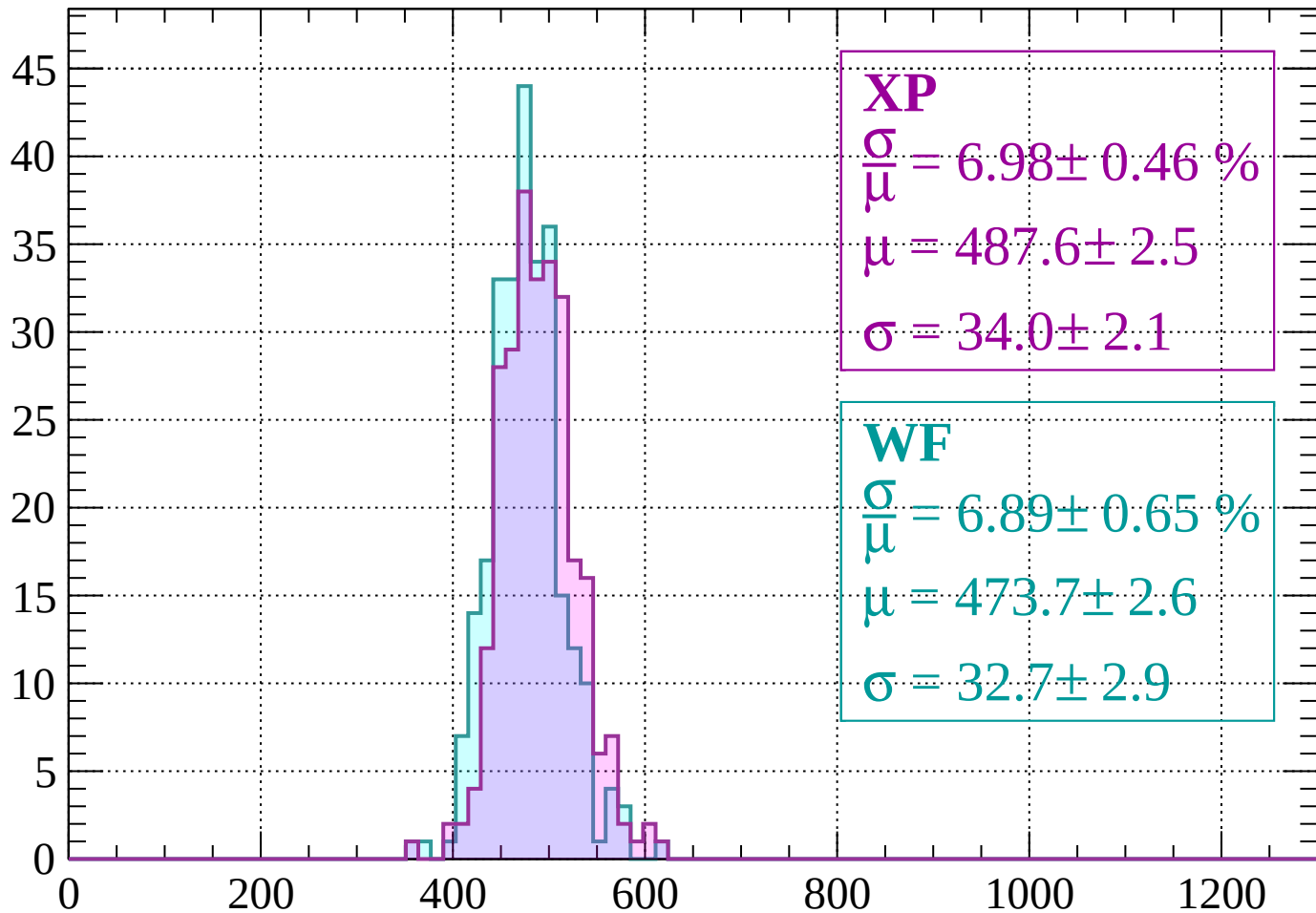
Energy loss | $-1760 < p < -1720$

Count



Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 6.98 \pm 0.46 \%$$

$$\mu = 487.6 \pm 2.5$$

$$\sigma = 34.0 \pm 2.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 6.89 \pm 0.65 \%$$

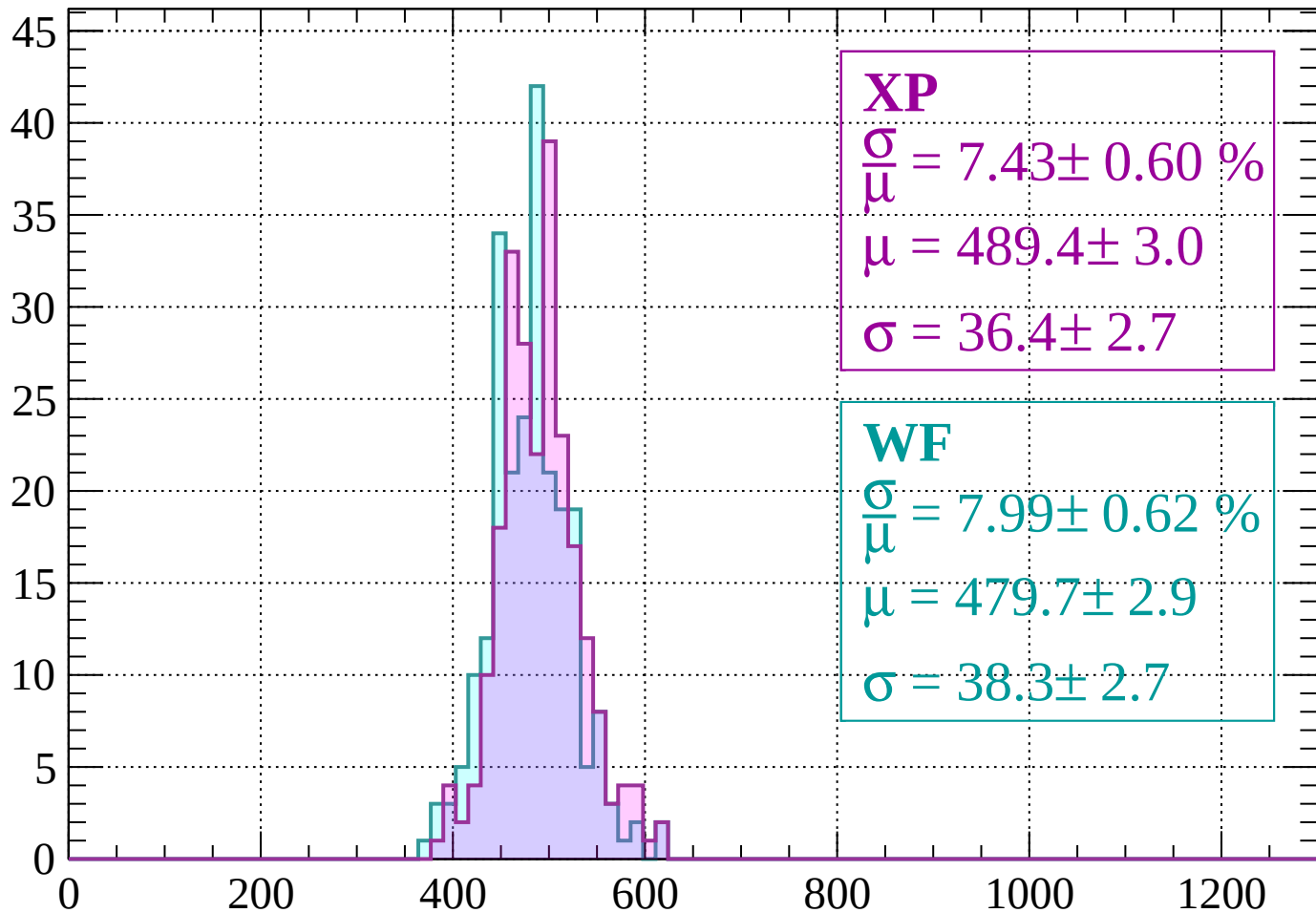
$$\mu = 473.7 \pm 2.6$$

$$\sigma = 32.7 \pm 2.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.60 \%$$

$$\mu = 489.4 \pm 3.0$$

$$\sigma = 36.4 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.99 \pm 0.62 \%$$

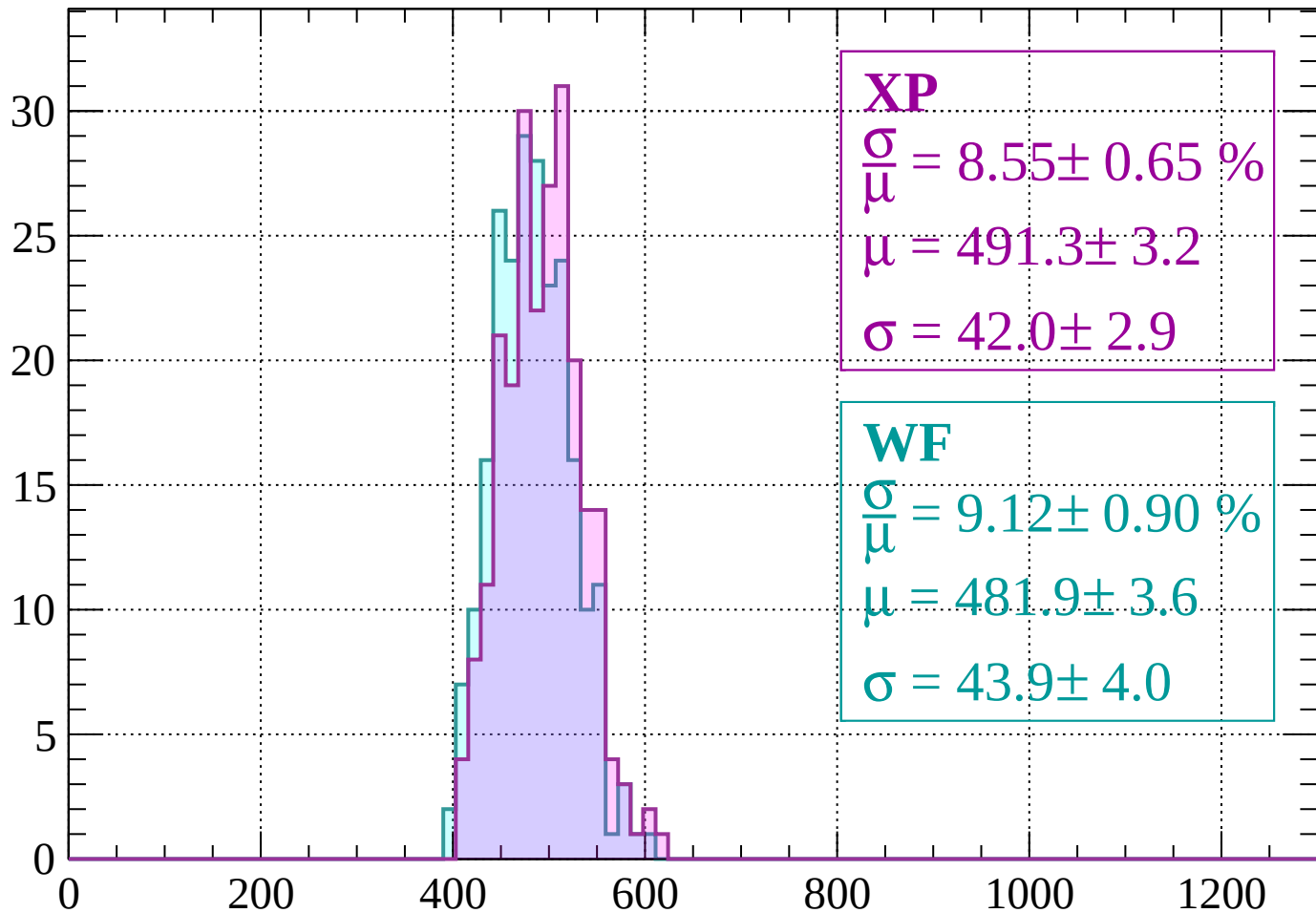
$$\mu = 479.7 \pm 2.9$$

$$\sigma = 38.3 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.65 \%$$

$$\mu = 491.3 \pm 3.2$$

$$\sigma = 42.0 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.90 \%$$

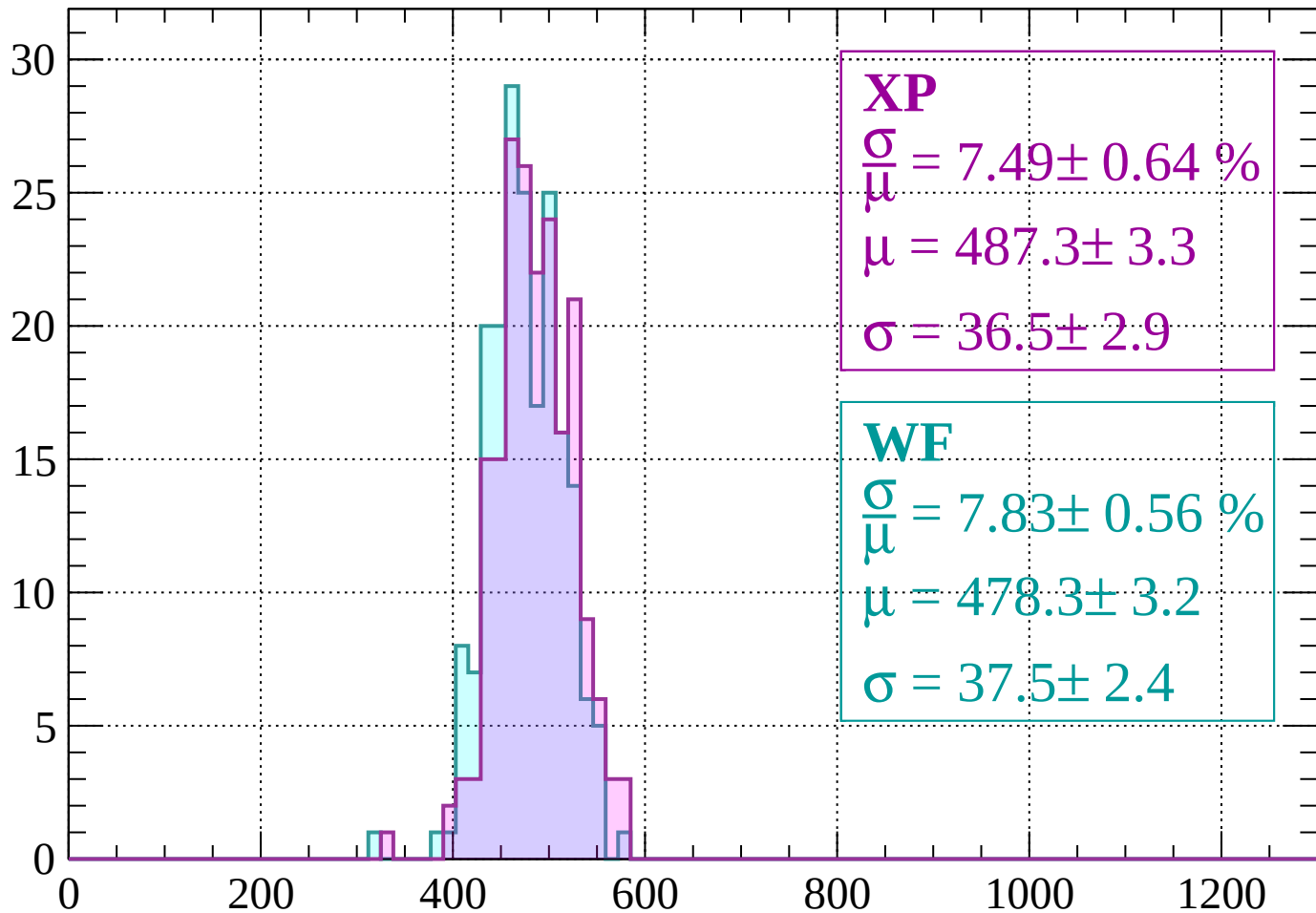
$$\mu = 481.9 \pm 3.6$$

$$\sigma = 43.9 \pm 4.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.64 \%$$

$$\mu = 487.3 \pm 3.3$$

$$\sigma = 36.5 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 7.83 \pm 0.56 \%$$

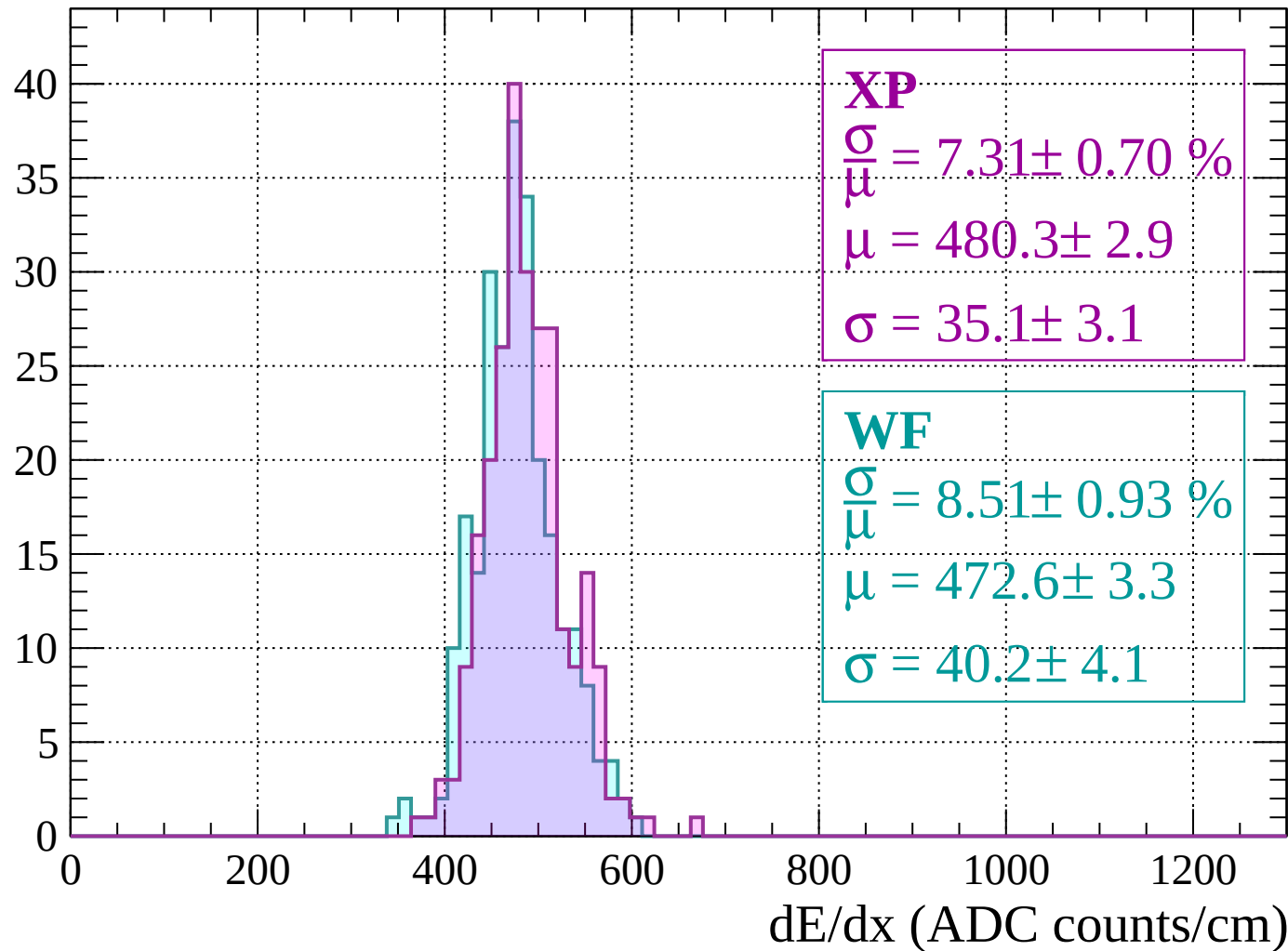
$$\mu = 478.3 \pm 3.2$$

$$\sigma = 37.5 \pm 2.4$$

dE/dx (ADC counts/cm)

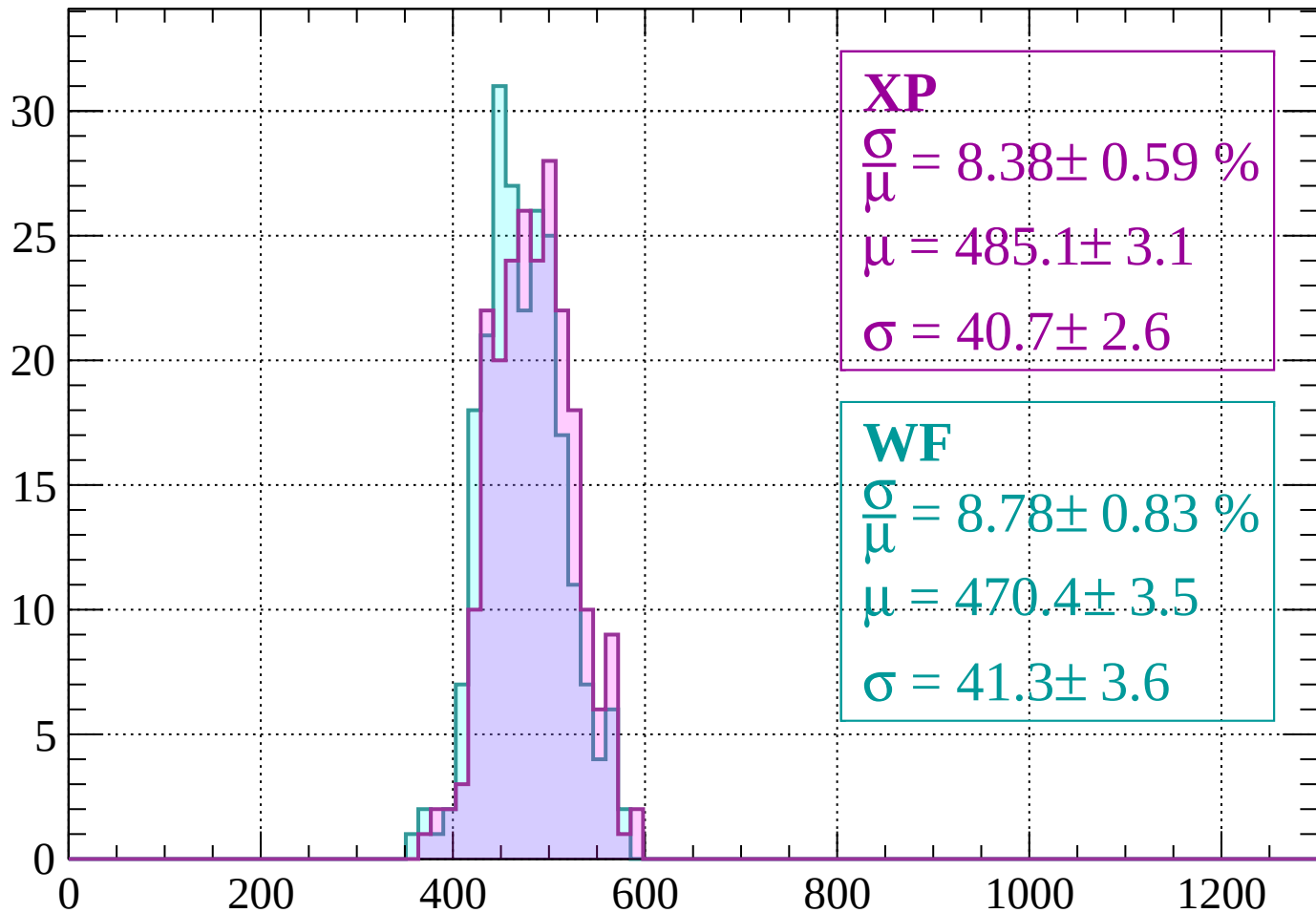
Energy loss | $-1560 < p < -1520$

Count



Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.59 \%$$

$$\mu = 485.1 \pm 3.1$$

$$\sigma = 40.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.83 \%$$

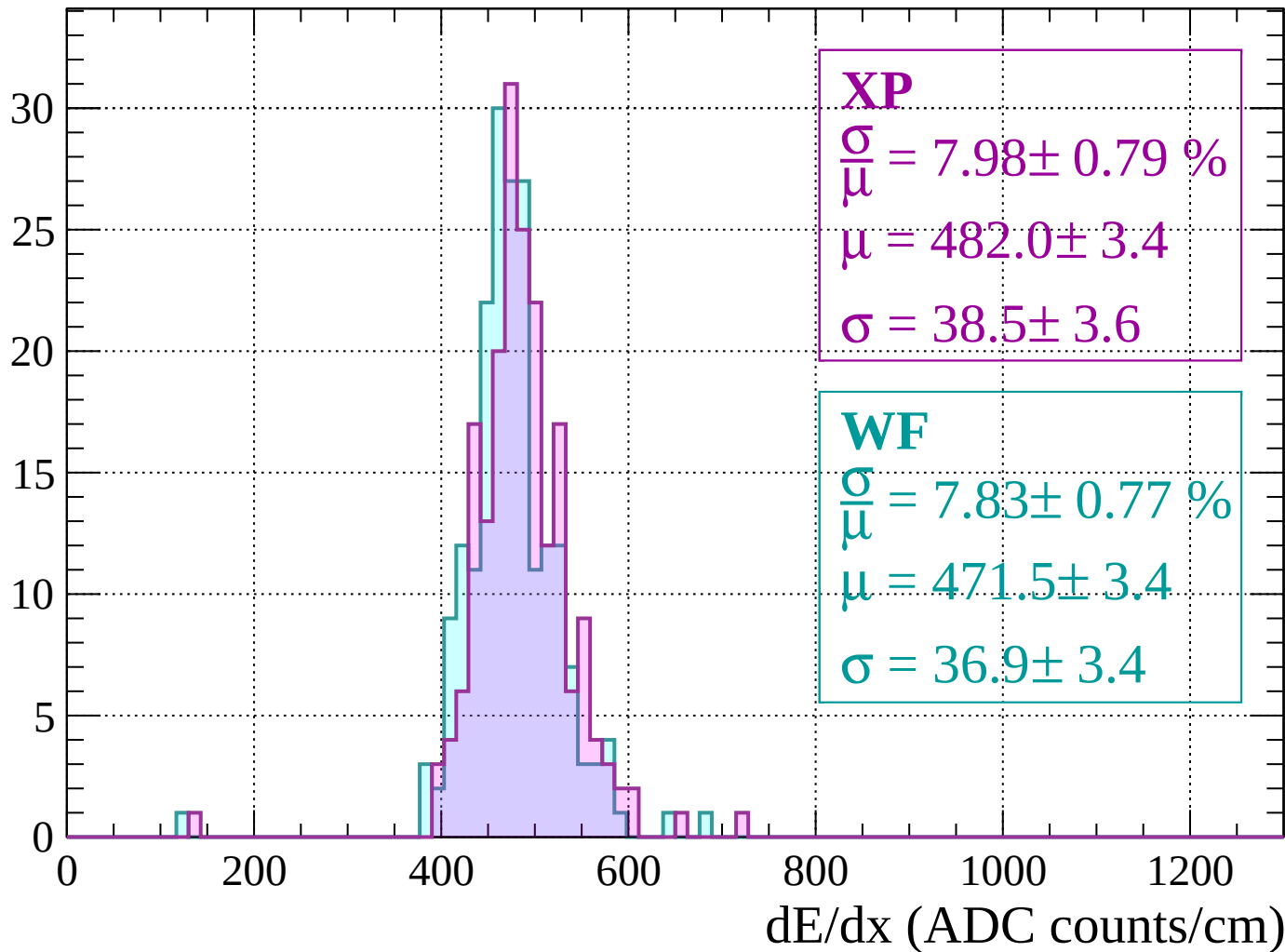
$$\mu = 470.4 \pm 3.5$$

$$\sigma = 41.3 \pm 3.6$$

dE/dx (ADC counts/cm)

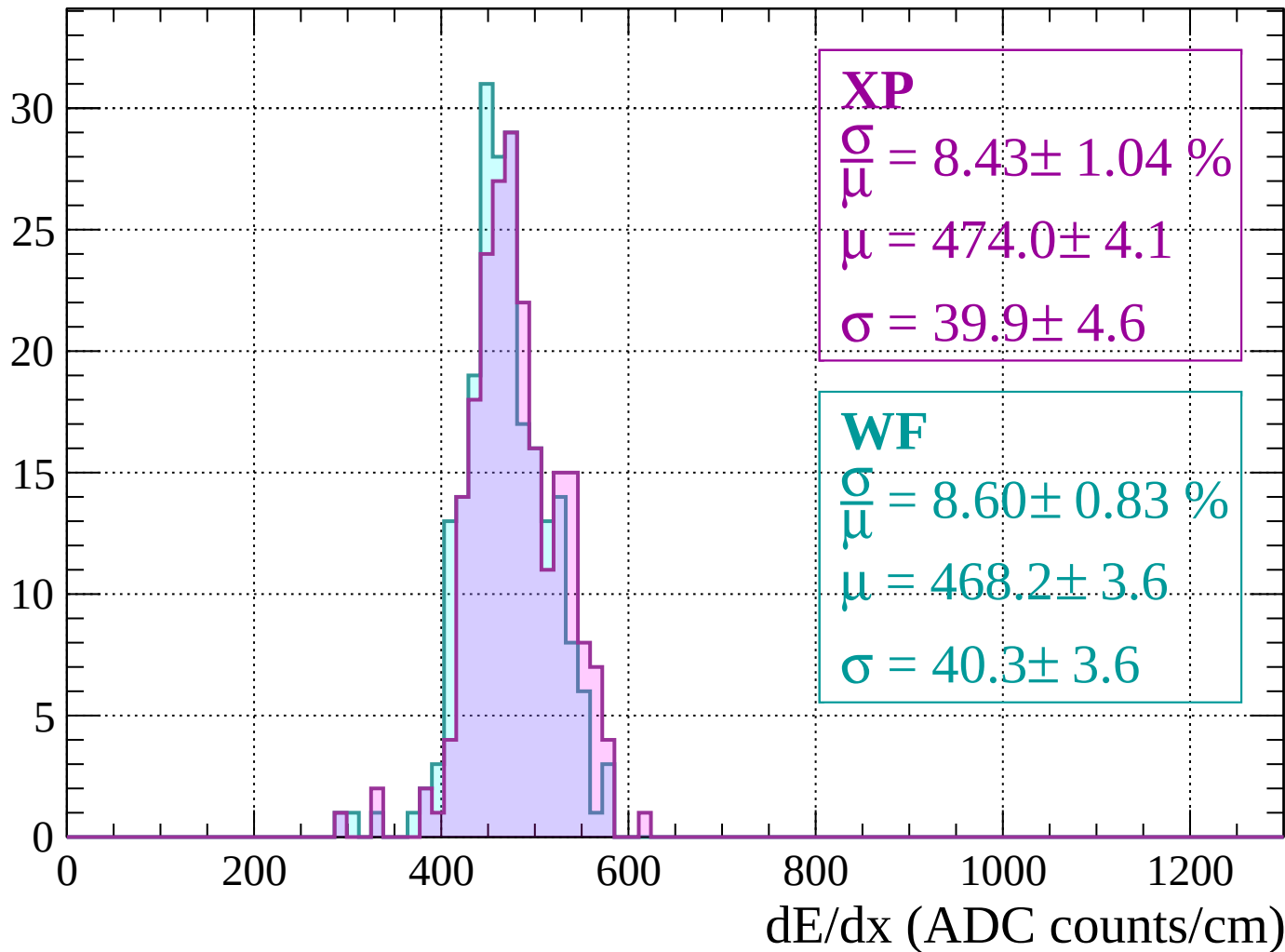
Energy loss | $-1480 < p < -1440$

Count



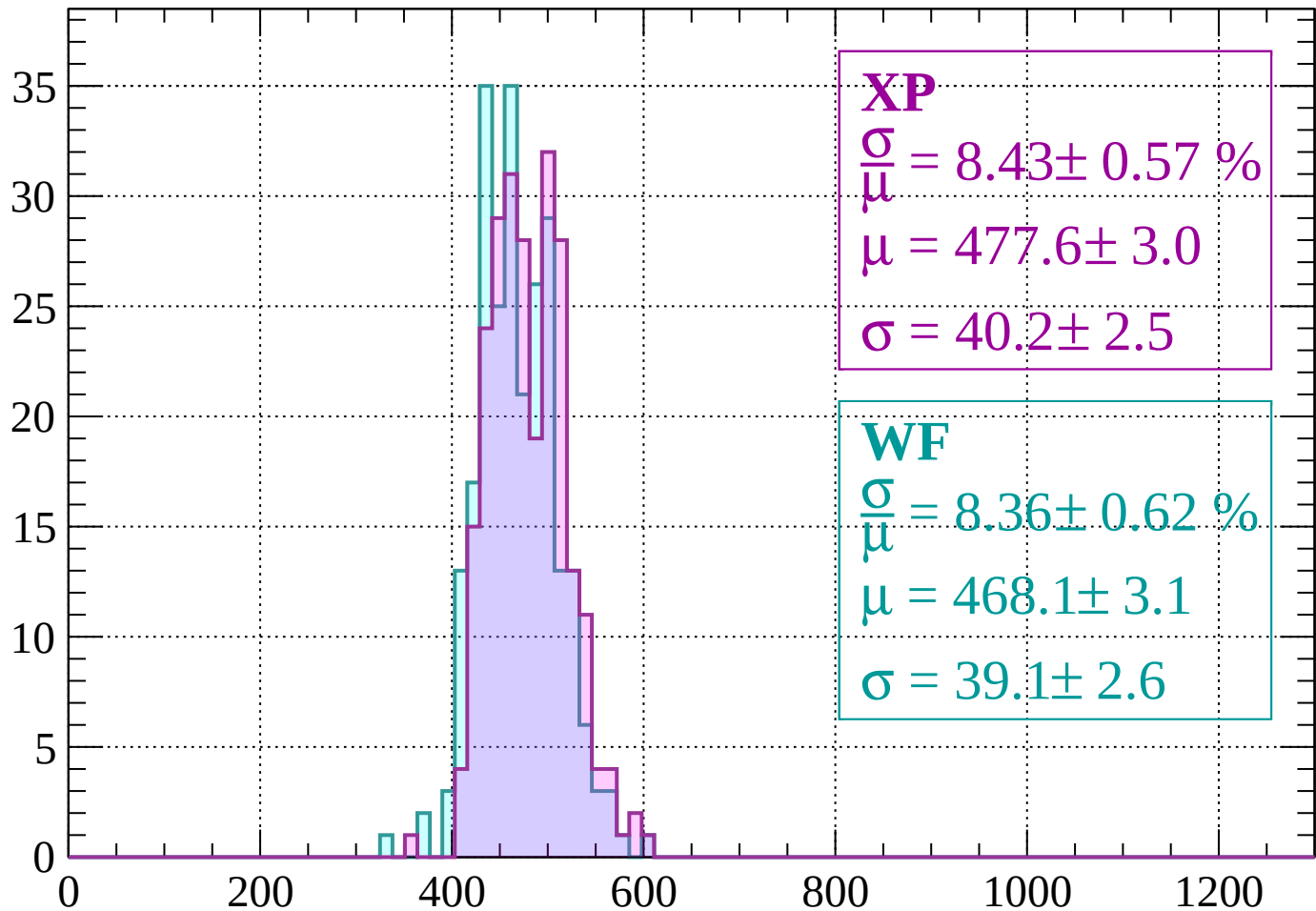
Energy loss | $-1440 < p < -1400$

Count



Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.43 \pm 0.57 \%$$

$$\mu = 477.6 \pm 3.0$$

$$\sigma = 40.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.36 \pm 0.62 \%$$

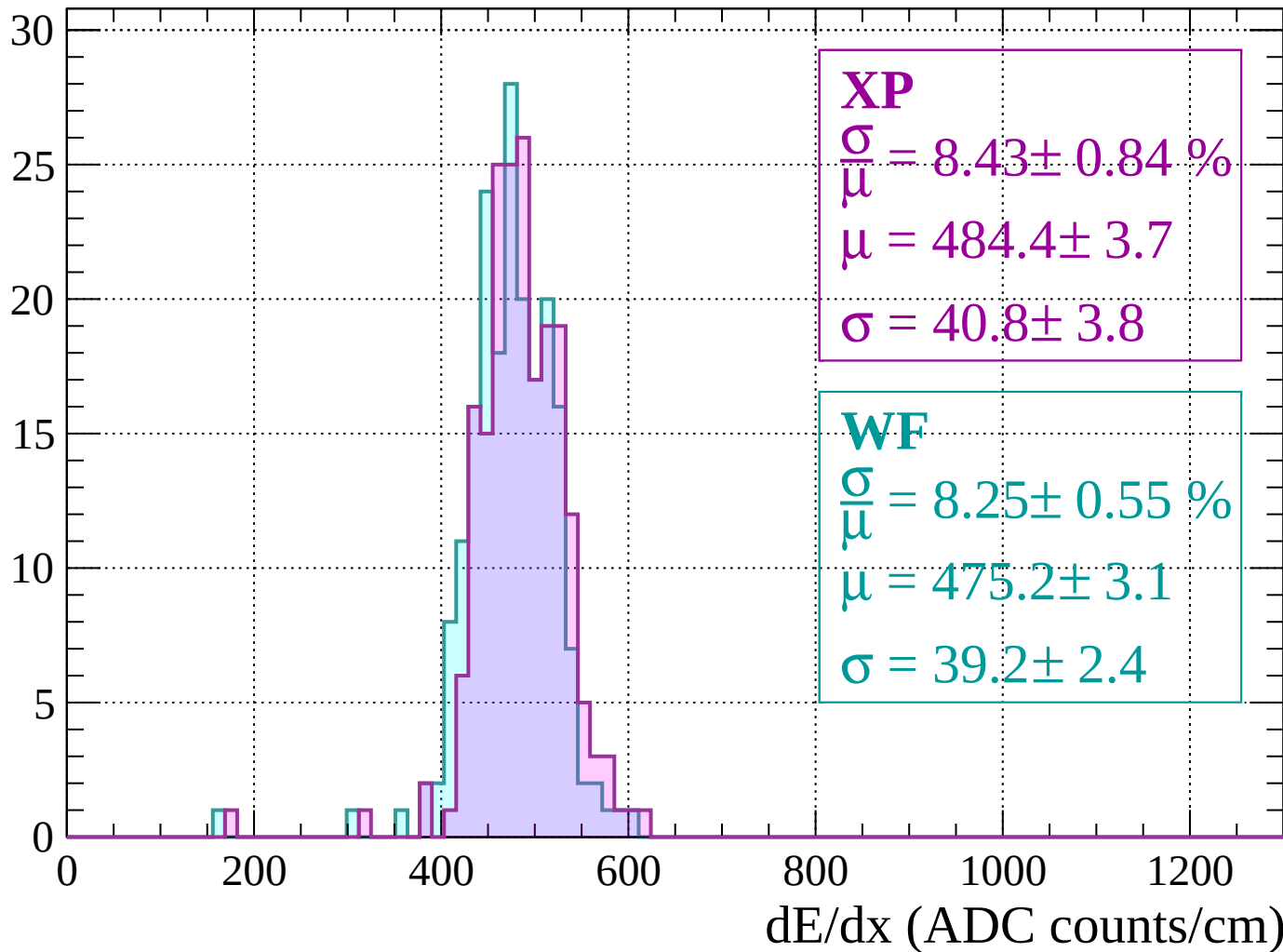
$$\mu = 468.1 \pm 3.1$$

$$\sigma = 39.1 \pm 2.6$$

dE/dx (ADC counts/cm)

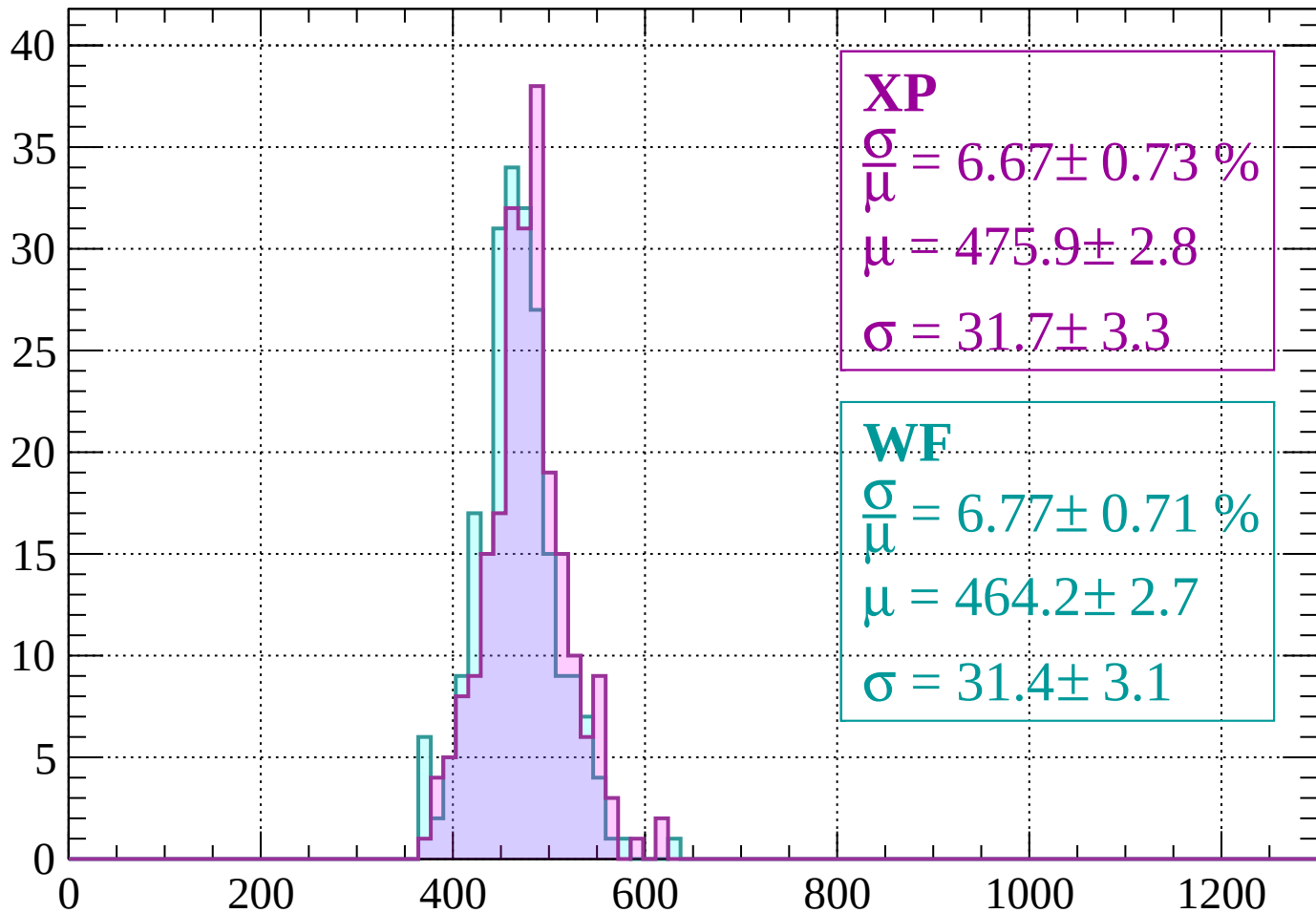
Energy loss | $-1360 < p < -1320$

Count



Energy loss | $-1320 < p < -1280$

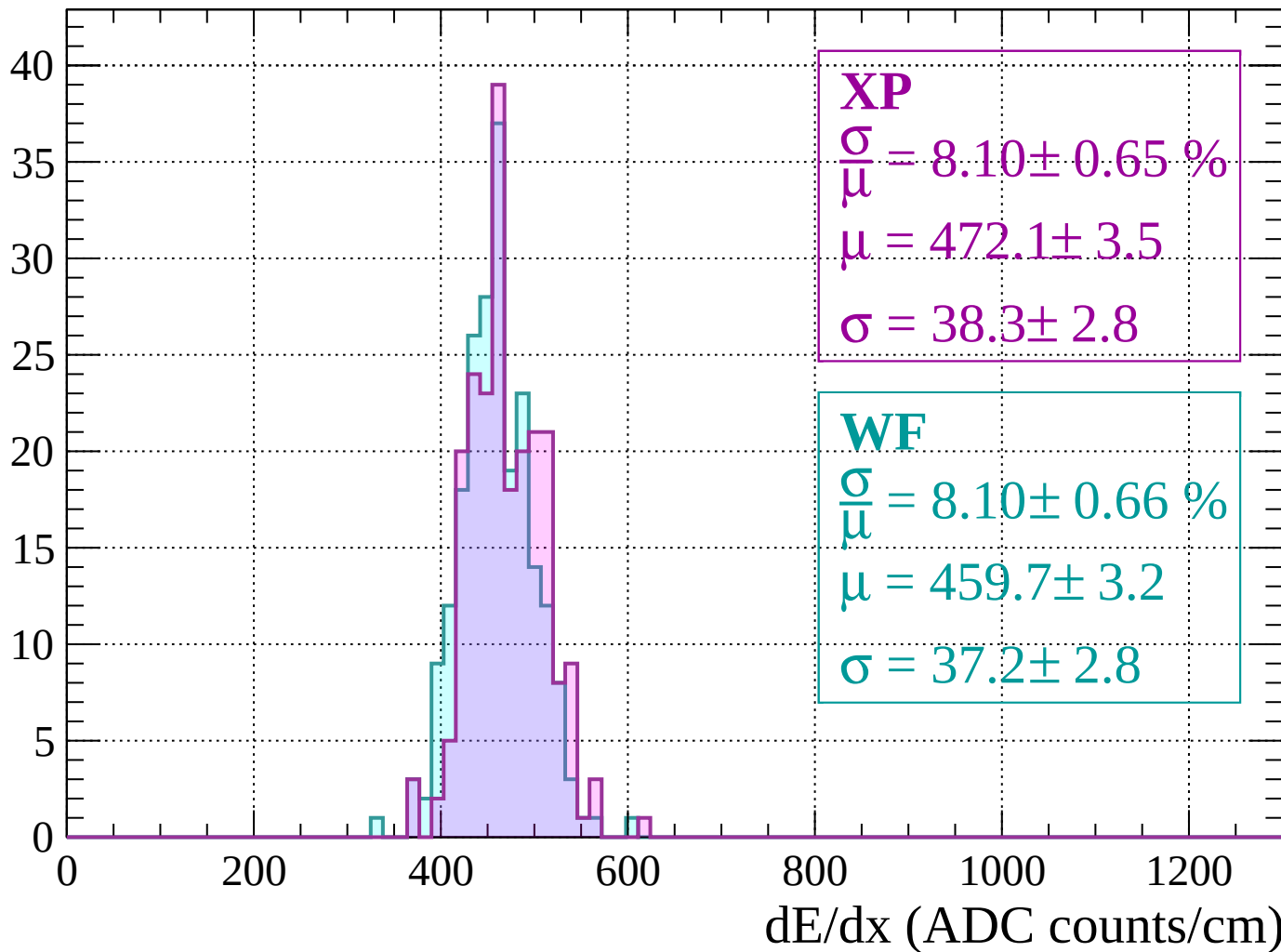
Count



dE/dx (ADC counts/cm)

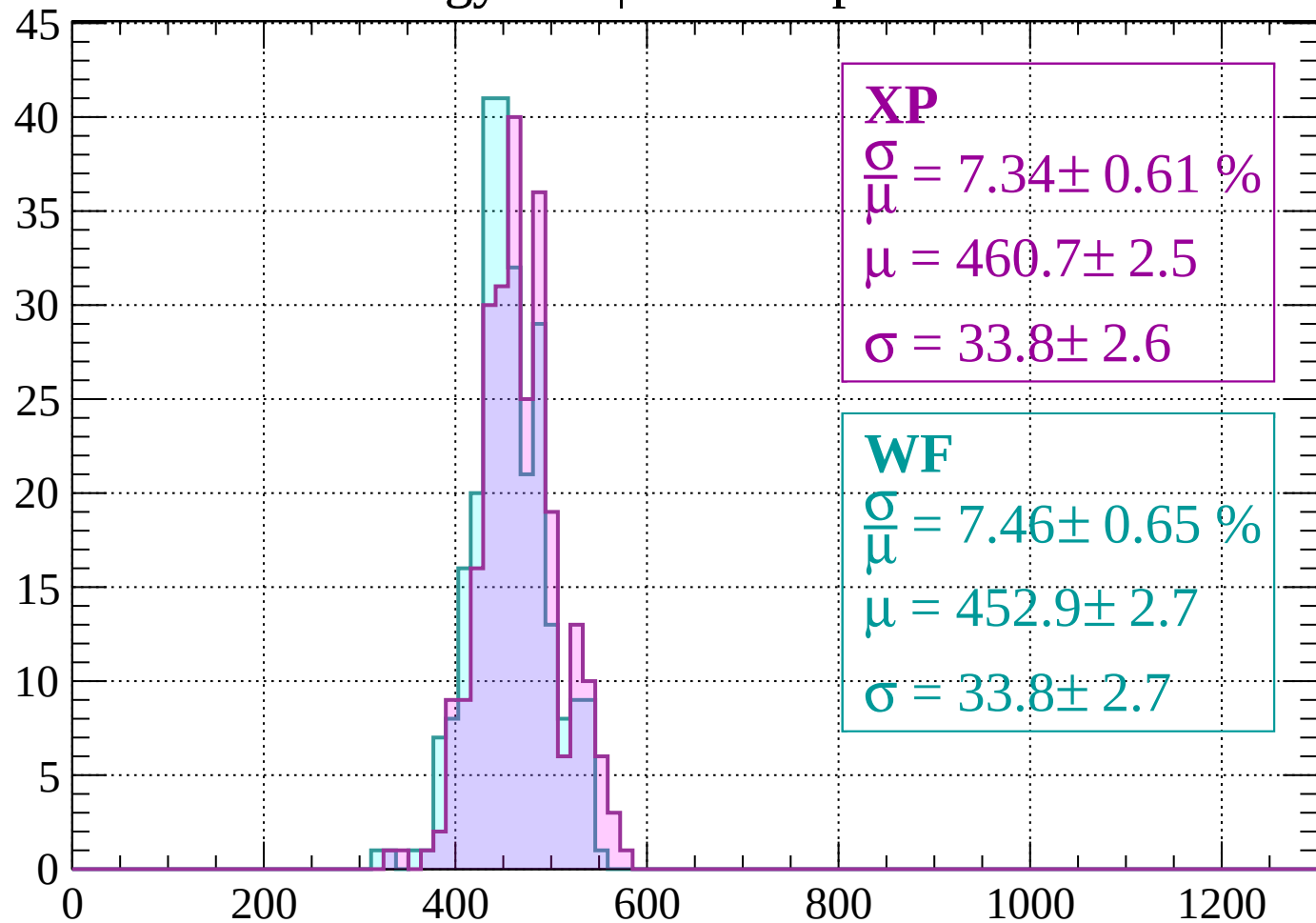
Energy loss | $-1280 < p < -1240$

Count



Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.61 \%$$

$$\mu = 460.7 \pm 2.5$$

$$\sigma = 33.8 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.46 \pm 0.65 \%$$

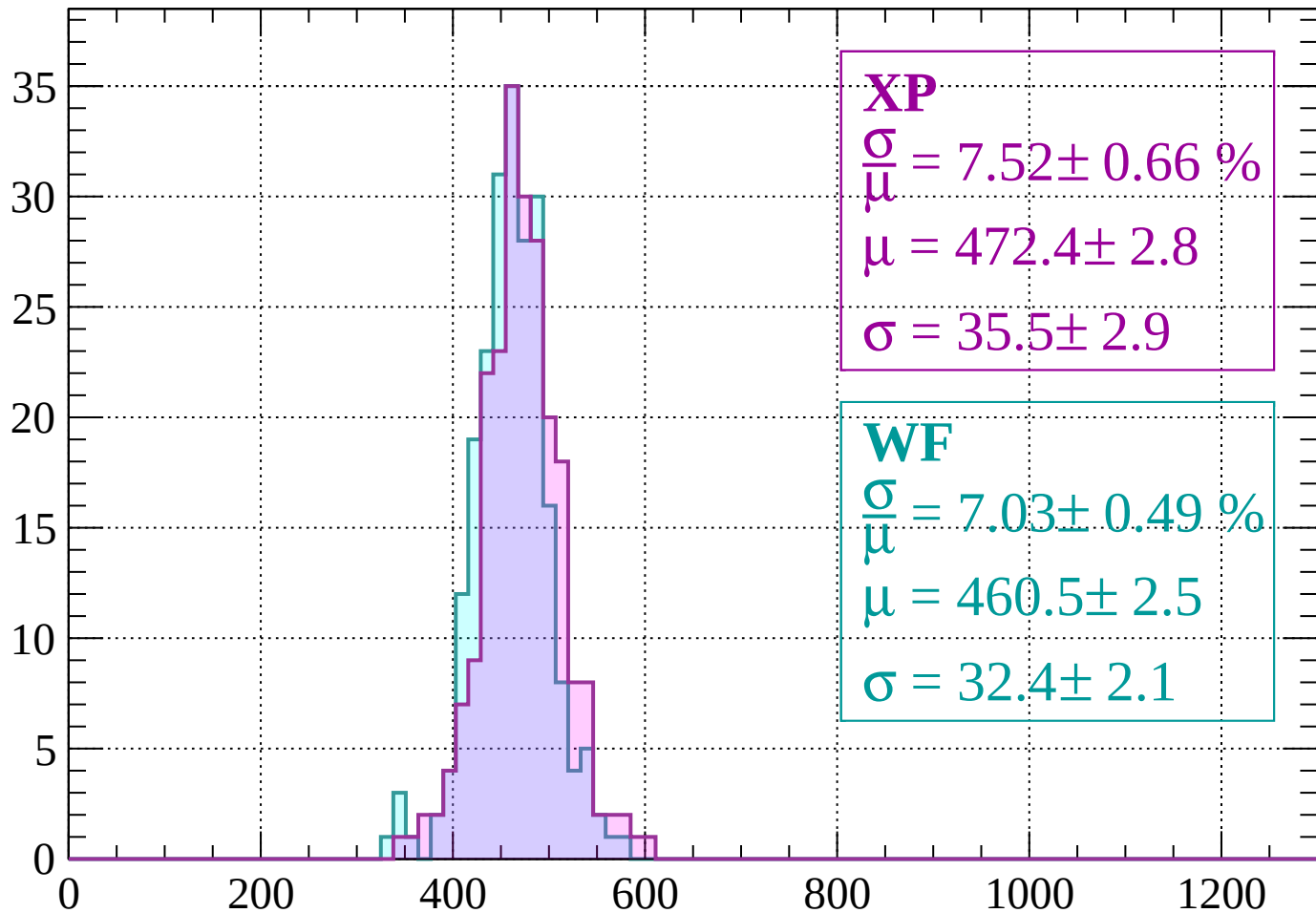
$$\mu = 452.9 \pm 2.7$$

$$\sigma = 33.8 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 7.52 \pm 0.66 \%$$

$$\mu = 472.4 \pm 2.8$$

$$\sigma = 35.5 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 7.03 \pm 0.49 \%$$

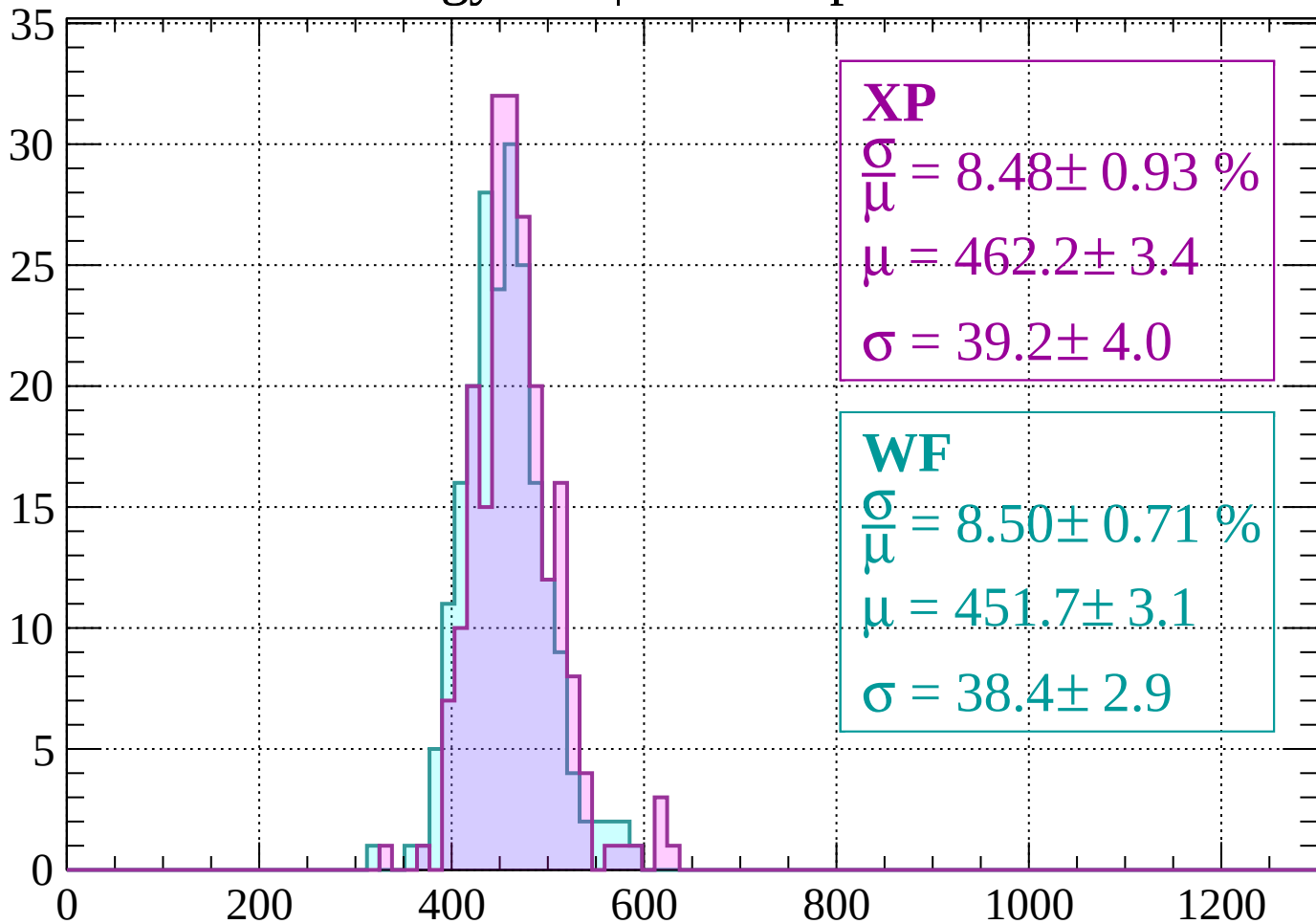
$$\mu = 460.5 \pm 2.5$$

$$\sigma = 32.4 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.93 \%$$

$$\mu = 462.2 \pm 3.4$$

$$\sigma = 39.2 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.71 \%$$

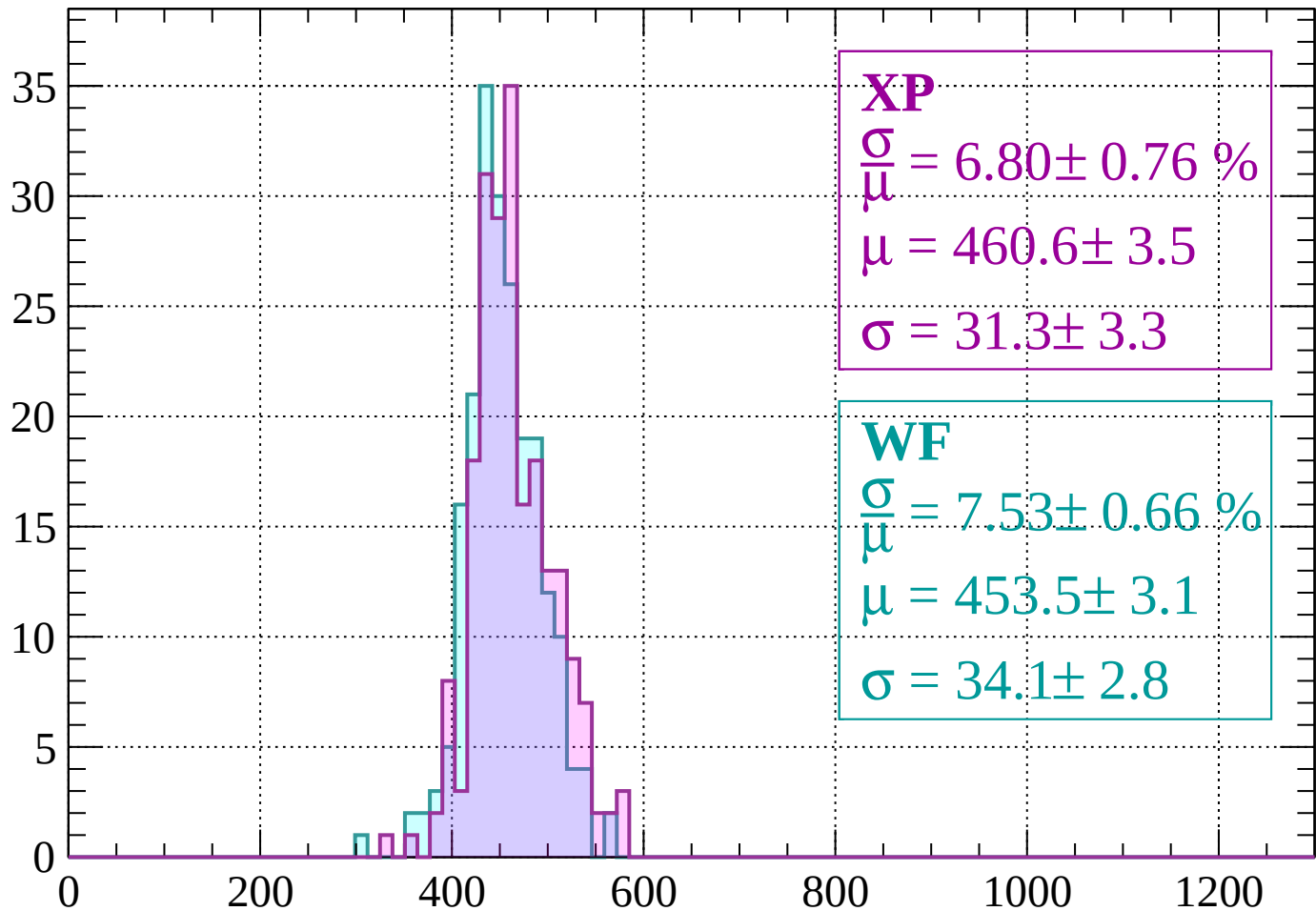
$$\mu = 451.7 \pm 3.1$$

$$\sigma = 38.4 \pm 2.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

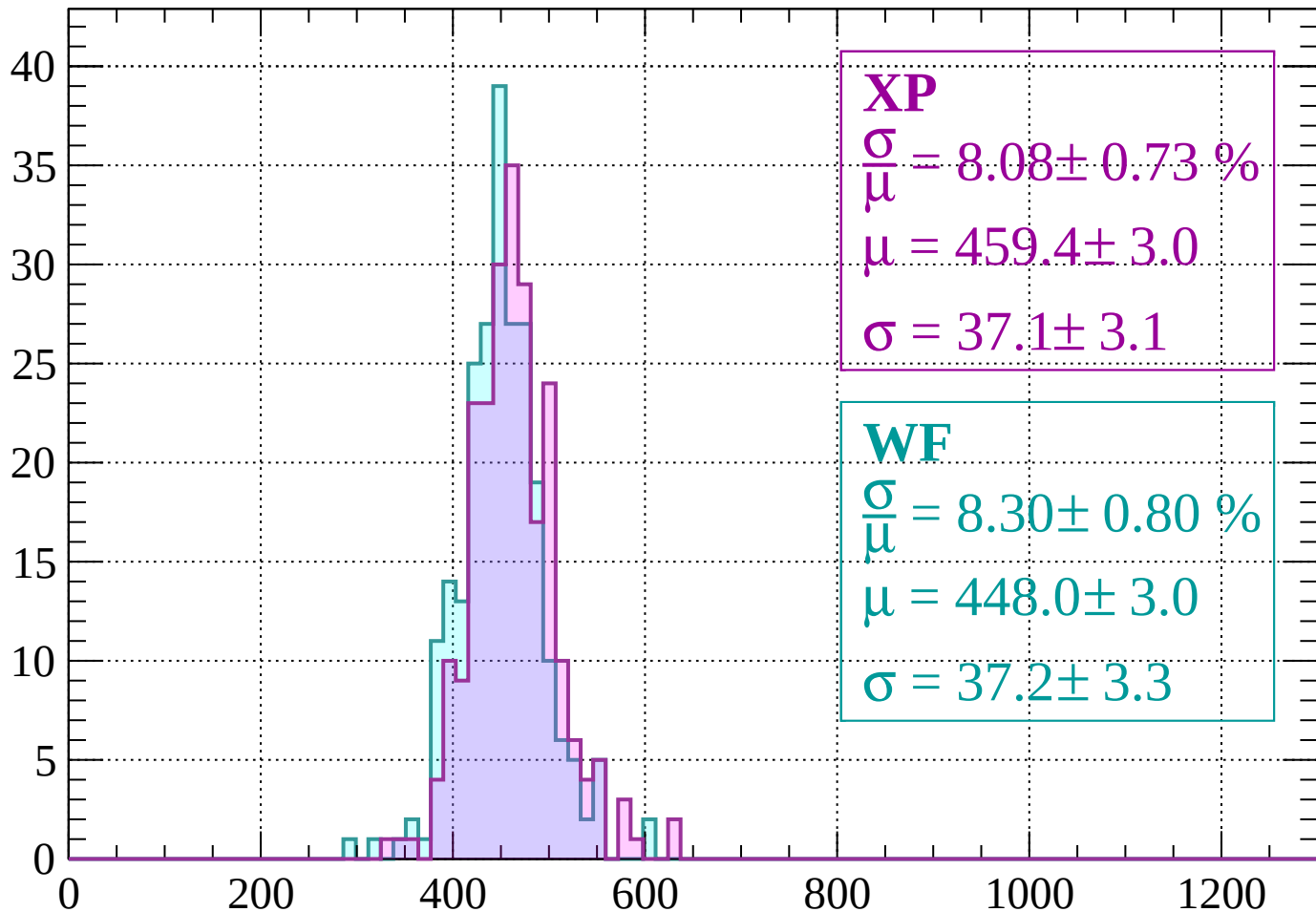
Count



dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.08 \pm 0.73 \%$$

$$\mu = 459.4 \pm 3.0$$

$$\sigma = 37.1 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.80 \%$$

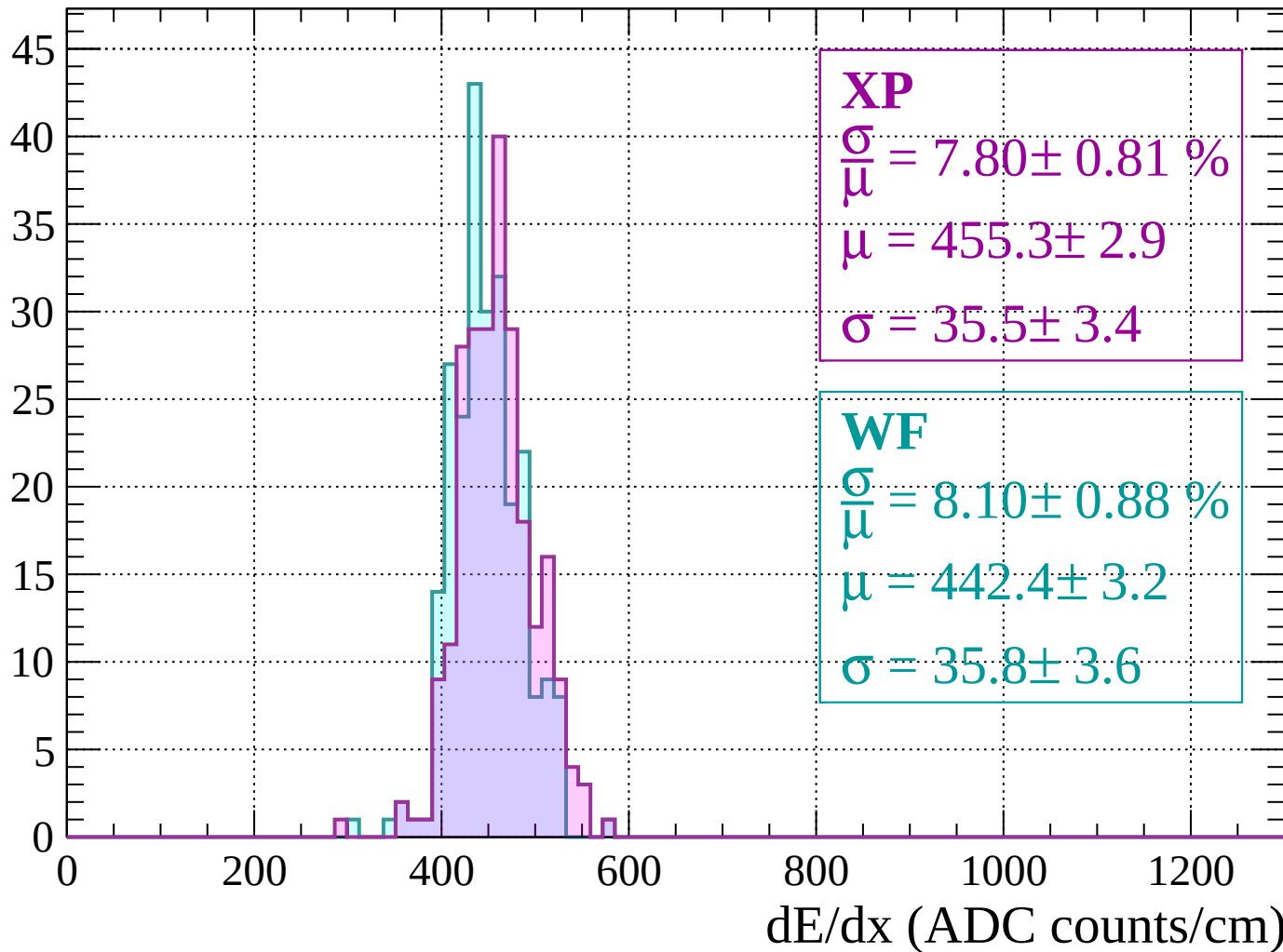
$$\mu = 448.0 \pm 3.0$$

$$\sigma = 37.2 \pm 3.3$$

dE/dx (ADC counts/cm)

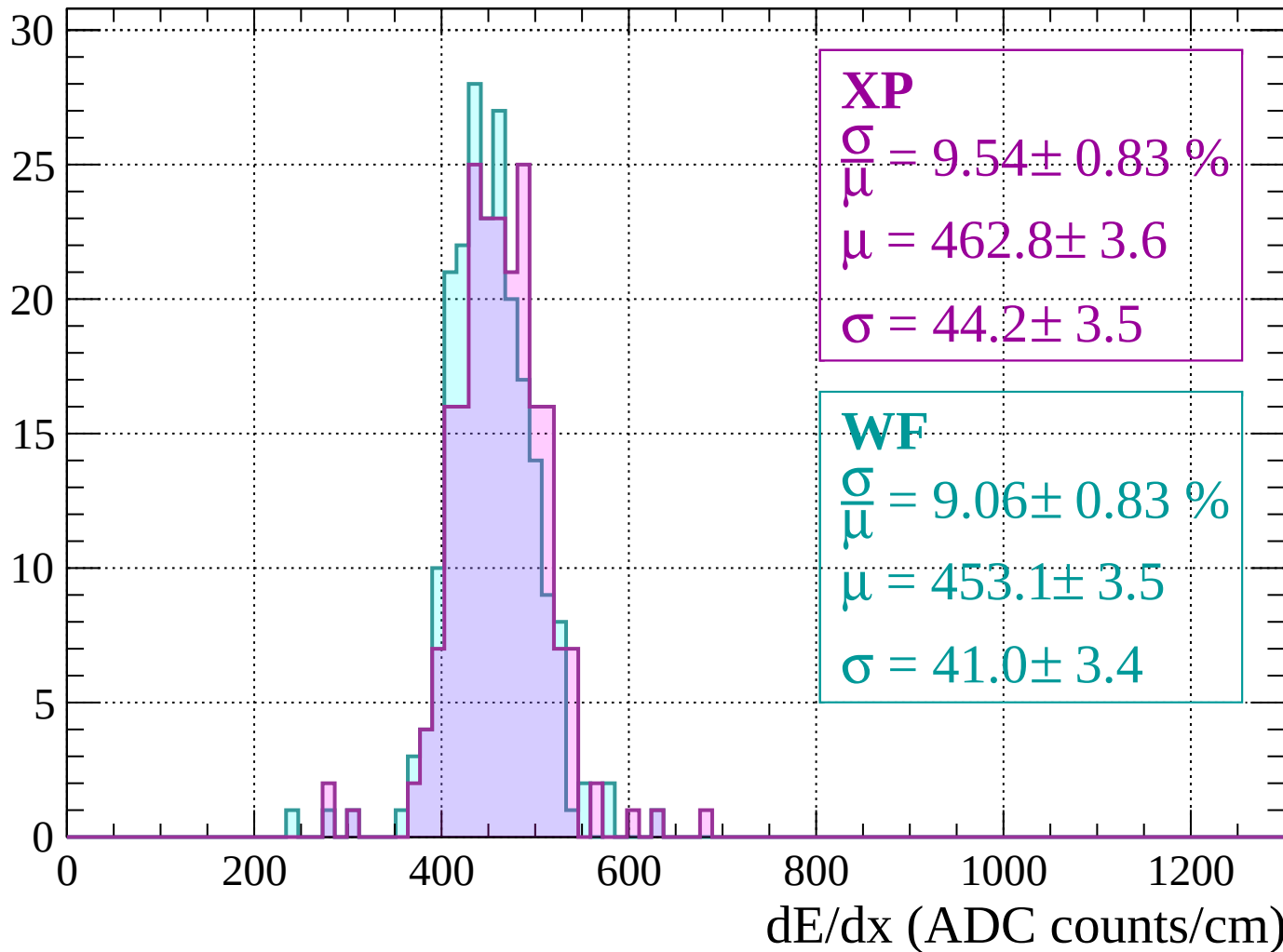
Energy loss | $-1040 < p < -1000$

Count



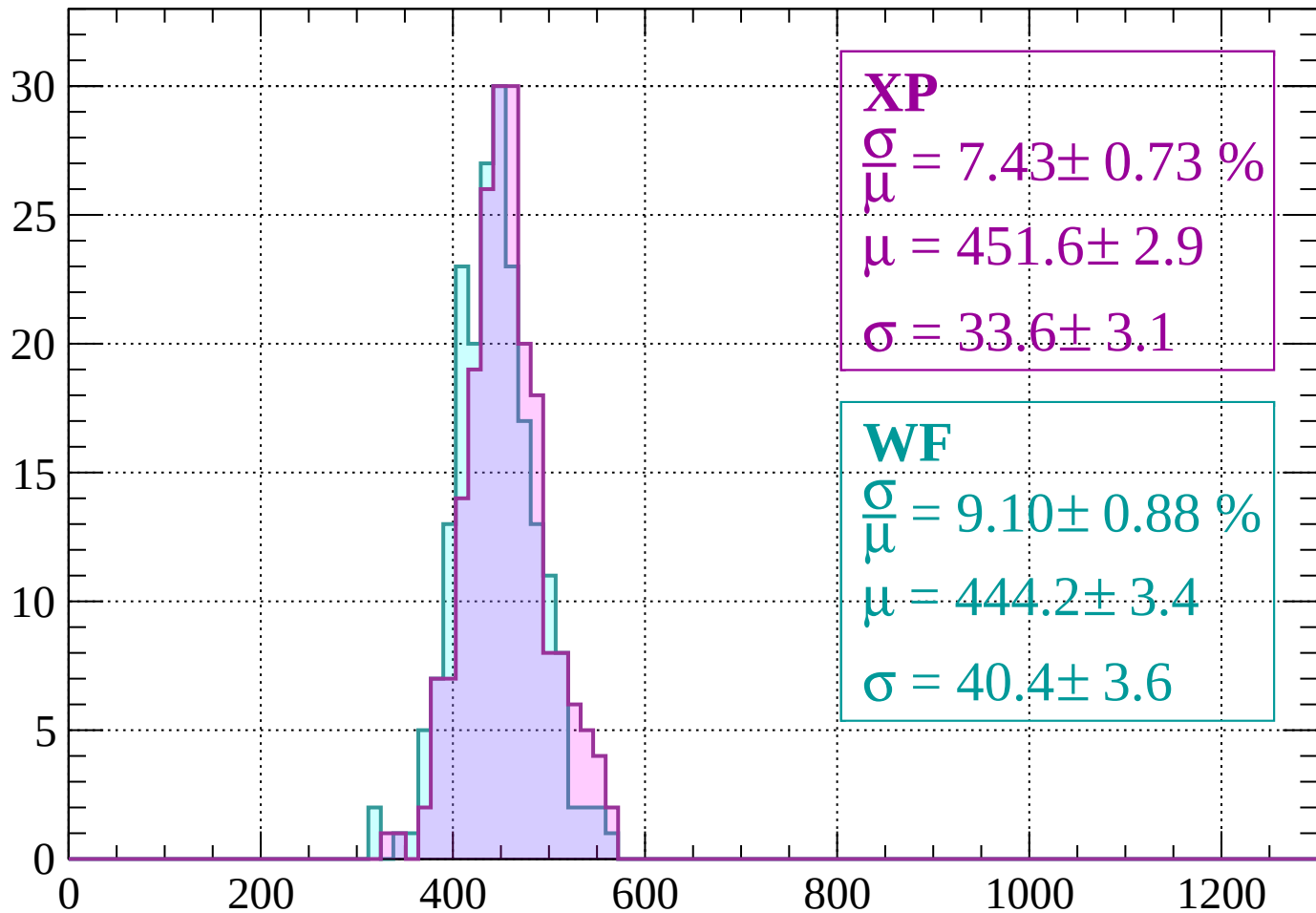
Energy loss | $-1000 < p < -960$

Count



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.43 \pm 0.73 \%$$

$$\mu = 451.6 \pm 2.9$$

$$\sigma = 33.6 \pm 3.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.10 \pm 0.88 \%$$

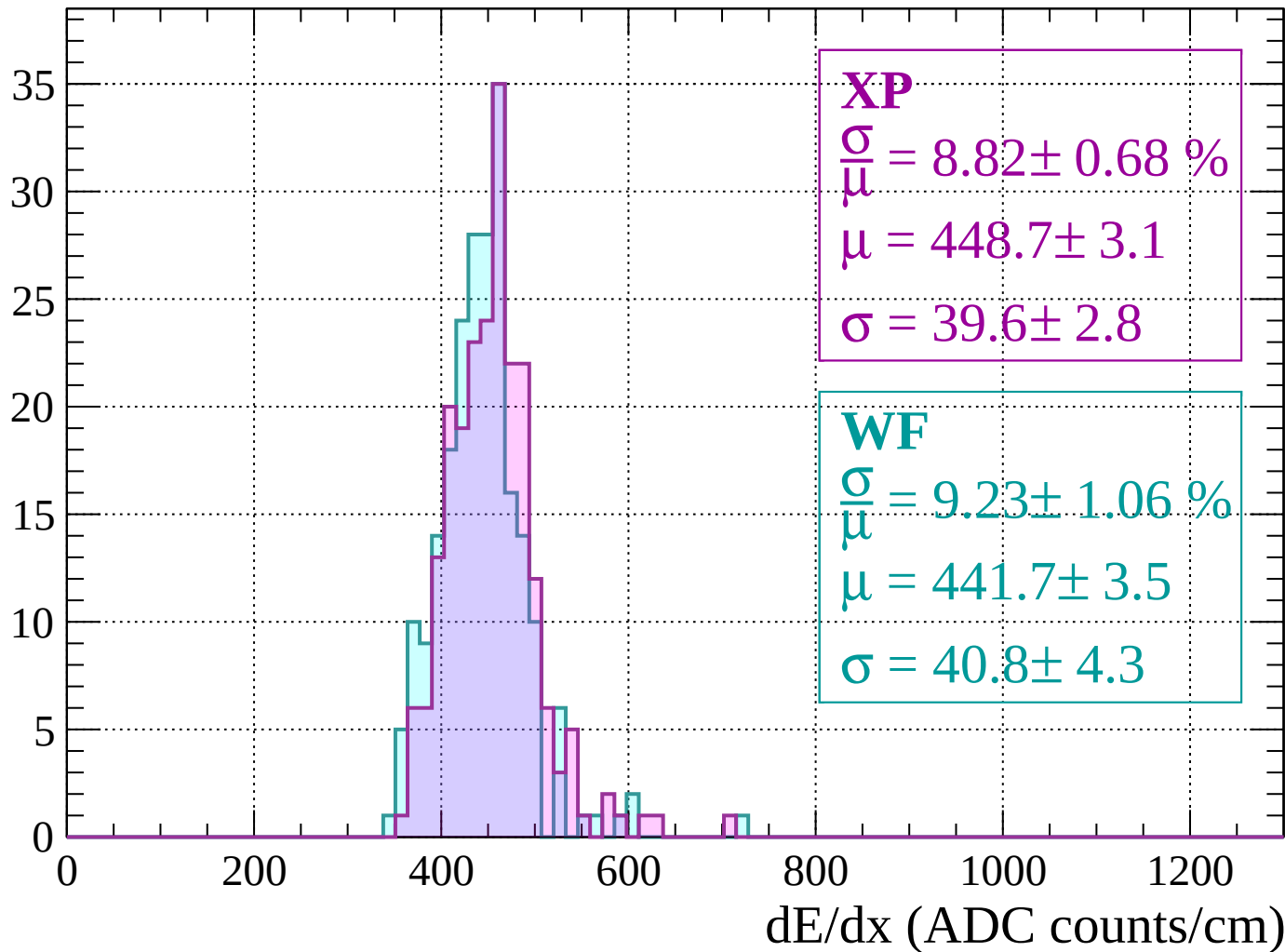
$$\mu = 444.2 \pm 3.4$$

$$\sigma = 40.4 \pm 3.6$$

dE/dx (ADC counts/cm)

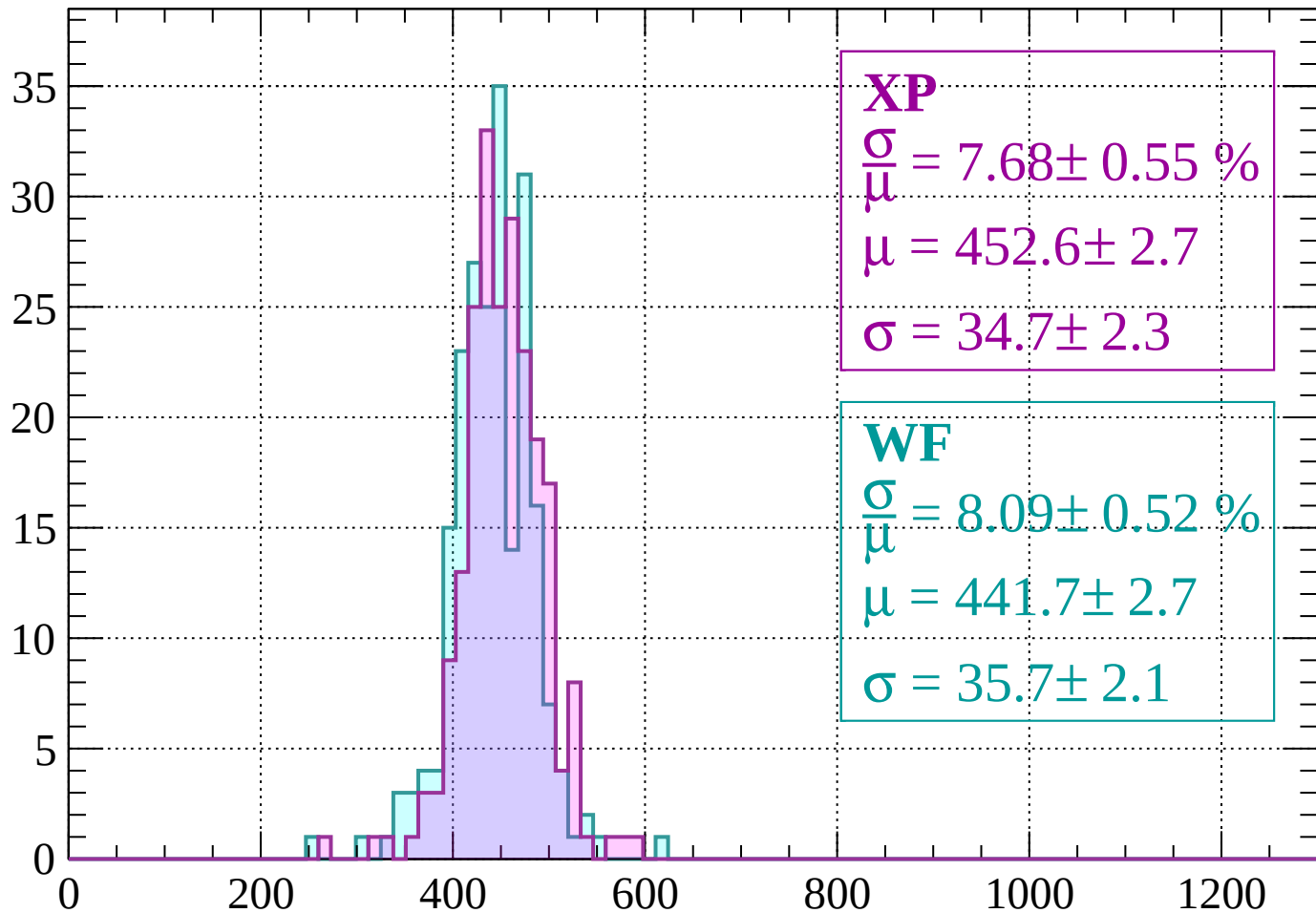
Energy loss | $-920 < p < -880$

Count



Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 7.68 \pm 0.55 \%$$

$$\mu = 452.6 \pm 2.7$$

$$\sigma = 34.7 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.52 \%$$

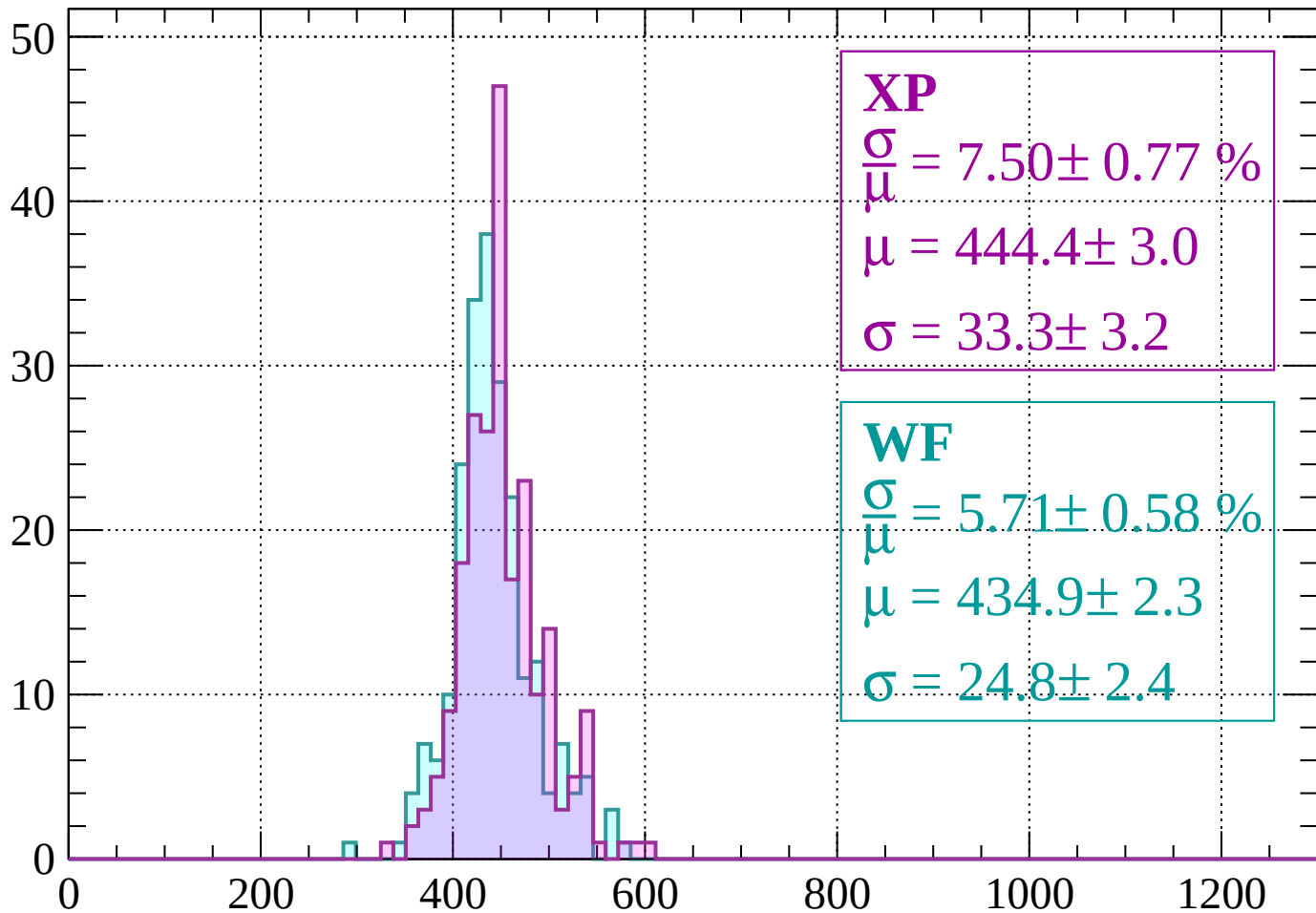
$$\mu = 441.7 \pm 2.7$$

$$\sigma = 35.7 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 7.50 \pm 0.77 \%$$

$$\mu = 444.4 \pm 3.0$$

$$\sigma = 33.3 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 5.71 \pm 0.58 \%$$

$$\mu = 434.9 \pm 2.3$$

$$\sigma = 24.8 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count

XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.66 \%$$

$$\mu = 440.9 \pm 2.6$$

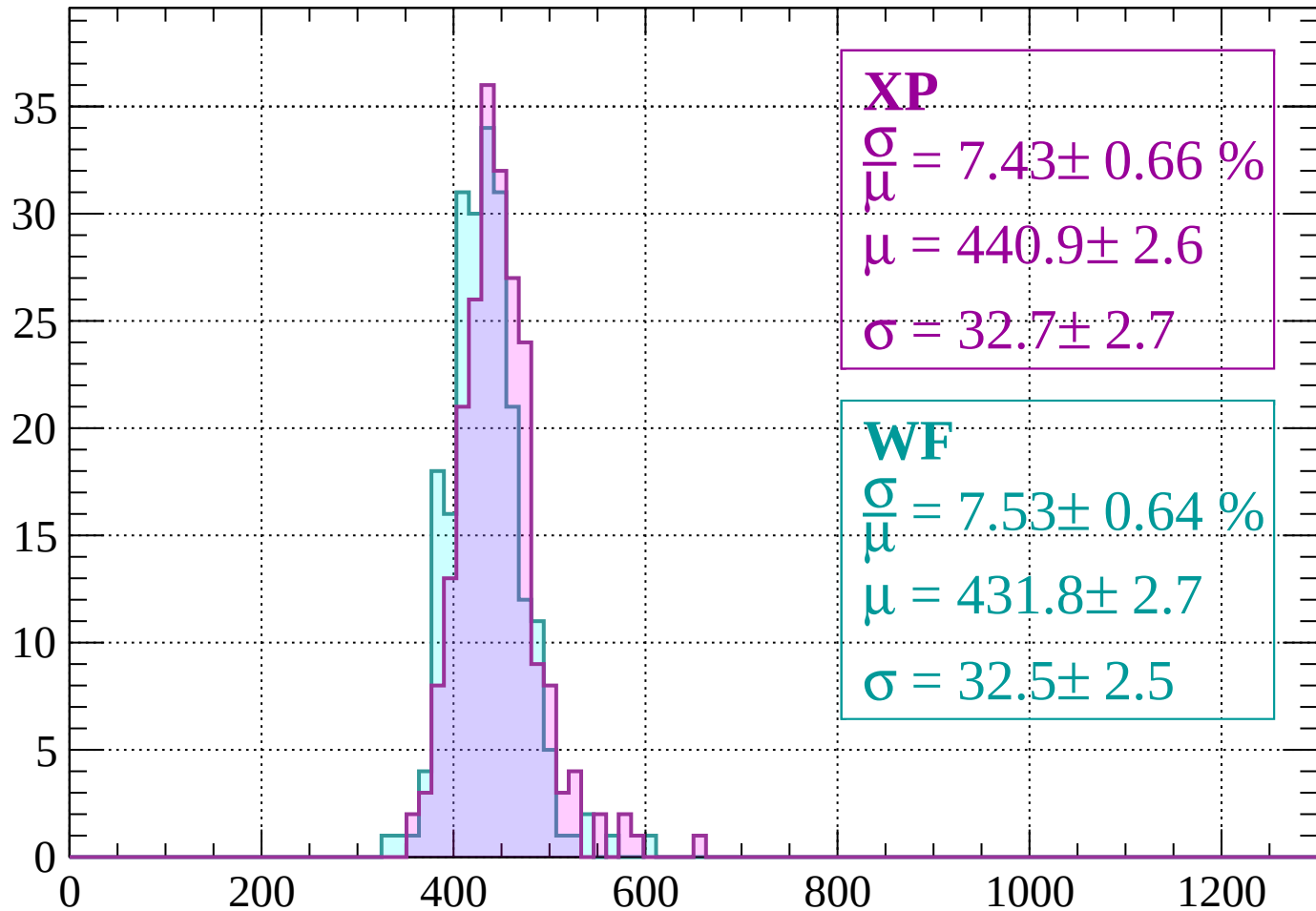
$$\sigma = 32.7 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.53 \pm 0.64 \%$$

$$\mu = 431.8 \pm 2.7$$

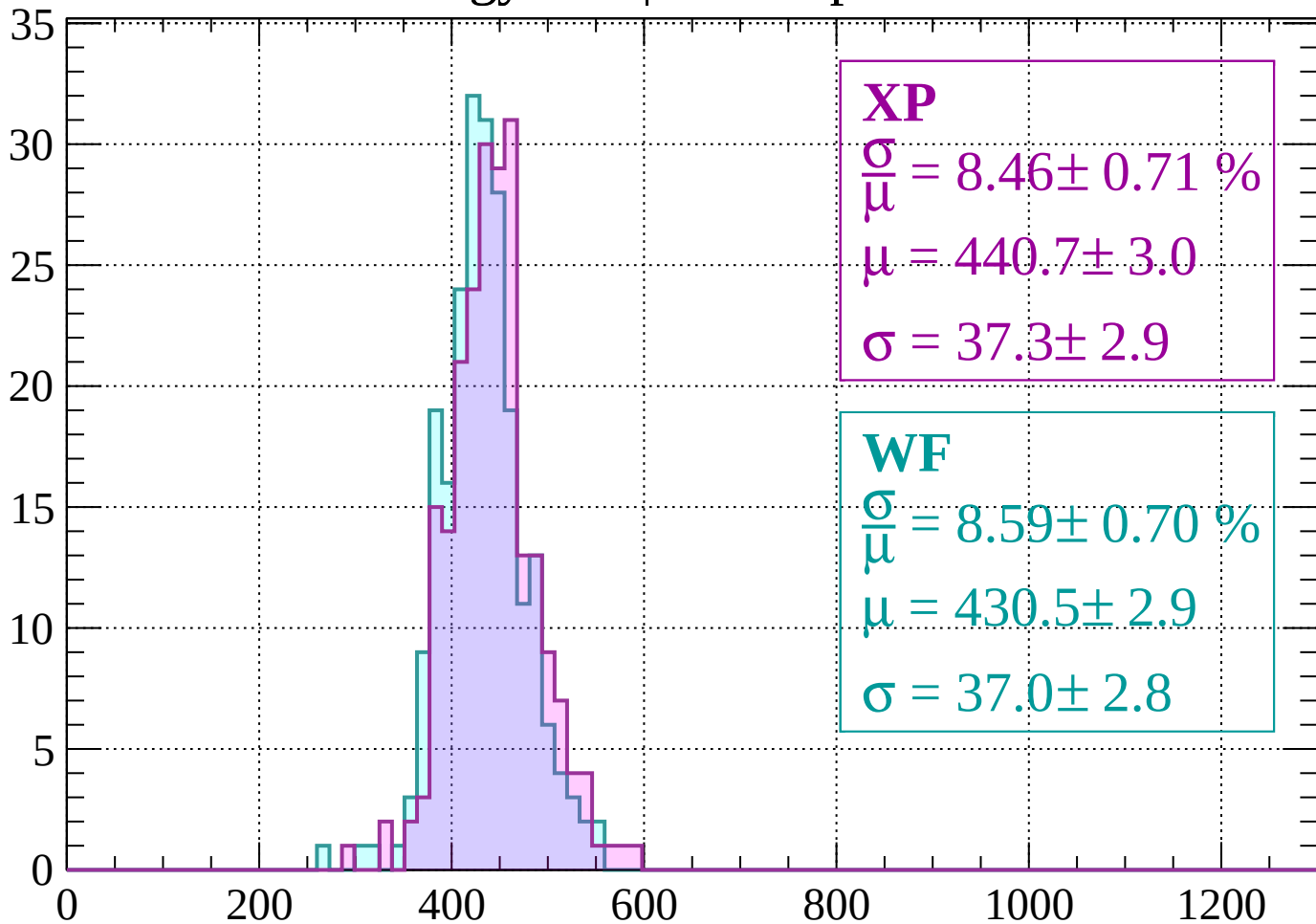
$$\sigma = 32.5 \pm 2.5$$



dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.71 \%$$

$$\mu = 440.7 \pm 3.0$$

$$\sigma = 37.3 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.70 \%$$

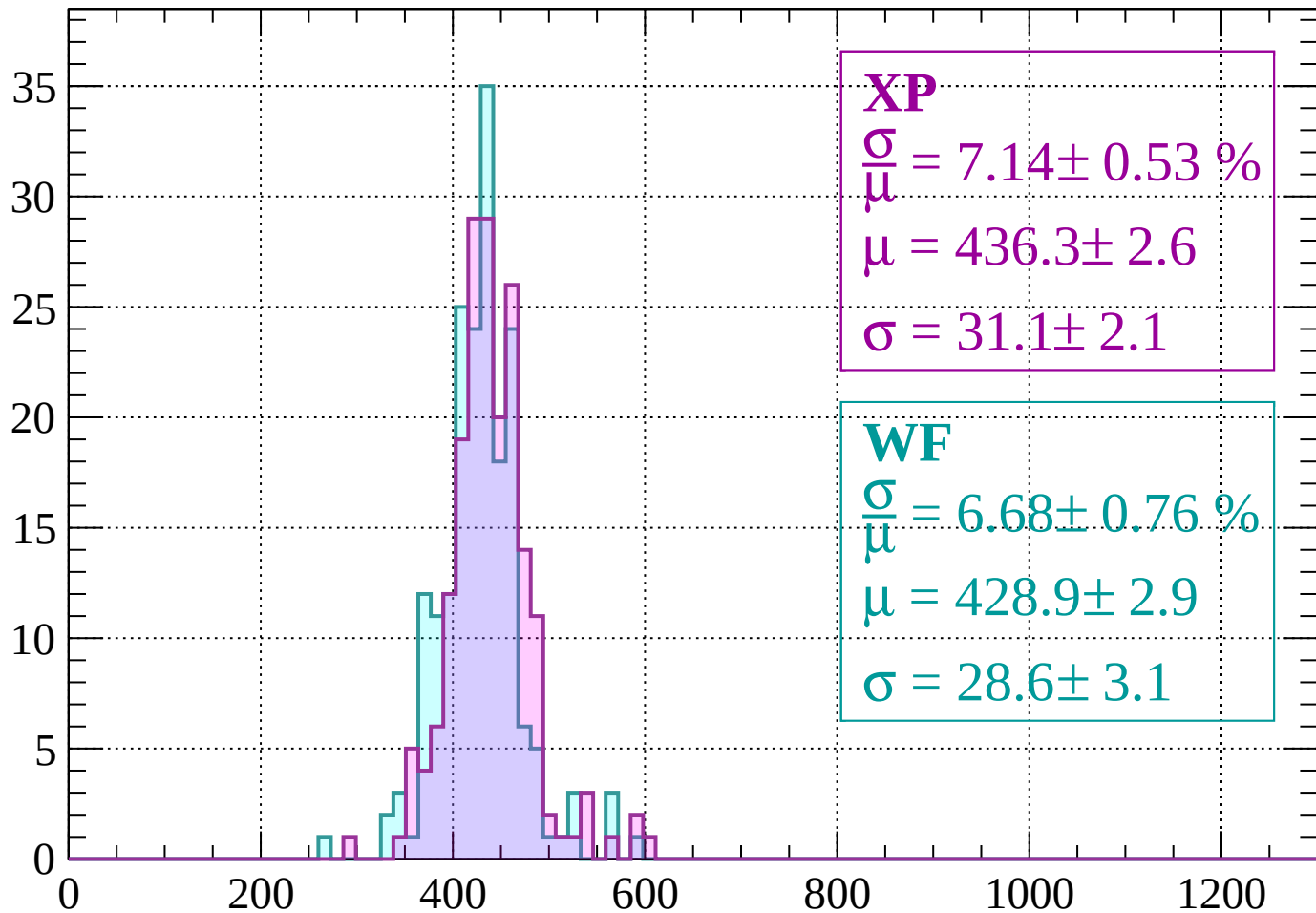
$$\mu = 430.5 \pm 2.9$$

$$\sigma = 37.0 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.53 \%$$

$$\mu = 436.3 \pm 2.6$$

$$\sigma = 31.1 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.76 \%$$

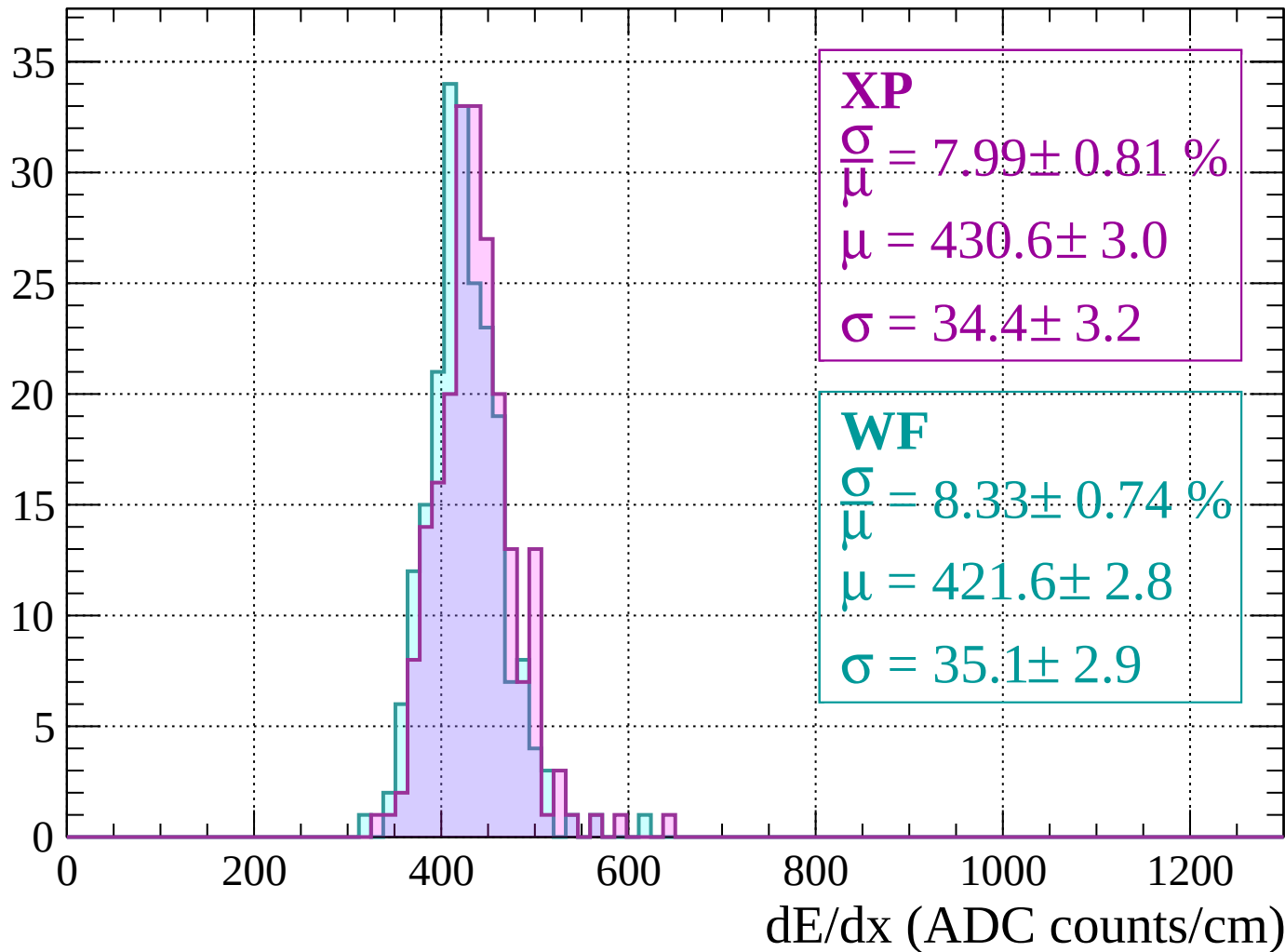
$$\mu = 428.9 \pm 2.9$$

$$\sigma = 28.6 \pm 3.1$$

dE/dx (ADC counts/cm)

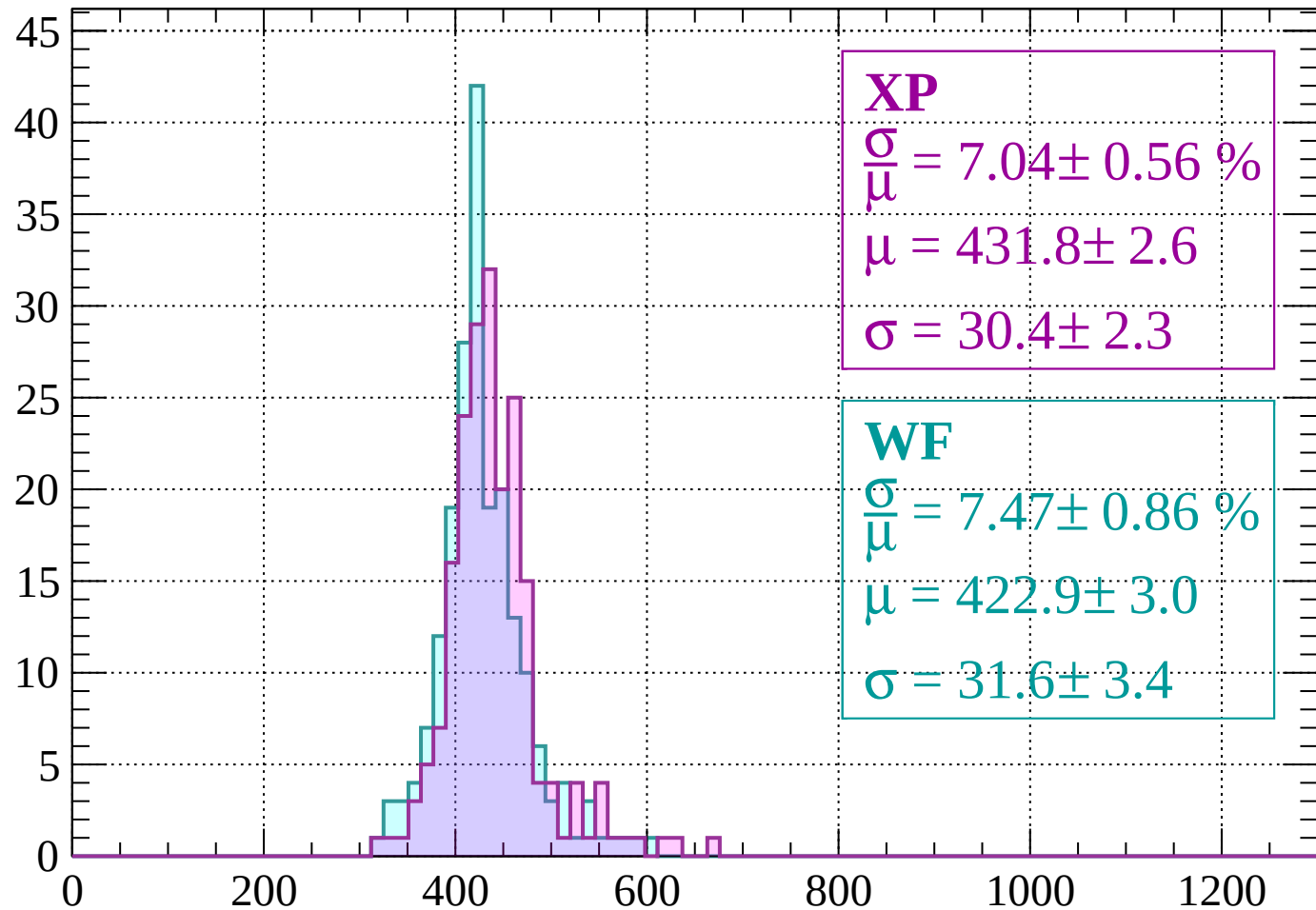
Energy loss | $-680 < p < -640$

Count



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.56 \%$$

$$\mu = 431.8 \pm 2.6$$

$$\sigma = 30.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.86 \%$$

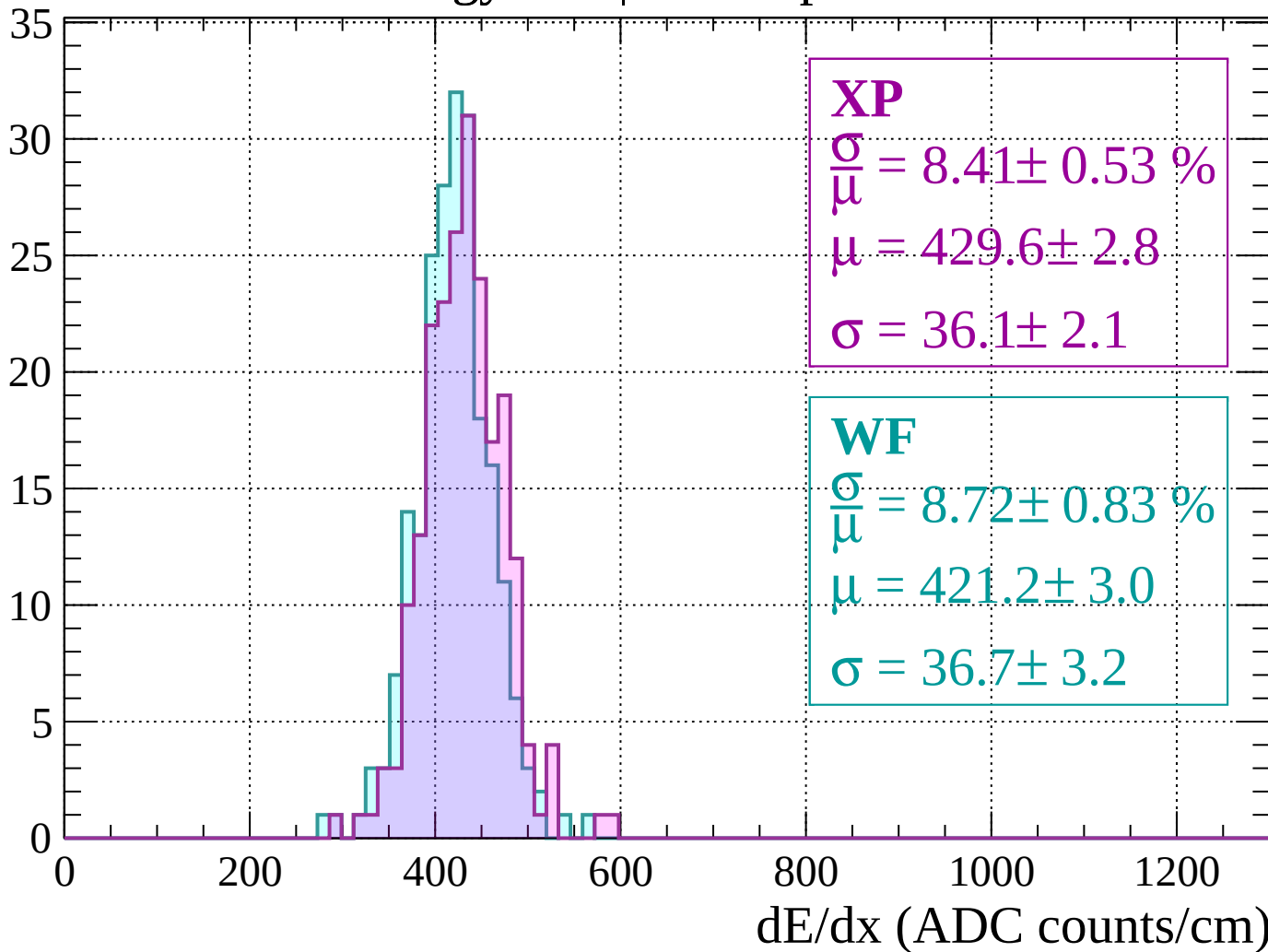
$$\mu = 422.9 \pm 3.0$$

$$\sigma = 31.6 \pm 3.4$$

dE/dx (ADC counts/cm)

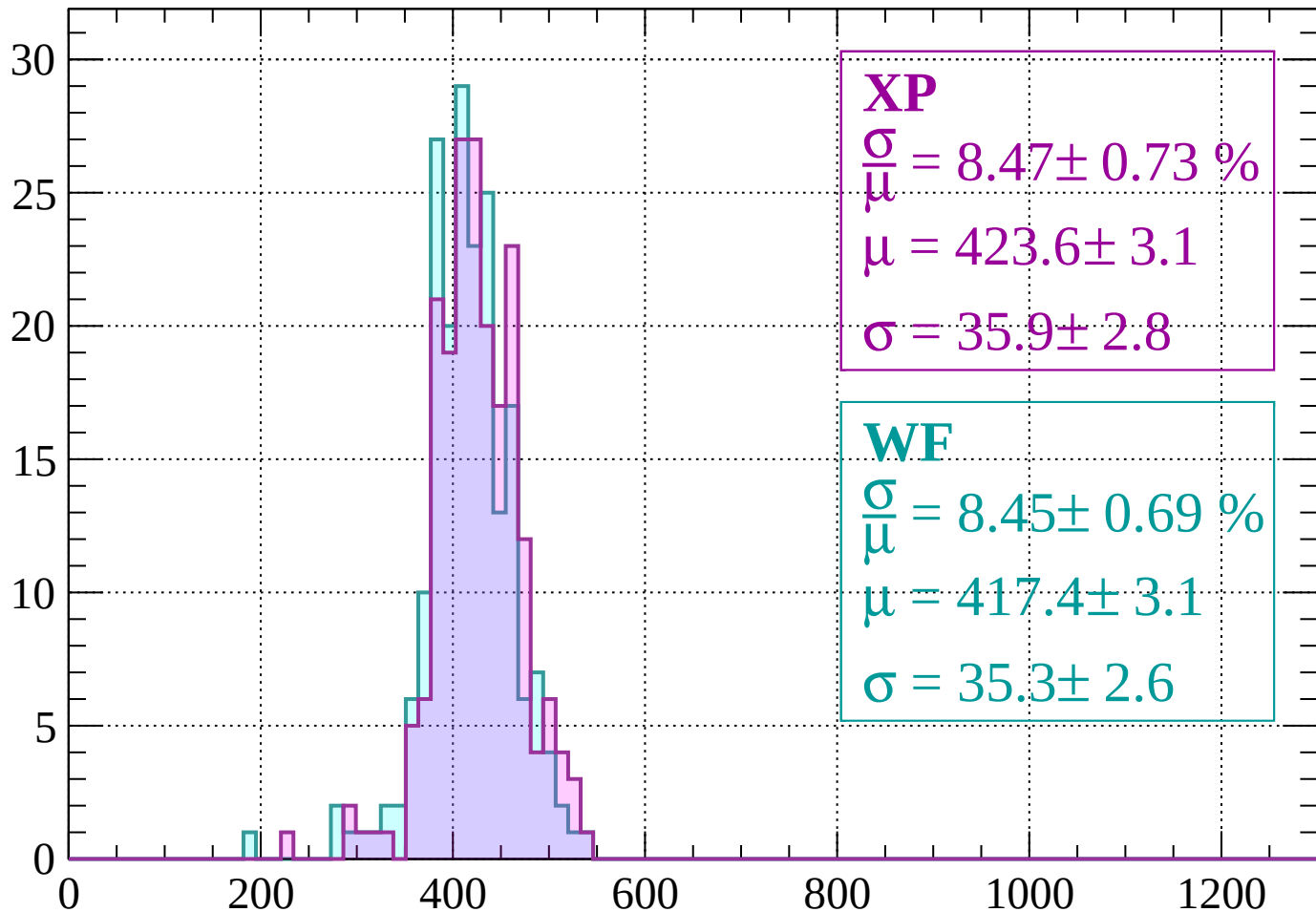
Energy loss | $-600 < p < -560$

Count



Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.73 \%$$

$$\mu = 423.6 \pm 3.1$$

$$\sigma = 35.9 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.69 \%$$

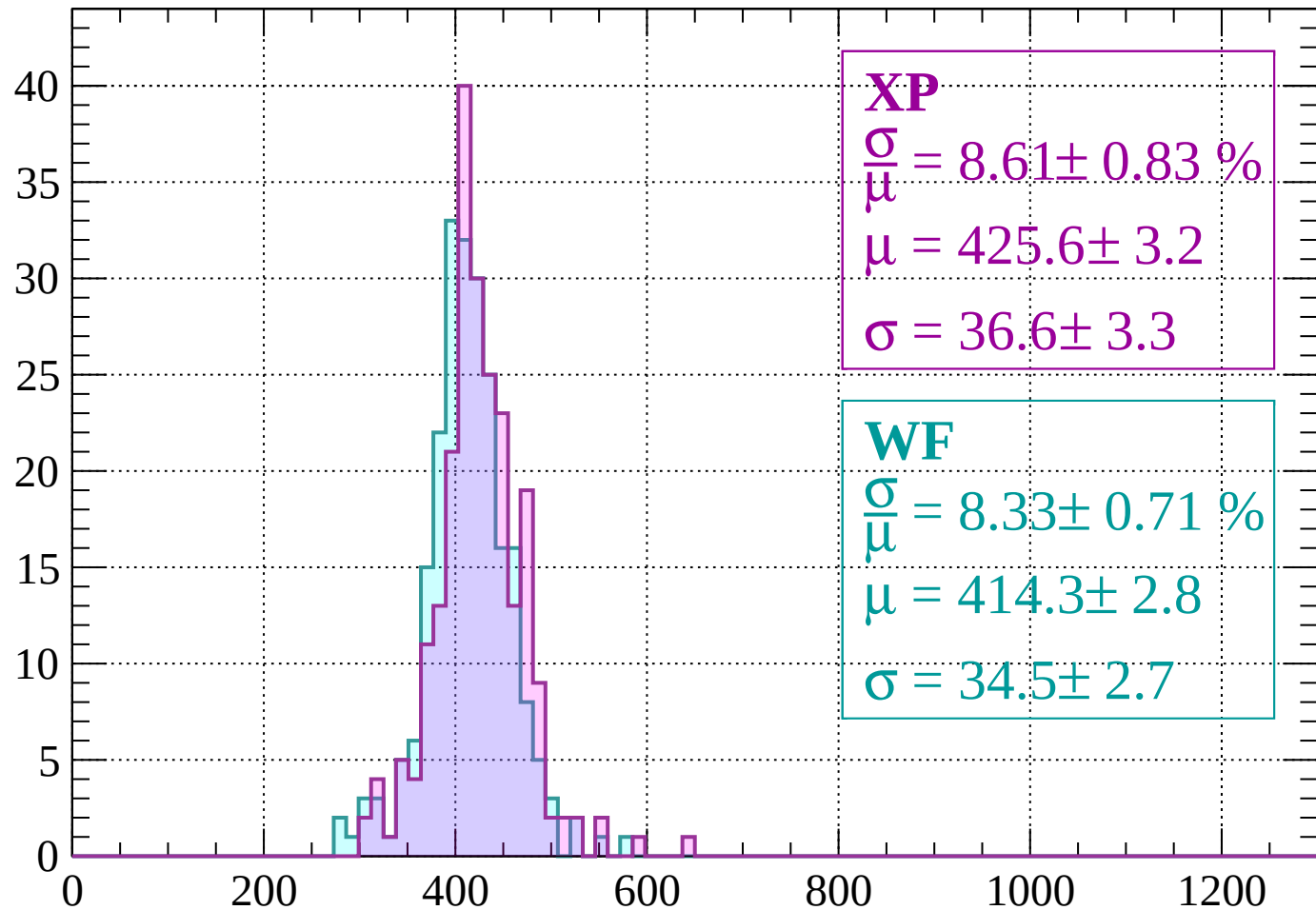
$$\mu = 417.4 \pm 3.1$$

$$\sigma = 35.3 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.83 \%$$

$$\mu = 425.6 \pm 3.2$$

$$\sigma = 36.6 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.71 \%$$

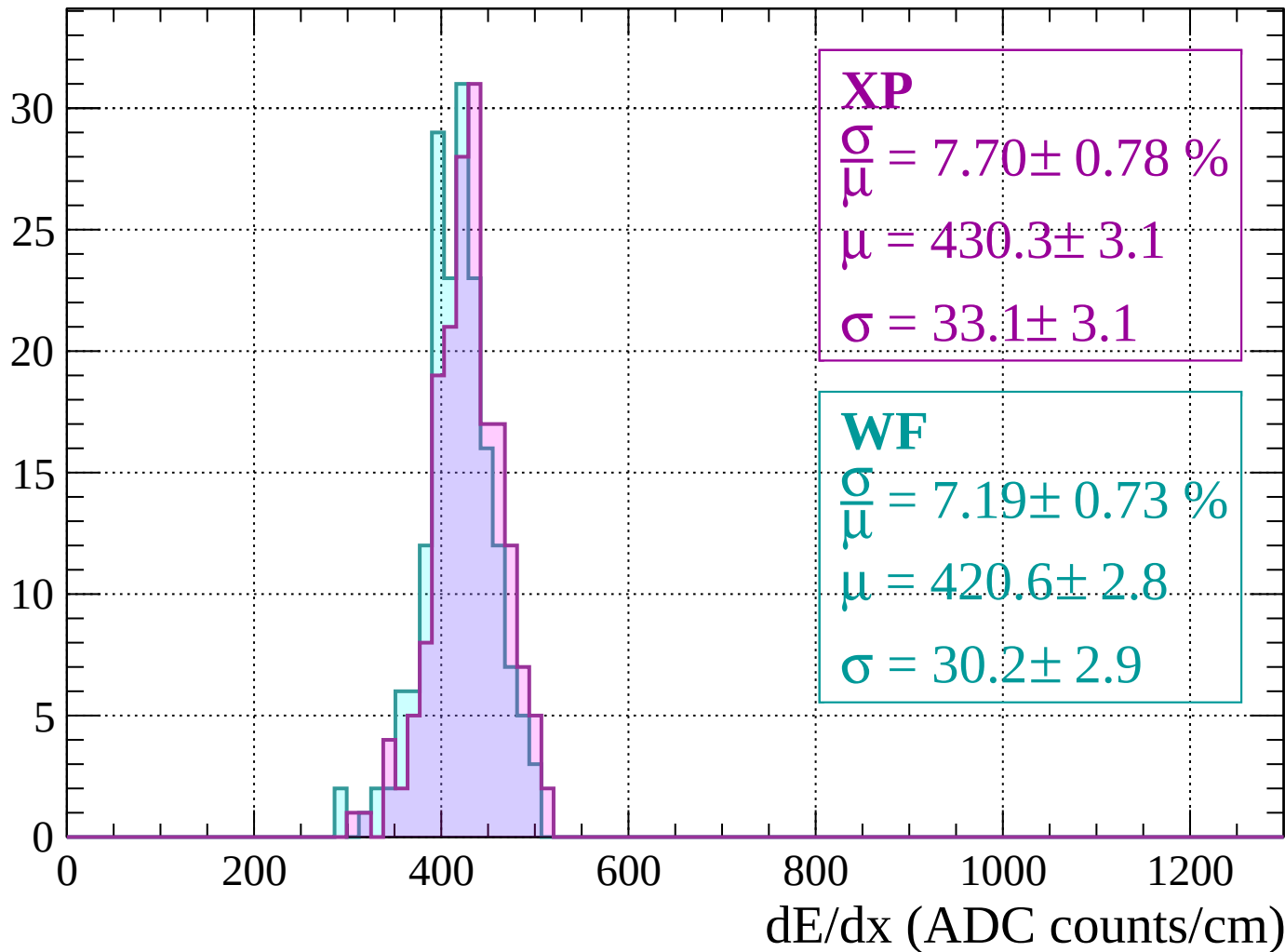
$$\mu = 414.3 \pm 2.8$$

$$\sigma = 34.5 \pm 2.7$$

dE/dx (ADC counts/cm)

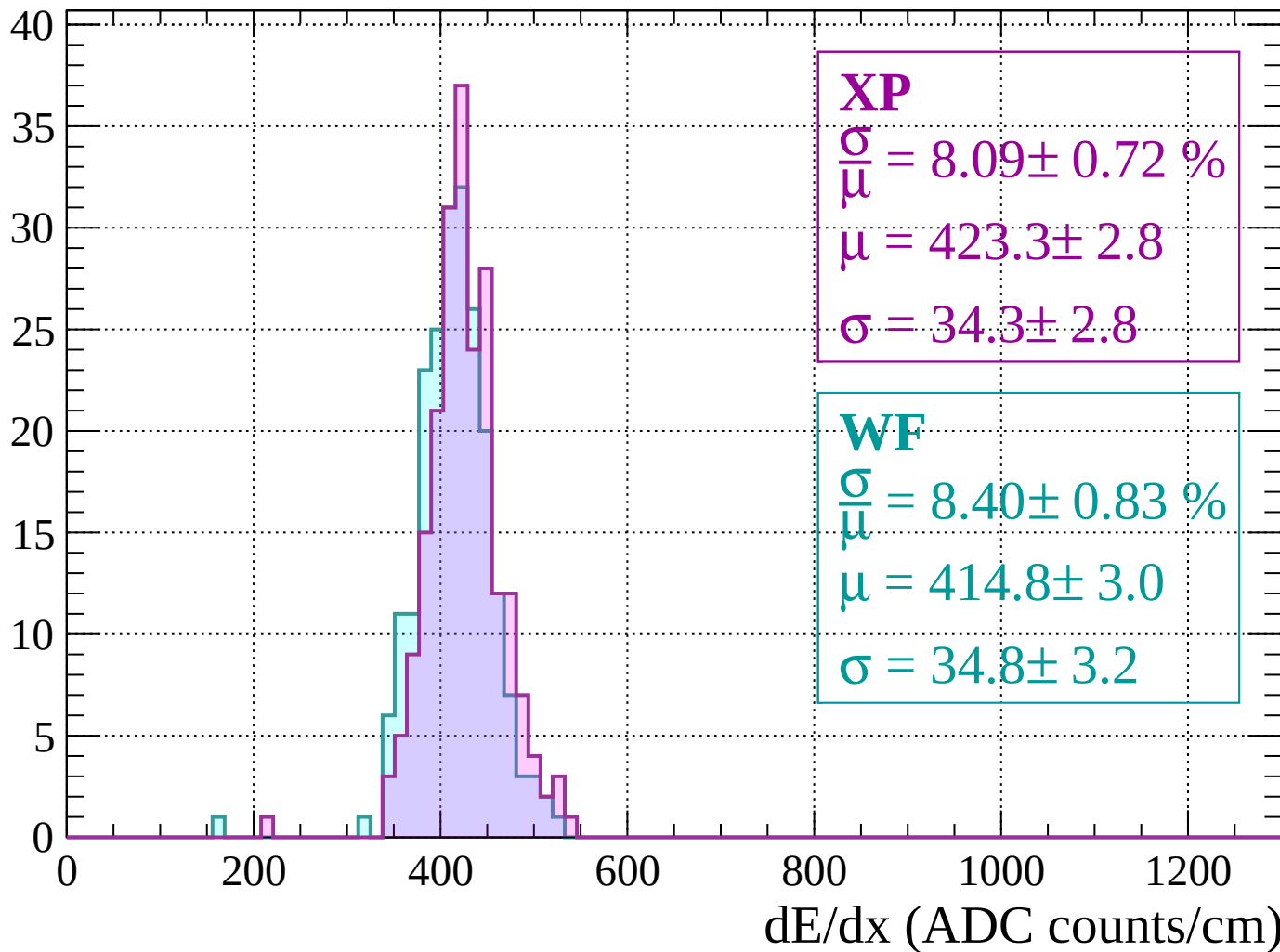
Energy loss | $-480 < p < -440$

Count



Energy loss | $-440 < p < -400$

Count



Energy loss | $-400 < p < -360$

Count

XP

$$\frac{\sigma}{\mu} = 6.94 \pm 0.76 \%$$

$$\mu = 416.4 \pm 2.7$$

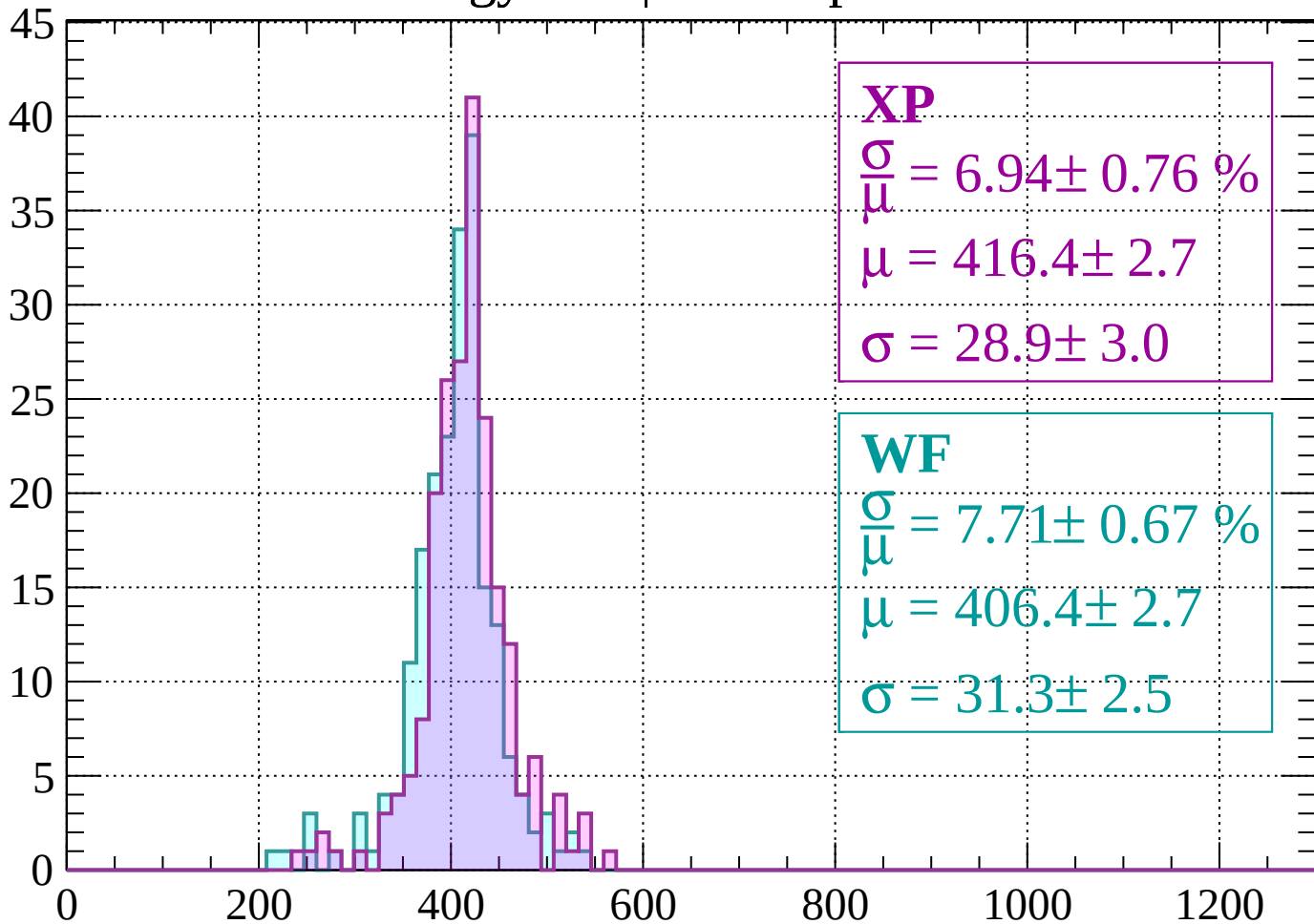
$$\sigma = 28.9 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 7.71 \pm 0.67 \%$$

$$\mu = 406.4 \pm 2.7$$

$$\sigma = 31.3 \pm 2.5$$



dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count

XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.55 \%$$

$$\mu = 422.8 \pm 2.5$$

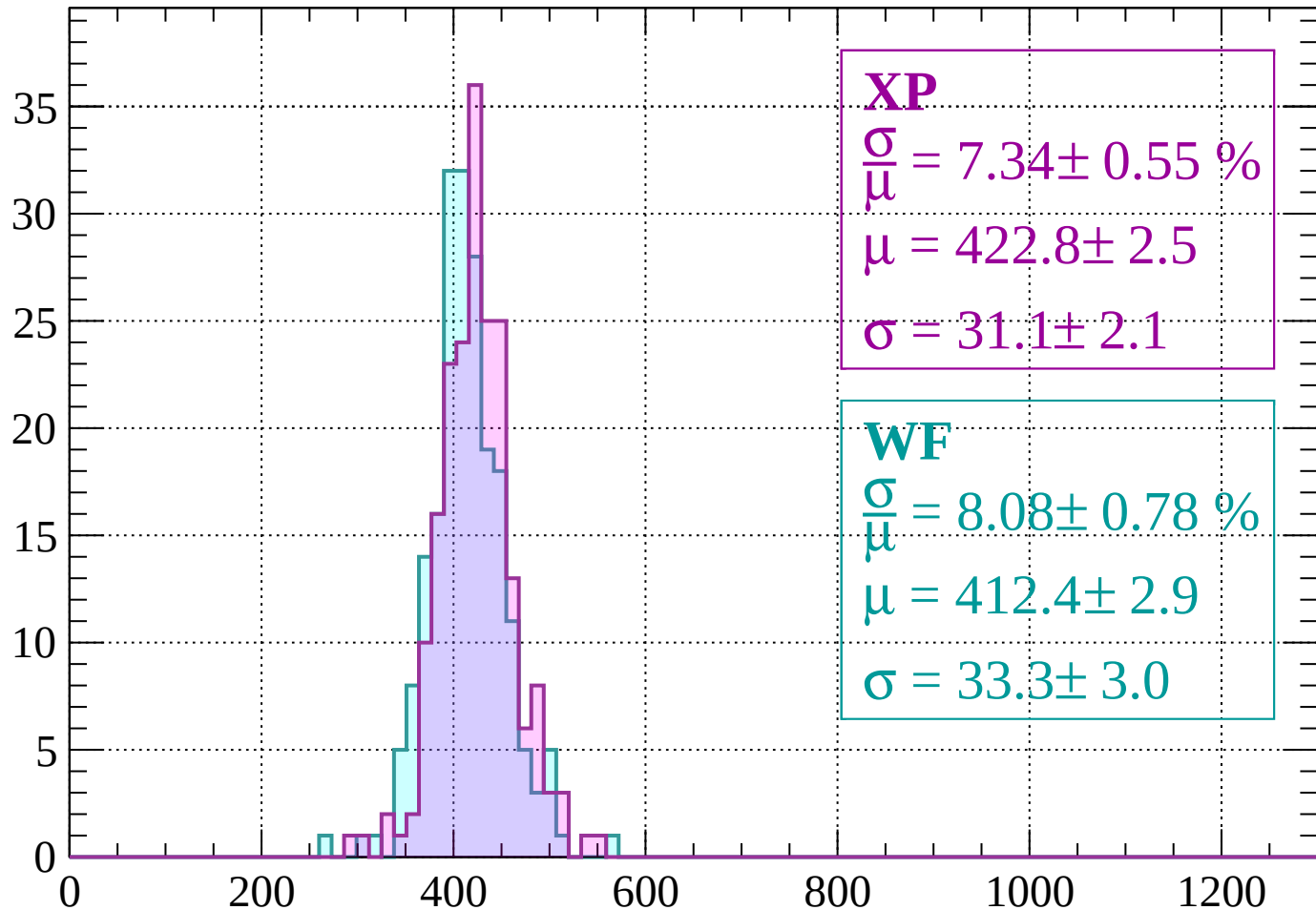
$$\sigma = 31.1 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.78 \%$$

$$\mu = 412.4 \pm 2.9$$

$$\sigma = 33.3 \pm 3.0$$



dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count

XP

$$\frac{\sigma}{\mu} = 7.79 \pm 0.53 \%$$

$$\mu = 426.5 \pm 2.8$$

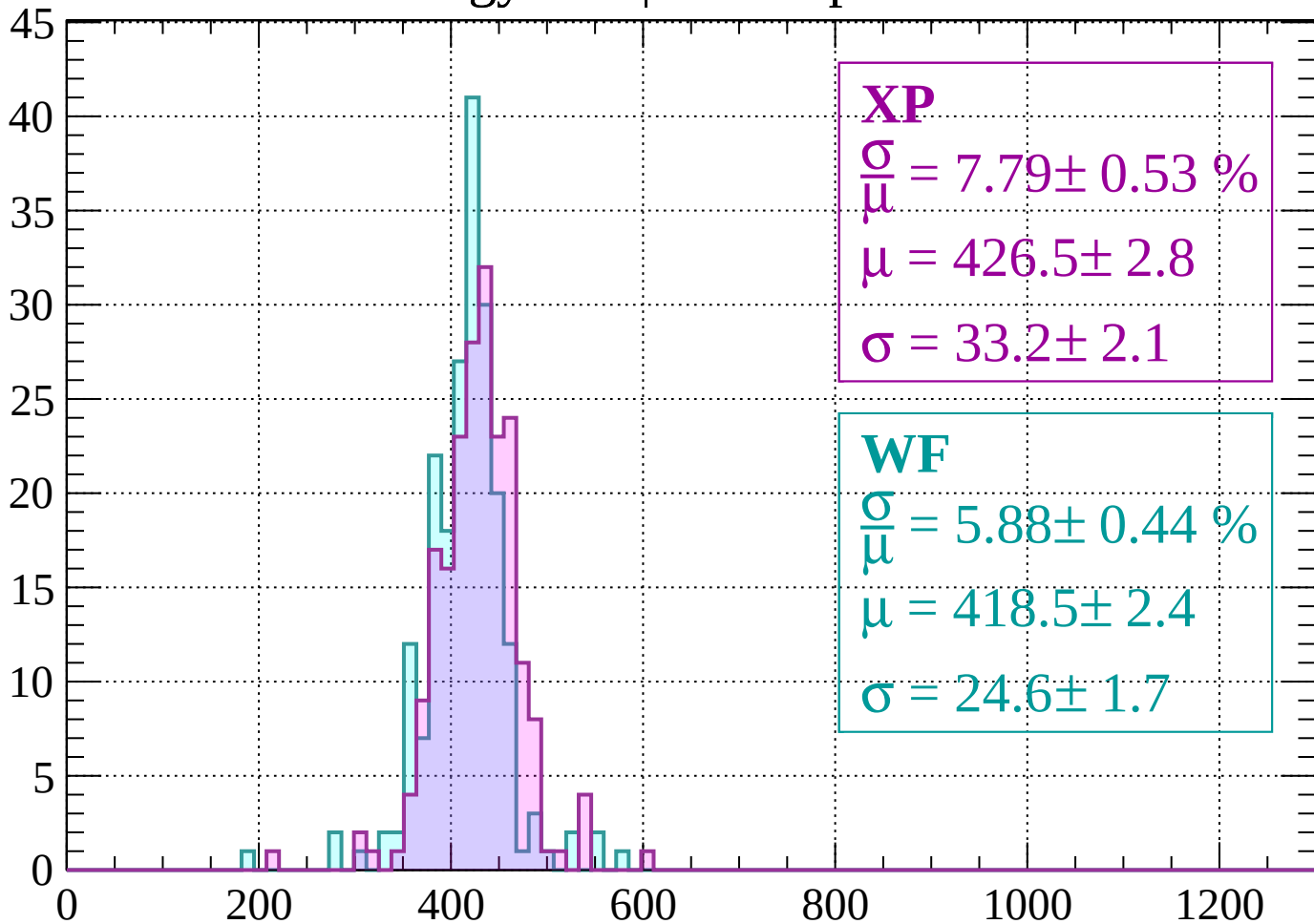
$$\sigma = 33.2 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.44 \%$$

$$\mu = 418.5 \pm 2.4$$

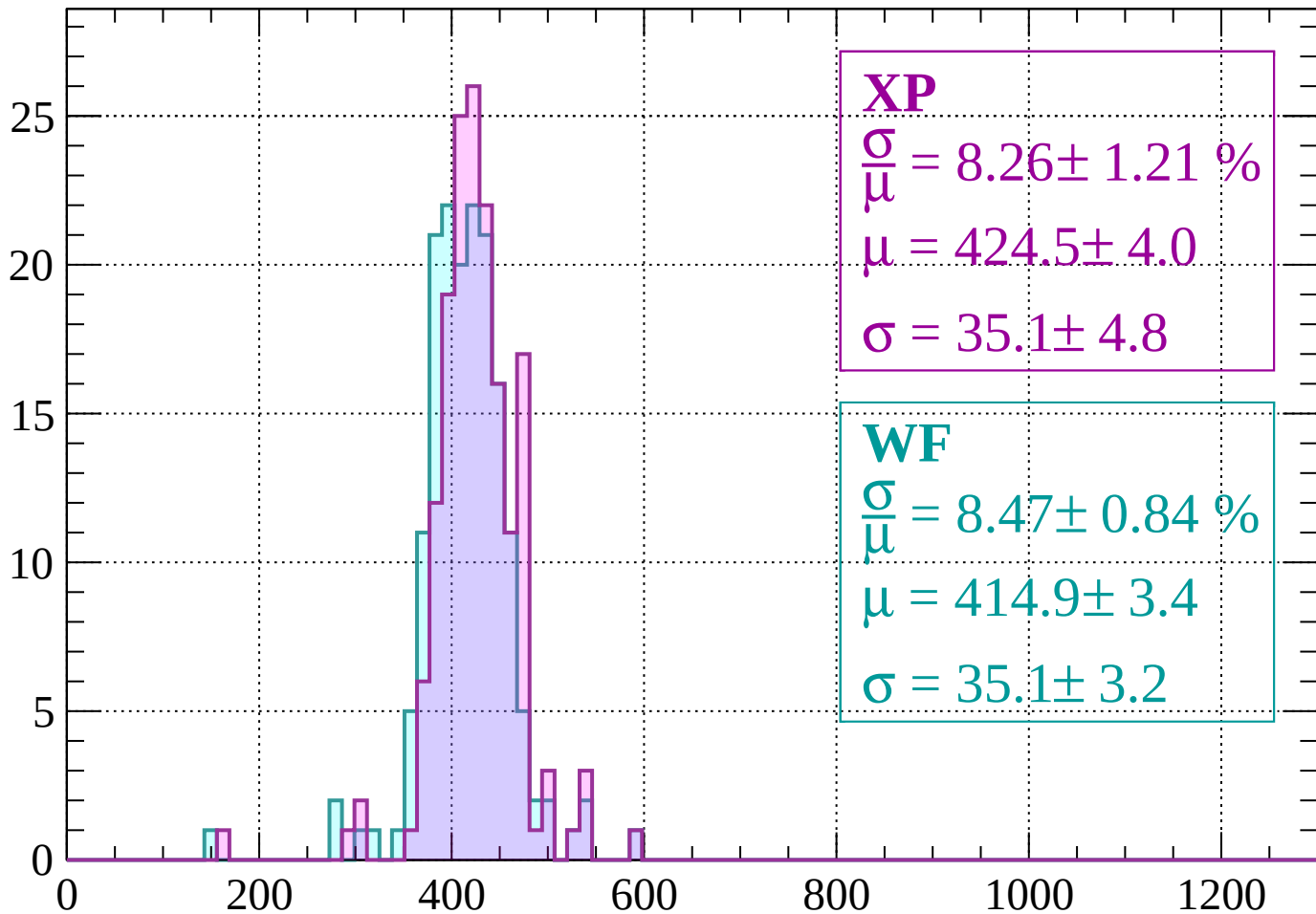
$$\sigma = 24.6 \pm 1.7$$



dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 1.21 \%$$

$$\mu = 424.5 \pm 4.0$$

$$\sigma = 35.1 \pm 4.8$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.84 \%$$

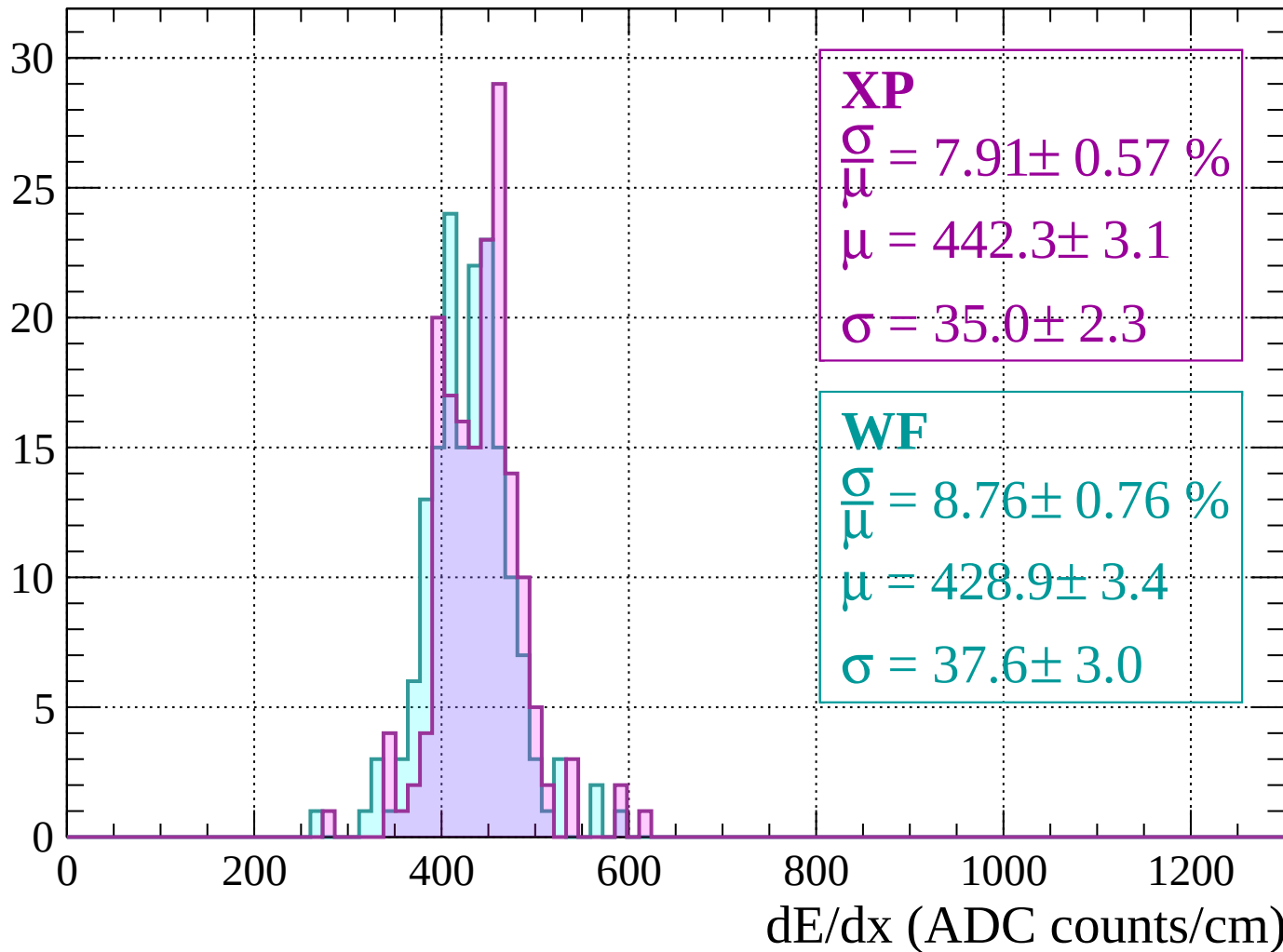
$$\mu = 414.9 \pm 3.4$$

$$\sigma = 35.1 \pm 3.2$$

dE/dx (ADC counts/cm)

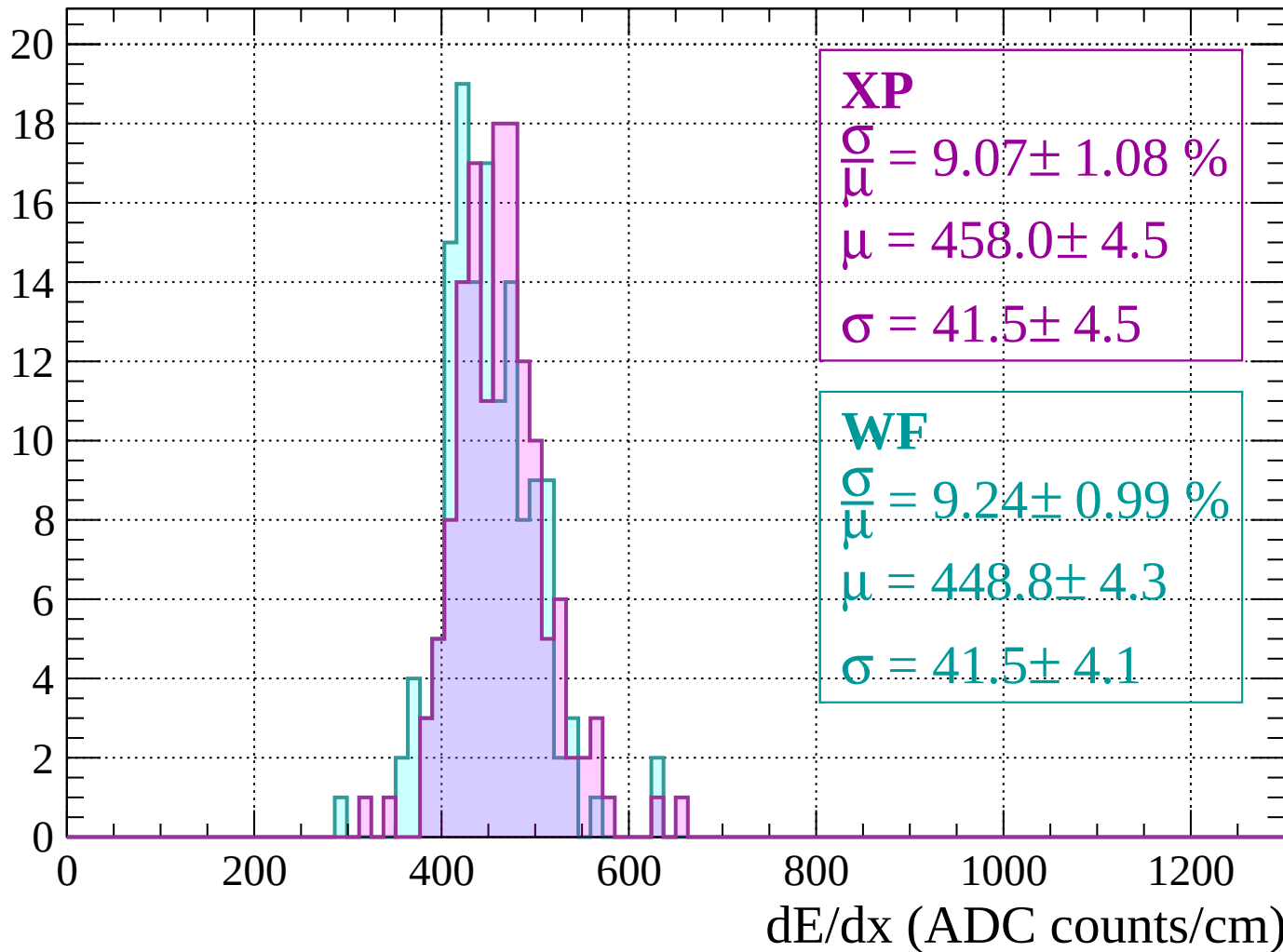
Energy loss | $-240 < p < -200$

Count



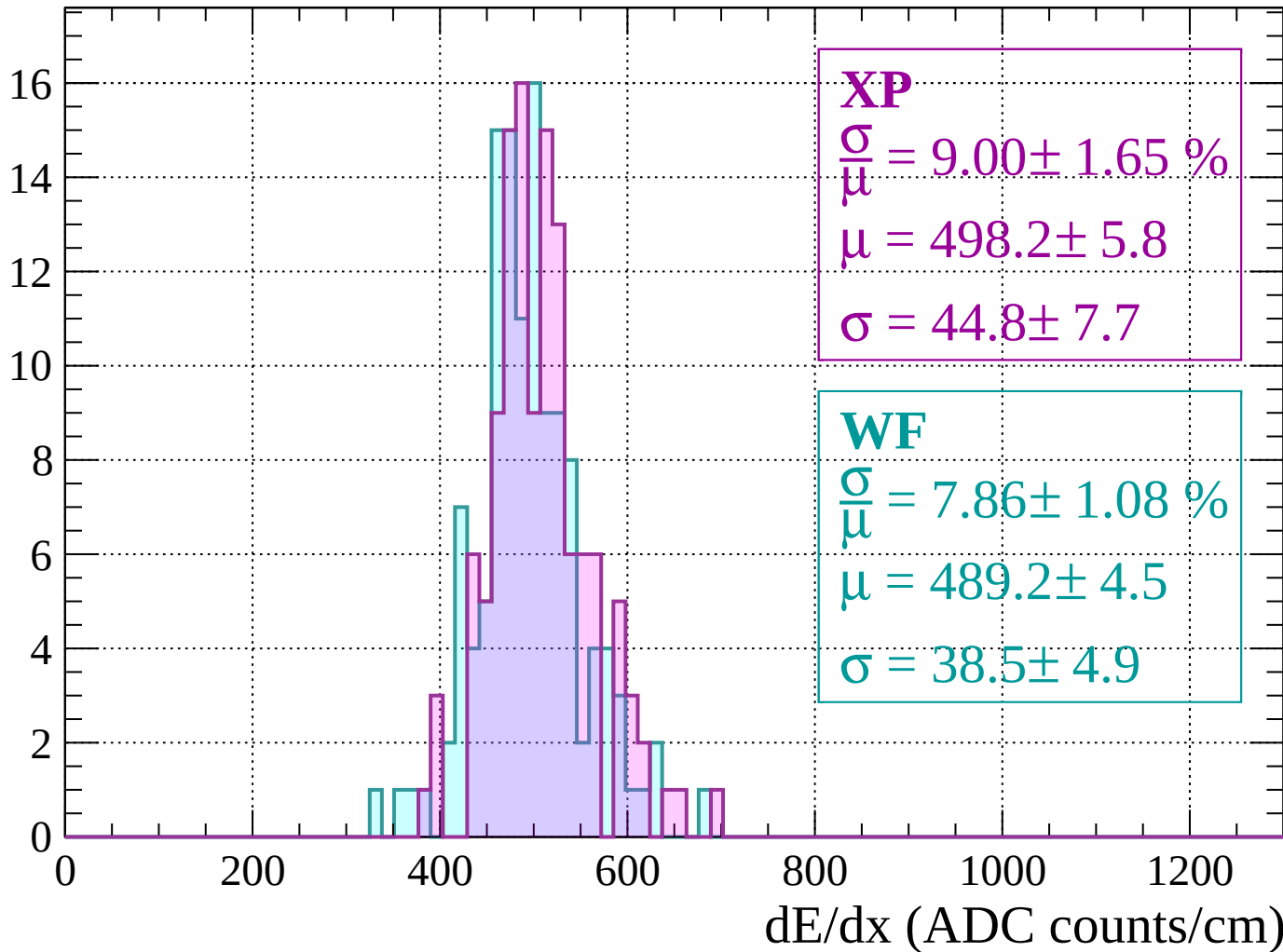
Energy loss | $-200 < p < -160$

Count



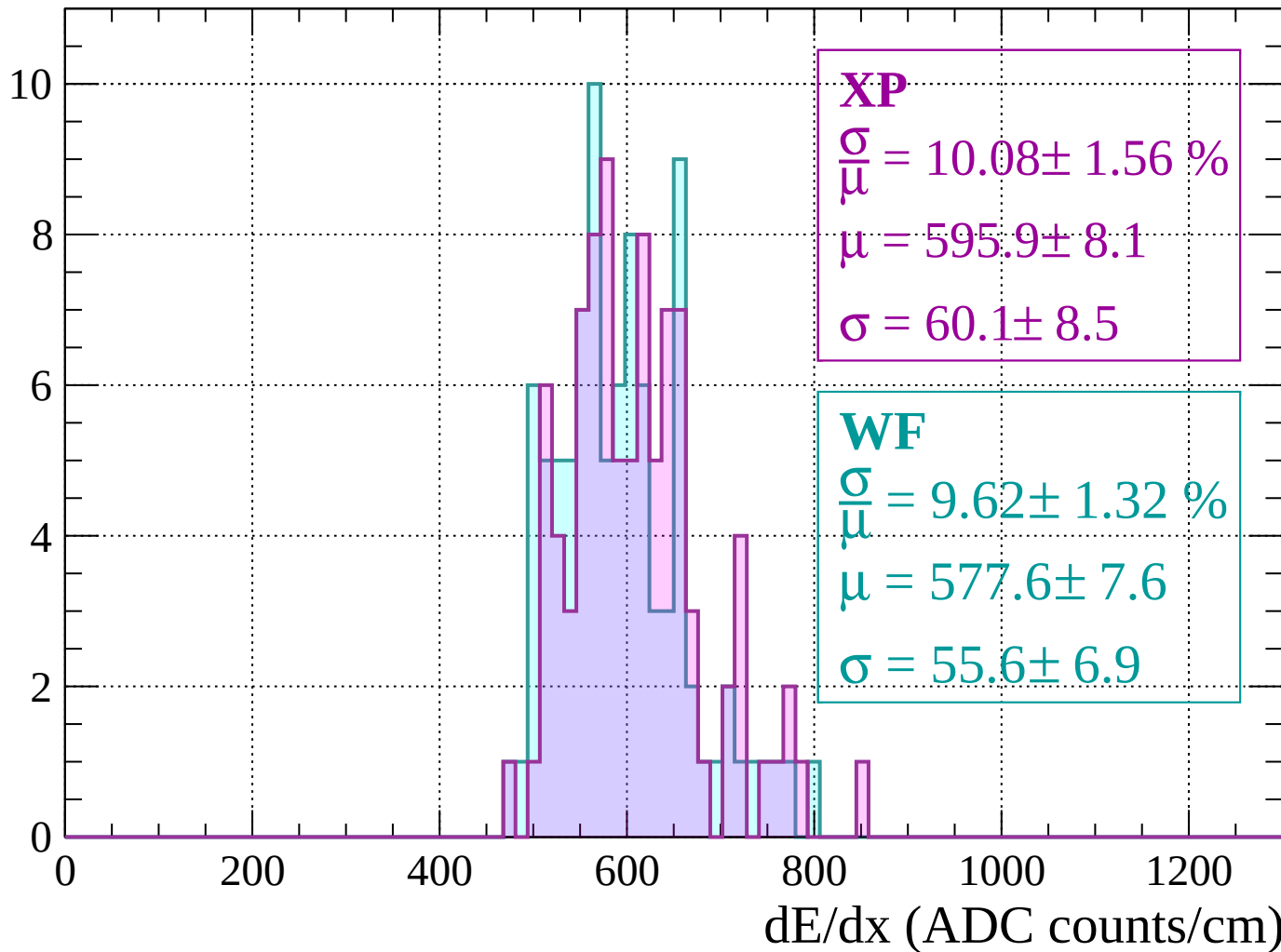
Energy loss | $-160 < p < -120$

Count



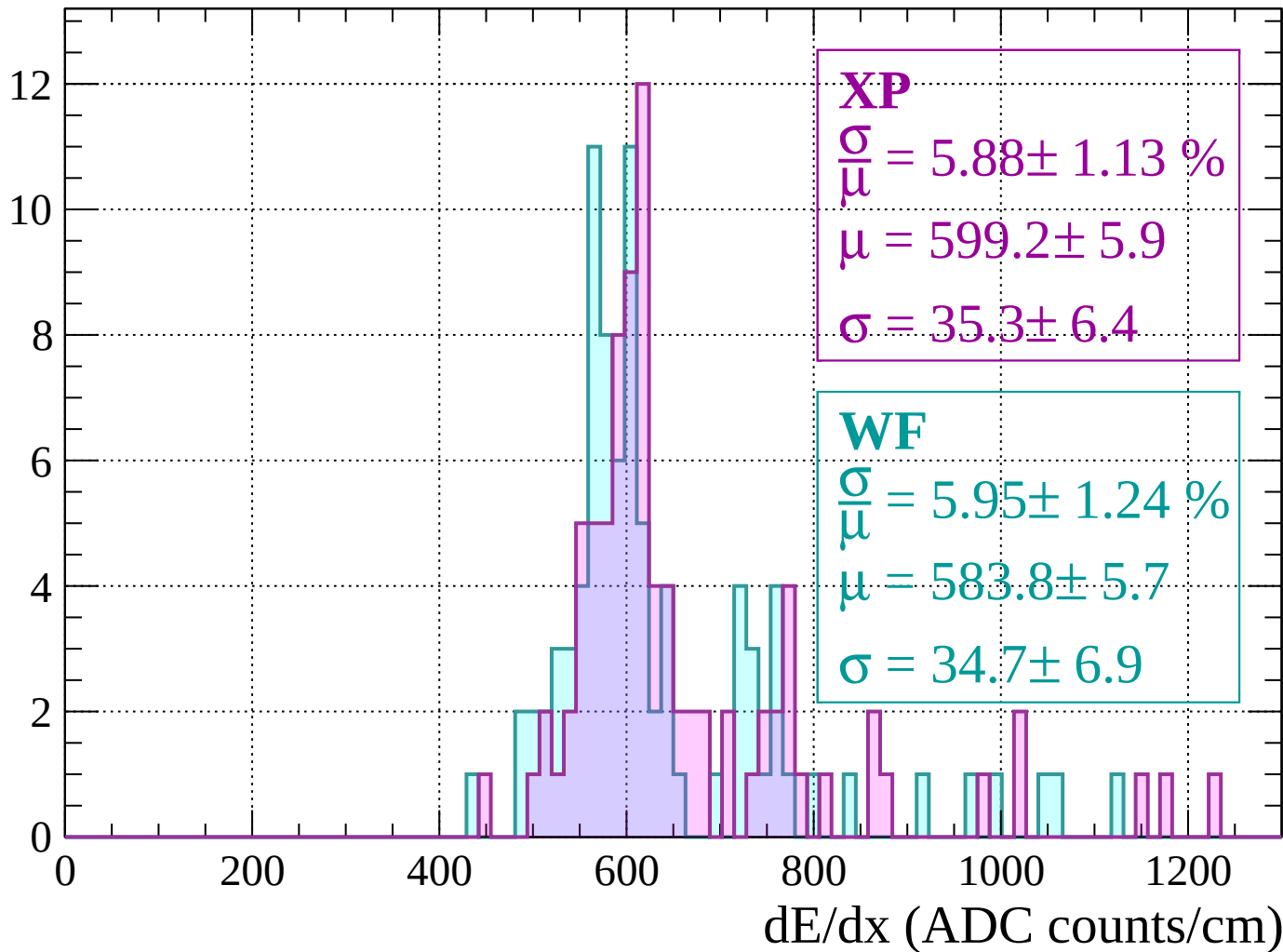
Energy loss | $-120 < p < -80$

Count



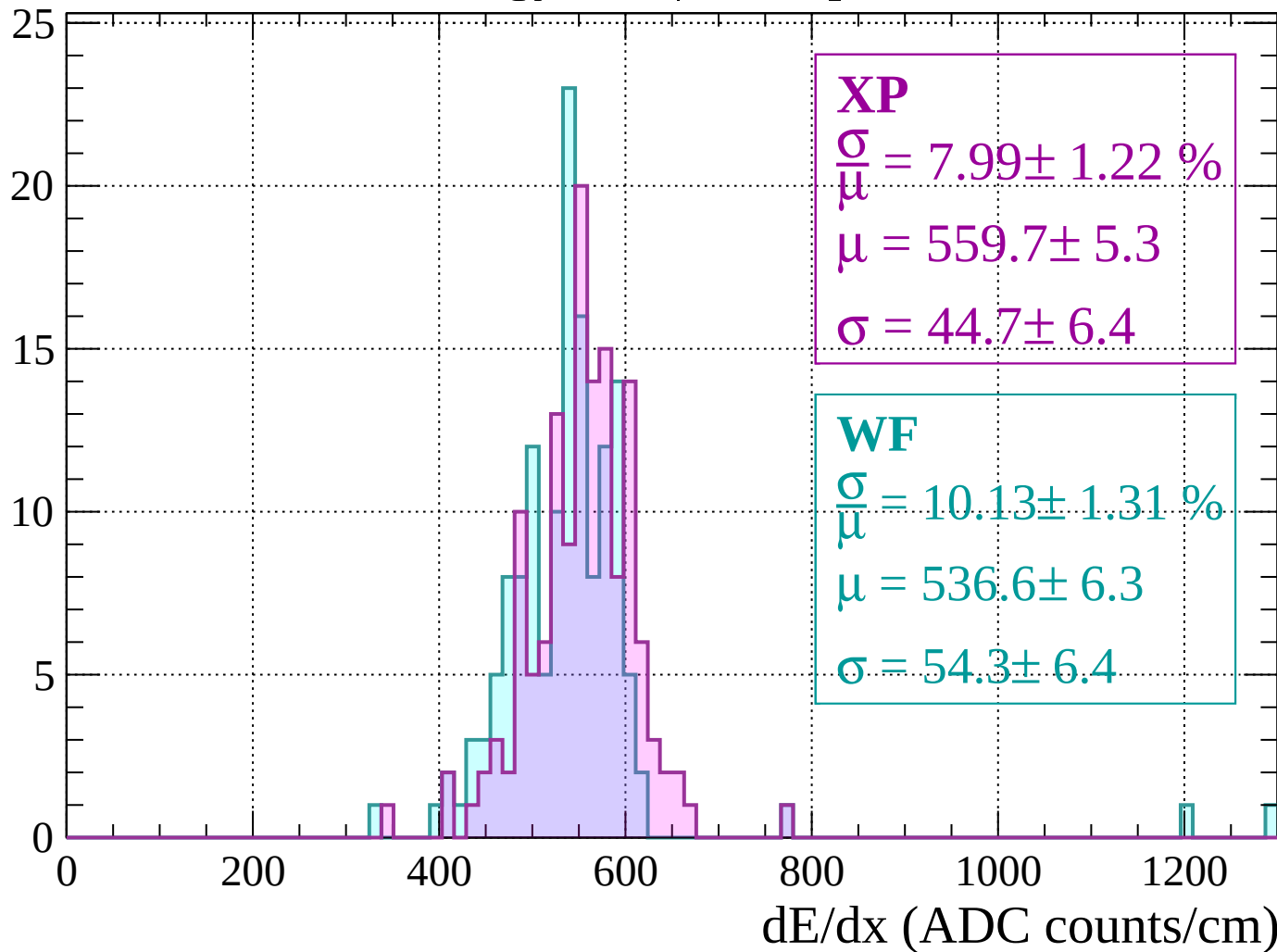
Energy loss | $-80 < p < -40$

Count



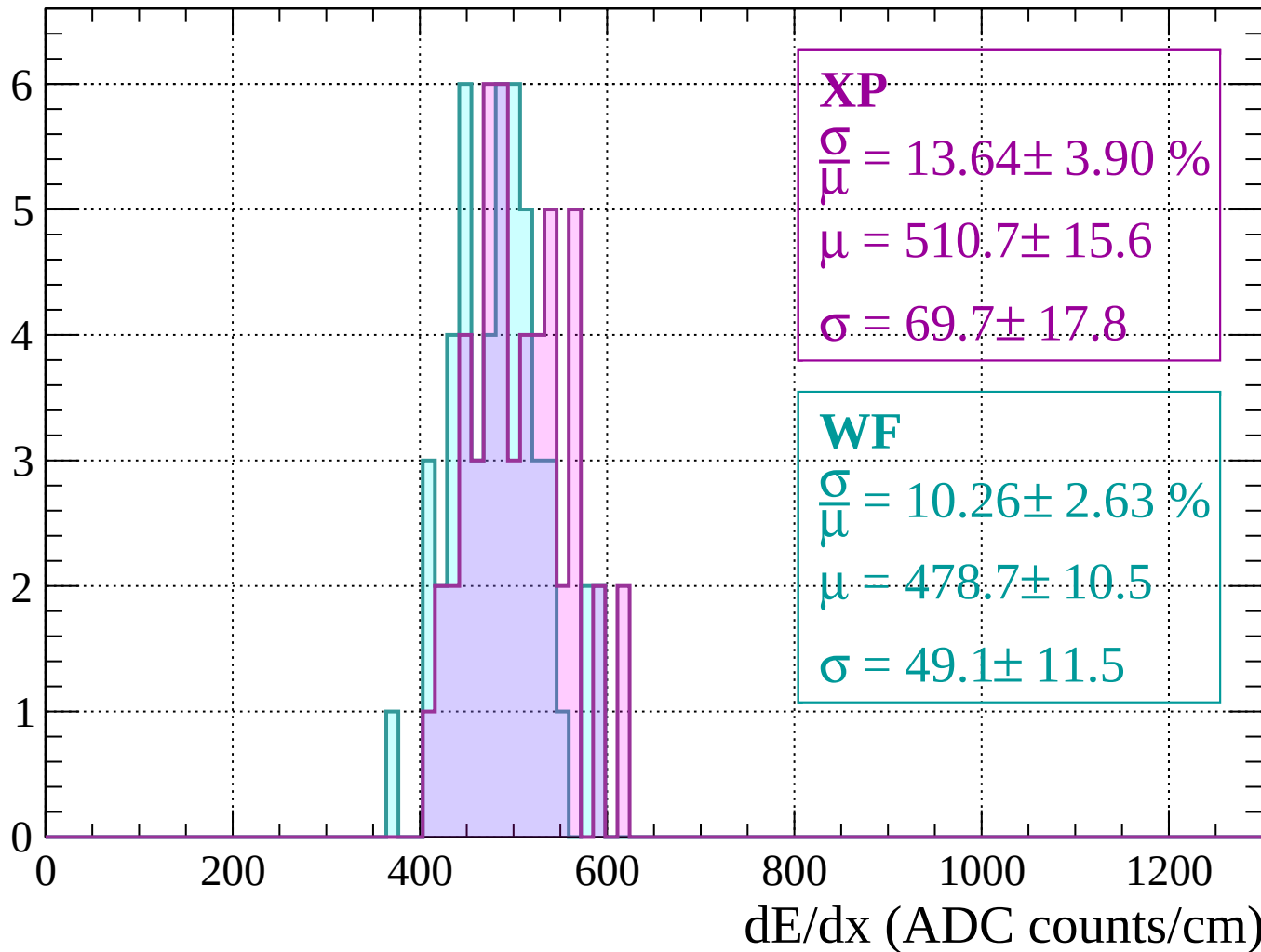
Energy loss | $-40 < p < 0$

Count



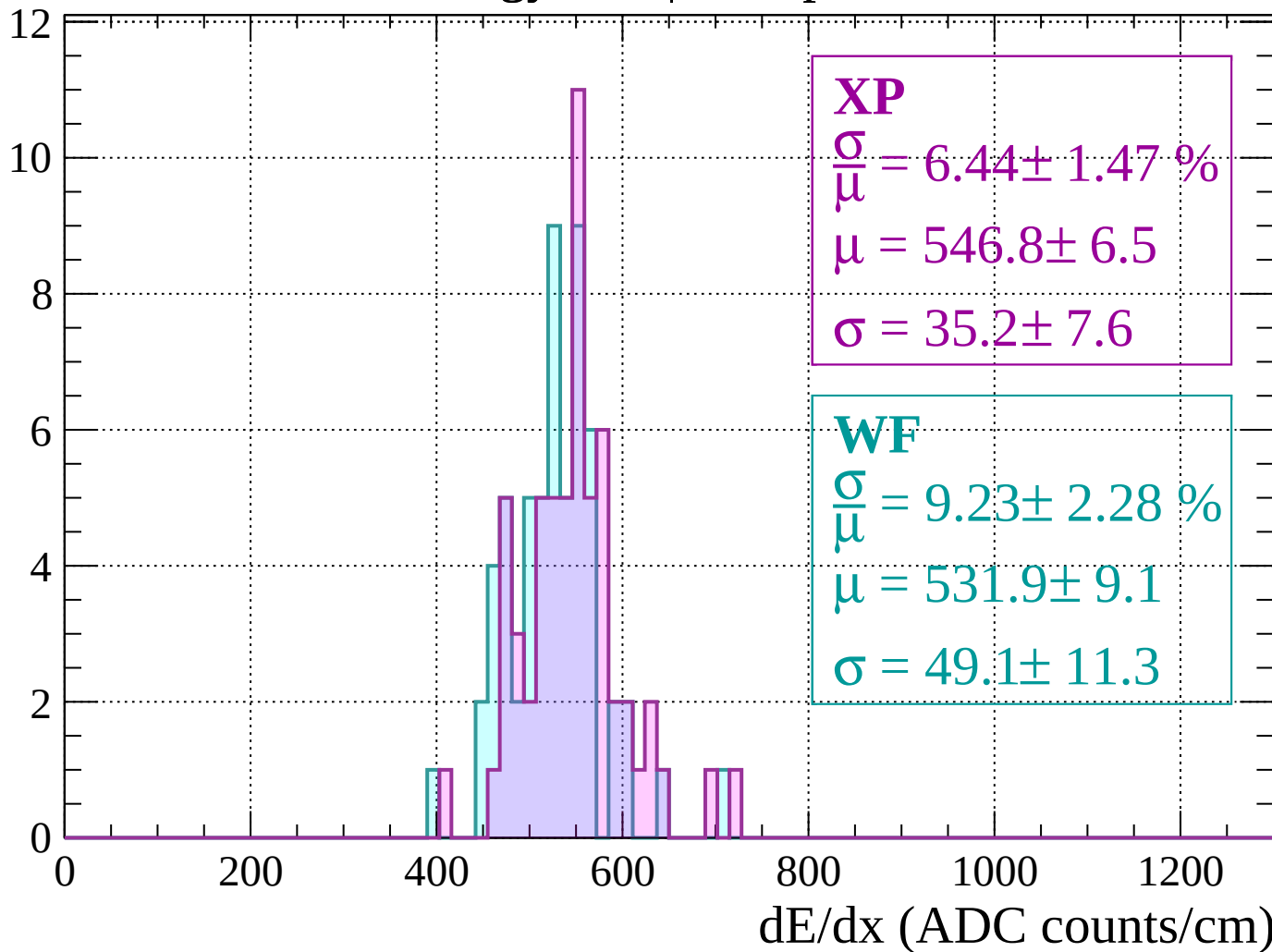
Energy loss | $0 < p < 40$

Count

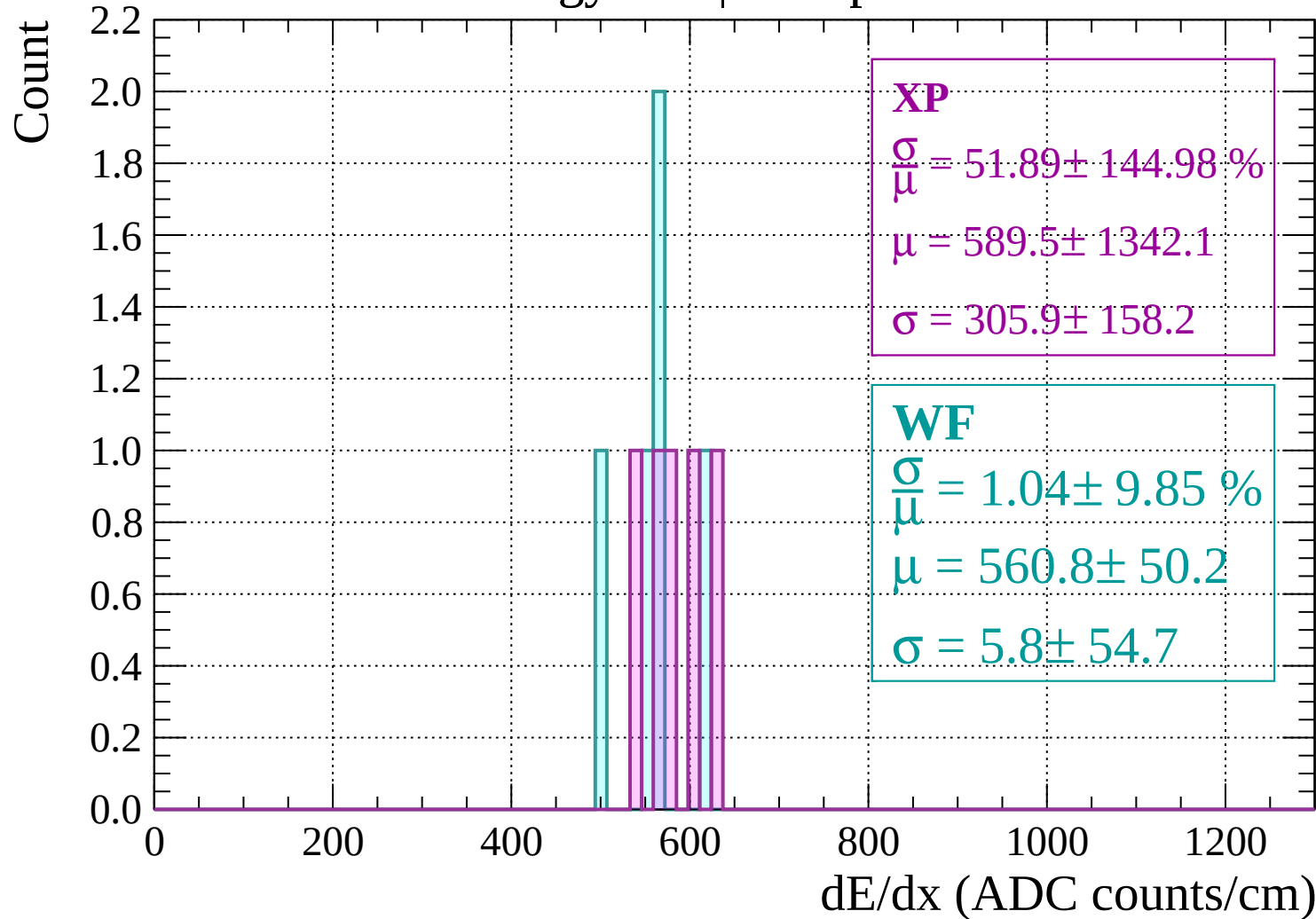


Energy loss | $40 < p < 80$

Count

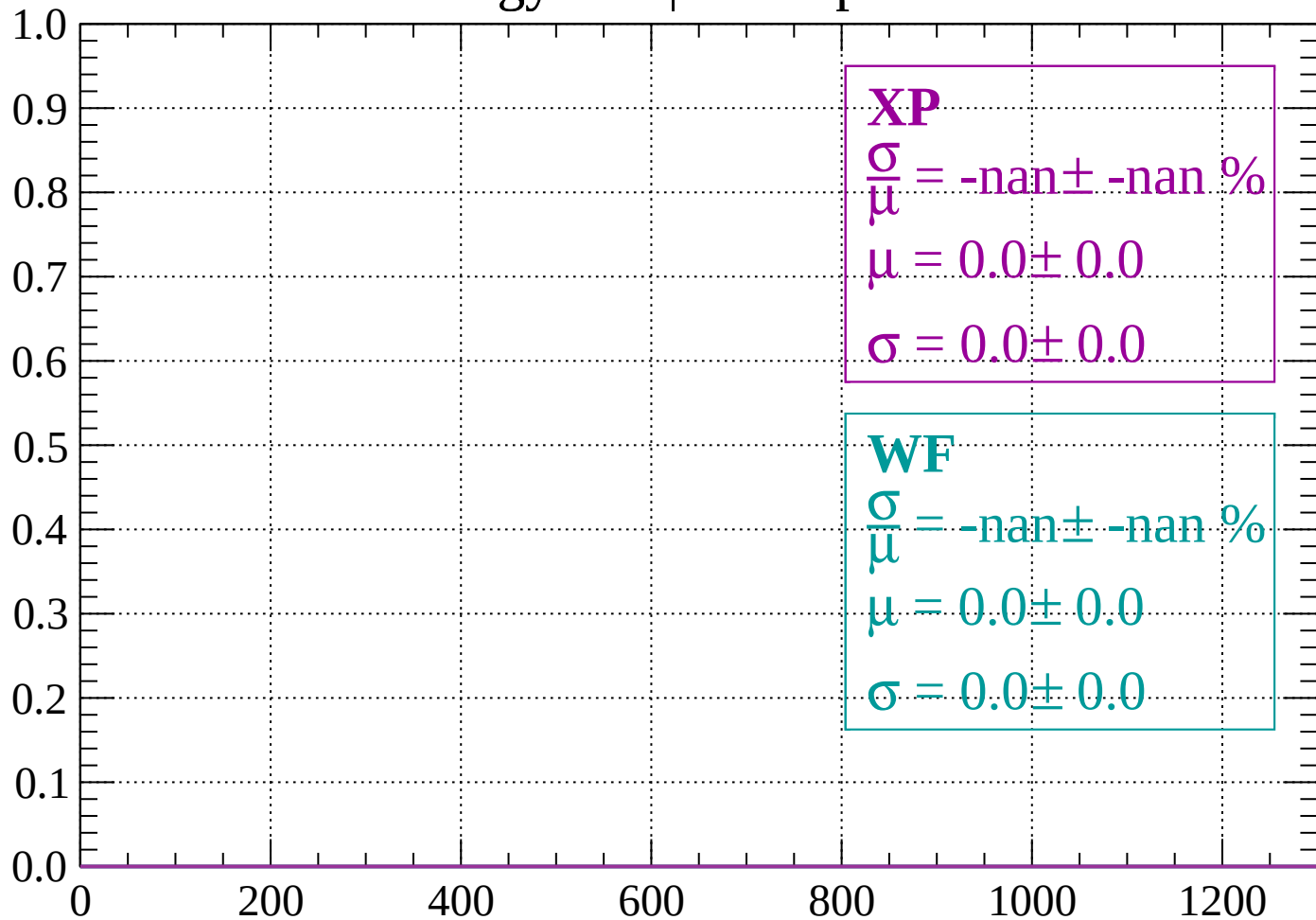


Energy loss | $80 < p < 120$



Energy loss | $120 < p < 160$

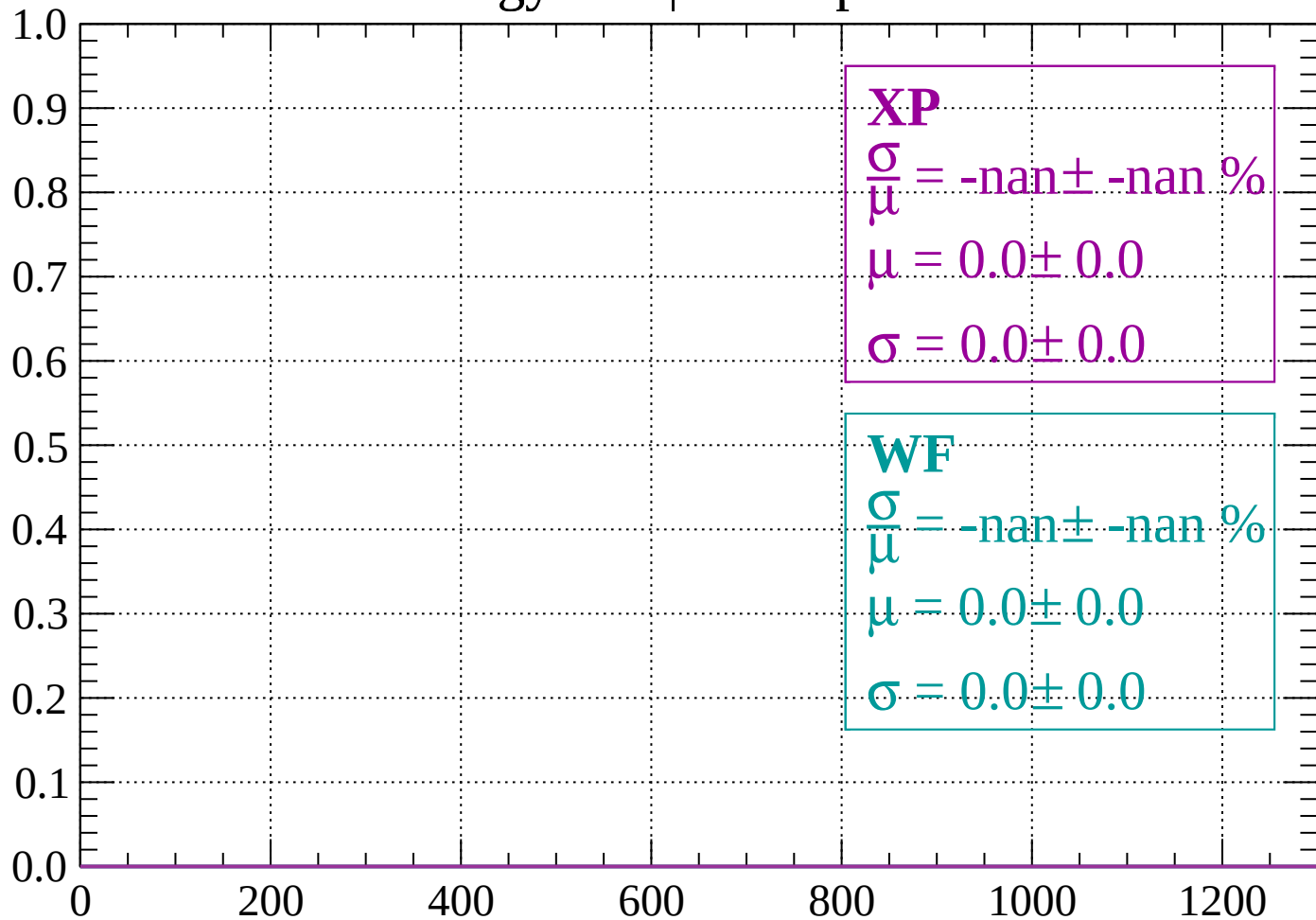
Count



dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

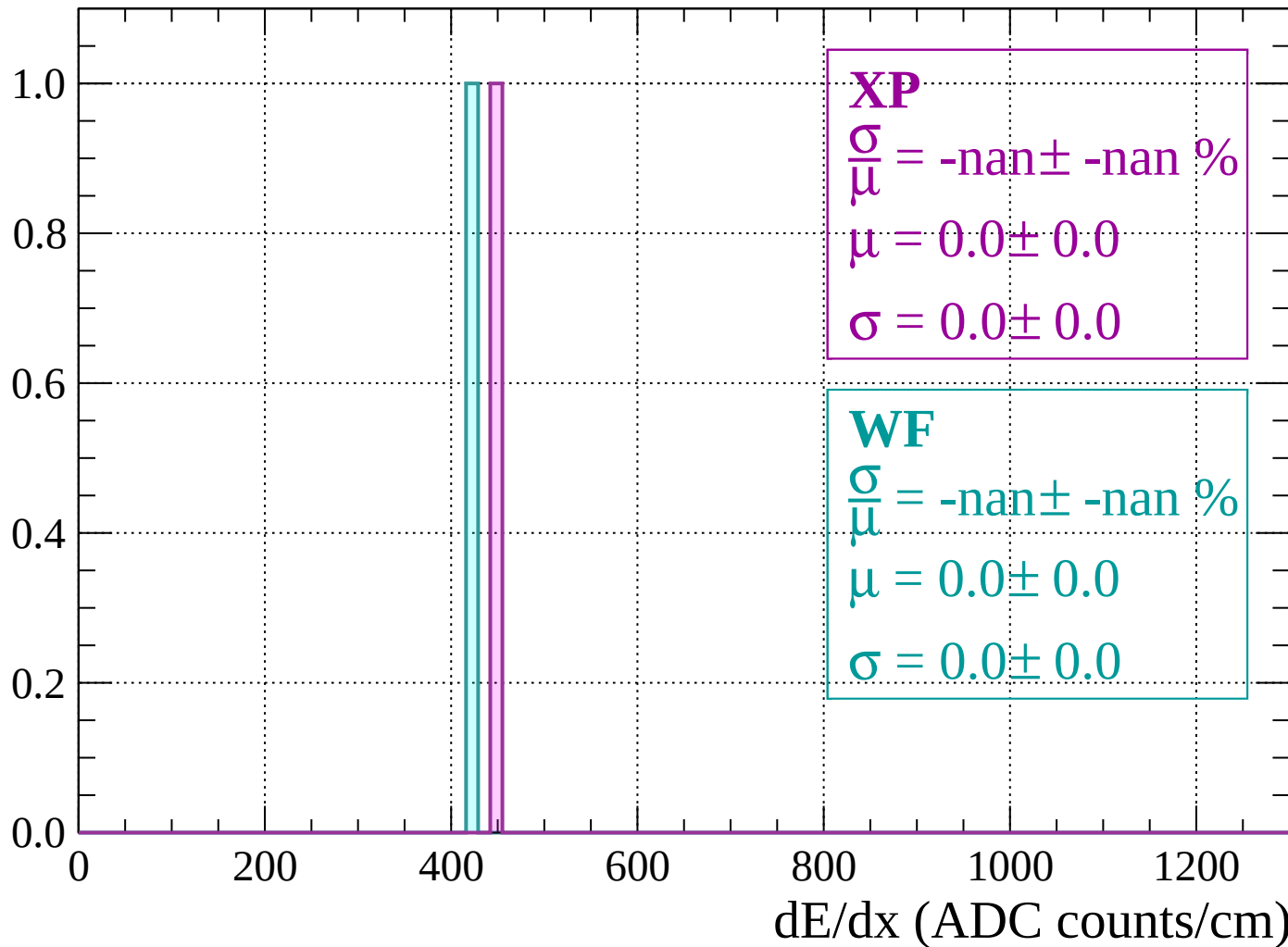
Count



dE/dx (ADC counts/cm)

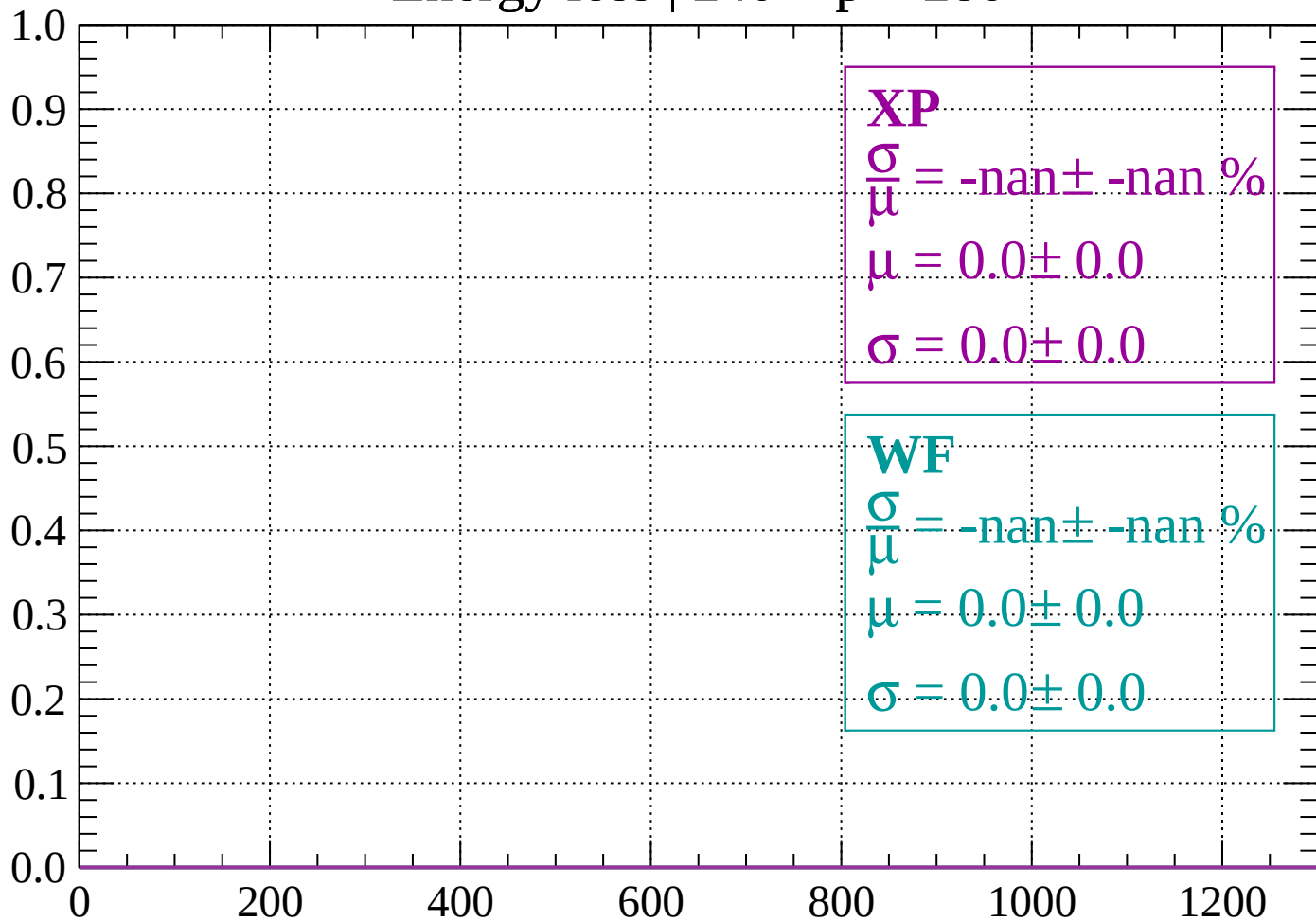
Energy loss | $200 < p < 240$

Count



Energy loss | $240 < p < 280$

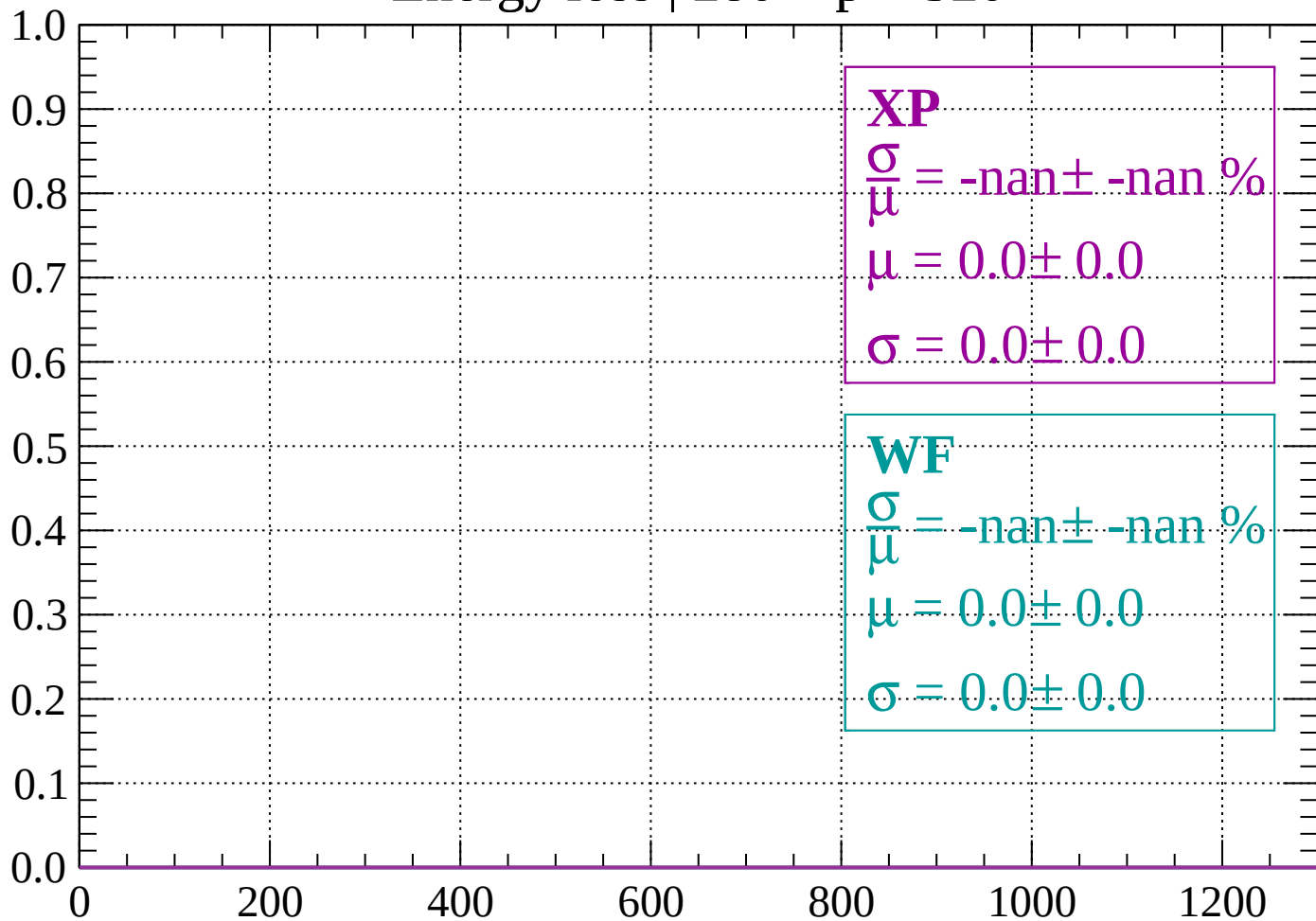
Count



dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

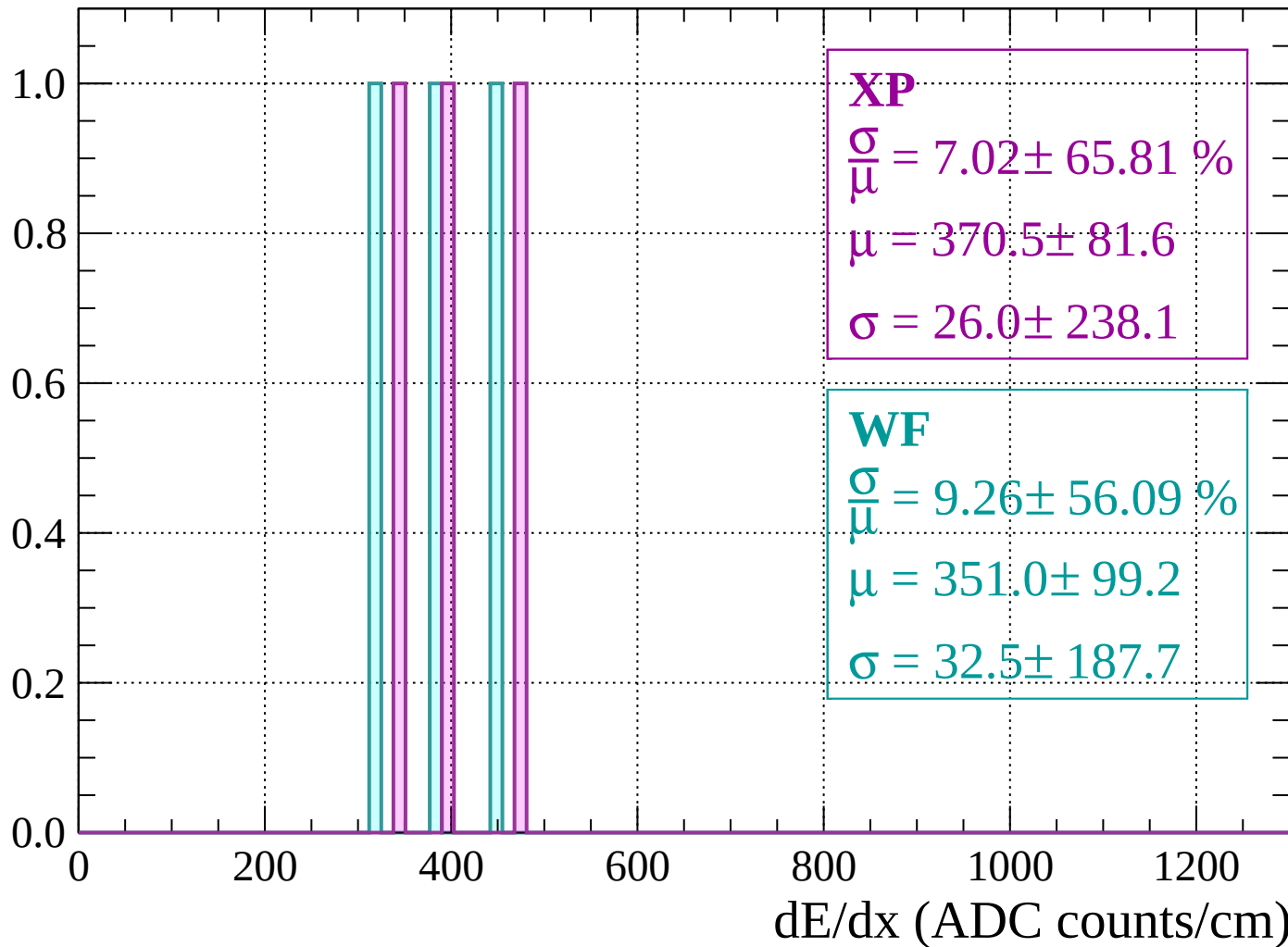
Count



dE/dx (ADC counts/cm)

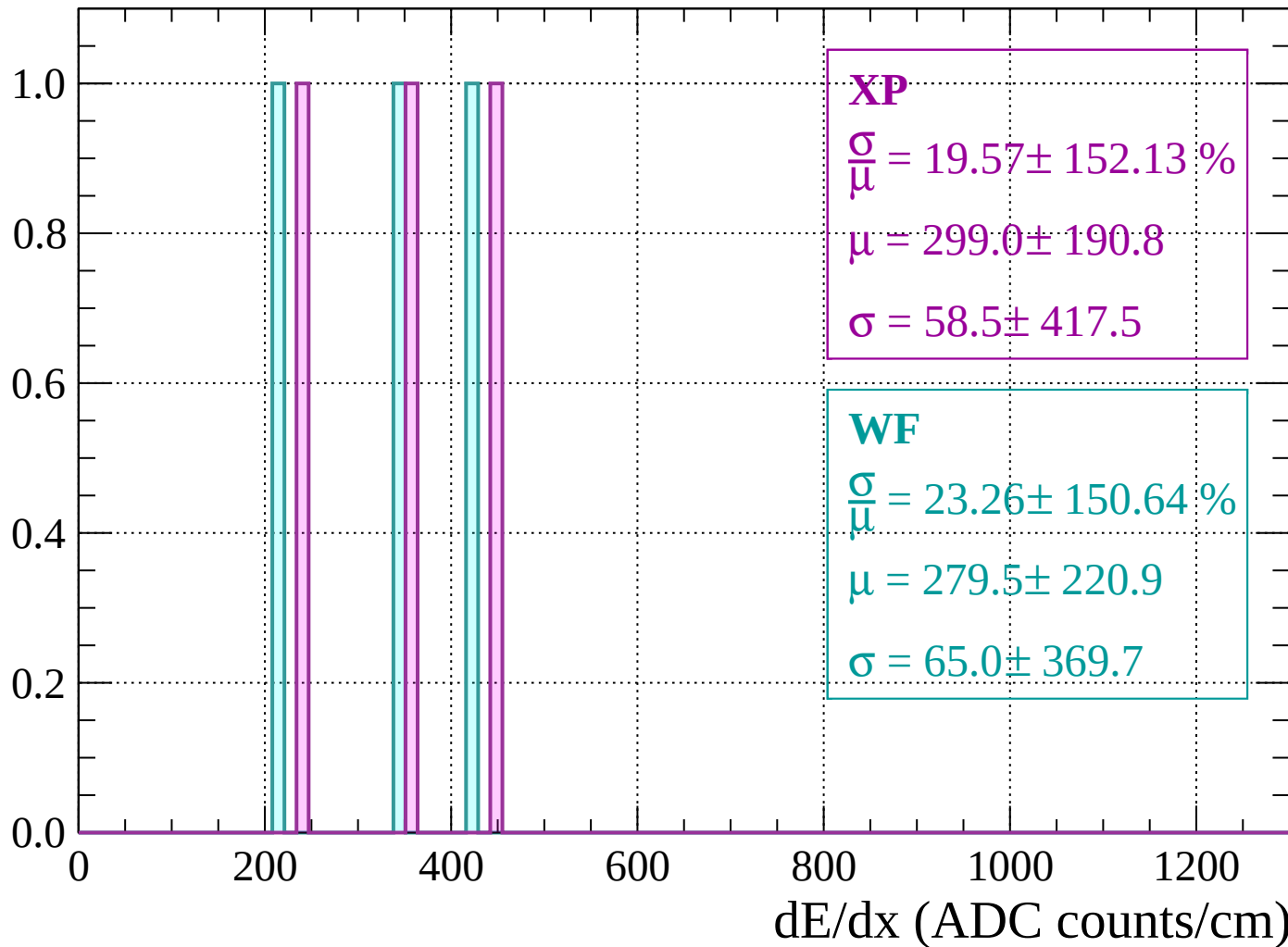
Energy loss | $320 < p < 360$

Count



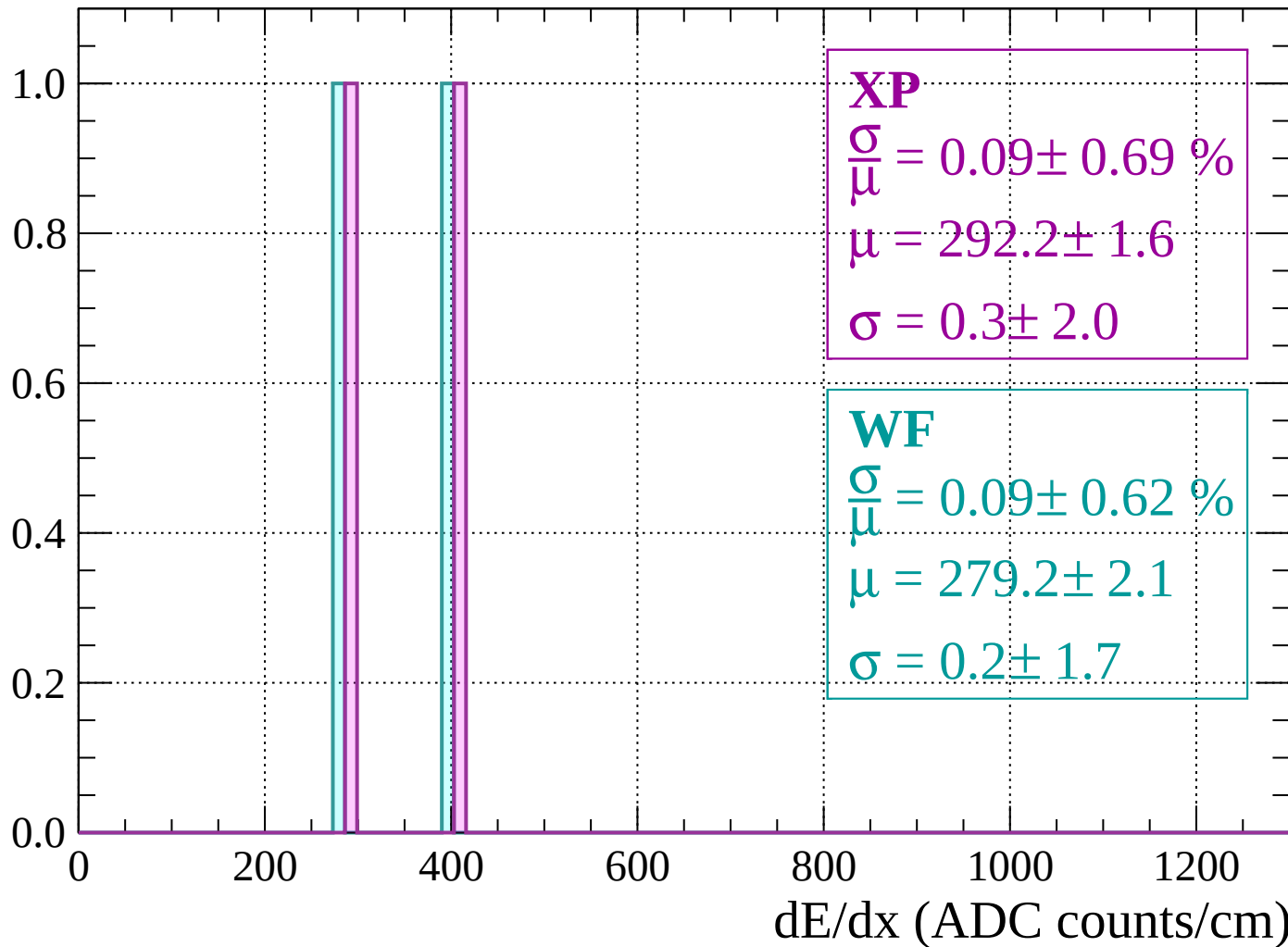
Energy loss | $360 < p < 400$

Count



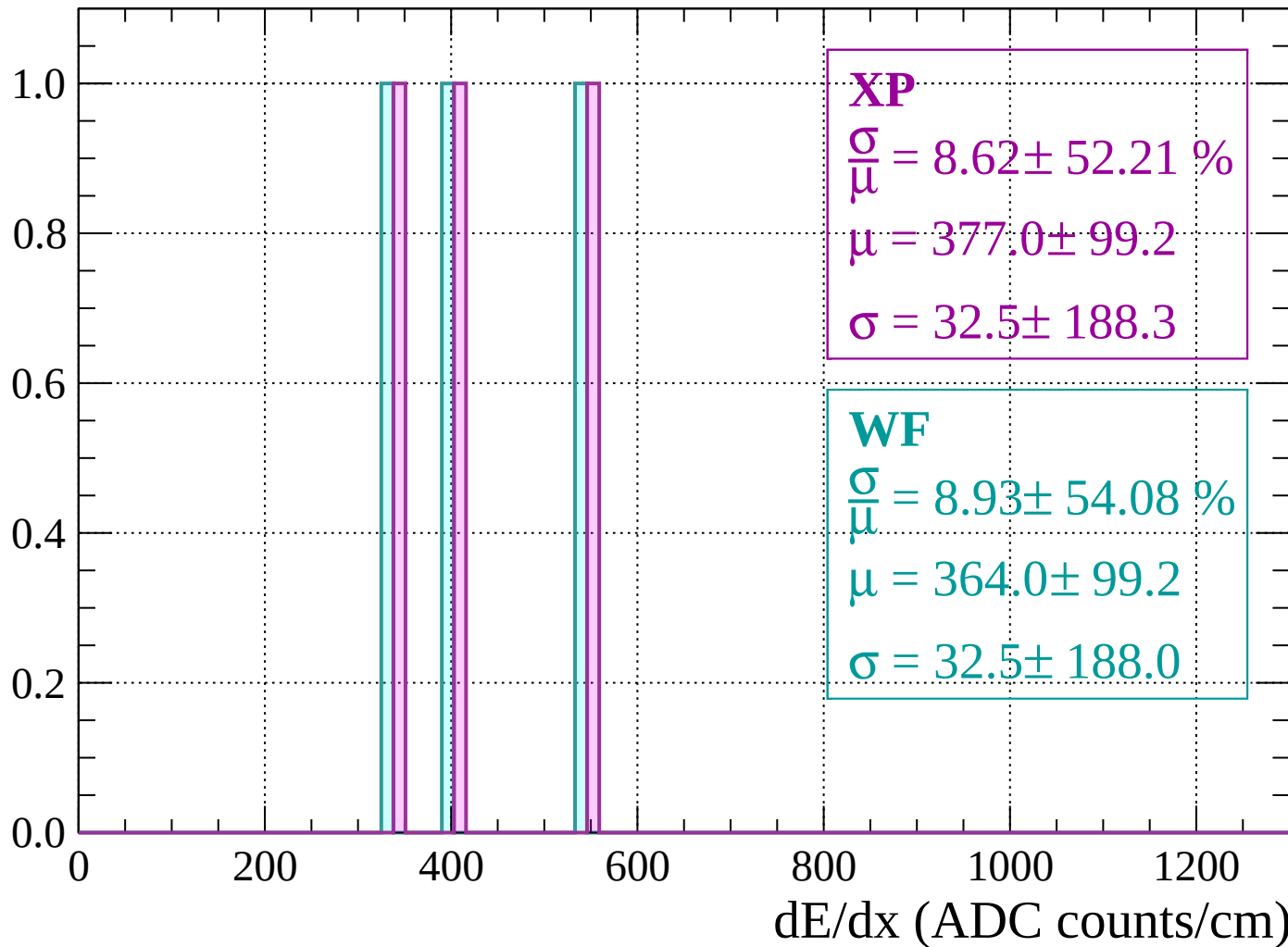
Energy loss | $400 < p < 440$

Count



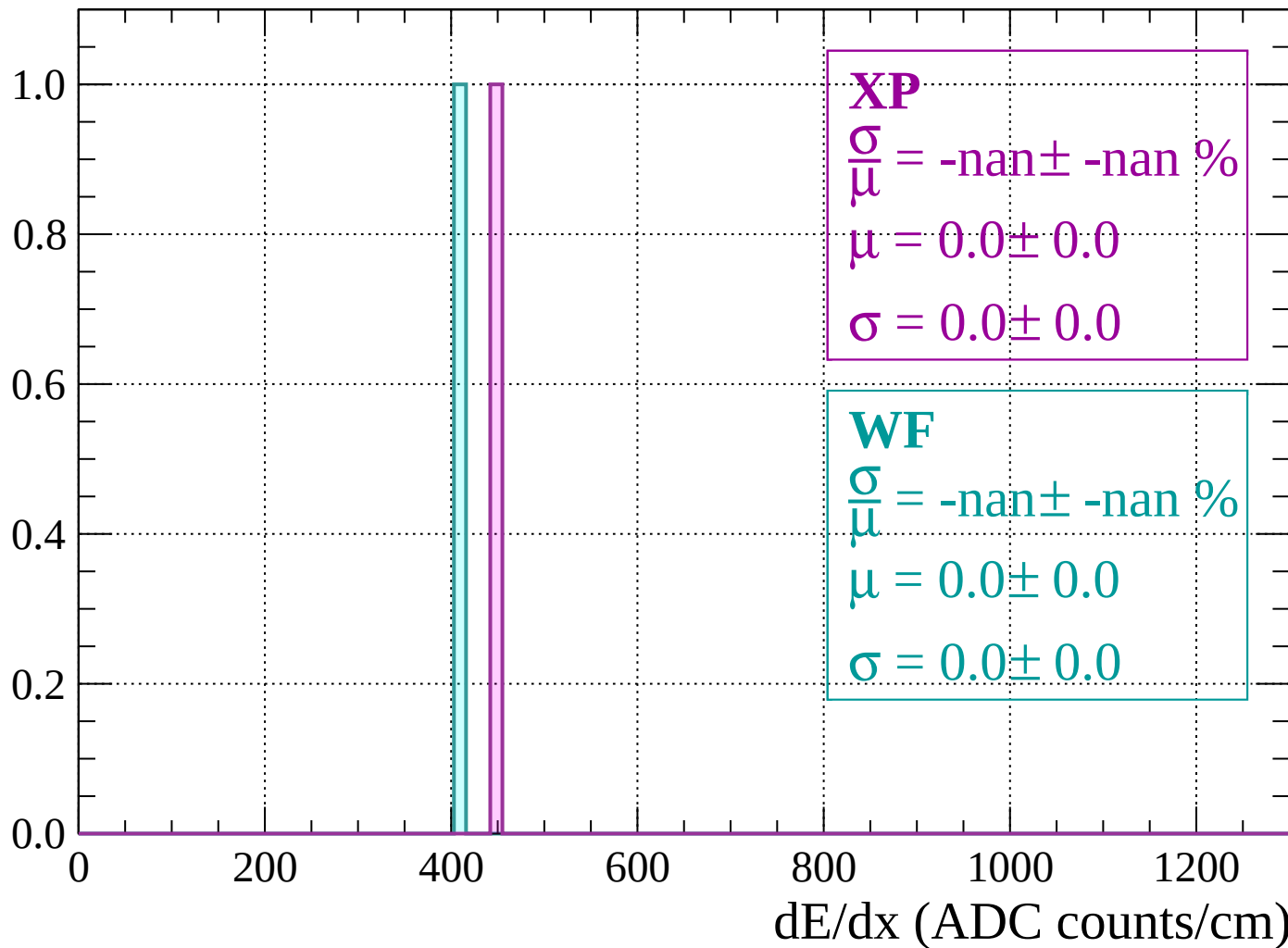
Energy loss | $440 < p < 480$

Count



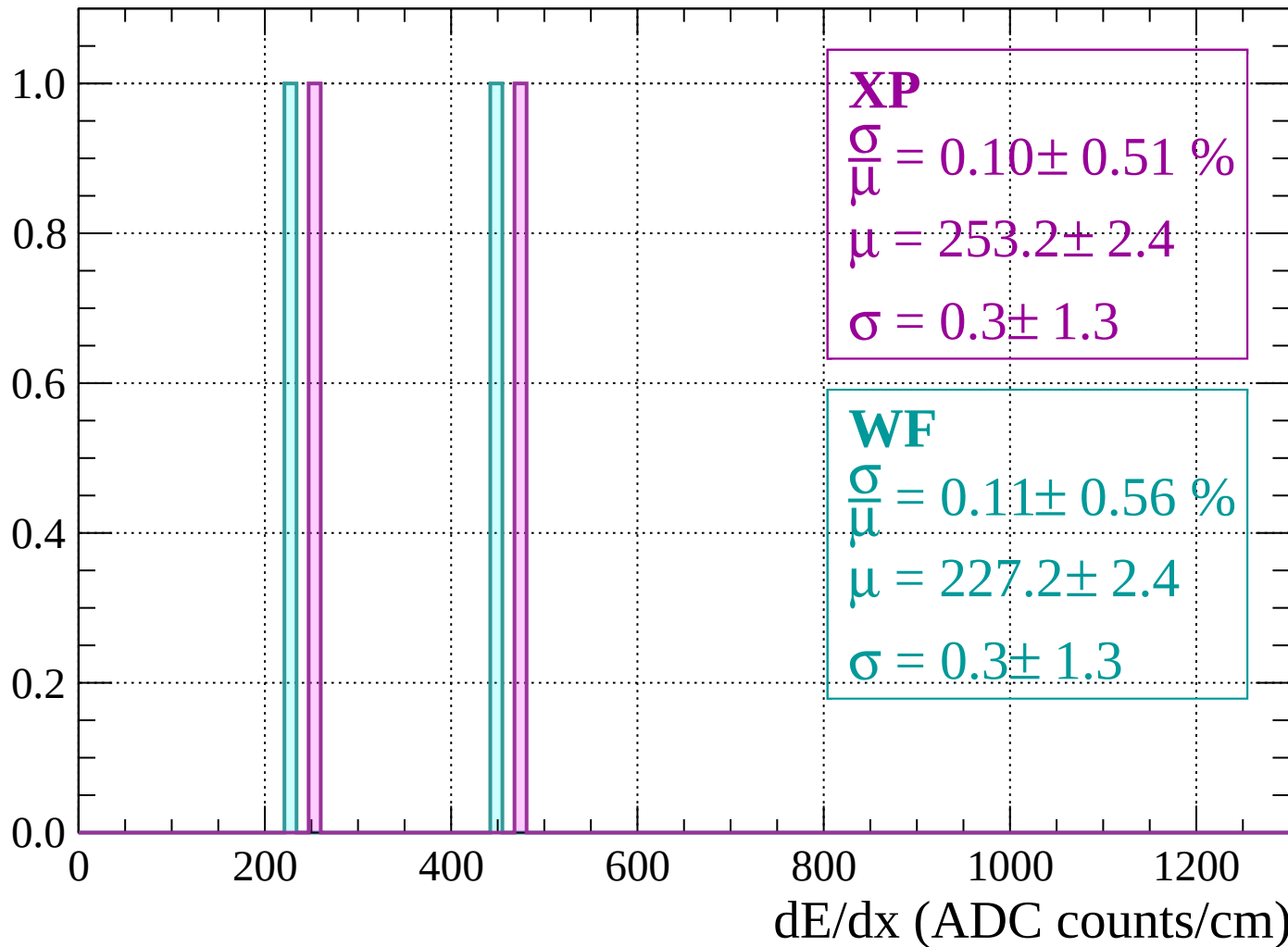
Energy loss | $480 < p < 520$

Count



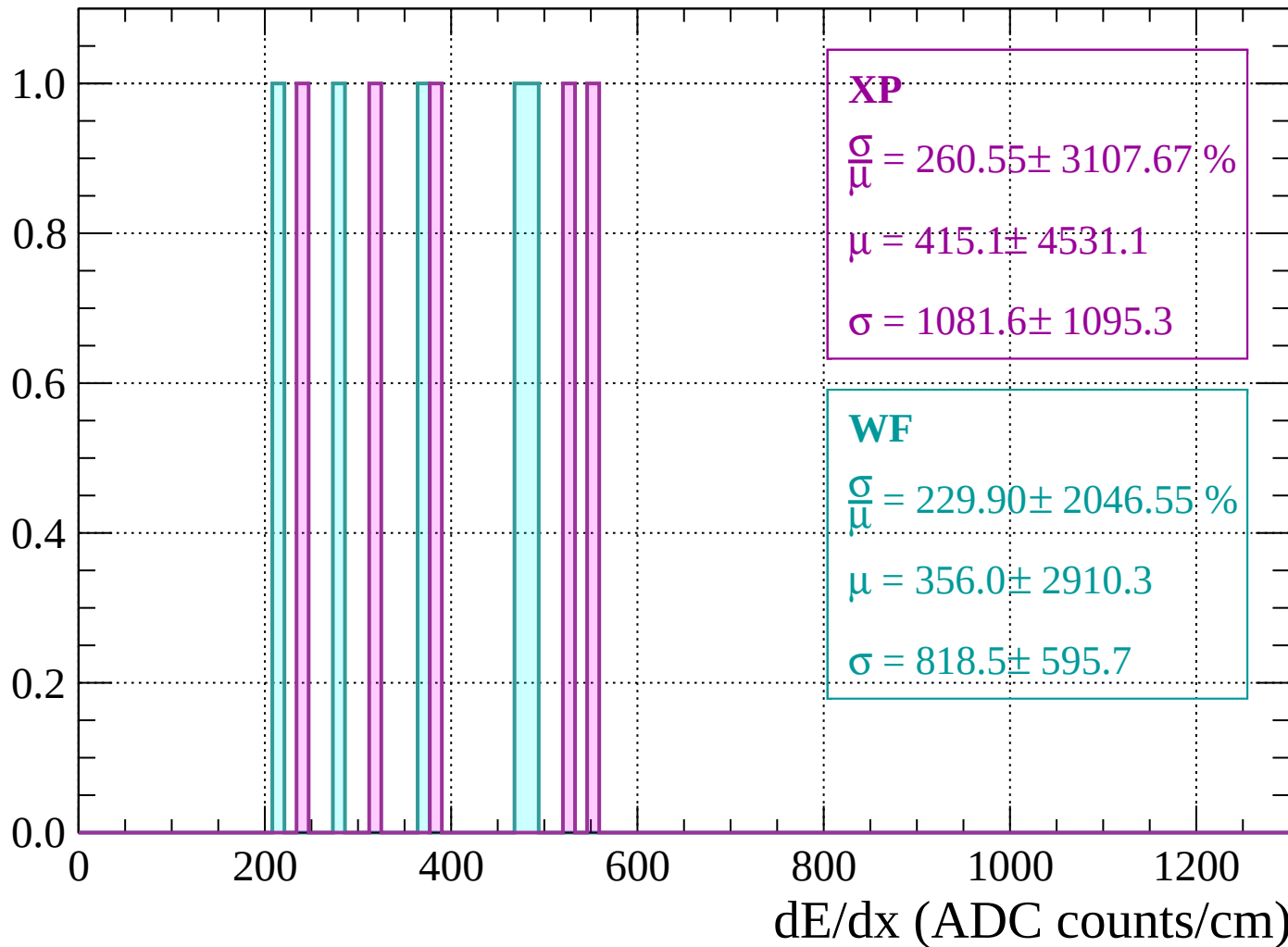
Energy loss | $520 < p < 560$

Count



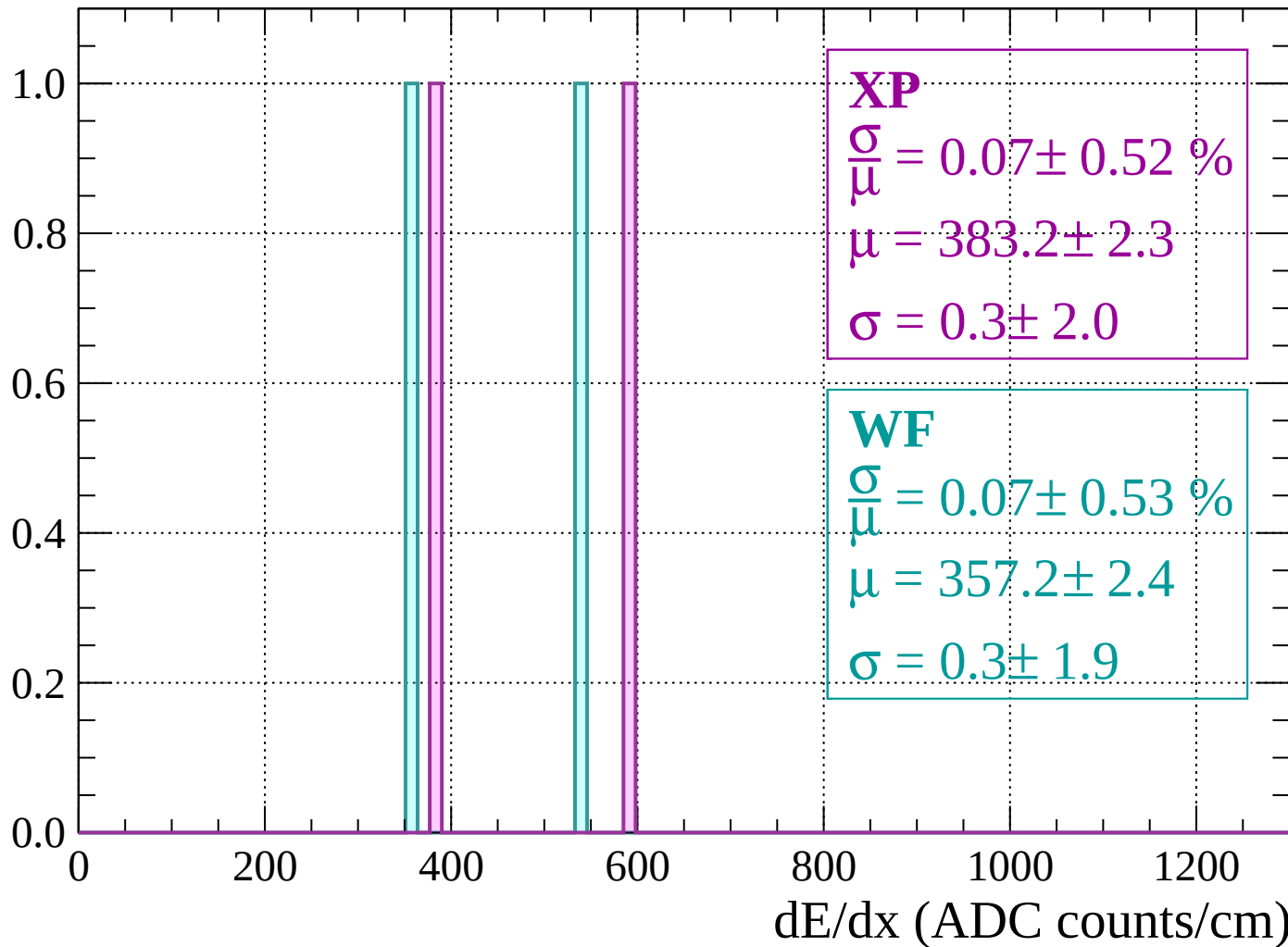
Energy loss | $560 < p < 600$

Count



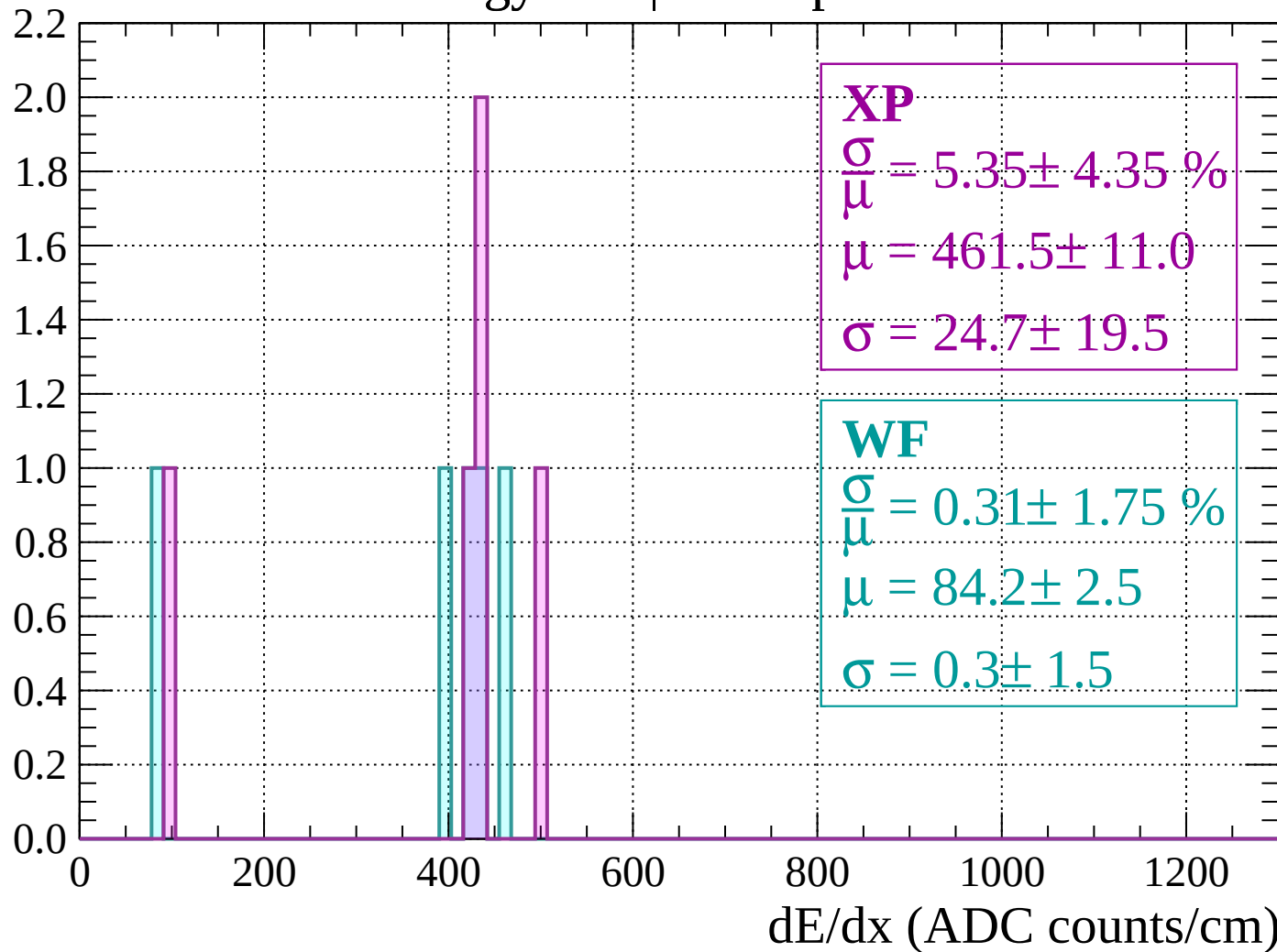
Energy loss | $600 < p < 640$

Count



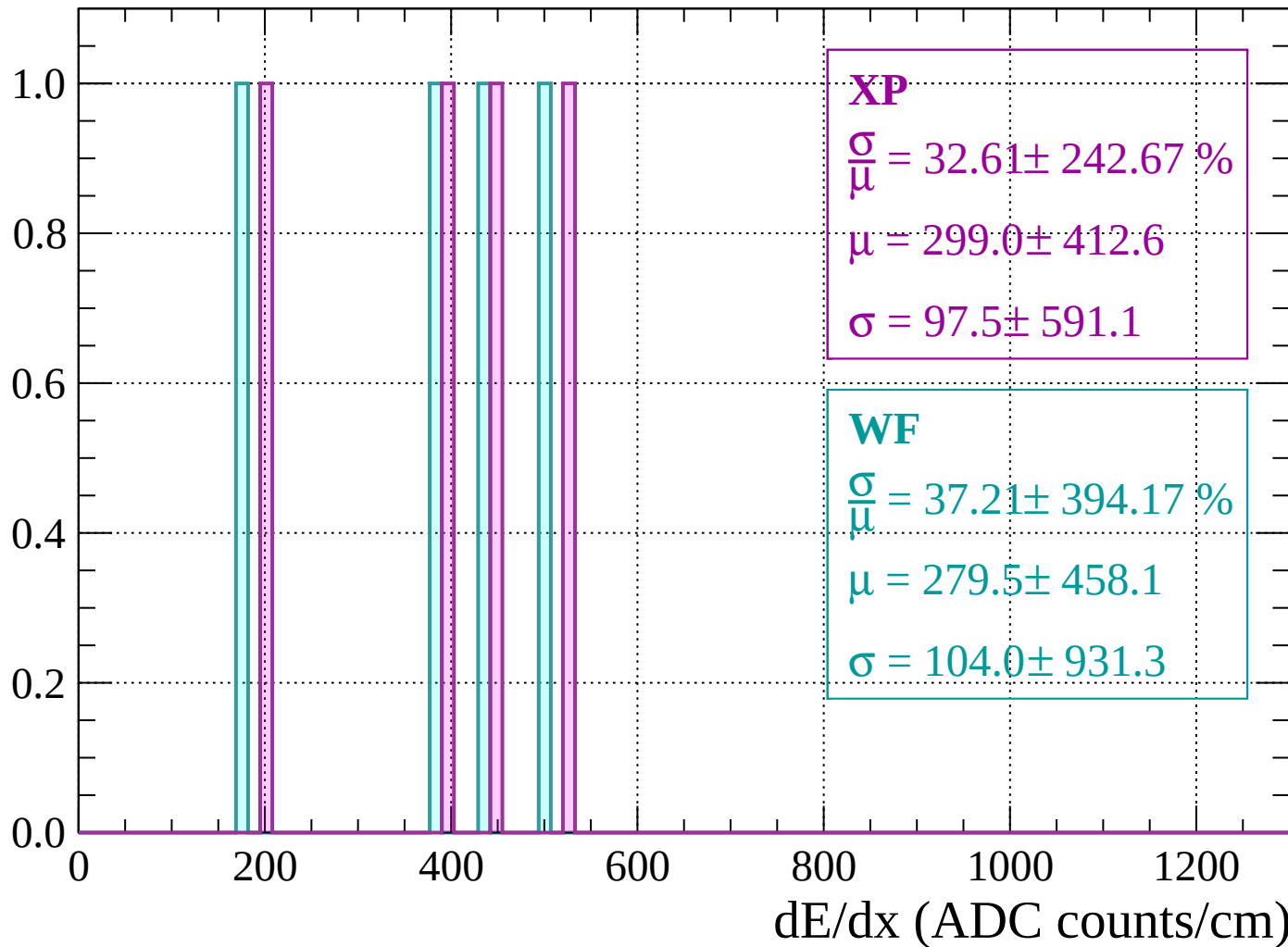
Energy loss | $640 < p < 680$

Count



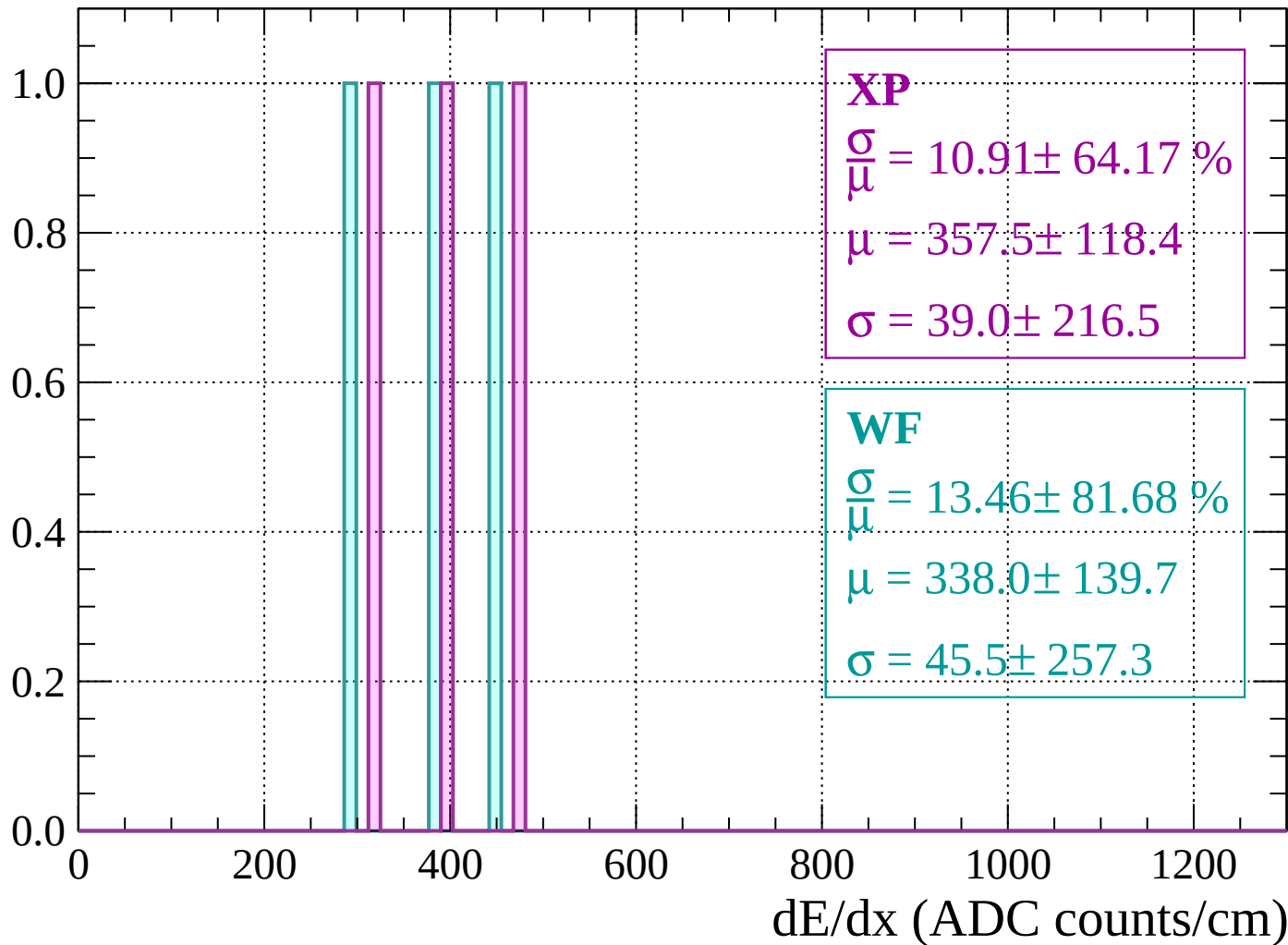
Energy loss | $680 < p < 720$

Count



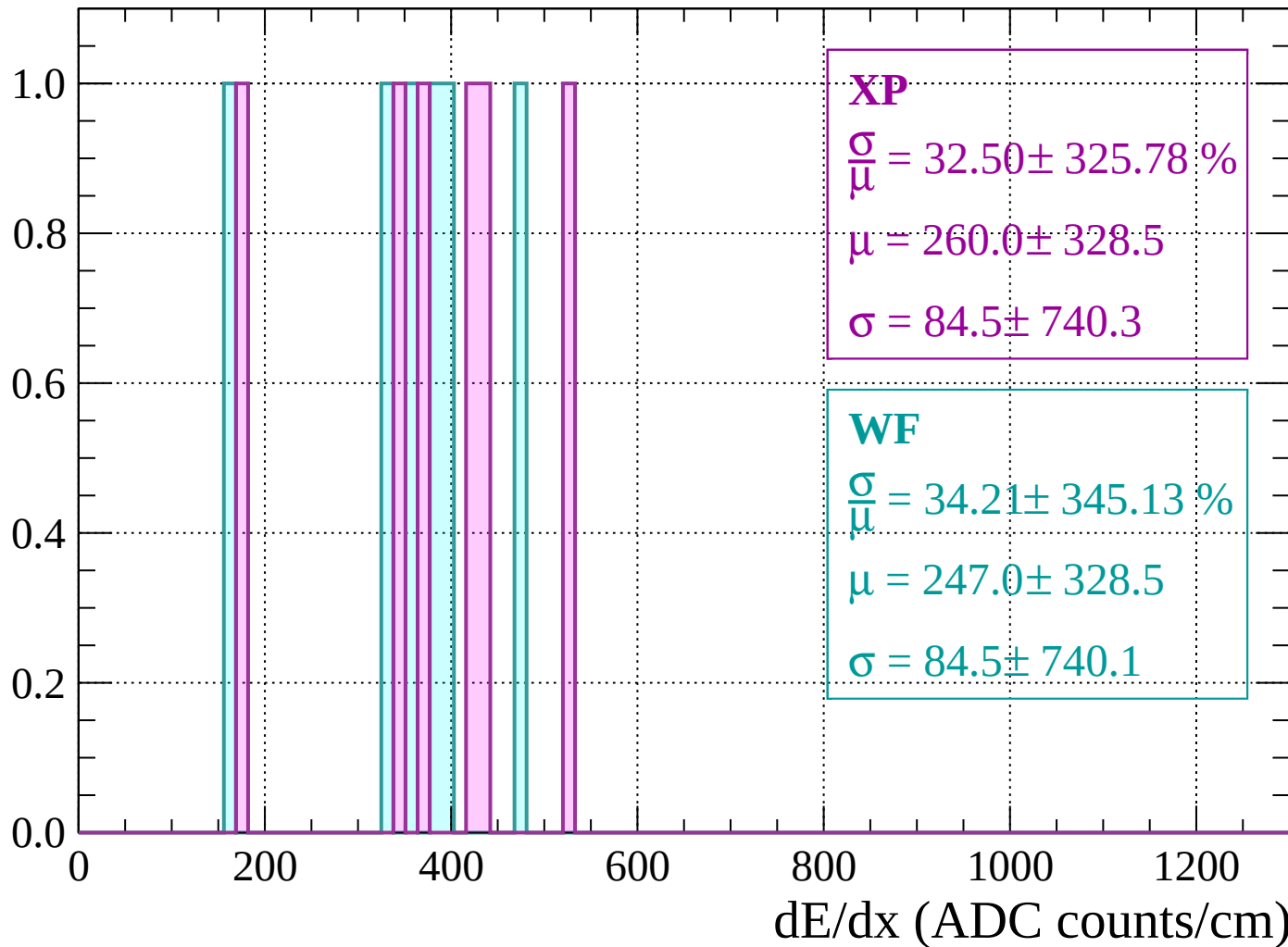
Energy loss | $720 < p < 760$

Count



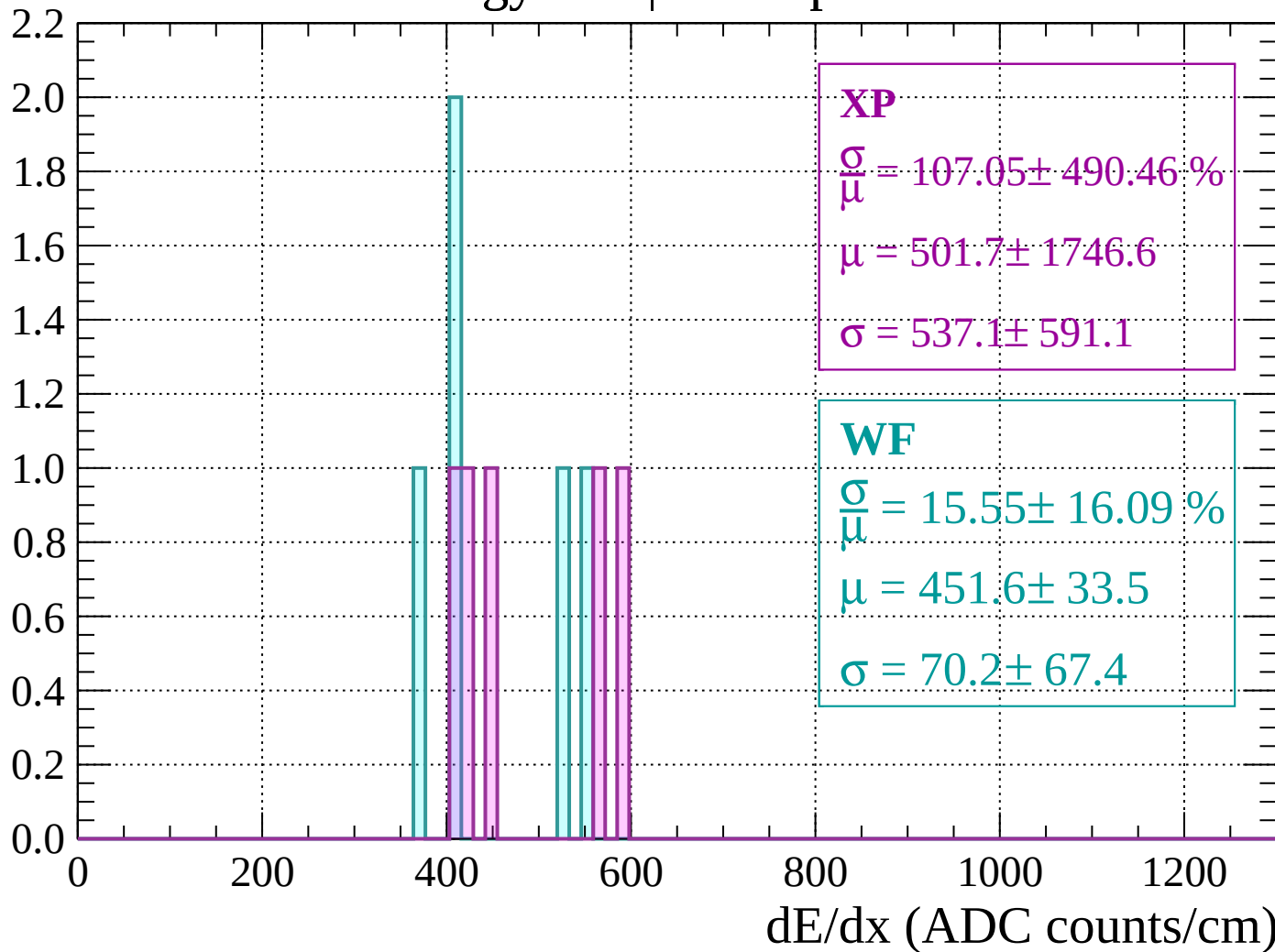
Energy loss | $760 < p < 800$

Count



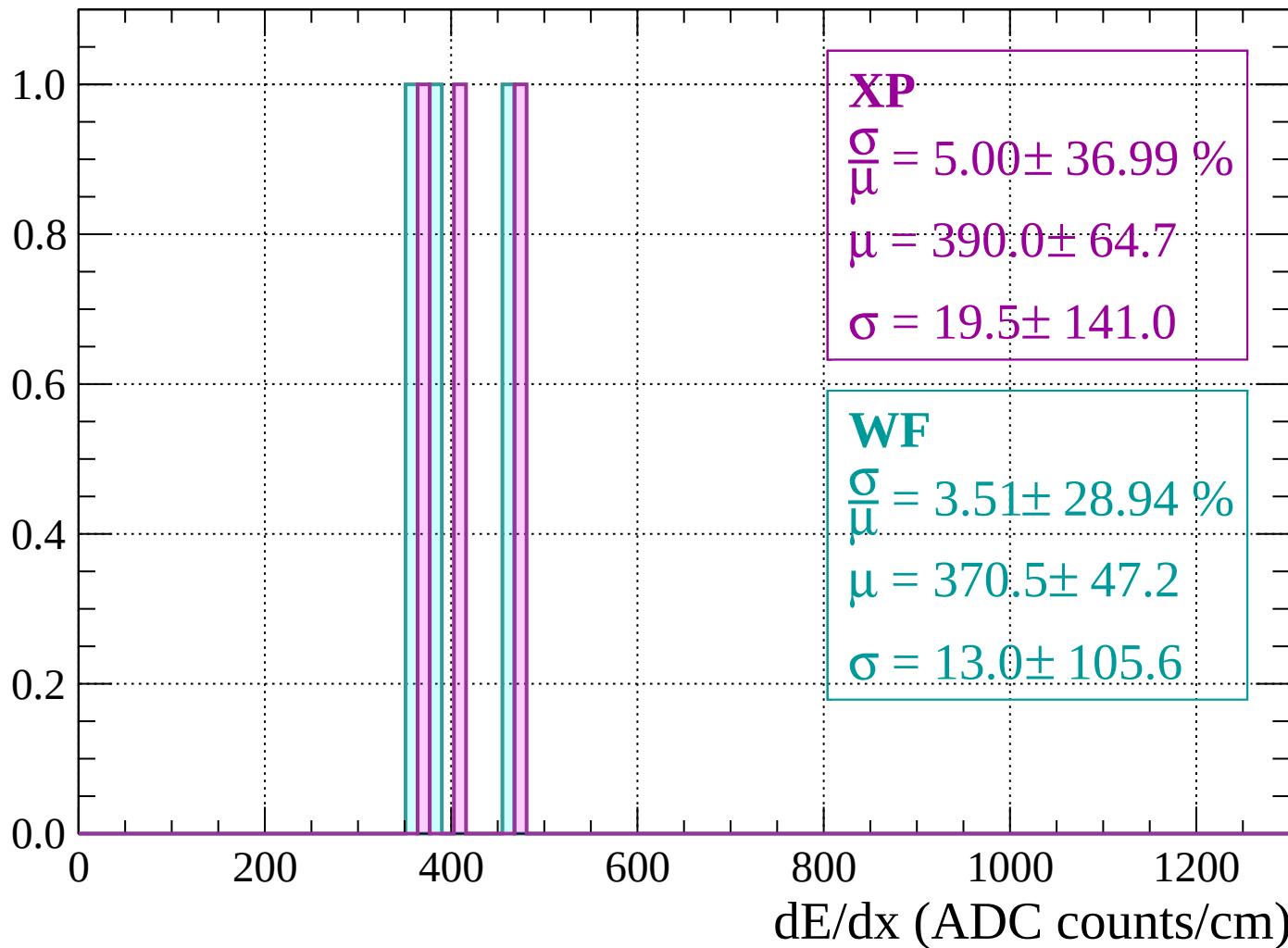
Energy loss | $800 < p < 840$

Count



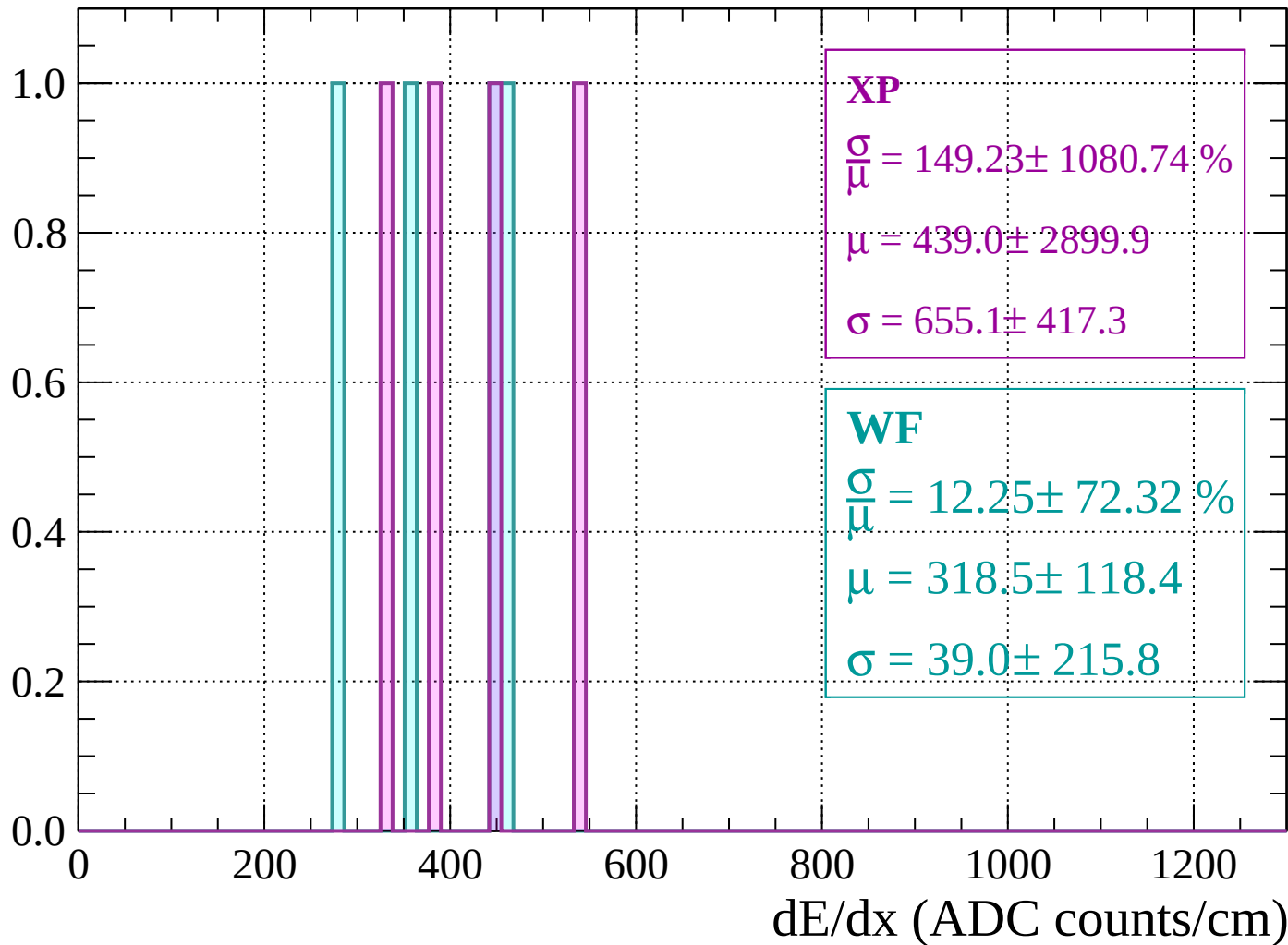
Energy loss | $840 < p < 880$

Count



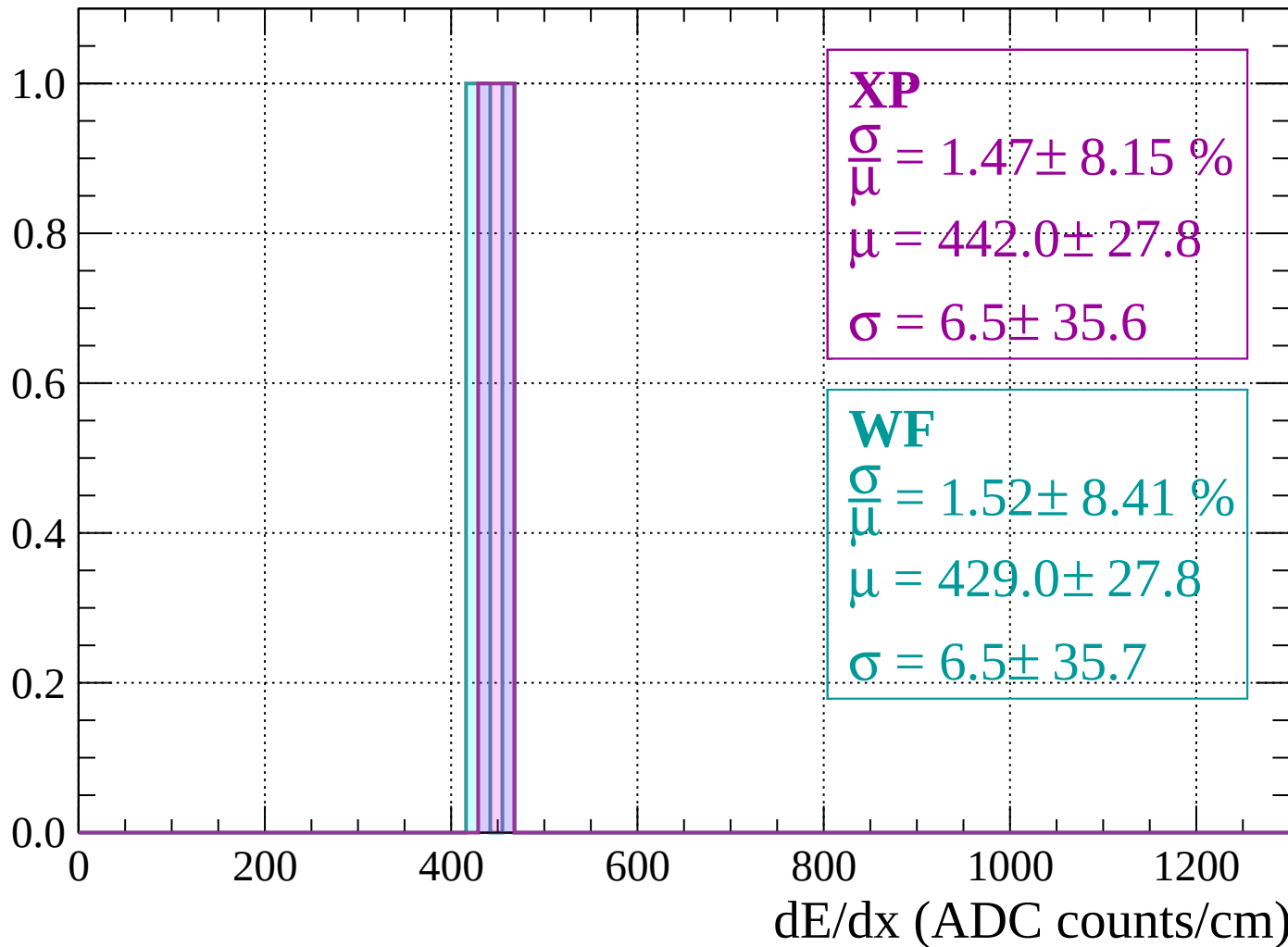
Energy loss | $880 < p < 920$

Count



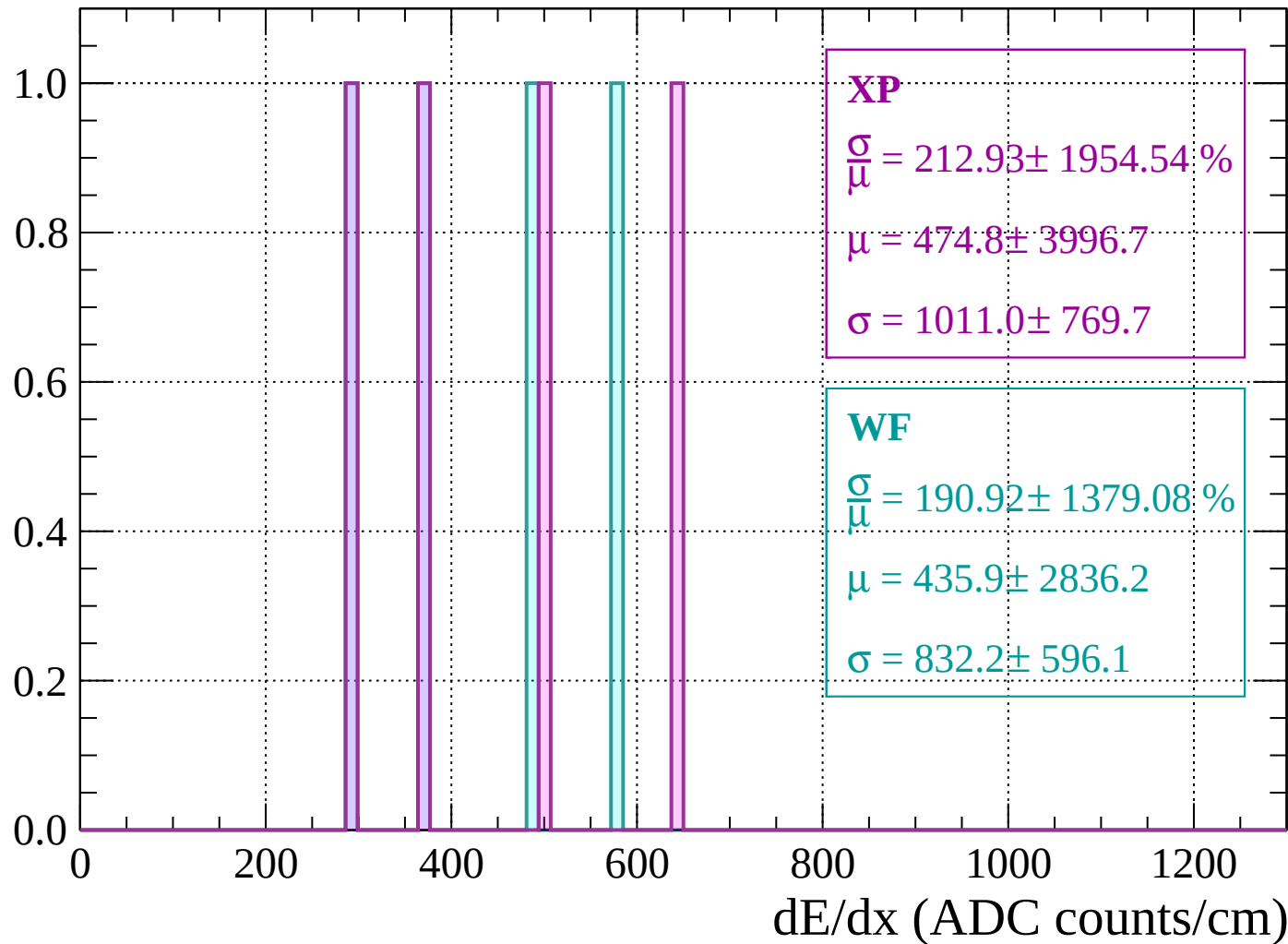
Energy loss | $920 < p < 960$

Count



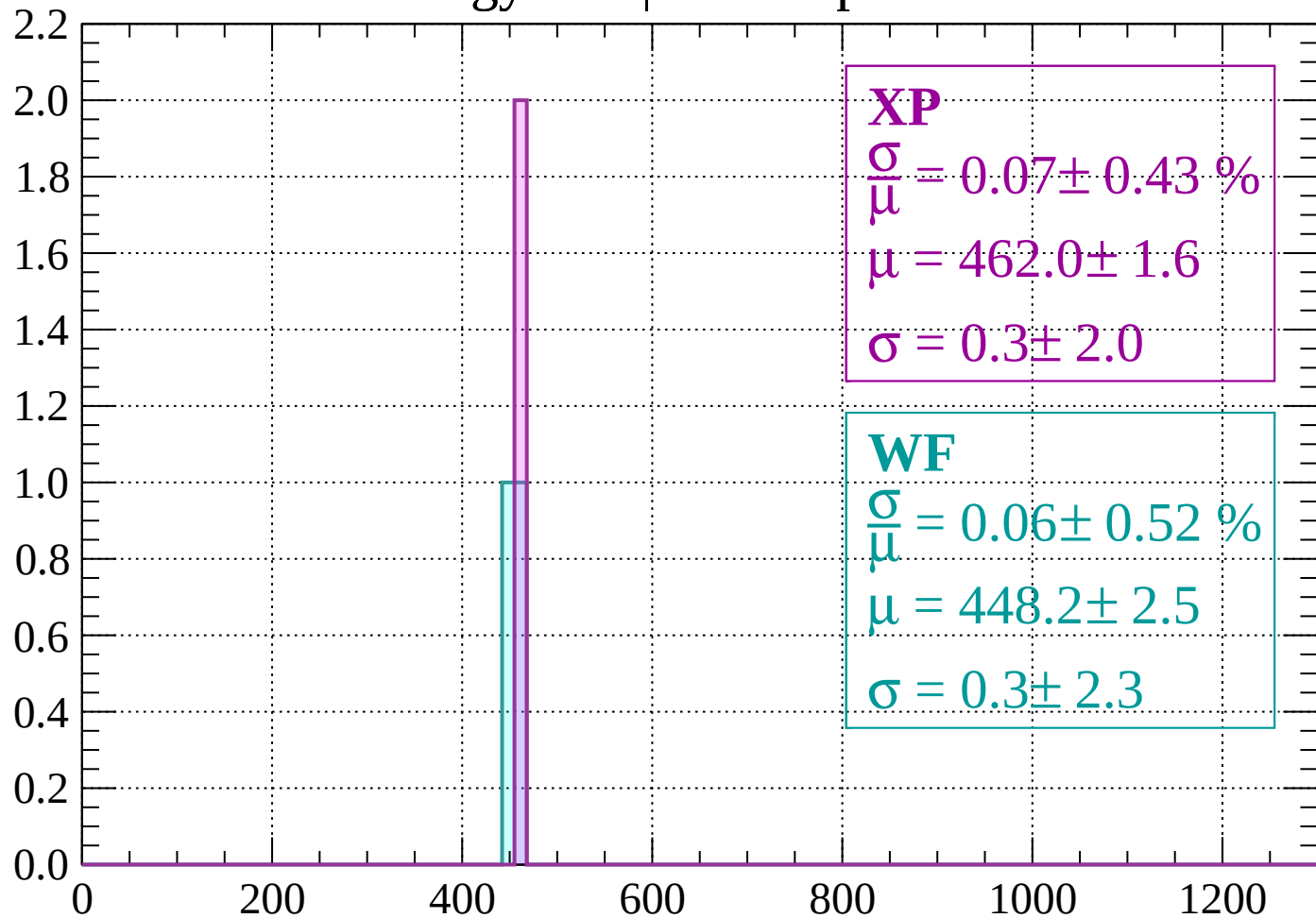
Energy loss | $960 < p < 1000$

Count



Energy loss | $1000 < p < 1040$

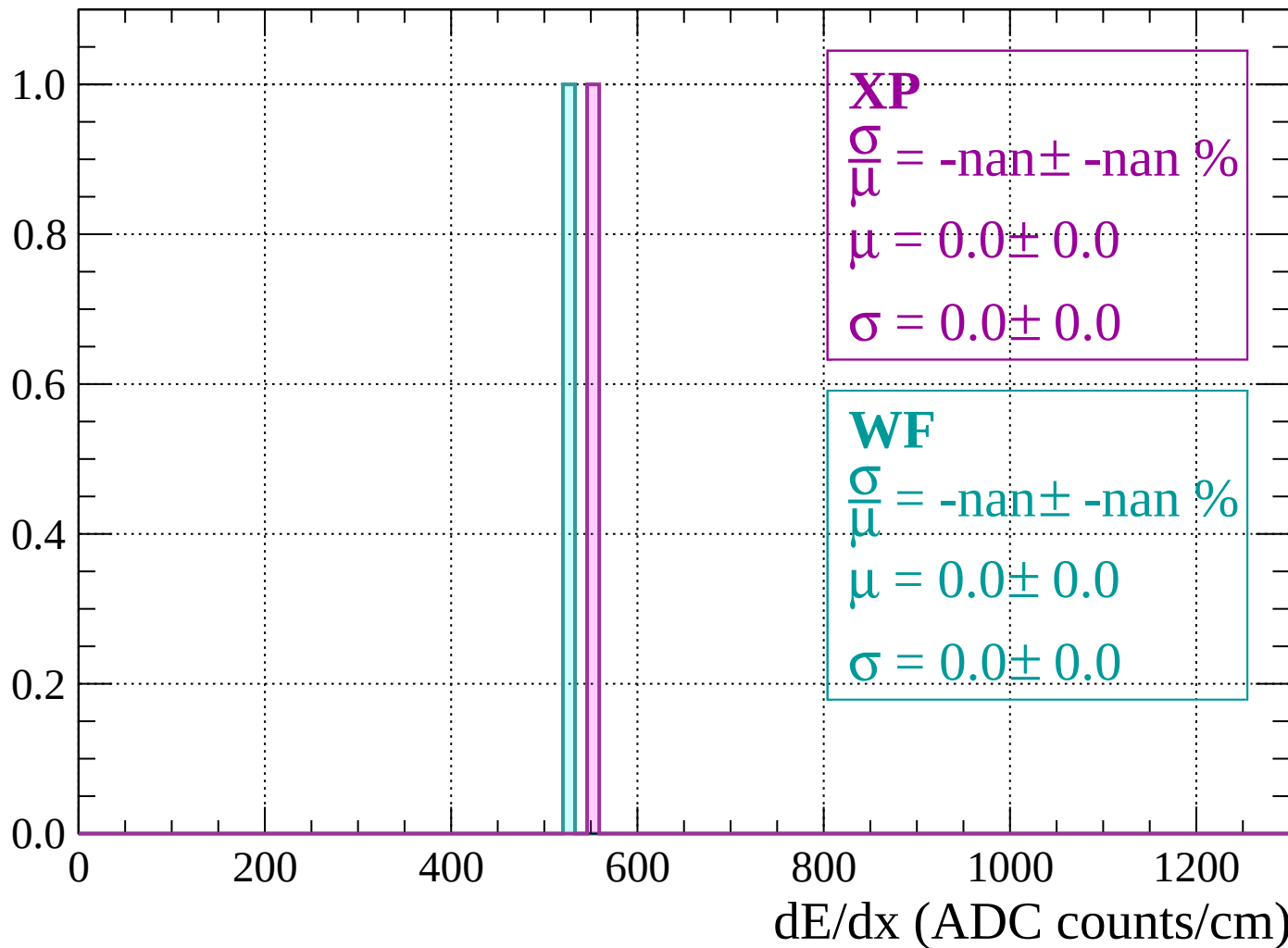
Count



dE/dx (ADC counts/cm)

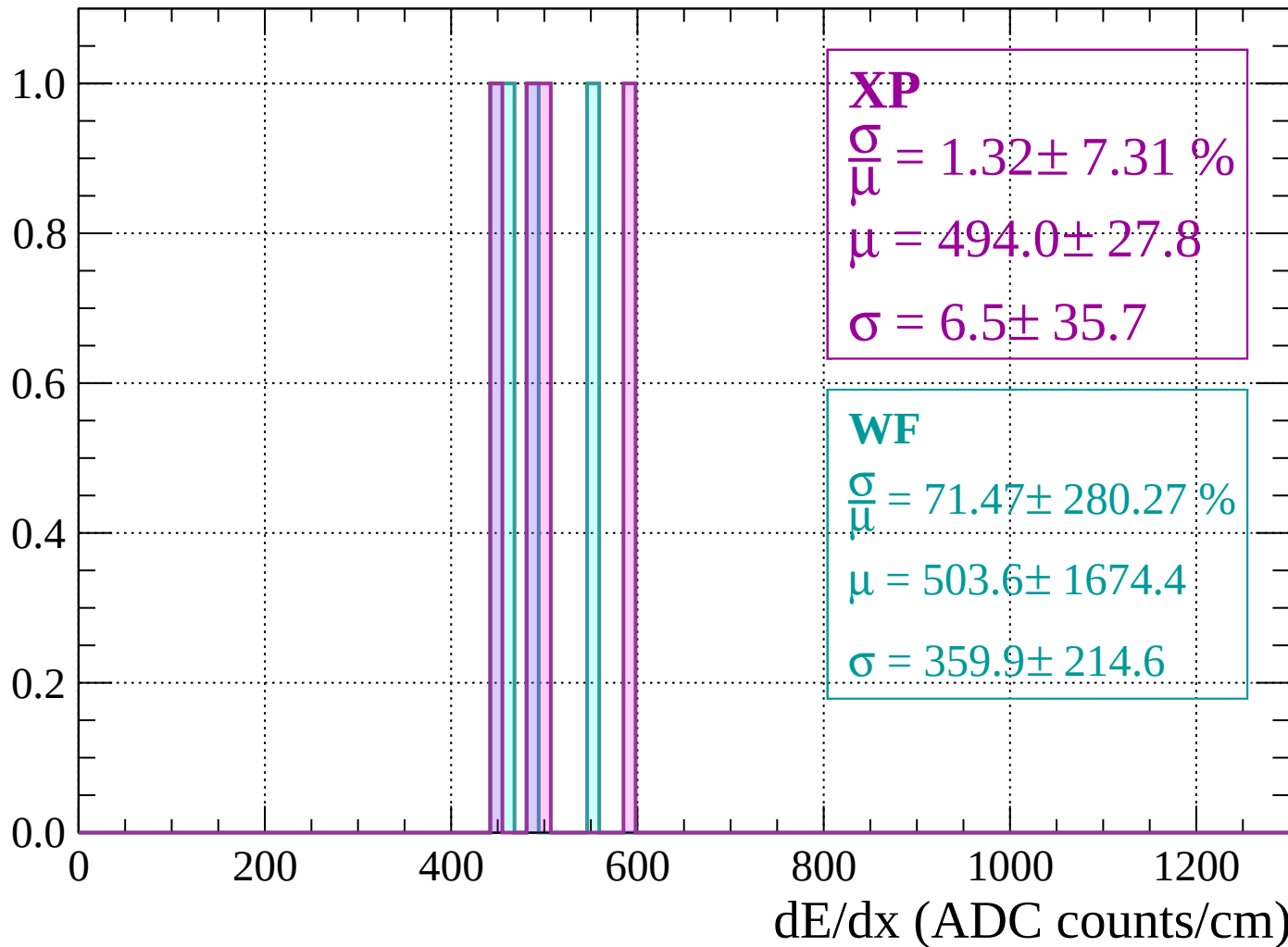
Energy loss | $1040 < p < 1080$

Count



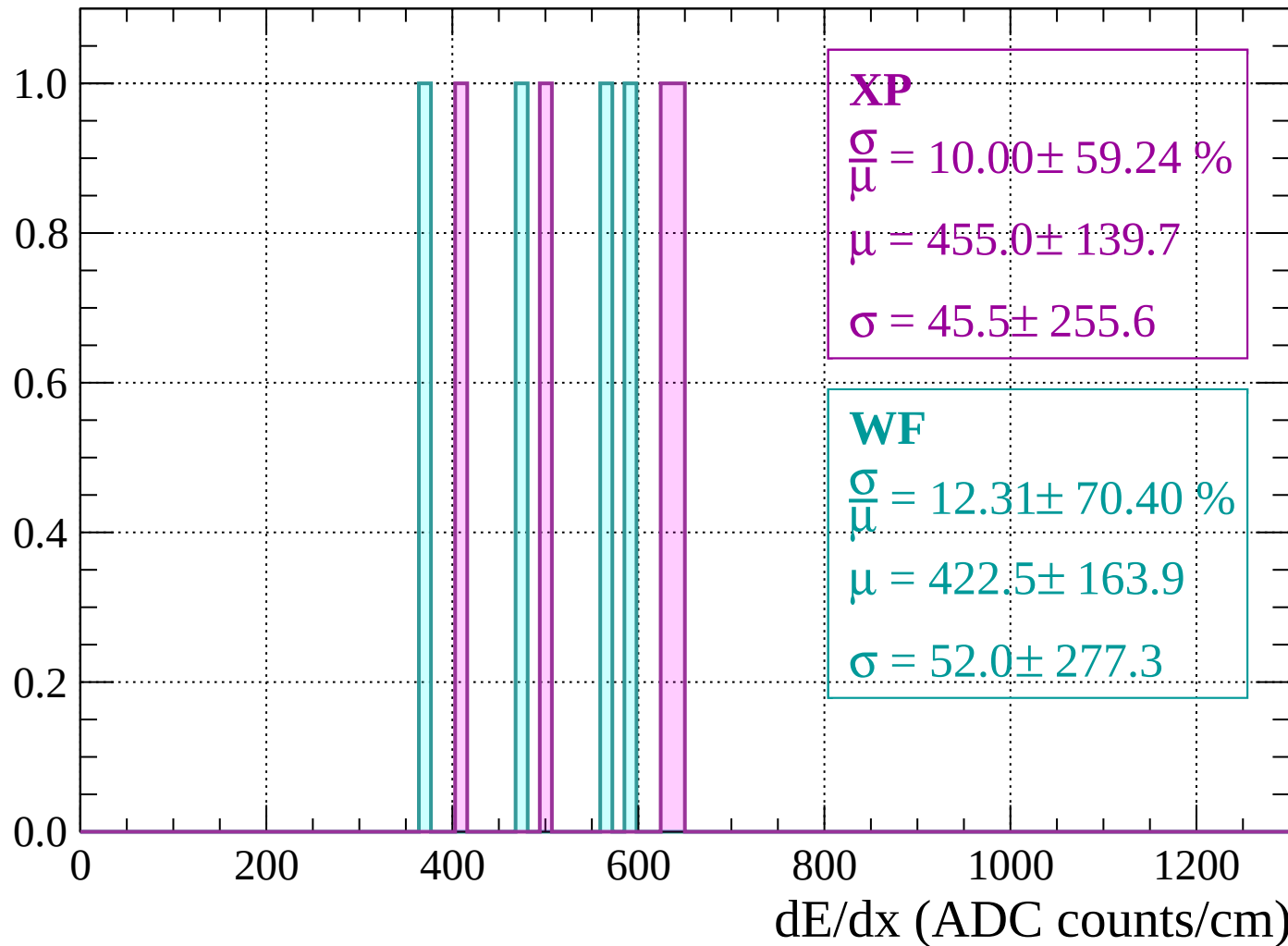
Energy loss | $1080 < p < 1120$

Count



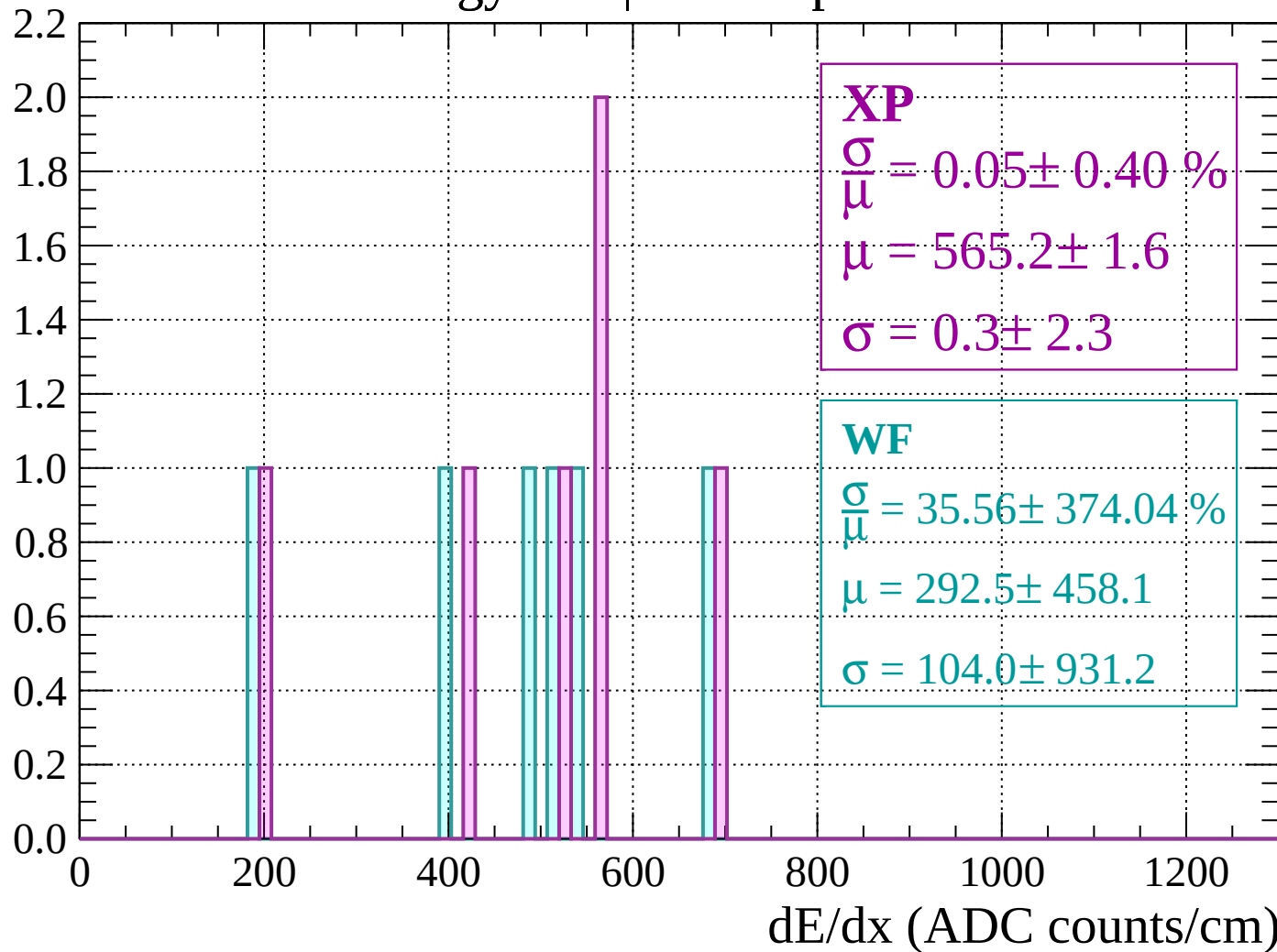
Energy loss | $1120 < p < 1160$

Count



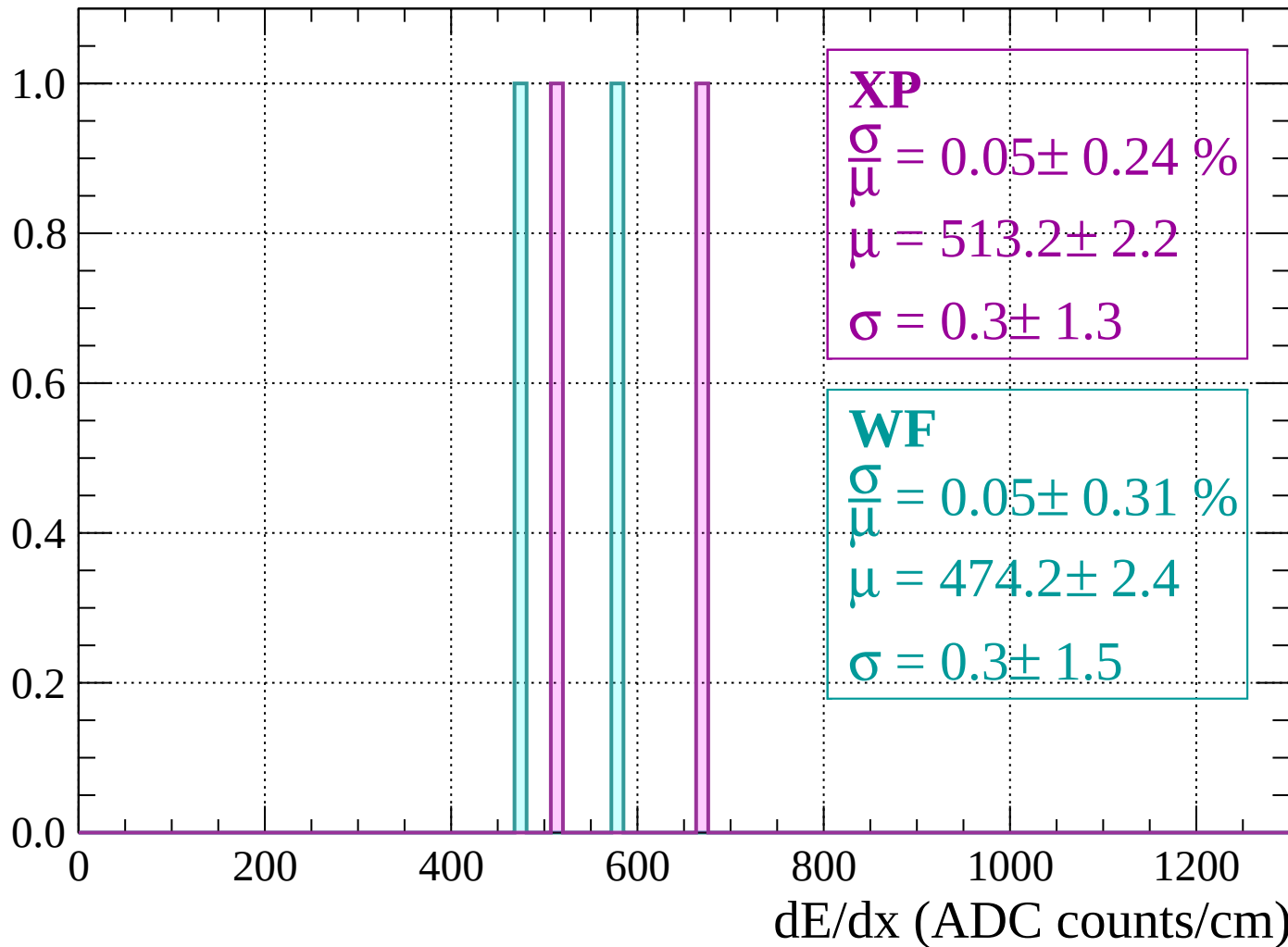
Energy loss | $1160 < p < 1200$

Count



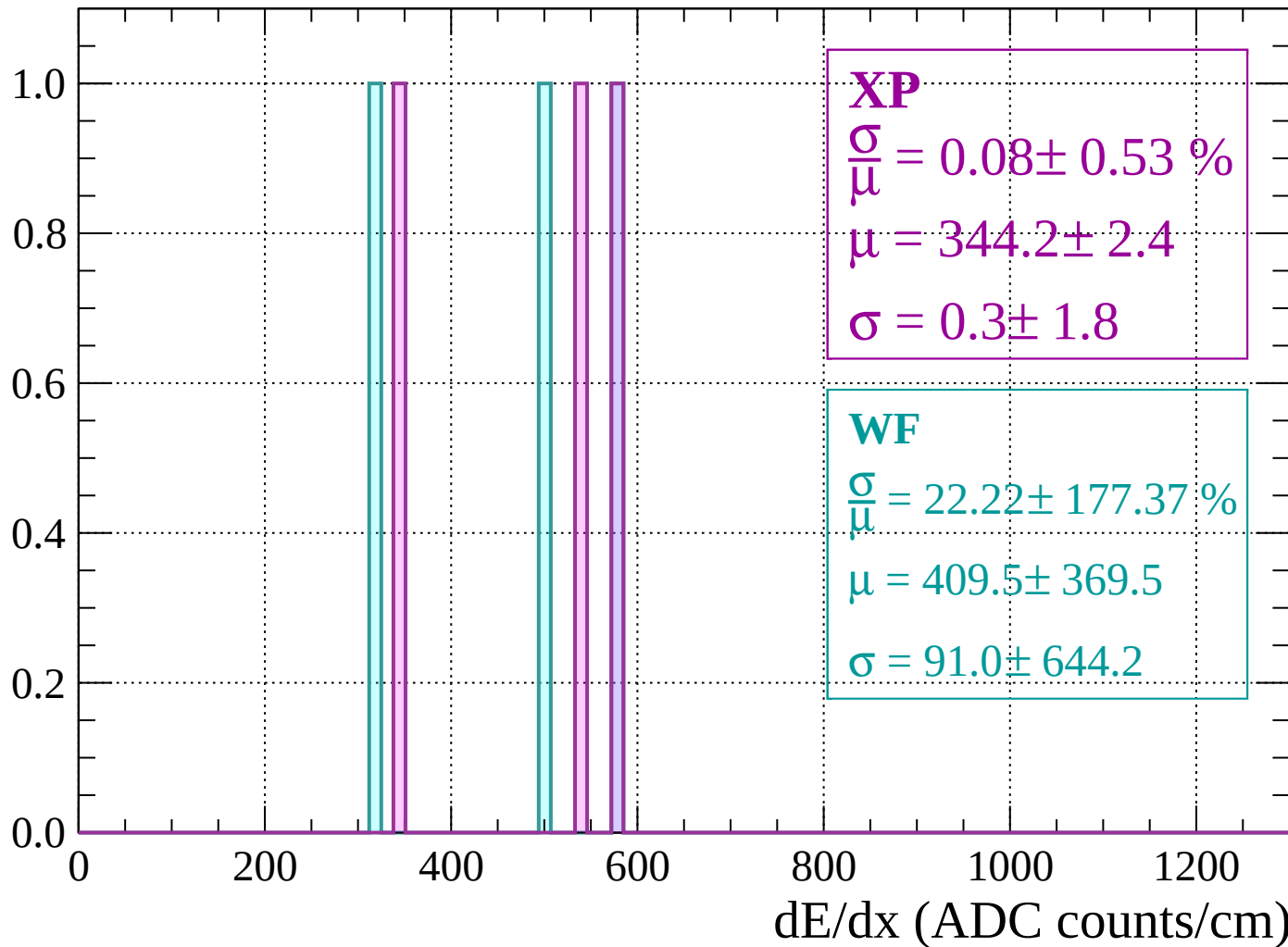
Energy loss | $1200 < p < 1240$

Count



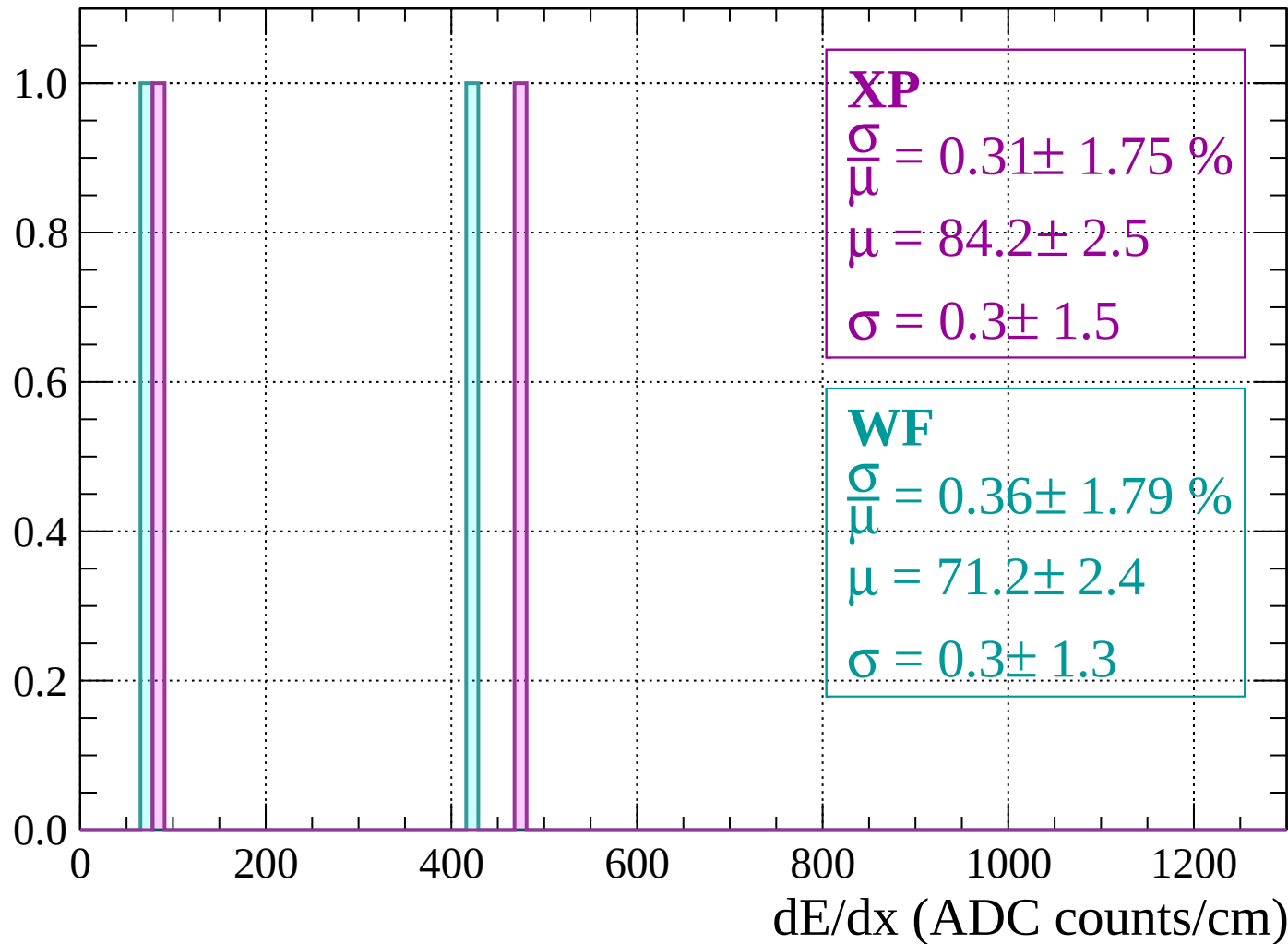
Energy loss | $1240 < p < 1280$

Count



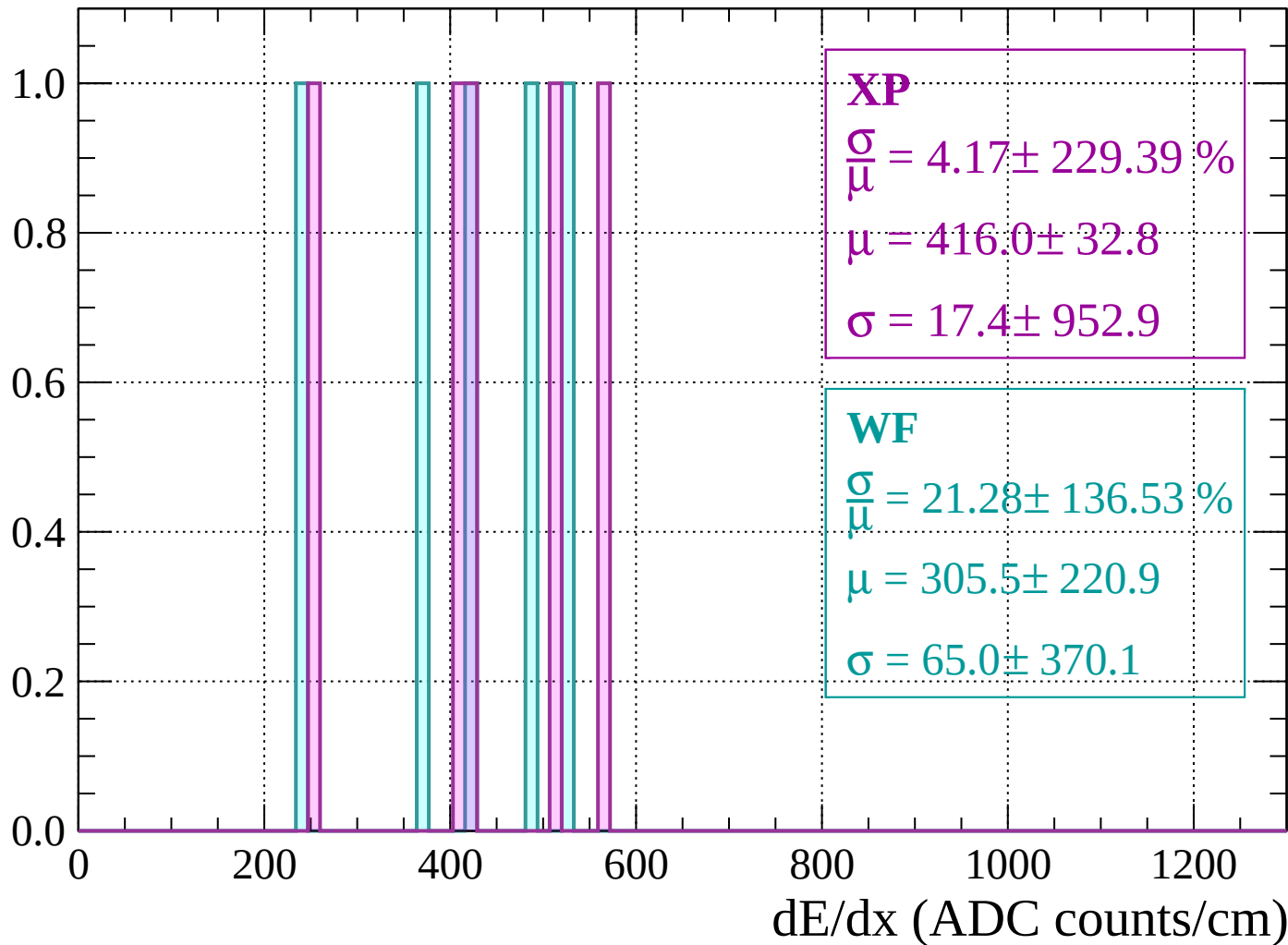
Energy loss | $1280 < p < 1320$

Count



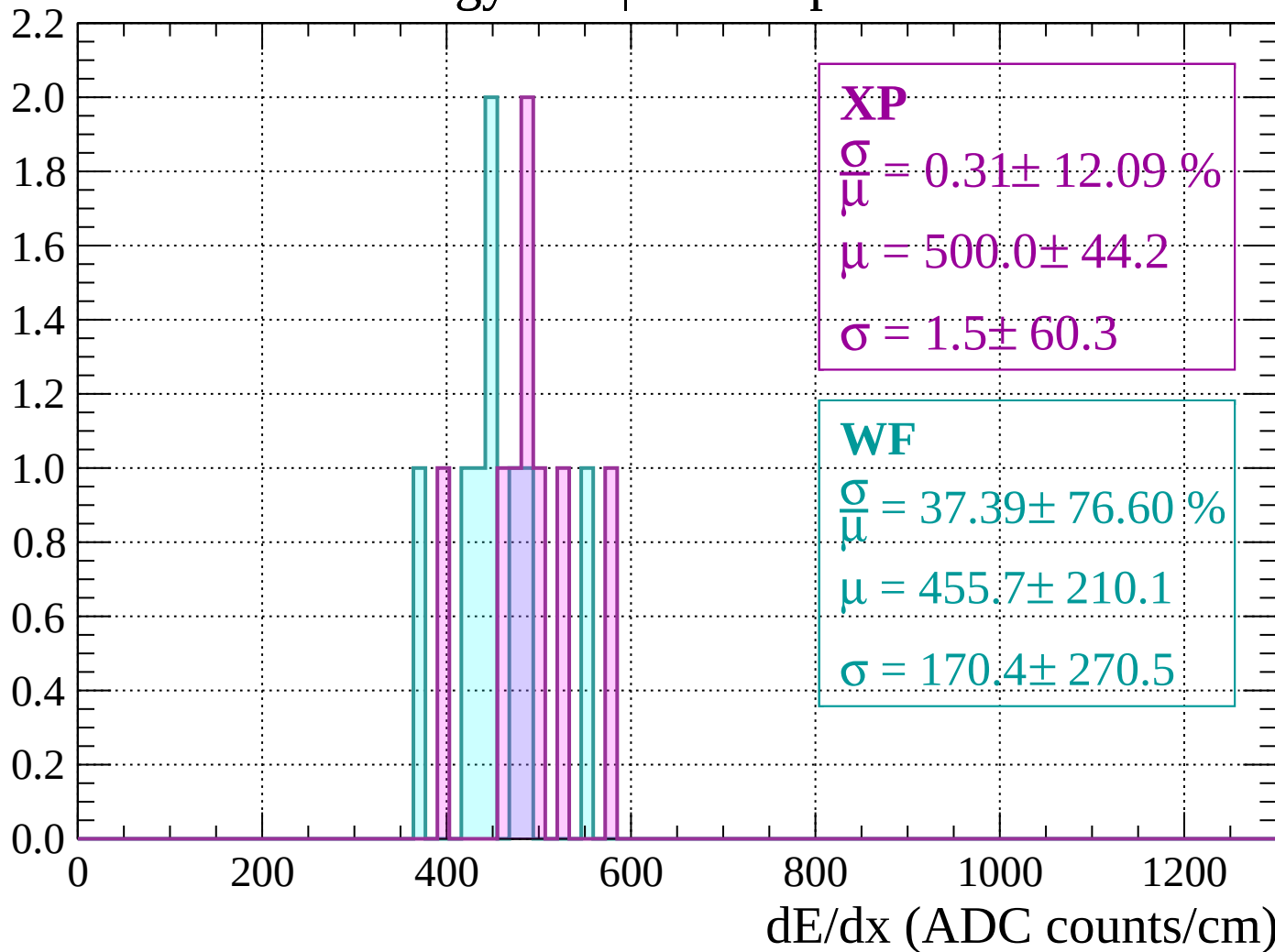
Energy loss | $1320 < p < 1360$

Count



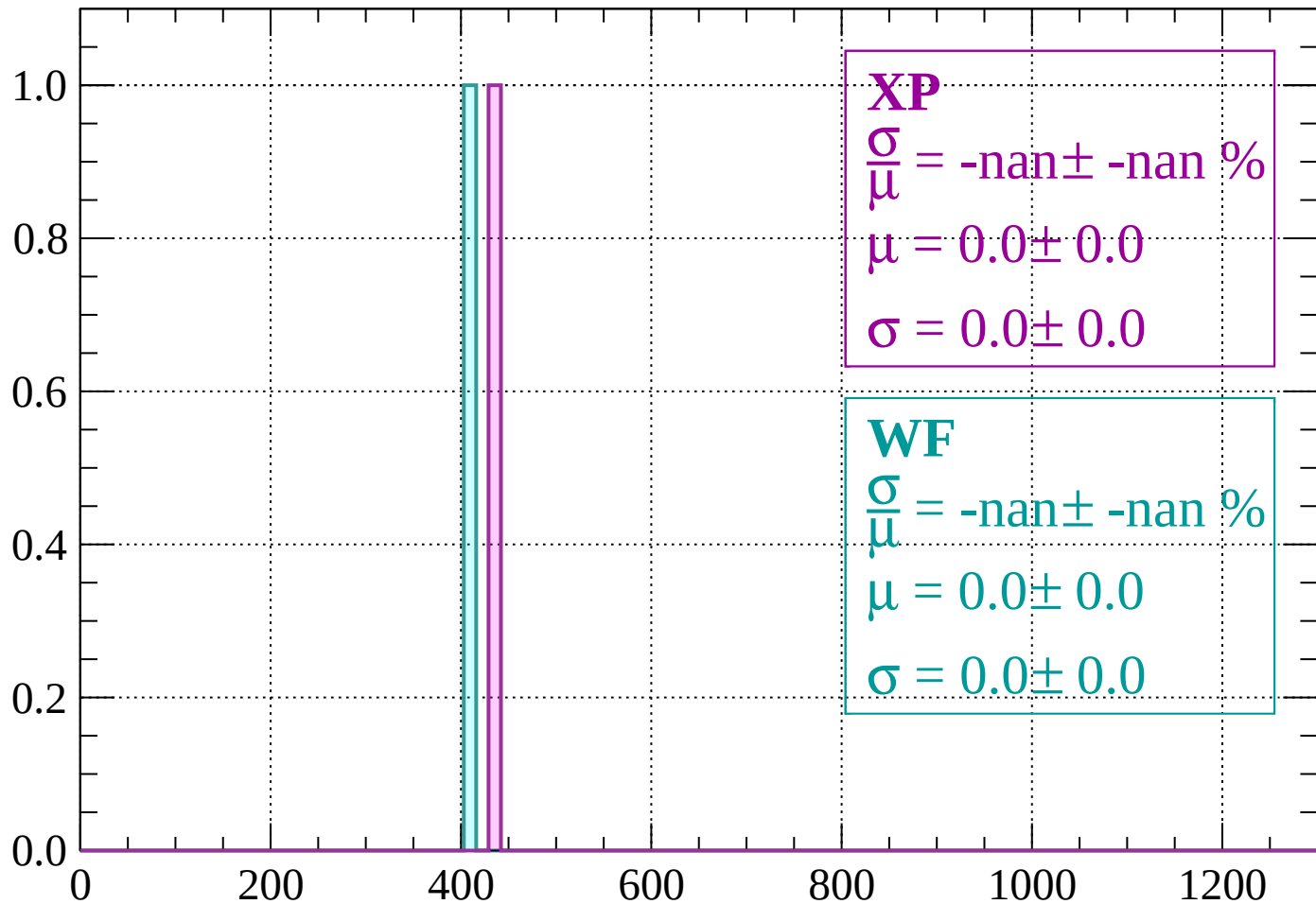
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

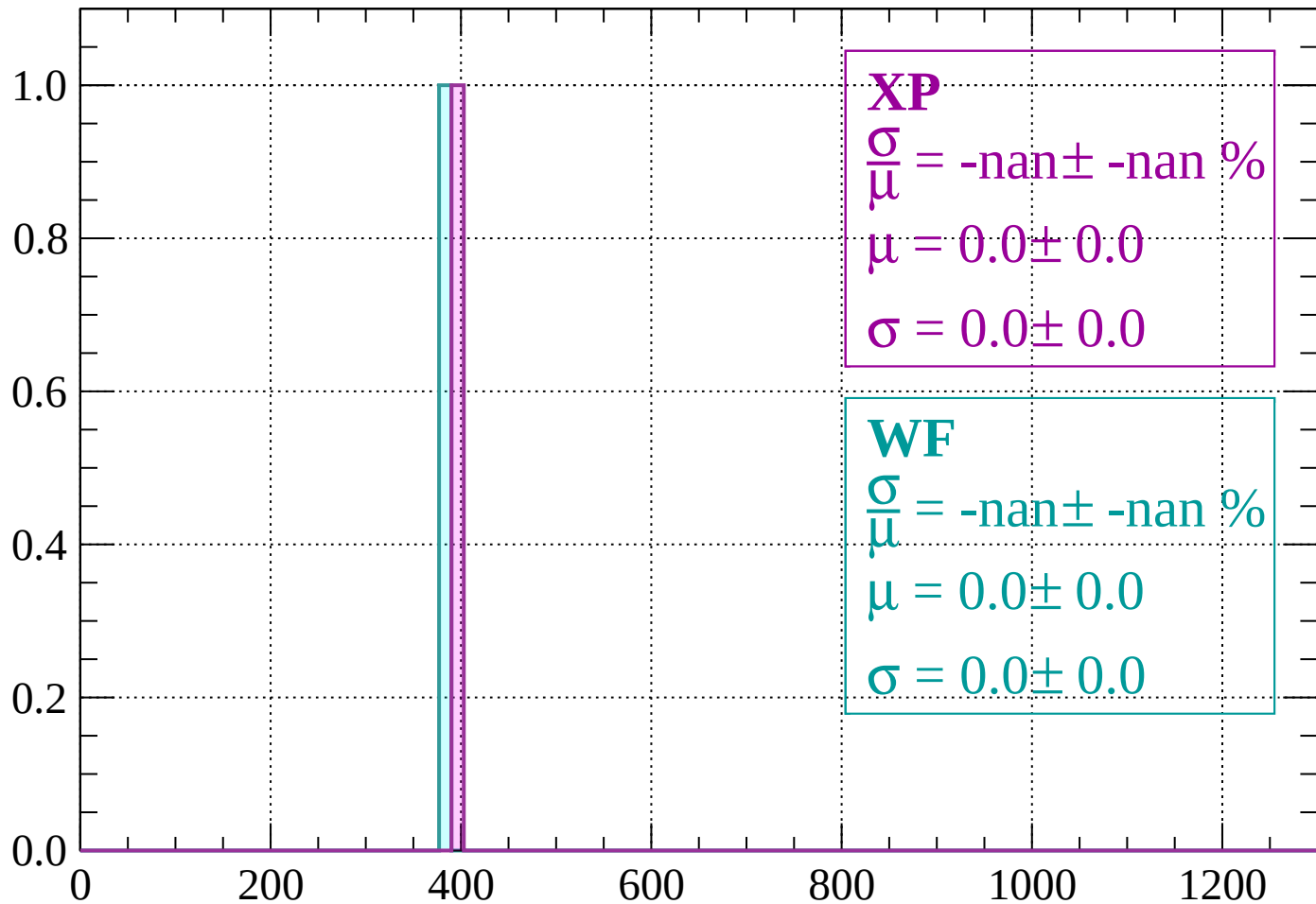
Count



dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

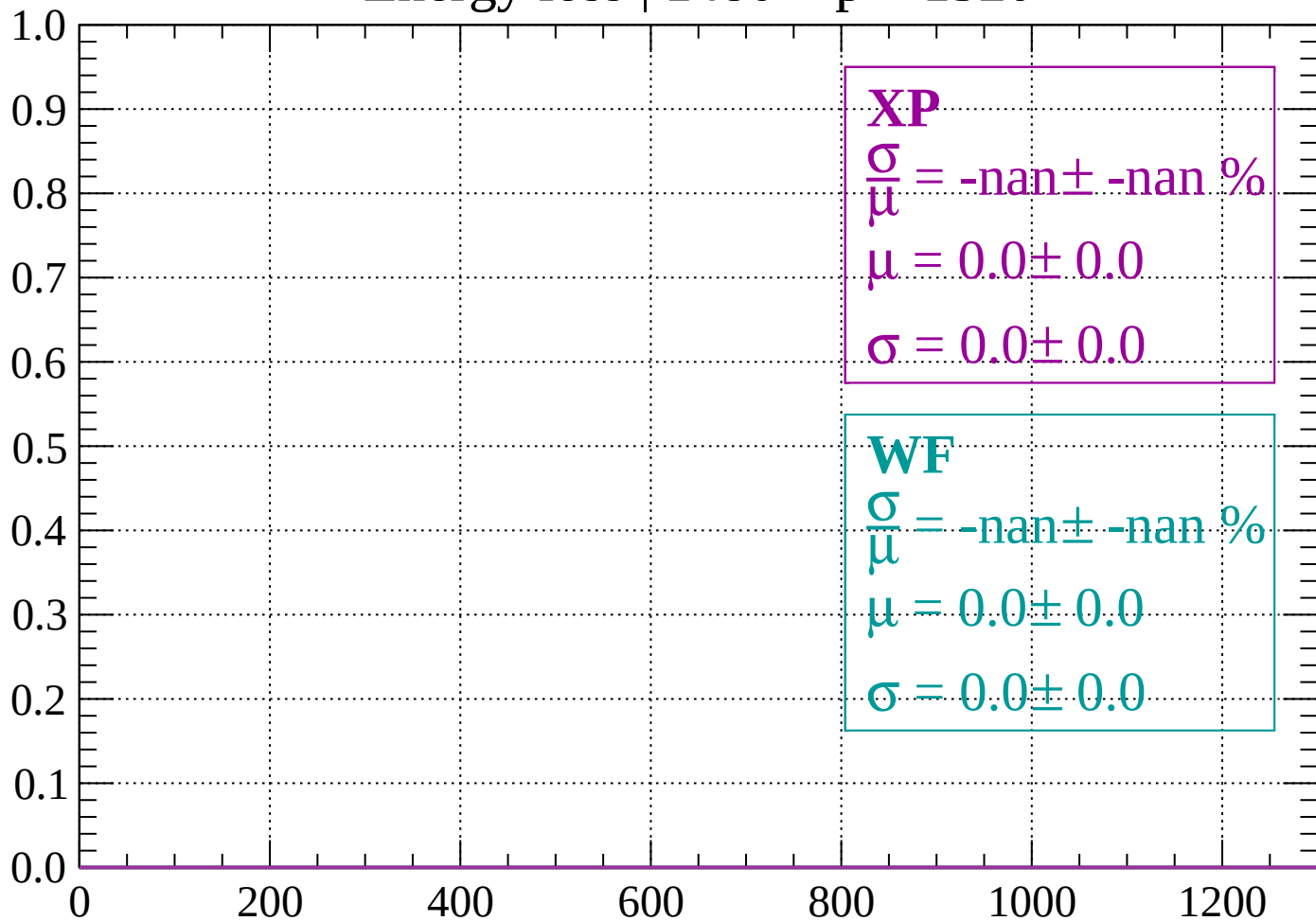
Count



dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

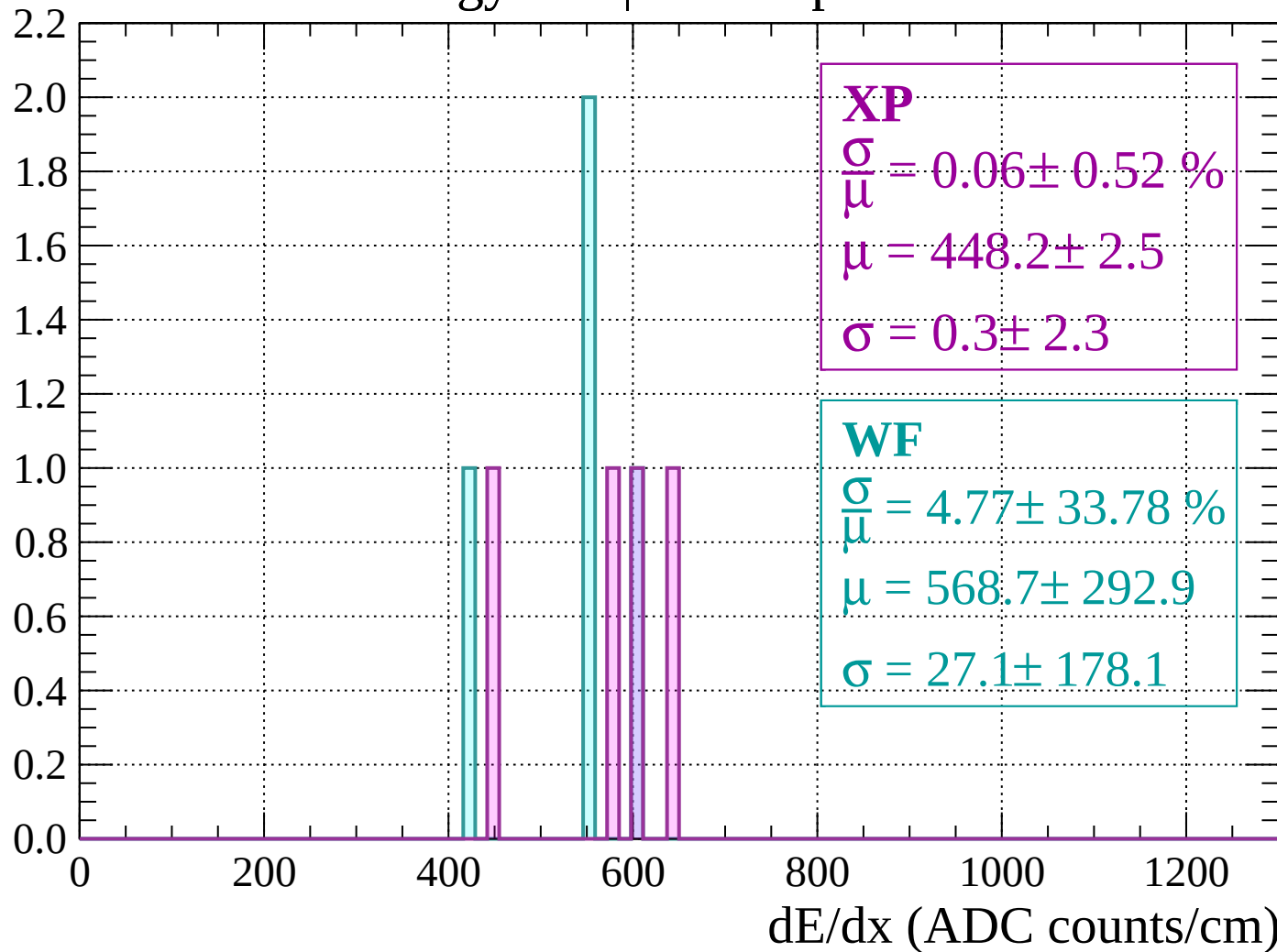
Count



dE/dx (ADC counts/cm)

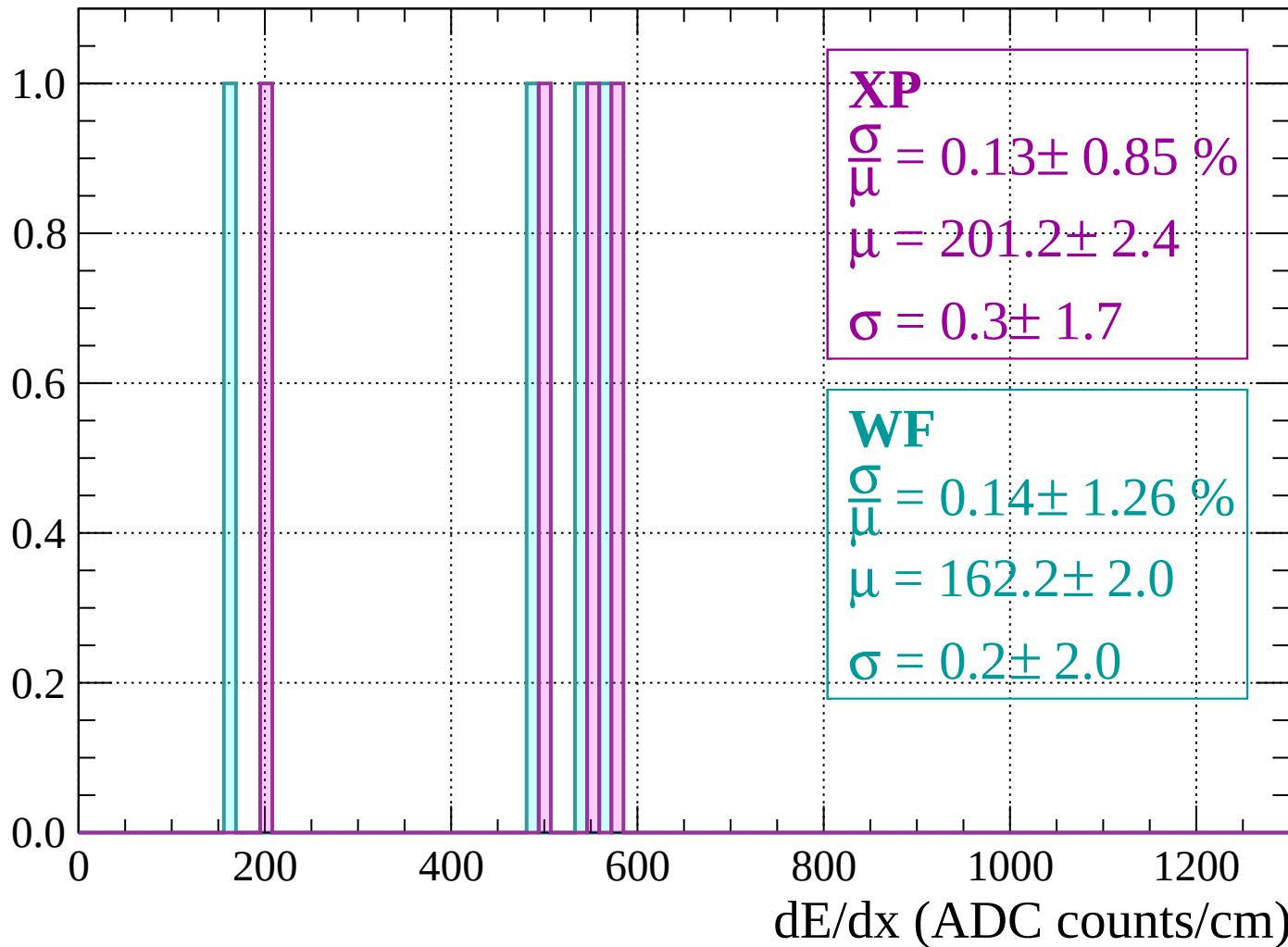
Energy loss | $1520 < p < 1560$

Count



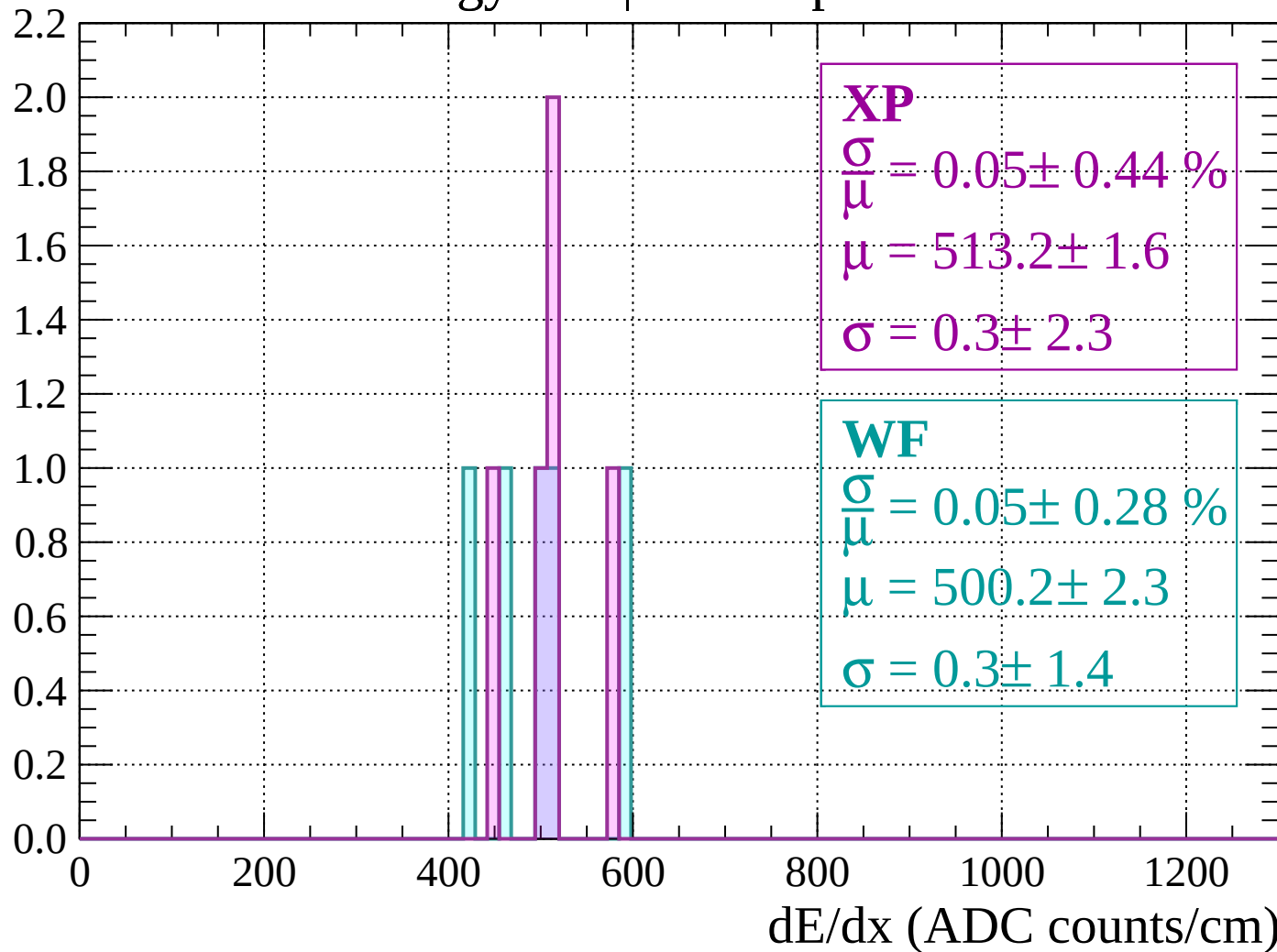
Energy loss | $1560 < p < 1600$

Count



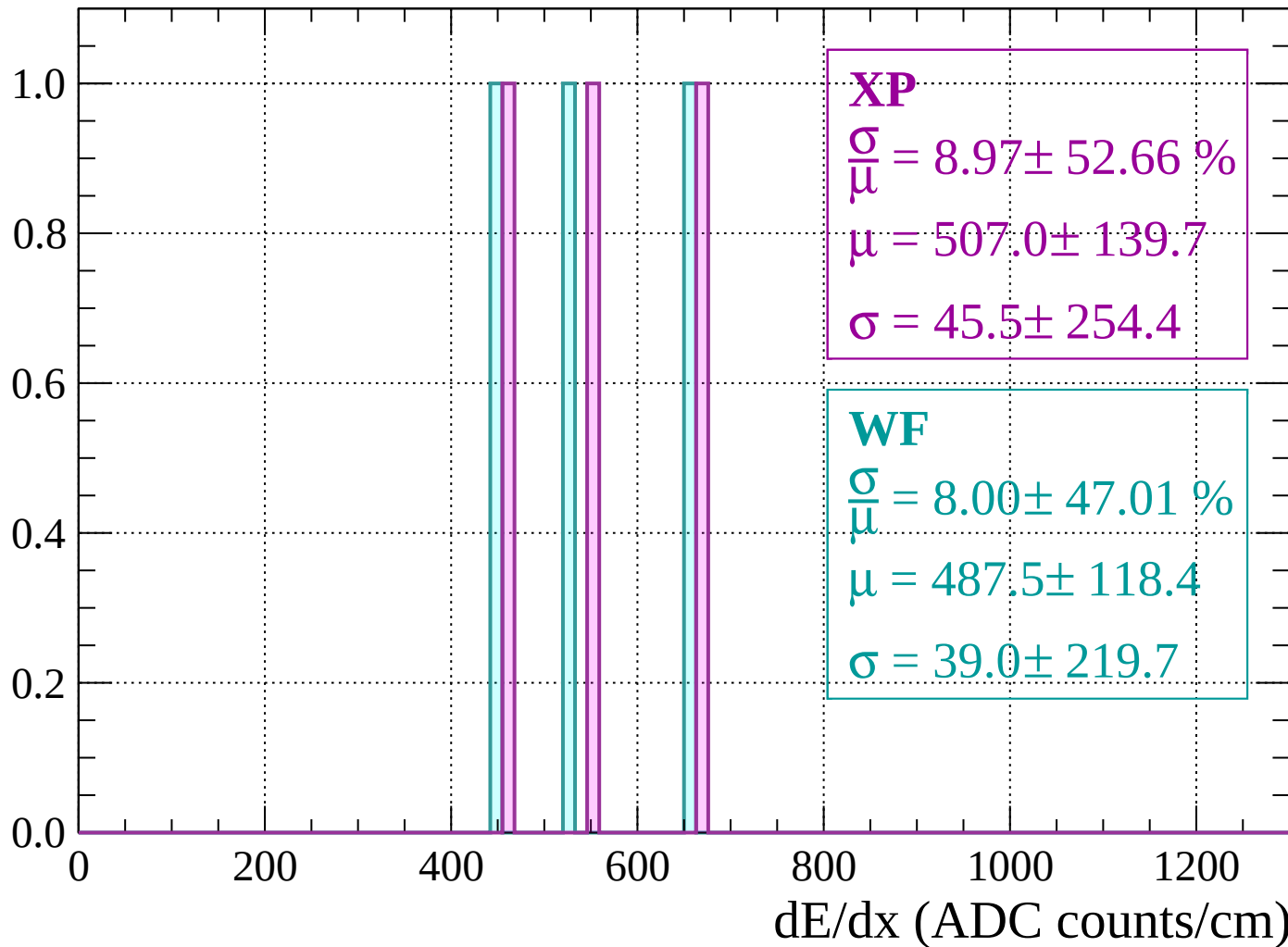
Energy loss | $1600 < p < 1640$

Count



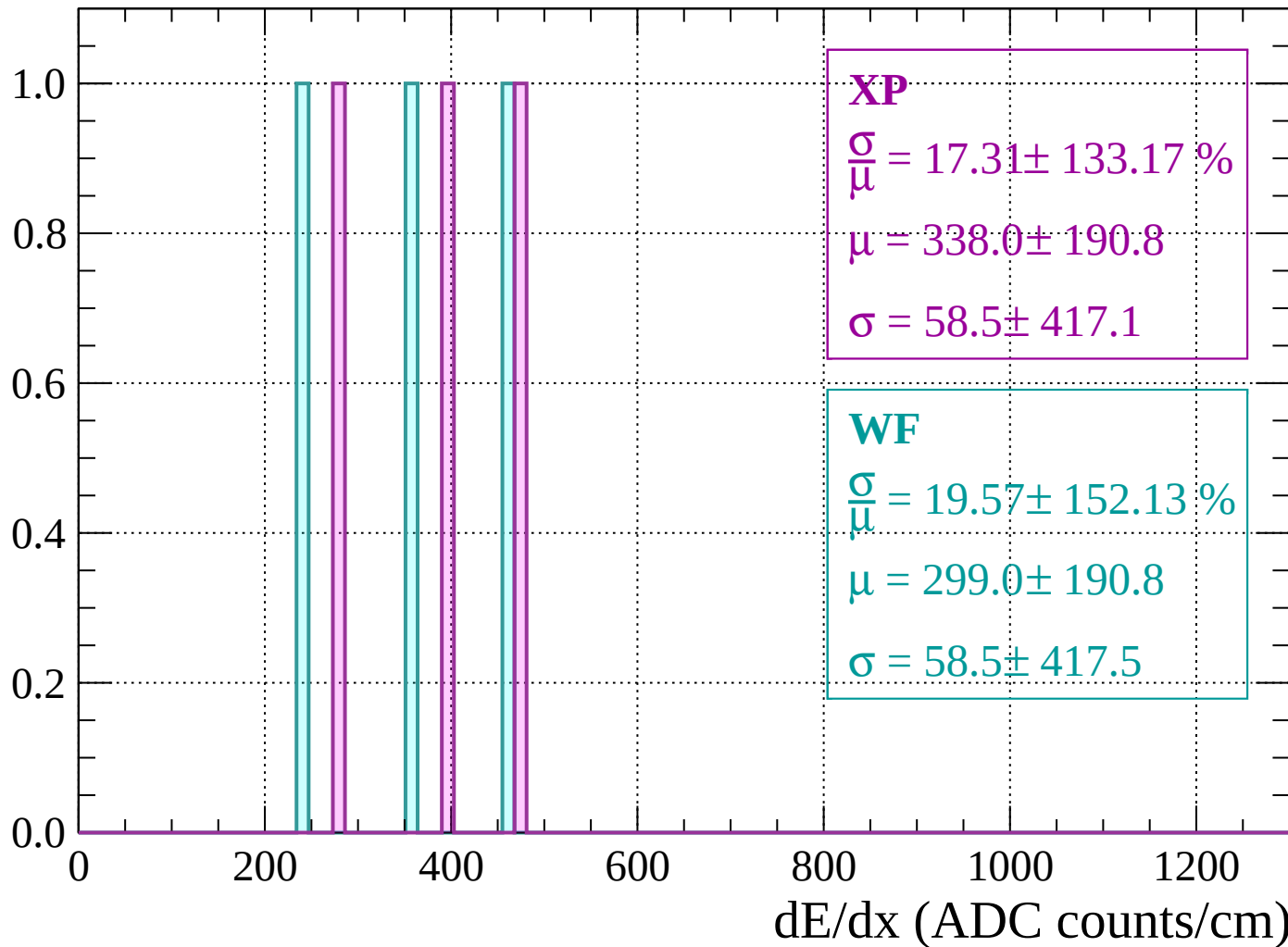
Energy loss | $1640 < p < 1680$

Count



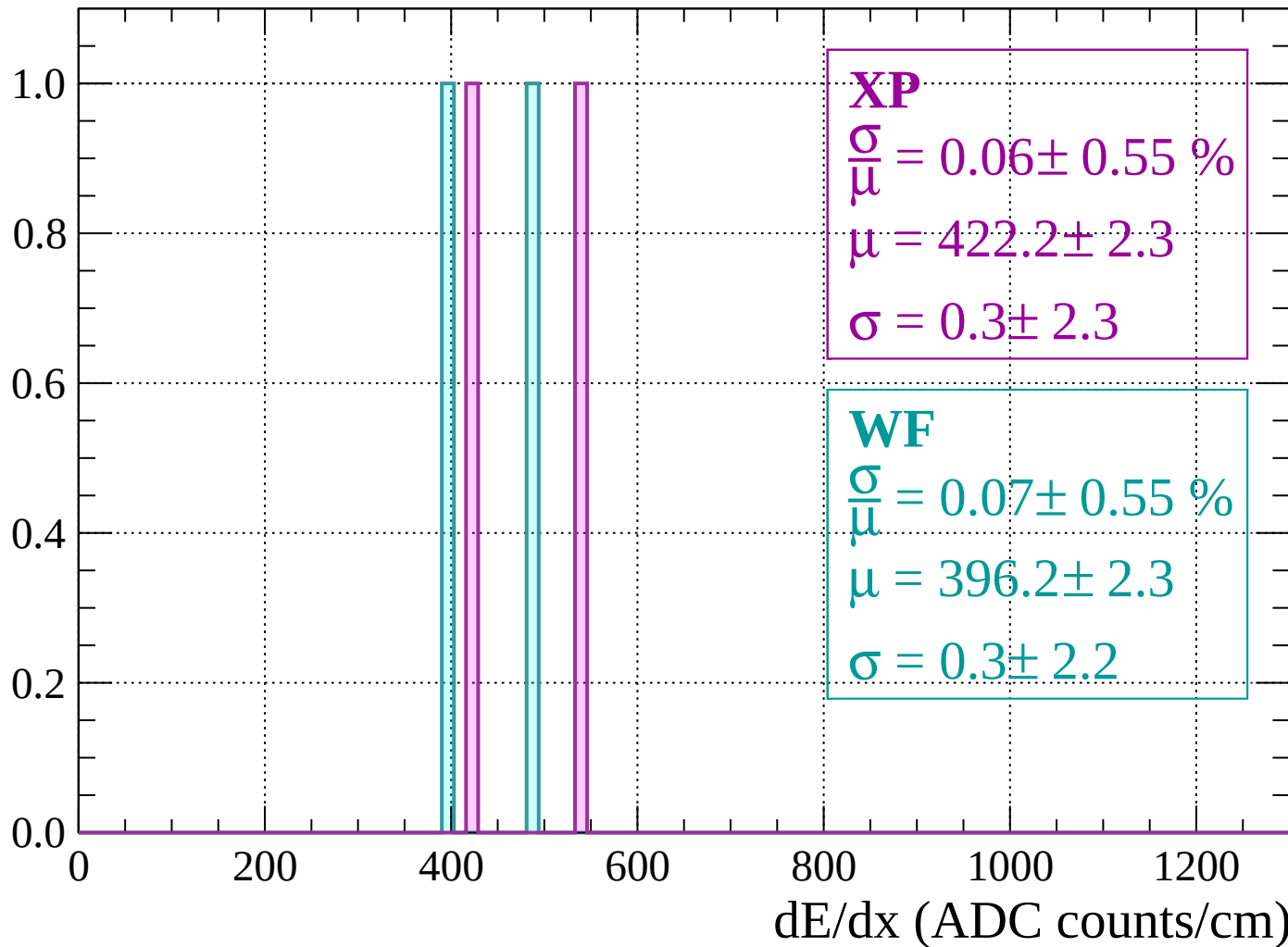
Energy loss | $1680 < p < 1720$

Count



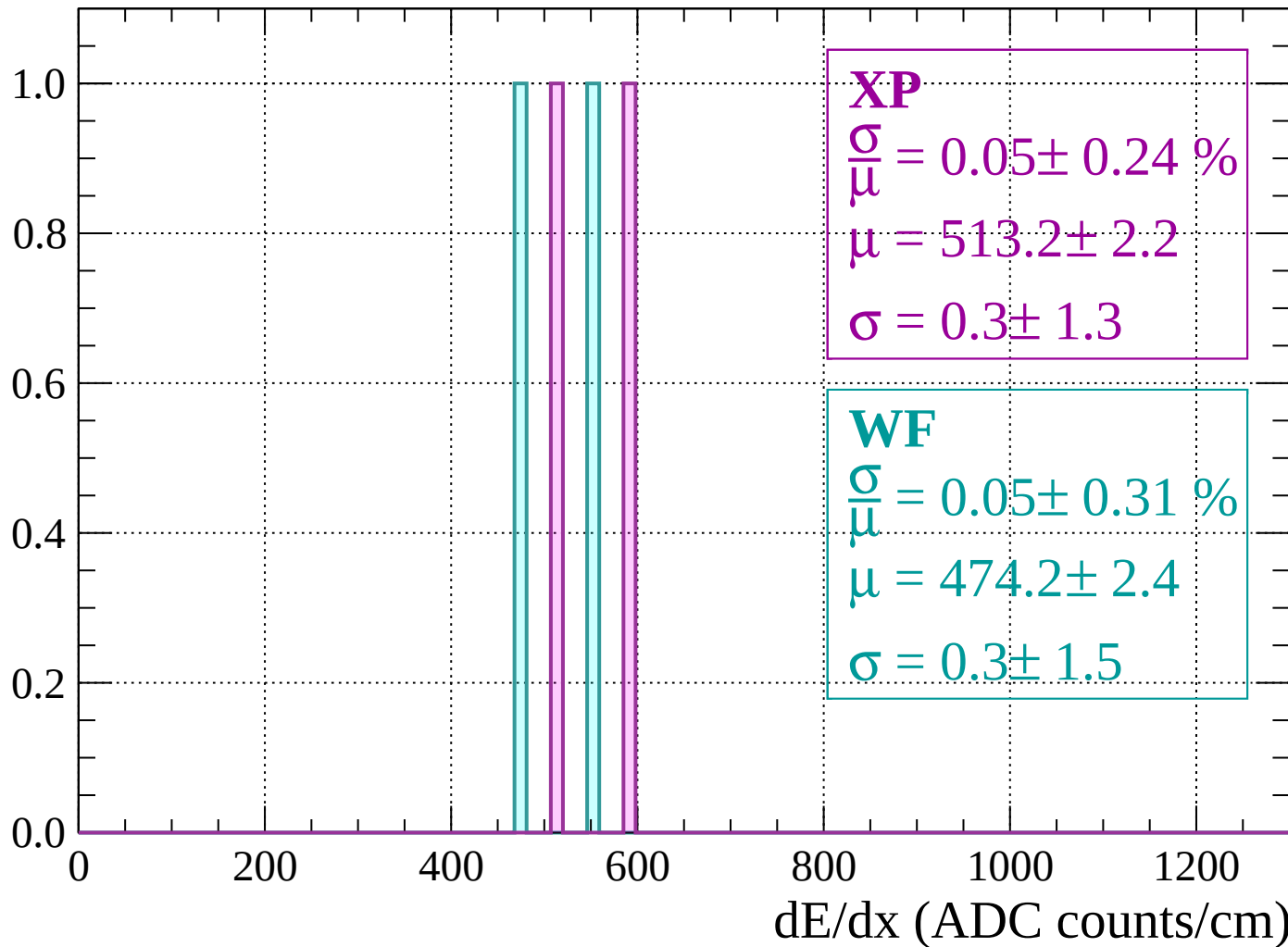
Energy loss | $1720 < p < 1760$

Count



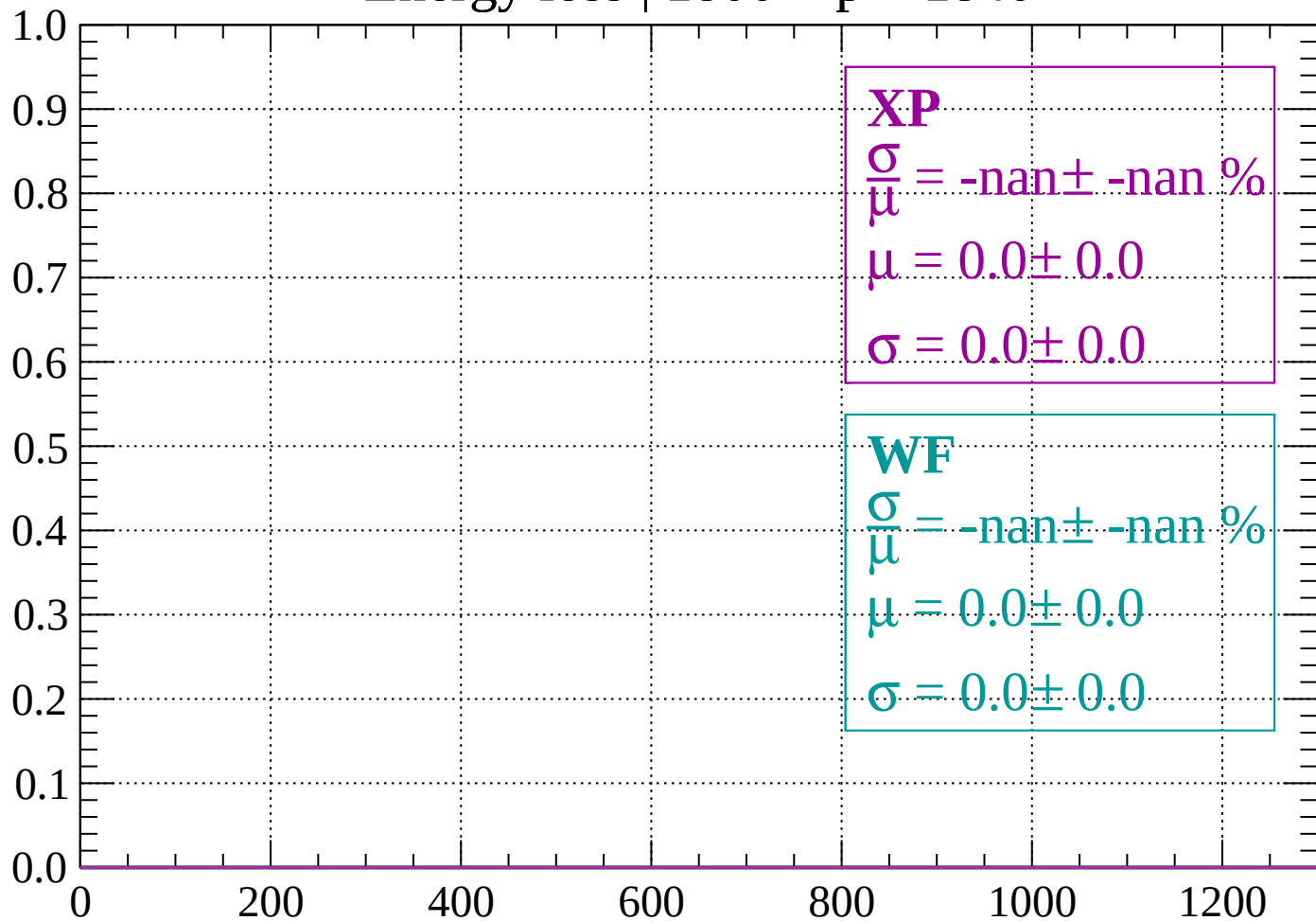
Energy loss | $1760 < p < 1800$

Count



Energy loss | $1800 < p < 1840$

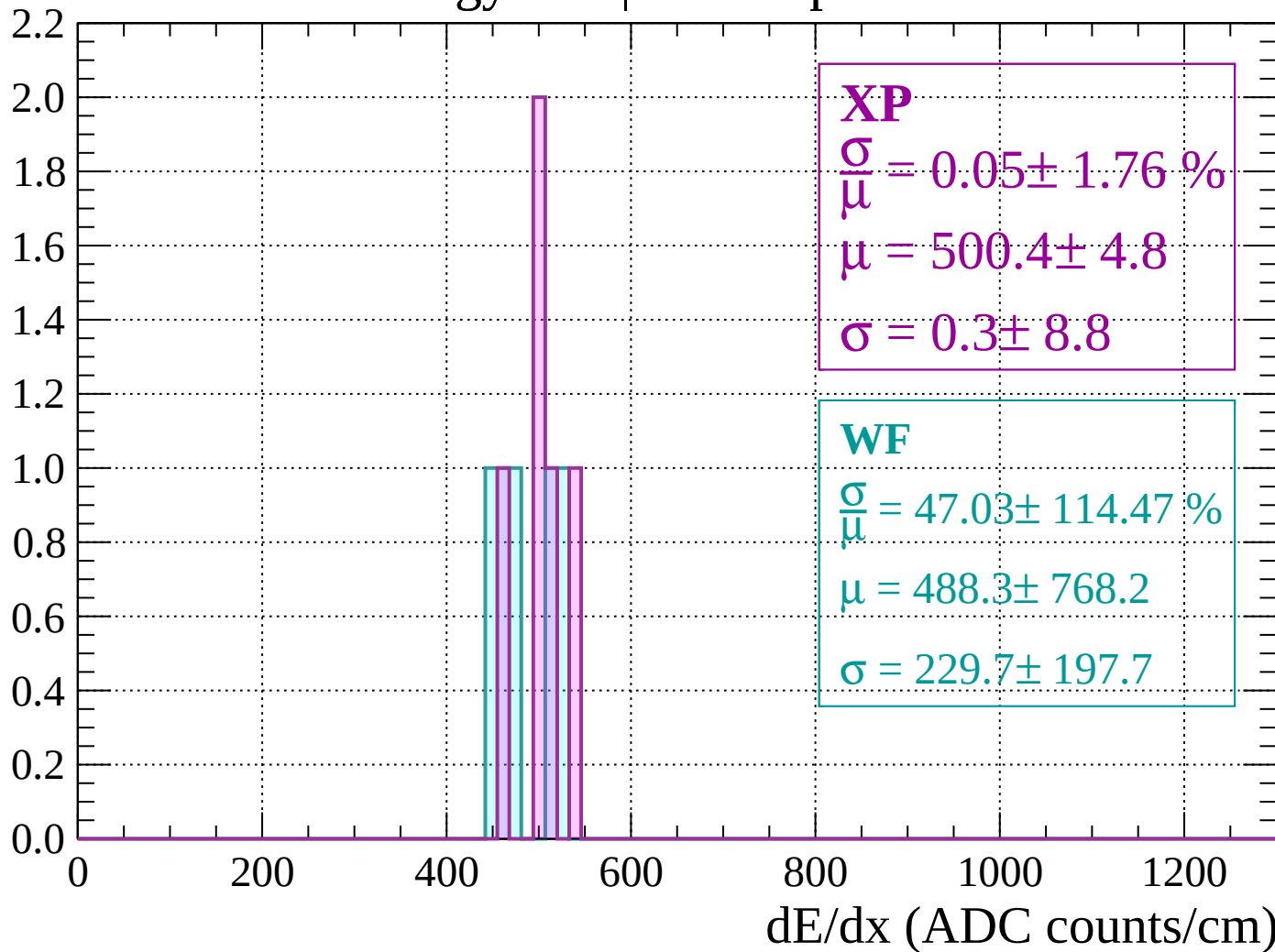
Count



dE/dx (ADC counts/cm)

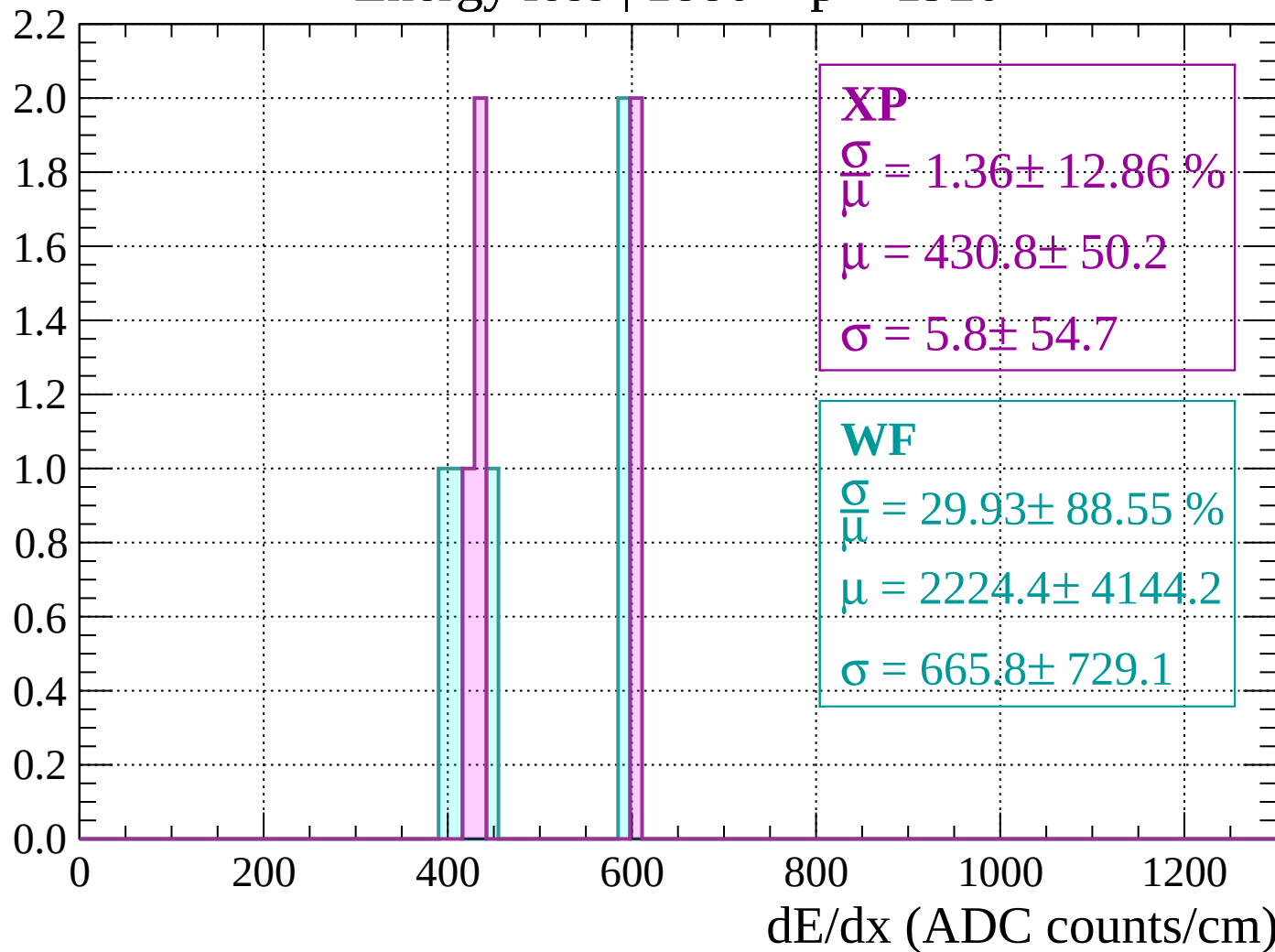
Energy loss | $1840 < p < 1880$

Count



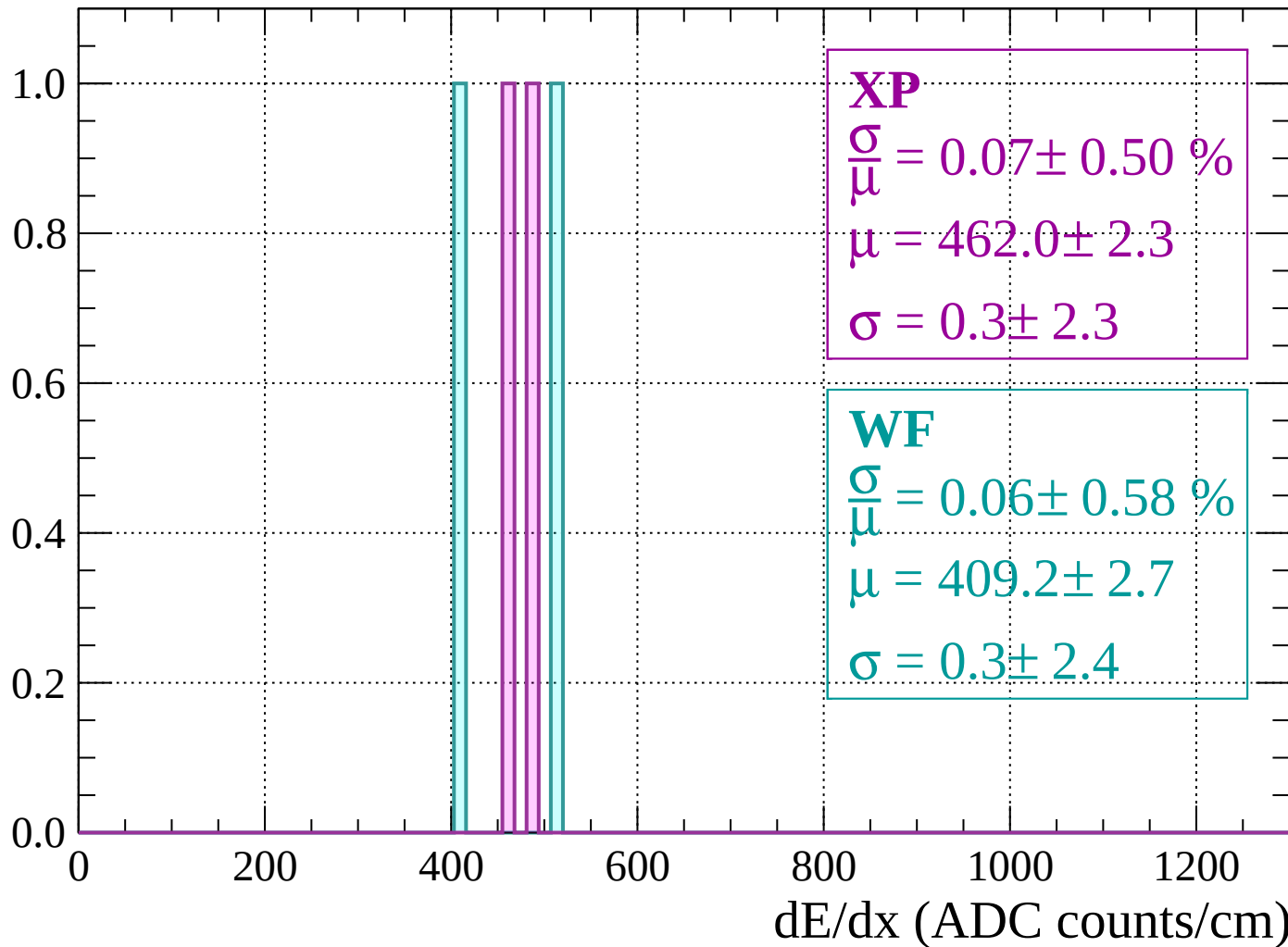
Energy loss | $1880 < p < 1920$

Count



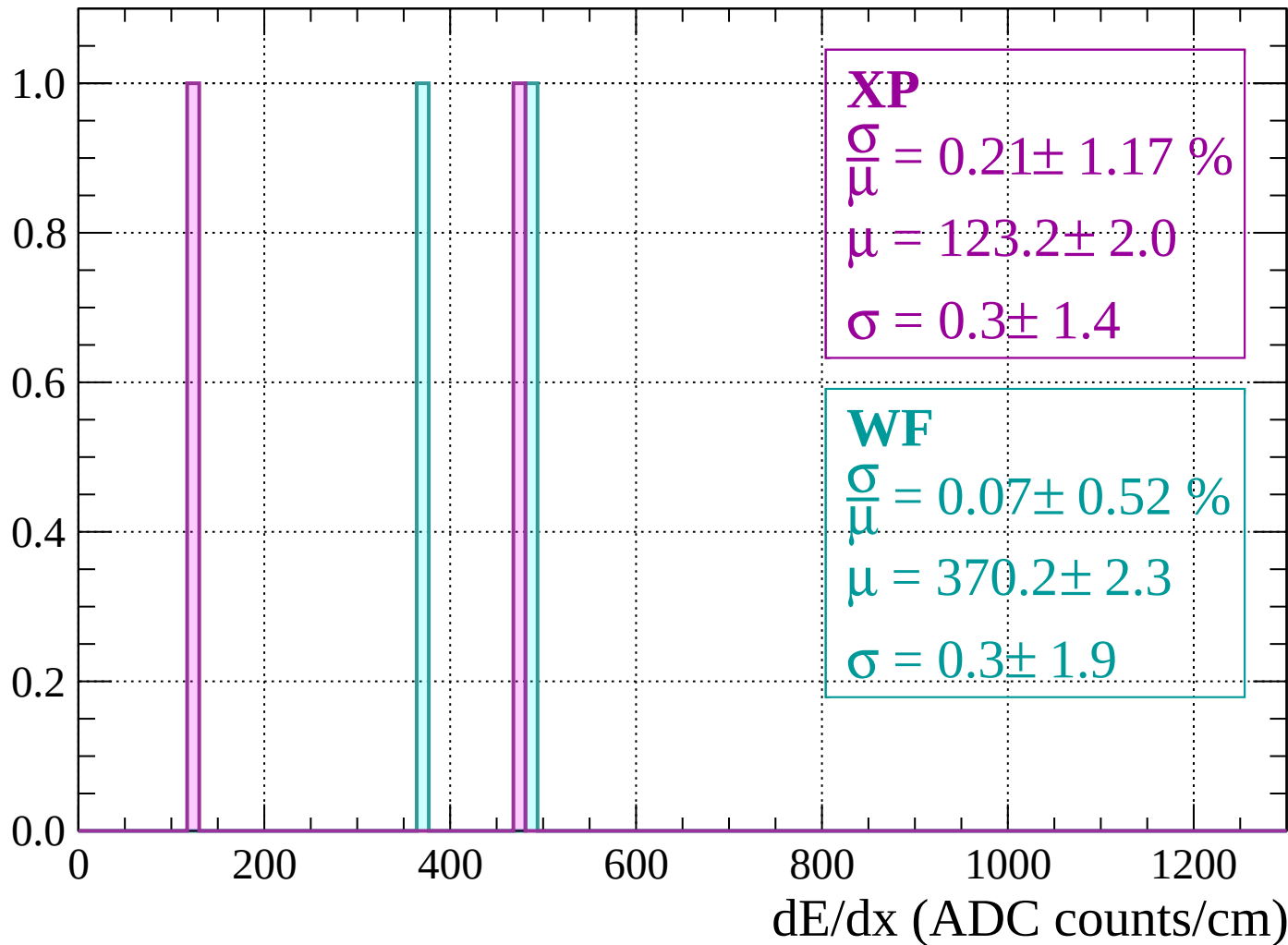
Energy loss | $1920 < p < 1960$

Count



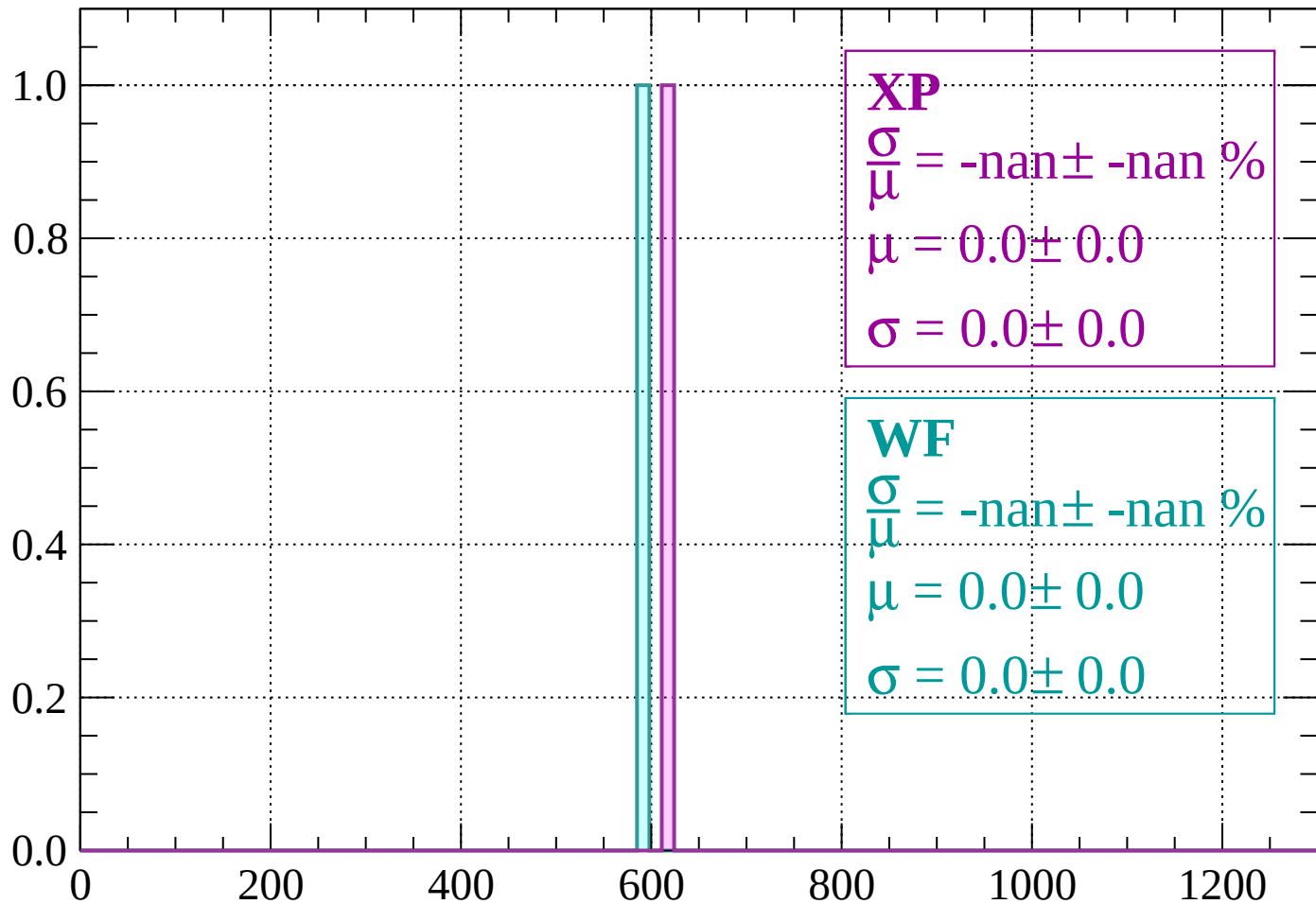
Energy loss | $1960 < p < 2000$

Count



Energy loss | 2000 < p < 2040

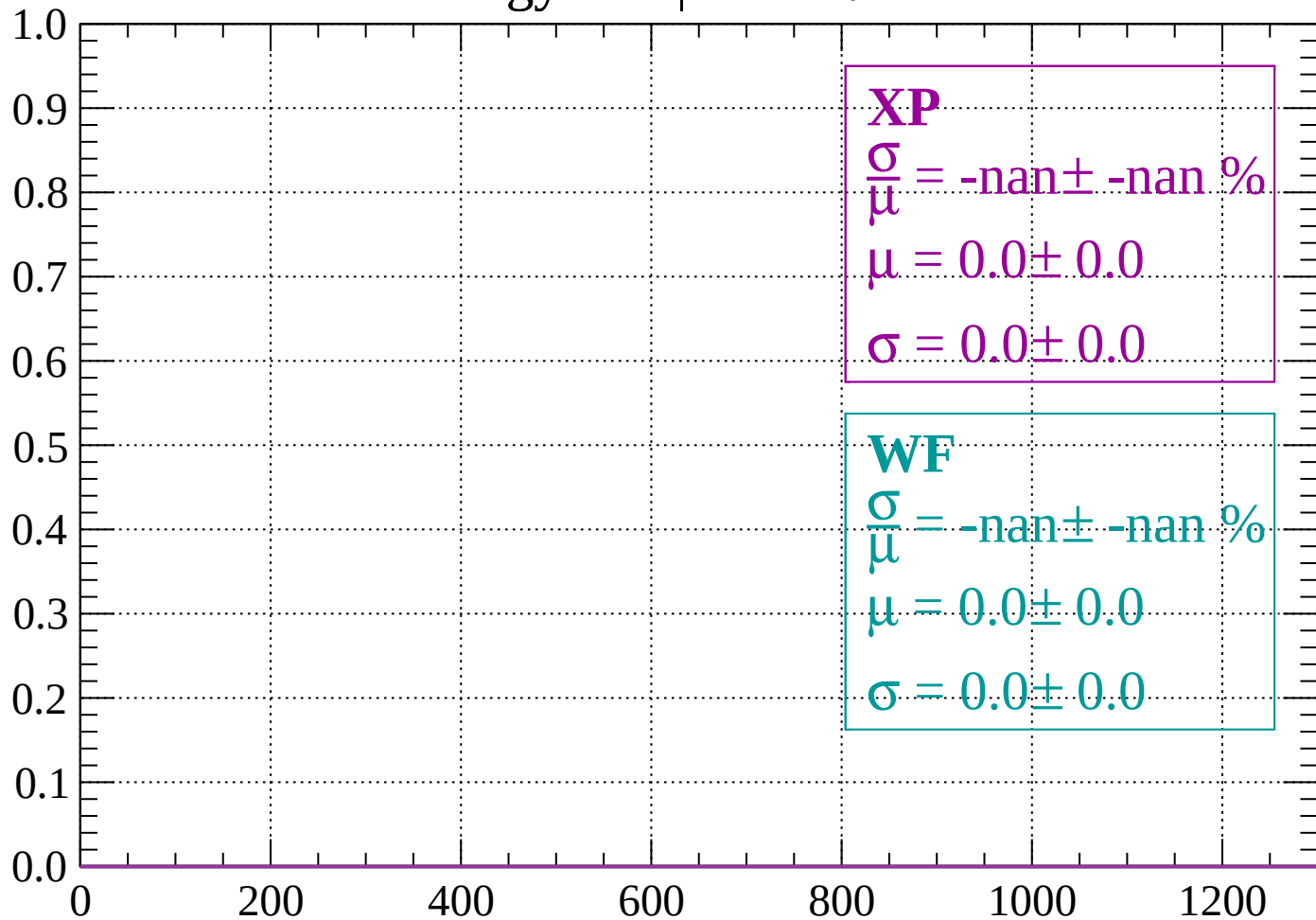
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

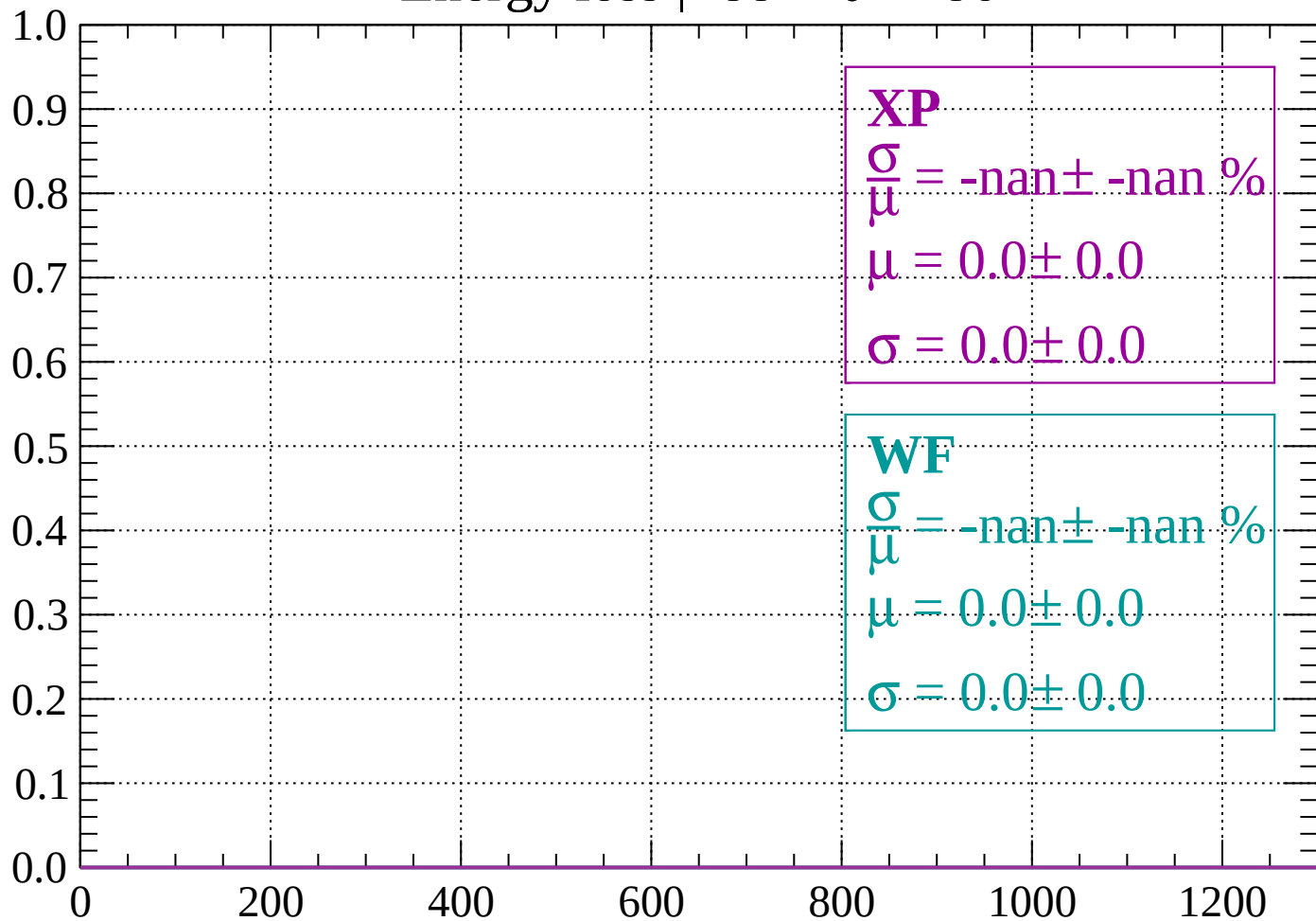
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

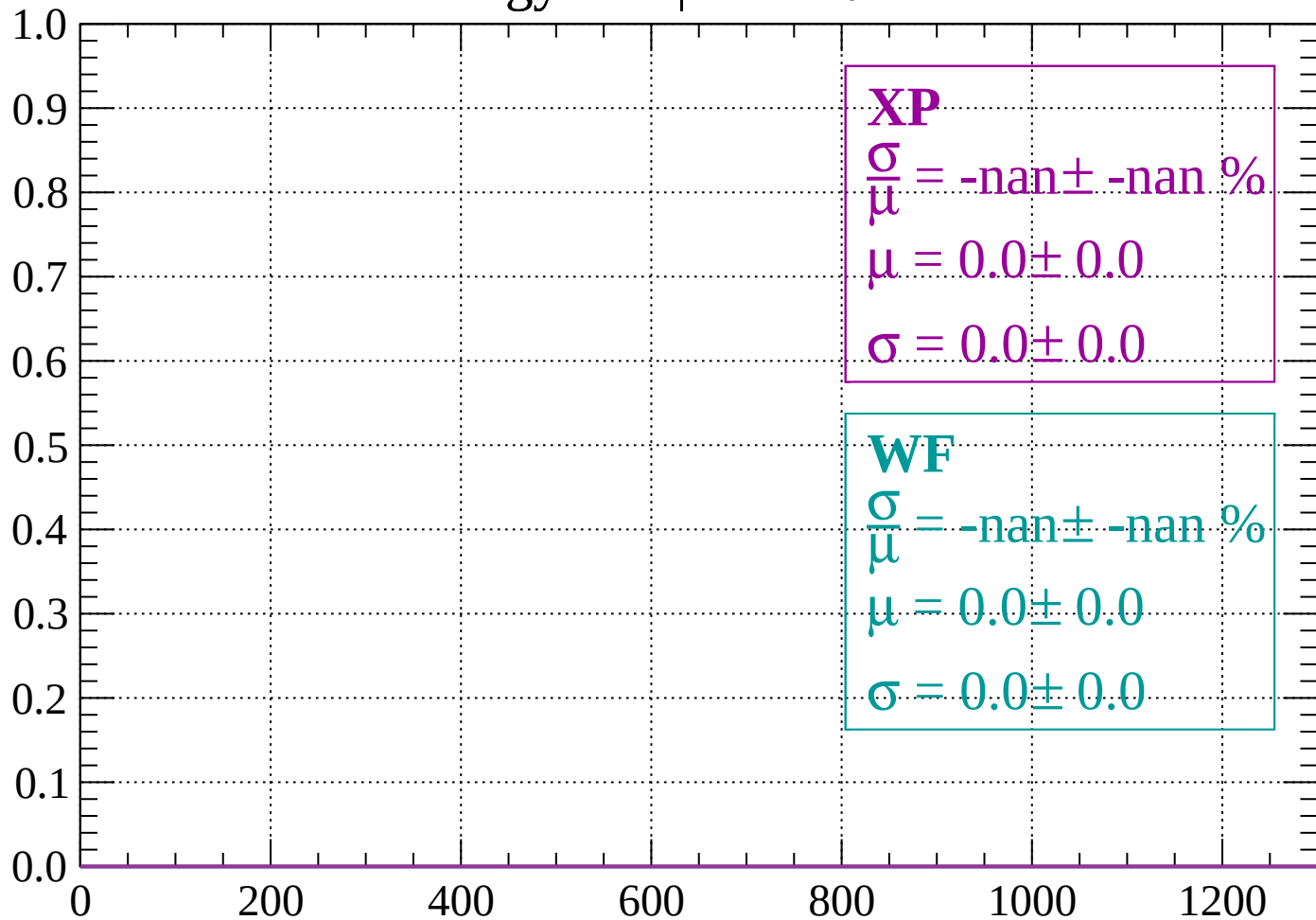
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

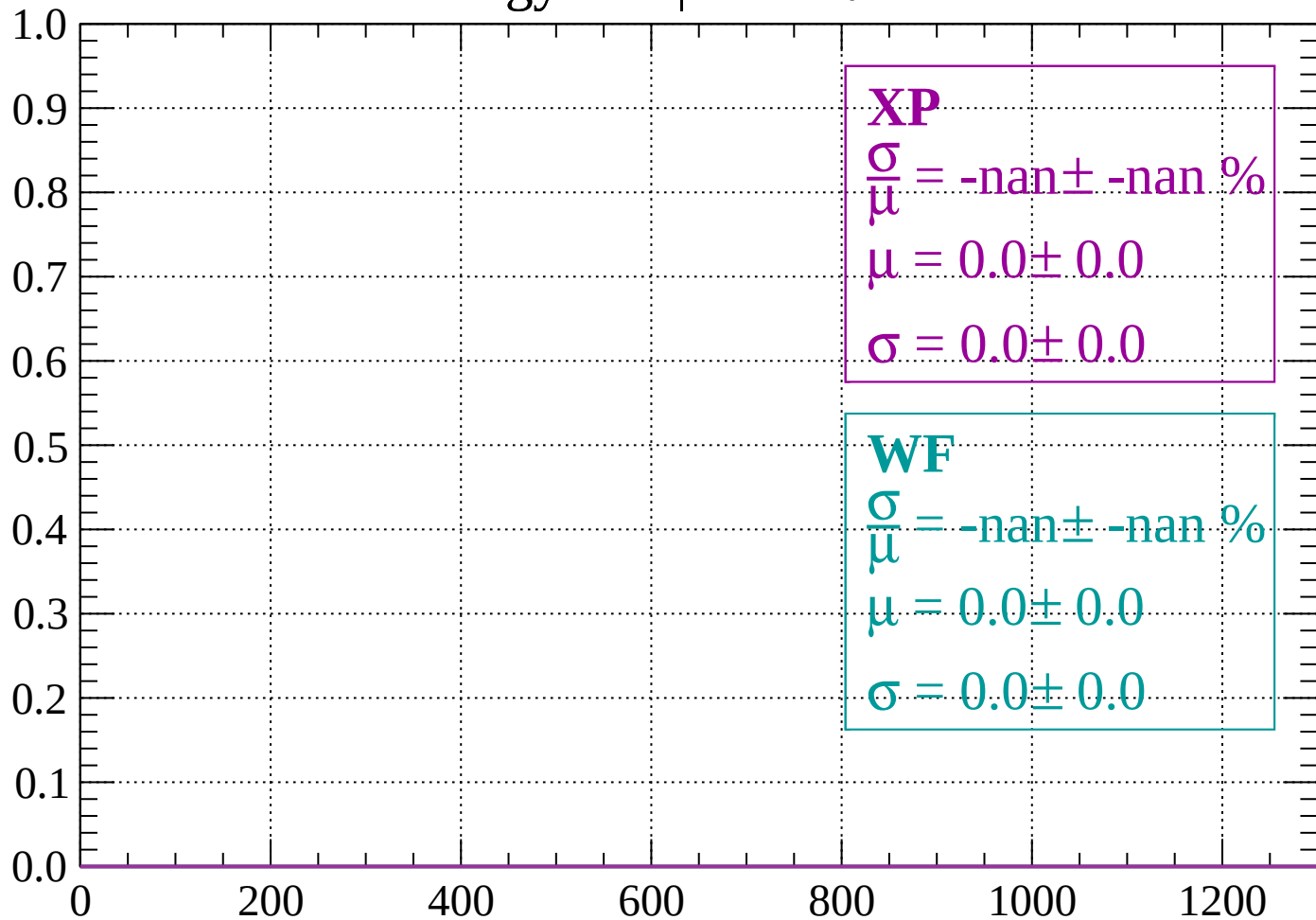
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

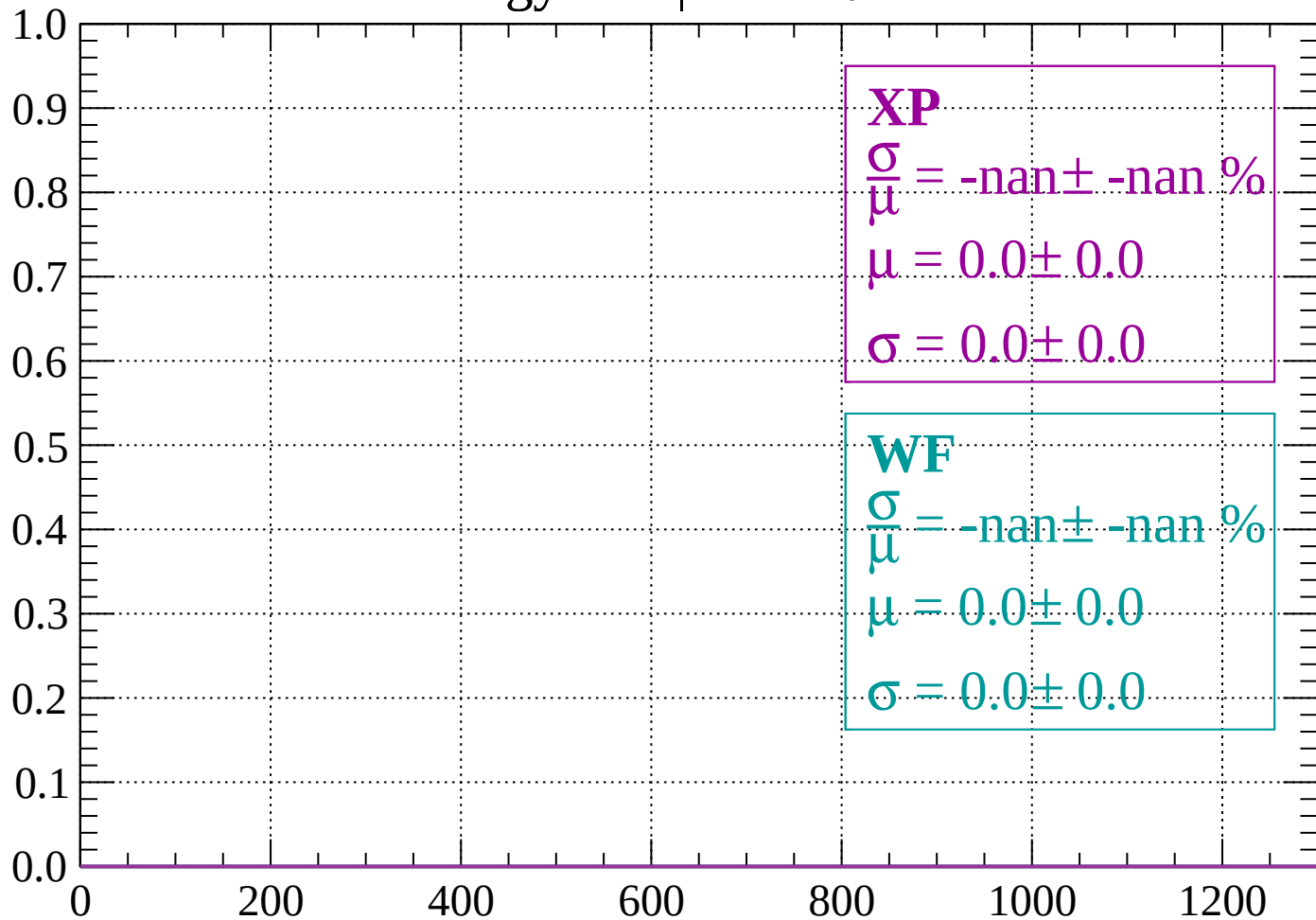
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

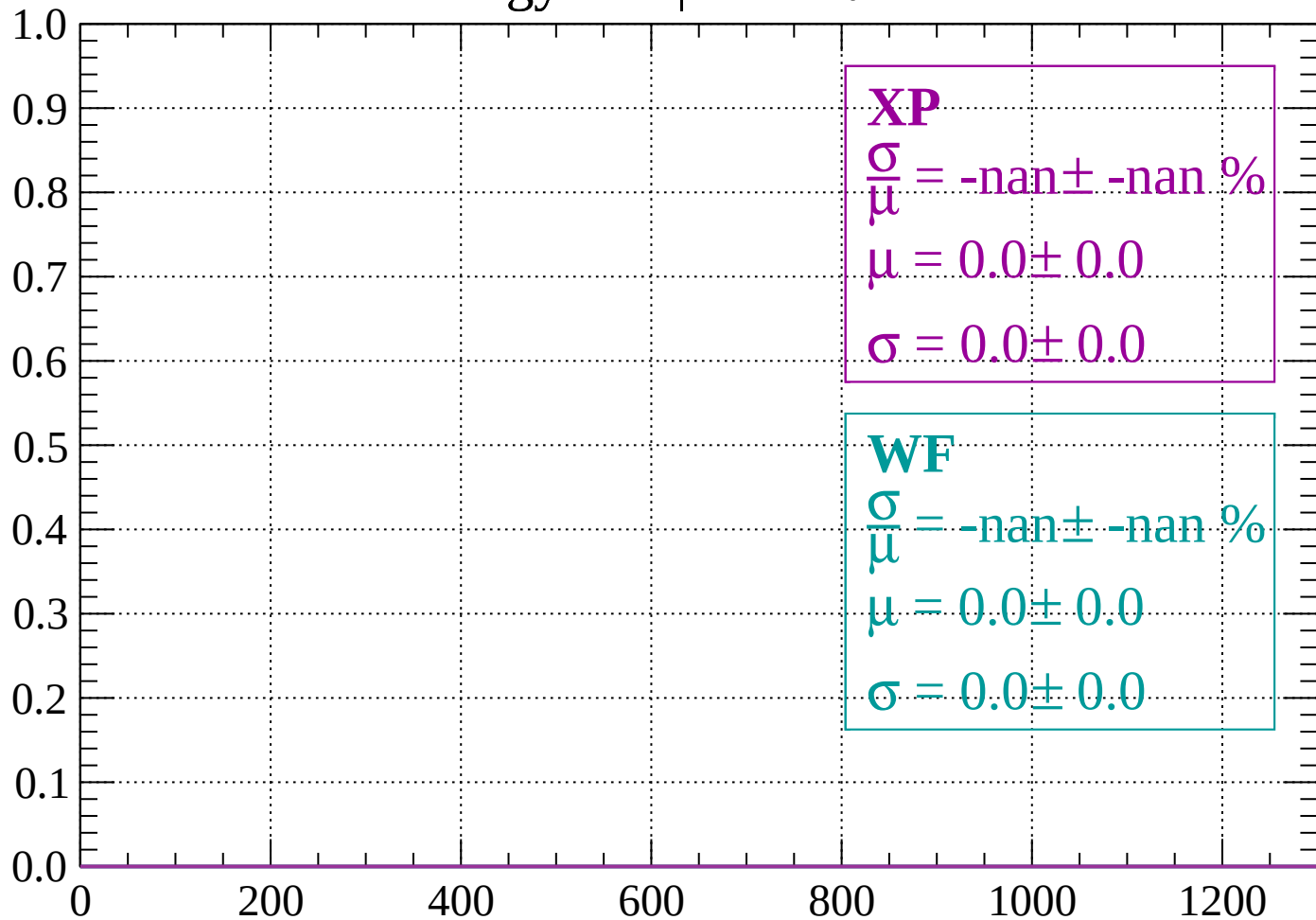
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

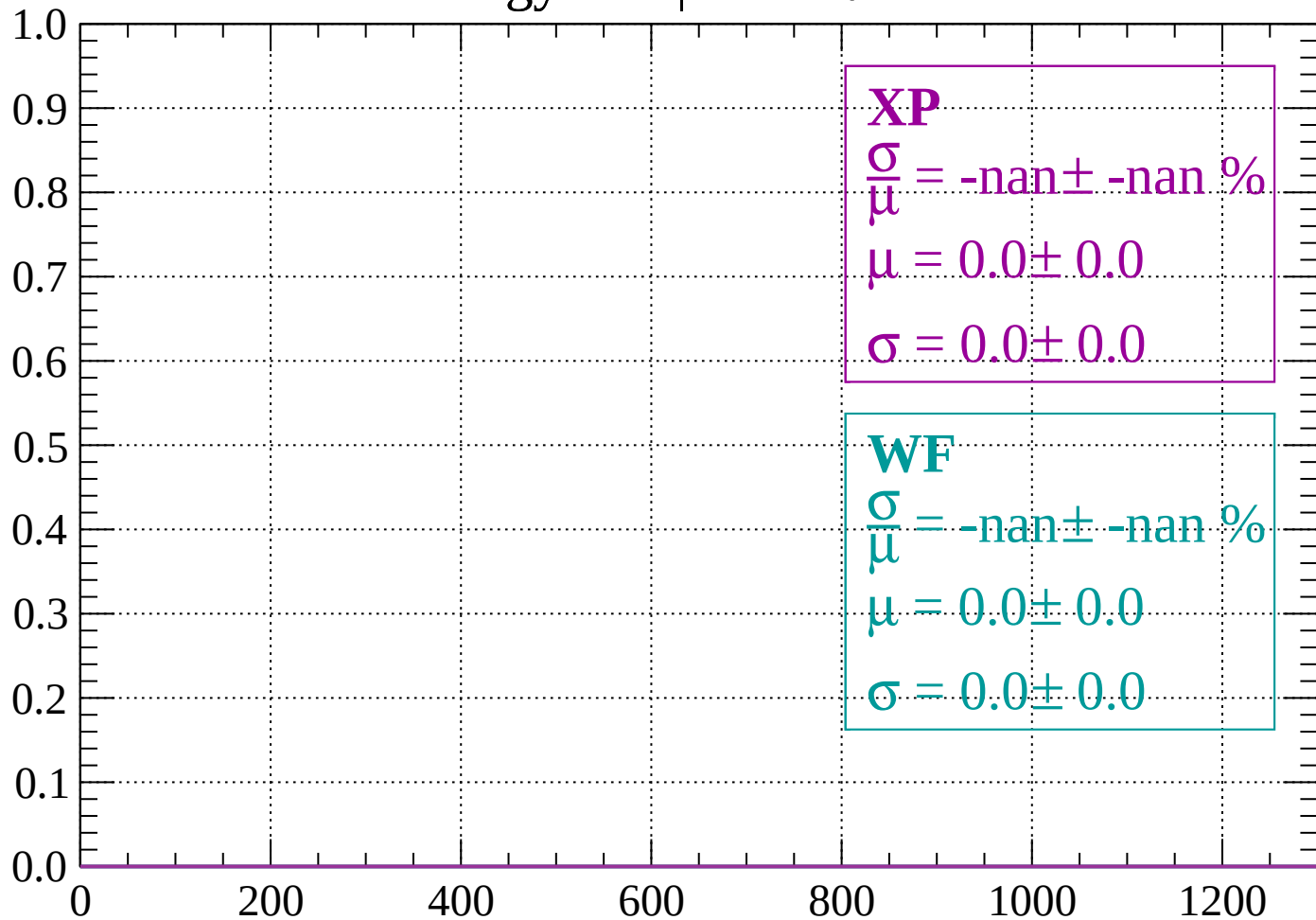
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

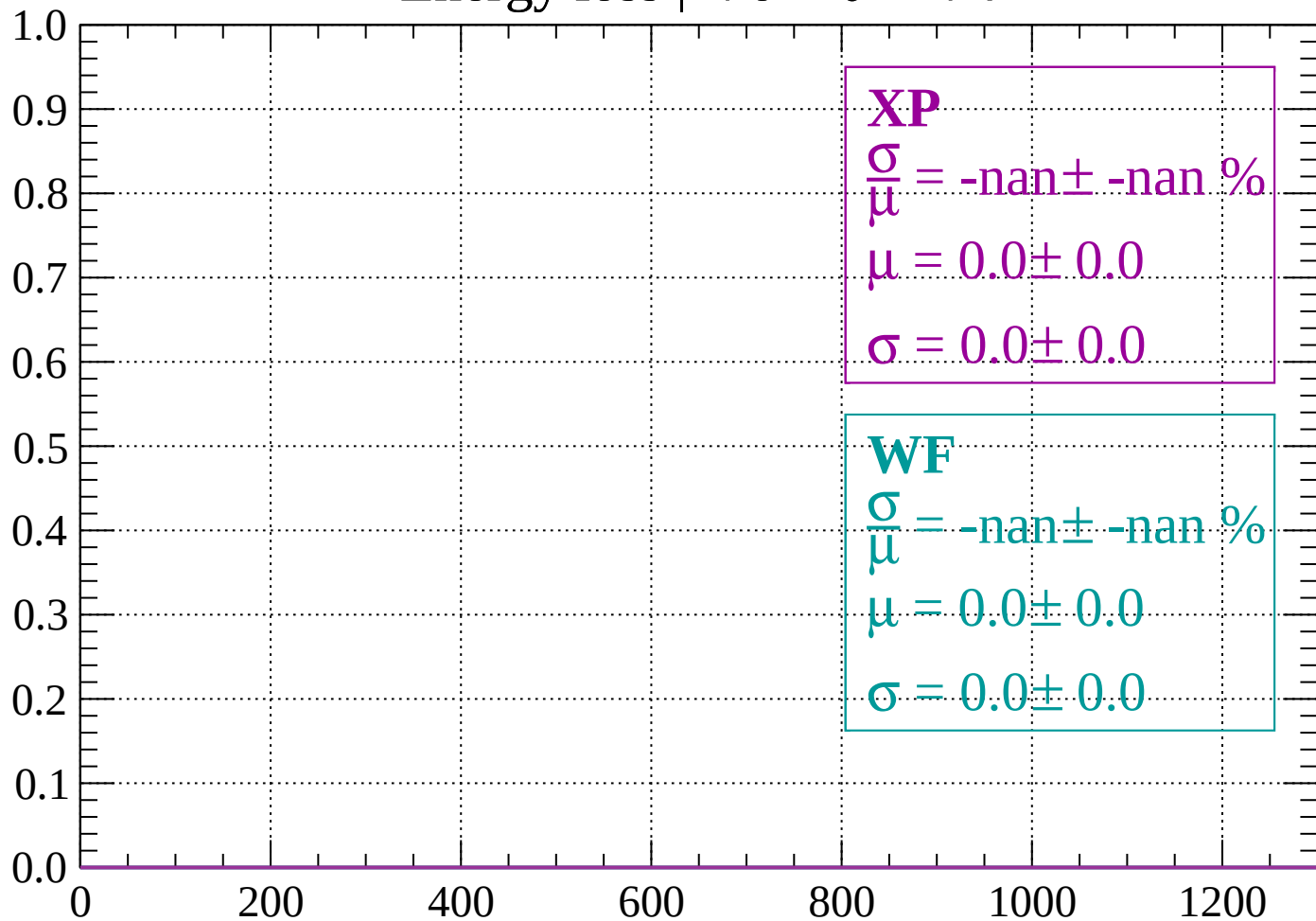
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

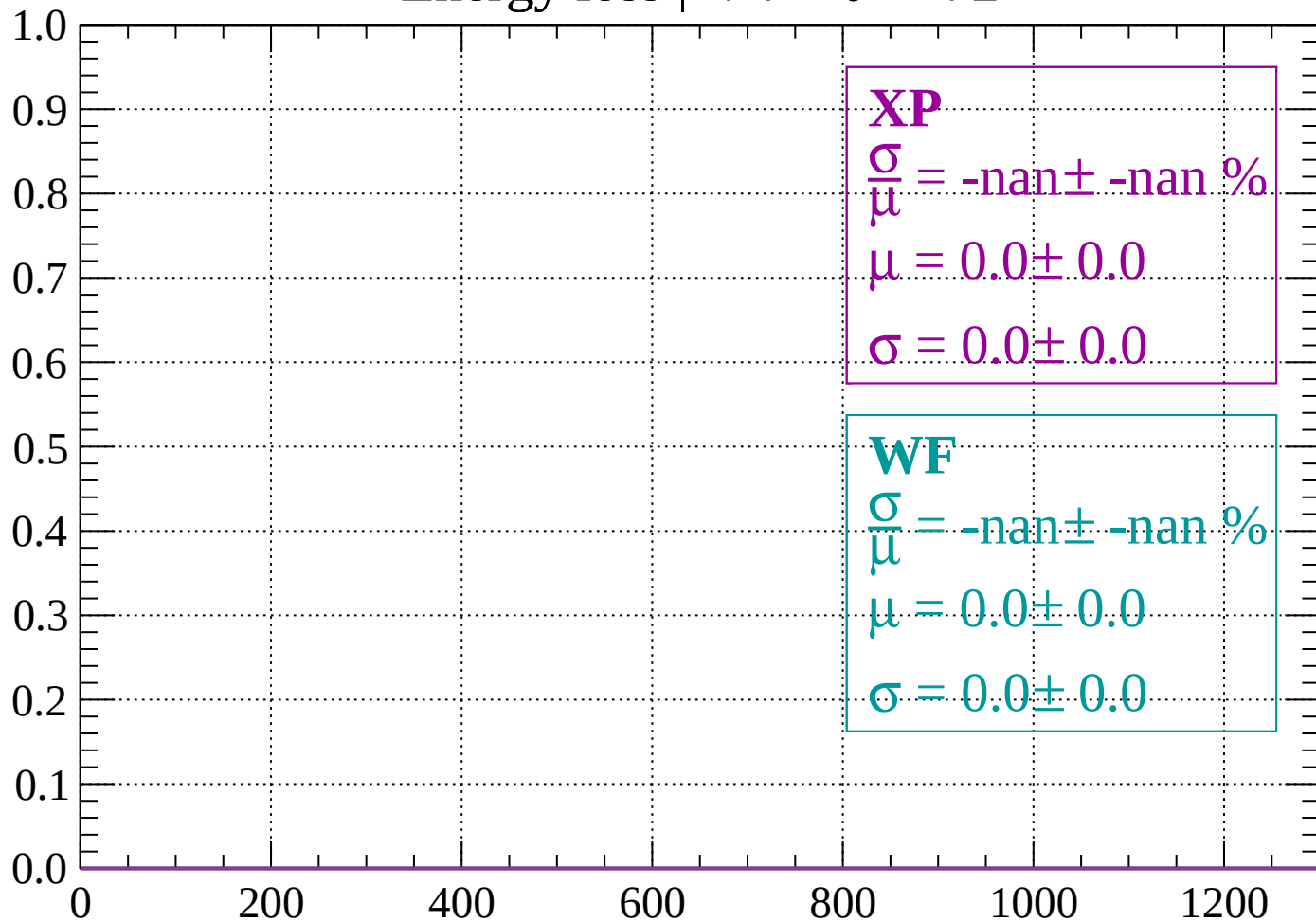
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

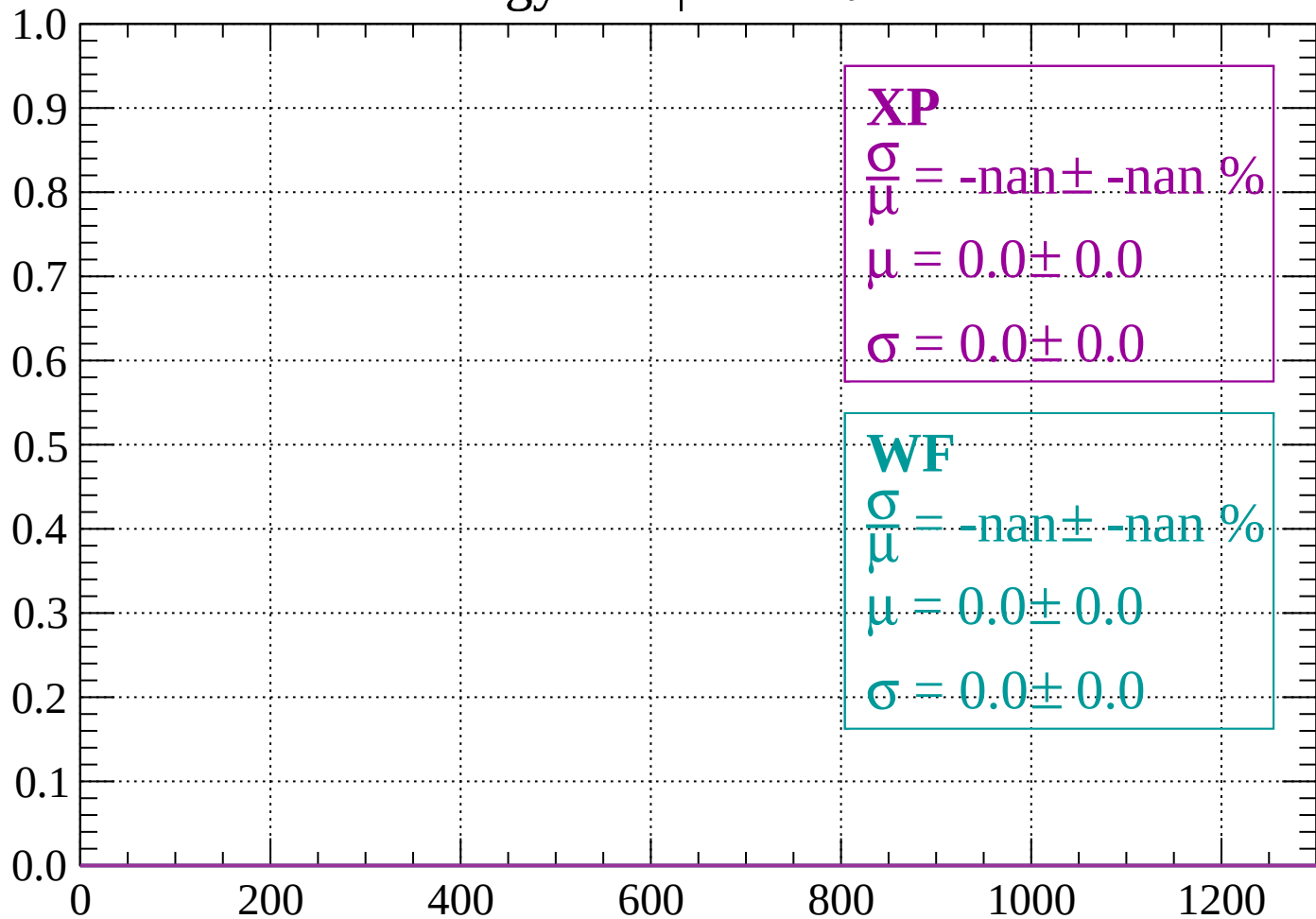
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

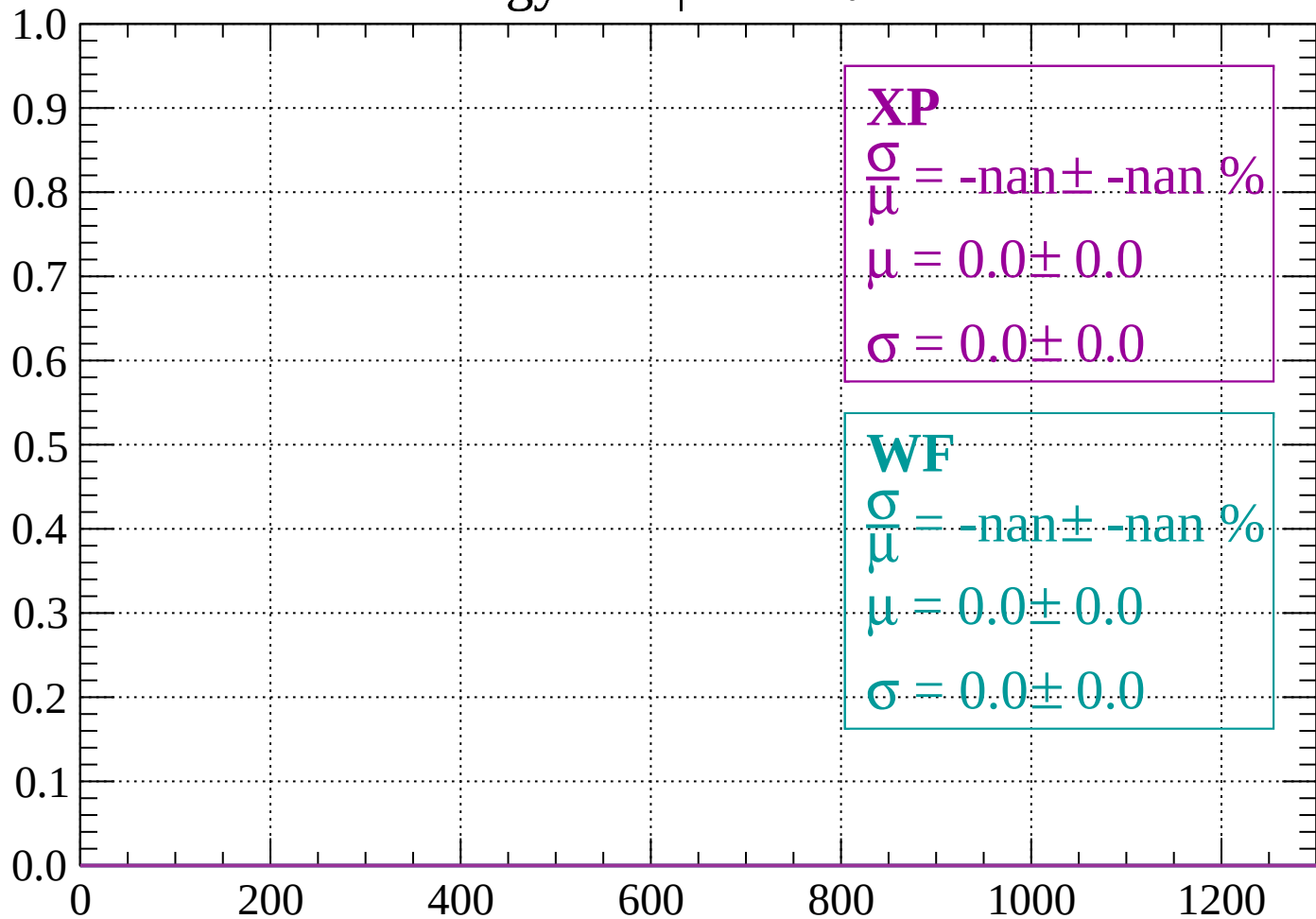
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

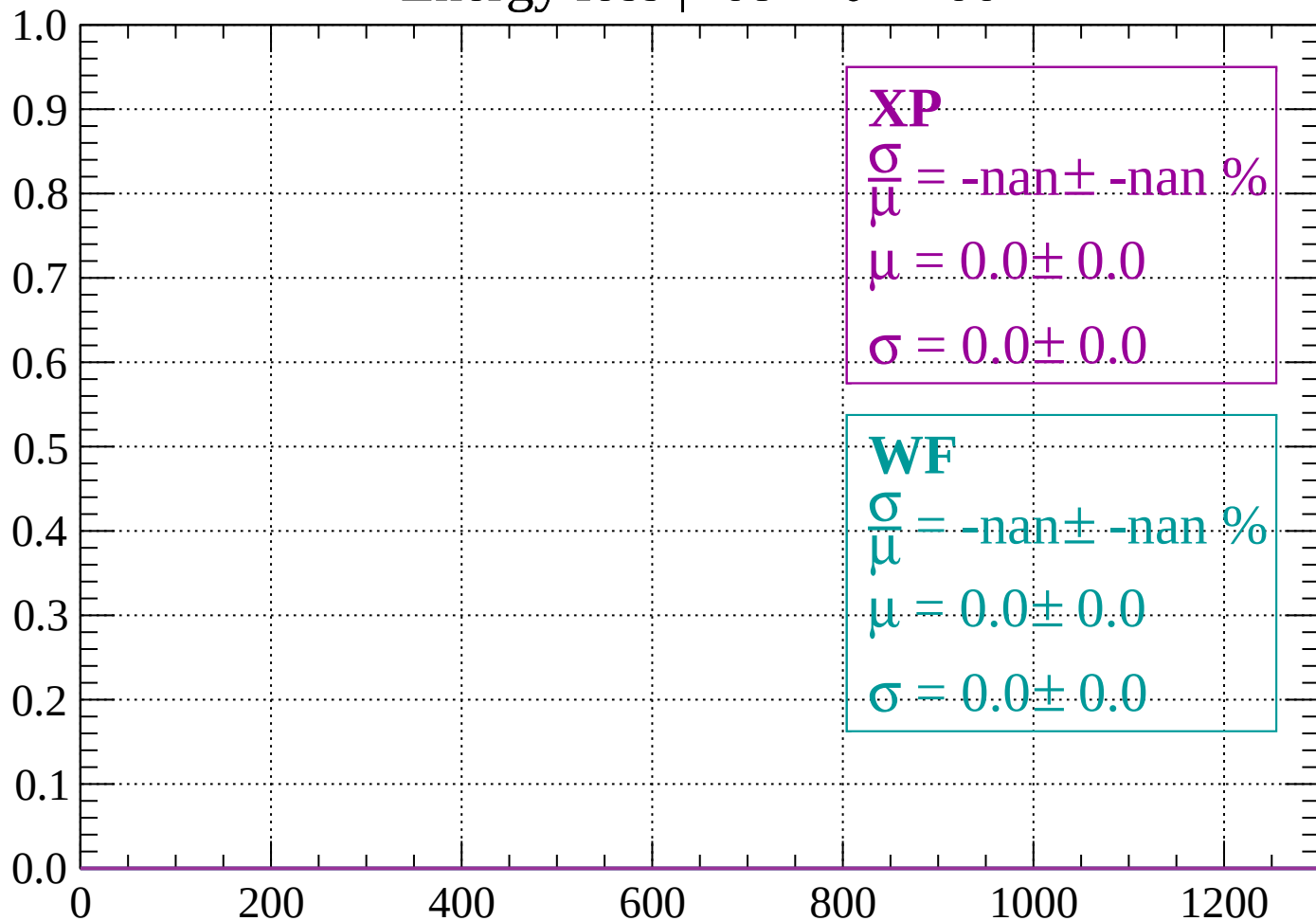
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

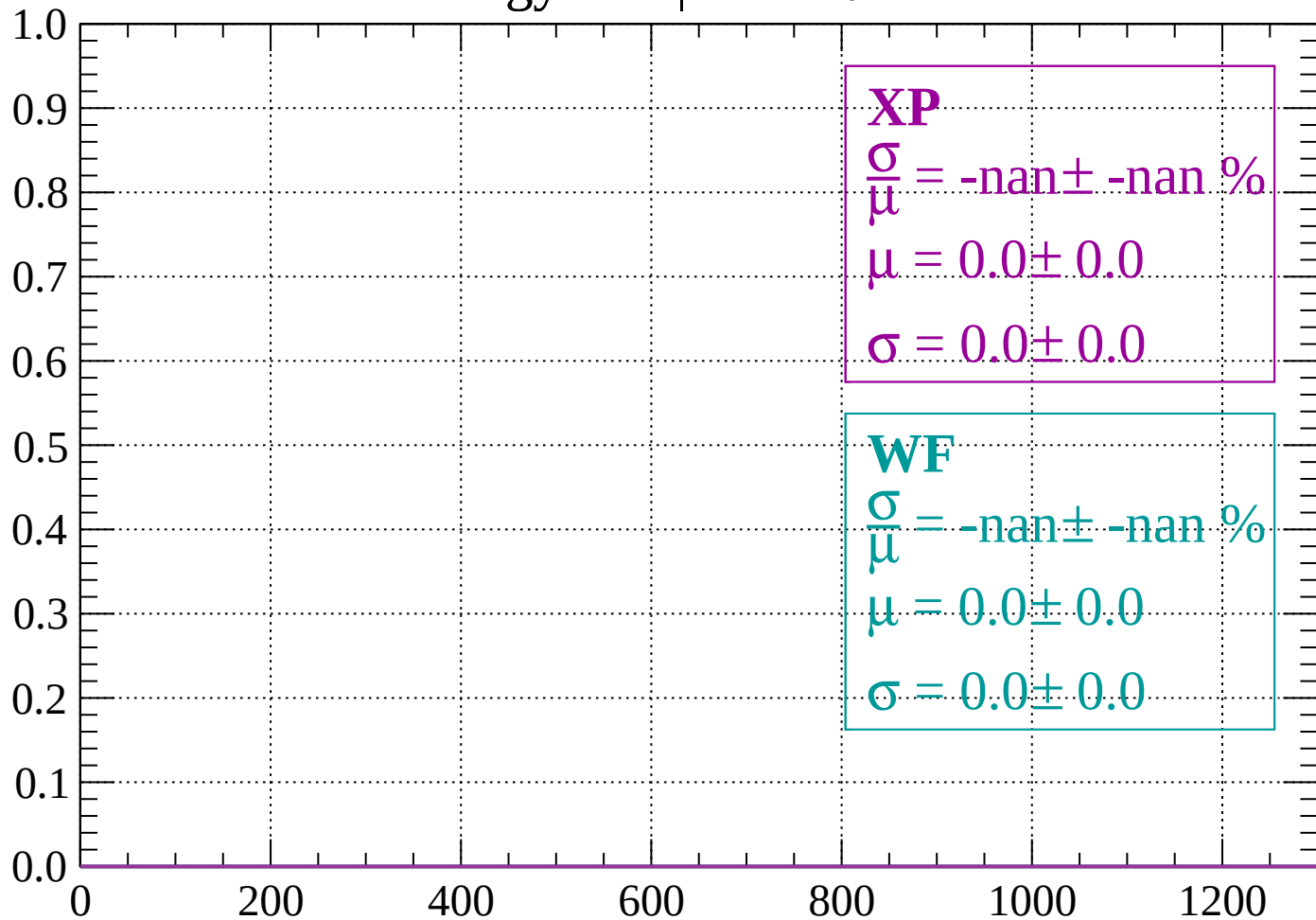
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

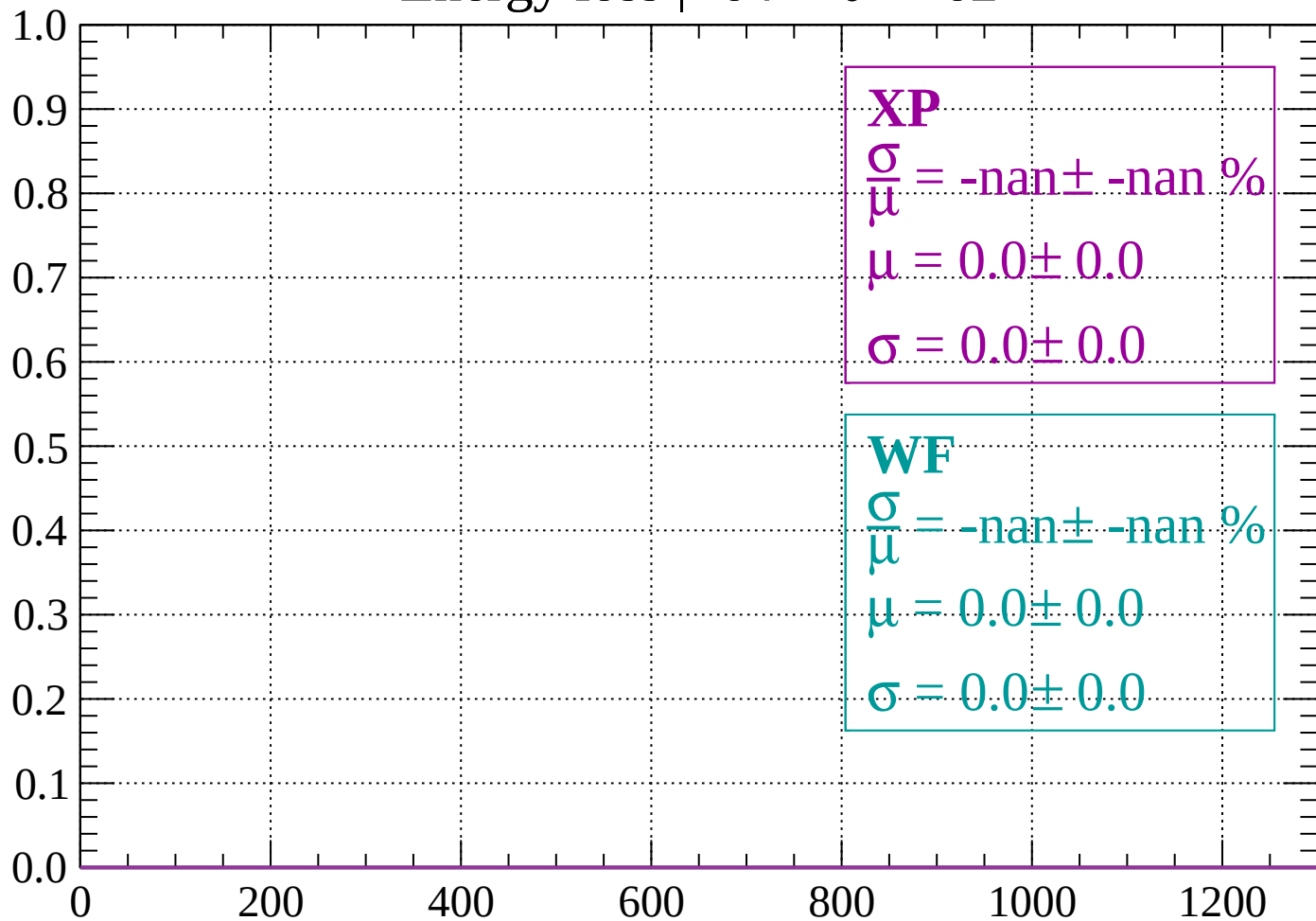
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

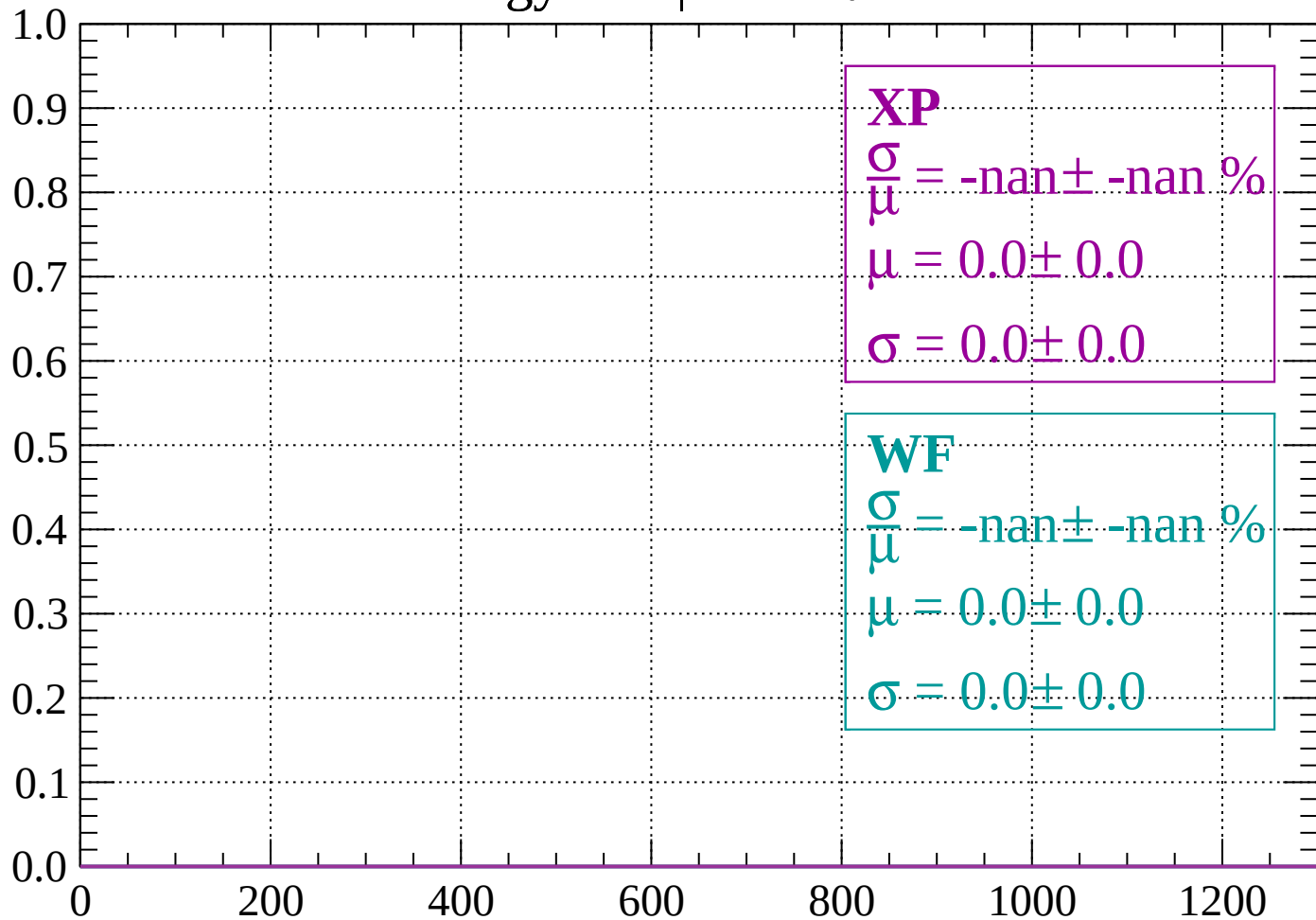
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

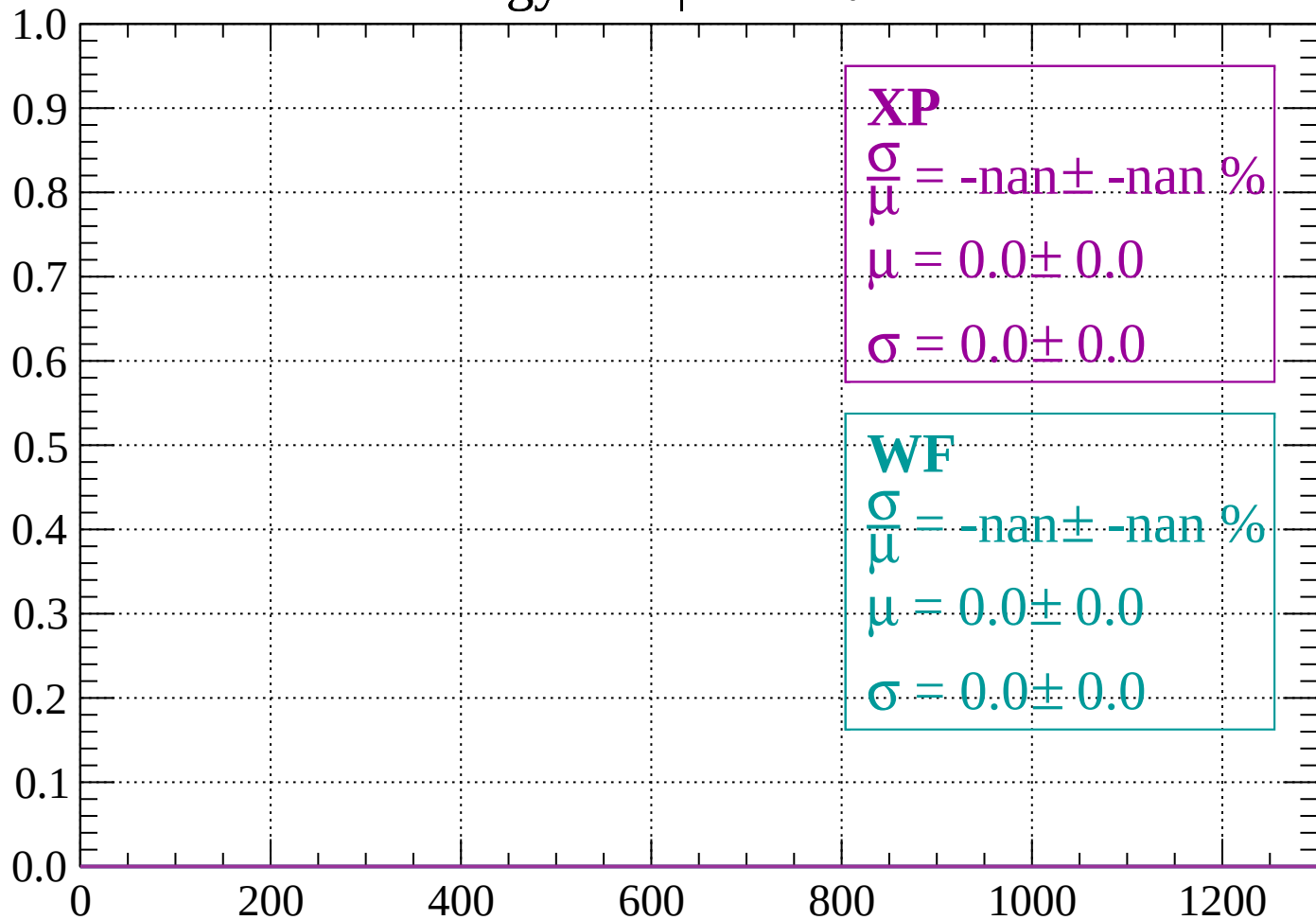
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

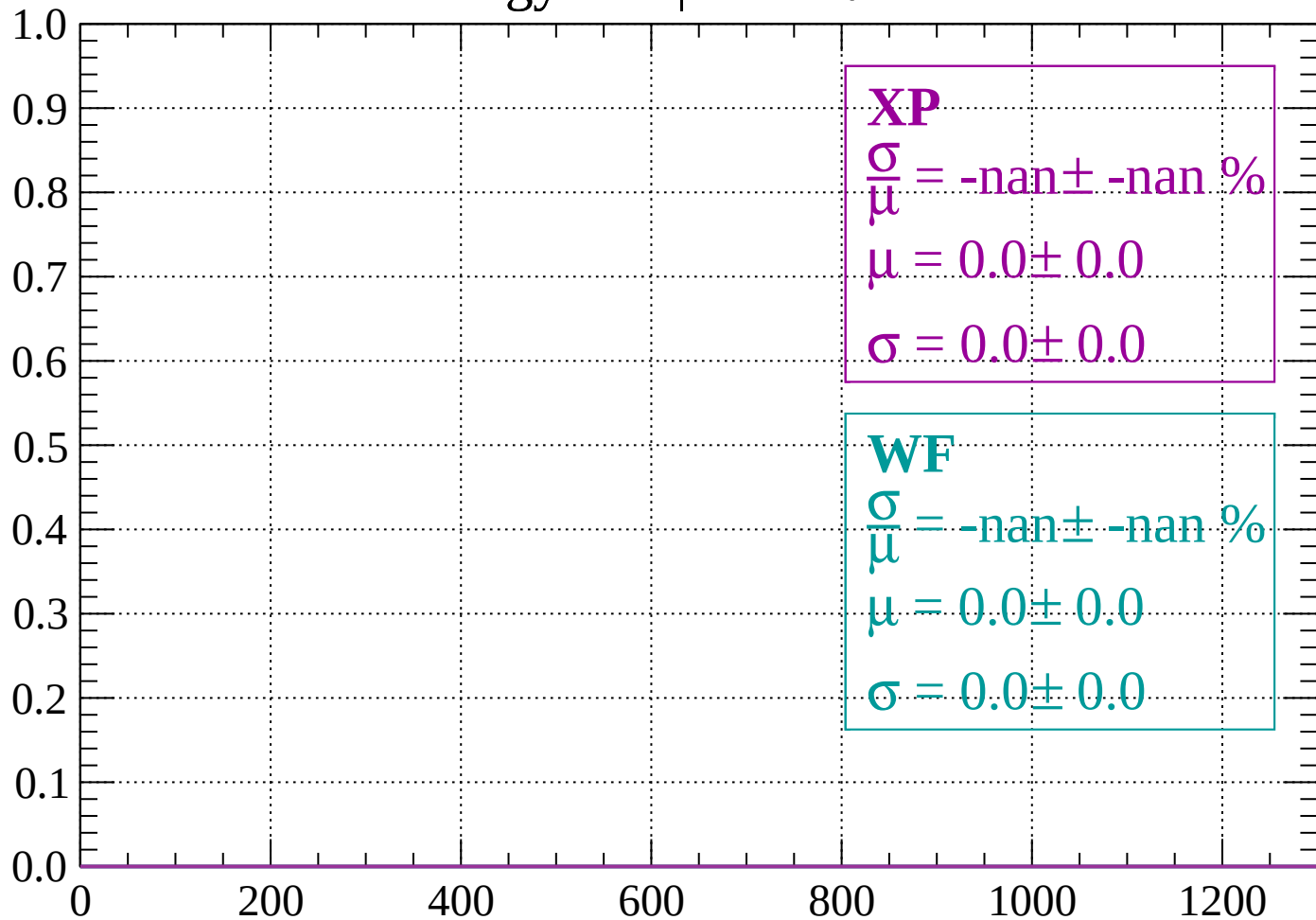
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

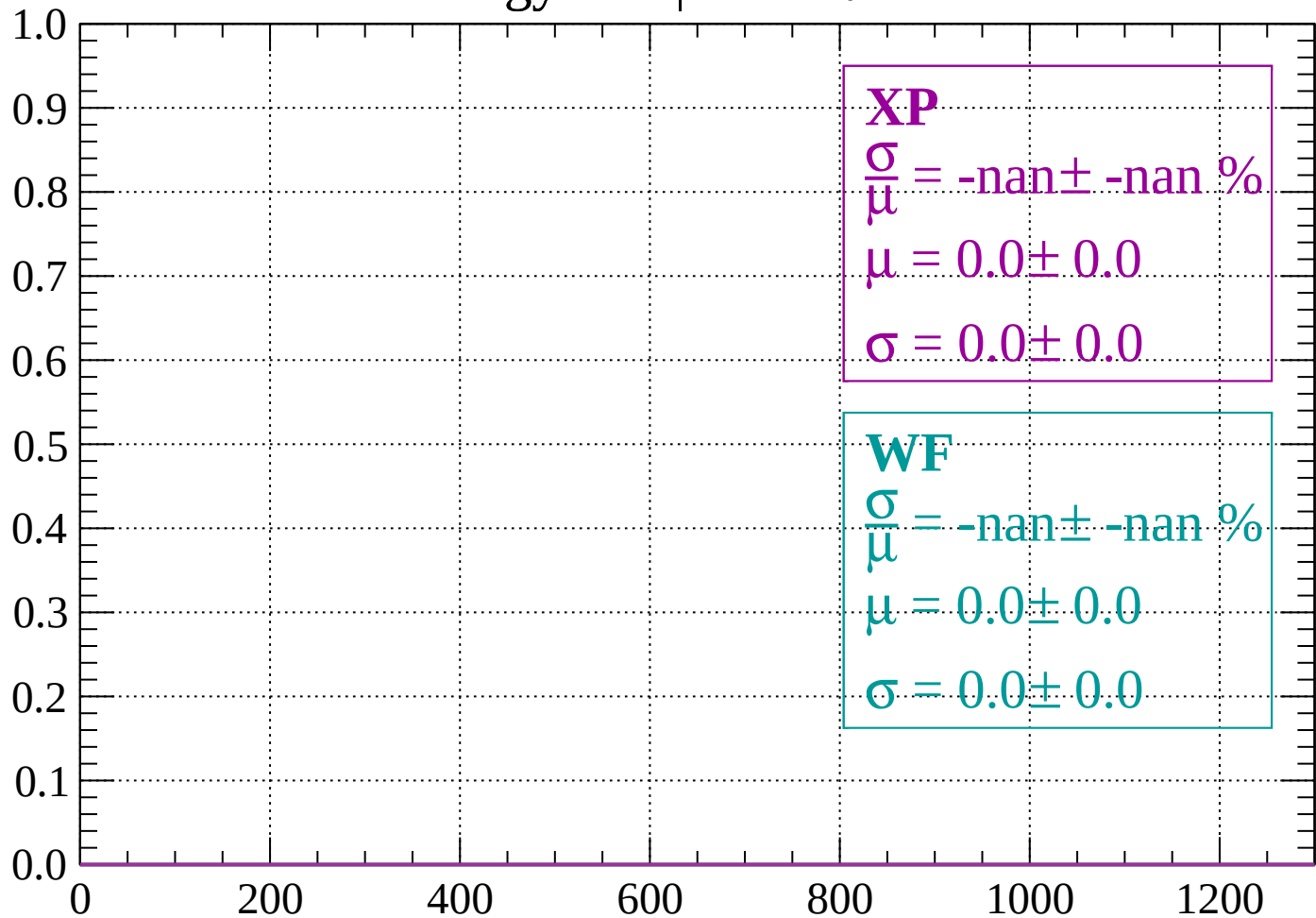
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

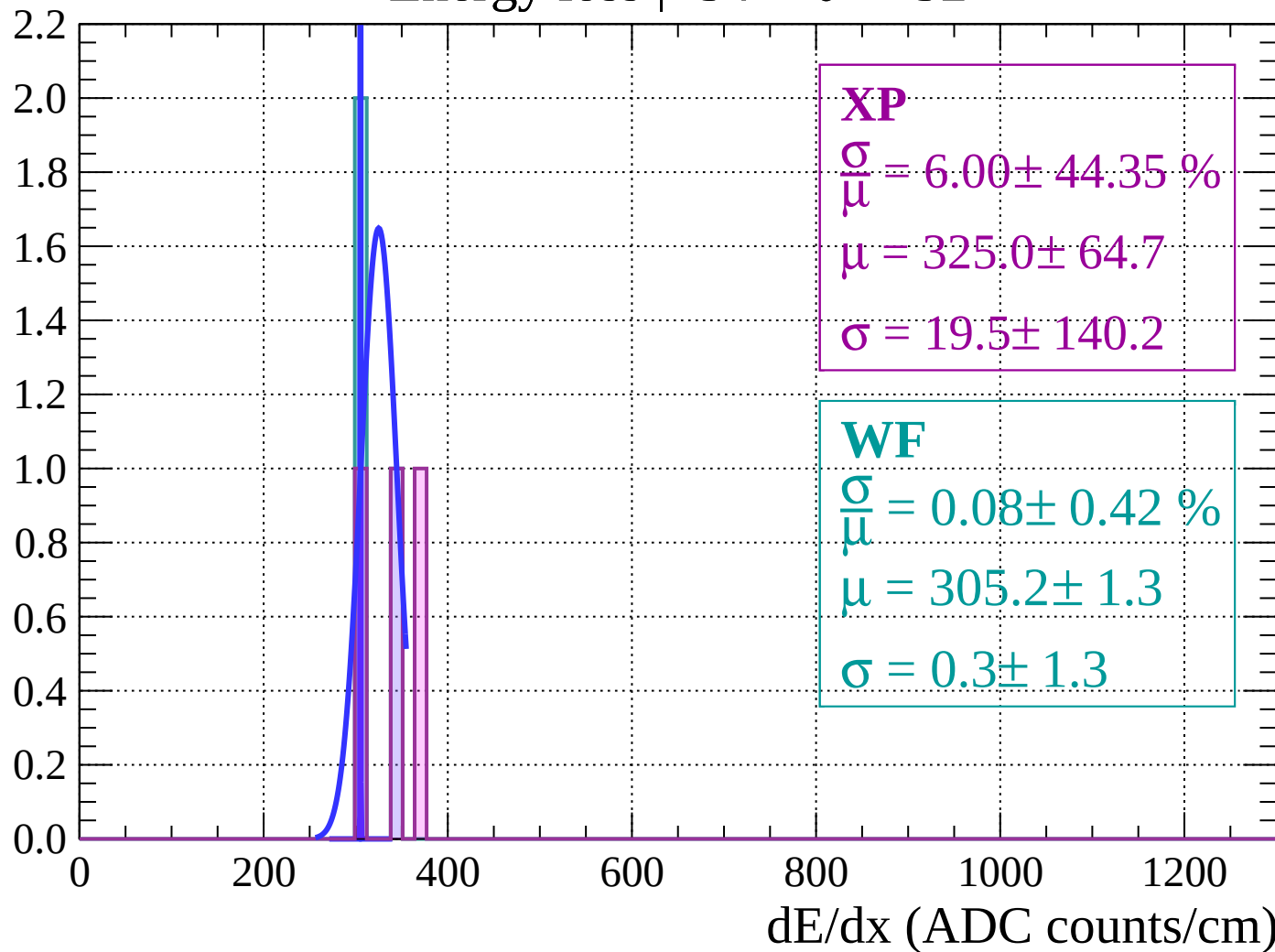
Count



dE/dx (ADC counts/cm)

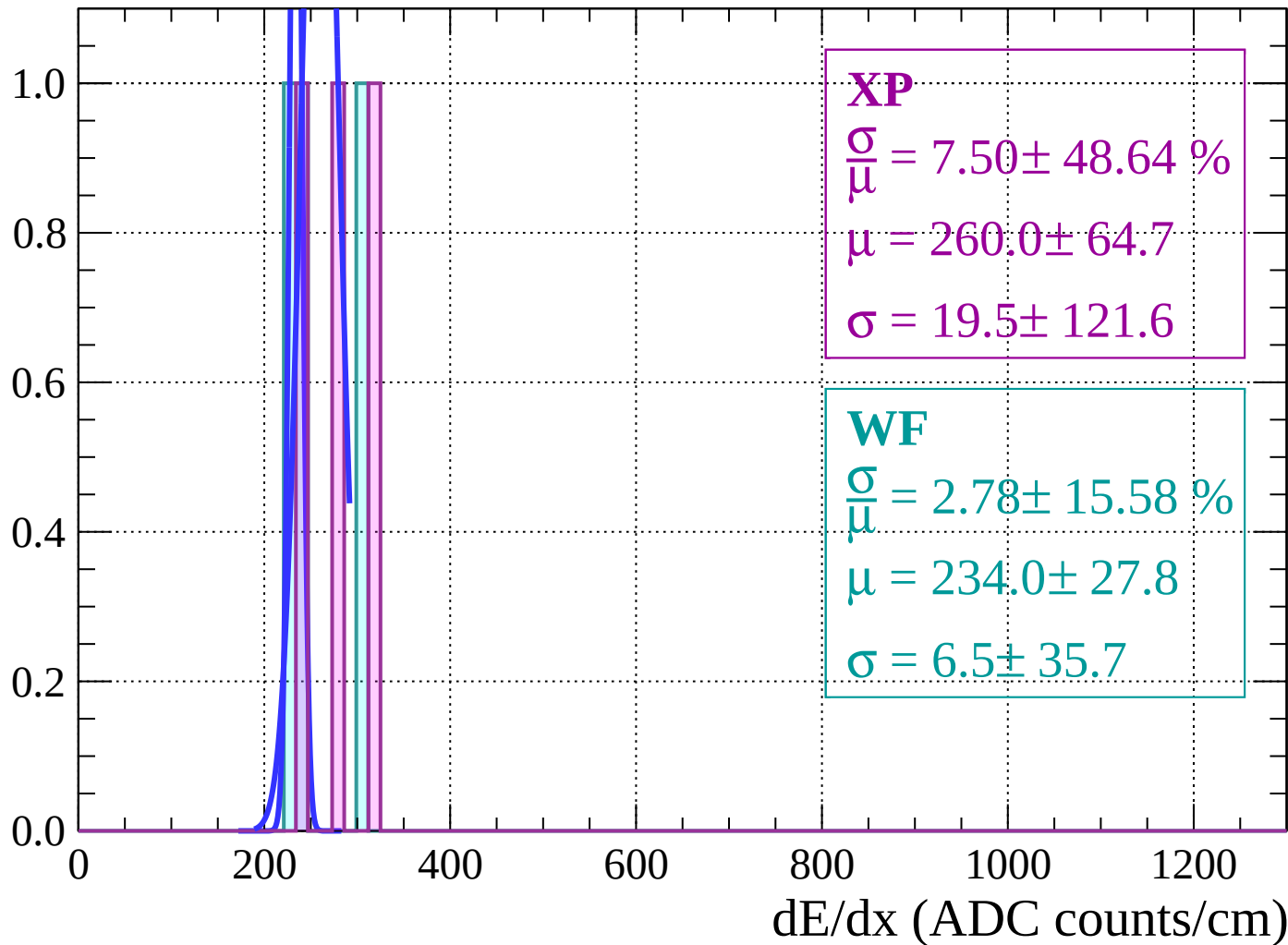
Energy loss | $-54 < \theta < -52$

Count



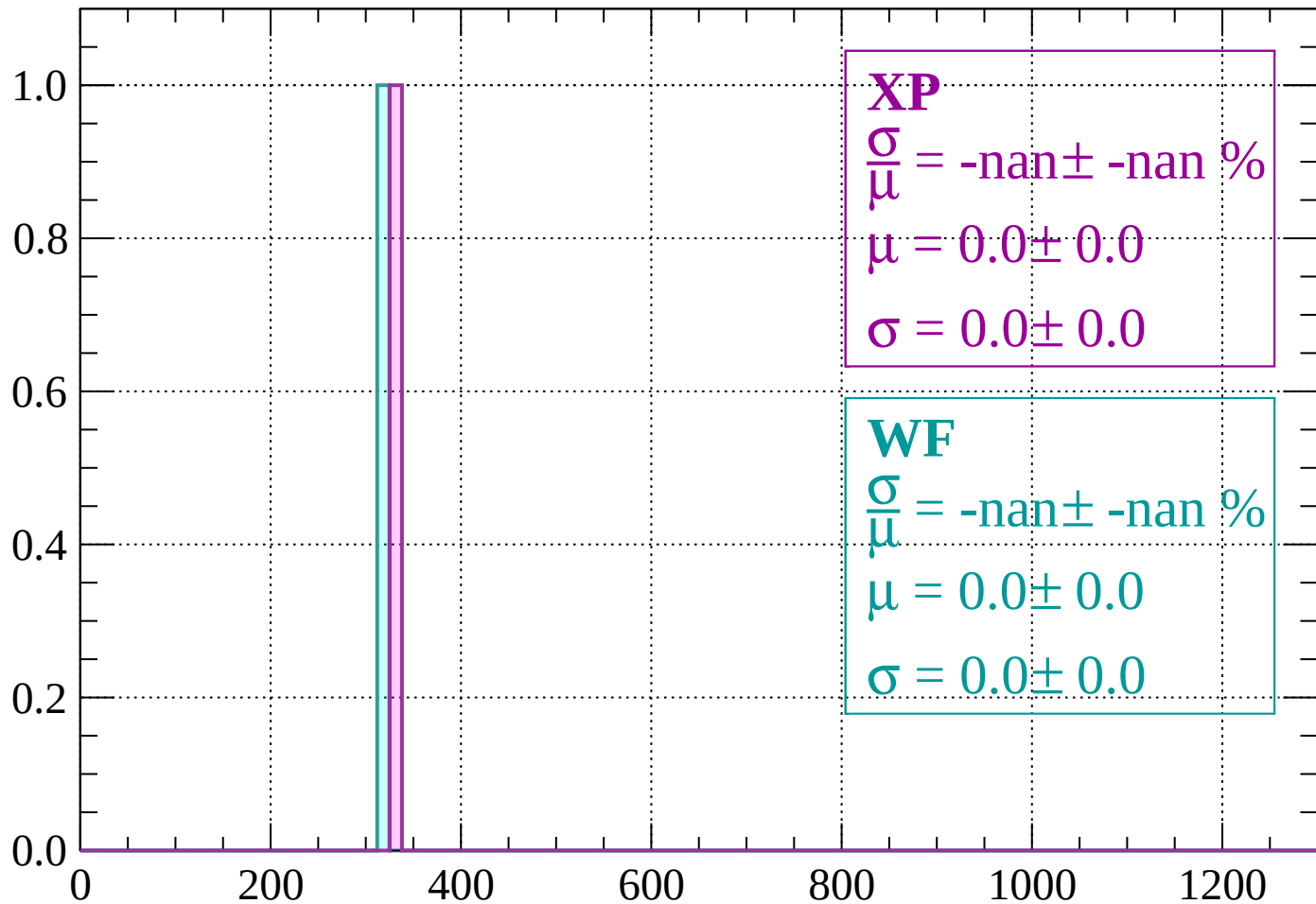
Energy loss | $-52 < \theta < -50$

Count



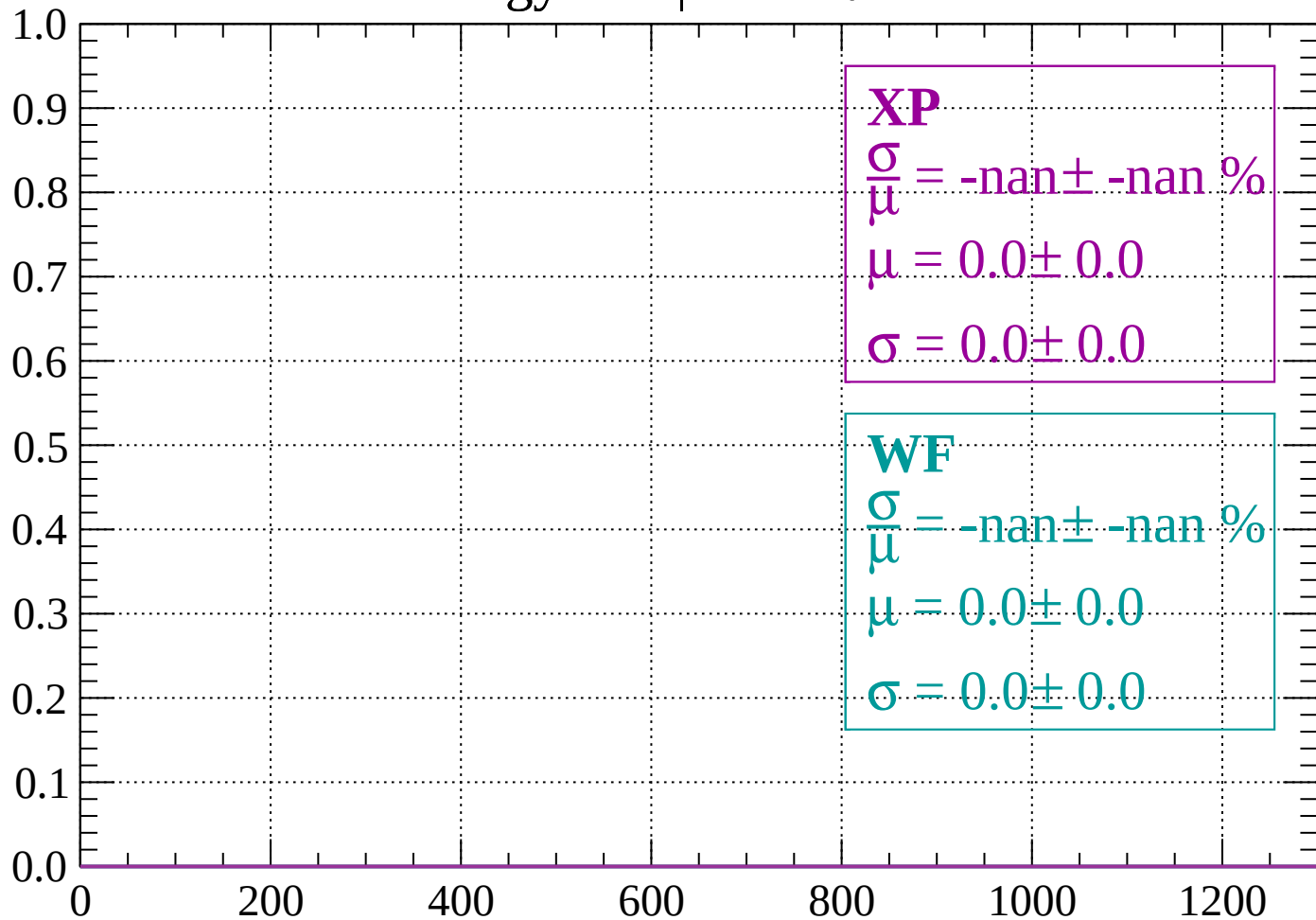
Energy loss | $-50 < \theta < -48$

Count



Energy loss | $-48 < \theta < -46$

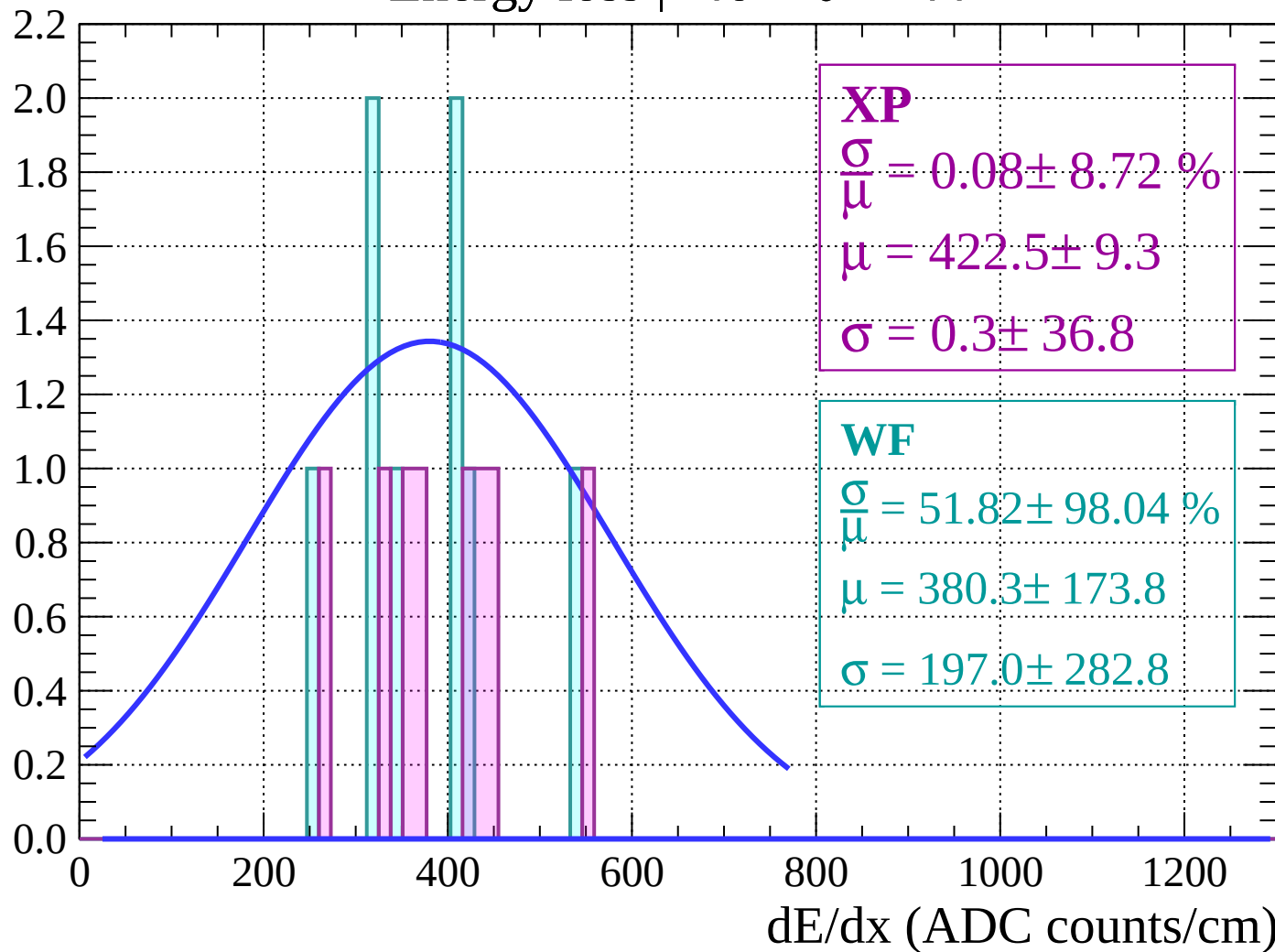
Count



dE/dx (ADC counts/cm)

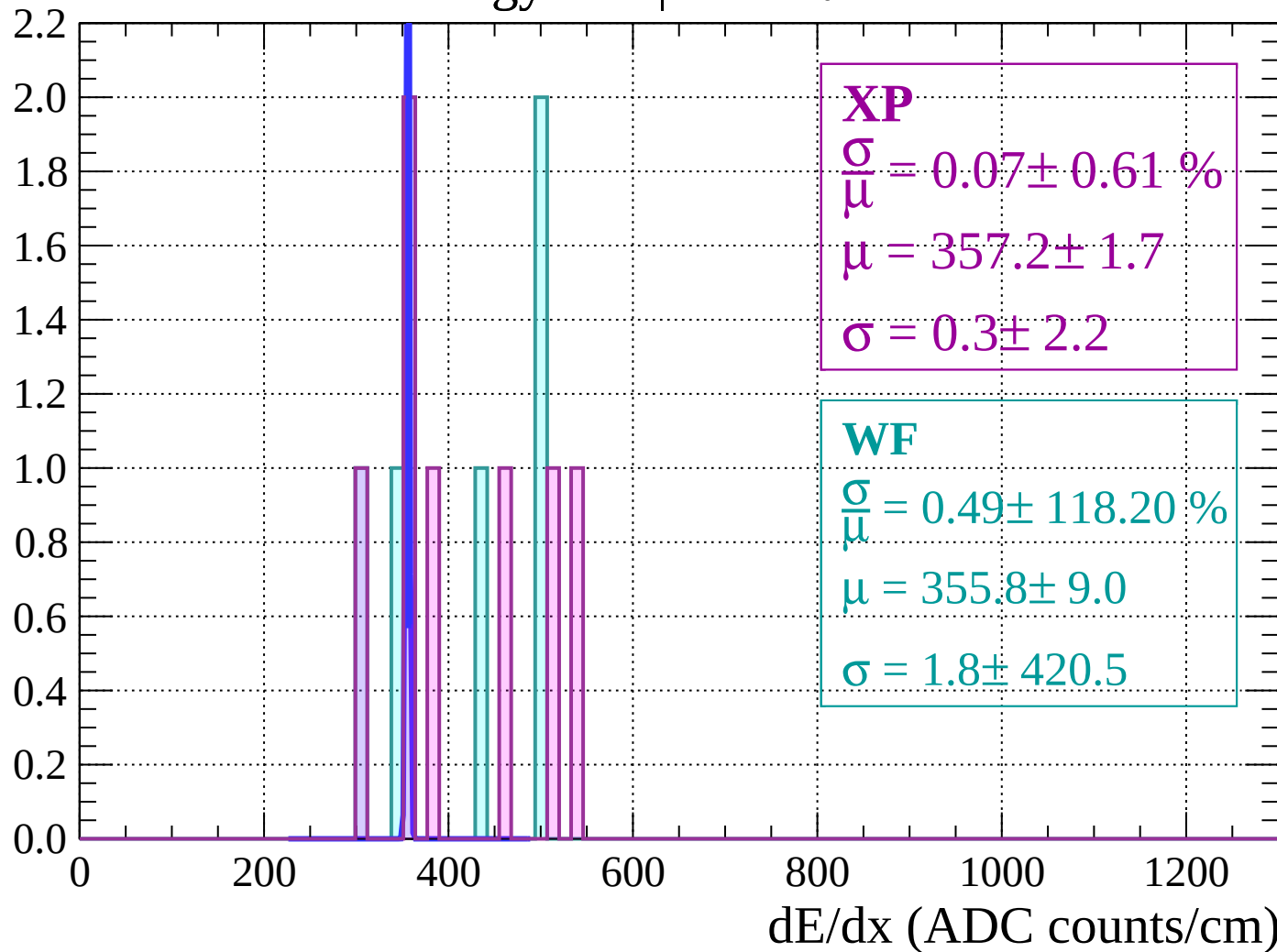
Energy loss | $-46 < \theta < -44$

Count



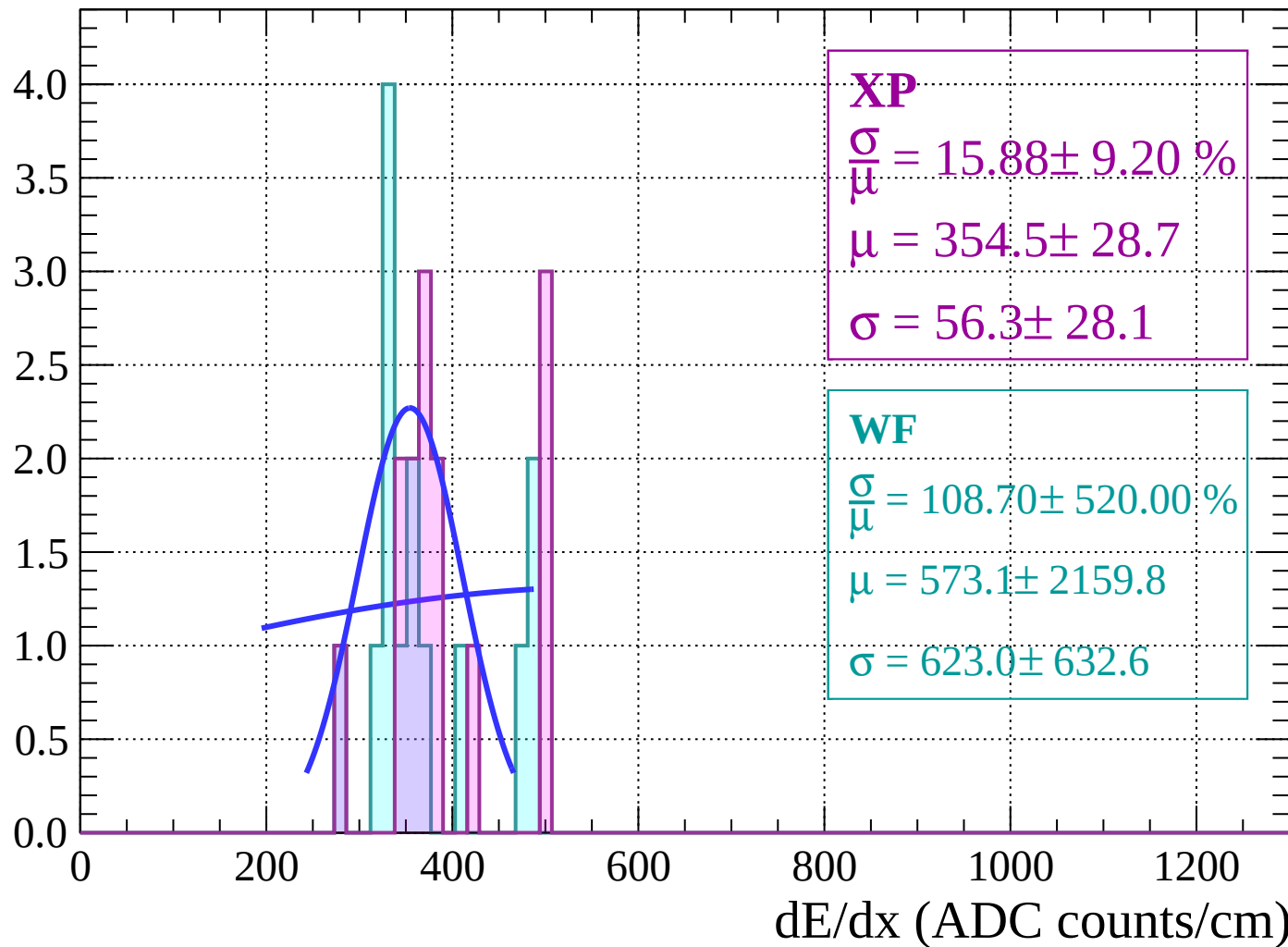
Energy loss | $-44 < \theta < -42$

Count



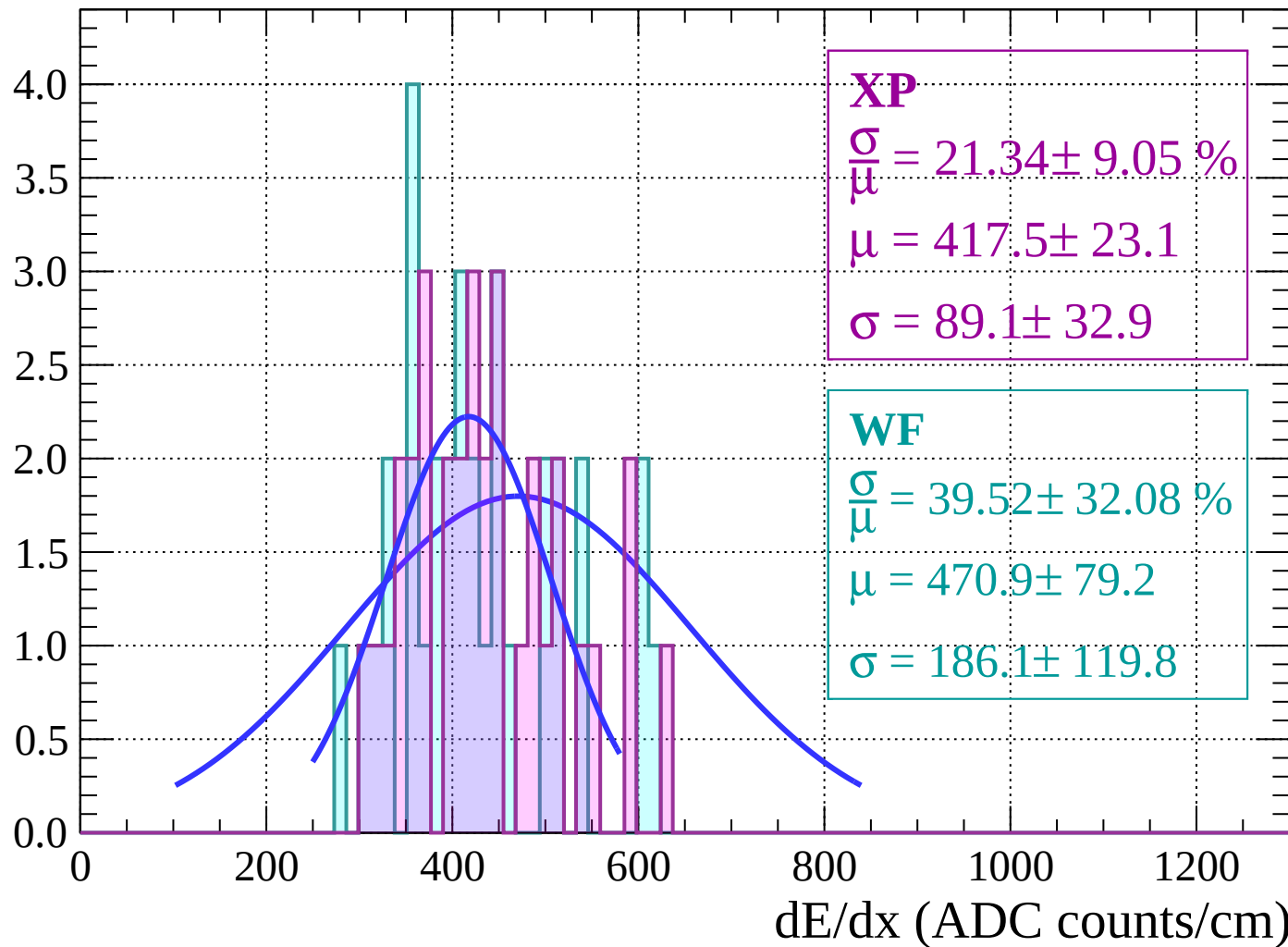
Energy loss | $-42 < \theta < -40$

Count



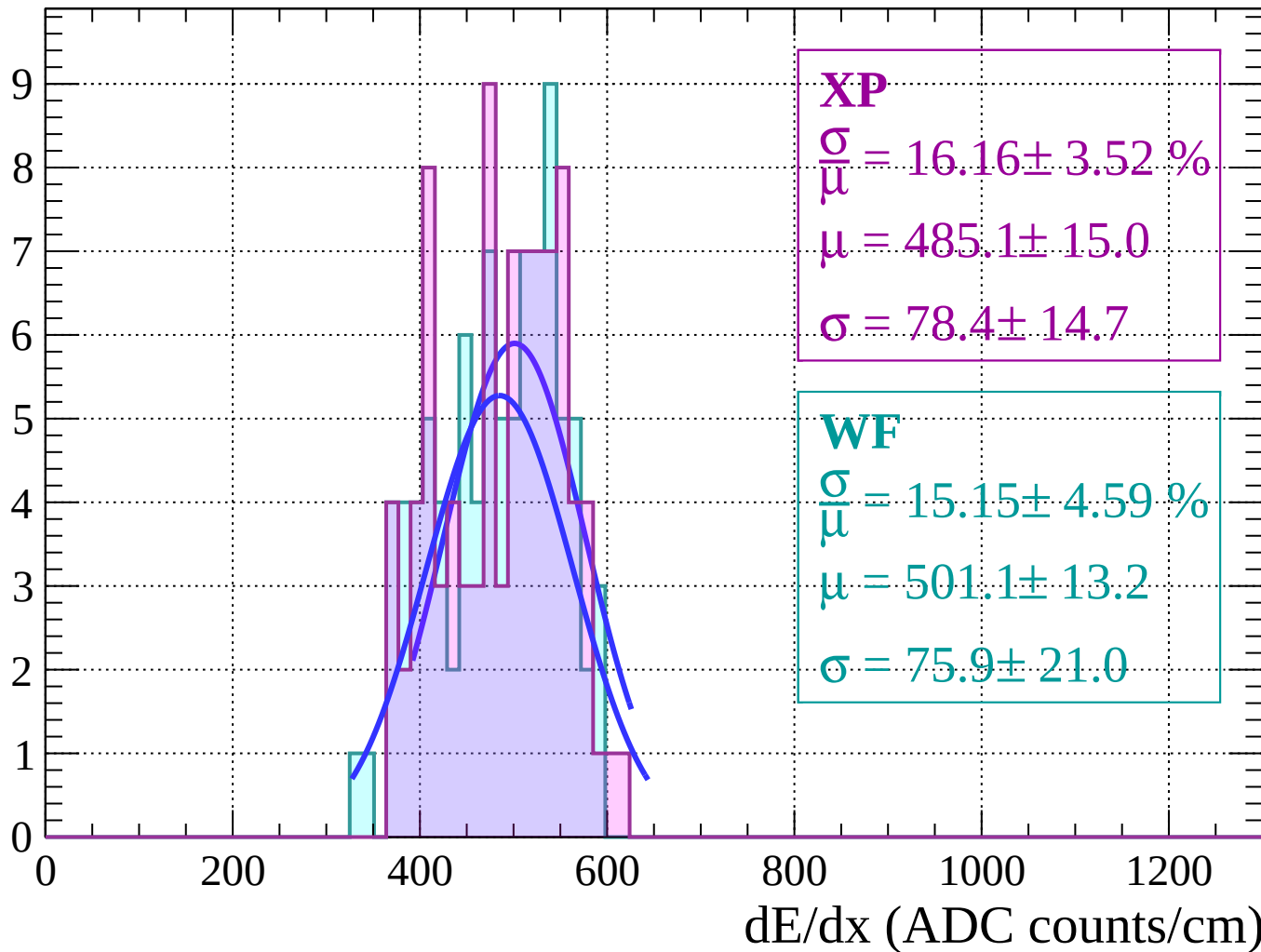
Energy loss | $-40 < \theta < -38$

Count



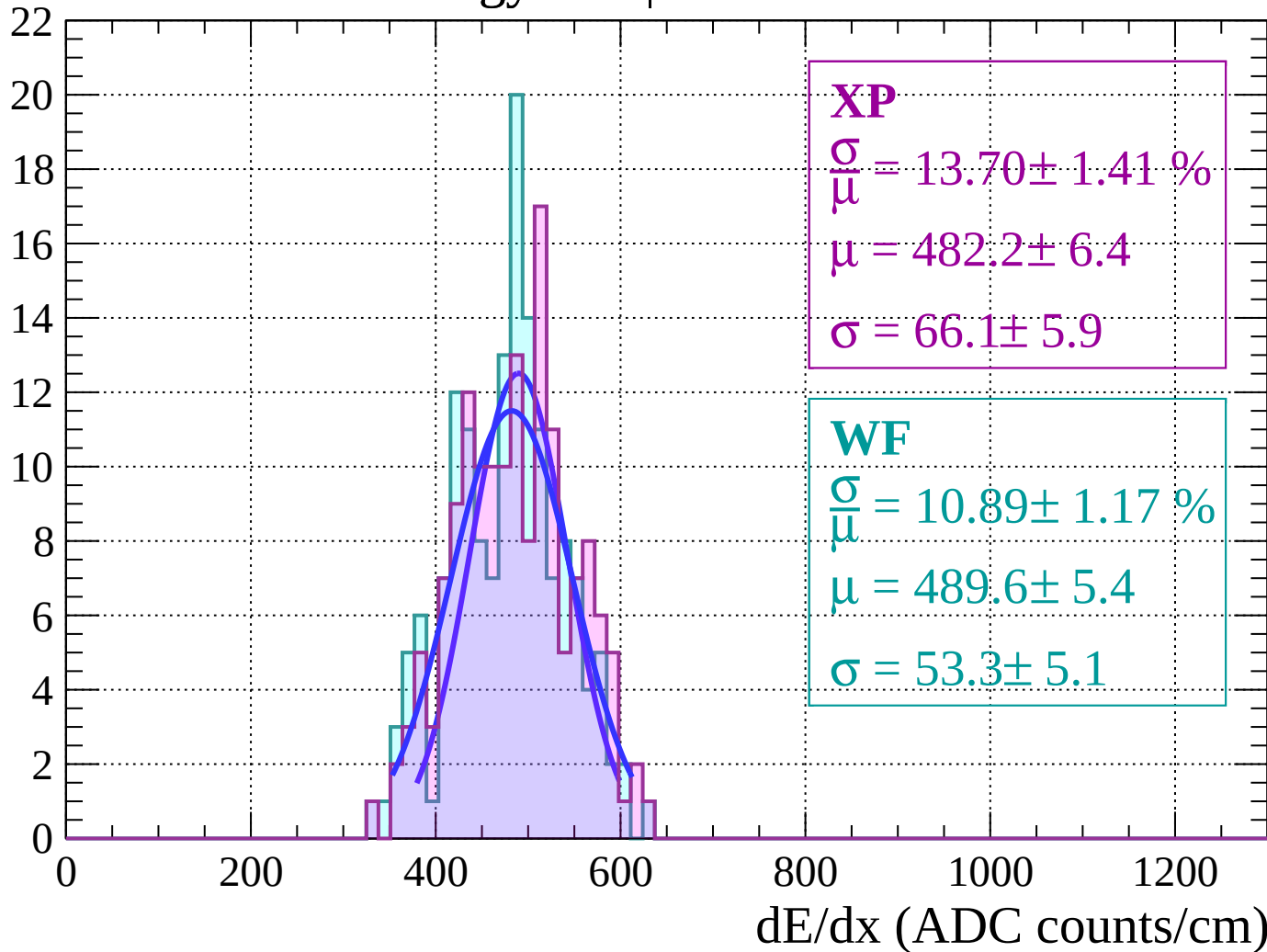
Energy loss | $-38 < \theta < -36$

Count



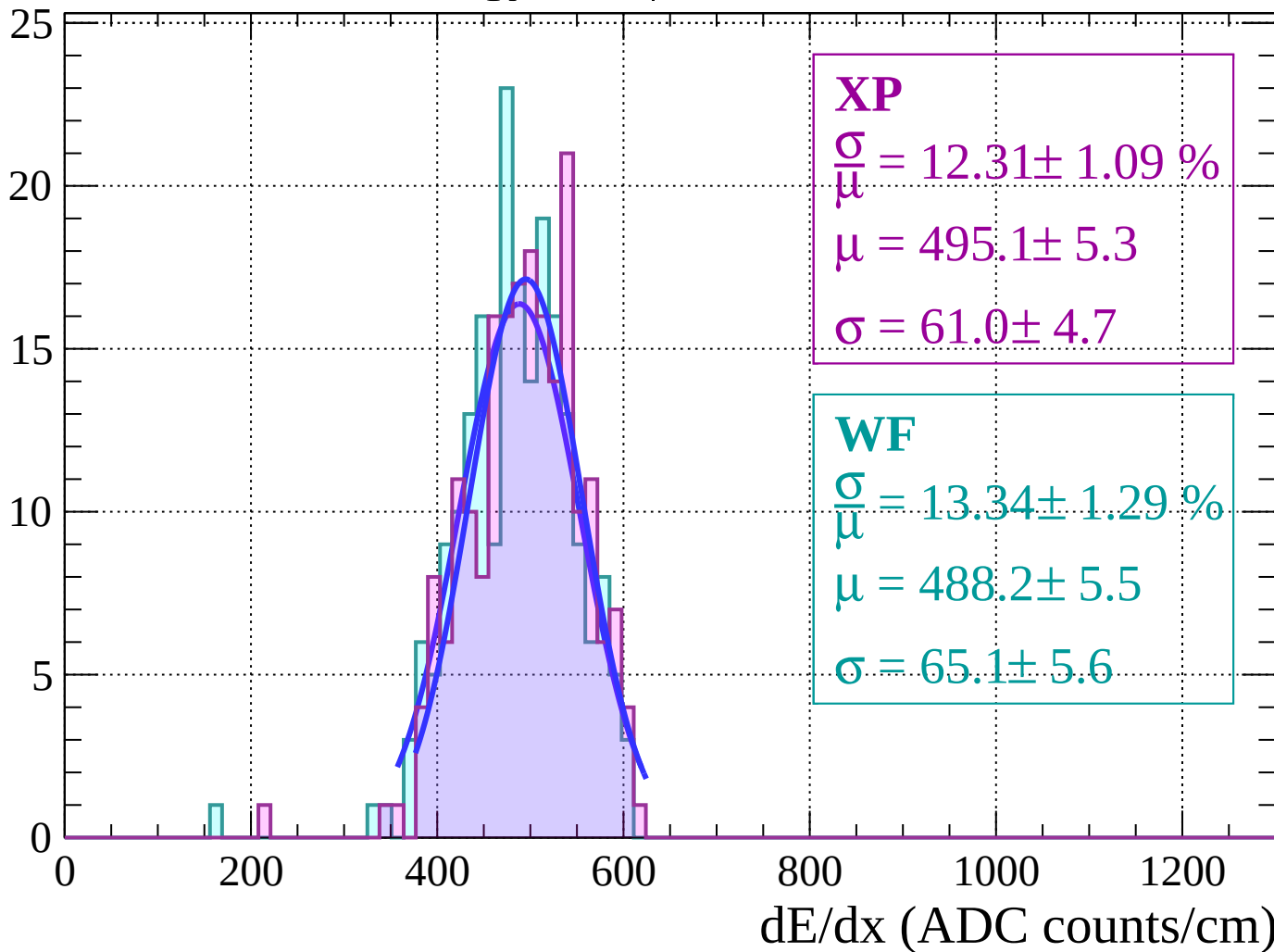
Energy loss | $-36 < \theta < -34$

Count



Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count

XP

$$\frac{\sigma}{\mu} = 11.84 \pm 0.81 \%$$

$$\mu = 495.8 \pm 3.9$$

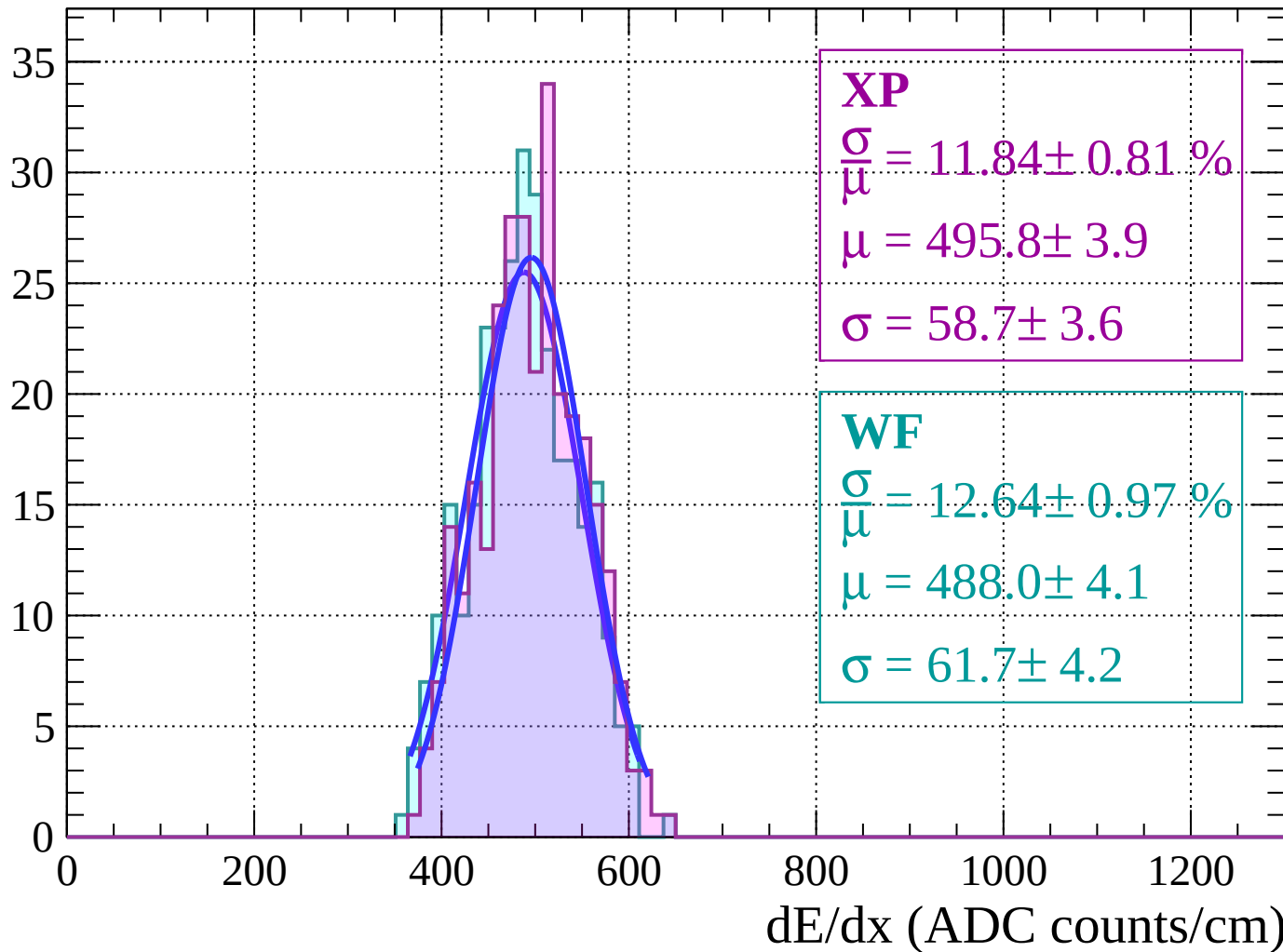
$$\sigma = 58.7 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 12.64 \pm 0.97 \%$$

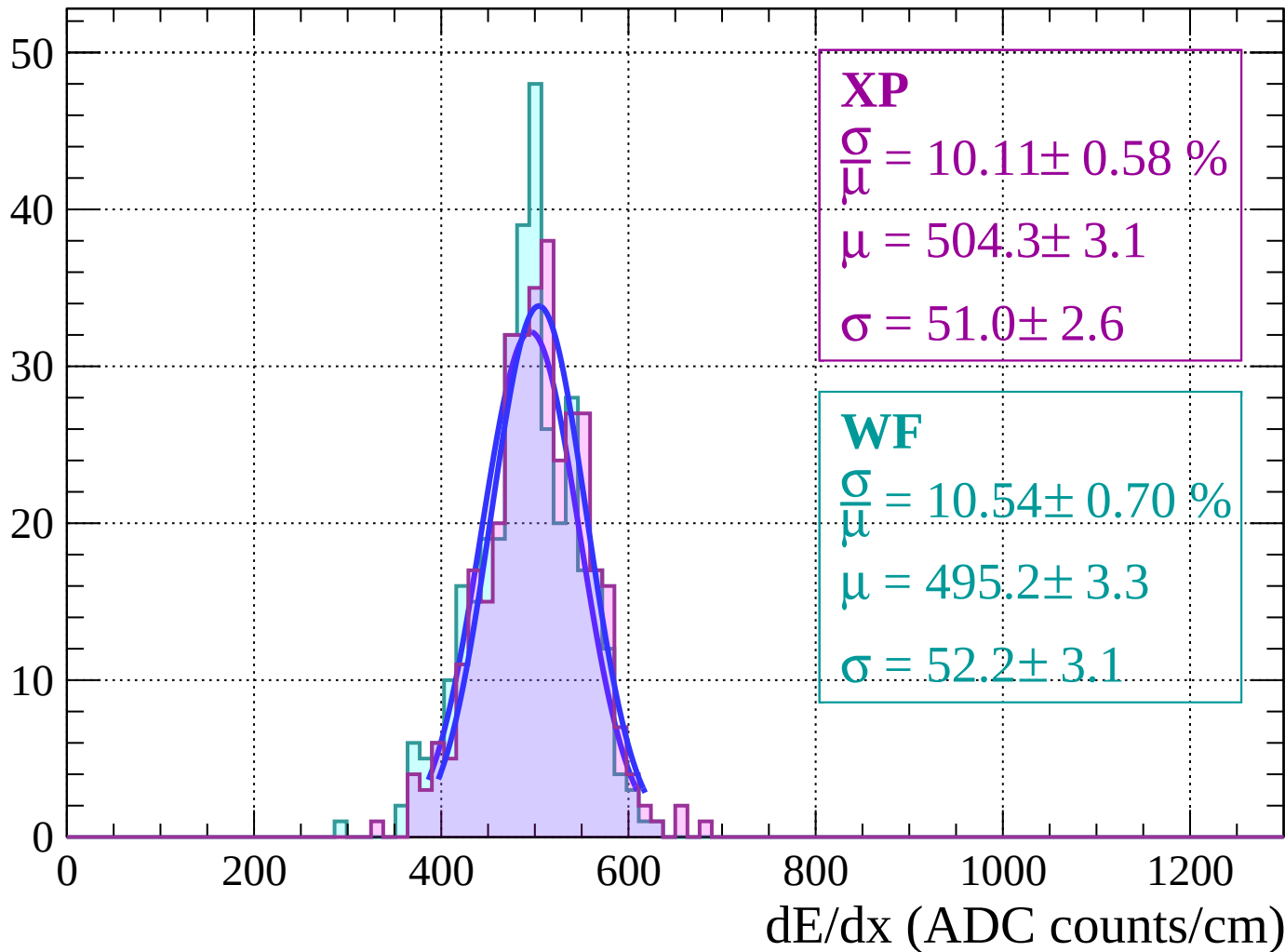
$$\mu = 488.0 \pm 4.1$$

$$\sigma = 61.7 \pm 4.2$$



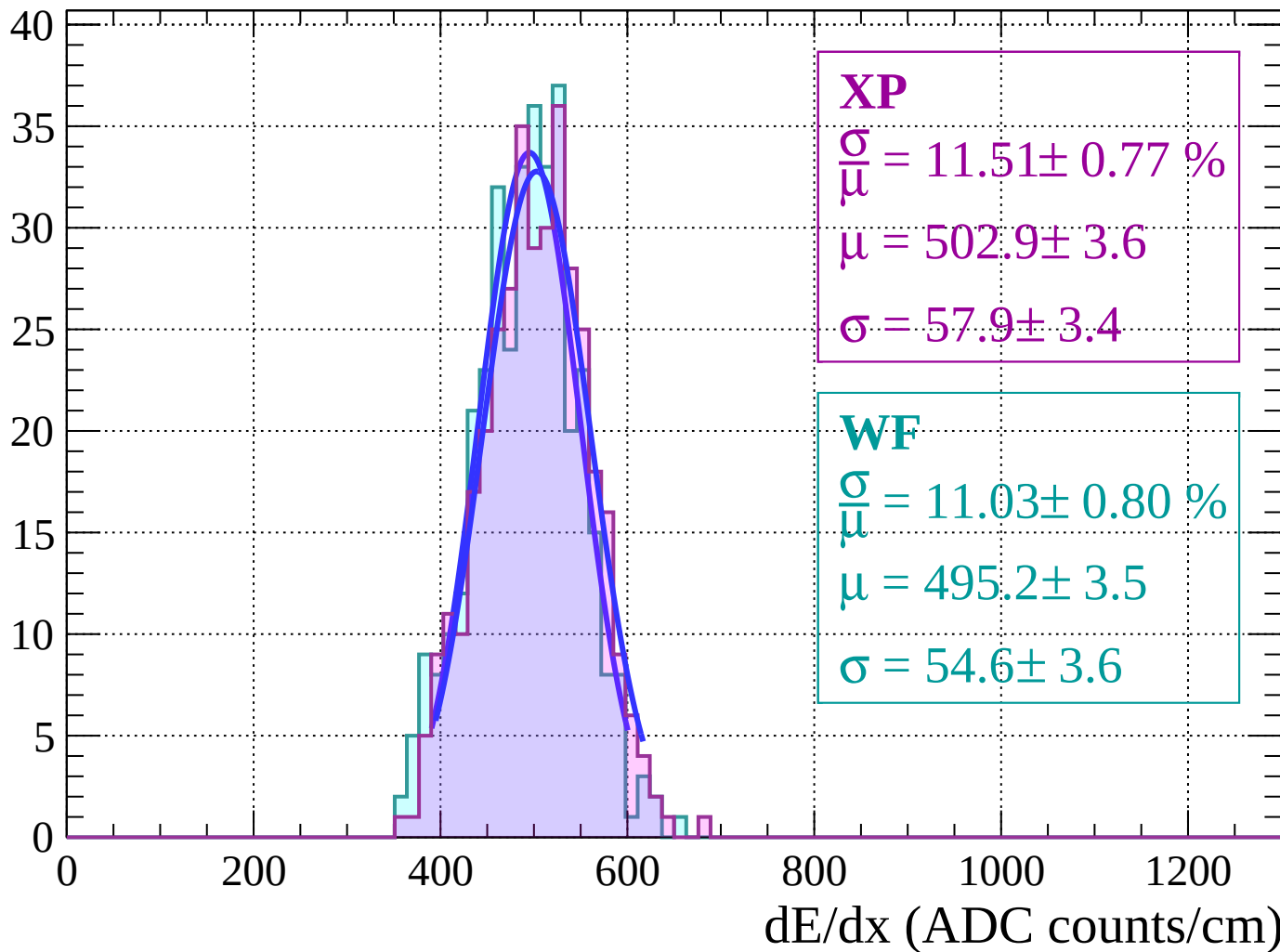
Energy loss | $-30 < \theta < -28$

Count



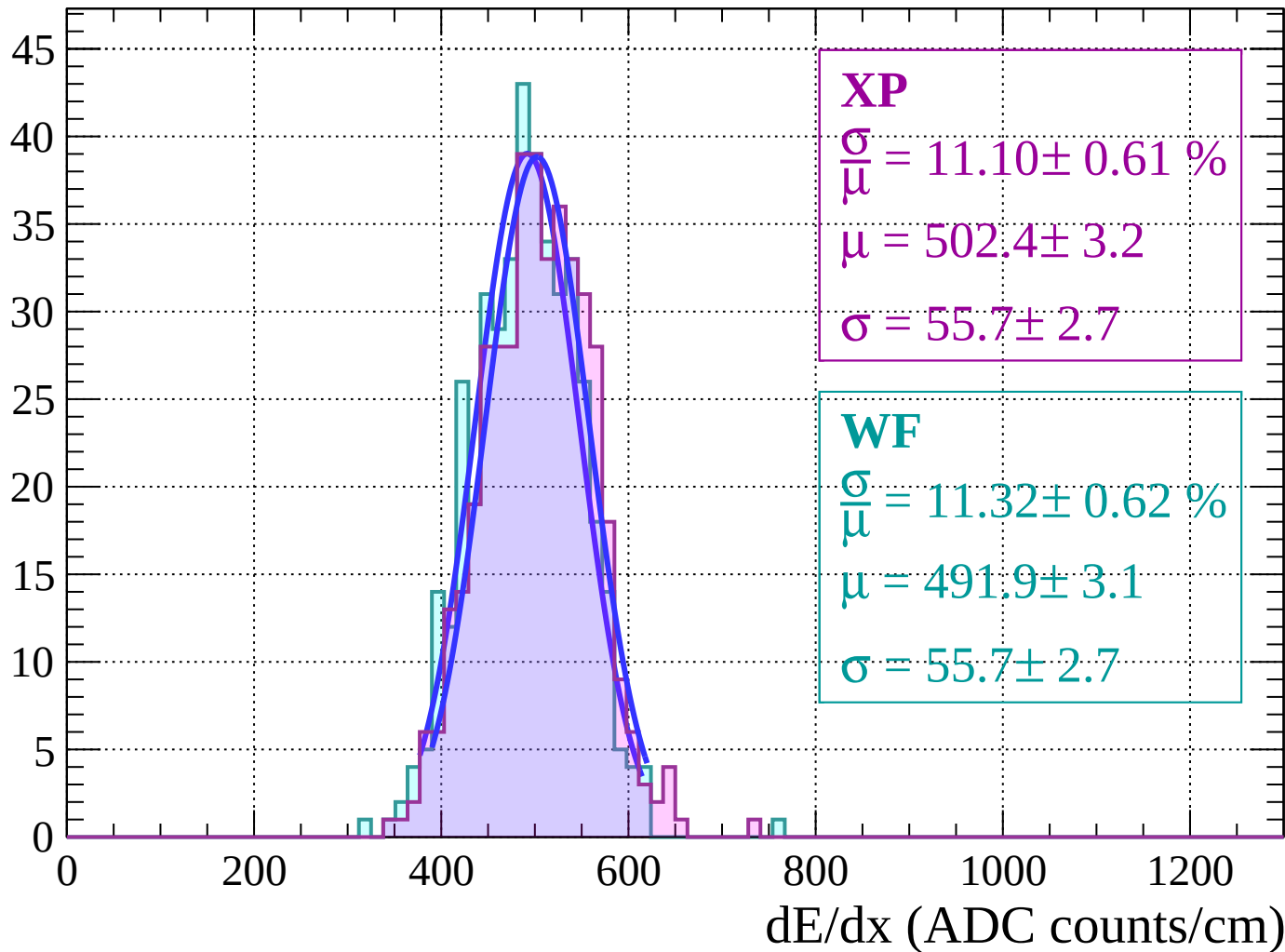
Energy loss | $-28 < \theta < -26$

Count



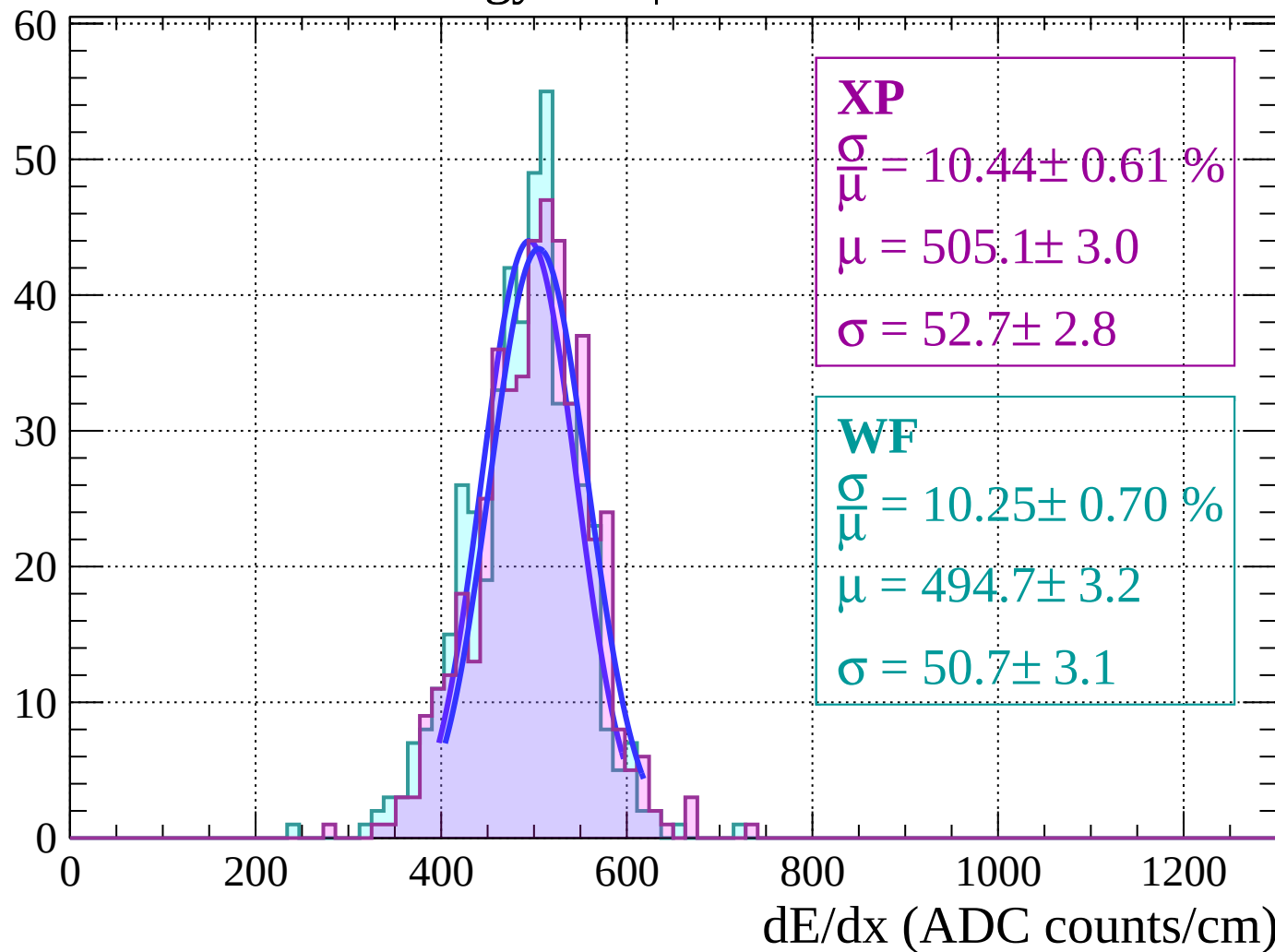
Energy loss $|-26 < \theta < -24$

Count



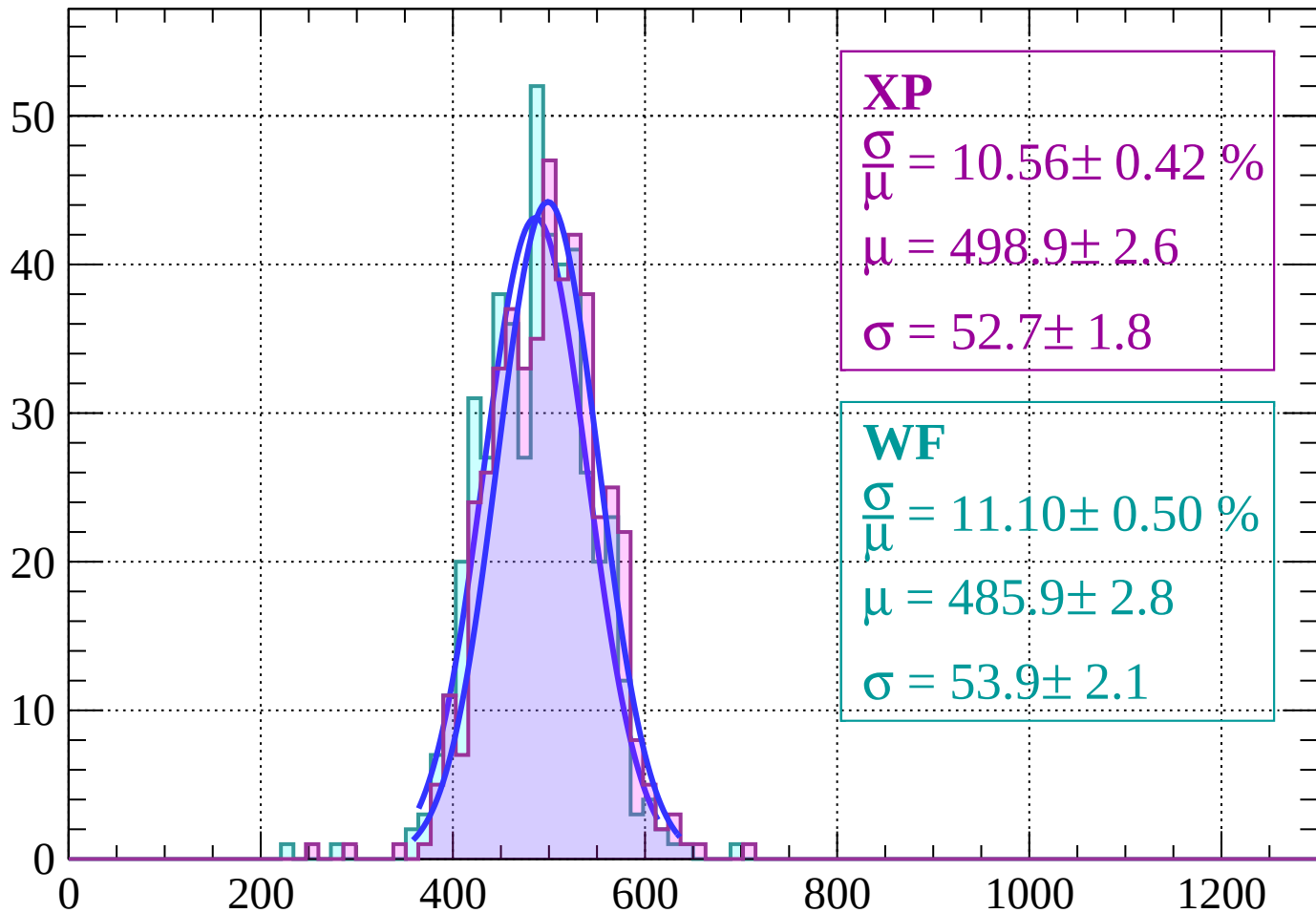
Energy loss $|-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$

Count



XP

$$\frac{\sigma}{\mu} = 10.56 \pm 0.42 \%$$

$$\mu = 498.9 \pm 2.6$$

$$\sigma = 52.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 11.10 \pm 0.50 \%$$

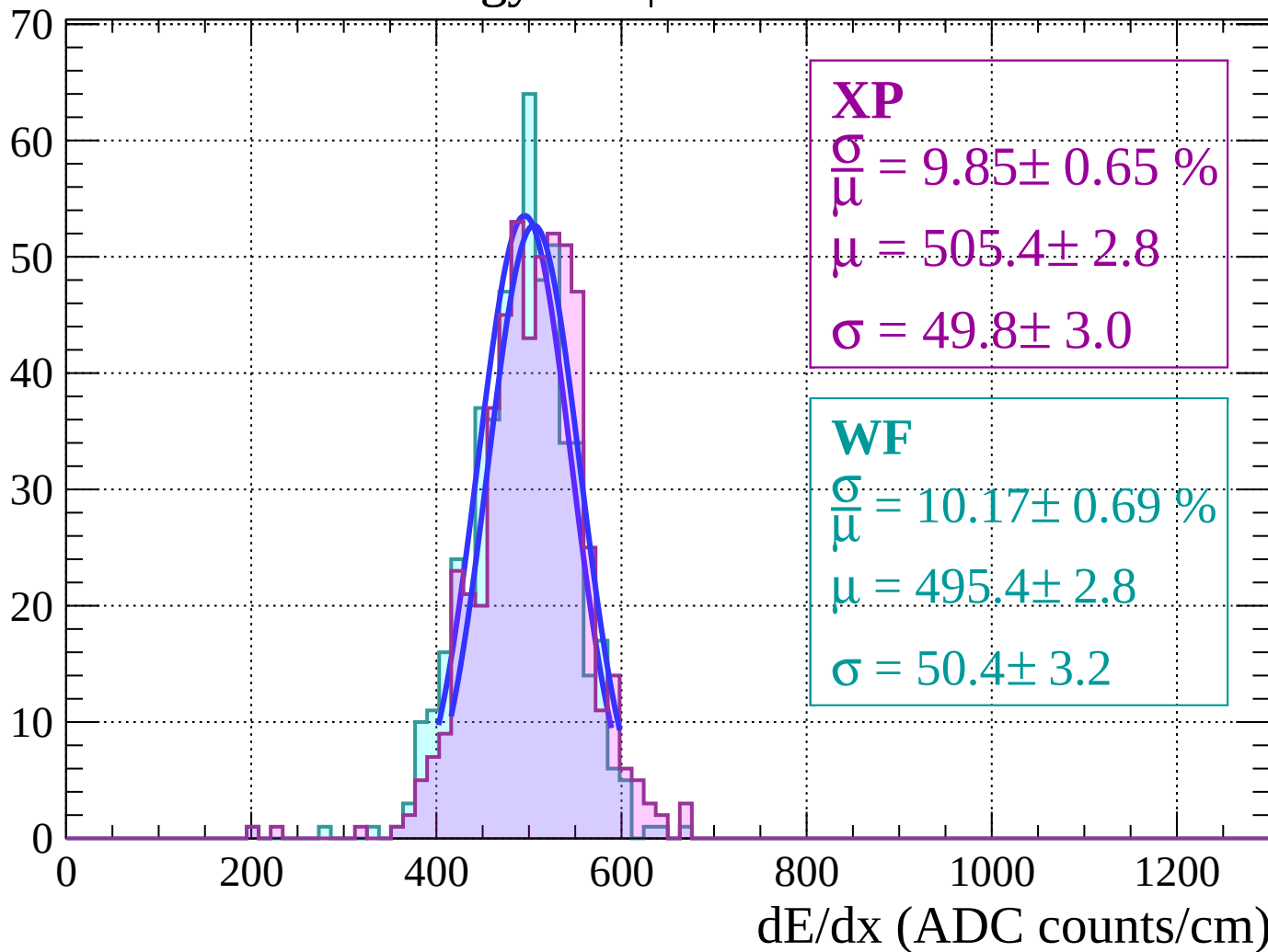
$$\mu = 485.9 \pm 2.8$$

$$\sigma = 53.9 \pm 2.1$$

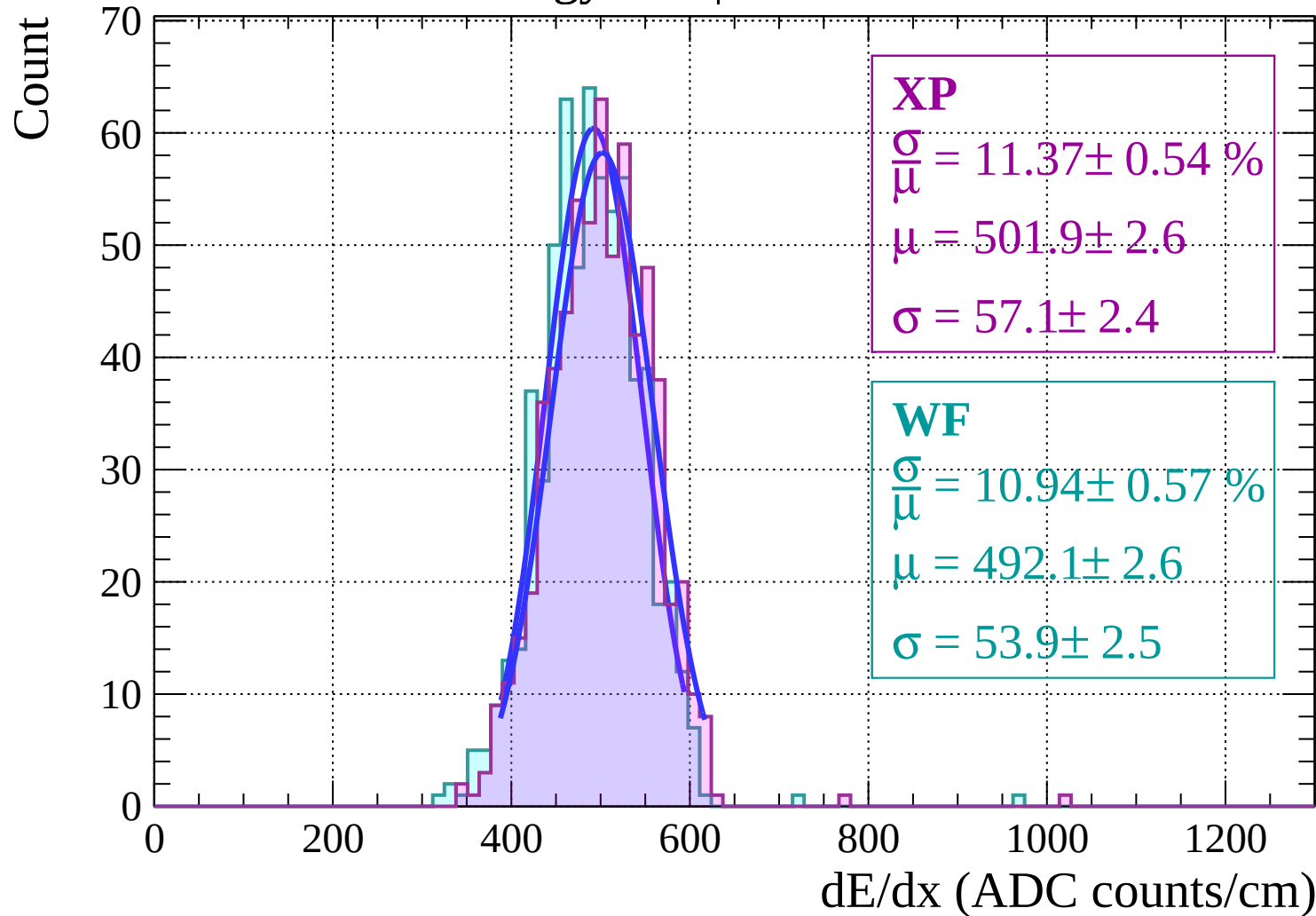
dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

Count



Energy loss | $-18 < \theta < -16$



Energy loss $|-16 < \theta < -14$

Count

XP

$$\frac{\sigma}{\mu} = 10.56 \pm 0.47 \%$$

$$\mu = 504.0 \pm 2.5$$

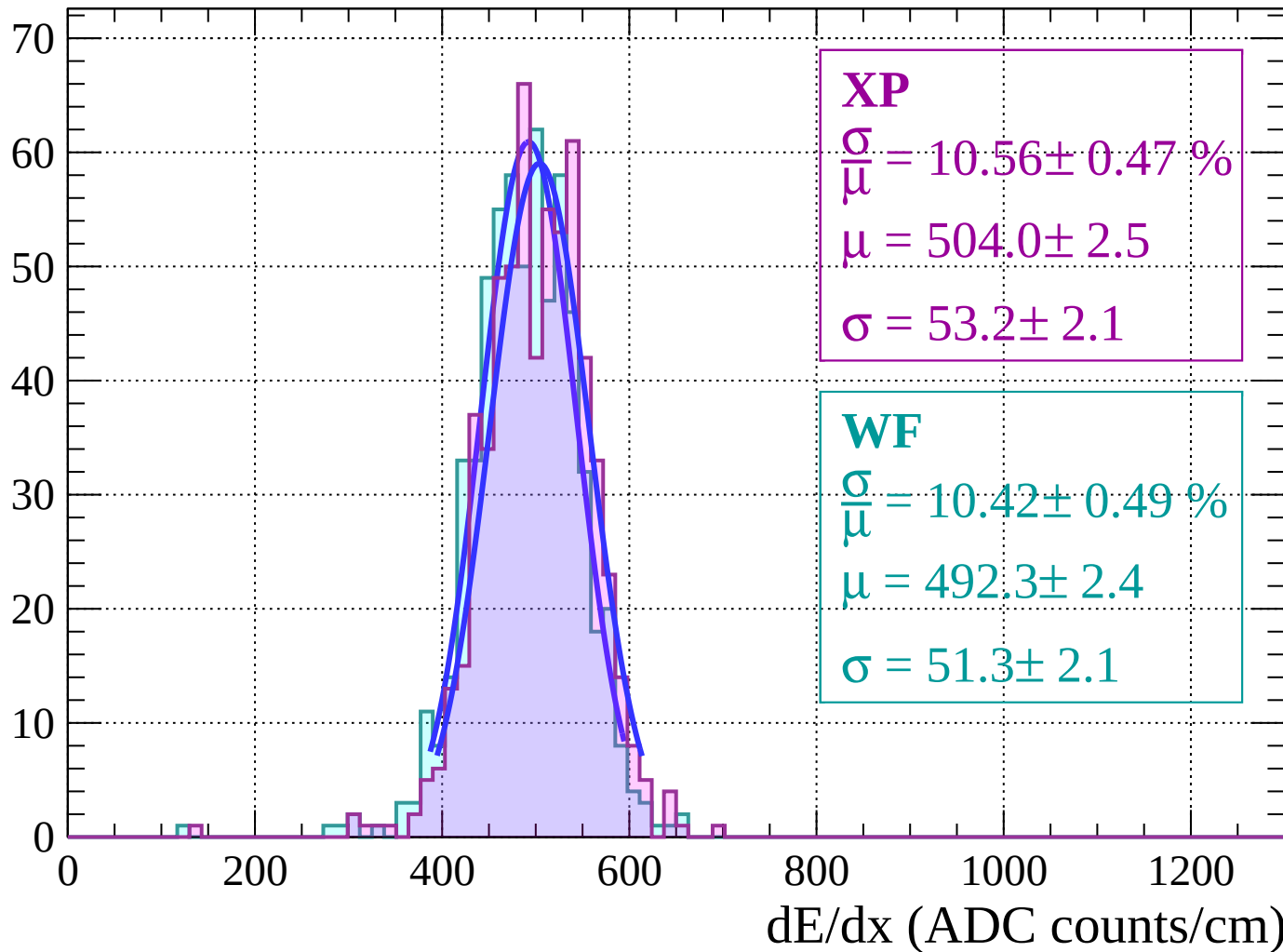
$$\sigma = 53.2 \pm 2.1$$

WF

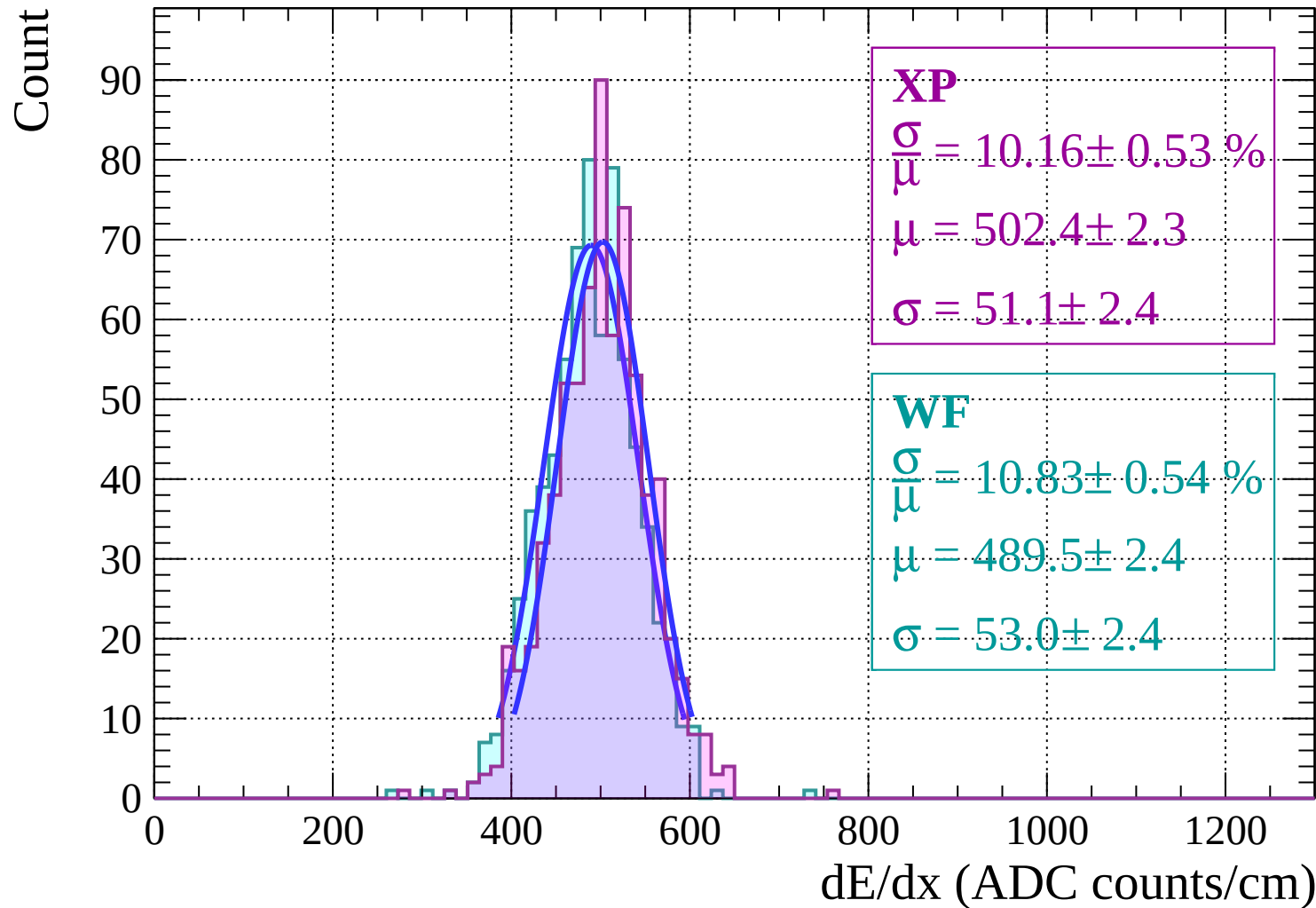
$$\frac{\sigma}{\mu} = 10.42 \pm 0.49 \%$$

$$\mu = 492.3 \pm 2.4$$

$$\sigma = 51.3 \pm 2.1$$

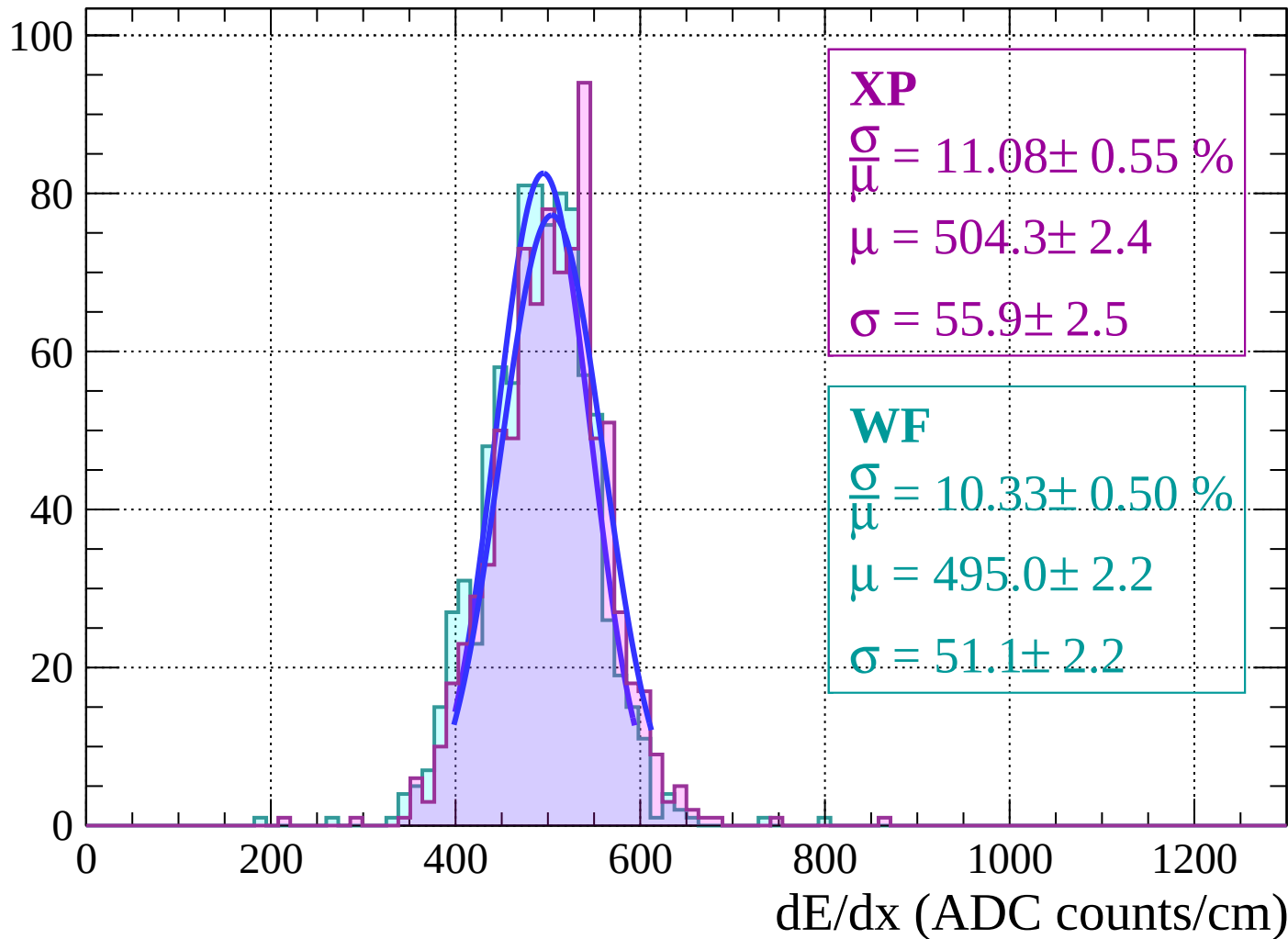


Energy loss | $-14 < \theta < -12$

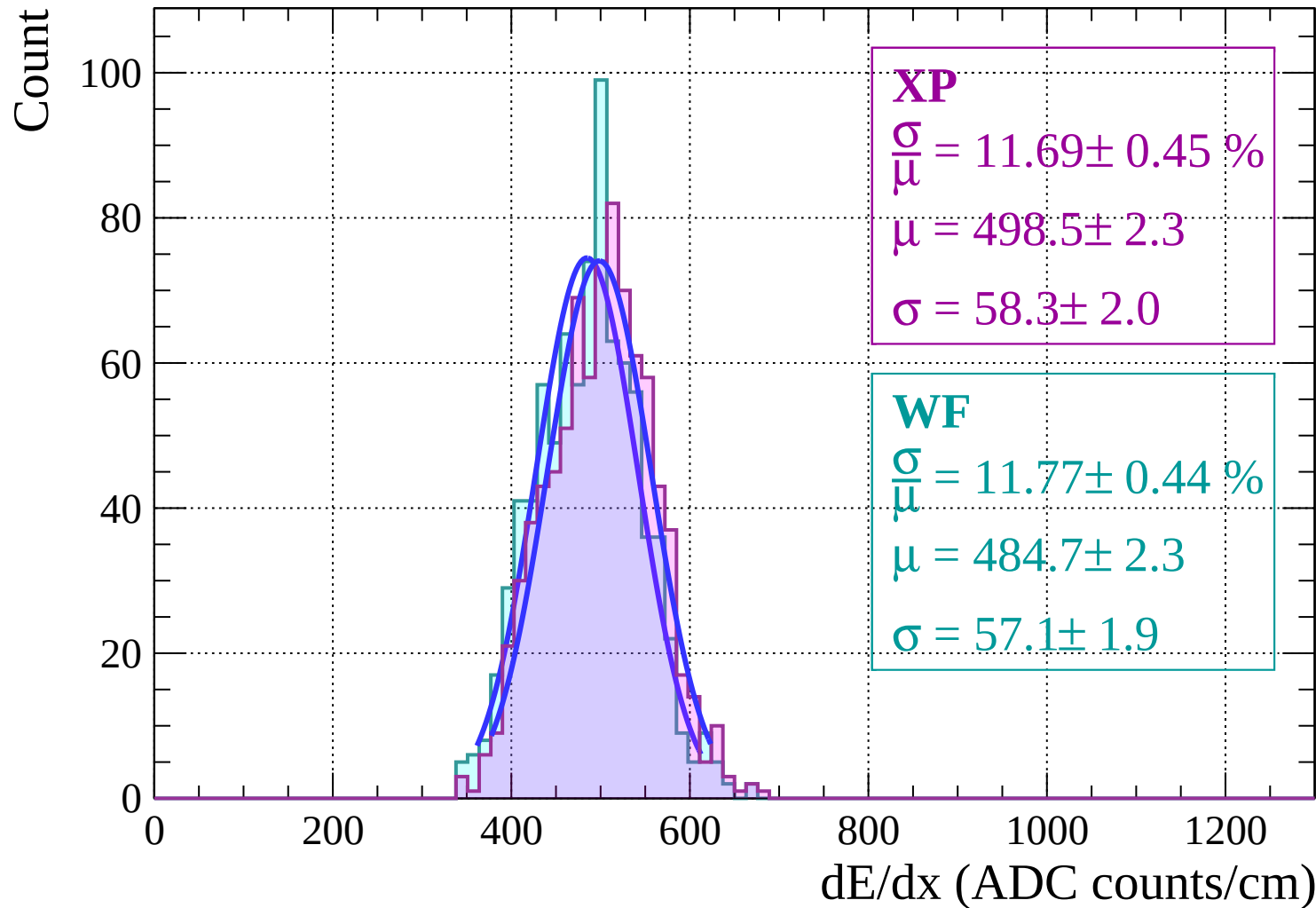


Energy loss | $-12 < \theta < -10$

Count

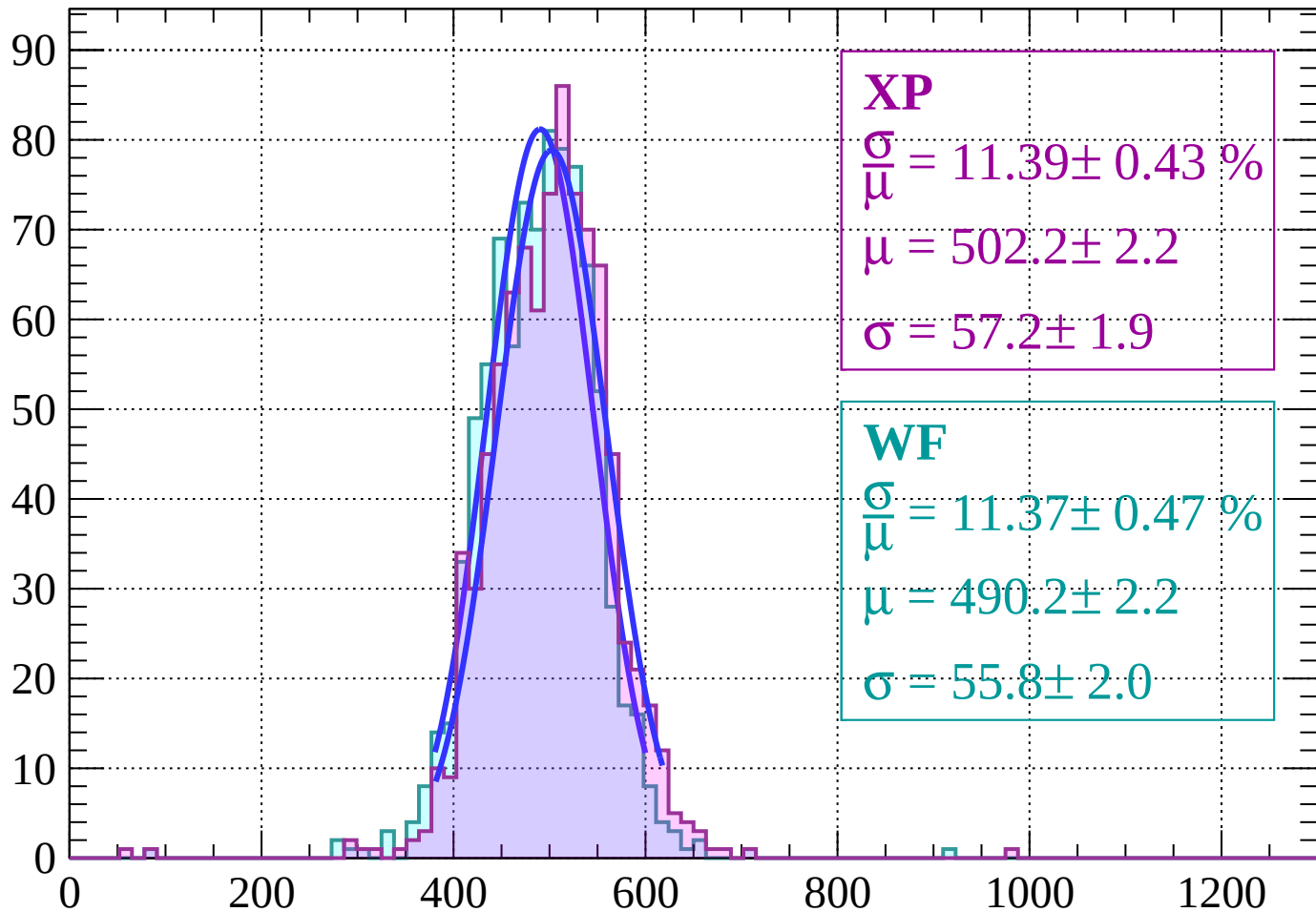


Energy loss | $-10 < \theta < -8$



Energy loss $|-8 < \theta < -6$

Count



XP

$$\frac{\sigma}{\mu} = 11.39 \pm 0.43 \%$$

$$\mu = 502.2 \pm 2.2$$

$$\sigma = 57.2 \pm 1.9$$

WF

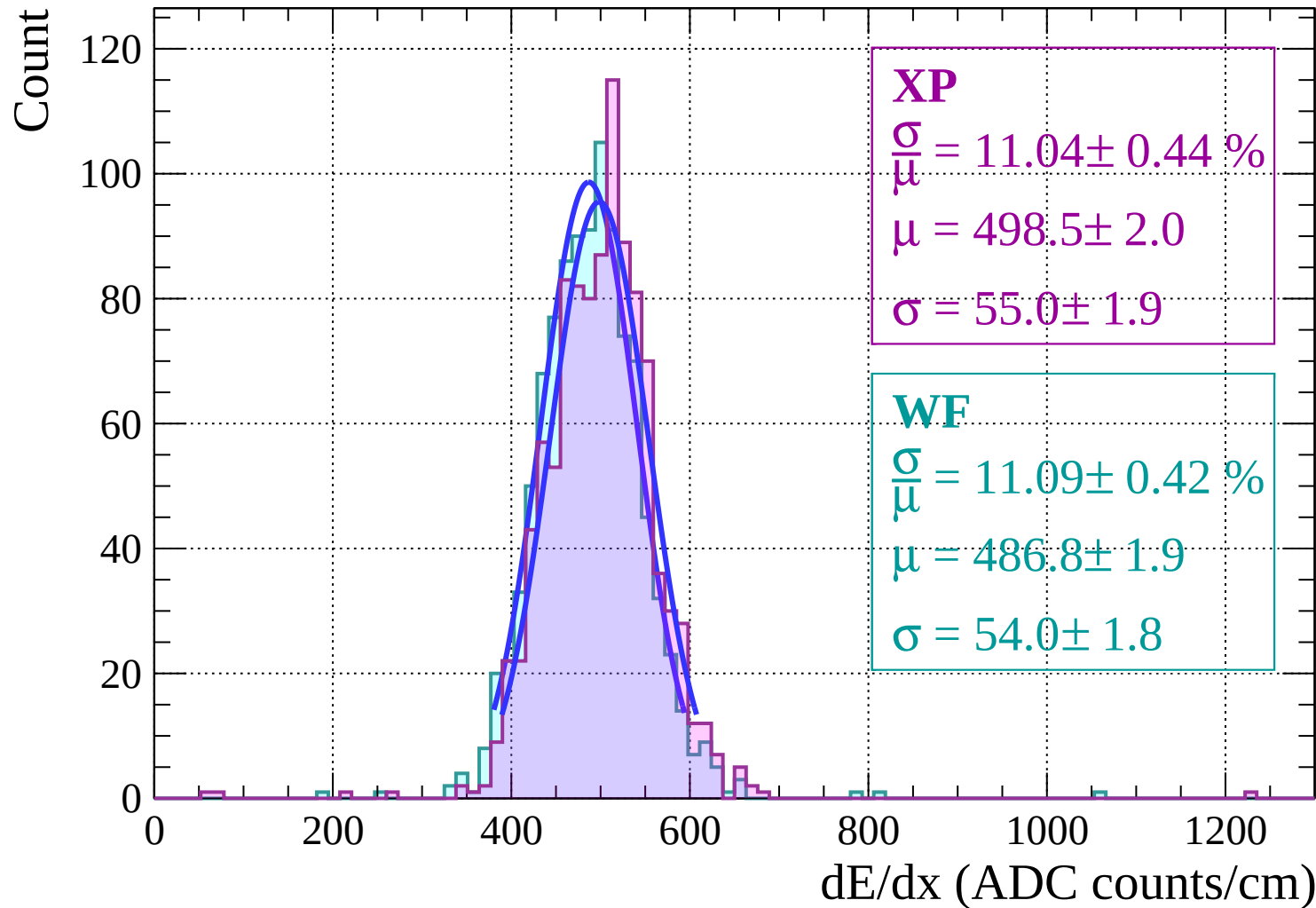
$$\frac{\sigma}{\mu} = 11.37 \pm 0.47 \%$$

$$\mu = 490.2 \pm 2.2$$

$$\sigma = 55.8 \pm 2.0$$

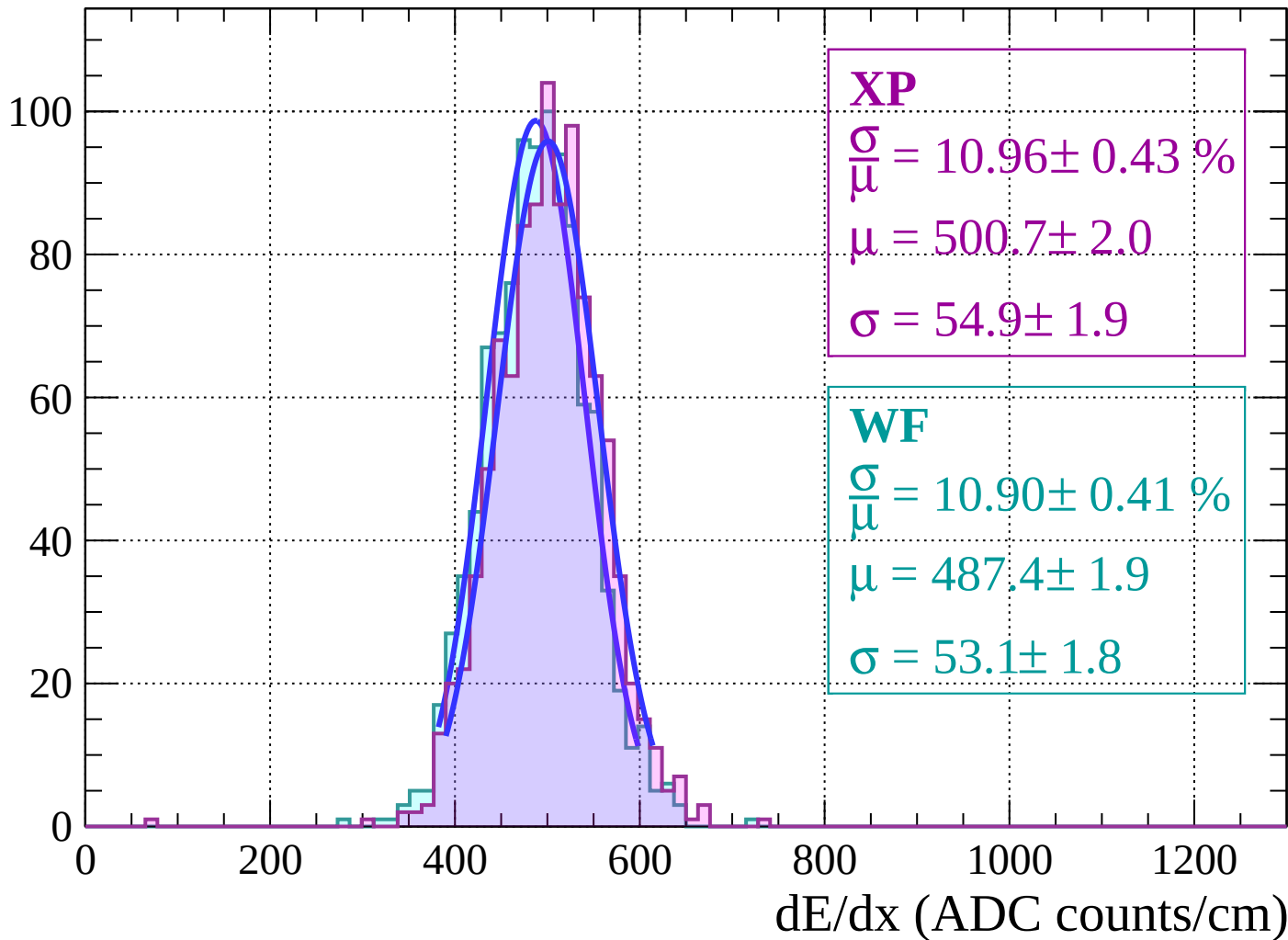
dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < -4$

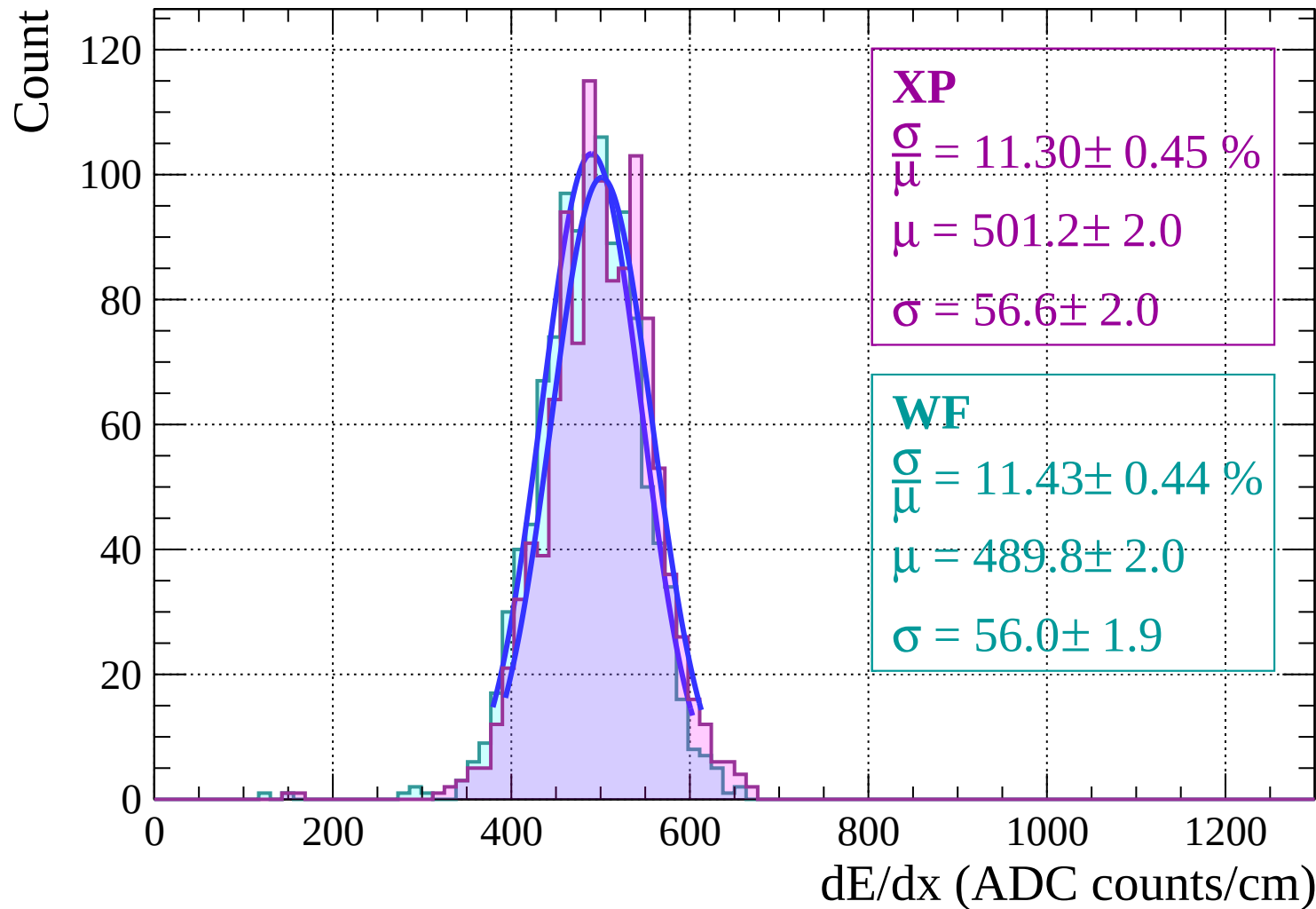


Energy loss | $-4 < \theta < -2$

Count

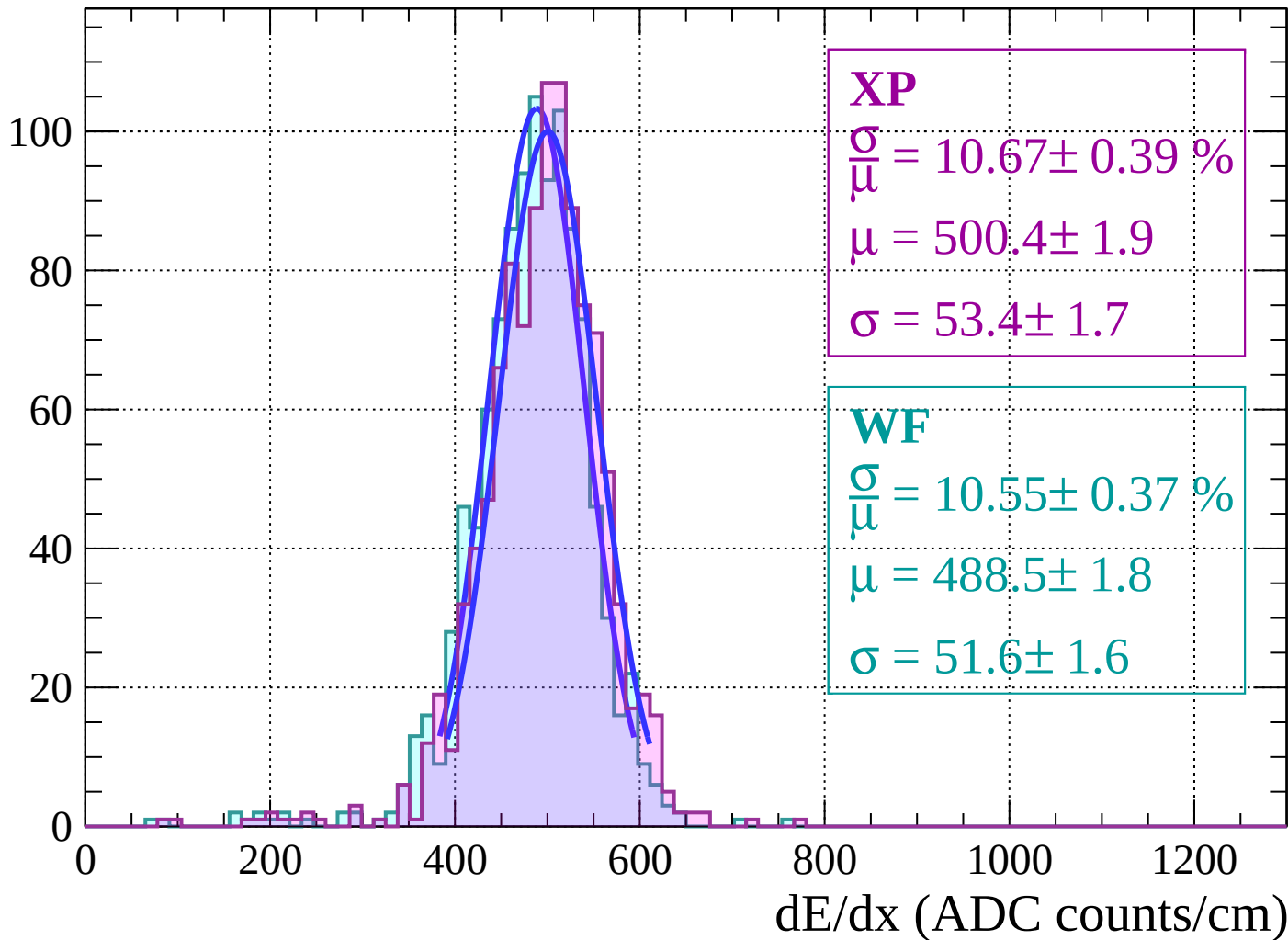


Energy loss | $-2 < \theta < 0$



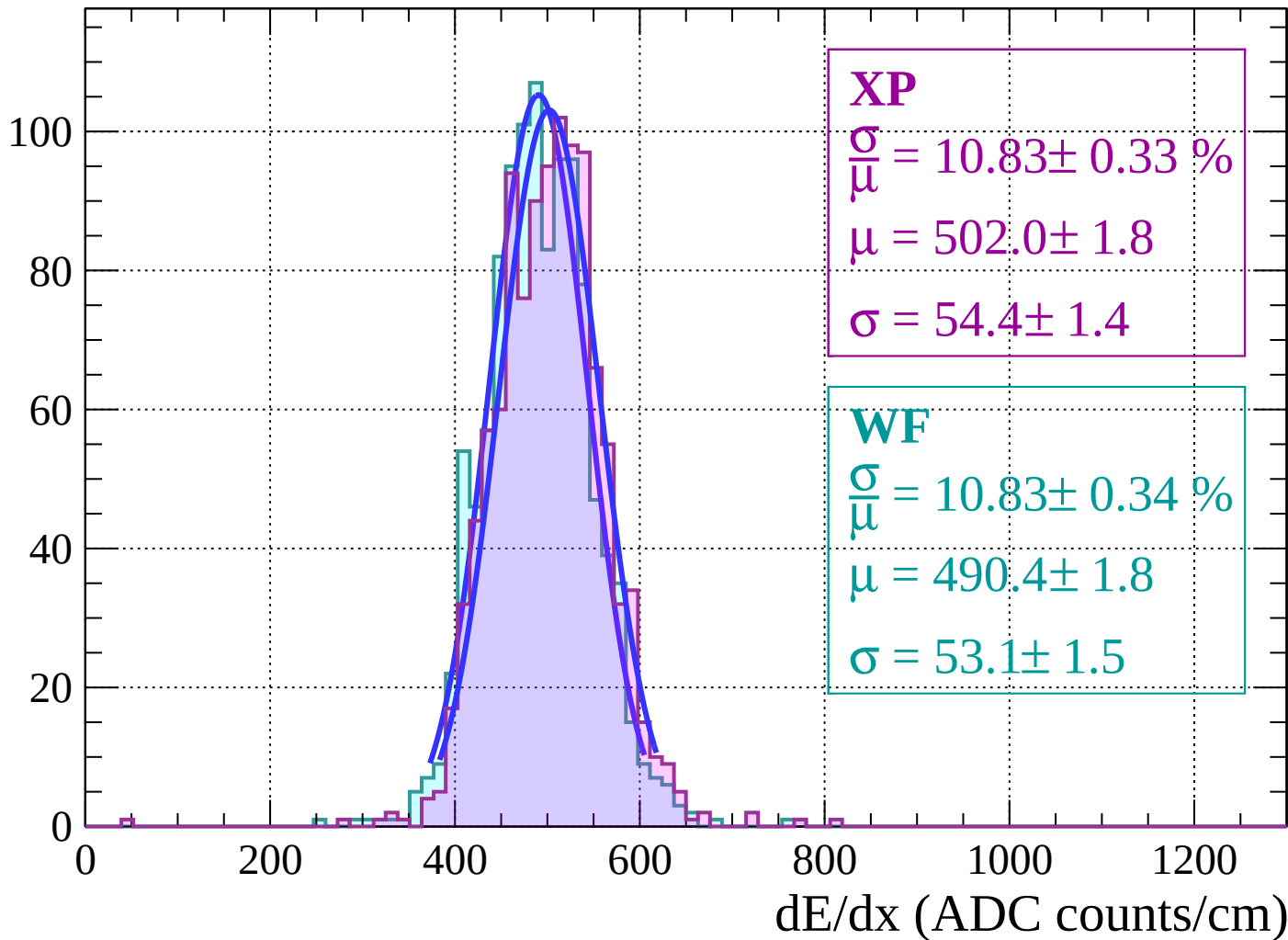
Energy loss | $0 < \theta < 2$

Count



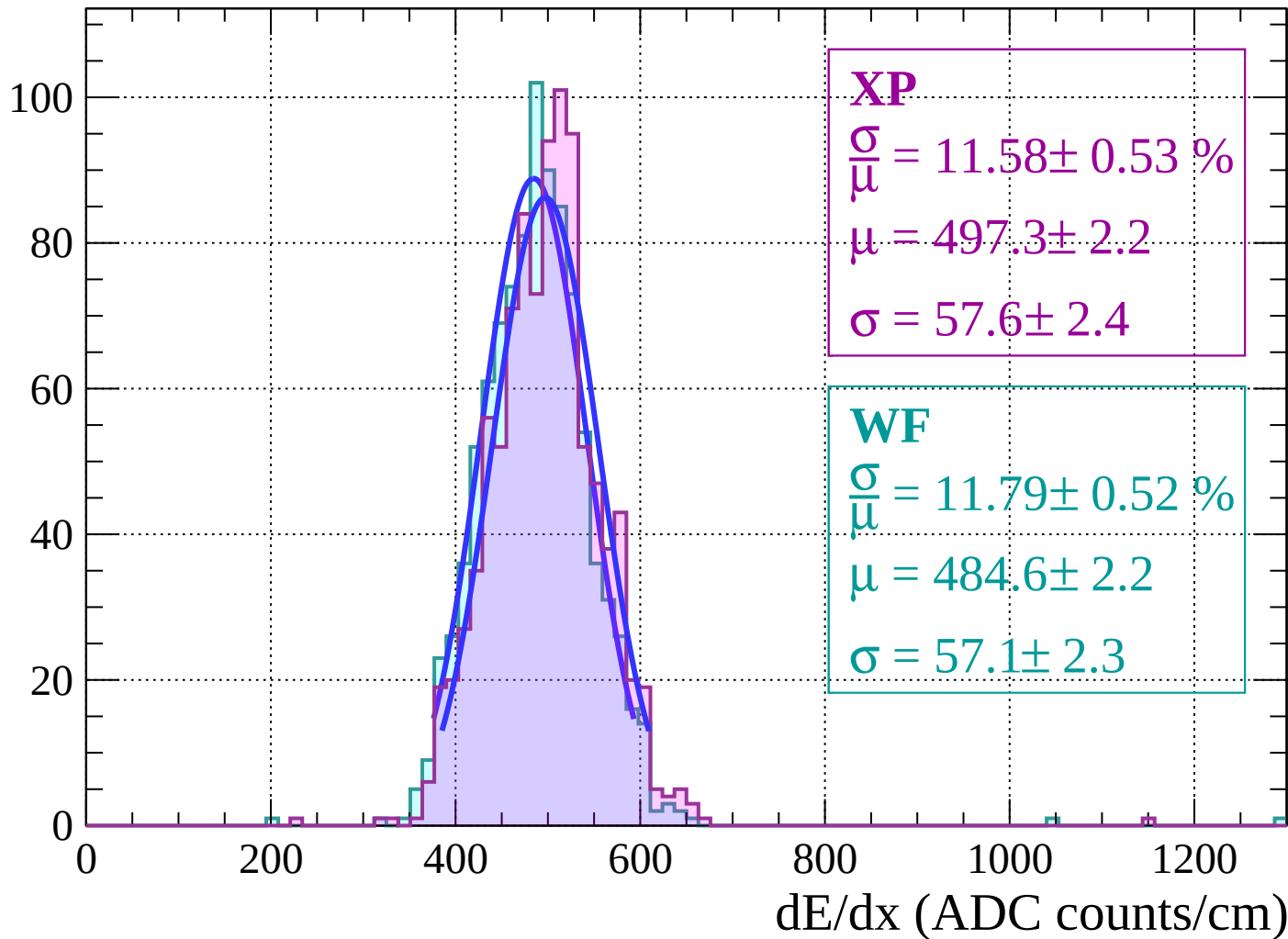
Energy loss | $2 < \theta < 4$

Count



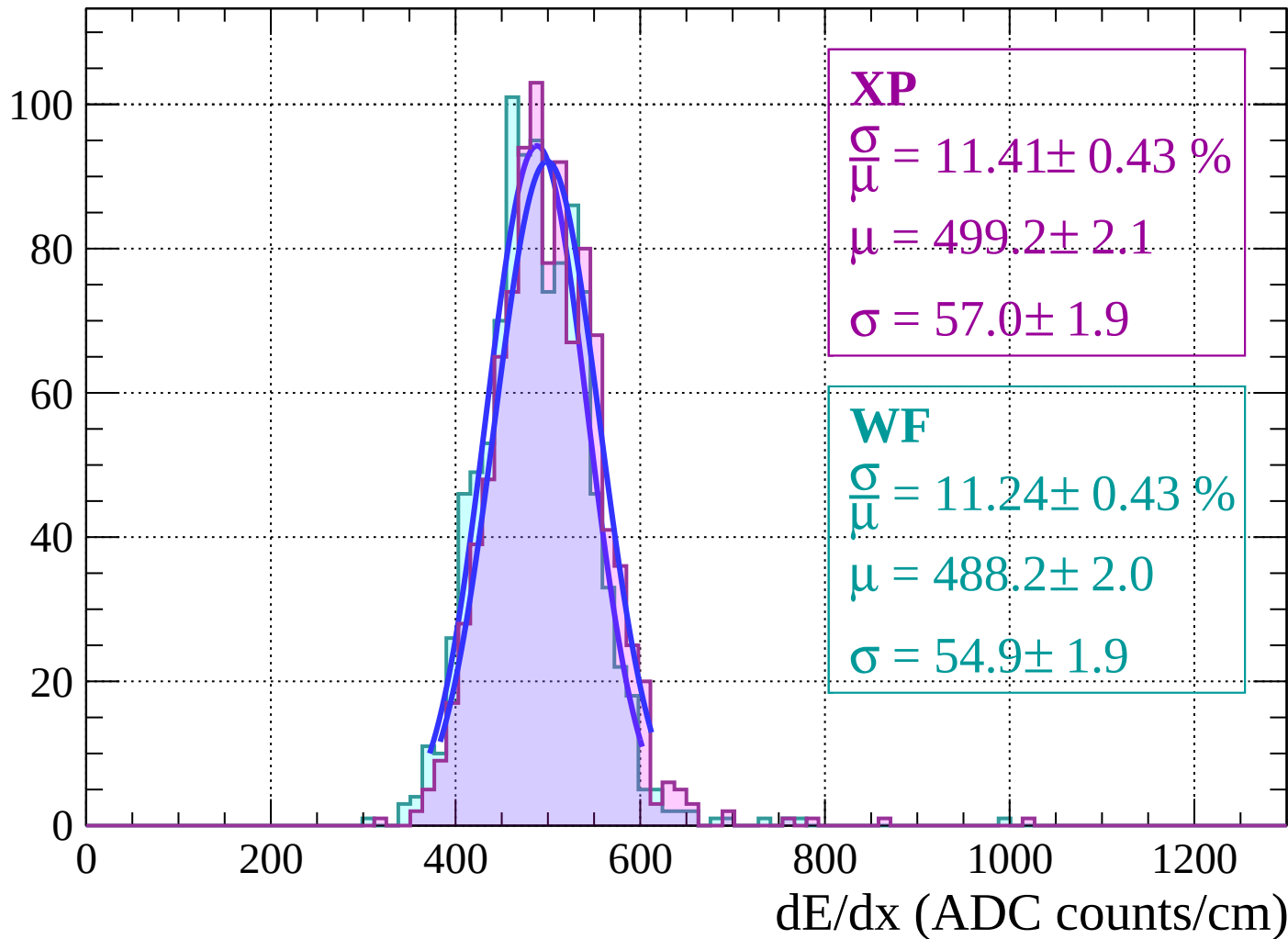
Energy loss | $4 < \theta < 6$

Count



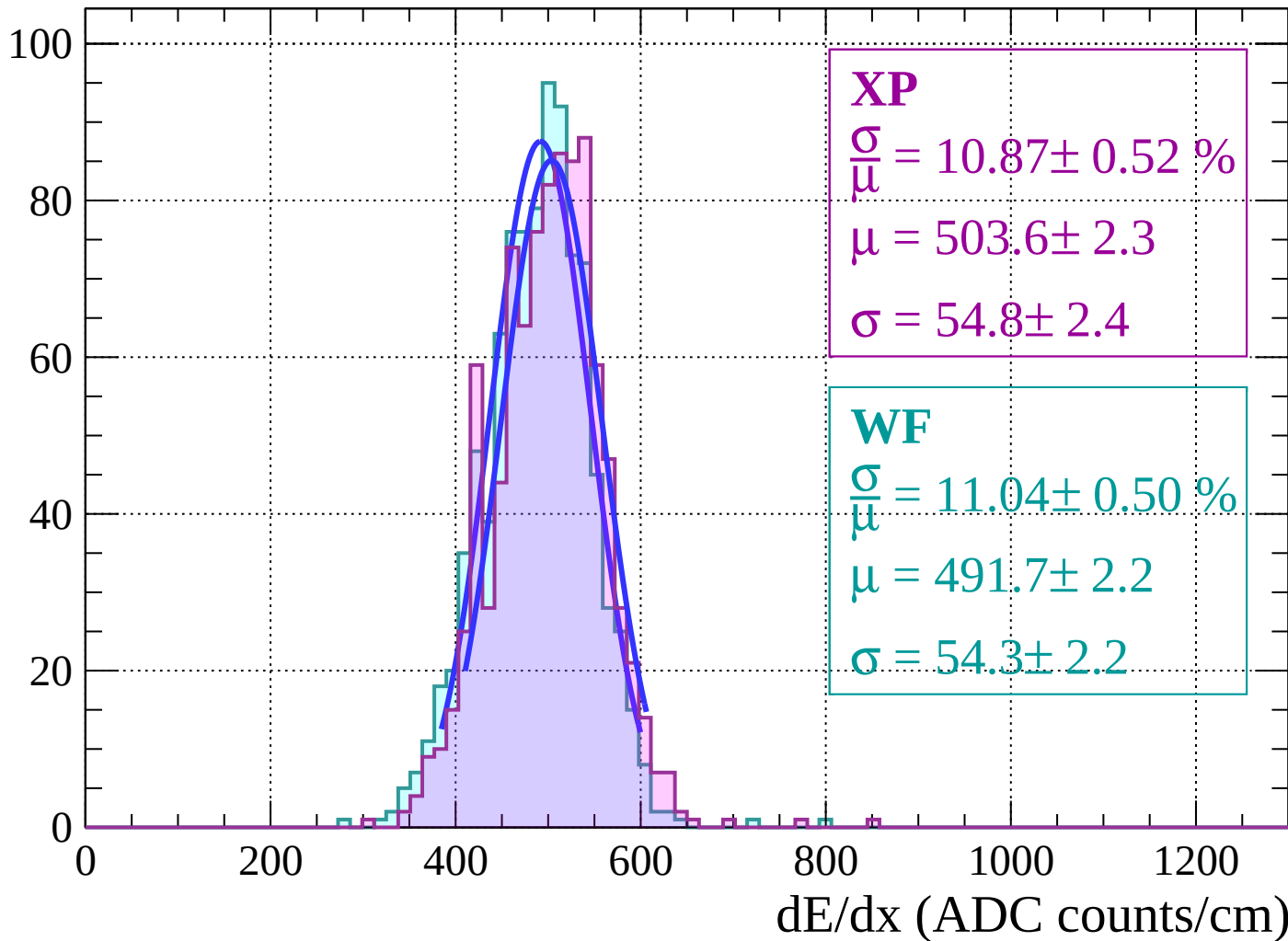
Energy loss | $6 < \theta < 8$

Count

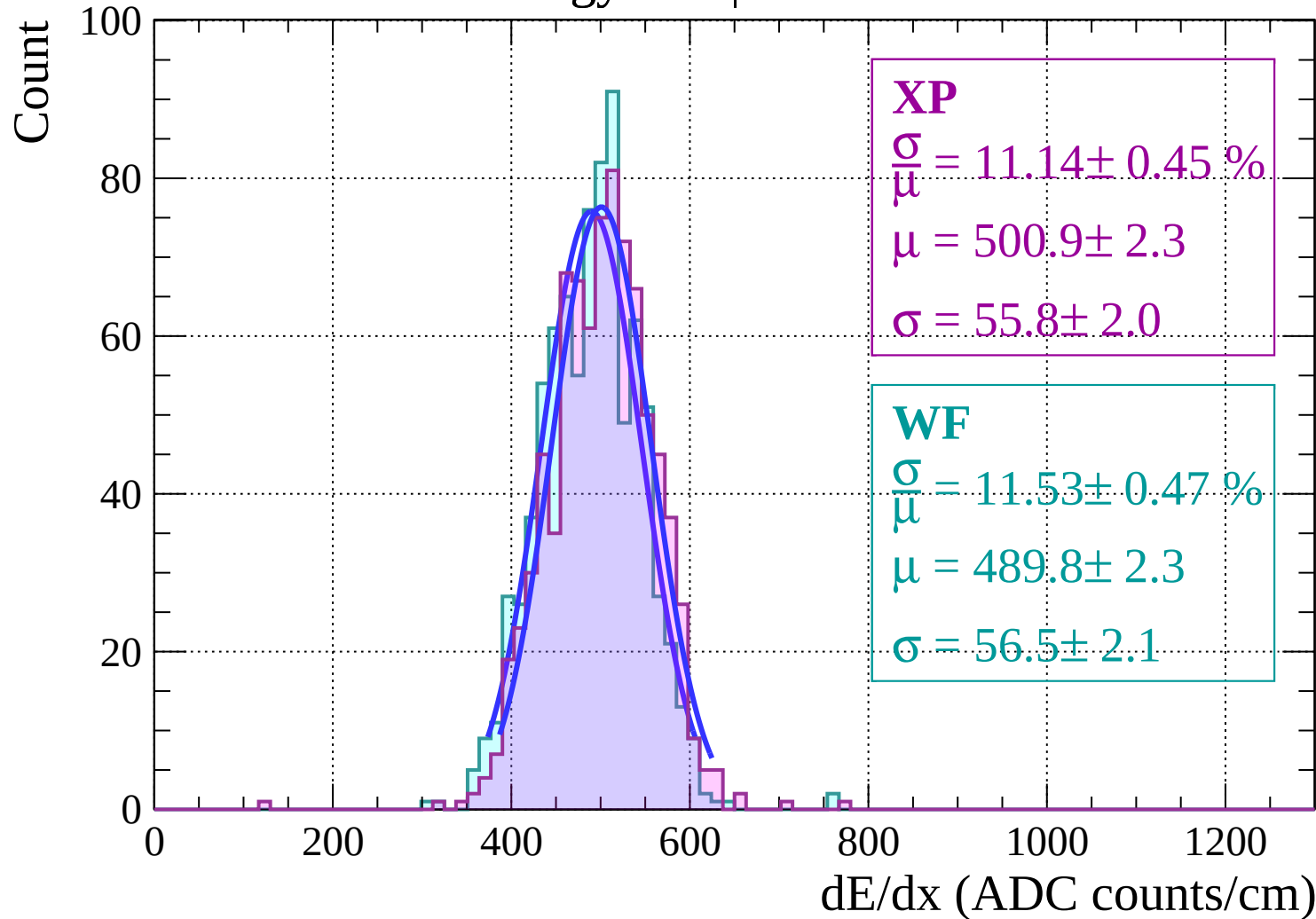


Energy loss | $8 < \theta < 10$

Count

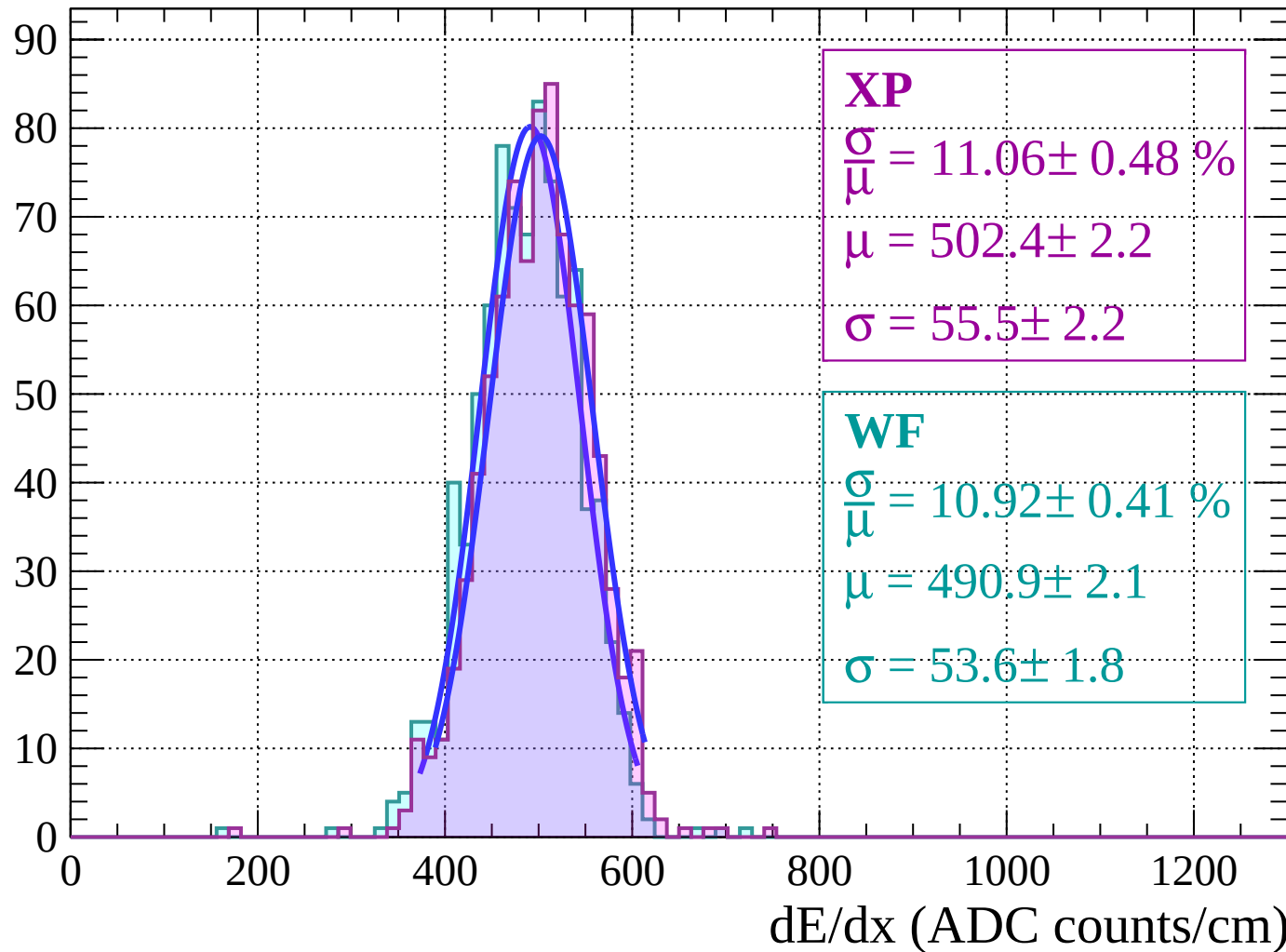


Energy loss | $10 < \theta < 12$



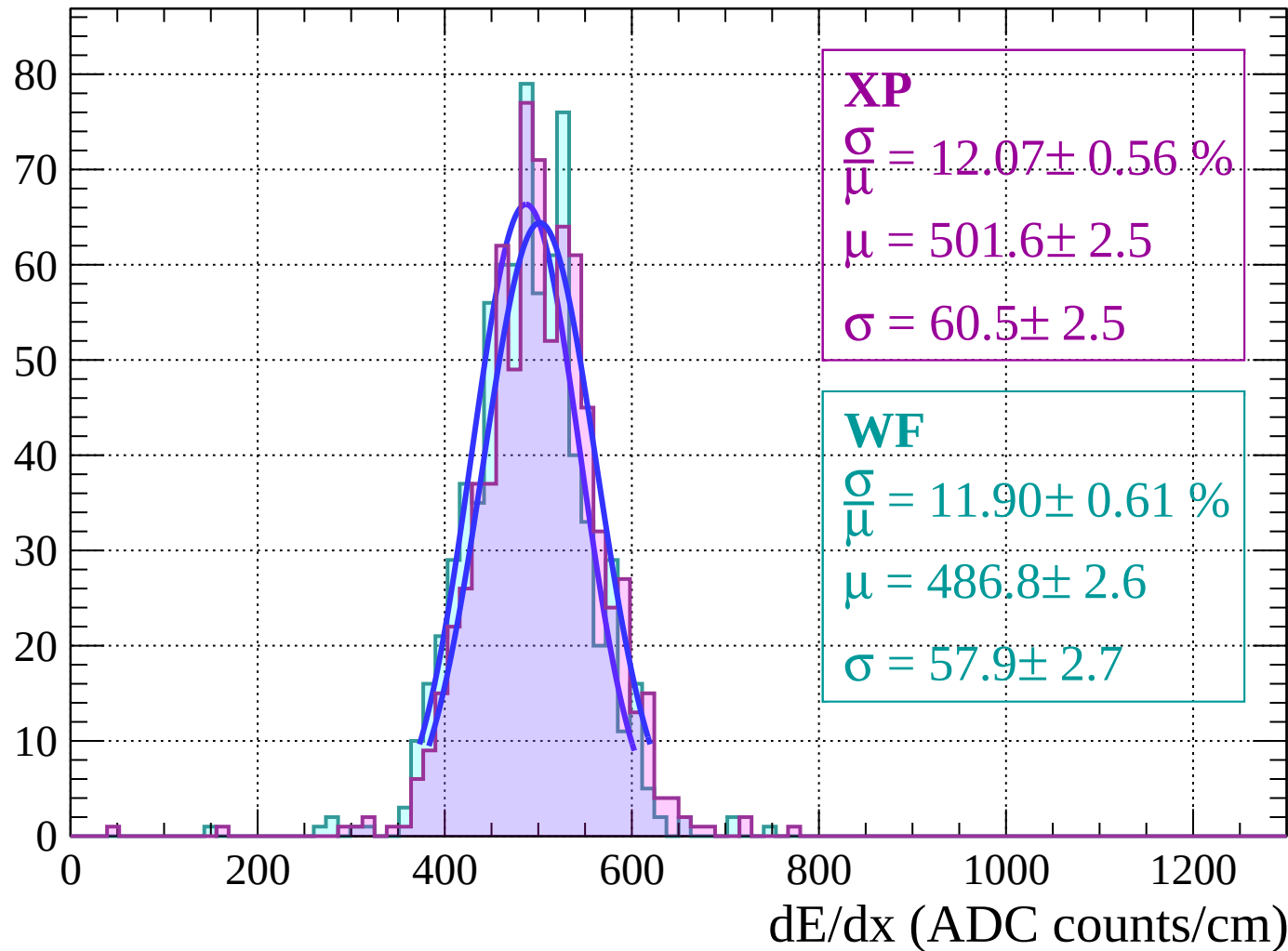
Energy loss | $12 < \theta < 14$

Count



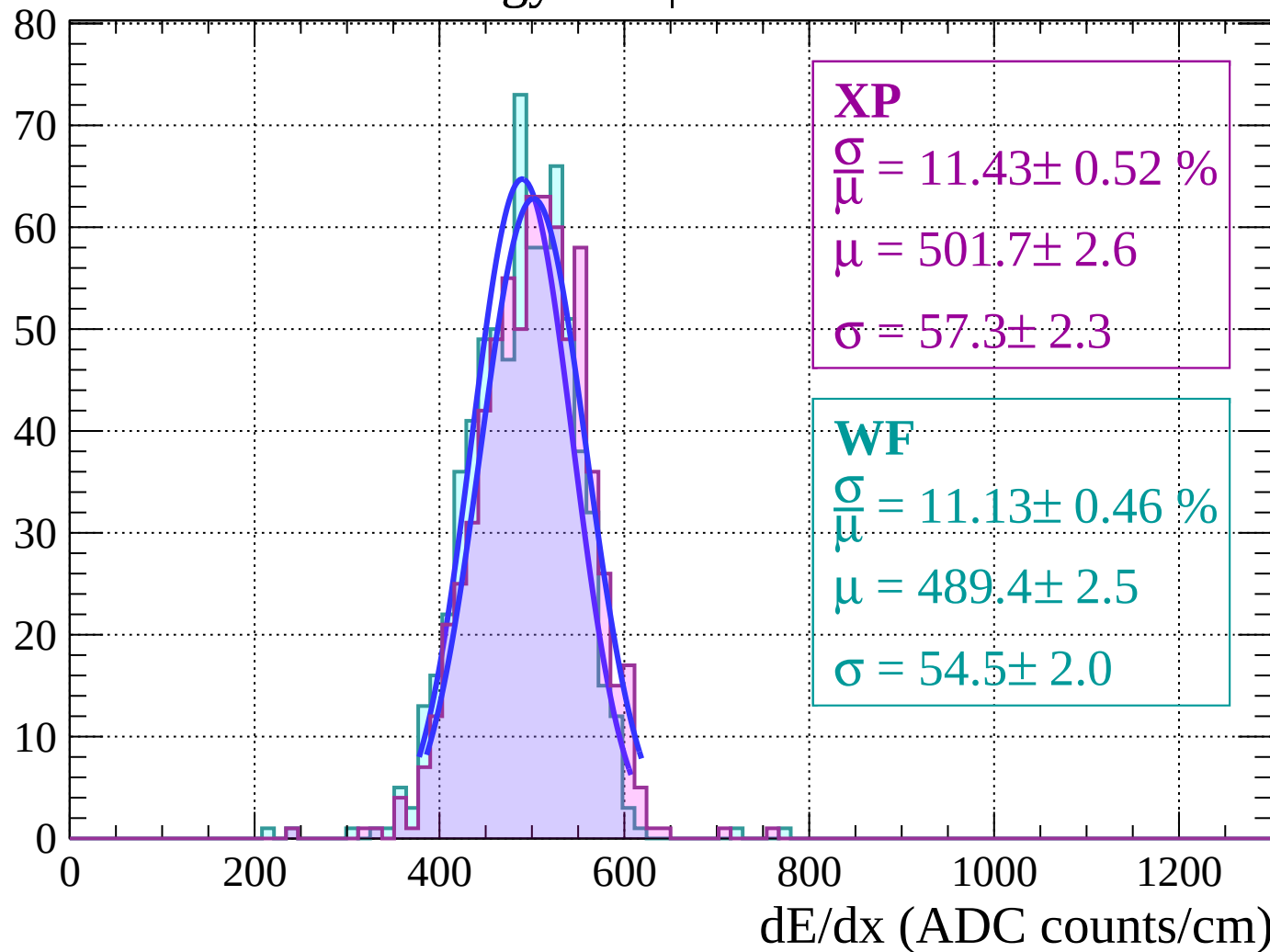
Energy loss | $14 < \theta < 16$

Count



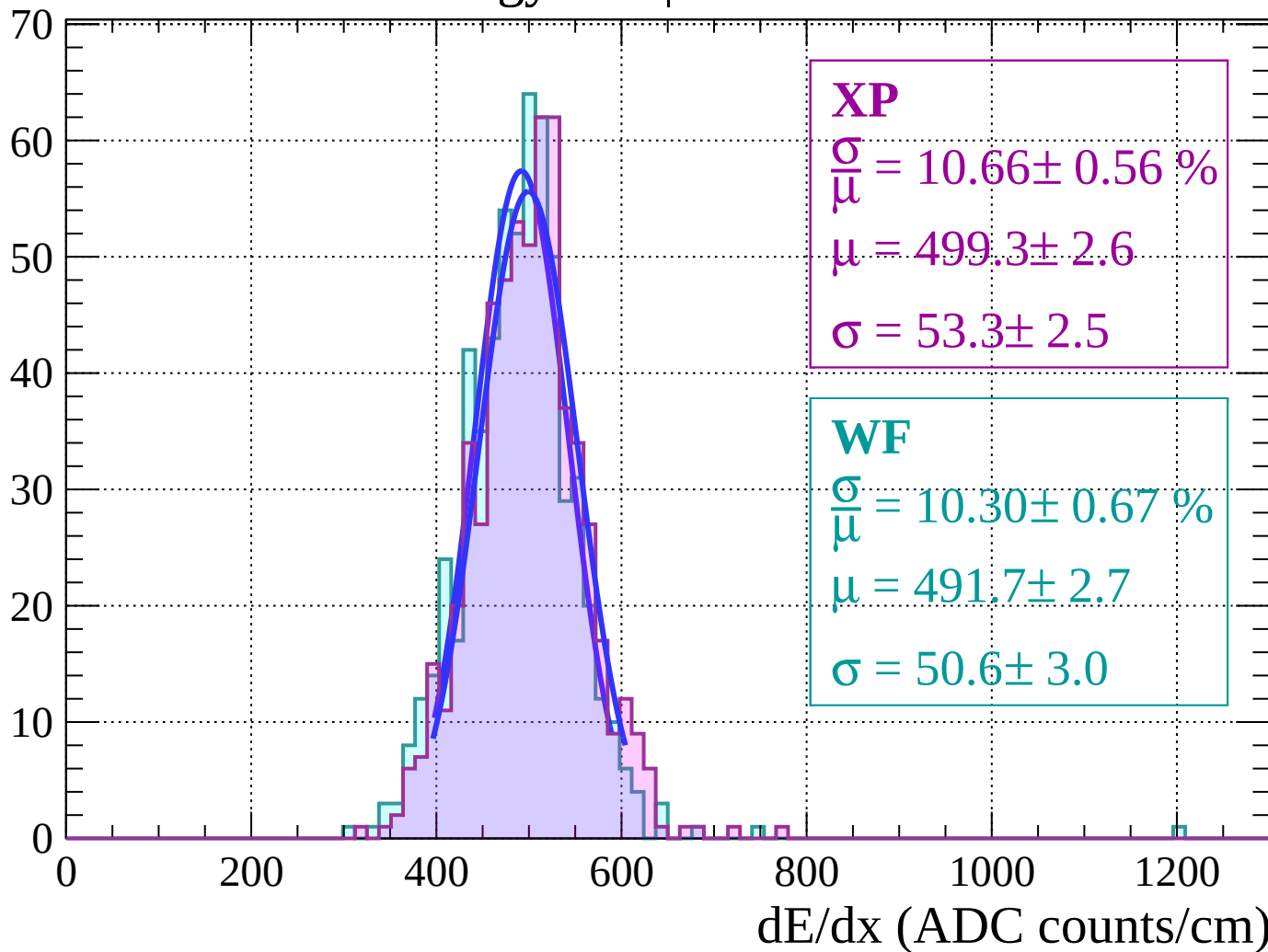
Energy loss | $16 < \theta < 18$

Count



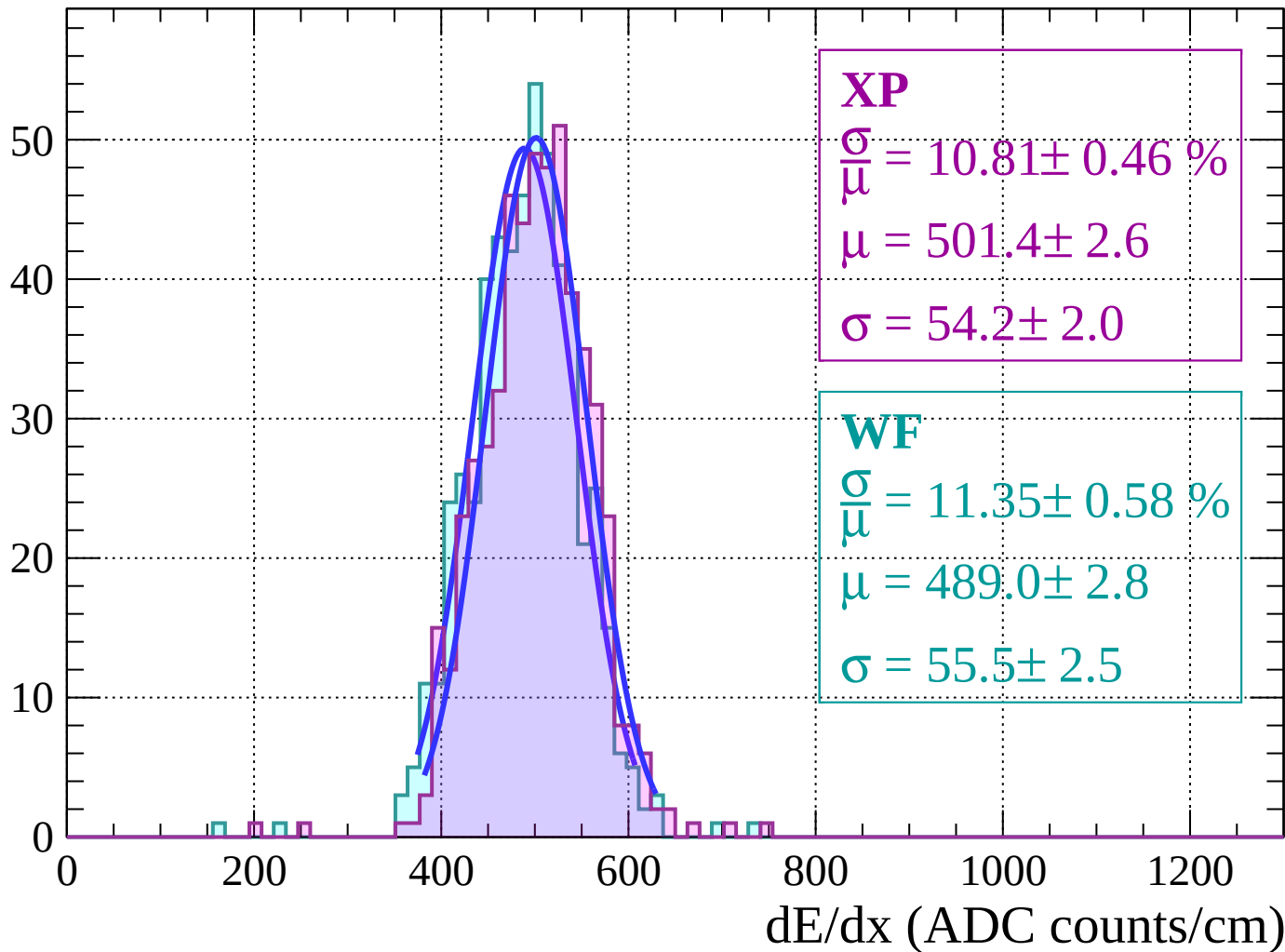
Energy loss | $18 < \theta < 20$

Count



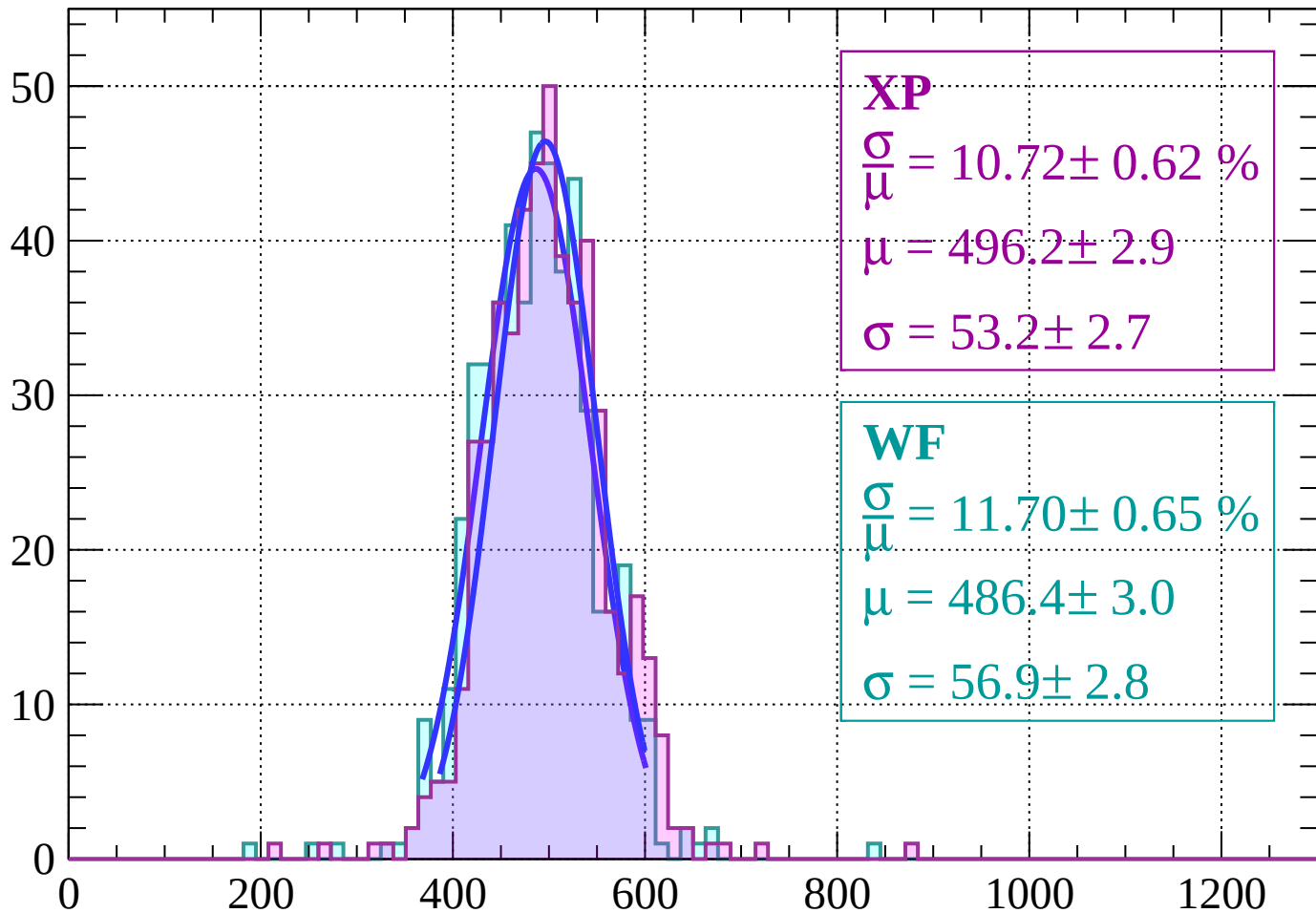
Energy loss | $20 < \theta < 22$

Count



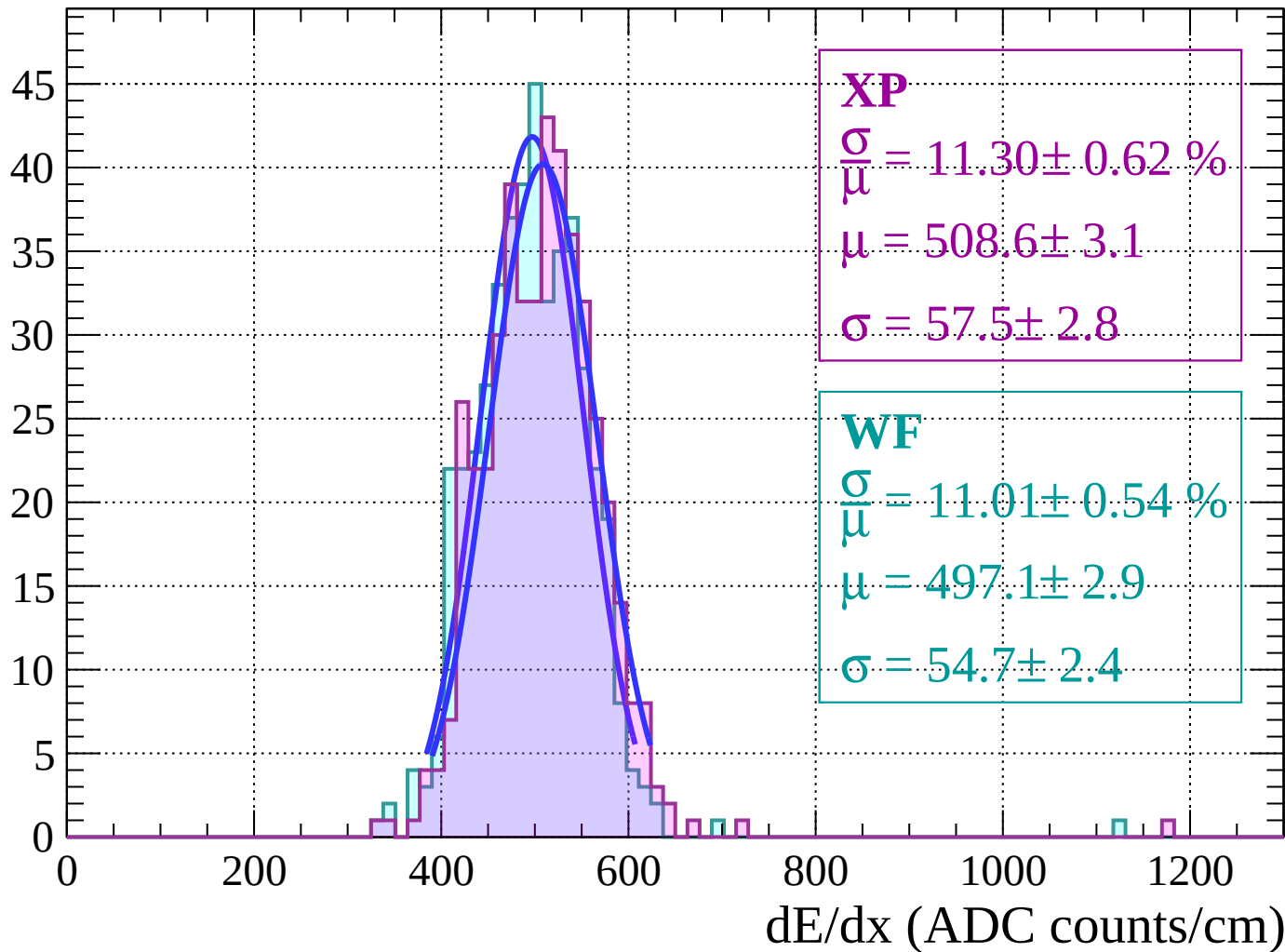
Energy loss | $22 < \theta < 24$

Count



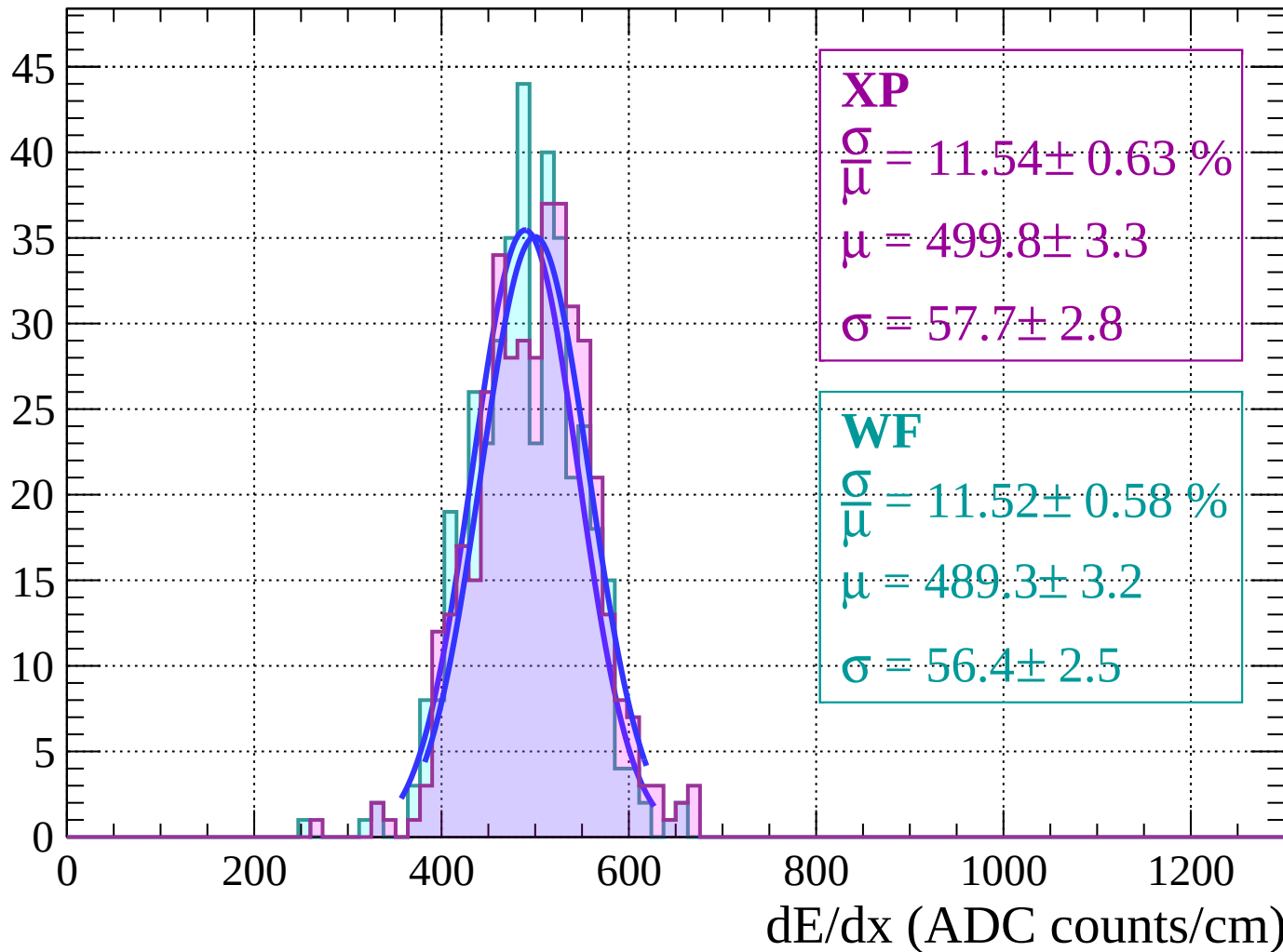
Energy loss | $24 < \theta < 26$

Count



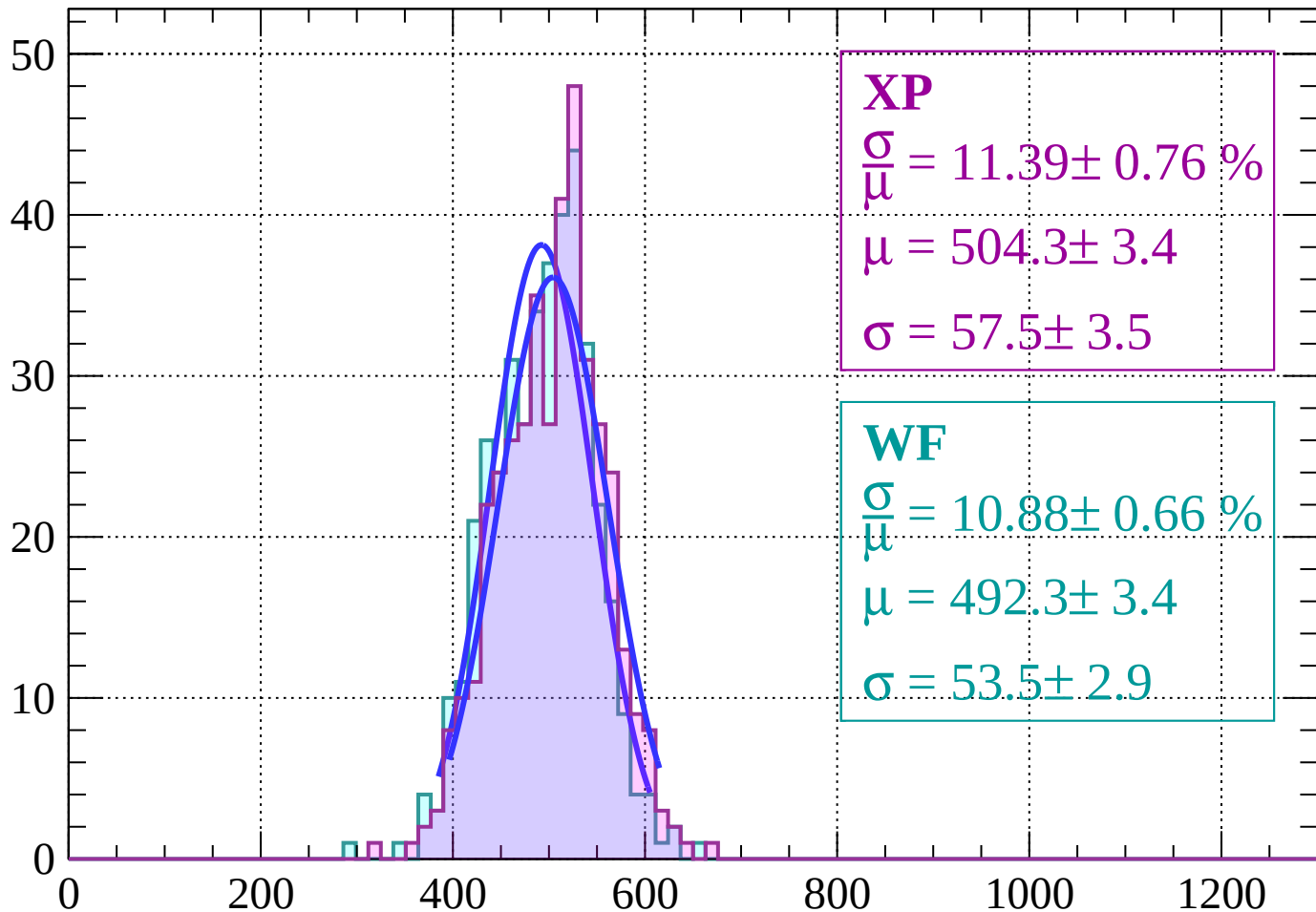
Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 11.39 \pm 0.76 \%$$

$$\mu = 504.3 \pm 3.4$$

$$\sigma = 57.5 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 10.88 \pm 0.66 \%$$

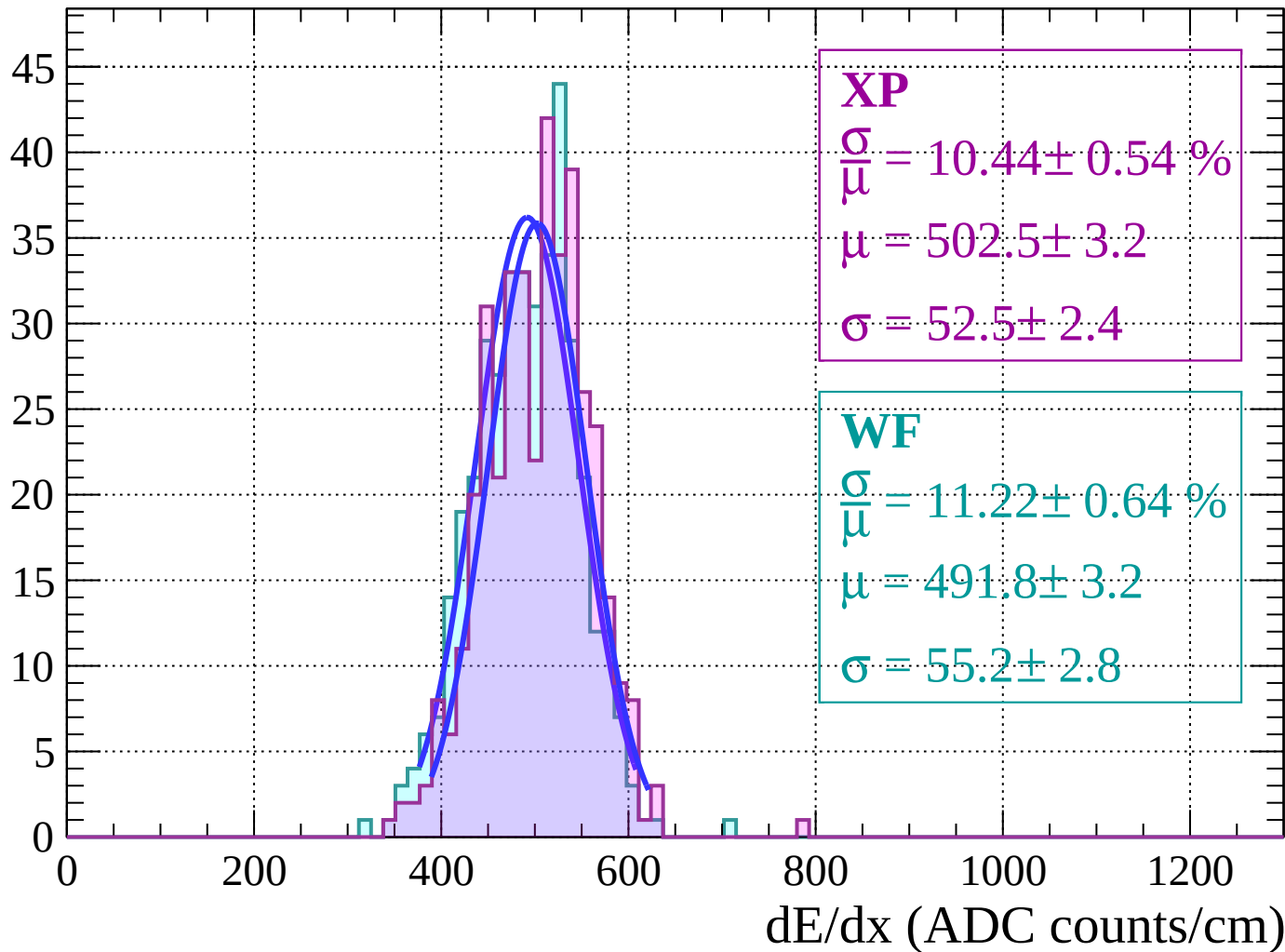
$$\mu = 492.3 \pm 3.4$$

$$\sigma = 53.5 \pm 2.9$$

dE/dx (ADC counts/cm)

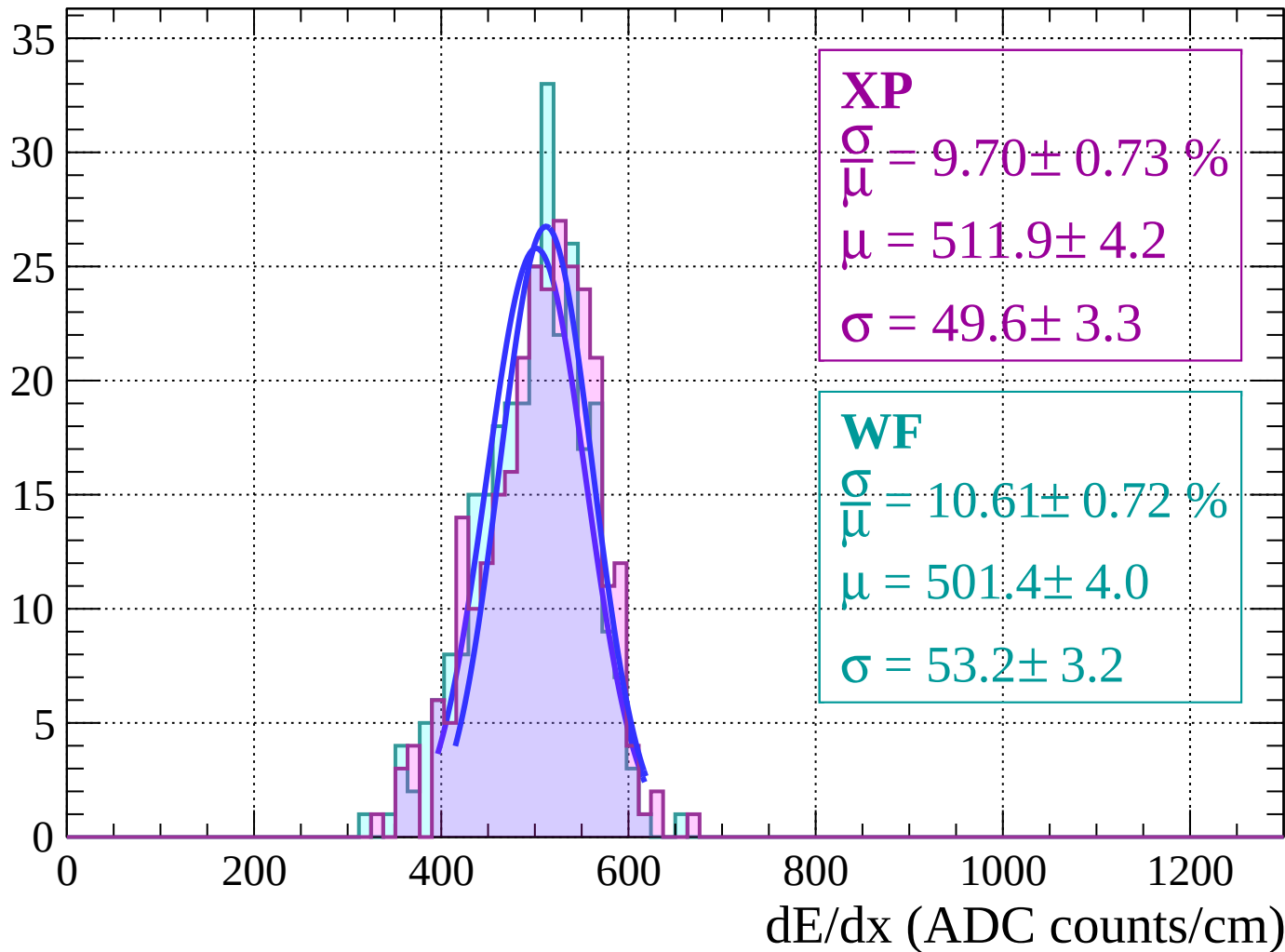
Energy loss | $30 < \theta < 32$

Count



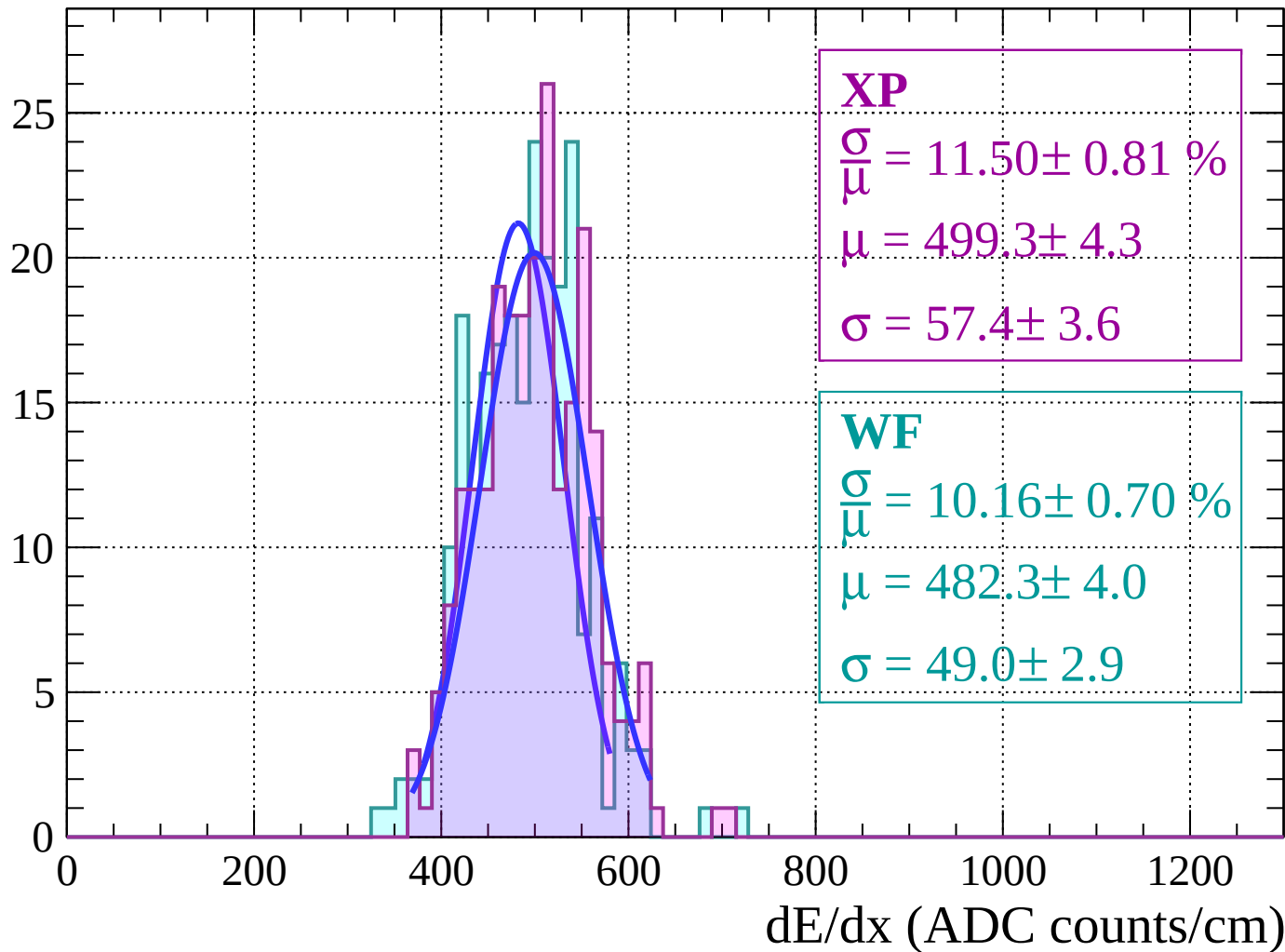
Energy loss | $32 < \theta < 34$

Count



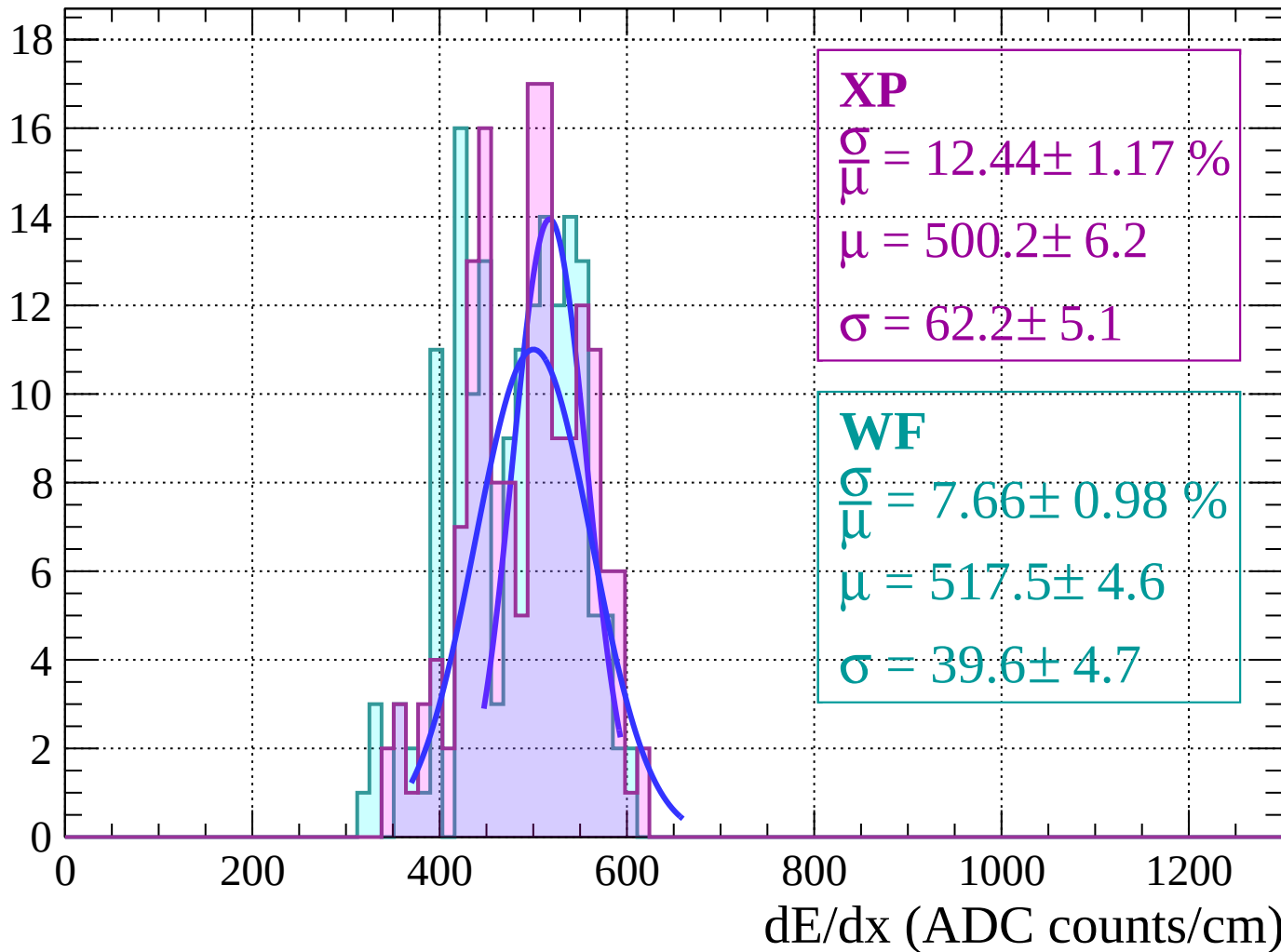
Energy loss | $34 < \theta < 36$

Count



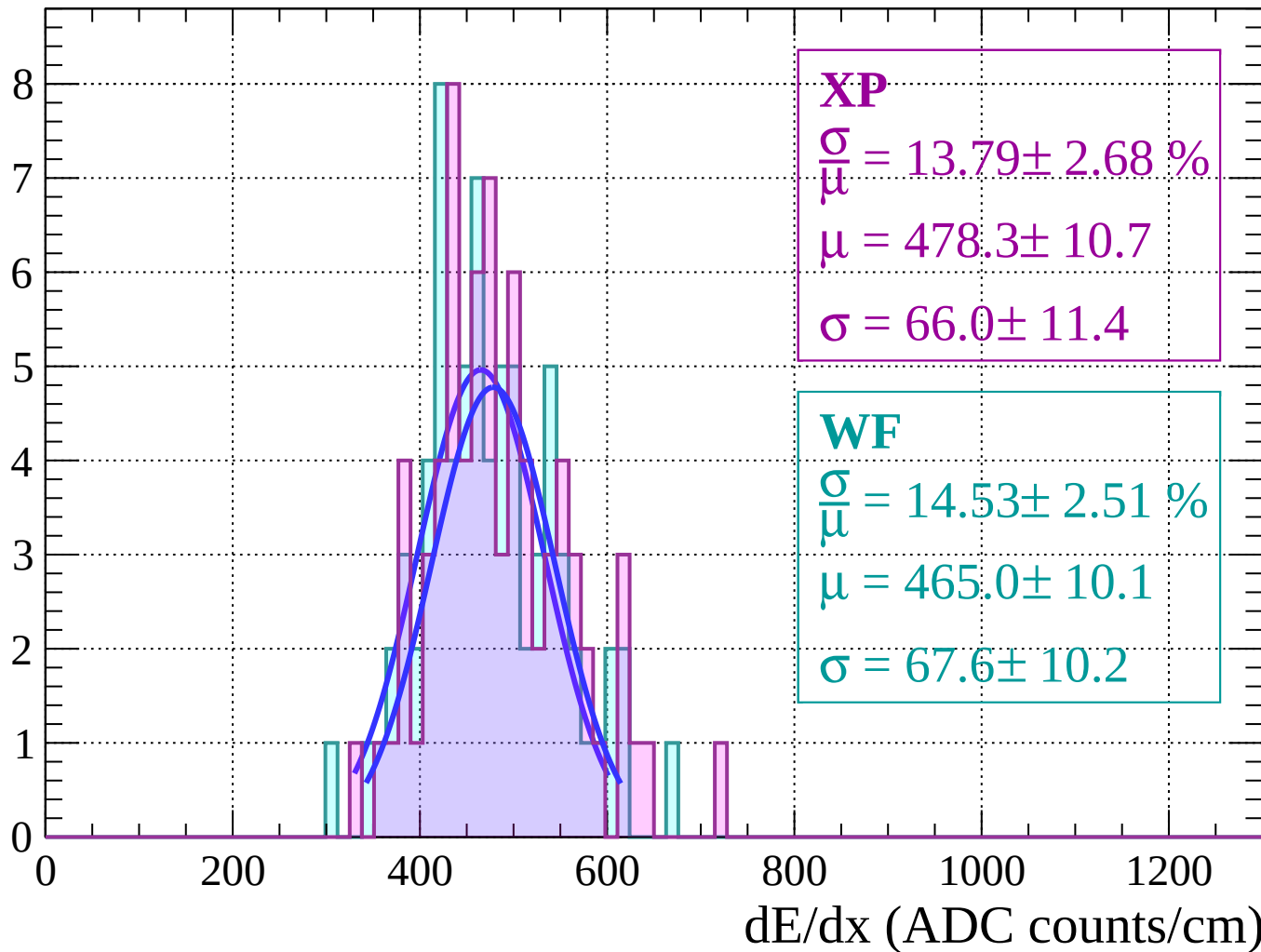
Energy loss | $36 < \theta < 38$

Count



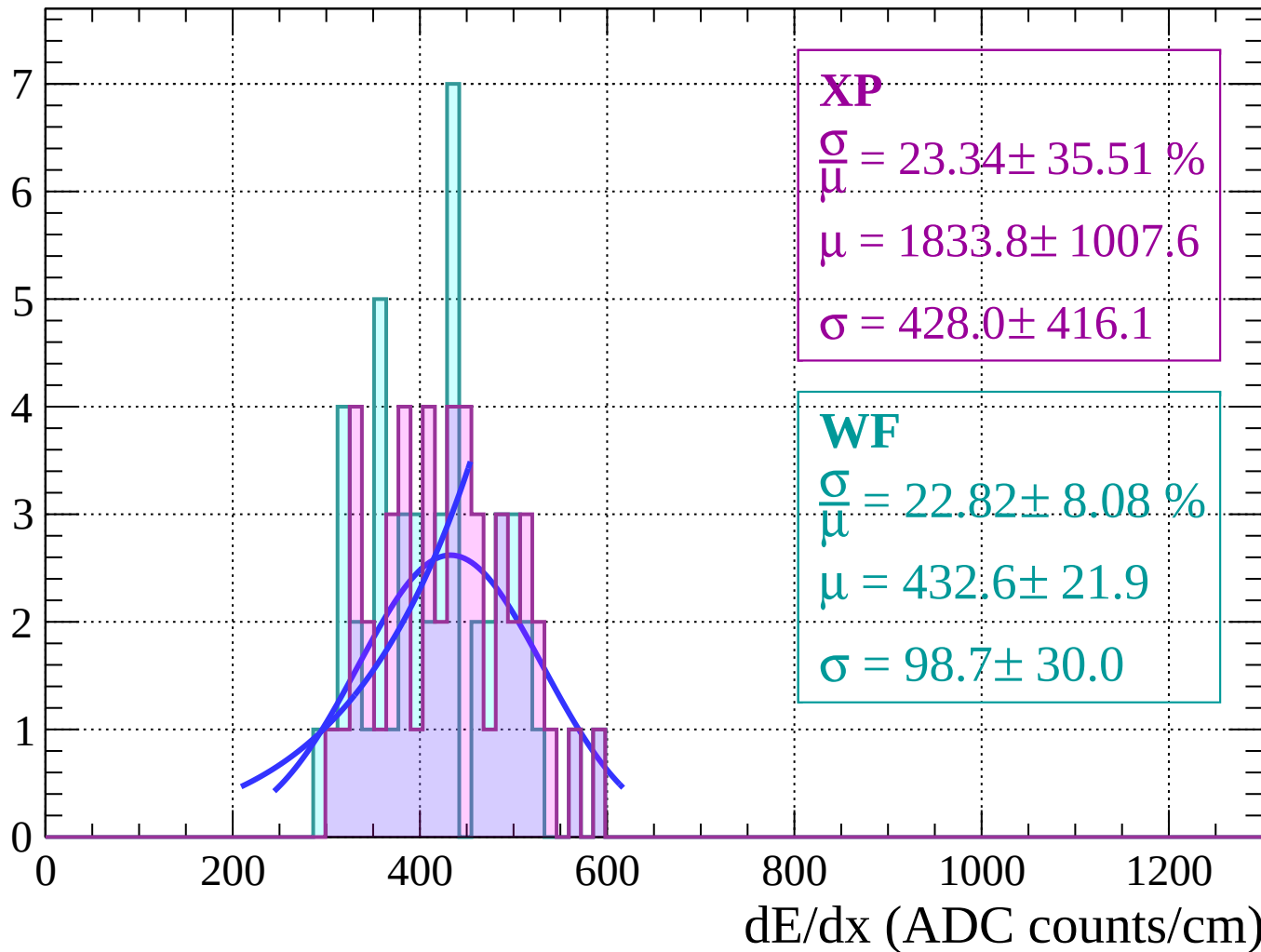
Energy loss | $38 < \theta < 40$

Count



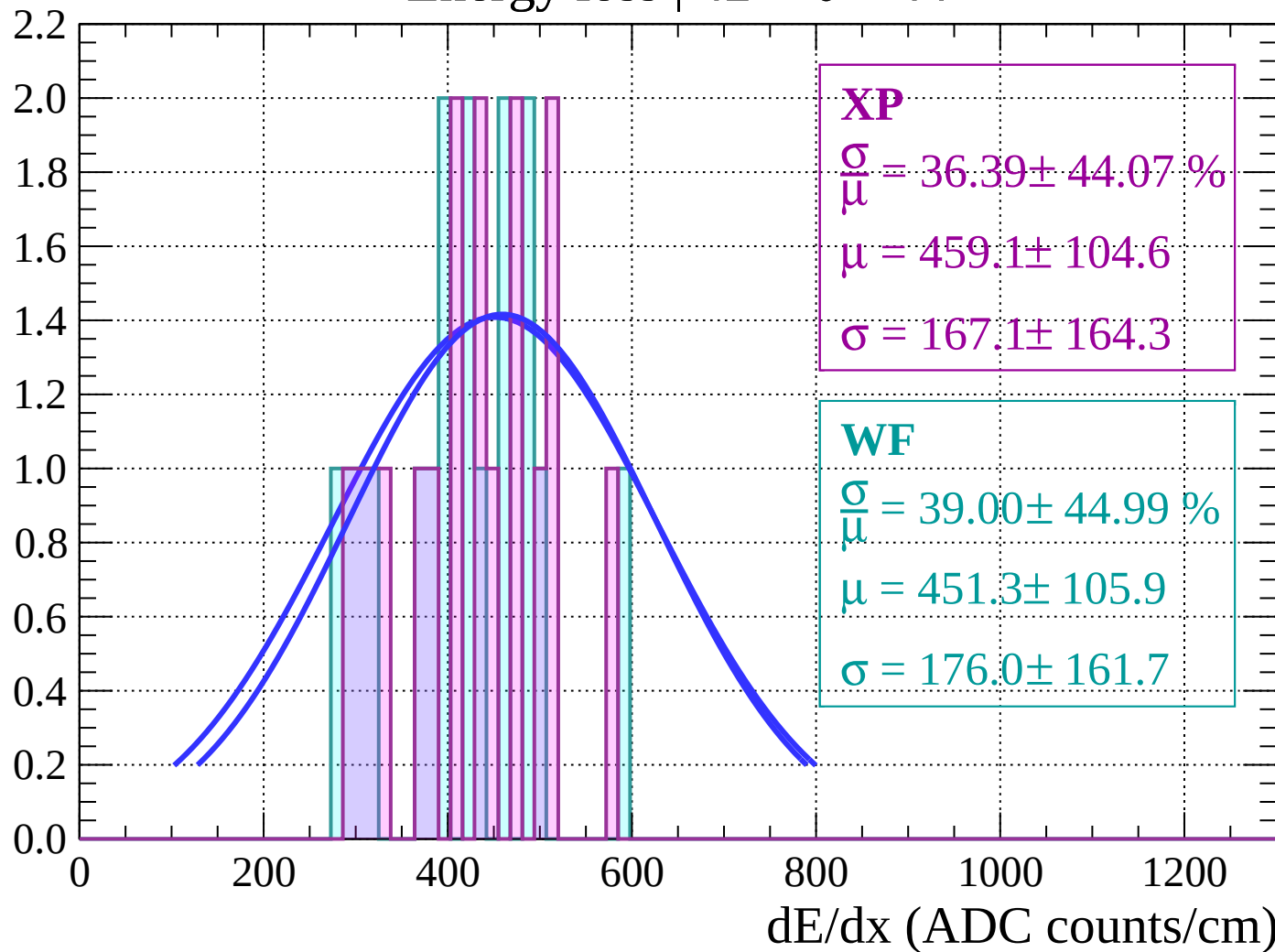
Energy loss | $40 < \theta < 42$

Count



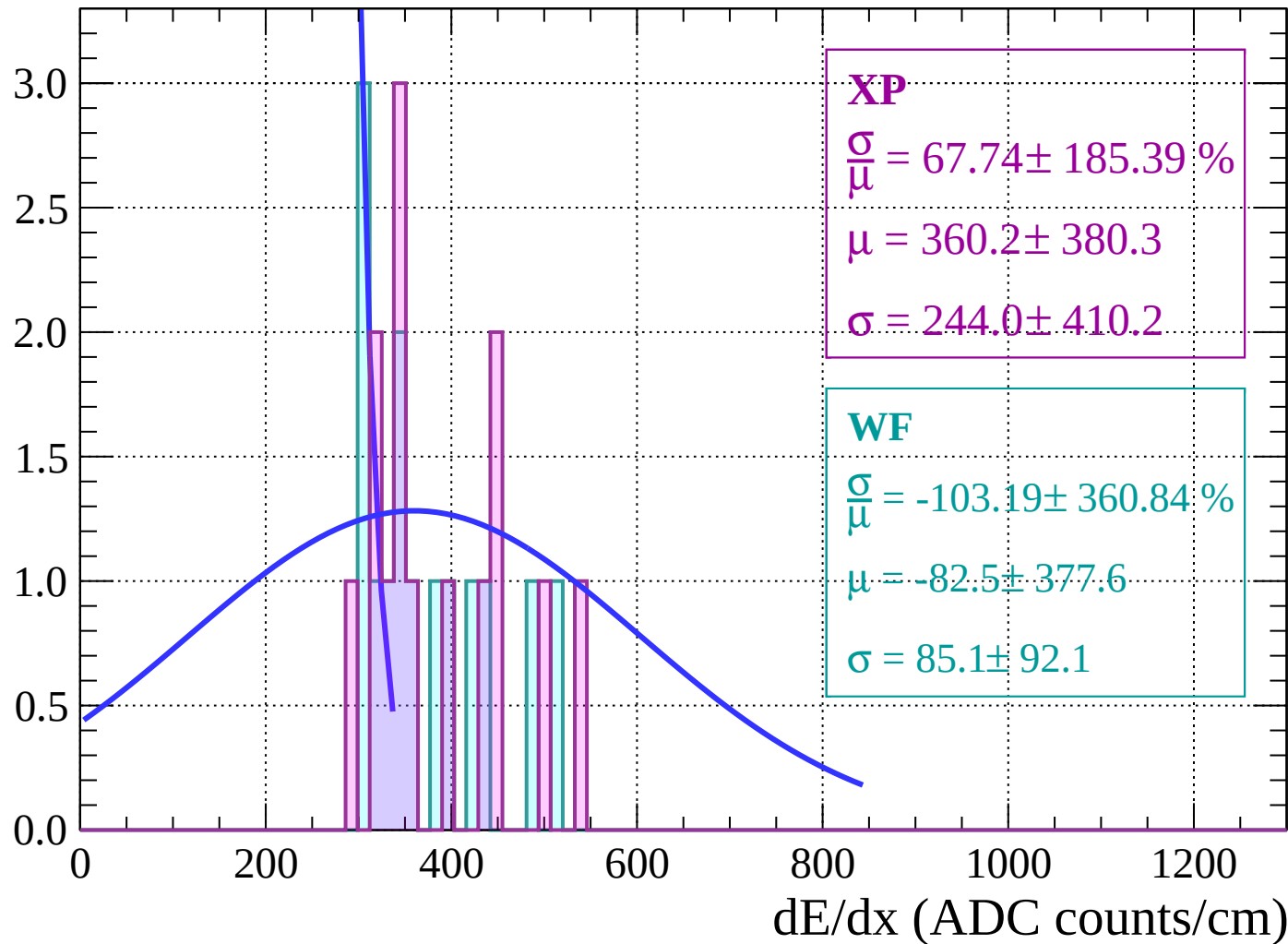
Energy loss | $42 < \theta < 44$

Count



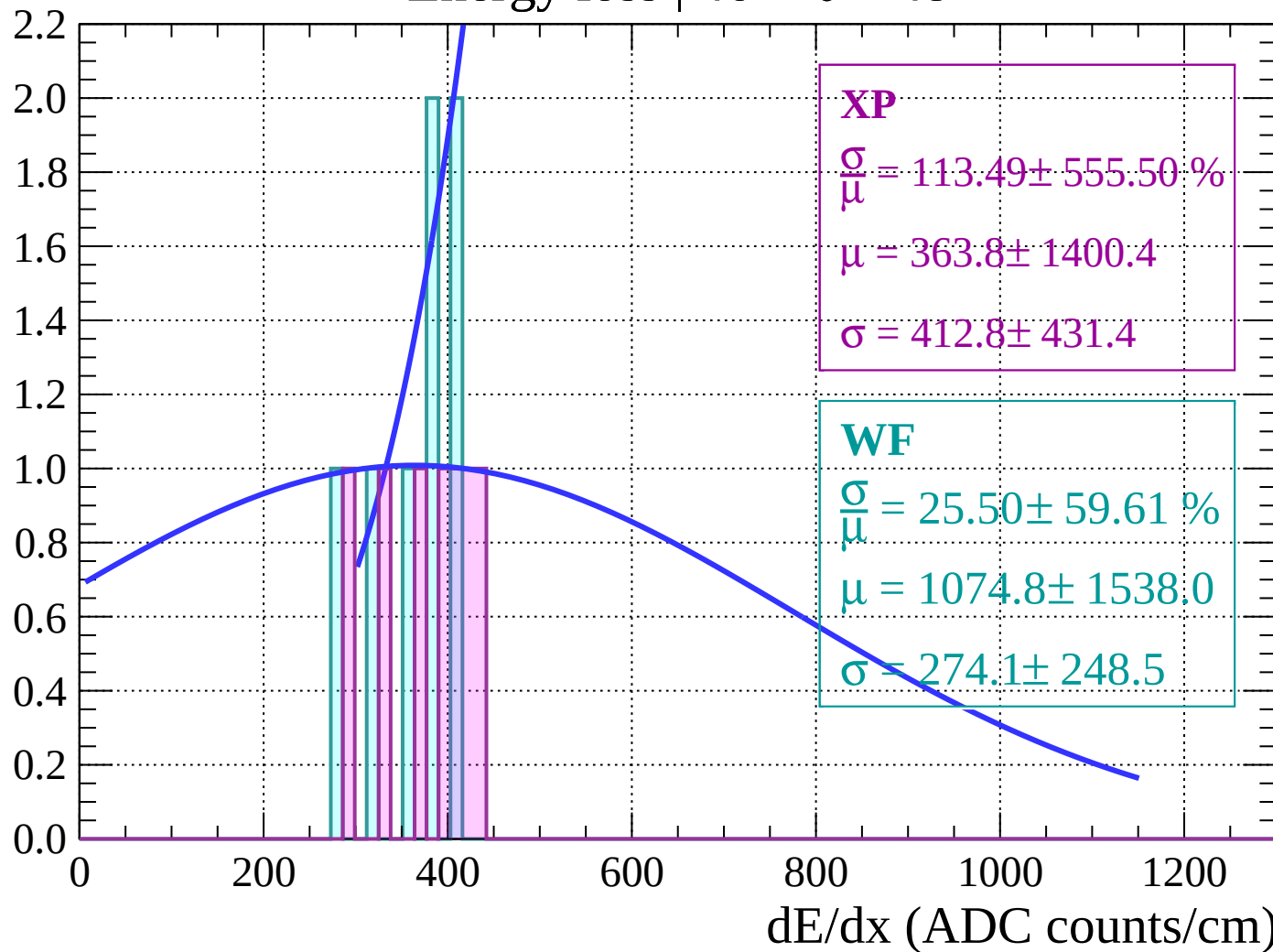
Energy loss | $44 < \theta < 46$

Count



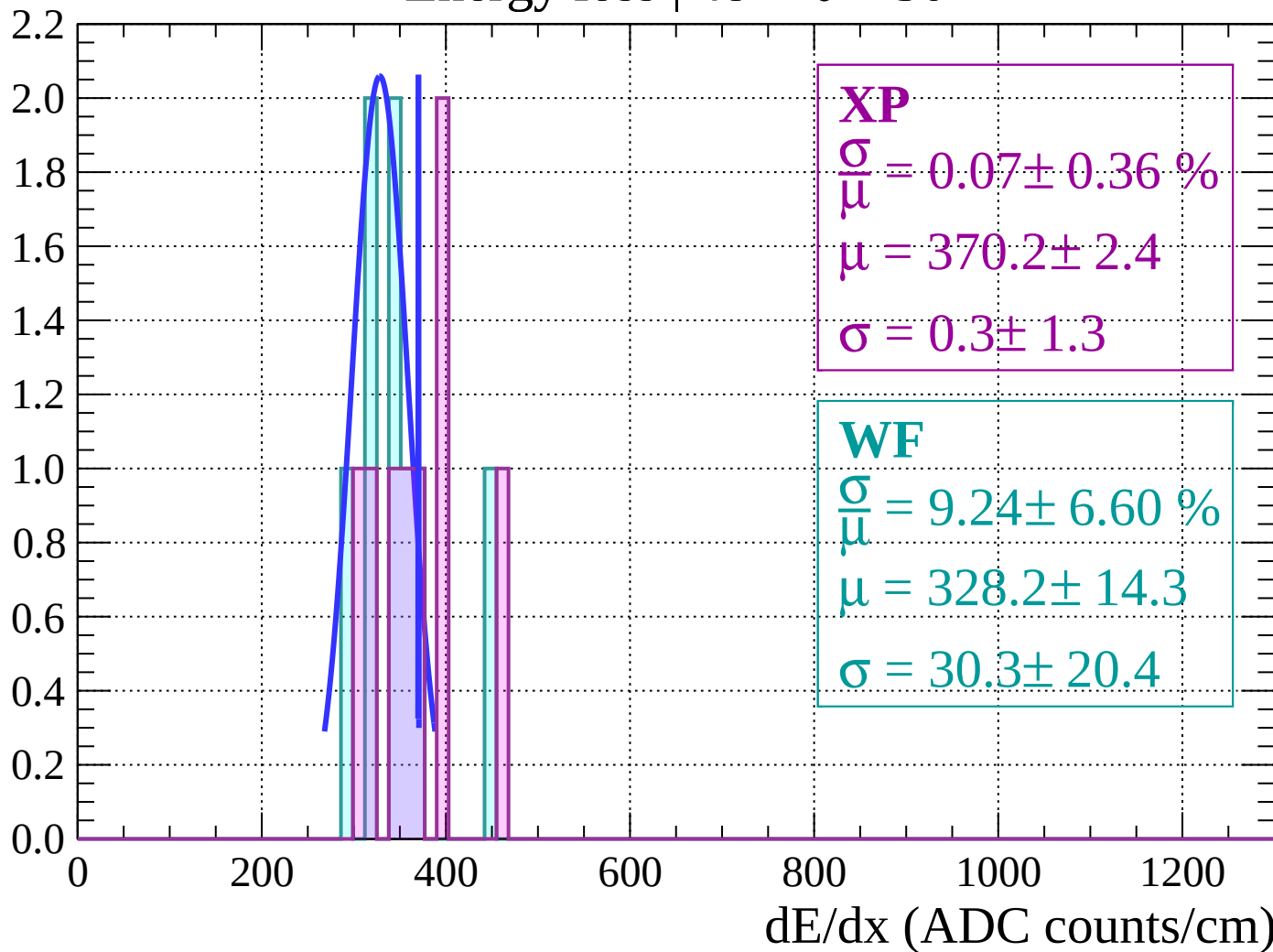
Energy loss | $46 < \theta < 48$

Count



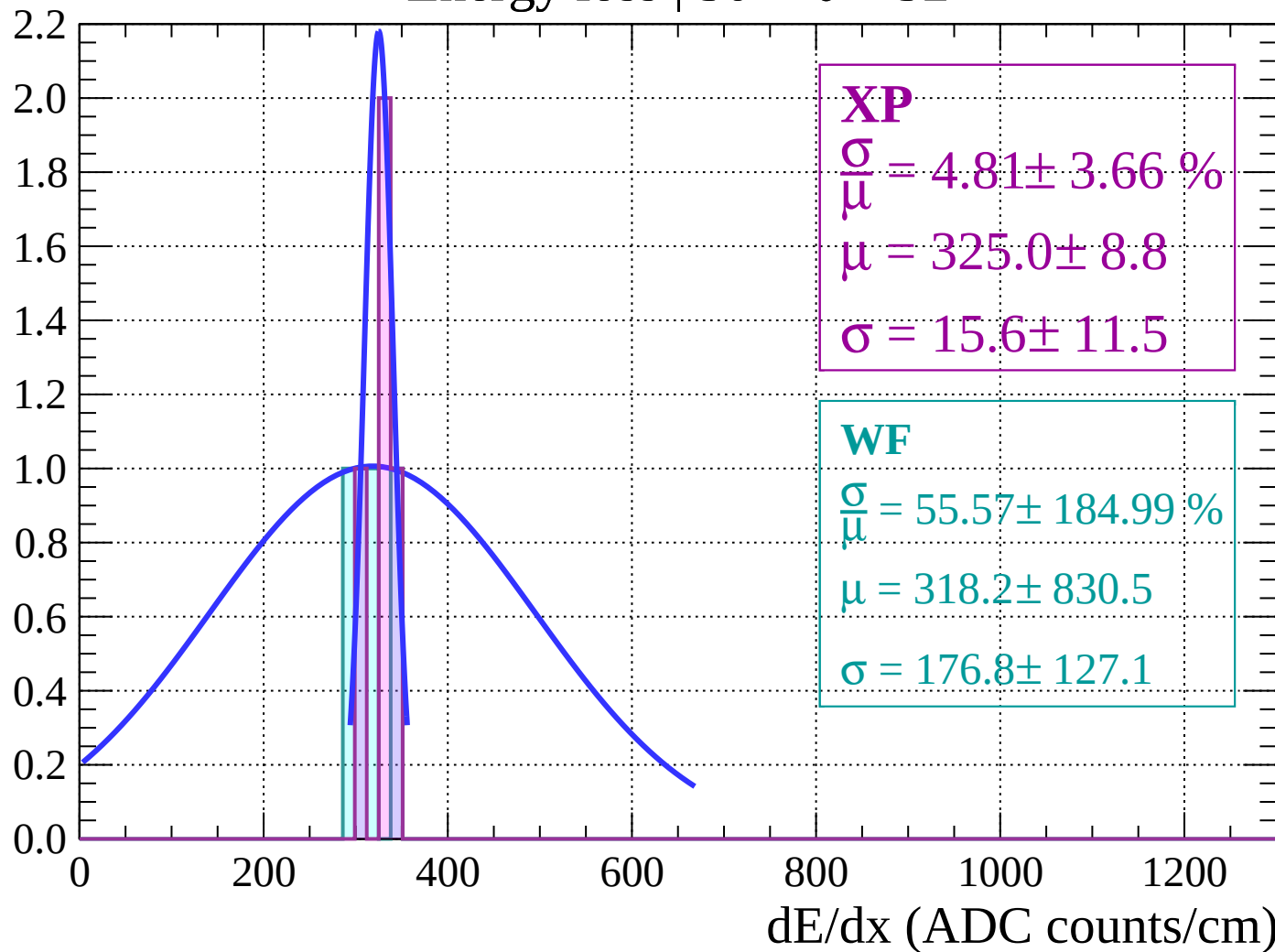
Energy loss | $48 < \theta < 50$

Count



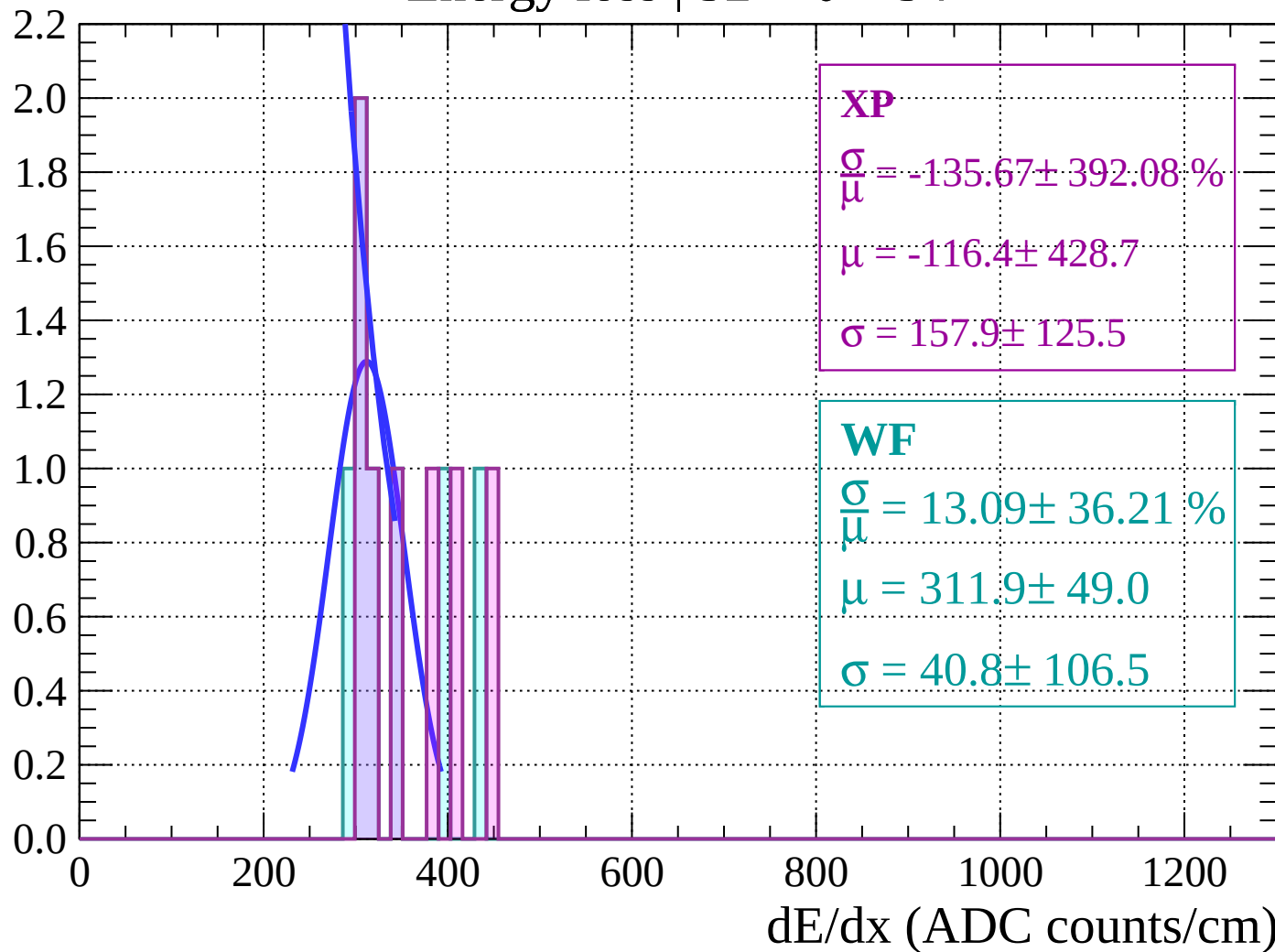
Energy loss | $50 < \theta < 52$

Count



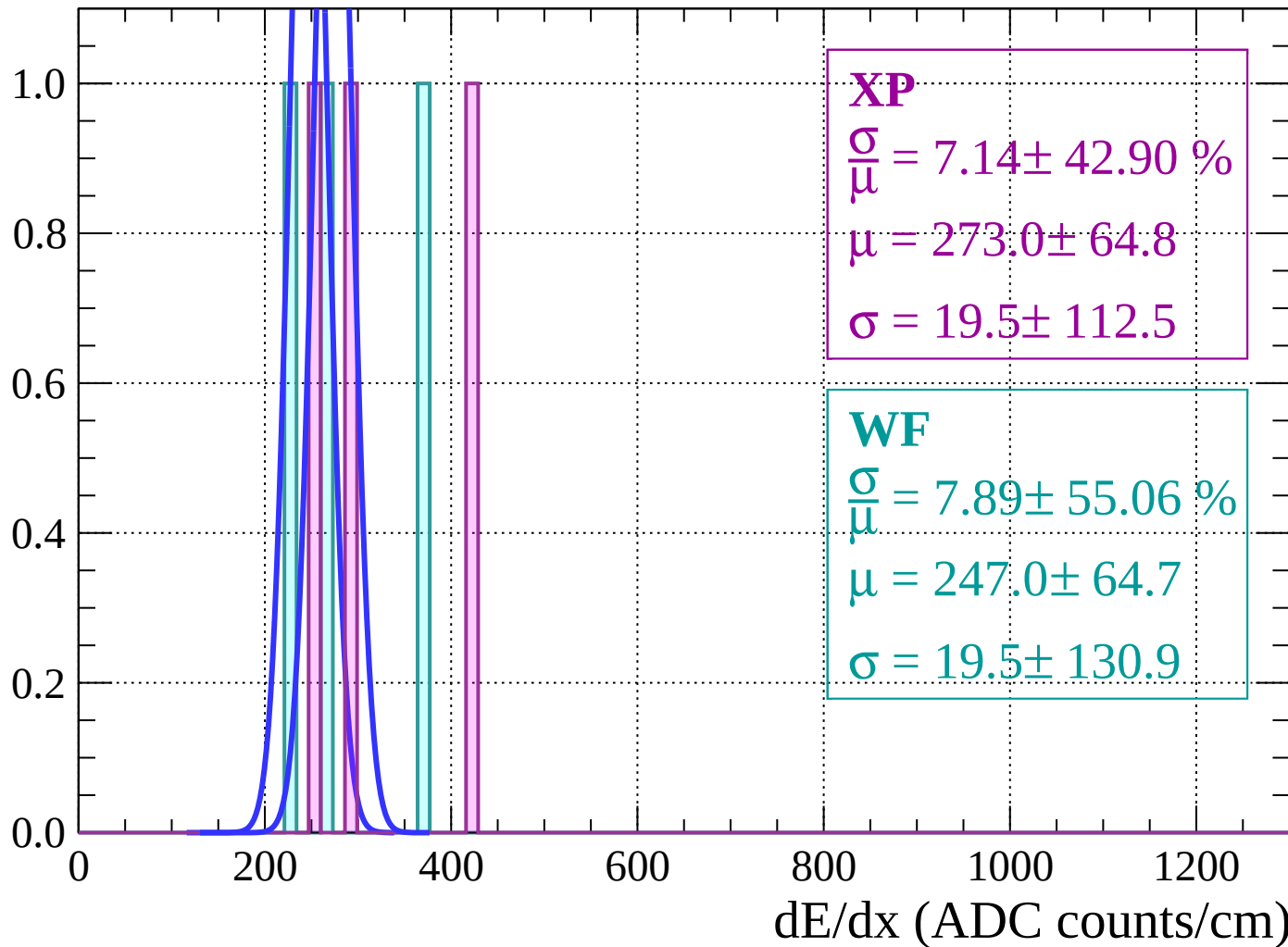
Energy loss | $52 < \theta < 54$

Count



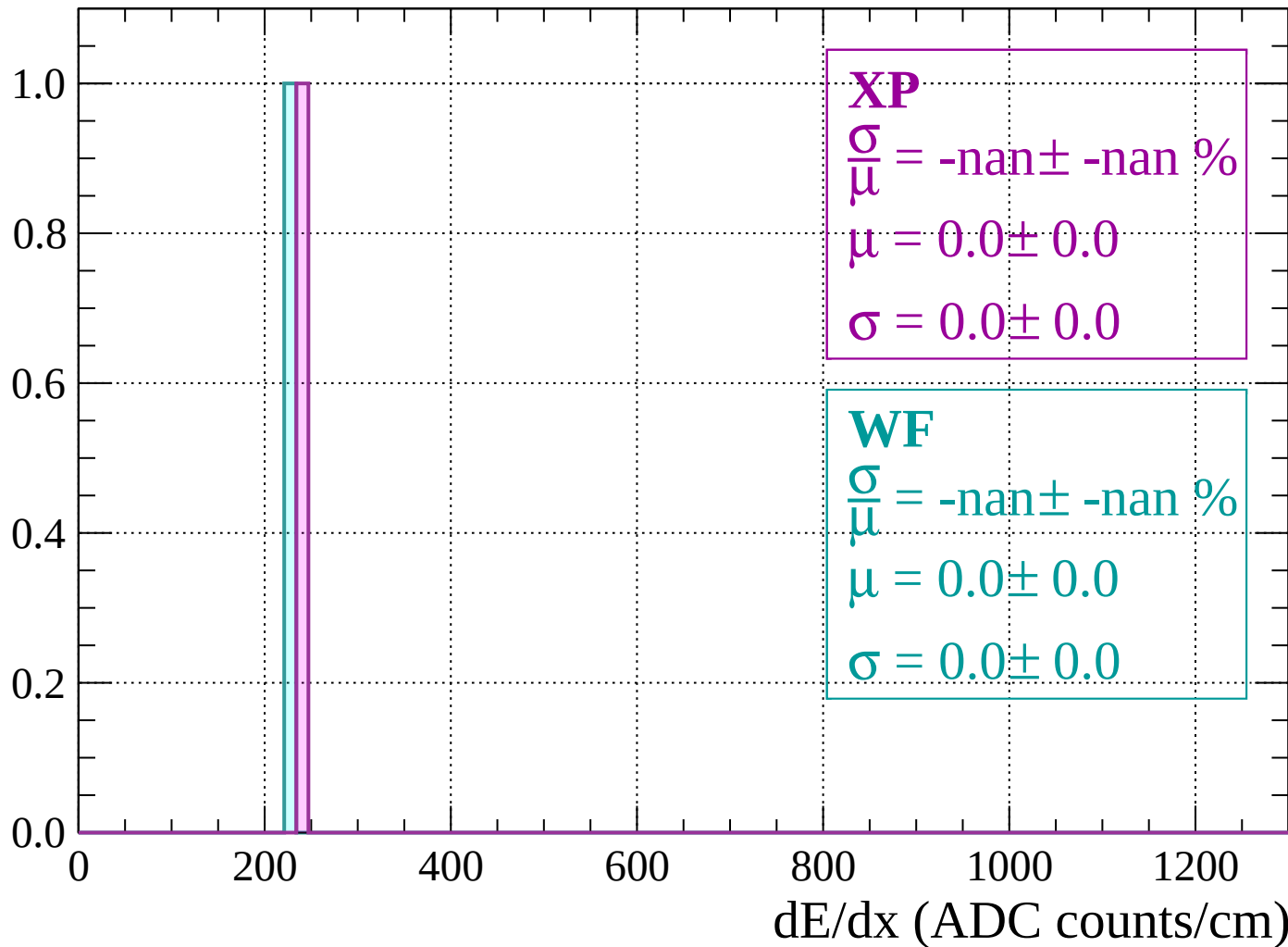
Energy loss | $54 < \theta < 56$

Count



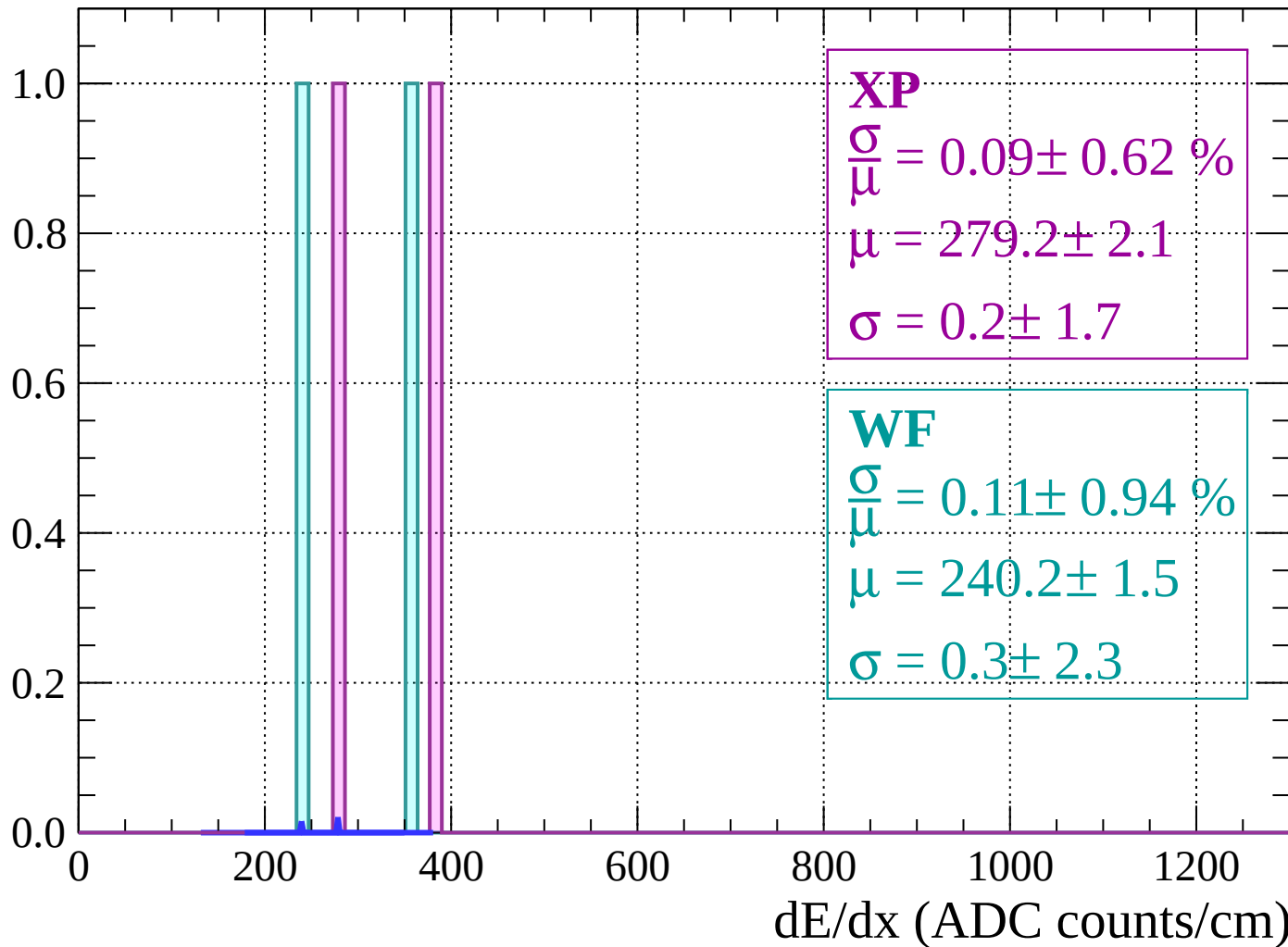
Energy loss | $56 < \theta < 58$

Count



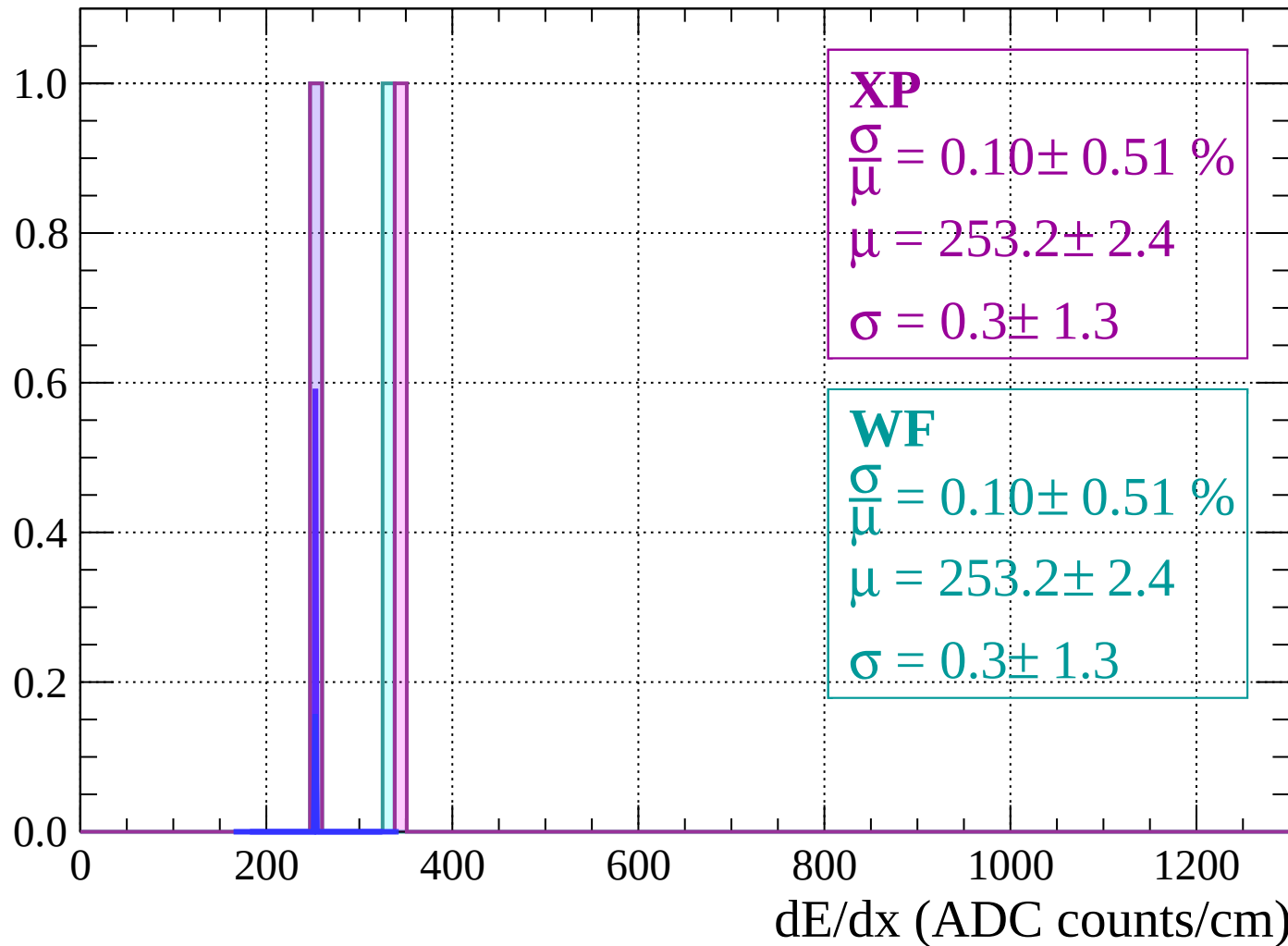
Energy loss | $58 < \theta < 60$

Count



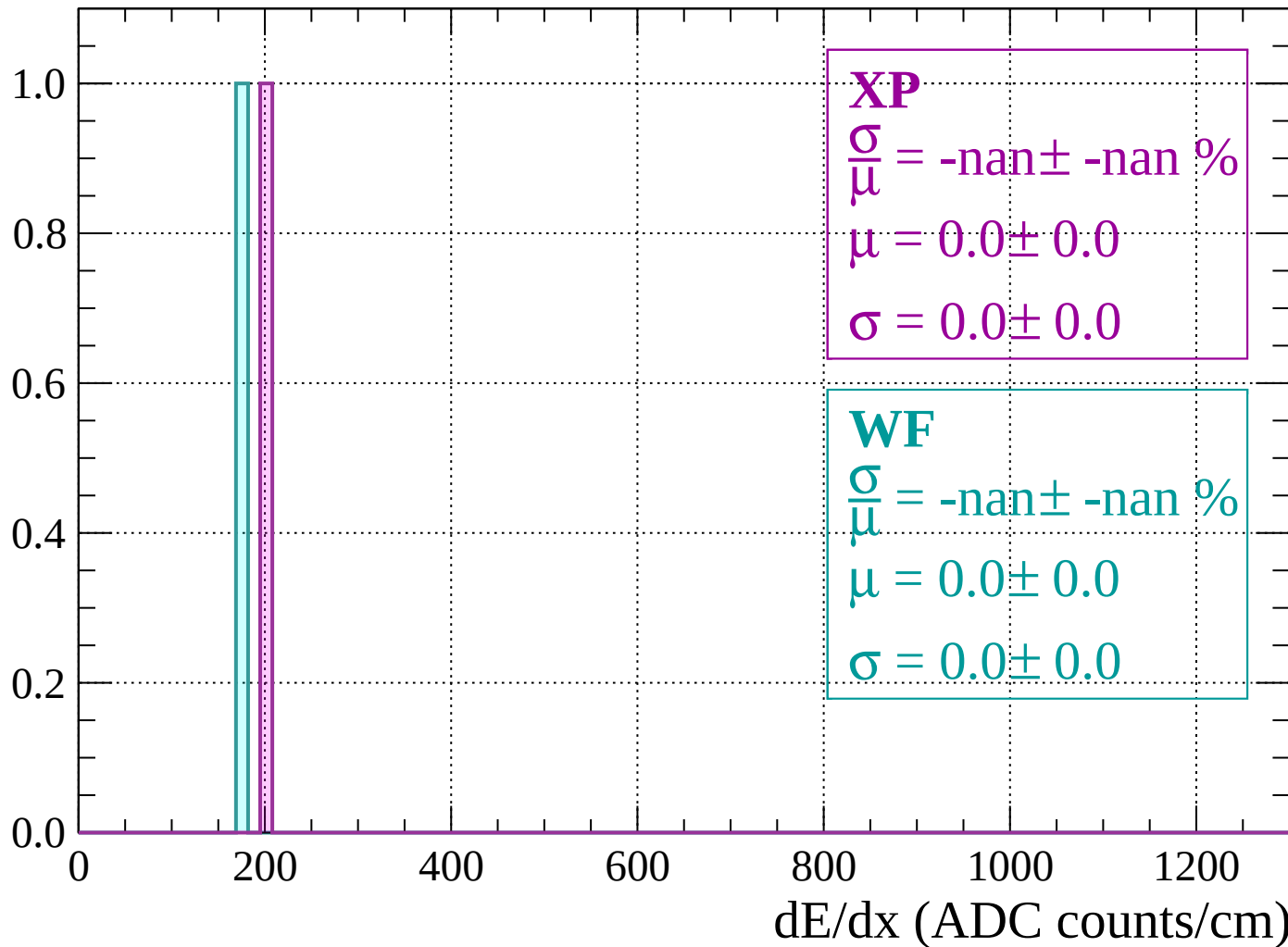
Energy loss | $60 < \theta < 62$

Count



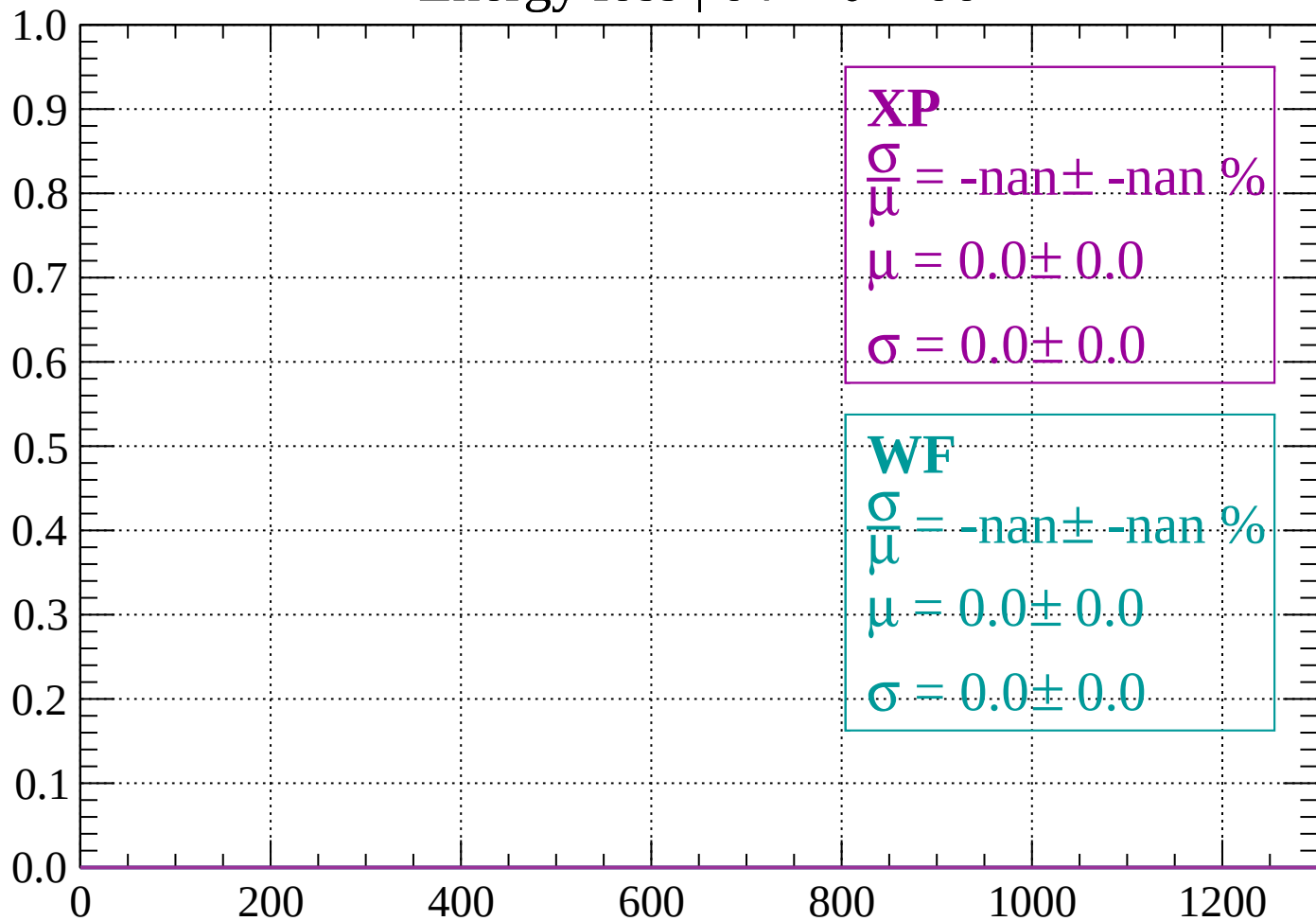
Energy loss | $62 < \theta < 64$

Count



Energy loss | $64 < \theta < 66$

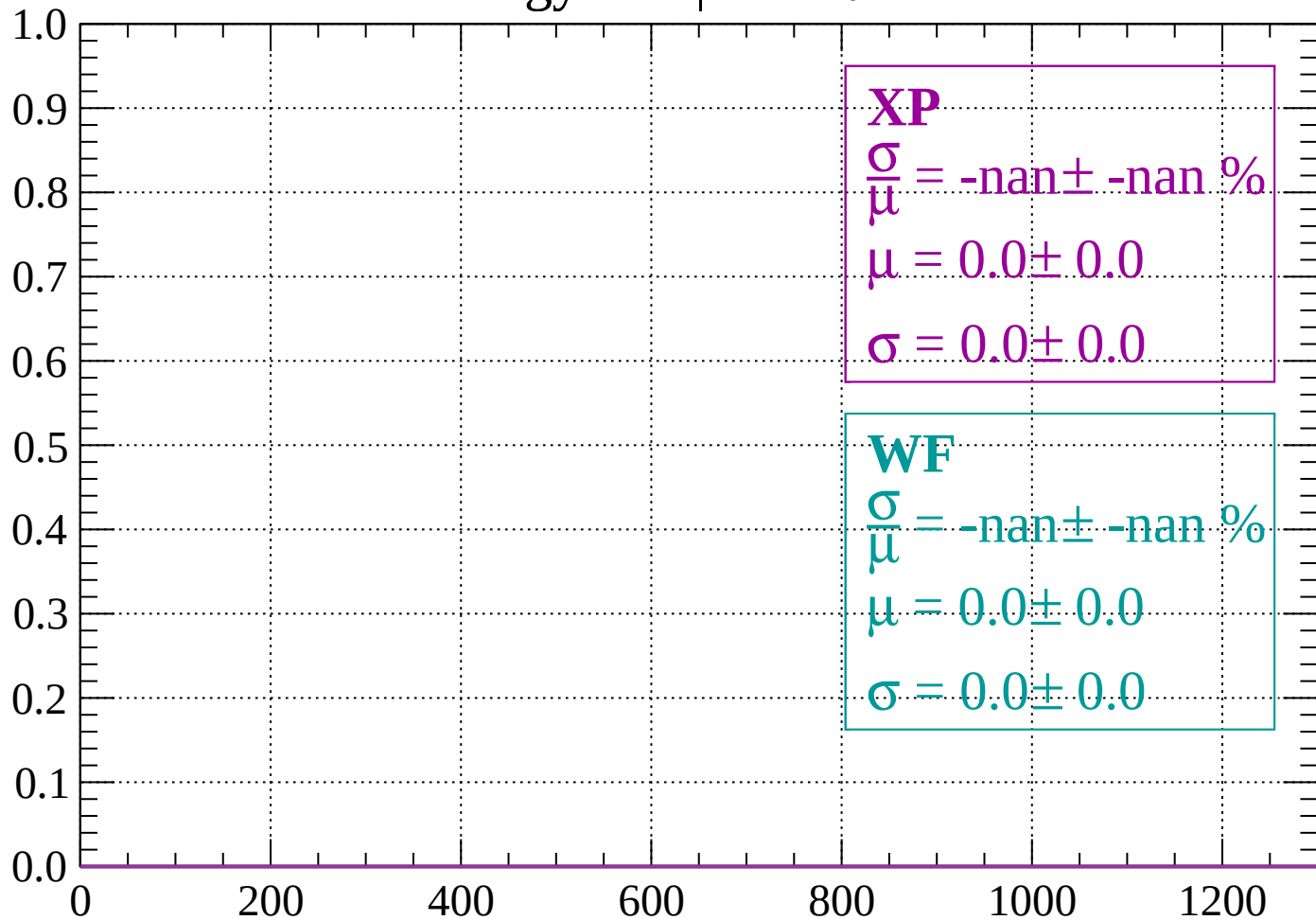
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

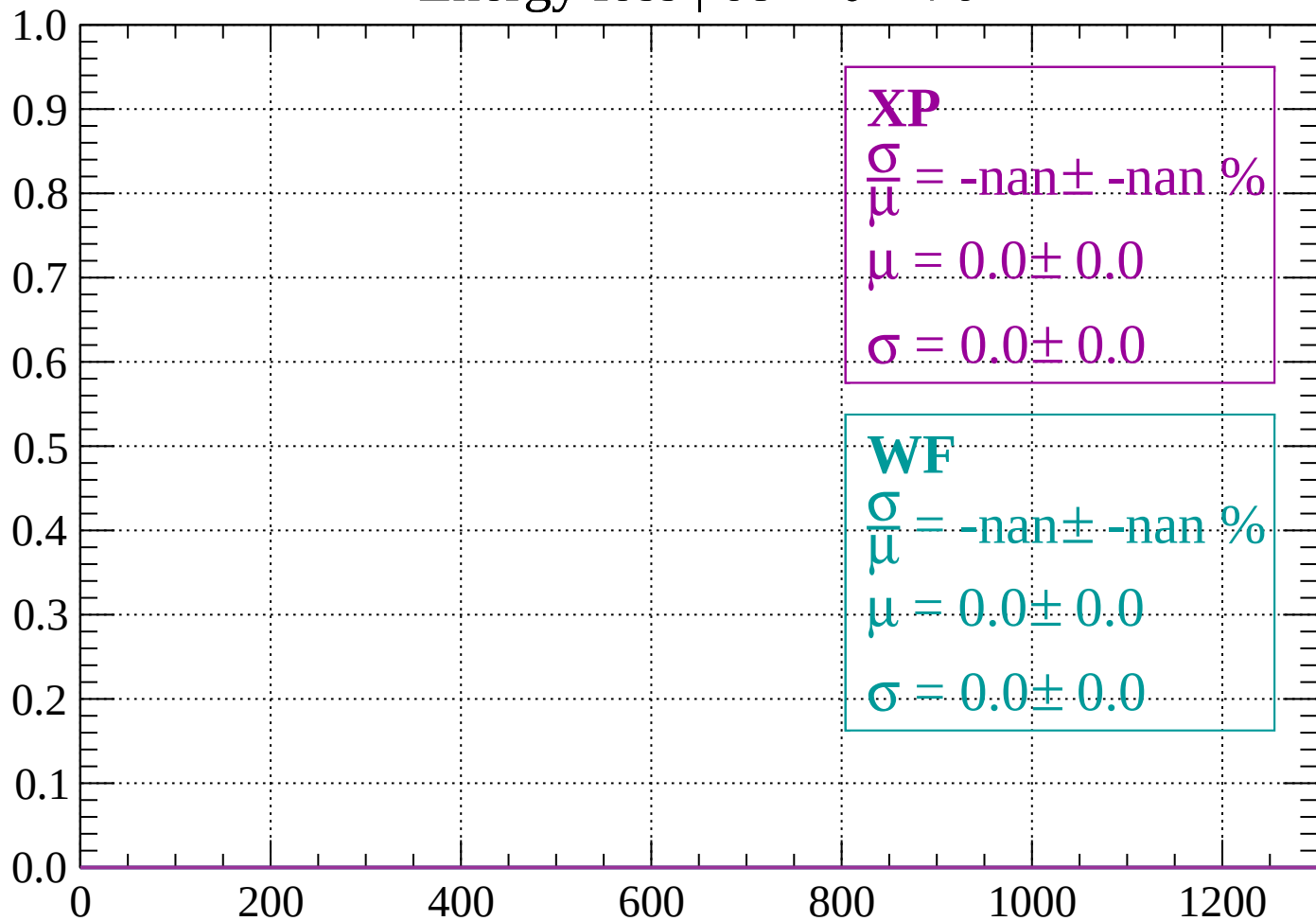
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

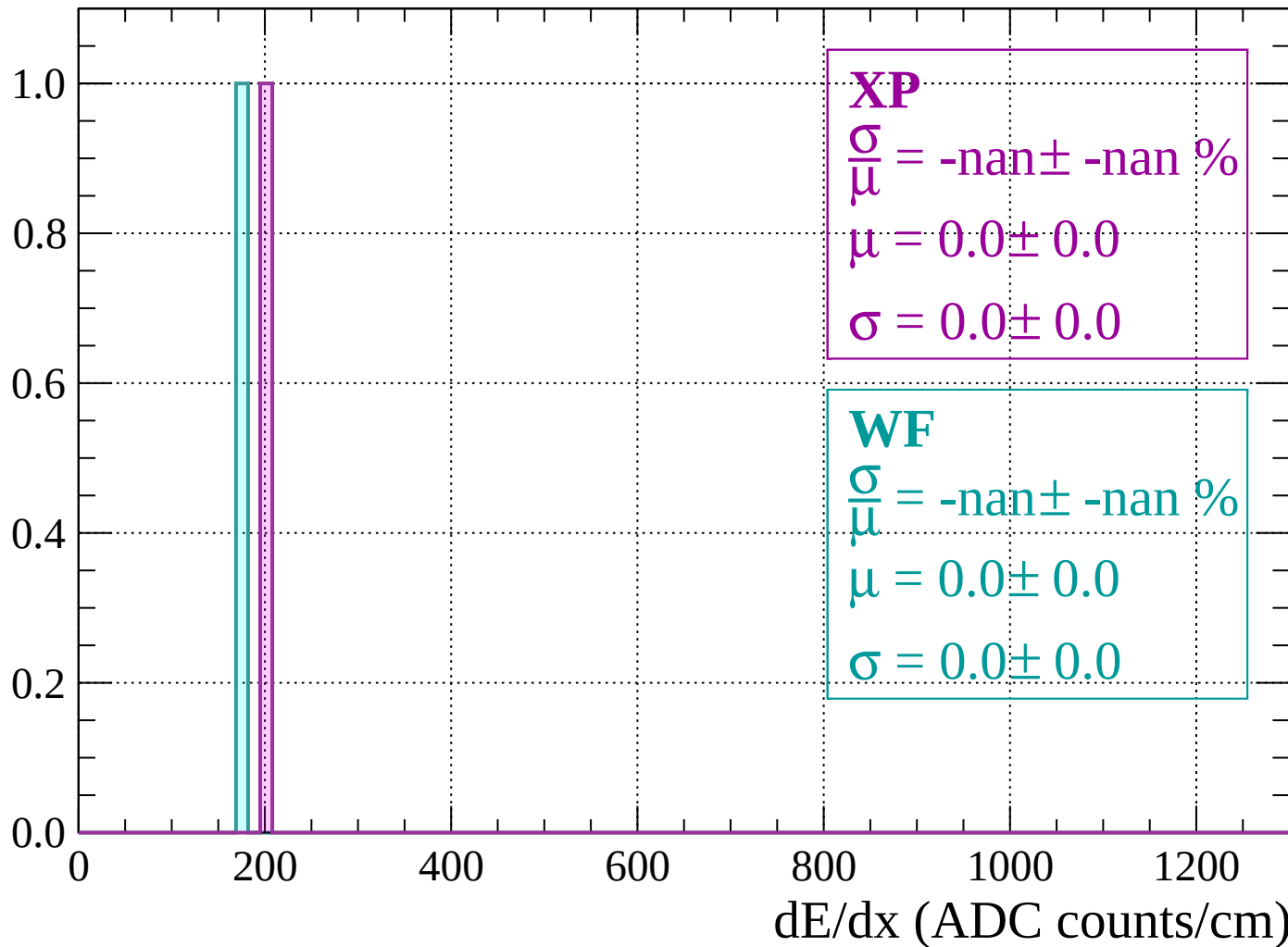
Count



dE/dx (ADC counts/cm)

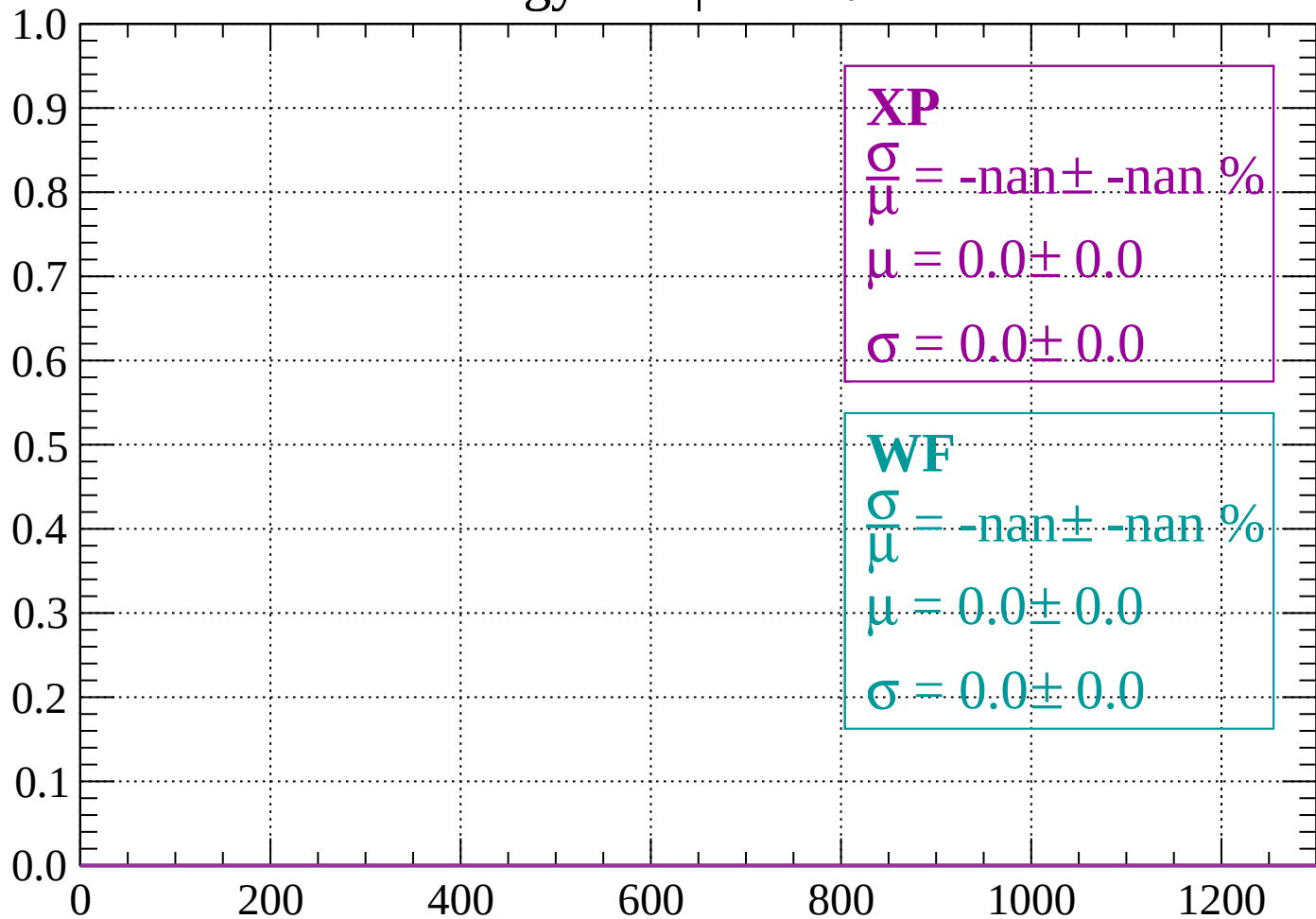
Energy loss | $70 < \theta < 72$

Count



Energy loss | $72 < \theta < 74$

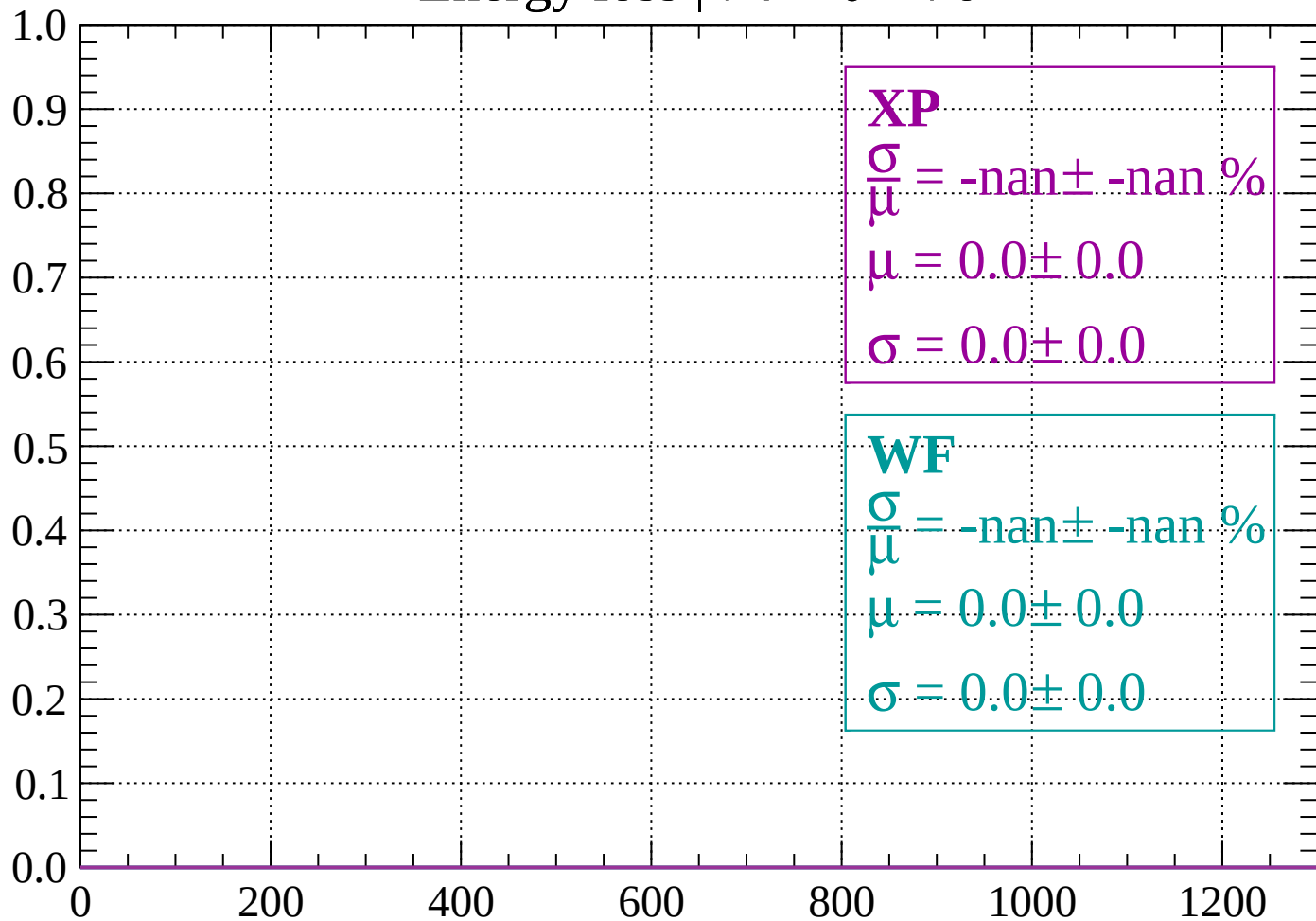
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

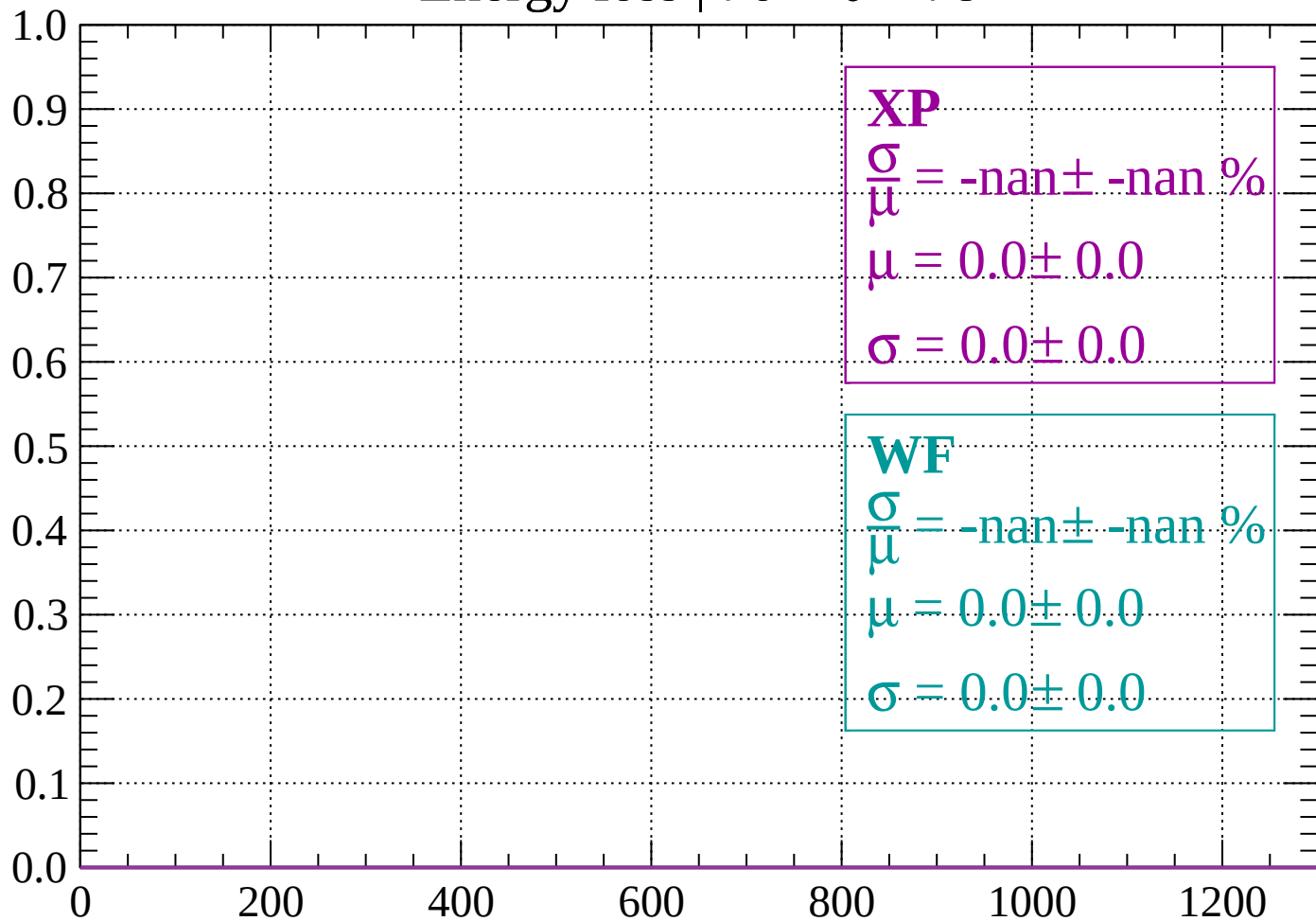
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

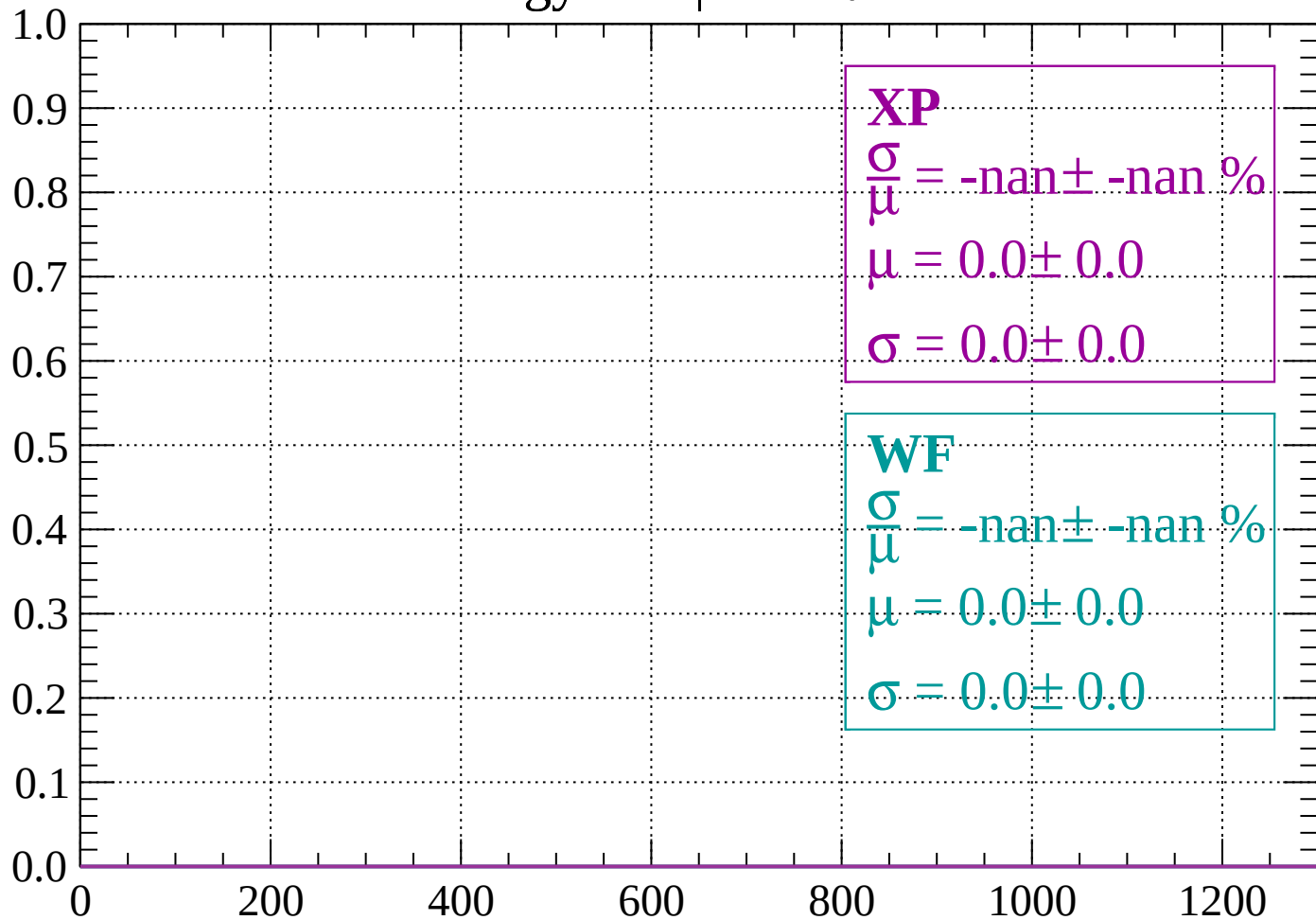
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

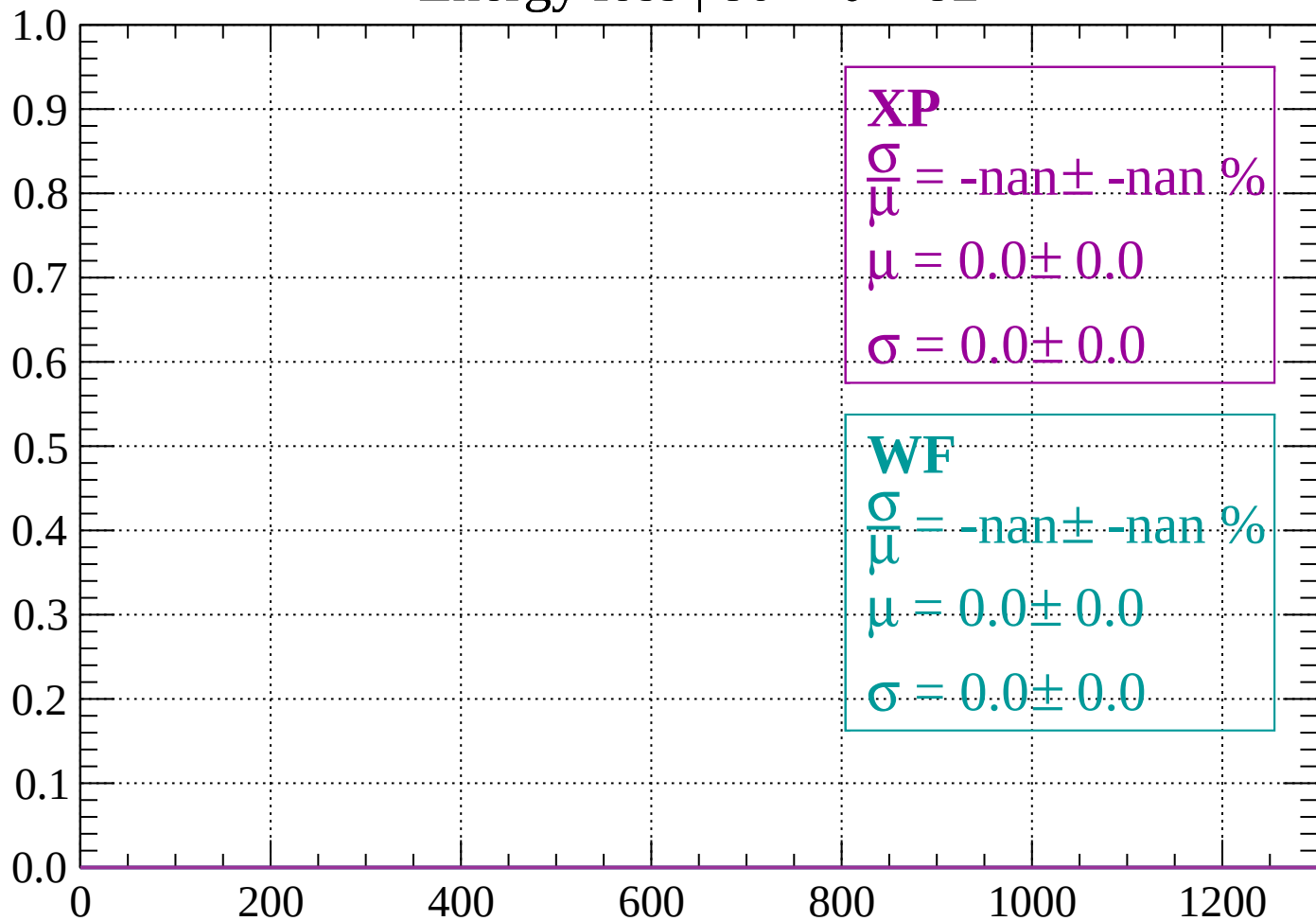
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

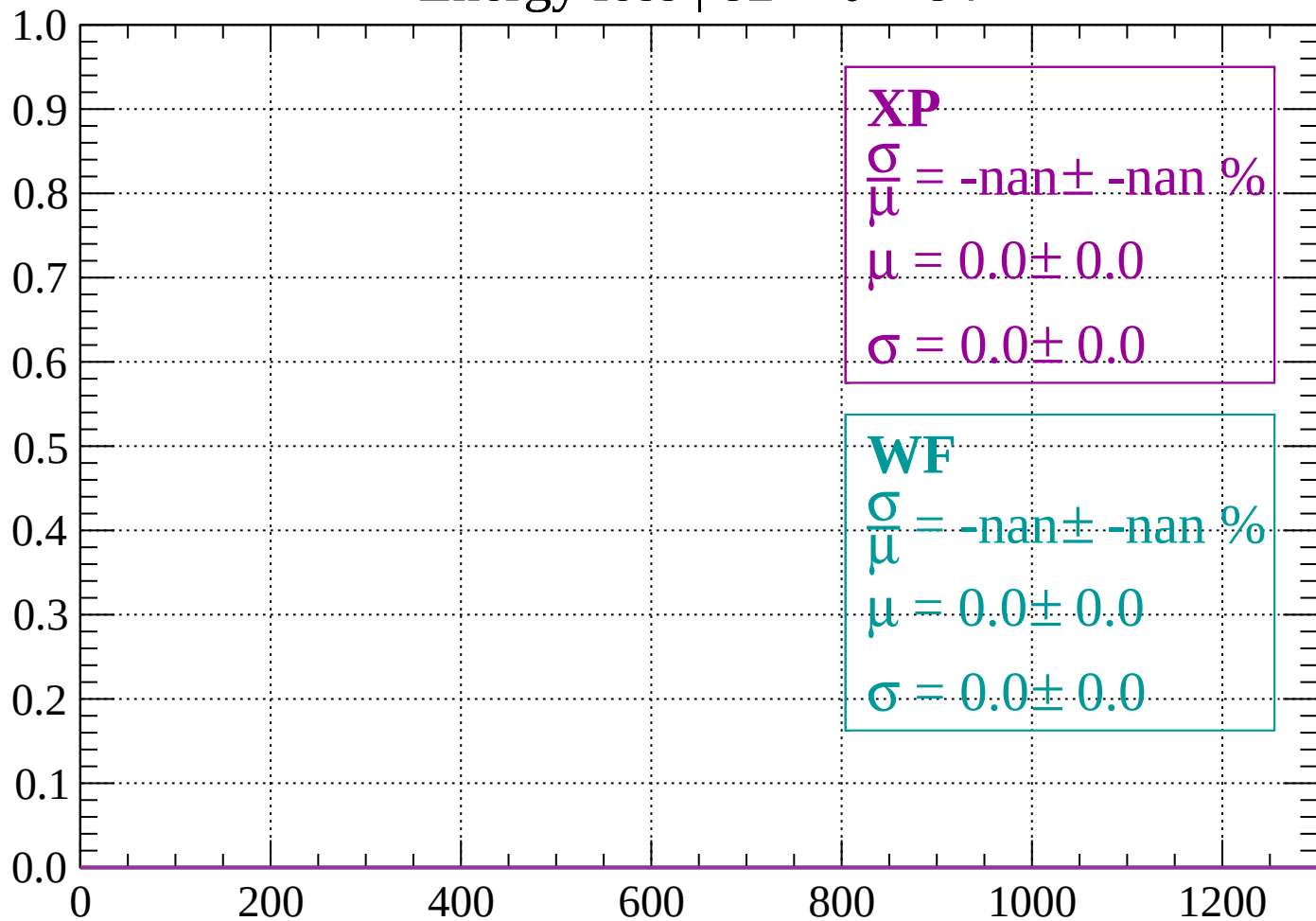
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

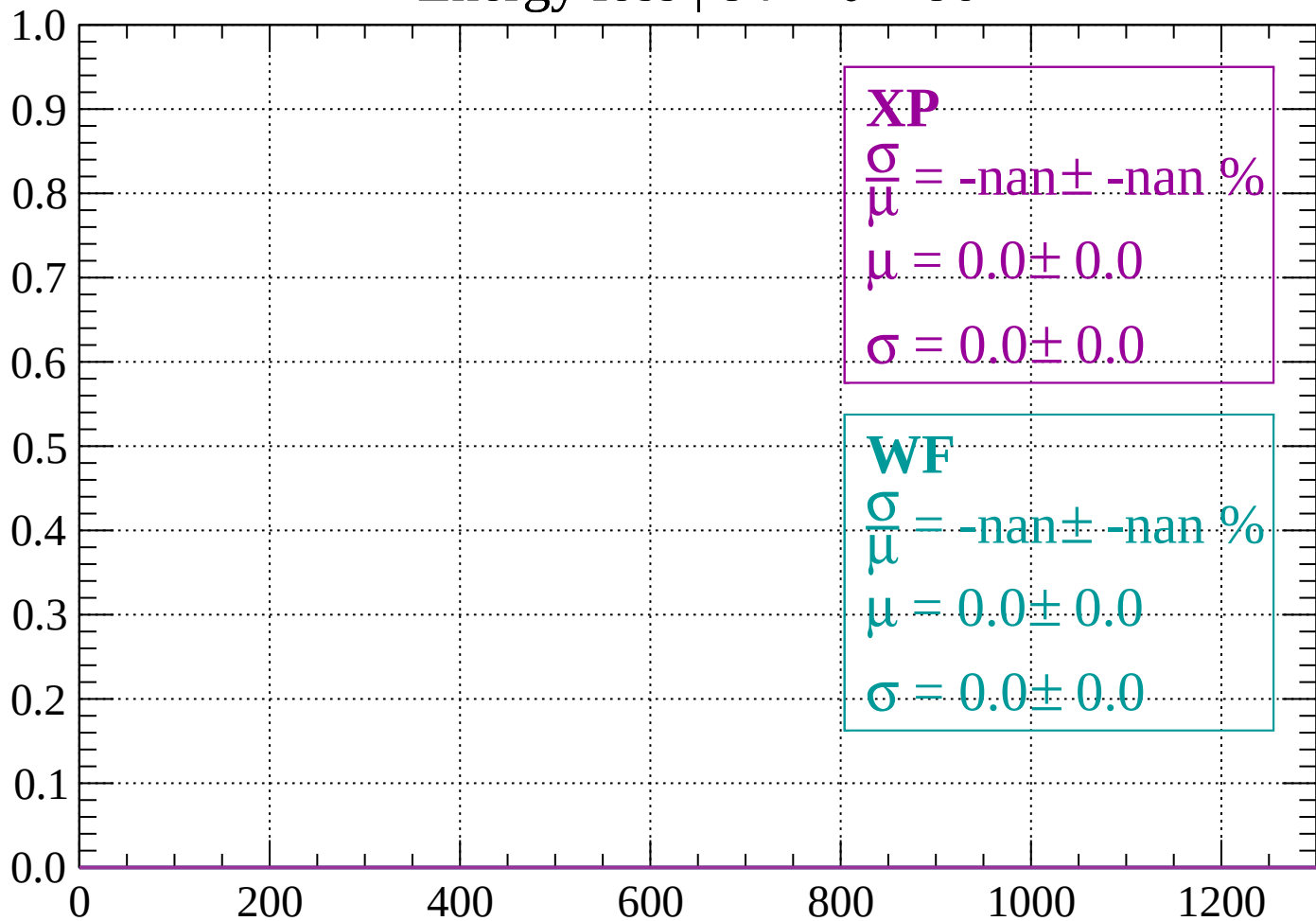
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

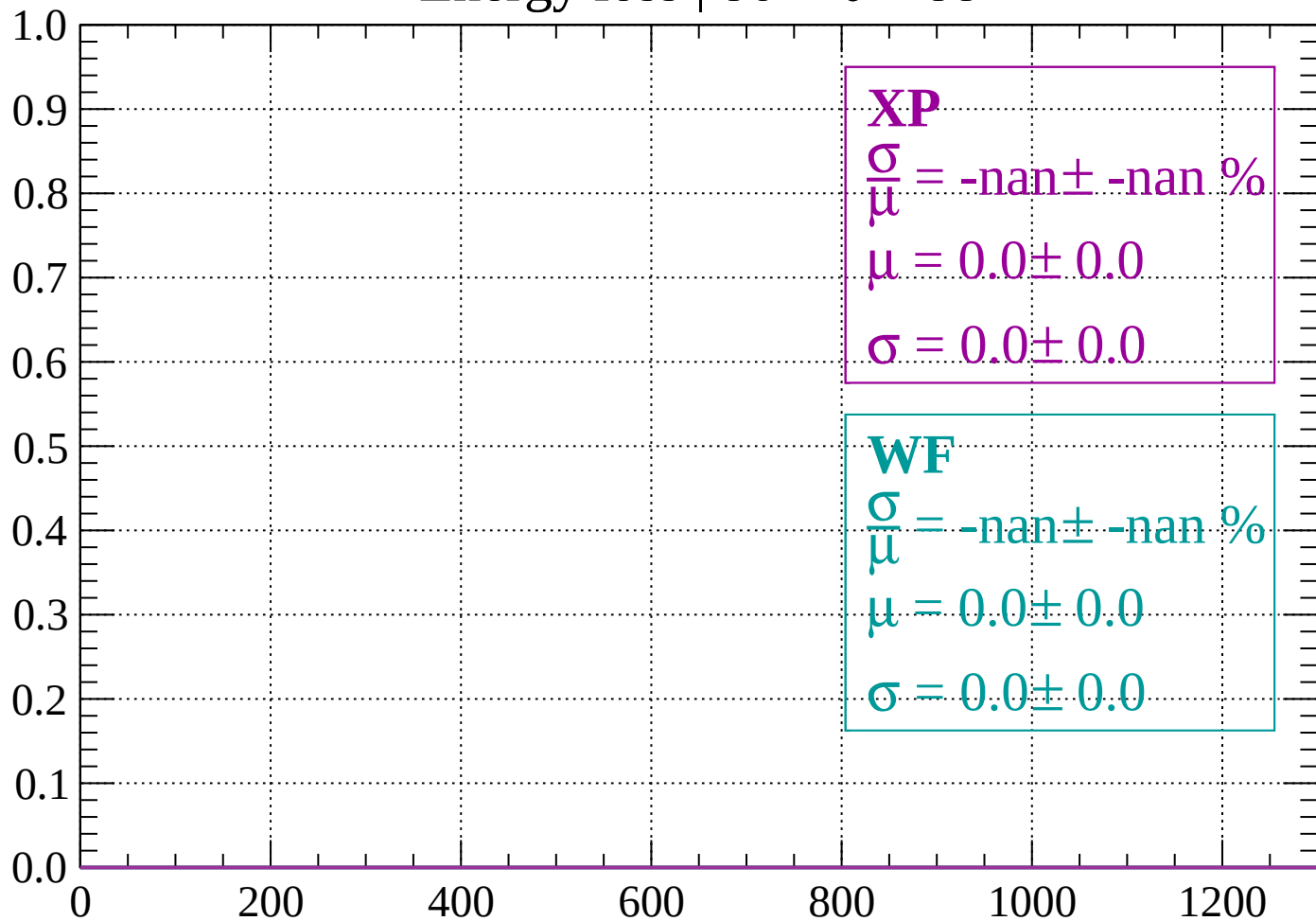
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

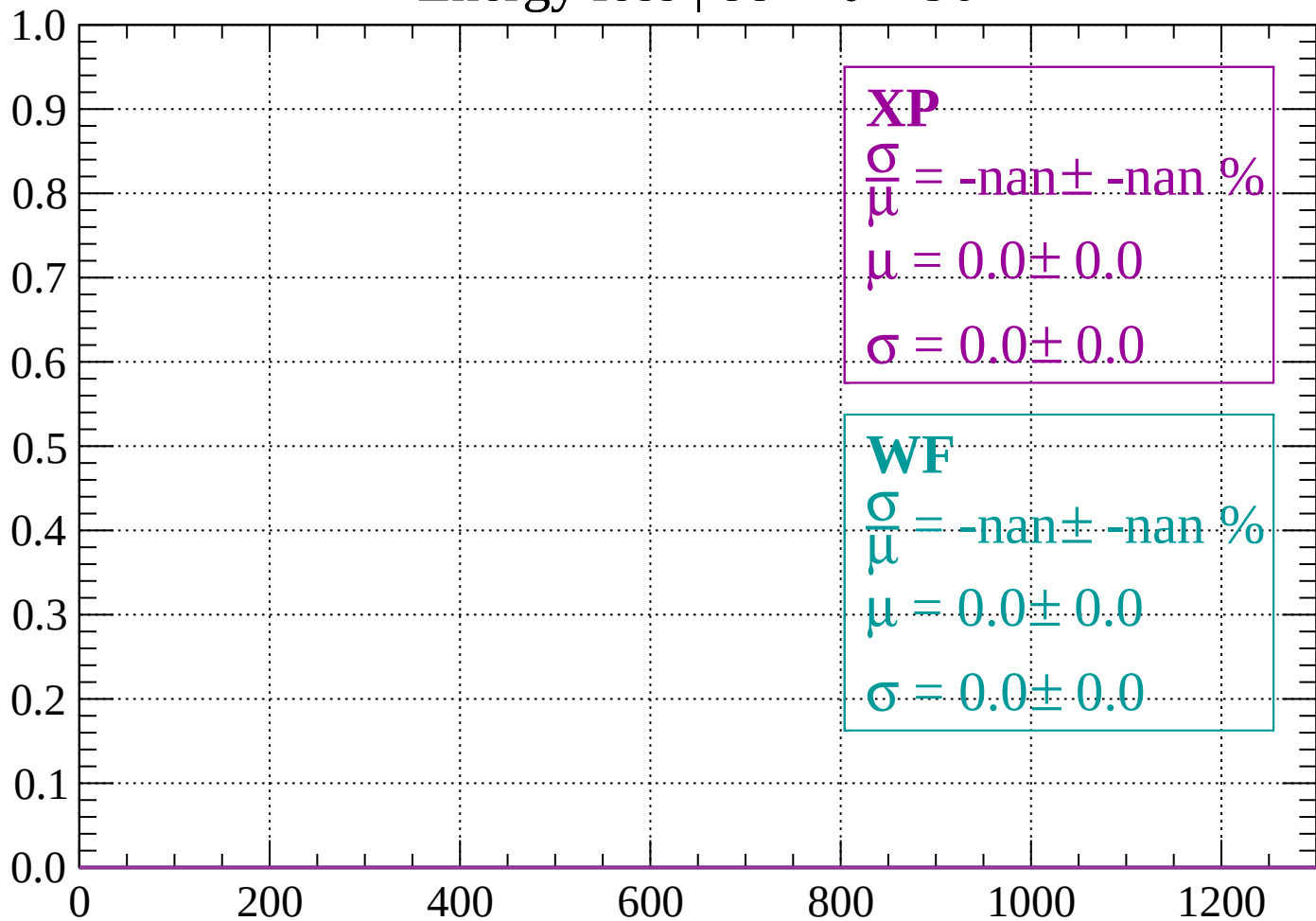
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

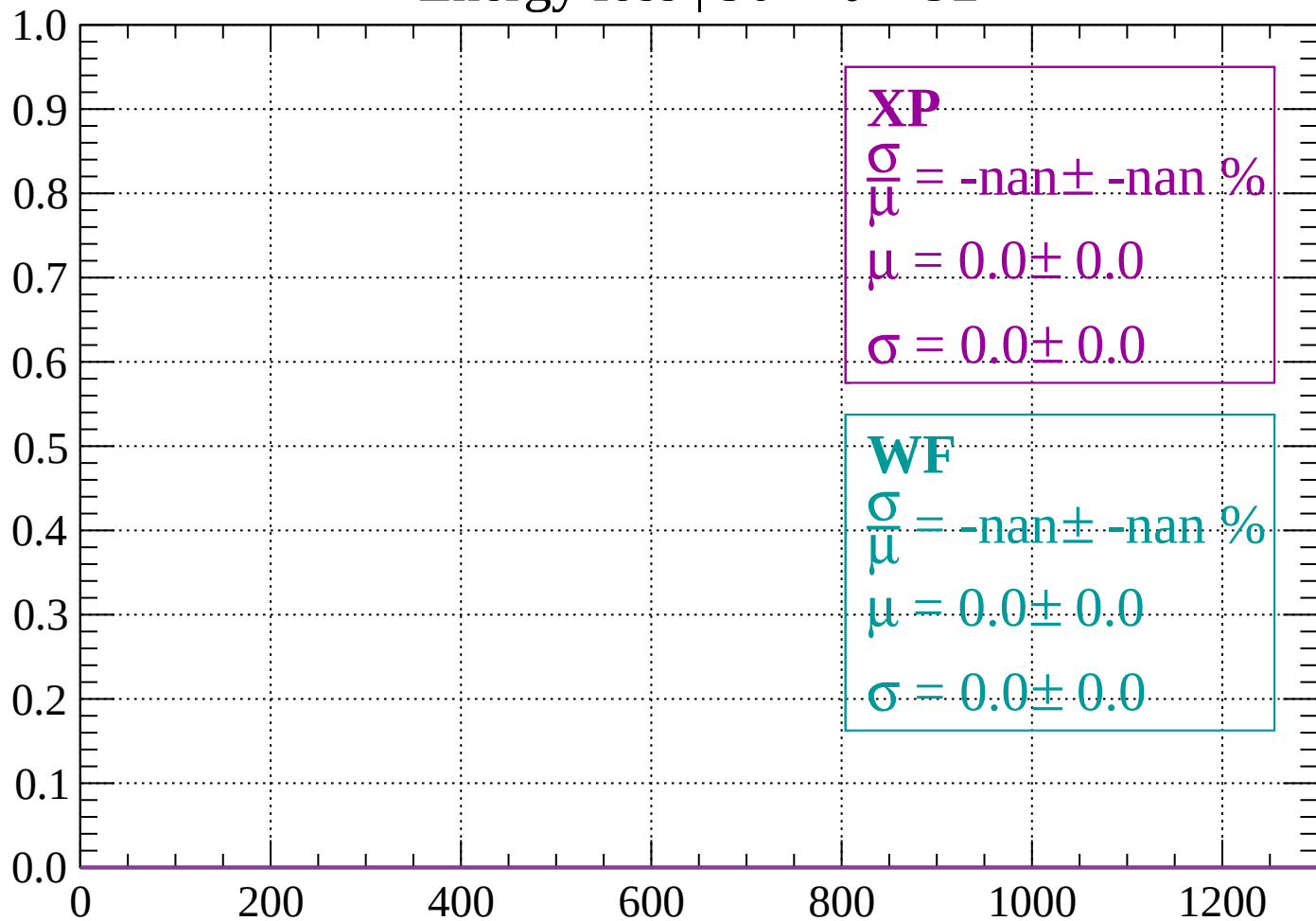
Count



dE/dx (ADC counts/cm)

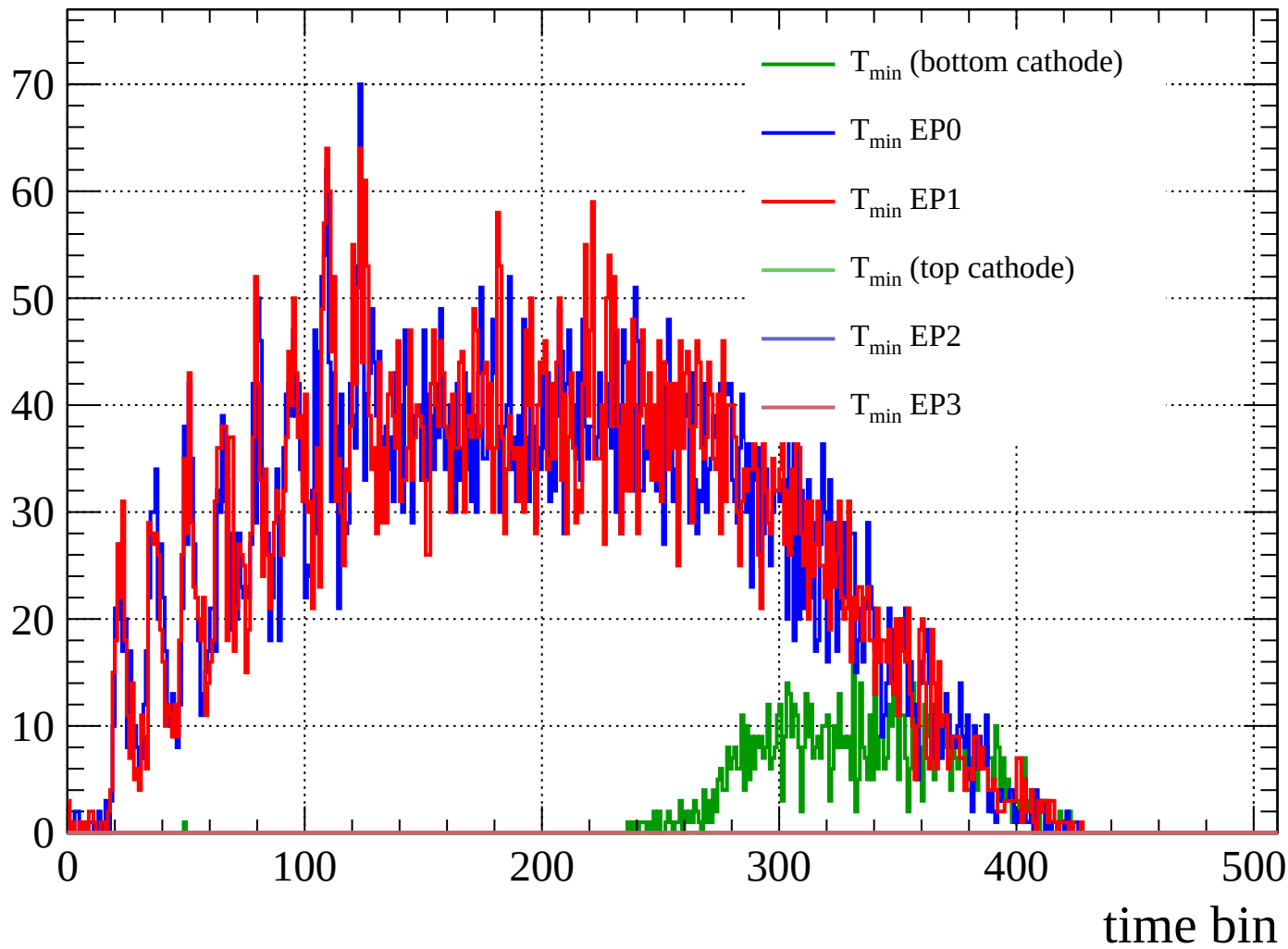
Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

Count



Count

